

BAGELS & CO.

799 W. GAINES STREET,
SUITE 121,
TALLAHASSEE FL 32310

ABBREVIATIONS

ACP. ADJ. AL.-ALUM. ALT. A.P. APPX. ARCH. BD. BLDG. BLK. BLKG. BM. BOT. CAB. CEM. CKT. CLG. CMJ. COL. CONC. CONST. CONT. CPT. CTR. DET. DIA. DIFF. DIM. DWG(S). EA. E.I.F.S.	ACOUSTICAL CEILING ADJACENT ALUMINUM ALTERNATE ACCESS PANEL APPROXIMATE ARCHITECTURAL BOARD BUILDING BLOCK BLOCKING BEAM BOTTOM CABINET CEMENT CIRCUIT CEILING CLEAR CONCRETE MASONRY UNIT COP. H.M. HR. HT. H.V.A.C. J.H.A. LAM. LAV. LL. LT. L.V.T. MAX. MECH.	ELEC. EM. EQ. EQUIP. EWC. EXH. EXIST. EXTIR. FIN. FIKT. FLR. FLOOR F.R. F.R.P. F.T.R. GA. GALV. GND,GRND GRP. H.M. HR. HT. H.V.A.C. J.H.A. LAM. LAV. LL. LT. L.V.T. MAX. MECH.	ELECTRIC, ELECTRICAL EMERGENCY EQUIPMENT ELECTRIC WATER COOLER EXHAUST EXISTING EXTINGUISHER FINISH FLOOR FIRE RATED/RETARDANT FIBERGLASS REINFORCED POLYESTER PANEL FIRE RETARDANT TREATED GAUGE GALVANIZED GROUND GYPSUM HOLLOW METAL HOUR HEIGHT HEATING, VENTILATING, AIR CONDITIONING JURISDICTION HAVING AUTHORITY LAMINATE LAVATORY LANDLORD LIGHT LUXURY VINYL TILE MAXIMUM MECHANICAL	MTL. MFR. MIN. MIR. MISC. MTD. MTG. N.I.C. NOM. N.T.S. O.C. O.D. OPP. PLYWD. PNL. PROP. P.T. PWR. QTY. R. REQ. R.O. S.C. SCHED. SECT. SQ.FT. SIM. SPEC. SQ. STD. STL. STRUCT. SYM.	METAL MANUFACTURER MINIMUM MIRROR MISCELLANEOUS MOUNTED MOUNTING NOT IN CONTRACT NOMINAL NOT TO SCALE ON CENTER OUTSIDE DIAMETER OPPOSITE PLYWOOD PANEL PROPERTY PRESSURE TREATED POWER QUANTITY RADIUS REQUIRED ROUGH OPENING SOLID CORE SCHEDULE SECTION SQUARE FEET SIMILAR SPECIFICATION SQUARE STANDARD STEEL STRUCTURAL SYMMETRICAL	TEL. TELCO. TWO. TYP. U.O.N. VCT. W/ WO. WITHOUT WP. W/O W.R. WT.	TELEPHONE TELE-COMMUNICATIONS T-MOBILE TYPICAL UNLESS OTHERWISE NOTED VINYL COMPOSITION WITH WOOD WITHOUT WATERPROOF WATER RESISTANT WEIGHT
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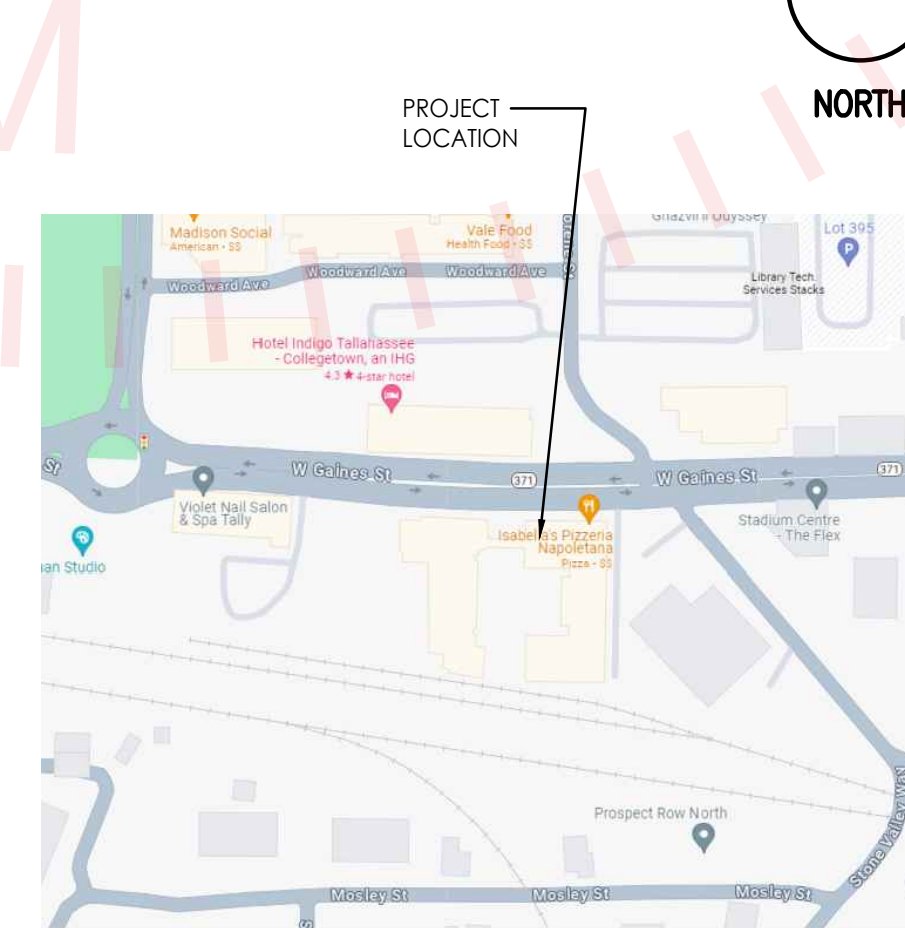
SYMBOLOLOGY / LEGEND

	ELEVATION NUMBER		NOTE NUMBER
	SHEET NUMBER		DOOR NUMBER REF SHEET A6.1
	DETAIL, SECTION OR PLAN NUMBER		REVISION NUMBER
	WALL SECTION NUMBER, BUILDING SECTION LETTER		INTERIOR FINISH NUMBER
	SHEET NUMBER		WALL TYPES
	DETAIL NUMBER		ROOM NAME
	SHEET NUMBER		ROOM NUMBER
	EXISTING COLUMN GRID NUMBER OR LETTER		FURNITURE/EQUIPMENT NUMBER RESTROOM ACCESSORY
			116'-0" TRUSS BRG. ELEV. DATUM POINT

VICINITY MAP



LOCATION MAP



GOVERNMENTAL AGENCIES

DEPT.: BUILDING DEPARTMENT
NAME: CITY OF TALLAHASSEE BUILDING INSPECTION DIVISION
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TALLAHASSEE, FL 32301
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PHONE: (850)891-7001 | (850)891-7099
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DEPT: FIRE MARSHAL
NAME: CITY OF TALLAHASSEE FIRE DEPARTMENT
ADDRESS: 911 EASTERWOOD DRIVE,
TALLAHASSEE, FL 32311
CONTACT: MONA PEARSON
PHONE: (850)891-7179
EMAIL: mona.pearson@talgov.com

DEPT: PLUMBING DEPARTMENT
NAME: CITY OF TALLAHASSEE BUILDING INSPECTION DIVISION
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CONTACT: ANN RANDOLPH - CHIEF PLUMBING INSPECTOR
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DEPT: HEALTH DEPARTMENT
NAME: FLORIDA DEPARTMENT OF AGRICULTURAL (FDOA)
ADDRESS: 3125 CONNER BOULEVARD, SUITE B
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CONTACT: LYDIA GONZALEZ
PHONE: (850)245-5520
EMAIL: FOODSAFETY@FDACS.GOV

OWNER & CONSULTANTS

OWNER
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1001 NORTH 2ND STREET
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ARCHITECT

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MECHANICAL/ PLUMBING ENGINEER

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ELECTRICAL ENGINEER

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GENERAL NOTES

- A. THE DRAWINGS ARE NOT A COMPLETE SET OF INSTRUCTIONS ON HOW TO CONSTRUCT THE PROJECT AND ARE ONLY INTENDED TO CONVEY THE DETAILS REQUIRED TO COMMUNICATE THE DESIGN INTENT FOR THE PROJECT. THE CONTRACTOR SHALL ACCOUNT FOR THE VARIABLES INVOLVED WITH THE CONSTRUCTION PROCESS (TOLERANCES, PRODUCT OPTIONS, LOCAL TRADE CUSTOMS, AND MANUFACTURERS PROPRIETARY REQUIREMENTS, ETC.) TO PROVIDE A COMPLETED PROJECT THAT CONFORMS TO THE DESIGN INTENT INFERRED BY THE DRAWINGS.
- B. CONTRACTOR SHALL VERIFY ALL DIMENSIONS AND EXISTING CONDITIONS PRIOR TO STARTING OF WORK AND SHALL NOTIFY, IN WRITING, OWNER OF ANY DISCREPANCIES.
- C. DO NOT SCALE DRAWINGS. WRITTEN DIMENSIONS TAKE PRECEDENCE OVER DRAWING SCALE.
- D. ALL DETAILS SHALL BE CONSIDERED TYPICAL AT SIMILAR CONDITIONS. DETAILS MARKED "NOT USED" INDICATES ENTIRE DETAIL NOT USED AND IS TO BE OMITTED.
- E. CONTRACTOR AND EACH SUBCONTRACTOR SHALL BE RESPONSIBLE FOR HAVING COMPLETE KNOWLEDGE OF ALL CONSTRUCTION DOCUMENTS AND THE RELEVANCE TO THE WORK. FAILURE TO BE ACQUAINTED WITH THIS KNOWLEDGE DOES NOT RELIEVE RESPONSIBILITY FOR PERFORMING ALL WORK PROPERLY. ADDITIONAL COMPENSATION SHALL NOT BE ALLOWED DUE TO THE FAILURE TO BECOME FAMILIAR WITH THE ENTIRE CONSTRUCTION DOCUMENT PACKAGE.
- F. THE FIRE ALARM SYSTEM AND SPRINKLER SYSTEM ARE EXISTING IN THIS PROJECT. THE CONTRACTOR SHALL MODIFY THE SYSTEMS AS NECESSARY TO COMPLY WITH LOCAL ORDINANCES, NFPA 72, & NFPA 13.
- G. FIRE SPRINKLER SYSTEM WORK AND FIRE ALARM SYSTEM WORK ARE DESIGN BUILD BY THE CONTRACTOR. CONTRACTOR SHALL SUBMIT FIRE SPRINKLER & FIRE ALARM DRAWINGS TO THE JURISDICTION (AND LANDLORD AS REQUIRED) AND OBTAIN APPROVAL PRIOR TO BEGINNING ANY WORK ON THE FIRE SPRINKLER OR ALARM SYSTEM. THE FIRE SPRINKLER AND ALARM WORK SHALL BE PERFORMED UNDER A SEPARATE PERMIT WHERE APPLICABLE.
- H. CONTRACTOR SHALL INSPECT ALL EXISTING FIRE PROOFING OF STRUCTURAL ELEMENTS, DEMISING WALLS, FLOOR/CEILING ASSEMBLIES, AND OTHER ELEMENTS WHICH ARE REQUIRED TO BE FIRE PROTECTED BY GOVERNING CODES. CONTRACTOR SHALL MAINTAIN, PATCH, AND REPAIR ALL DAMAGED OR REMOVED FIREPROOFING AND SHALL REPLACE ALL MISSING FIREPROOFING TO MAINTAIN ALL FIRE RATINGS.
- I. OWNER OR REP HAS FILED FOR AND OBTAINED APPROVAL OF THE BUILDING PERMIT AND HEALTH DEPARTMENT. CONTRACTOR SHALL BE RESPONSIBLE FOR COMPLETING ANY OUTSTANDING BUILDING PERMIT ITEMS AND PICKING UP THE PERMIT. CONTRACTOR SHALL BE RESPONSIBLE FOR APPLYING FOR AND OBTAINING ALL TRADE PERMITS AND OTHER PERMITS AS MAY BE REQUIRED BY THE JURISDICTIONS HAVING AUTHORITY OVER THE PROJECT.
- J. CONTRACTOR SHALL BE RESPONSIBLE FOR SUBMITTING ANY REVISIONS OF THE APPROVED PERMIT DOCUMENTS AND PROCESSING THE APPROVAL OF THE REVISED DOCUMENTS WITH THE JURISDICTIONS HAVING AUTHORITY.
- K. CONTRACTOR SHALL VERIFY THAT ALL EXISTING DEMISING WALLS EXTEND TO THE STRUCTURE ABOVE AND SHALL NOTIFY THE ARCHITECT OF ANY DEFICITS IDENTIFIED.
- L. CONTRACTOR SHALL PATCH AND REPAIR ALL EXISTING SUBSTRATES TO RECEIVE NEW FINISHES AND ALL EXISTING SURFACES AND FINISHES AS NECESSARY FOR A COMPLETE AND PROPER INSTALLATION.
- M. CONTRACTOR SHALL ENGAGE A STRUCTURAL ENGINEER TO REVIEW, DESIGN, AND SEAL ALL CHANGES NECESSARY TO THE BUILDING STRUCTURE FOR THE INSTALLATION OF OR REVISION OF ALL CONTRACTOR INSTALLED MECHANICAL UNITS OR OTHER CONTRACTOR INSTALLED ELEMENTS SUPPORTED BY OR ANCHORED TO THE EXISTING STRUCTURE. THE ENGINEER SHALL SUBSTANTIATE THE DESIGN MODIFICATIONS NECESSARY TO MAINTAIN THE INTEGRITY OF THE EXISTING BUILDING STRUCTURE.
- N. CONTRACTOR SHALL BE RESPONSIBLE FOR THE PROPER, SAFE AND ADEQUATE DESIGN FOR THE INSTALLATION AND REMOVAL OF ALL SHORING AND BRACING REQUIRED TO CARRY OUT THE WORK.
- O. CONTRACTOR SHALL MAINTAIN WORK AREA IN A NEAT AND ORDERLY CONDITION. PROVIDE TRASH CONTAINERS IN THE WORK AREA AND EMPTY ON A DAILY BASIS. PROVIDE A GENERAL CLEANUP IN ACCORDANCE WITH LANDLORD'S STANDARDS AT THE END OF EACH WORK DAY.

CODES

BUILDING CODE: FLORIDA BUILDING CODE - 8TH EDITION (2023)
PLUMBING CODE: FLORIDA PLUMBING CODE - 8TH EDITION (2023)
MECHANICAL CODE: FLORIDA MECHANICAL CODE - 8TH EDITION (2023)
ELECTRICAL CODE: NATIONAL ELECTRIC CODE - 2020 EDITION
ACCESSIBILITY CODE: FLORIDA ACCESSIBILITY CODE - 8TH EDITION (2023)
ENERGY CODE: FLORIDA ENERGY CONSERVATION CODE - 8TH EDITION (2023)
FIRE CODE: FLORIDA FIRE PREVENTION CODE - 8TH EDITION (2023)

CODE ANALYSIS:

OCCUPANT LOAD CALCULATIONS:

(FBC 1004.5)
KITCHEN 929 SF @ 200G SF/PERSON = 5 PERSONS
STORAGE 110 SF @ 300G SF/PERSON = 1 PERSONS
UN-CONCENTRATED SEATING 163 SF @ 15N SF/PERSON = 11 PERSONS
CONCENTRATED SEATING 37 SF @ 7N SF/PERSON = 5 PERSONS
STANDING 123 SF @ 15N SF/PERSON = 9 PERSONS

TOTAL ALLOWABLE OCCUPANT LOAD = 31 PERSONS

(FFPA 101.7.3.1.2)

KITCHEN 929 SF @ 100G SF/PERSON = 9 PERSONS
STORAGE 110 SF @ 300G SF/PERSON = 1 PERSONS
UN-CONCENTRATED SEATING 163 SF @ 15N SF/PERSON = 11 PERSONS
CONCENTRATED SEATING 37 SF @ 7N SF/PERSON = 5 PERSONS
STANDING 123 SF @ 3N SF/PERSON = 67 PERSONS

TOTAL ALLOWABLE OCCUPANT LOAD = 67 PERSONS

SEATING CALCULATIONS:
MAIN DINING: 9 SEATS
BAR DINING: 6 SEATS
TOTAL PROPOSED: 15 SEATS

ACCESSIBLE SEATS REQUIRED 5% OF 15 SEATS = 1 TOTAL
ACCESSIBLE SEATS PROVIDED 2 TOTAL

TOILET/PLUMBING FIXTURES (FPC CHP 4, TABLE 403.1):

OCCUPANT LOAD @ 67
33.5 MALE/ 33.5 FEMALE
REQUIRED PROVIDED
E M E M
WATER CLOSET @ 1/75 1 1 1 1
LAVATORY @ 1/200 1 1 1 1

INTERPLAN
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CA 8660

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SEAL:

THIS DOCUMENT IS NOT
FOR REGULATORY
APPROVAL, PERMITTING,
OR CONSTRUCTION.

NO DATE REMARKS
REVISIONS

BAGELS
& CO.

BAGEL & CO
TALLAHASSEE

799 W. GAINES STREET
SUITE 121
TALLAHASSEE, FL 32310

PROJECT NO: 2023.0719
DATE: 04/12/2024

G1.1
COVER SHEET

CHECKED: DC DRAWN: WC

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RESPONSIBILITY MATRIX										
ITEM	OWNER		LANDLORD		GENERAL CONTRACTOR		EXISTING	NOT APPLICABLE	REQUIRES SUBMITTAL OR SHOP DRAWINGS	REMARKS
	FURNISH	INSTALL	FURNISH	INSTALL	FURNISH	INSTALL				
DIVISION 01000 GENERAL CONDITIONS										
BUILDER'S RISK - PROPERTY INSURANCE					■					
CONSTRUCTION FACILITIES & TEMPORARY CONTROLS					■	■				
TEMPORARY POWER AND WATER					■					ELECTRICIAN AND PLUMBER
PROGRESS & FINAL CLEANUP (INCL'D DUMPSTER & TRASH REMOVAL)					■	■				
SUB-CONTRACTOR PERFORMANCE AND MATERIAL BONDS					■					
PERMITS & FEES					■					
CERTIFICATE OF OCCUPANCY					■	■				IF REQUIRED BY LOCAL JURISDICTION
DIVISION 02000 SITE CONSTRUCTION										
SELECTIVE DEMOLITION (IF APPLICABLE)					■	■				
SAWCUT SLAB FOR NEW IN-FLOOR PLUMBING FIXTURES					■	■				
DIVISION 03000 CONCRETE										
REINFORCED CONCRETE SLAB / SLAB INFILL (IF APPLICABLE)					■	■				
DIVISION 04000 MASONRY										
STRUCTURAL ELEMENTS & CMU VENEER					■	■				
DIVISION 05000 METALS										
STRUCTURAL ELEMENTS (BEAMS, COLUMNS, ETC.)							■			
PATIO RAILING								■		
STAIR / RAMP HANDRAILS								■		
METAL STUDS					■	■				
DIVISION 06000 WOOD & COMPOSITES (INCL'D FINISH CARPENTRY & CASEWORK)										
STRUCTURAL ELEMENTS					■	■			■	
WOOD FRAMING					■	■				
RUNNING TRIM AND MOULDING					■	■				
CASEWORK & SOLID SURFACE COUNTERS					■	■			■	
MILLWORK					■	■			■	
INTERIOR / EXTERIOR WOOD BLOCKING FOR ALL ACCESSORIES					■	■				
DECOR, WALL MOUNTED PICTURES	■					■				
DECOR, WALL MOUNTED MIRROR	■					■				
DECOR, WALL / DOOR MOUNTED SIGNS	■					■				
DIVISION 07000 THERMAL & MOISTURE PROTECTION										
ROOFING								■		
INSULATION					■	■				
EIFS								■		
CAULKING AND SEALANTS					■	■				
STAINLESS STEEL WALL PANELS & CORNER GUARDS					■	■				
DIVISION 08000 DOORS & WINDOWS										
ALUMINUM STOREFRONT SYSTEM (FOR CHANGES AT EXISTING STOREFRONT)									■	
EXTERIOR DOORS, FRAMES & HARDWARE									■	
INTERIOR DOORS, FRAMES & HARDWARE					■	■			■	
EXTERIOR WINDOWS (IF APPLICABLE)								■		
DOOR, HM FRAME & HARDWARE					■	■			■	
DIVISION 09000 FINISHES										
GYPSUM BOARD, CORNER & J-BEADS, TAPE & MUD (INCL. WALLS & CEILINGS)					■	■				
FLOOR TILES					■	■			■	PROVIDE SAMPLES FOR APPROVAL.
WALL TILE					■	■			■	PROVIDE SAMPLES FOR APPROVAL.
LAY-IN SUSPENDED CEILING GRID SYSTEM					■	■			■	
CEILING TILES					■	■			■	
PAINTING / STAINING					■	■			■	
ALL ITEMS ATTACHED TO THE CEILING TO BE PAINTED					■	■				
FRP WALL PANELS, TRIM, ADHESIVES & SEALANT					■	■			■	
CONCRETE STAINS & EPOXYS (IF APPLICABLE)					■	■			■	
FLOOR MATS	■	■								
DIVISION 10800 TOILET ACCESSORIES										
SOAP DISPENSERS	■					■			■	
HAND TOWEL DISPENSERS	■								■	
TOILET PAPER HOLDERS	■					■			■	
NAPKIN HOLDERS								■		
BABY CHANGING STATIONS								■		
GRAB BARS					■	■			■	
TOILET PARTITIONS								■		
RESTROOM MIRRORS					■	■			■	
DIVISION 10900 SPECIALTIES										
MENU BOARDS					■	■				TO BE PROVIDED WITH MILLWORK PACKAGE
OFFICE SAFE								■		
WINDOW TREATMENTS								■		
POINT OF SALE: CONDUITS AND PULL STRINGS					■	■				ELECTRICIAN; 100% CONDUIT ALL HOME RUNS
CABLING	■	■								
SPEAKERS	■	■								
SIGNAGE: INTERIOR	■				■	■				G.C. TO PROVIDE AND INSTALL ALL SIGNAGE REQUIRED BY CODE.
SIGNAGE: EXTERIOR (SEPARATE PERMIT)	■	■								ELECTRICIAN TO PROVIDE POWER CONNECTION
C02 EQUIPMENT								■		
INTERIOR OR EXTERIOR GREASE TRAP					■	■	■			
FIRE EXTINGUISHERS					■	■				FINAL MOUNTING HEIGHT & QUANTITY BY LOCAL FIRE INSP. - VERIFY W/ OWNER
WALL CAPS & CORNER GUARDS					■	■				STAINLESS STEEL

CONSTRUCTION MATRIX									
ITEM	OWNER		LANDLORD		GENERAL CONTRACTOR		NOT APPLICABLE REQUIRES SUBMITTAL OR SHOP DRAWINGS	REMARKS	
	FURNISH	INSTALL	FURNISH	INSTALL	FURNISH	INSTALL			
DIVISION 10900 SPECIALTIES (CONT'D)									
EXTERIOR AWNINGS							■		
PLANTERS / PLANTS / GRASSES							■		
EXTERIOR LANDSCAPING - PLANTS @ PATIO							■		
ROLL-DOWN ENCLOSURES @ PATIO							■		
DIVISION 11000 EQUIPMENT									
WALK-IN COOLERS					■	■	■	PLUMBER & ELECTRICIAN TO PROVIDE UTILITY CONNECTIONS	
KITCHEN EQUIPMENT UNLESS NOTED OTHERWISE ON EQUIPMENT PLAN					■	■	■	PLUMBER & ELECTRICIAN TO PROVIDE UTILITY CONNECTIONS	
OFFICE EQUIPMENT							■	ELECTRICIAN TO PROVIDE UTILITY CONNECTIONS.	
DIVISION 12000 FURNISHINGS									
BOOTHES AND BANQUETTES							■		
TABLES (TOPS AND BASES)	■					■			
CHAIRS	■					■			
HIGH CHAIRS							■		
PATIO FURNITURE AND UMBRELLAS							■		
SHELVING IN OFFICE							■		
OFFICE COPIER							■		
OFFICE FAX MACHINE							■		
OFFICE CHAIRS							■		
OFFICE FILE CABINET							■		
ART WORK							■	COORDINATE LOCATION WITH OWNER. ARTWORK TO BE HUNG W/ LOCKING HARDWARE.	
DIVISION 13000 SPECIAL CONSTRUCTION									
ANSUL SYSTEMS W/ PULLBOXES						■	■	G.C. TO COORDINATE LOCATION W/ OWNER.	
DIVISION 14000 CONVEYING SYSTEMS									
N/A									
DIVISION 15000 MECHANICAL									
KITCHEN HOODS					■	■	■	G.C. AND MECHANICAL SUB - RECEIVE & INSTALL (TYPICAL)	
EXHAUST FANS					■	■	■		
EXHAUST FAN CURBS					■	■	■		
MAKE-UP AIR FANS					■	■	■		
MAKE-UP AIR FAN CURBS					■	■	■		
MAKE-UP AIR & EXHAUST DUCTWORK & FIRE CHASES					■	■		MECHANICAL SUB	
HVAC PACKAGED UNITS							■		
CURBS FOR PACKAGED UNITS							■		
AIR DISTRIBUTION DUCTWORK					■	■		MECHANICAL SUB	
HVAC CONTROL WIRING / CONDUIT & START UP							■	MECHANICAL SUB TO SUPPLY REFRIGERANT	
REFRIGERATION INSTALL, ICE MACH AND WALK-INS					■	■		PLUMBER & ELECT. TO PROVIDE UTILITY CONNECTION /CHASES & CONDENSATE	
PLUMBING SYSTEMS AND FIXTURES					■	■	■		
WATER HEATERS					■	■	■		
WATER SOFTENERS					■	■		PLUMBER TO PROVIDE ALL PIPING AND CONNECTIONS	
WATER FILTERS					■	■		PLUMBER INSTALL	
FIRE PROTECTION SYSTEMS, SPRINKLERS					■	■	■	SUBCONTRACTOR (SUBMITTAL FOR HEADS)	
DIVISION 16000 ELECTRICAL									
TRANSFORMERS							■	PROVIDED & INSTALLED BY POWER COMPANY	
PANELS, BREAKERS, AND SWITCHES						■	■	ELECTRICIAN	
DIMMER PANELS						■	■	ELECTRICIAN INSTALL	
DECORATIVE LIGHTING						■	■	ELECTRICIAN INSTALL	
PENDANT LIGHTS						■	■	ELECTRICIAN INSTALL	
ALL REMAINING LIGHTS						■	■	ELECTRICIAN INSTALL	
LAMPS & EXTRA LAMPS						■	■	ELECTRICIAN INSTALL	
LAMPS FOR COOLER AND FREEZER LIGHT FIXTURES						■	■	ELECTRICIAN	
EMERGENCY LIGHTS AND EXIT SIGNS						■	■		
HOOD INTERCONNECTS AND FAN STARTER MOTORS						■	■		
HVAC CONTROL WIRING						■	■		
FIRE ALARM						■	■		
EXTERIOR SIGNAGE (BACKLIT)							■	TO BE INSTALLED BY SIGN VENDOR. ELECTRICIAN TO PROVIDE POWER TO SIGN.	
EXTERIOR LIGHTING (BUILDING) IF REQUIRED.							■		
SECURITY ALARM	■	■				■	■	ELECTRICIAN TO PROVIDE POWER AS REQUIRED - JBOX & CONDUIT TO ABOVE CLG.	
CONDUIT & PULL STRING FOR PHONES, POS AND AV					■	■		ELECTRICIAN, 100% CONDUIT & HOME RUNS FOR TELEPHONE & P.O.S.	
POS SURGE PROTECTOR	■	■					■		
AV EQUIPMENT	■	■					■		
DIVISION 17000 FINISHES									
DIVISION 18000 ROADS, PAVEMENTS, AND RELATED WORK									
DIVISION 19000 FENCES, GUARDS, AND RAILROADS									
DIVISION 20000 STRUCTURAL STEEL ERECTION									
DIVISION 21000 CAST-IN-PLACE CONCRETE									
DIVISION 22000 PRECAST/CAST-IN-PLACE CONCRETE									
DIVISION 23000 MASONRY									
DIVISION 24000 METAL DECKING									
DIVISION 25000 METAL FABRICATED BUILDINGS									
DIVISION 26000 WOODWORK, MILLWORK, AND JOINERY									
DIVISION 27000 PLASTER, GYPSONUM BOARD, AND ACoustIC CEILING									
DIVISION 28000 PAINTS, COATINGS, AND RELATED MATERIALS									
DIVISION 29000 GLASS, GLAZING, AND RELATED MATERIALS									
DIVISION 30000 MECHANICAL, ELECTRICAL, AND PLUMBING									
DIVISION 31000 HEATING, VENTILATION, AND AIR CONDITIONING									
DIVISION 32000 REFRIGERATION									
DIVISION 33000 RADIANT HEATING									
DIVISION 34000 AIR PURIFICATION									
DIVISION 35000 EXHAUST SYSTEMS									
DIVISION 36000 FAN SYSTEMS									
DIVISION 37000 DUCTWORK									
DIVISION 38000 PIPING									
DIVISION 39000 VALVES									
DIVISION 40000 PUMP AND MOTOR									

FIRE DEPARTMENT REQUIRED NOTES

A. 2A10B-C FIRE EXTINGUISHERS SHALL BE PROVIDED IN THE DINING AREAS AND "KITCHEN" EXTINGUISHERS SHALL BE WITHIN 30 FT OF PREP AREA, STORAGE AND SALES AREA. MOUNTING HEIGHT SHALL NOT EXCEED 48" AFF TO THE TOP OF EXTINGUISHER AND TRAVEL DISTANCE TO REACH ONE SHALL NOT EXCEED 75 FEET.

B. DECORATIVE MATERIALS, INCLUDING DRAPES AND HANGINGS, MUST BE NONCOMBUSTIBLE, FLAME RESISTANT OR TREATED WITH A FLAME RETARDANT.

C. GENERAL CONTRACTOR IS RESPONSIBLE FOR ENTIRE FIRE SPRINKLER SYSTEM, INCLUDING SUBMITTAL AND APPROVAL FROM THE LOCAL FIRE DEPARTMENT.

INTERIOR GENERAL NOTES

- A. INTERIOR TRIM; MATERIALS, OTHER THAN FOAM PLASTICS USING AS TRIM SHALL HAVE A MIN CLASS A FLAME SPREAD (0-25) & SMOKE-DEVELOPED INDEX (0-450) PER ASTM E-84 OR UL 723.
- B. THE FACE OF AN EXIT SIGN ILLUMINATED FROM AN EXTERNAL SOURCE SHALL HAVE AN INTENSITY OF NOT LESS THAN 5 FOOT - CANDLES (54 LUX) & SHALL BE ILLUMINATED AT ALL TIMES & FOR NOT LESS THAN 90 MINUTES IN THE EVENT OF TEMPORARY POWER LOSS USING BATTERIES OR GENERATORS.
- C. TEMPERED WINDOWS/GLASS WHERE APPLICABLE.
- D. MILLWORK TO MEET ADA REQUIREMENTS.
- E. PANIC HARDWARE REQUIRED ON ALL EGRESS PATH LATCHING DOORS.

BID NOTES

- A. NO SUBCONTRACTOR SHOULD BE PROVIDED WITH A PARTIAL SET OF PLANS FOR EITHER BIDDING OR CONSTRUCTION PURPOSES WITHOUT FIRST HAVING AMPLE TIME TO OBTAIN A COMPLETE SET OF CONSTRUCTION DOCUMENTS. ONLY HAVING A PARTIAL SET OF DRAWINGS WILL NOT BE ACCEPTED AS AN EXCUSE FOR DELAYS, INCOMPLETE WORK, OR CHANGE ORDERS AS THERE ARE MANY CROSS REFERENCES IN A SET OF CONSTRUCTION DOCUMENTS.
- B. BIDDERS ARE REQUIRED TO VISIT SITE AND VERIFY CONDITIONS PRIOR TO SUBMITTING BIDS.
- C. SUBCONTRACTORS WORK MAY INCLUDE WORK IN CONJUNCTION WITH OTHER TRADES. IT SHALL BE THE G.C.'S RESPONSIBILITY TO MAKE SURE THAT THE SUBCONTRACTORS ARE FAMILIAR WITH ALL DRAWINGS AND SPECS. FOR THIS PROJECT AND SHALL BE REFLECTED IN THE G.C.'S BID.
- D. ANY CONFLICTS SHALL BE BROUGHT TO THE ATTENTION OF THE ARCHITECT & OWNER PRIOR TO SUBMISSION OF BIDS. CONTRACTOR SHALL TAKE FULL RESPONSIBILITY FOR ANY CONFLICTS NOT IDENTIFIED PRIOR TO BID.

DBPR NOTES

- A. ALL MILLWORK USED AS PART OF FOOD SERVICE SHALL BE CONSTRUCTED WITH 3/4" MARINE GRADE PLYWOOD SUBSTRATE.
- B. ALL LIGHT FIXTURES IN FOOD PREP, STORAGE, AND UTENSIL WASH AREAS MUST HAVE SHATTERPROOF BULBS AND/OR ACRYLIC LENSES.
- C. ACOUSTICAL CEILING TILE IN FOOD PREP AREA SHALL BE VINYL BACKED TILE WHICH IS WASHABLE AND SCRUBBABLE. SEE FINISH FOR SPECIFICATION.
- D. ALL WALLS IN FOOD PREP AREA TO HAVE FR PANELING OR STAINLESS STEEL FROM FLOOR TO UNDERSIDE OF CEILING UNLESS NOTED OTHERWISE. SEE FINISH SCHEDULE FOR SPECIFICATION.
- E. ALL WALLS BEHIND HOOD SHALL BE STAINLESS STEEL FULL WIDTH OF HOOD FROM UNDERSIDE OF HOOD TO TOP OF BASE, UNLESS NOTED OTHERWISE.
- F. ALL MOUNTED SHELVING TO BE MOUNTED 3" AWAY FROM ADJACENT WALLS AND HAVE BLOCKING IN WALL FOR SUPPORT.
- G. ALL EQUIPMENT SHALL BE SEALED OR INSTALLED A MINIMUM OF 3" AWAY FROM THE NEXT PIECE OF EQUIPMENT OR WALL SURFACE TO ALLOW FOR EASY CLEANING.
- H. SEE PLAN FOR SPLASH GUARD LOCATIONS (IF REQUIRED).
- I. ALL EQUIPMENT IS PROVIDED BY THE FOOD SERVICE EQUIPMENT VENDOR (NOT GC). GC INSTALLS ALL ELECTRICAL, GAS, & PLUMBING AND ALL FINAL EQUIPMENT CONNECTIONS.
- J. FLOOR WILL BE CLEAR OF EXPOSED PIPES & LINES & NO UTILITY LINES WILL BE UNNECESSARILY EXPOSED.
- K. GARBAGE CONTAINERS WILL DURABLE, CLEANABLE, AND RODENT RESISTANT WITH SUFFICIENT STORAGE SPACE BOTH INSIDE & OUTSIDE.

LANDLORD COORDINATION NOTES

A. A PRE-CONSTRUCTION MEETING BETWEEN G.C. BAGEL & CO, AND LANDLORD'S REPRESENTATIVE MUST BE SCHEDULED PRIOR TO COMMENCING CONSTRUCTION.

B. G.C. IS REQUIRED TO VERIFY ALL EXISTING CONDITIONS AND DIMENSIONS PRIOR TO COMMENCING WORK.

[illegible]

REFERENCED STANDARD: 2023 FLORIDA BUILDING CODE, 8th EDITION (FBC) & ACCESSIBILITY, 8th EDITION (FBCA)

3. PROVIDE METALLIC SIGN OVER EACH STOREFRONT DOOR STATING: "THIS DOOR MUST REMAIN UNLOCKED DURING BUSINESS HOURS". LETTERS SHALL NOT BE LESS THAN 1" HIGH ON A CONTRASTING BACKGROUND. THE SIGN SHALL BE INSTALLED BY THE GENERAL CONTRACTOR ON THE STOREFRONT ALUMINUM HEADER FRAME. IF PLACED ABOVE 70" AFF NEEDS TO BE 2" HIGH.

5. WHEN NOT ALL ENTRANCES ARE ACCESSIBLE, ALL ACCESSIBLE ENTRANCES SHALL BE IDENTIFIED WITH AT LEAST ONE STANDARD SIGN AND WITH ADDITIONAL DIRECTIONAL SIGNS, AS REQUIRED, VISIBLE FROM APPROACHING PEDESTRIAN WAYS.
6. EVERY REQUIRED ENTRANCE OR PASSAGE DOORWAY SHALL BE OF A SIZE AS TO PERMIT THE INSTALLATION OF A DOOR NOT LESS THAN 3 FEET IN WIDTH AND NOT LESS THAN 6 FEET 8 INCHES IN HEIGHT. DOORS SHALL BE CAPABLE OF OPENING AT LEAST 90 DEGREES AND SHALL BE SO MOUNTED THAT THE CLEAR WIDTH OF DOORWAY IS NOT LESS THAN 32". IF DOORS HAVE CLOSERS, BOTTOM OF CLOSER OR STOP MUST BE 6"-6" ABOVE FLOOR.
7. WHERE PAIR OF DOORS IS UTILIZED AT LEAST ONE OF THE DOORS SHALL PROVIDE A CLEAR UNOBSTRUCTED OPENING WIDTH OF 32" WITH THE LEAF POSITIONED AT AN ANGLE OF 90 DEGREES FROM ITS CLOSED POSITION.
8. LATCHING AND LOCKING DOORS THAT ARE HAND ACTIVATED AND WHICH ARE IN PATH OF TRAVEL SHALL BE OPERABLE WITH A SINGLE EFFORT BY LEVER TYPE HARDWARE THAT DOES NOT REQUIRE TIGHT GRASPING, PINCHING OR TWISTING OF WRIST TO OPERATE. PANIC BARS, PUSH-PULL ACTIVATING BARS, OR OTHER HARDWARE DESIGNED TO PROVIDE PASSAGE WITHOUT REQUIRING THE ABILITY TO GRASP THE OPENING HARDWARE ARE ALLOWABLE.
9. HAND ACTIVATED DOOR OPENING HARDWARE SHALL BE BETWEEN 34" MIN. AND 48" MAX ABOVE THE FLOOR.
10. THE FLOOR AND LANDING ON EACH SIDE OF AN ENTRANCE OR PASSAGE DOOR SHALL BE LEVEL AND CLEAR. SEE DIAGRAM 2" MANEUVERING CLEARANCE AT SWINGING DOORS. FOR DEPTH AND WIDTH REQUIREMENTS OF LEVEL LANDING.
11. THE FLOOR OR LANDING SHALL BE NOT MORE THAN 1/2-INCH LOWER THAN THE THRESHOLD OF THE DOORWAY. CHANGE IN LEVEL BETWEEN 1/4- INCH AND 1/2-INCH SHALL BE BEVELED WITH A SLOPE NO GREATER THAN 1:2.
12. THE BOTTOM 10" OF ALL DOORS EXCEPT AUTOMATIC AND SLIDING DOORS SHALL HAVE A SMOOTH UNINTERRUPTED SURFACE TO ALLOW THE DOOR TO BE OPENED BY A WHEELCHAIR FOOTREST WITHOUT CREATING A TRAP OR HAZARDOUS CONDITION. WHERE NARROW FRAME DOORS ARE USED, A 10-INCH HIGH SMOOTH PANEL SHALL BE INSTALLED ON THE PUSH SIDE OF THE DOOR, WHICH WILL ALLOW THE DOOR TO BE OPENED BY A WHEELCHAIR FOOTREST WITHOUT CREATING A TRAP OR HAZARDOUS CONDITION. CAVITIES CREATED BY PANEL MUST BE CAPPED.
13. A NARROW FRAME WITH A BEVELED TOP (30 DEGREES MAX. BEVEL TO VERTICAL PLANE) INSTALLED AT THE BOTTOM OF THE GLASS DOOR (WITH NO SIDE FRAMES) MAY BE USED IN LIEU OF PROVIDING THE REQUIRED 10 INCH UNINTERRUPTED SURFACE AT THE BOTTOM OF THE DOOR.
14. MAXIMUM EFFORT TO OPERATE DOORS SHALL NOT EXCEED 5 LBS. FOR INTERIOR SWING DOORS AND ALL SLIDING AND FOLDING DOORS. MAXIMUM EFFORT TO SWING EXTERIOR SWING DOORS SHALL NOT EXCEED 8.5 LBS. SUCH PULL OR PUSH EFFORT BEING APPLIED AT RIGHT ANGLES TO HINGED DOORS AND AT THE CENTER PLANE OF SLIDING OR FOLDING DOORS. COMPENSATING DEVICES OR AUTOMATIC DOOR OPERATORS MAY BE UTILIZED TO MEET THE ABOVE STANDARDS. WHEN FIRE DOORS ARE REQUIRED, THE EFFORT TO OPERATE THE DOOR MAY BE MODIFIED PER AUTHORITY HAVING JURISDICTION.
15. DOOR CLOSERS SHALL BE ADJUSTED SO THAT THE CLOSING TIME FROM 90DEGREES TO 12DEGREES (APPROXIMATELY 6") FROM THE LATCH IS 5 SECONDS MINIMUM. SPRING HINGES SHALL BE ADJUSTED SO THAT THE TIME FROM THE OPEN POSITION OF 70DEGREES FROM THE LATCH TO CLOSED POSITION IS 1.5SECONDS MINIMUM.
16. ACCESSIBLE SIGN CONTAINING TACTILE CHARACTERS SHALL BE PROVIDED AT DOORS TO PERMANENT SPACES, THE SIGN SHALL BE ALONGSIDE THE DOOR ON LATCH SIDE AND AT DOUBLE DOORS, THE SIGN SHALL BE RIGHT OF THE RIGHT HANDED DOOR. THE SIGN CONTAINING TACTILE CHARACTERS SHALL HAVE 18" MIN. BY 18" MIN. SPACE ON THE FLOOR CENTERED ON SIGN. THE SIGN TACTILE CHARACTER SHALL BE 48" MIN. AND 60" MAX ABOVE FLOOR.

16. FLOOR SURFACES SHALL BE SLIP-RESISTANT.
17. EVERY PORTION OF EVERY BUILDING IN WHICH ARE INSTALLED SEATS, TABLES, MERCHANDISE, EQUIPMENT OR SIMILAR MATERIALS SHALL BE PROVIDED WITH AISLES LEADING TO AN EXIT.
18. AISLE WIDTH SHALL BE NOT LESS THAN 3 FEET PER ACCESSIBILITY REQUIREMENTS. LIFE SAFETY REQUIREMENTS MAY DICTATE INCREASED AISLE WIDTH DEPENDENT UPON OCCUPANCY CLASSIFICATION, OCCUPANT LOAD AND PLAN LAYOUT. REFER TO LIFE SAFETY PLAN FOR ADDITIONAL INFORMATION.

19. CLEARANCE AROUND THE WATER CLOSET SHALL BE 60" MIN. MEASURED PERPENDICULAR FROM THE SIDEWALL, AND 56" MIN. MEASURED PERPENDICULAR FROM REAR WALL. NO OTHER FIXTURES OR OBSTRUCTION SHALL BE WITHIN WATER CLOSET CLEARANCE
20. TOILET ROOMS SHALL BE EQUIPPED WITH A DOOR THAT HAS AN AUTOMATIC CLOSING DEVICE, AND SHALL HAVE CLEAR UNOBSTRUCTED OPENING WIDTH OF 32" WHEN L THE DOOR POSITIONED AT AN ANGLE OF 90 DEGREES FROM ITS CLOSED POSITION.
21. ACCESSIBLE LAVATORIES ARE REQUIRED WITHIN ACCESSIBLE TOILET ROOMS. IF LAVATORIES ARE PROVIDED OUTSIDE THE ACCESSIBLE TOILET ROOM, AT LEAST ONE LAVATORY OUTSIDE THE TOILET ROOM MUST BE ACCESSIBLE.
22. THE HEIGHT OF ACCESSIBLE WATER CLOSERS SHALL BE A MINIMUM OF 17 INCHES AND A MAXIMUM OF 19 INCHES MEASURED TO THE TOP OF TOILET SEAT.
23. TOILET FLUSH CONTROLS SHALL BE OPERABLE WITH ONE HAND, AND SHALL NOT REQUIRE TIGHT GRASPING, PINCHING, OR TWISTING OF THE WRIST. CONTROLS FOR THE FLUSH VALVES SHALL BE MOUNTED ON THE WIDE SIDE OF THE TOILET AREAS, NO MORE THAN 48" ABOVE THE FLOOR. THE FORCE REQUIRED TO ACTIVATE CONTROLS SHALL BE NO GREATER THAN 5 POUNDS.
24. WHERE TWO OR MORE URINALS ARE PROVIDED, AT LEAST ONE SHALL HAVE A CLEAR SPACE 30" WIDE X 48" LONG IN FRONT OF THE URINAL.
25. WHERE TWO OR MORE URINALS ARE PROVIDED, AT LEAST ONE WITH A RIM PROJECTING A MINIMUM OF 13-1/2" FROM THE WALL AND 17" MAX. ABOVE THE FLOOR SHALL BE PROVIDED.
26. A CLEAR FLOOR SPACE 30" WIDE X 48" LONG SHALL BE PROVIDED IN FRONT OF A LAVATORY TO ALLOW A FORWARD APPROACH. SUCH CLEAR FLOOR SPACE SHALL ADJOIN OR OVERLAP AN ACCESSIBLE ROUTE AND SHALL EXTEND INTO KNEE AND TOE SPACE UNDERNEATH THE LAVATORY.

27. LAVATORIES SHALL BE MOUNTED TO MAXIMUM 34" TO THE TOP, WITH A CLEARANCE OF AT LEAST 27" FROM THE FLOOR TO THE BOTTOM OF THE APRON/BOWL WITH KNEE CLEARANCE UNDER THE FRONT LIP EXTENDING A MINIMUM OF 30" IN WIDTH WITH 8" MINIMUM DEPTH AT THE 27" AND 11" MINIMUM DEPTH AT 9" FROM THE FLOOR. TOE CLEARANCE SHALL BE THE SAME WIDTH AND SHALL BE A MINIMUM OF 9" HIGH FROM THE FLOOR AND A MINIMUM OF 17" DEEP FROM THE FRONT OF THE LAVATORY. SEE FIGURE 14.
28. SUPPLY AND DRAIN PIPES UNDER LAVATORIES SHALL BE INSULATED OR OTHERWISE COVERED. THERE SHALL BE NO SHARP OR ABRASIVE SURFACES UNDER LAVATORIES.
29. FAUCET CONTROLS AND OPERATING MECHANISMS SHALL BE OPERABLE WITH ONE HAND AND SHALL NOT REQUIRE TIGHT GRASPING, PINCHING OR TWISTING OF THE WRIST. THE FORCE REQUIRED TO ACTIVATE CONTROLS SHALL BE NO GREATER THAN 5 LBS. LEVER-OPERATED, PUSH-TYPE AND ELECTRONICALLY CONTROLLED MECHANISMS ARE EXAMPLES OF ACCEPTABLE DESIGN. SELF-CLOSING VALVES ARE ALLOWED IF THE FAUCET REMAINS OPEN FOR AT LEAST 10 SECONDS.
30. MIRRORS SHALL BE MOUNTED WITH THE BOTTOM REFLECTIVE EDGE NOT MORE THAN 40" FROM THE FLOOR.
31. LOCATE TOWEL, SANITARY NAPKIN, AND WASTE RECEPTACLES WITH ALL OPERABLE PARTS NOT MORE THAN 48" FROM THE FLOOR.
32. LOCATE TOILET TISSUE DISPENSERS ON THE WALL WITHIN 7-9 INCHES IN FRONT OF THE WALL CLOSET MEASURED TO THE CENTERLINE OF THE DISPENSER. THE OUTFLET SHALL BE 15 INCHES MINIMUM AND 48 INCHES MAXIMUM ABOVE THE FLOOR, AND SHALL NOT BE LOCATED BEHIND THE GRAB BARS. DISPENSERS SHALL NOT BE OF A TYPE THAT CONTROL DELIVERY, OR DO NOT ALLOW CONTINUOUS PAPER FLOW.

33. GRAB BARS, FASTENERS AND MOUNTING DEVICES SHALL BE DESIGNED FOR 250 LBS, APPLIED IN ANY DIRECTION.

34. GRAB BARS SHALL BE LOCATED ON EACH SIDE (AMBULATORY STALLS), OR ON THE SIDE AND THE BACK OF THE ACCESSIBLE TOILET ROOM OR COMPARTMENT. THEY SHALL BE SECURELY ATTACHED 33" MIN. AND 36" MAXIMUM ABOVE AND PARALLEL TO THE FLOOR FROM THE TOP OF THE GRIPPING SURFACE WITH A MINIMUM 1 1/2" CLEAR BELOW THE GRAB BAR AND 12" CLEAR ABOVE.
35. GRAB BARS AT THE SIDE SHALL BE AT LEAST 42" LONG MOUNTED AT A MAX. 12" FROM THE REAR WALL EXTENDING A MINIMUM OF 54" FROM THE REAR WALL.
36. GRAB BARS AT THE BACK SHALL BE NOT LESS THAN 36" LONG. AND EXTEND 12" MIN. TO ONE SIDE & 24" TO THE OTHER FROM THE CENTERLINE OF THE WATER CLOSET.
37. THE DIAMETER OR WIDTH OF THE GRIPPING SURFACES OF A GRAB BAR SHALL BE 1-1/4 INCHES TO 2 INCHES OR THE SHAPE SHALL PROVIDE AN EQUIVALENT GRIPPING SURFACE.
38. THE GAP BETWEEN THE WALL AND THE GRAB BAR SHALL BE 1-1/2 INCHES.
39. A GRAB BAR AND ANY WALL OR OTHER SURFACE ADJACENT SHALL BE FREE OF ANY SHARP OR ABRASIVE ELEMENTS.
40. GRAB BARS SHALL NOT ROTATE WITHIN THEIR FITTINGS AND EDGES SHALL HAVE MINIMUM RADIUS OF 1/8-INCH.

41. THE CENTER OF RECEPTACLE OUTLETS SHALL BE NOT LESS THAN 15- INCHES ABOVE THE FLOOR OR WORKING PLATFORMS.
42. THE CENTER OF ANY OPERATING HANDLE/ SWITCH INTENDED TO CONTROL OR DISPENSE, SHALL BE PER DETAIL TO THIS SHEET.
43. THE INTERNATIONAL SYMBOL OF ACCESSIBILITY SHALL BE THE STANDARD USED TO IDENTIFY FACILITIES THAT ARE ACCESSIBLE TO AND USABLE BY PHYSICALLY DISABLED PERSONS. THE SYMBOL SPECIFIED PER DIAGRAM 1 SHALL CONSIST OF A WHITE FIGURE ON A BLUE BACKGROUND OR THE REVERSE. THE BLUE SHALL BE EQUAL TO COLOR NO. 15090 IN FEDERAL STANDARD 595A.

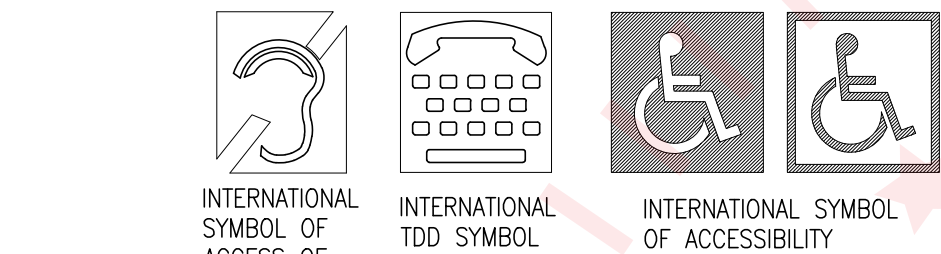
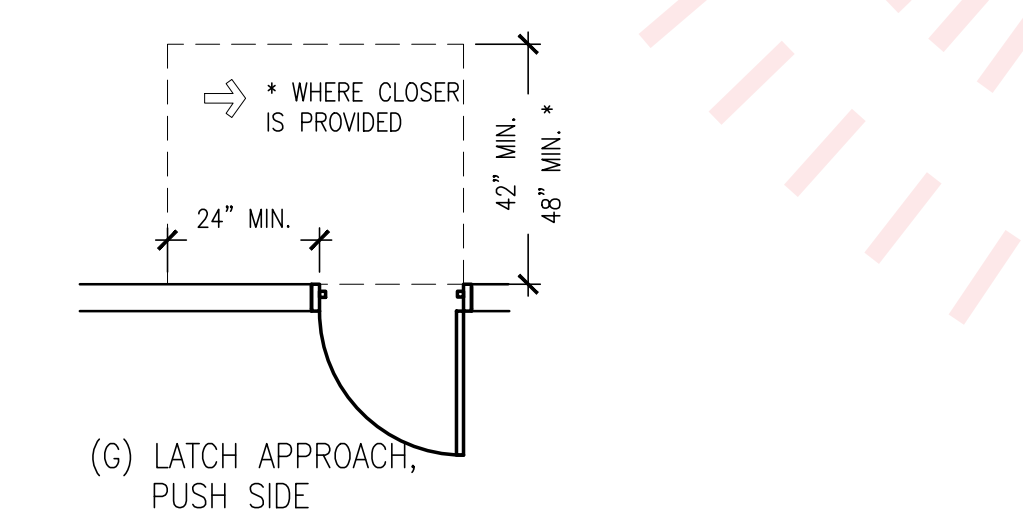
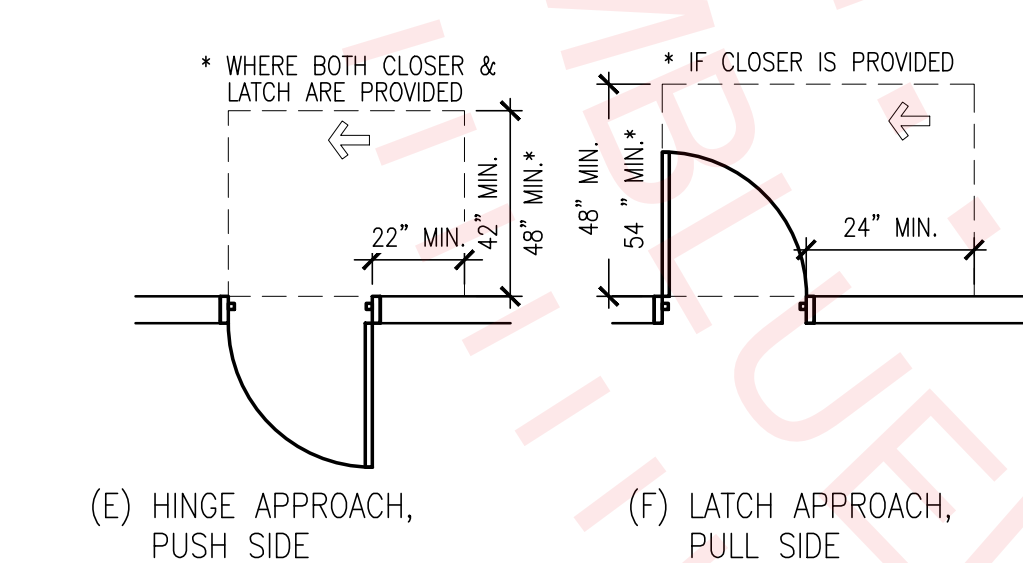
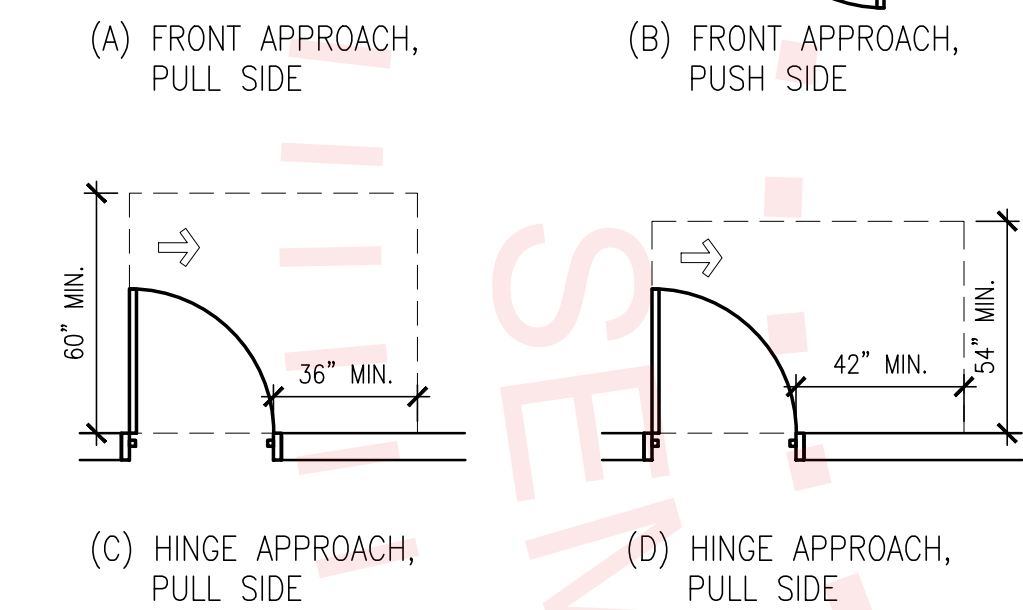


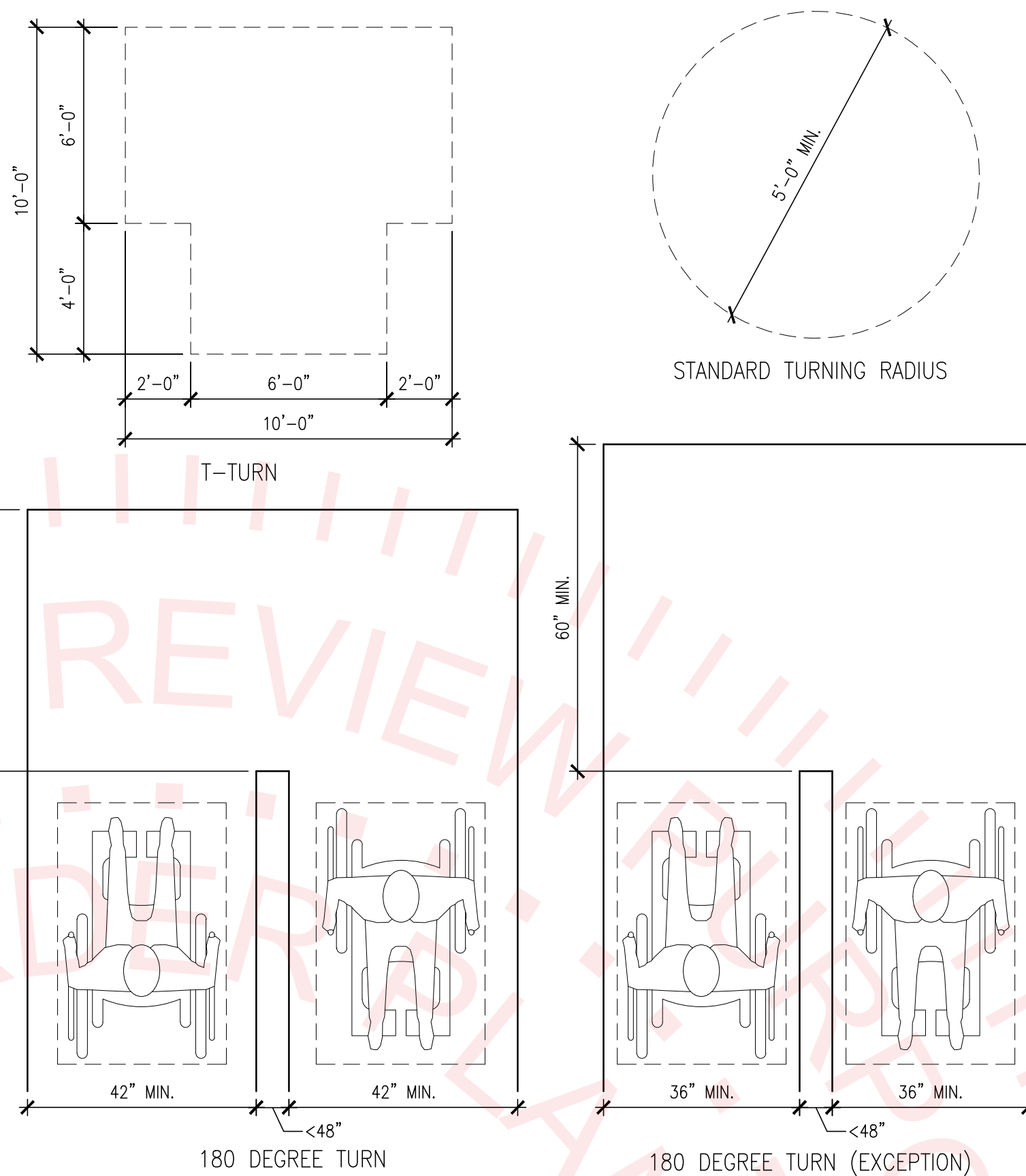
Diagram 1 illustrates the detail of a door and frame assembly. The door is shown with a top edge dimension of 60" MIN. and a bottom edge dimension of 18" MIN. The door is hinged on the left side. A note indicates: "WHERE BOTH CLOSER & LATCH ARE PROVIDED".



(H) FRONT APPROACH

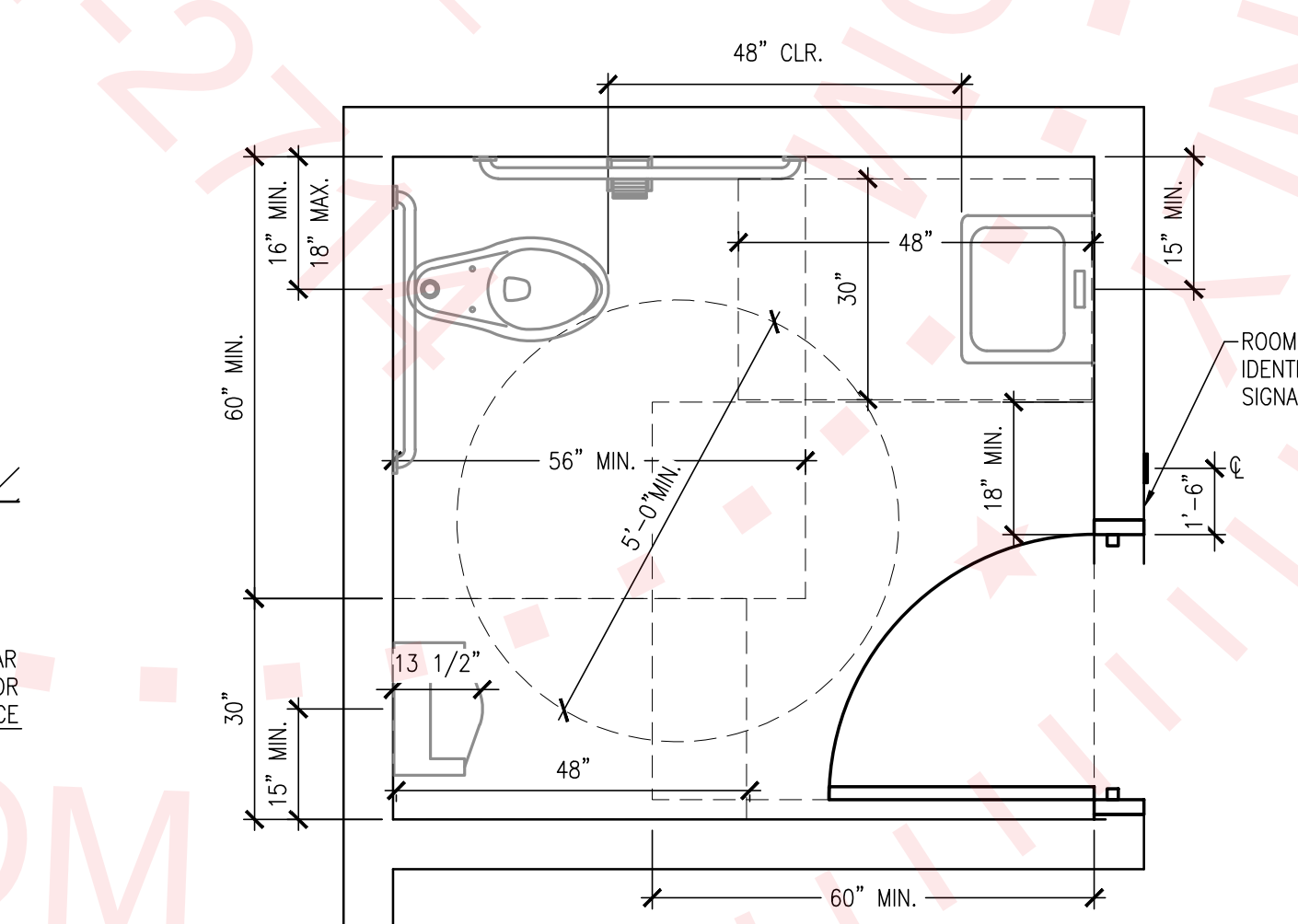
(I) POCKET OR HINGED APPROACH

Diagram illustrating a parallel approach for a vehicle. The vehicle is shown with a minimum clear area of 30' x 48'. The approach is labeled "PARALLEL APPROACH". The dimensions are indicated as 8'-0" for the width and 30' MIN. CLEAR for the length.



The diagrams illustrate the required clearances and dimensions for ADA-compliant toilet stalls:

- TOILET SIDE ELEVATION:** Shows the side view of a toilet. Key dimensions include:
 - Grab bars w/ blocking: 54" MIN. (center-to-center)
 - Clearance from wall to grab bar: 12" (MAX.)
 - Clearance from grab bar to toilet: 42" MIN.
 - Toilet paper dispenser: 7"-9" (height), 1 1/2" MIN. (width)
 - Clearance from floor to toilet seat: 15" MIN.
 - Overall stall width: 33"-36"
- TOILET REAR ELEVATION:** Shows the rear view of a toilet. Key dimensions include:
 - Overall stall width: 36" MIN.
 - Clearance from side wall to grab bar: 24" MIN.
 - Clearance from grab bar to rear wall: 12" MIN.
 - Clearance from floor to toilet seat: 17"-19"
 - Clearance from floor to rear wall: 15" MIN.
 - Overall stall depth: 33"-36"
 - Clearance from toilet to rear wall: 16"-18"
- ELECTRONIC FLUSH VALVE URINAL:** Shows a side view of a urinal. Key dimensions include:
 - Height from floor to urinal: 1'-5" MAX.
 - Label: ADA URINAL



A. 5/8" HIGH BY 1/32" RAISED LETTERING (UNLESS NOTED OTHERWISE)

B. GRADE 2 BRAILLE POSITIONED DIRECTLY BELOW TEXT

C. 1/32" RAISED PICTOGRAM WITH 70% CONTRASTING BETWEEN BACKGROUND AND TEXT. FINISH TO BE EGGSHELL.

D. COLOR TO BE BLUE BACKGROUND WITH WHITE LETTERS.

The diagram shows a double door with a sign above it. The sign is a rectangle with a black border. Inside the sign, the text 'ROOM NAME' is written in a sans-serif font. Above the text, the word 'VARIES' is written. Below the text, there is a small rectangular area with a black background and white text. The sign is mounted on a wall. To the right of the sign, there is a vertical dimension line with arrows at both ends, labeled '9\"

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OR CONSTRUCTION.**

GENERAL CONDITIONS:

- THE FOLLOWING NOTES SHALL APPLY TO ALL STRUCTURAL DRAWINGS.
- ALL DESIGN AND CONSTRUCTION SHALL BE BASED ON AND IN ACCORDANCE WITH THE FLORIDA BUILDING CODE 8TH EDITION 2023 AND ASCE/SEI 7-22.
- THE GENERAL CONTRACTOR SHALL VERIFY ALL DIMENSIONS AND SITE CONDITIONS AND NOTIFY THE ARCHITECT PRIOR TO FABRICATION AND CONSTRUCTION. FOR DIMENSIONS NOT SHOWN ON THE STRUCTURAL DRAWINGS, SEE THE ARCHITECTURAL DRAWINGS.
- IF MATERIAL QUANTITIES, STRENGTHS OR SIZES INDICATED BY THE DRAWINGS OR SPECIFICATIONS ARE NOT IN AGREEMENT WITH THESE NOTES, THE BETTER QUALITY AND/OR QUANTITY, STRENGTH OR NOTED SHALL BE PROVIDED.
- IT IS THE SOLE RESPONSIBILITY OF THE CONTRACTOR TO DETERMINE ERECTION PROCEDURE AND SEQUENCE TO ENSURE THE SAFETY OF THE STRUCTURE AND ITS COMPONENT PARTS DURING ERECTION. THIS INCLUDES, BUT IS NOT LIMITED TO, THE ADDITION OF TEMPORARY WHATEVER BRACING, GUYS, TIE-DOWNS OR SHORING MAY BE NECESSARY. SUCH MATERIAL SHALL BE REMOVED AND SHALL REMAIN THE PROPERTY OF THE CONTRACTOR AFTER THE COMPLETION OF THE PROJECT.
- IT SHALL BE THE SOLE RESPONSIBILITY OF THE CONTRACTOR FOR INITIATING, MAINTAINING AND SUPERVISING ALL SAFETY PROCEDURES. THE STRUCTURAL ENGINEER OF RECORD IS NOT RESPONSIBLE FOR MEANS AND METHODS OF CONSTRUCTION OR FOR RELATED SAFETY PROCEDURES.
- THE STRUCTURAL DRAWINGS ARE ONE DISCIPLINE OF THE CONTRACT DOCUMENTS AND DO NOT BY THEMSELVES CONTAIN ALL THE INFORMATION REQUIRED TO PROPERLY COMPLETE THE PROJECT STRUCTURE. THE GENERAL CONTRACTOR SHALL REFER TO THE ARCHITECTURAL, MECHANICAL, ELECTRICAL AND PLUMBING DRAWINGS AND COORDINATE THE INFORMATION INDICATED IN THESE CONSTRUCTION DOCUMENTS WITH THE STRUCTURAL DRAWINGS TO PROPERLY CONSTRUCT THE PROJECT.
- ALL DETAILS, SECTIONS AND NOTES INDICATED ON THE CONSTRUCTION DOCUMENTS ARE INTENDED TO BE TYPICAL AND SHALL APPLY TO SIMILAR SITUATIONS ELSEWHERE UNLESS OTHERWISE SHOWN.
- SPECIALTY ENGINEERED PRODUCTS
 - LIVE LOADS
 - HOOD AND MECHANICAL UNITS = SEE PLAN
 - DURAVENT GREASE DUCT = 21.8 PLF
 - DESIGN WIND LOAD:
 - ULTIMATE DESIGN WIND SPEED (3 SECOND GUST), V_{ult} = 120 MPH. NOMINAL DESIGN WIND SPEED (3 SECOND GUST), V_{ASD} = 93 MPH.
 - RISK CATEGORY: II
 - WIND EXPOSURE CATEGORY: C
 - COMPONENTS AND CLADDING WIND PRESSURE: SEE WIND LOAD SCHEDULE THIS SHEET.
 - INTERNAL PRESSURE COEFFICIENTS +/- 0.18
 - DESIGN SNOW LOAD:
 - GROUND SNOW LOAD s PSF
 - C_e EXPOSURE FACTOR = 1.0
 - C_t THERMAL FACTOR = 1.0
 - I_s SNOW IMPORTANCE FACTOR = 1.0
 - FLAT ROOF SNOW LOAD = s PSF
 - SEISMIC DESIGN DATA:
 - S_s = 0.110g S_1 = 0.059g
 - SITE CLASS D
 - S_M = 0.140g S_M1 = 0.130g
 - S_D = 0.097g S_D1 = 0.084g
 - RISK CATEGORY II
 - SEISMIC IMPORTANCE FACTOR = 1.0
 - SEISMIC DESIGN CATEGORY B

MASONRY WALL CONSTRUCTION:

- HOLLOW LOAD BEARING UNITS SHALL BE NORMAL WEIGHT, GRADE N, TYPE 2, CONFORMING TO ASTM C90, WITH A MINIMUM NET COMPRESSIVE STRENGTH OF 2000 PSI (f_m = 1500 PSI).
- MORTAR SHALL BE TYPE M OR S, CONFORMING TO ASTM C270.
- COARSE GROUT SHALL CONFORM TO ASTM C475 WITH A MAX. AGGREGATE SIZE OF 3/8" AND A MINIMUM COMPRESSIVE STRENGTH OF 3,000 PSI.
- VERTICAL REINFORCEMENT SHALL BE AS NOTED ON THE DRAWINGS. FILL ALL CELLS CONTAINING REINFORCING WITH COARSE GROUT.
- REINFORCING STEEL SHALL BE LAPPED MINIMUM 48 BAR DIAMETERS.

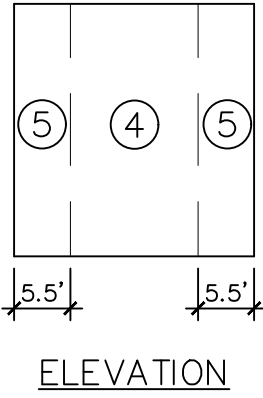
STRUCTURAL STEEL:

- CONFORM TO AISC MANUAL OF STEEL CONSTRUCTION, 16TH EDITION.
- MATERIALS:
 - ASTM A 992, FY=50 KSI FOR W-SHAPES.
 - ASTM A 500, GRADE B, FY=46 KSI FOR STRUCTURAL TUBING.
 - ASTM A 501, FY=36 KSI FOR PIPES.
 - ASTM A 325, TYPE I, FOR HIGH STRENGTH BOLTS.
 - F1554, GRADE 36, ANCHOR BOLTS
 - ASTM A 36, FY=36 KSI FOR PLATES, BARS, RODS, AND ANGLES.
 - ASTM A 563, HEAVY HEX NUTS.
 - ASTM A 436, HARDENED STEEL WASHERS.
- STRUCTURAL STEEL SHALL RECEIVE ONE SHOP COAT OF RUST INHIBITIVE PRIMER UNLESS THE STEEL IS TO RECEIVE SPRAY ON CEMENTITIOUS FIREPROOFING. SEE ARCHITECTURAL DRAWINGS. ALL EXPOSED STEEL SHALL BE HOT DIPPED GALVANIZED OR COATED WITH CORROSION INHIBITED PAINT.

CONCRETE:

- CONCRETE MEMBERS TO HAVE THE FOLLOWING MINIMUM COMPRESSIVE STRENGTHS AT 28 DAYS:
 - SLAB-ON-GRADE 3000 PSIALL OTHER CONCRETE TO BE 3000 PSI UNLESS NOTED OTHERWISE.
- ALL CONCRETE SHALL BE READY MIX AND MEET THE FOLLOWING REQUIREMENTS:
 - CONCRETE SLUMP SHALL BE 4 INCHES PLUS OR MINUS 1".
 - CONCRETE SHALL HAVE 2 TO 4 PERCENT AIR ENTRAINMENT.
 - ALL CONCRETE TO HAVE MAXIMUM WATER/CEMENT RATIO OF 0.54.
- CONCRETE MIX SHALL BE IN ACCORDANCE WITH THE LATEST EDITION OF ACI 301-16 SECTION 4. CONTRACTOR SHALL SUBMIT BACKUP DATA PER CHAPTER 26 SECTION 26.4.4.1 OF ACI 318-14.
- ALL REINFORCING STEEL SHALL BE NEW DOMESTIC DEFORMED BILLET STEEL CONFORMING TO ASTM A-615 GRADE 60. SUBMIT ALL REINFORCING STEEL SUBMITTAL SHOP DRAWINGS FOR APPROVAL PRIOR TO ANY FABRICATION.
- CONCRETE COVER FOR REINFORCING STEEL SHALL BE AS REQUIRED BY ACI SPECIFICATIONS.
- WELDED WIRE FABRIC SHALL CONFORM WITH ASTM A 185. WELDED WIRE FABRIC SHALL BE LAPPED AT LEAST 12 INCHES UNLESS NOTED OTHERWISE.
- LAP ALL BARS MINIMUM 48 DIAMETERS UNLESS OTHERWISE NOTED ON DRAWINGS.
- PROVIDE ACI STANDARD HOOKS UNLESS NOTED OTHERWISE ON DRAWINGS.
- ALL CONCRETE WORK SHALL CONFORM TO ACI 318-14
"THE BUILDING CODE REQUIREMENTS FOR STRUCTURAL CONCRETE", AND ACI 301-10 "SPECIFICATIONS FOR STRUCTURAL CONCRETE."
- ALL CONCRETE DETAILS SHALL CONFORM TO ACI 315R-18 "GUIDE TO PRESENTING REINFORCING STEEL DESIGN DETAILS" UNLESS NOTED OTHERWISE ON THE DRAWINGS.
- CONTRACTOR SHALL REFER TO ARCHITECTURAL AND MECHANICAL DRAWINGS FOR LOCATIONS AND SIZES OF SLEEVES, OPENINGS, EMBEDDED ITEMS, SLAB RECESSES, SLOPES, ECT. THESE ITEMS SHALL BE COORDINATED WITH OTHER TRADES AND INSTALLED PRIOR TO CONCRETE PLACEMENT.
- BAR LENGTHS PROVIDED ON DRAWINGS DO NOT INCLUDE HOOK LENGTH. HOOKS SHALL BE PROVIDED AT TOP BARS AT BEAM ENDS AND SLAB EDGES.
- CONTRACTOR SHALL PROVIDE CHAIRS, BOLSTERS, SPACERS, ECT. AS REQUIRED TO SECURELY SUPPORT REINFORCEMENT. SUPPORT ITEMS ON EXPOSED CONCRETE SHALL BE PLASTIC SUPPORT REINFORCEMENT. TIPPED OR STAINLESS STEEL. IN HIGHLY CORROSIVE ENVIRONMENTS, SUPPORT ITEM SHALL BE PLASTIC.
- ONE COPY OF ALL THE CONCRETE TEST RESULTS SHALL BE SUBMITTED BY THE TESTING AGENCY DIRECTLY TO THE ENGINEER OF RECORD.
- THE CONTRACTOR SHALL SEAL ALL HAIRLINE CRACKING IN CONCRETE SLAB ON GRADE WITH A CRACK SUPPRESSION KIT SUCH AS LATICRETE OR APPROVED EQUAL.
- A COPY OF THE "FIELD REFERENCE MANUAL" ACI SP-15 LATEST EDITION SHALL BE KEPT BY THE CONTRACTOR ON SITE.

COMPONENT AND CLADDING WIND PRESSURES				
EFFECTIVE WIND AREA WALL ZONES	V_{ult} = 120 MPH, EXP. C, RISK CATEGORY II, C_{pi} = +/- 0.18			
	10	20	50	100
(4)	+42.0/-45.3	+40.0/-43.4	+37.5/-40.9	+35.6/-39.1
(5)	+42.0/-55.9	+40.0/-52.2	+37.5/-47.3	+35.6/-43.4

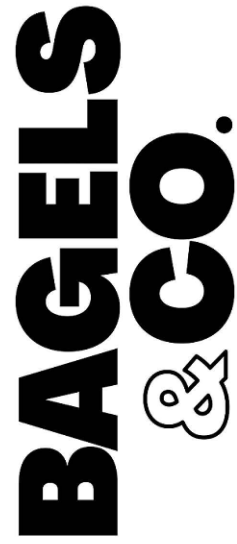


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HAROLD G. HIGGENBOTHAM
PROFESSIONAL ENGINEER
FL. NO. 56467

NO DATE REMARKS
REVISIONS



BAGEL & CO
TALLAHASSEE

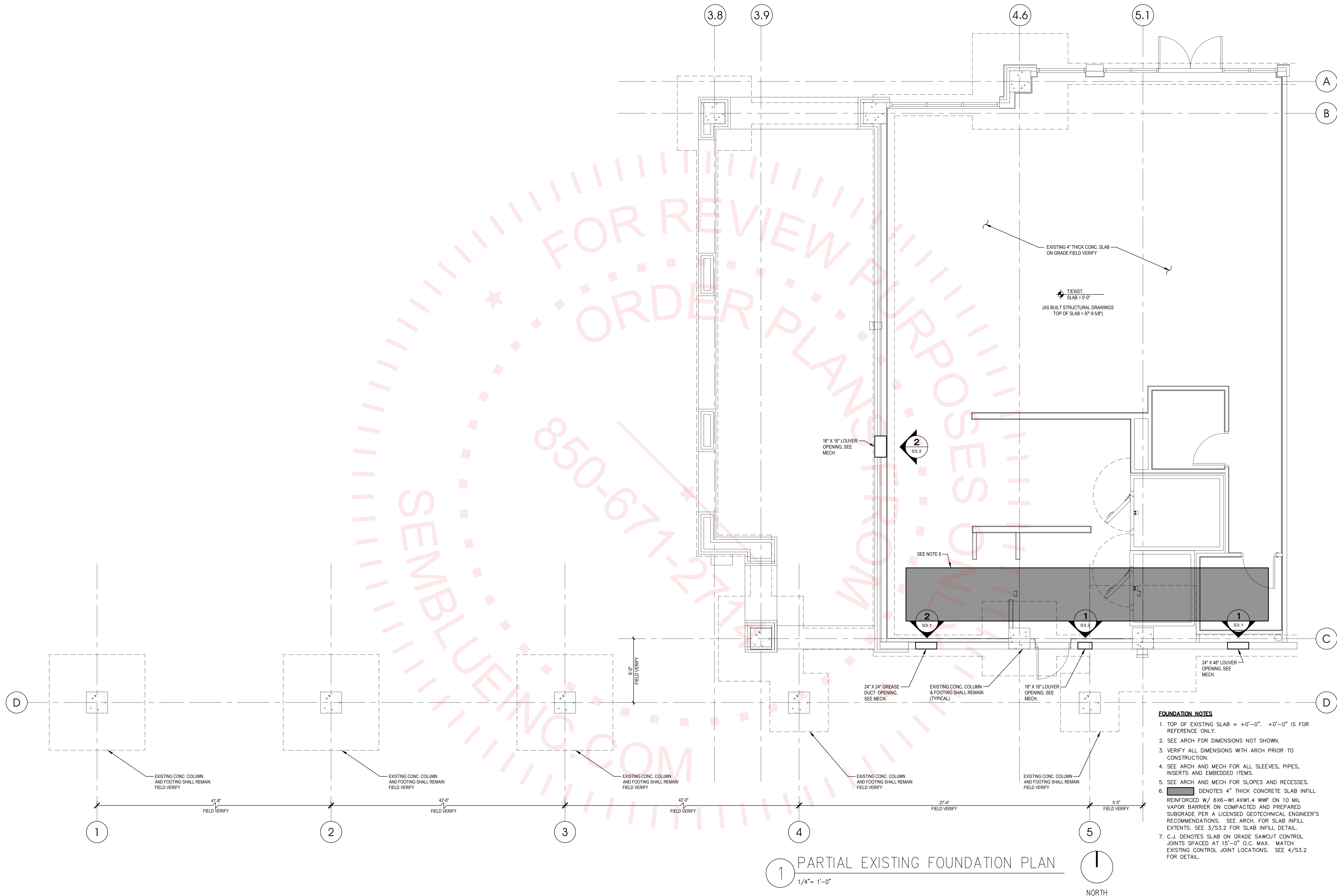
799 W. GAINES STREET
SUITE 121
TALLAHASSEE, FL 32310

PROJECT NO: 2023.0719
DATE: 04/12/2024

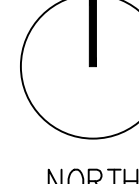
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STRUCTURAL NOTES

CHECKED: HGH DRAWN: KAH



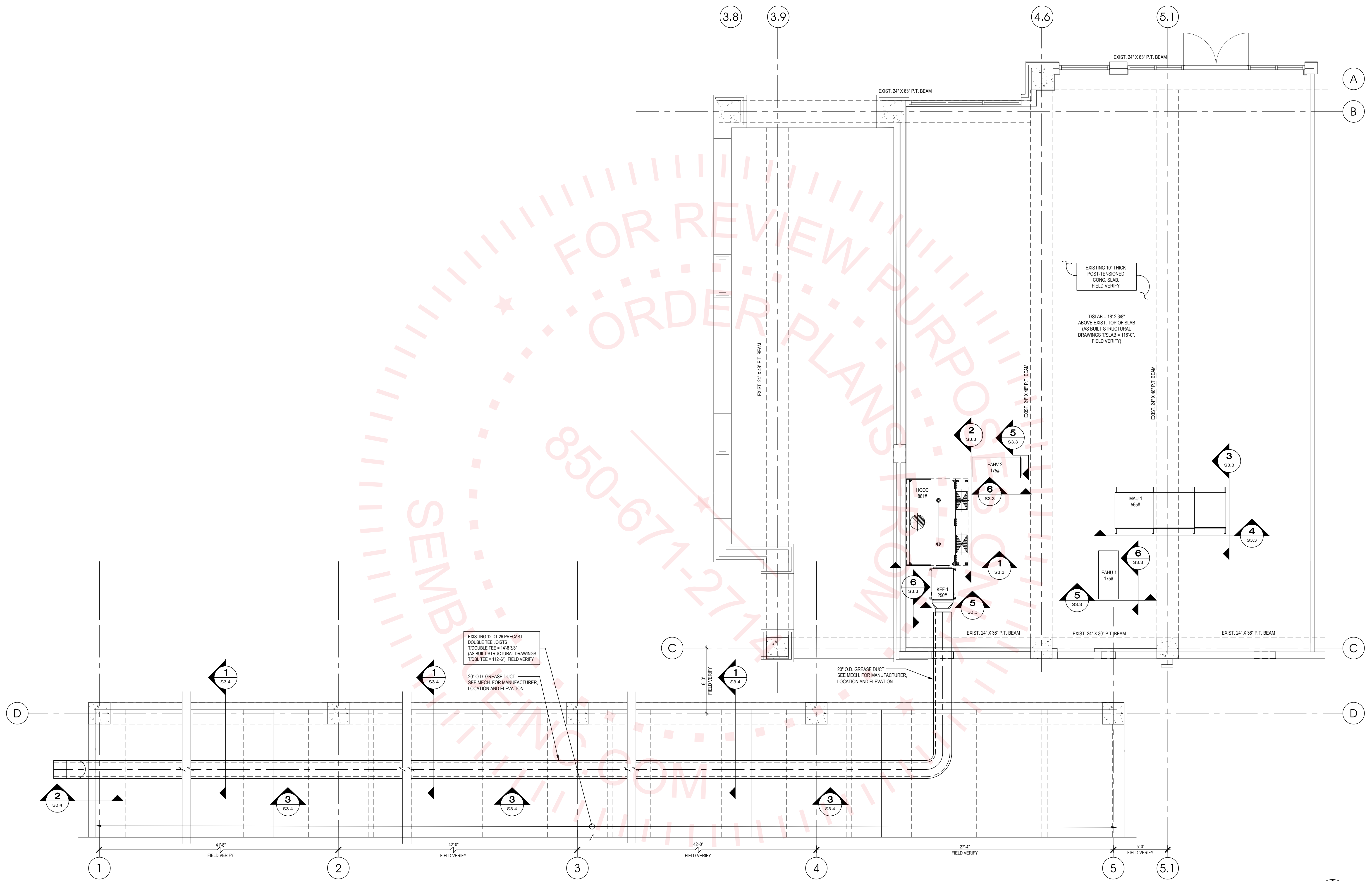
1 PARTIAL EXISTING FOUNDATION PLAN
1/4" = 1'-0"



SHORING NOTE:
THE GENERAL CONTRACTOR SHALL PROVIDE SHORING FOR EXISTING ROOF FRAMING MEMBERS PRIOR TO REMOVAL OF EXISTING SUPPORTS. SHORING SHALL REMAIN IN PLACE UNTIL ALL NEW STRUCTURAL SUPPORTS ARE INSTALLED.

EXISTING STRUCTURAL MEMBERS:
THE GENERAL CONTRACTOR SHALL FIELD VERIFY LOCATIONS, DIMENSIONS, ELEVATIONS, AND CONDITION OF ALL PERTINENT EXISTING STRUCTURAL MEMBERS, (CONCEALED AND EXPOSED) PRIOR TO FABRICATION AND CONSTRUCTION. THE GENERAL CONTRACTOR SHALL NOTIFY THE ARCHITECT AND STRUCTURAL ENGINEER OF RECORD OF ANY DISCREPANCIES A MINIMUM OF 48 HOURS IN ADVANCE PRIOR TO CONSTRUCTION, TO ALLOW FOR MODIFICATION OF DESIGN PER UNFORSEEN FIELD CONDITIONS

HIGGENBOTHAM
Engineering, Inc.
Structural Engineers
3222 Corrine Drive, Suite F
Orlando, Florida 32803
Phone: 407 286 1787
Fax: 407 286 2281
FLORIDA C.A. NO. 28850



1 PARTIAL EXISTING 2ND FLOOR FRAMING PLAN
1/4" = 1'-0"

SHORING NOTE:
THE GENERAL CONTRACTOR SHALL PROVIDE SHORING FOR EXISTING ROOF FRAMING MEMBERS PRIOR TO REMOVAL OF EXISTING SUPPORTS. SHORING SHALL REMAIN IN PLACE UNTIL ALL NEW STRUCTURAL SUPPORTS ARE INSTALLED.

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BAGELS & CO.

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TALLAHASSEE

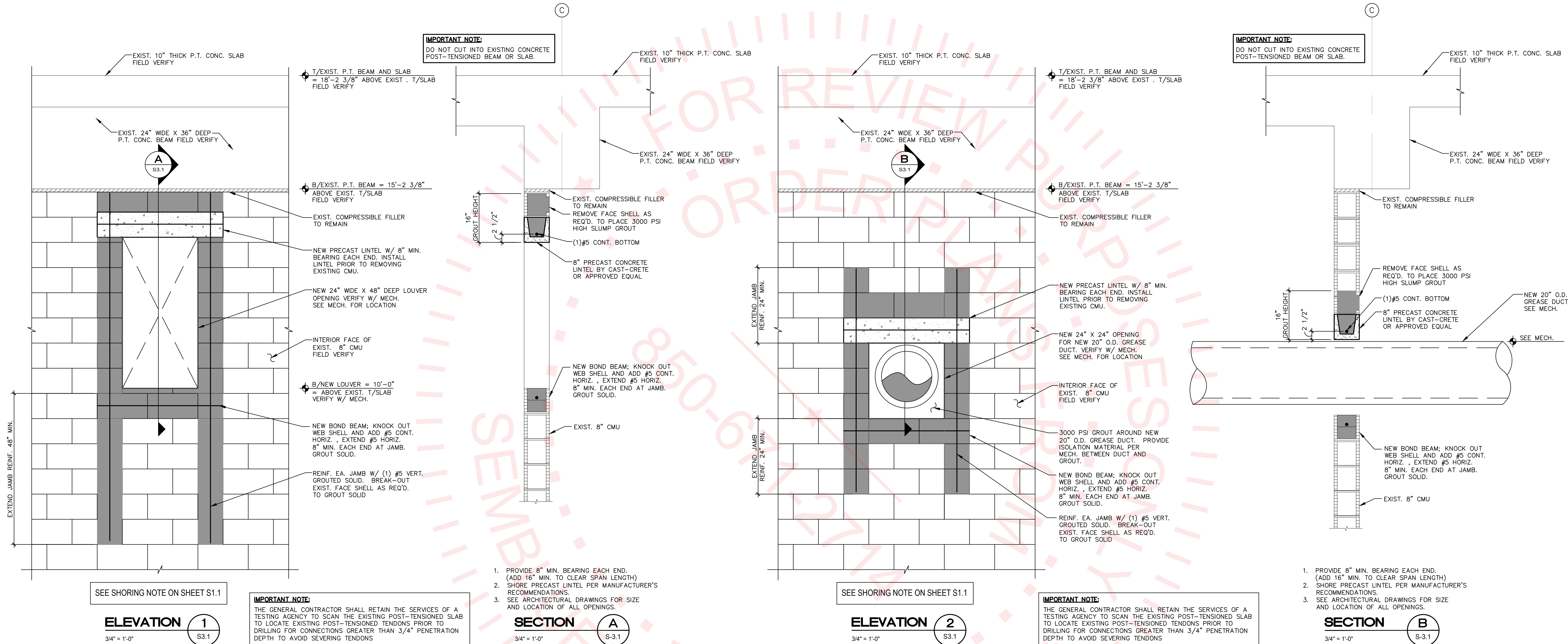
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S2.1

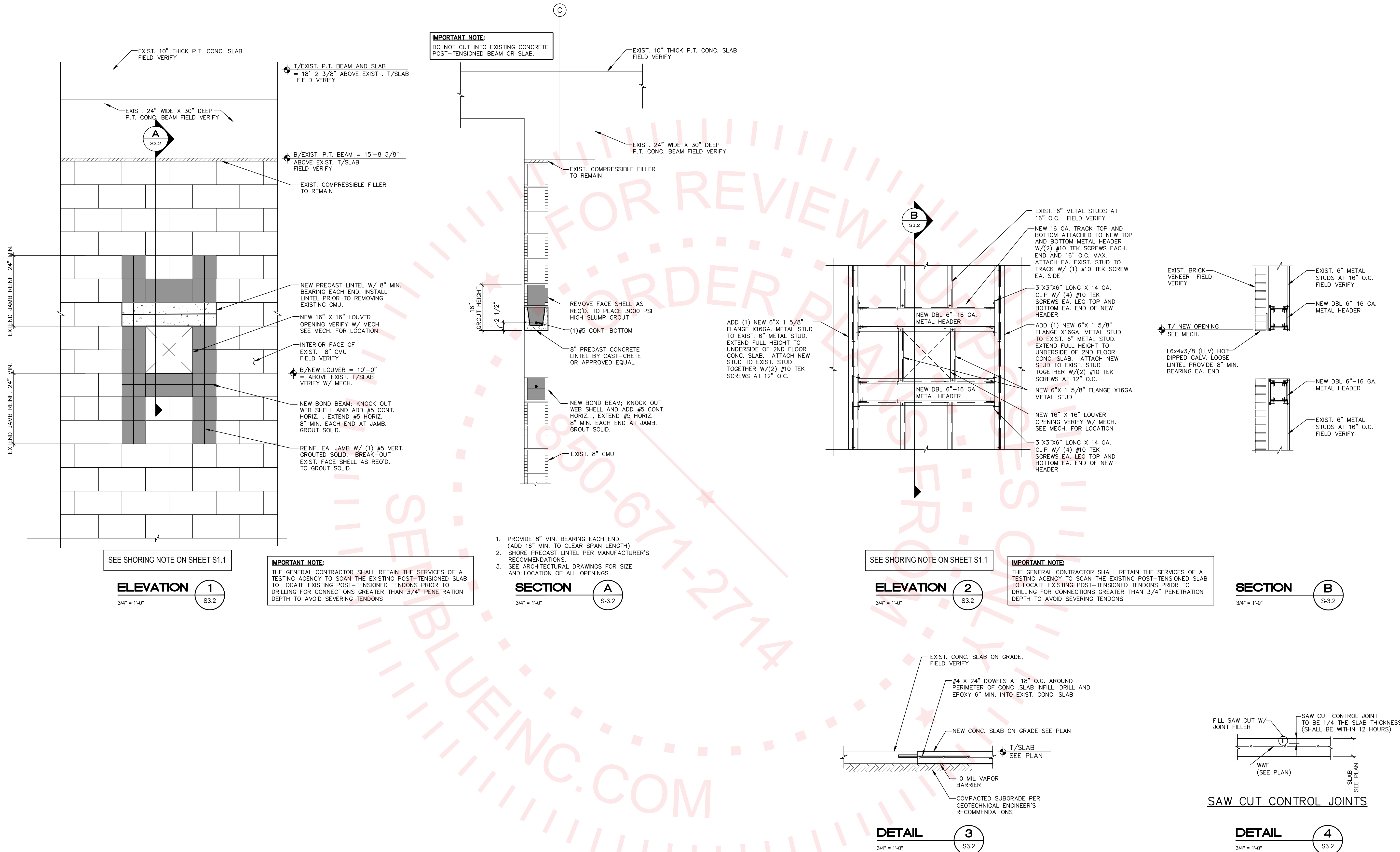
PARTIAL EXISTING
2ND FLOOR FRAMING PLAN

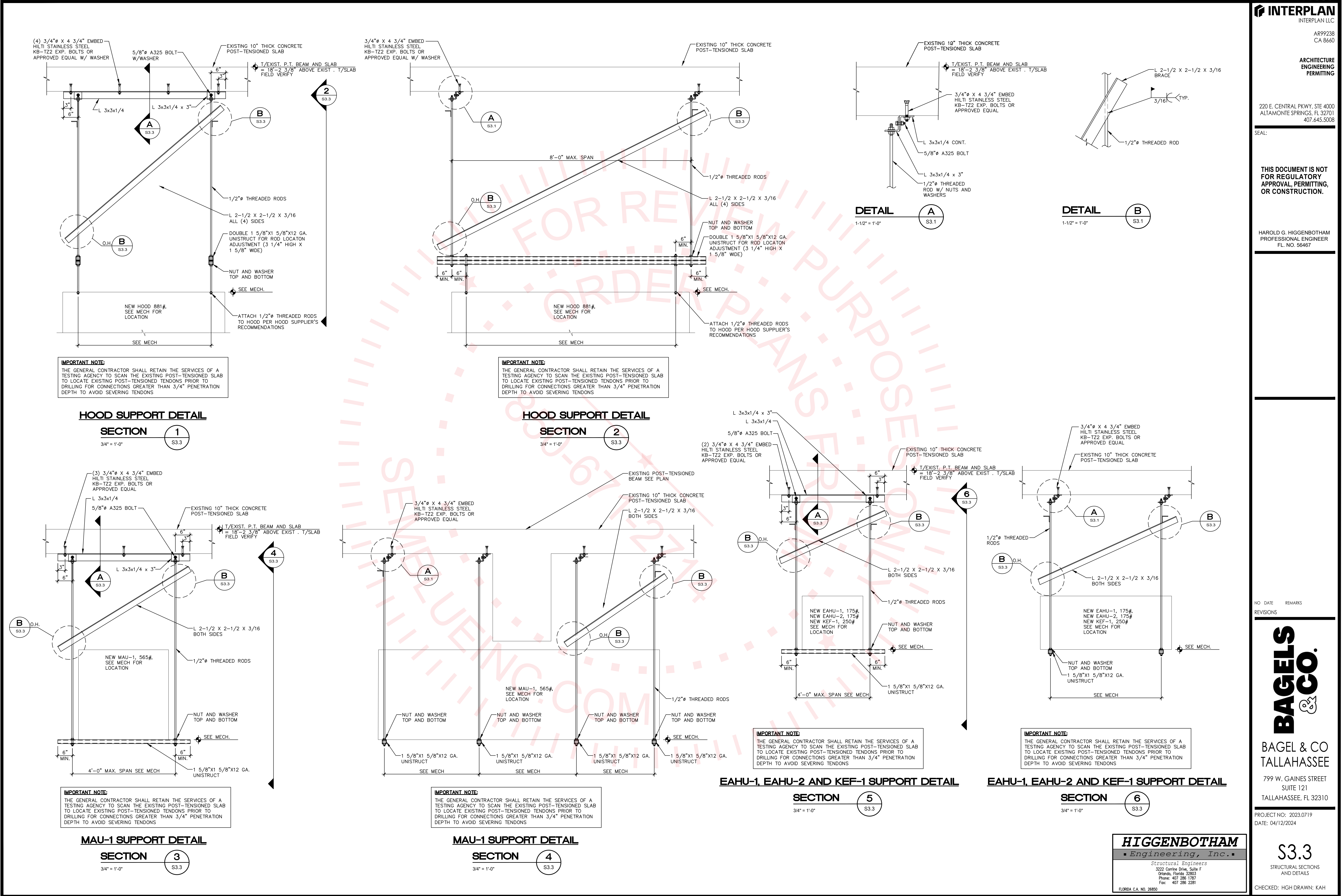
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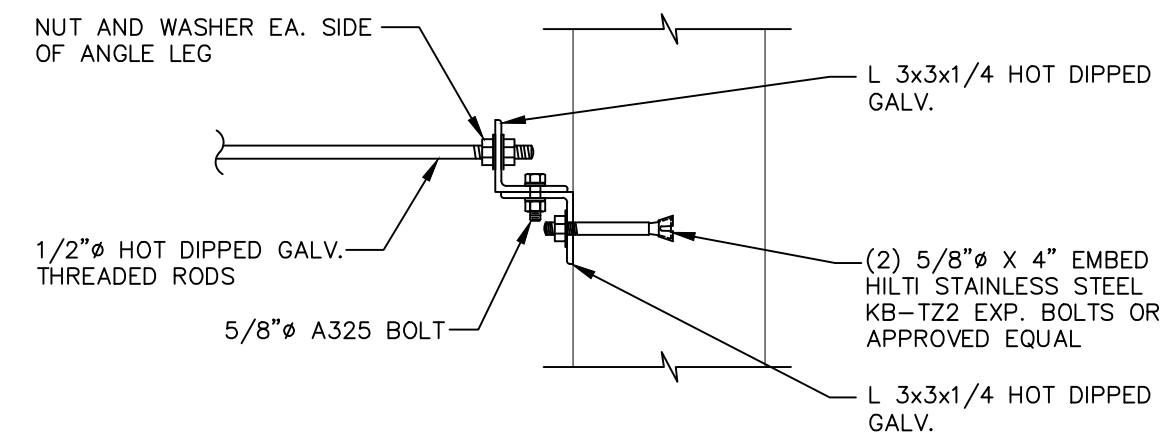


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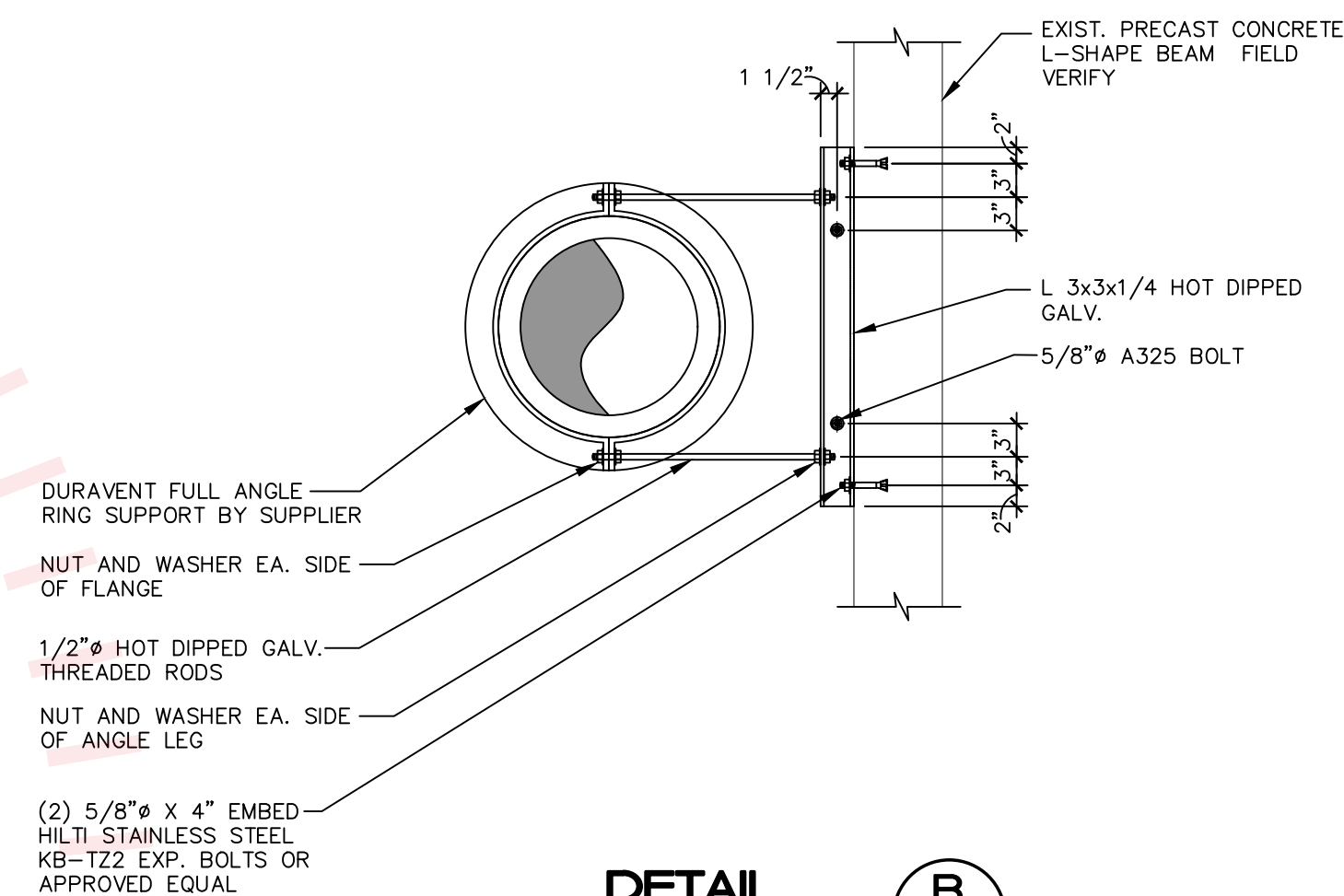
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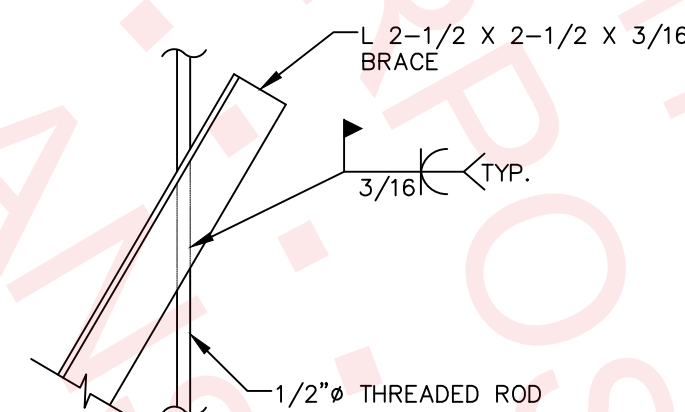




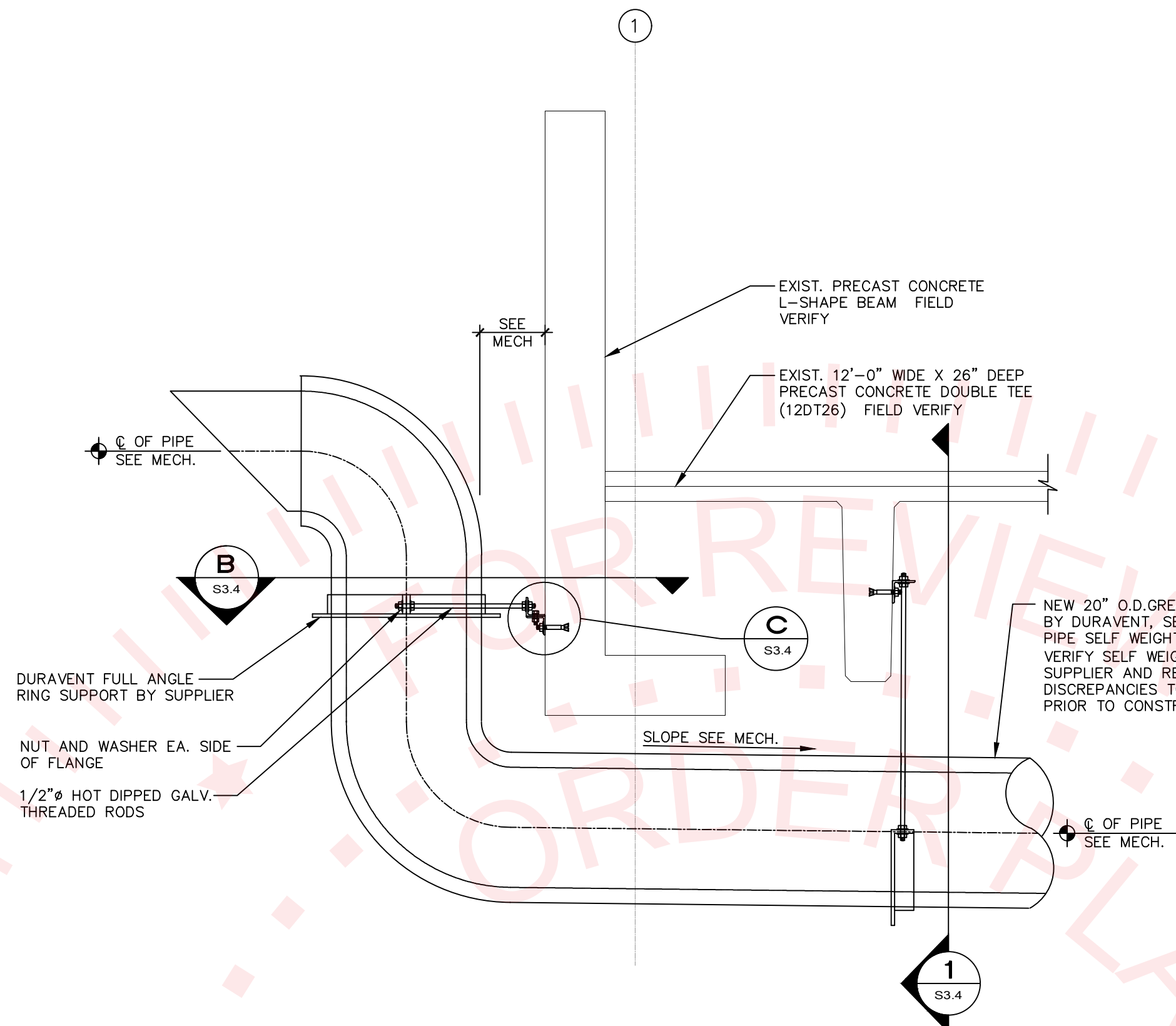
DETAIL C
1-1/2" = 1'-0"



DETAIL B
3/4" = 1'-0"

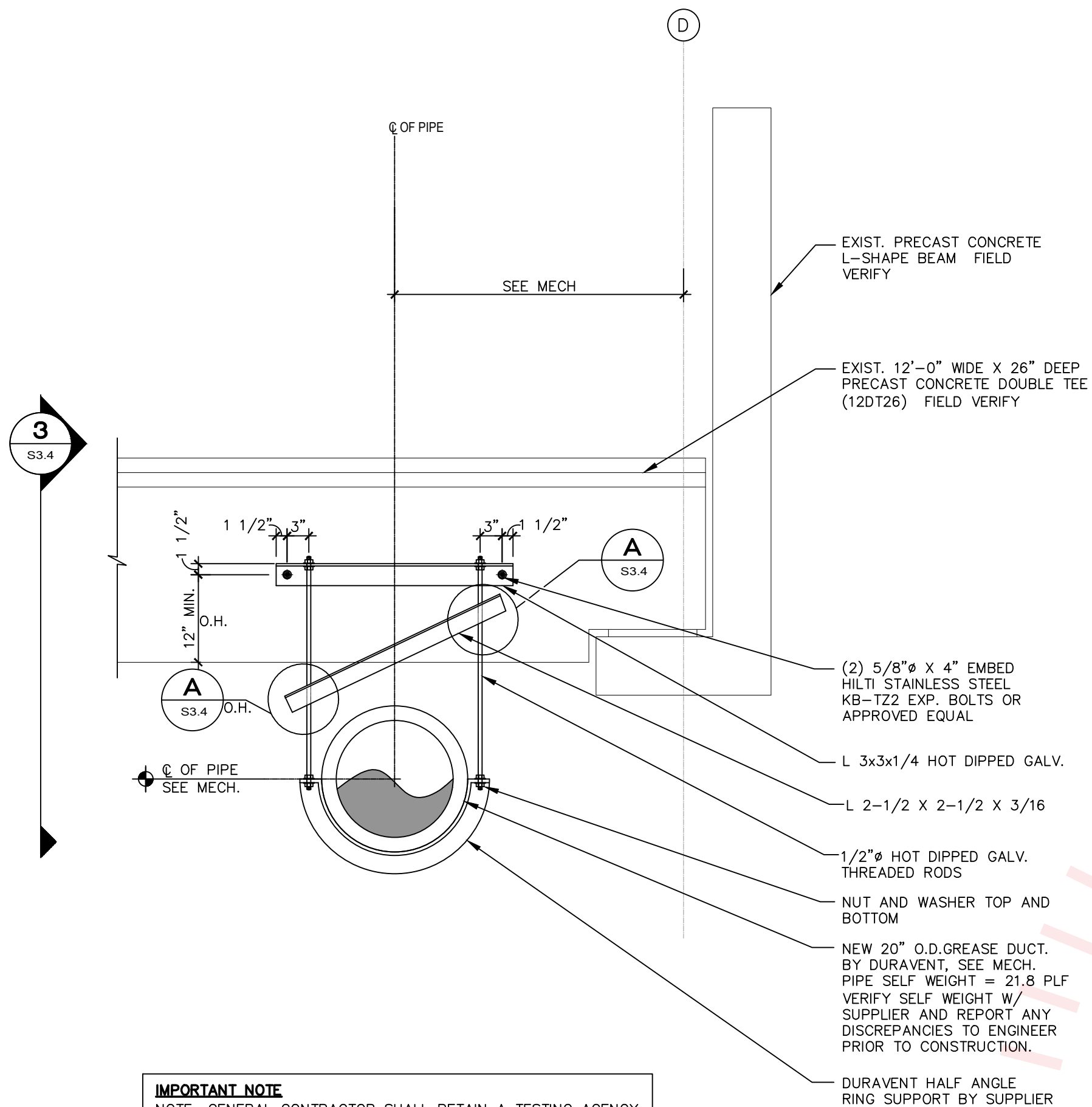


DETAIL A
1-1/2" = 1'-0"



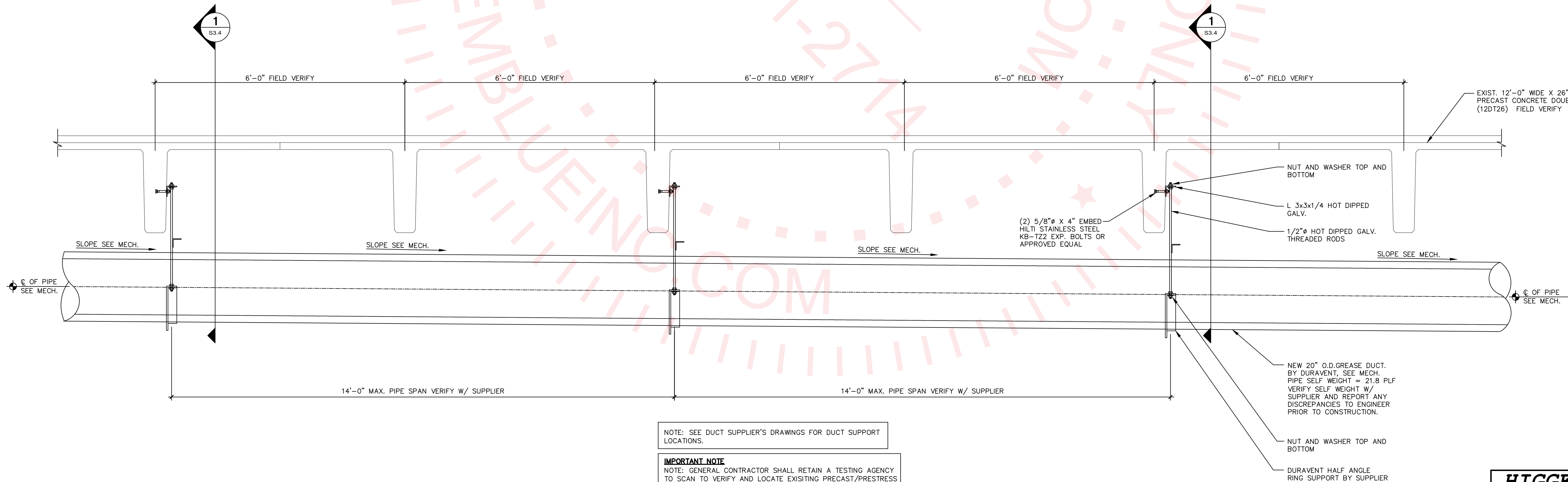
IMPORTANT NOTE
NOTE: GENERAL CONTRACTOR SHALL RETAIN A TESTING AGENCY TO SCAN TO VERIFY AND LOCATE EXISTING PRECAST/PRESTRESS MEMBER PRESTRESS STRANDS PRIOR TO DRILLING TO AVOID DAMAGE TO STRANDS

SECTION 2
3/4" = 1'-0"



IMPORTANT NOTE
NOTE: GENERAL CONTRACTOR SHALL RETAIN A TESTING AGENCY TO SCAN TO VERIFY AND LOCATE EXISTING PRECAST/PRESTRESS MEMBER PRESTRESS STRANDS PRIOR TO DRILLING TO AVOID DAMAGE TO STRANDS

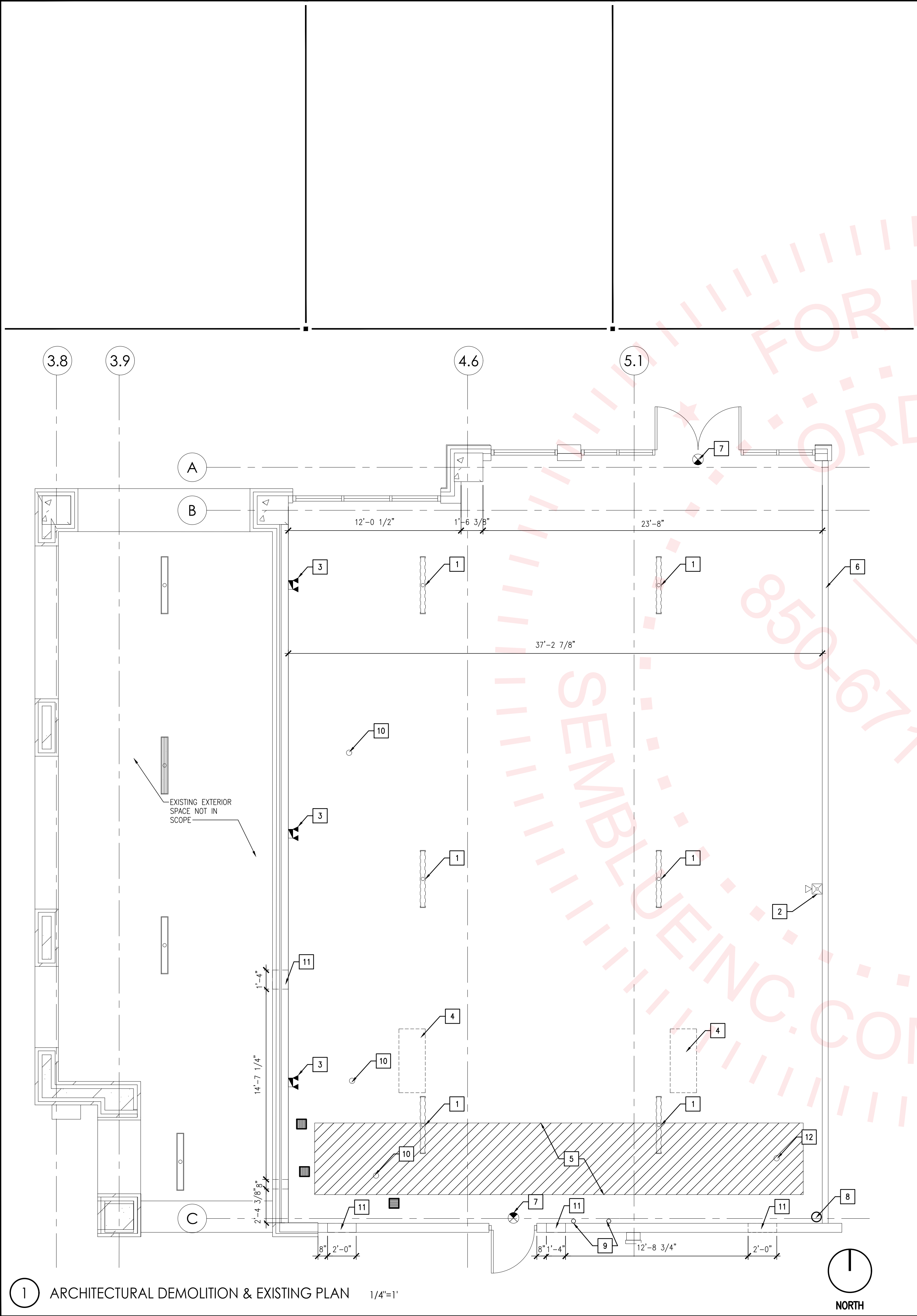
SECTION 1
3/4" = 1'-0"



NOTE: SEE DUCT SUPPLIER'S DRAWINGS FOR DUCT SUPPORT LOCATIONS.

IMPORTANT NOTE
NOTE: GENERAL CONTRACTOR SHALL RETAIN A TESTING AGENCY TO SCAN TO VERIFY AND LOCATE EXISTING PRECAST/PRESTRESS MEMBER PRESTRESS STRANDS PRIOR TO DRILLING TO AVOID DAMAGE TO STRANDS

ELEVATION 3
3/4" = 1'-0"



WALL TYPE LEGEND

- EXISTING TO REMAIN
- - - EXISTING TO BE DEMOLISHED

GENERAL NOTES

- A. THIS SECTION IS INCLUDED FOR GENERAL REFERENCE WORK TO BE PERFORMED BY THE GENERAL CONTRACTOR ("G.C."), APPLIED AS NEEDED TO STORE.
- B. G.C. IS TO REMOVE ALL EXISTING LEASEHOLD IMPROVEMENTS NOT REQUIRED TO THE REMOVAL OF SPECIFIED INTERIOR WALLS PARTITIONS, CEILINGS, FLOOR COVERINGS, PLATFORMS, DISPLAY SHOW WINDOWS, CABINETS, ETC. DEMOLITION ALSO INCLUDES THE REMOVAL OF EXISTING ELECTRICAL, MECHANICAL EQUIPMENT AND PLUMBING FIXTURES BOTH WITHIN THE LEASE SPACE AND ON THE ROOF. G.C. IS TO CONTACT OWNER'S CONSTRUCTION MANAGER IF THERE ARE ANY QUESTIONS CONCERNING THE EXTENT OF DEMOLITION.
- C. G.C. IS RESPONSIBLE FOR PATCHING AND/OR REPAIRING EXISTING FLOOR SLAB, DEMISING WALLS, AND ROOFING AFTER COMPLETION OF ALL DEMOLITION AND PRIOR TO THE COMMENCEMENT OF NEW CONSTRUCTION. G.C. IS TO COORDINATE WITH LANDLORD REPRESENTATIVE IN PERFORMING ANY AND ALL ROOF WORK. G.C. IS TO PATCH, REPAIR AND LEVEL EXISTING FLOOR SLAB AND INFORM OWNER OF ALL SEVERE IRREGULARITIES IN EXISTING SLAB IMMEDIATELY.
- D. G.C. TO FIELD VERIFY EXISTING CONDITIONS. REPORT ANY DISCREPANCIES TO ARCHITECT/TENANT'S CONSTRUCTION PROJECT MANAGER BEFORE DEMOLITION BEGINS.
- E. G.C. IS TO PAY SPECIAL ATTENTION TO AVAILABLE HEIGHT FOR INTENDED CEILING ELEVATION, MECHANICAL DUCTWORK, RECESSED LIGHTING AND ACCESS TO SIGNAGE.
- F. ALL DEBRIS TO BE REMOVED FROM THE SITE IN ACCORDANCE WITH ALL APPLICABLE ORDINANCES & THE REQUIREMENTS OF THE LANDLORD.
- G. G.C. TO NOTE ALL UTILITIES THAT WILL BE AFFECTED BY CONSTRUCTION & INCLUDE IN BID THE NECESSARY CHARGES & FEES INCURRED FROM UTILITY COMPANY(S) FOR SAID WORK.
- H. G.C. IS TO EXERCISE EXTREME CARE DURING DEMOLITION SO AS NOT TO DAMAGE OR DISRUPT ALL UTILITIES OR SPRINKLER LINES WHICH ARE NOT TO BE RELOCATED AND MIGHT PASS THROUGH THIS SPACE TO SERVICE OTHER SPACES OR OTHER TENANTS. G.C. WILL BEAR SOLE RESPONSIBILITY OF ANY SUCH DISRUPTION OR DAMAGE.
- I. IF HAZARDOUS MATERIALS ARE ENCOUNTERED DURING DEMOLITION OPERATIONS, NOTIFY OWNER AND COMPLY WITH APPLICABLE REGULATIONS, LAWS, AND ORDINANCE CONCERNING REMOVAL, HANDLING, AND PROTECTION AGAINST EXPOSURE OR ENVIRONMENTAL POLLUTION.
- J. IN ORDER TO CAUSE THE LEAST AMOUNT OF INCONVENIENCE, G.C. IS TO COORDINATE WITH LANDLORD A CONVENIENT TIME TO PERFORM DEMOLITION WORK. G.C. IS TO PROVIDE AND INSTALL ALL TEMPORARY PROTECTED MEANS OF EGRESS, TEMPORARY LIGHTING, AND ALL SAFETY DEVICES IN ACCORDANCE WITH ALL STATE AND LOCAL CODES AND LANDLORD REQUIREMENTS. DEMOLITION SHALL NOT INTERFERE WITH ADJACENT TENANTS DAILY OPERATION OR DAILY OPERATION OF BUILDING.
- K. PRIOR TO DEMOLITION, ELECTRICAL SUBCONTRACTOR SHALL DISCONNECT ALL ELECTRICAL CIRCUITS SERVING THE EXISTING SPACE EXCEPT FIRE ALARM &/OR SMOKE DETECTORS. MARK ALL CIRCUITS THAT REMAIN.
- L. G.C. TO REMOVE ALL WIRING & CONDUIT BACK TO SOURCE THAT IS NOT BEING USED.
- M. G.C. TO CAP &/OR REMOVE ALL WATER SUPPLY & SANITARY WASTE LINES BACK TO SOURCE NOT INDICATED TO BE RE-USED. INSPECT SITE TO DETERMINE EXTENT OF EXISTING TO BE REMOVED/CAPPED.
- N. WHERE APPLICABLE, G.C. IS TO COORDINATE THE REMOVAL, DISPOSAL AND/OR RETURN OF ALL EXISTING SECURITY SYSTEMS WITH OWNER'S CONSTRUCTION MANAGER PRIOR TO ITS REMOVAL. FAILURE TO DO SUCH COORDINATION WILL RESULT IN G.C.'S MONETARY FINE IN THE EQUIVALENCE OF THE COST OF REPLACEMENT VALUE OF THE EXISTING SECURITY SYSTEM.
- O. REFER TO MEP SHEETS THIS SET FOR ADDITIONAL SPECIFIC INFORMATION REGARDING ELECTRICAL EQUIPMENT AND HVAC EQUIPMENT THAT SHALL BE REMOVED OR REUSED.
- P. PROVIDE ALL BRACING AND SHORING REQUIRED FOR PROPOSED CONSTRUCTION.
- Q. SCARIFY OR GRIND FLOORS AND FILL CRACKS AS NECESSARY TO PROVIDE SMOOTH, CLEAN SURFACE FOR NEW FLOORING (TYP.)
- R. G.C. TO PROVIDE, UNIFORM, LEVEL CONCRETE FLOOR AT A SINGLE ELEVATION WITHOUT DEPRESSIONS AND A MAXIMUM SURFACE VARIATION OF 1/4" OVER 10 FT. THROUGHOUT THE PREMISES READY TO ACCEPT FLOOR COVERING.
- S. G.C. TO PROVIDE TERMITE TREATMENT TO SLAB ON GRADE AT AREAS OF TRENCHING FOR NEW UTILITIES AND SLAB INFILL.
- T. THE GENERAL CONTRACTOR SHALL BE RESPONSIBLE FOR ENSURING THAT ALL SUBCONTRACTORS PLACE ALL CONSTRUCTION DEBRIS DIRECTLY IN TO CONSTRUCTION DUMPSTERS OR DUMP TRUCKS FOR REMOVAL. ALL DEMO'D MATERIALS ARE TO BE REMOVED FROM THE SITE.

KEYED NOTES

- REMOVE EXISTING LIGHT FIXTURE; EXISTING LIGHT FIXTURE, WIRING AND CONDUIT ABOVE TO BE REMOVED BACK TO PANEL.
- REMOVE FIRE ALARM WALL PACK. COORDINATE WITH ELECTRICAL DRAWINGS FOR NEW FIXTURES AND LOCATIONS.
- REMOVE EMERGENCY LIGHTING. COORDINATE WITH ELECTRICAL DRAWINGS AND LIGHTING PLAN FOR NEW FIXTURES AND LOCATIONS.
- EXISTING AHU ABOVE TO BE RELOCATED, SEE MECHANICAL DRAWINGS FOR NEW LOCATION AND RELOCATION DETAILS.
- EXISTING OPENING IN FLOOR SLAB. REFER TO STRUCTURAL DRAWINGS FOR NEW SLAB INFORMATION.
- EXISTING DEMISING WALL. REFER TO CONSTRUCTION PLAN FOR NEW WORK.
- RELOCATE EMERGENCY EXIT LIGHTING. COORDINATE WITH ELECTRICAL DRAWINGS AND LIGHTING PLAN FOR NEW LOCATIONS.
- EXISTING WATER PIPE TO REMAIN. SEE PLUMBING DRAWINGS FOR INFORMATION.
- EXISTING ELECTRICAL STUB IN, SEE ELECTRICAL DRAWINGS FOR INFORMATION.
- EXISTING SANITARY STUB IN, SEE PLUMBING DRAWINGS FOR INFORMATION.
- NEW OPENINGS FOR HVAC, SEE MECHANICAL AND STRUCTURAL DRAWINGS.
- EXISTING GREASE STUB IN, SEE PLUMBING DRAWINGS FOR INFORMATION.

1 ARCHITECTURAL DEMOLITION & EXISTING PLAN 1/4"=1'

SEAL:

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REVISIONS

**BAGELS
& CO.**

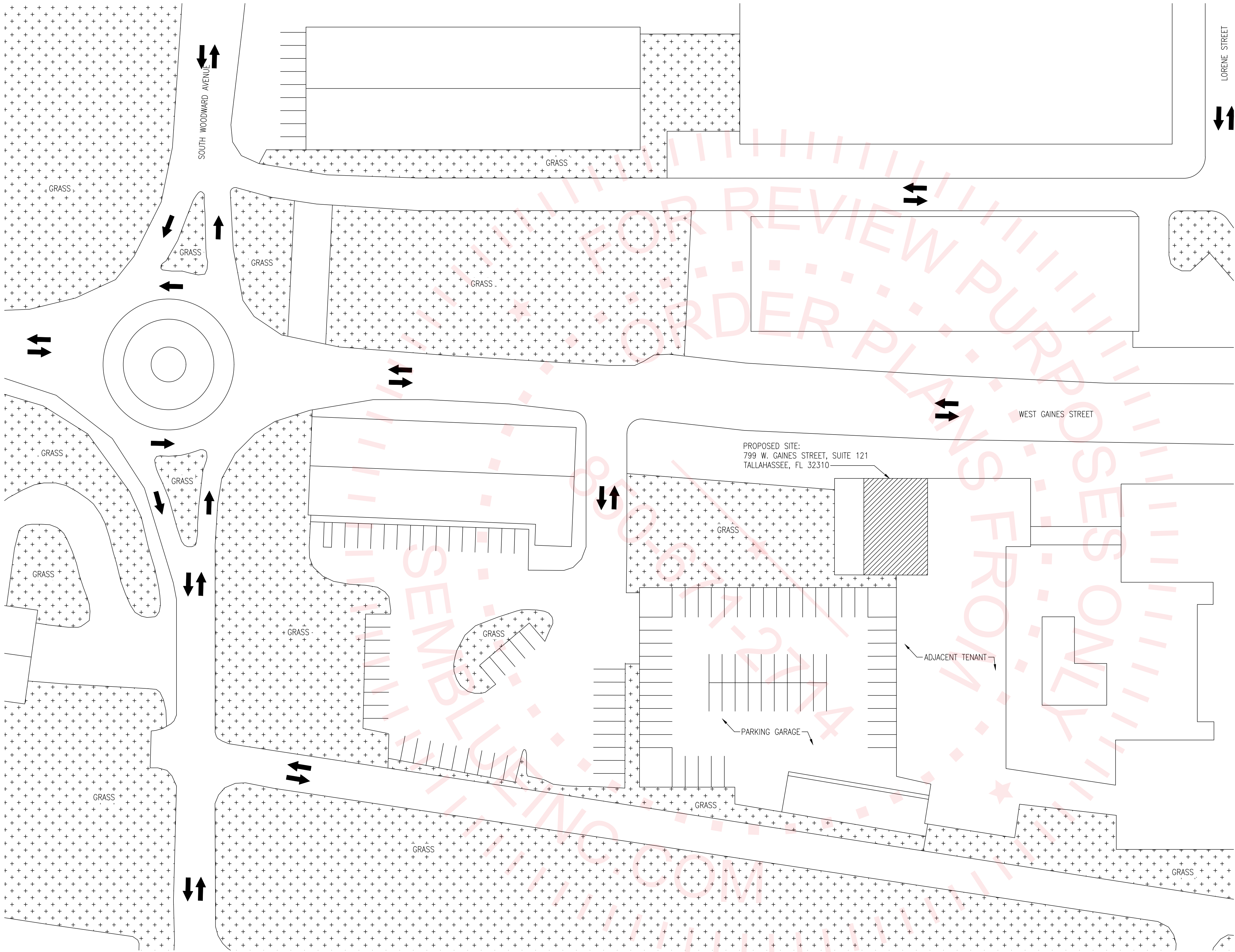
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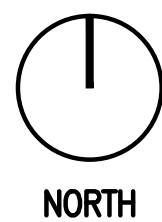
D1.0
ARCH. DEMO. & EXIST.
FLOOR PLAN

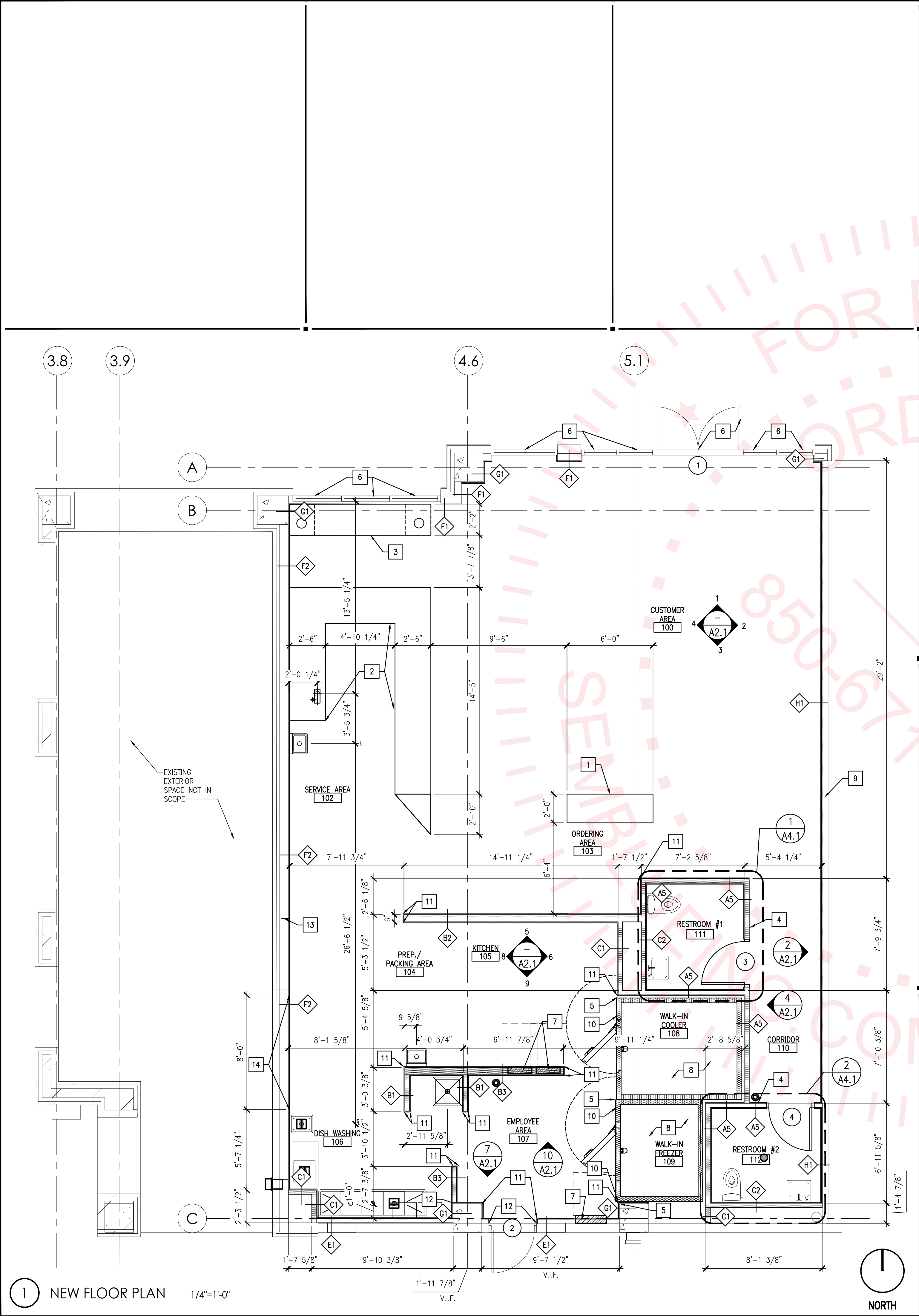
CHECKED: DC DRAWN: EC/WC



PROPERTY DATA:
LANDLORD
NAME: DISTRICT JOINT VENTURE LLC
ADDRESS: PO BOX 530292
MTN BROOK, AL 35253-0292
PARCEL ID: 2135230000030

NOTE:
THE GENERAL CONTRACTOR SHALL PHOTOGRAPH OR
VIDEOTAPE EXISTING CONDITIONS IN AND AROUND
THE STORE PRIOR TO BEGINNING CONSTRUCTION.
PARTICULAR ATTENTION SHOULD BE DIRECTED TO
THE ROOF AND ALL ASPHALT DRIVES AND PARKING
AREAS.





SYMBOLS LEGEND

- KEYED NOTES
- WALL TYPES - REF. SHEET A5.1
- DOOR TAG - REF. SHEET A6.1
- ROOM TAG
- ELEVATION TAG
- DETAIL TAG

WALL TYPE LEGEND

- EXISTING TO REMAIN
- NEW STUD WALL
- NEW COOLER / CHILLER WALL PANEL

GENERAL NOTES

- IN THE CASE OF DISCREPANCY BETWEEN THE ARCHITECT'S AND ENGINEER'S DRAWINGS, THE MORE STRINGENT, HIGHER QUALITY SHALL PREVAIL. CONTRACTOR IS TO INFORM ARCHITECT OF ANY DISCREPANCIES PRIOR TO SUBMITTING BID.
- VERIFY ALL DIMENSIONS OF WALLS, MILLWORK & FIXTURES PRIOR TO LAYOUT.
- G.C. IS RESPONSIBLE TO FIELD VERIFY ALL EXISTING CONDITIONS AND DIMENSIONS IN THE SPACE. FAILURE TO PROPERLY VERIFY EXISTING CONDITIONS DOES NOT RELIEVE THE G.C. OF ANY EXPENSES OR RESPONSIBILITIES RESULTING FROM THAT FAILURE.
- G.C. TO PATCH AND PAINT ALL G.W.B. EXPOSED AREAS AT EXISTING WINDOW WALLS & ALL EXISTING GYP. BD.
- ALL PLYWOOD BACKING/BLOCKING TO BE FIRE RESISTANT. G.C. IS RESPONSIBLE FOR PROVIDING ALL BRACING /BLOCKING REQUIRED BY CODE AND STRUCTURAL NECESSITY FOR ALL FIXTURES AFFIXED TO WALLS AND CEILINGS, DETAILED OR NOT. THIS INCLUDES WALL HUNG CABINETS, COUNTER TOPS, SHELVING, PLUMBING FIXTURES, ETC.
- SEE MEP DRAWINGS FOR ADDITIONAL NOTES AND DETAILS.
- ALL WORK TO COMPLY WITH ALL APPLICABLE LOCAL, STATE, AND FEDERAL BY-LAWS AND CODE REQUIREMENTS.
- G.C. TO PROTECT AND MAINTAIN EXISTING FIRE PROOFING AROUND COLUMN AND BEAMS, AND ALL STRUCTURAL MEMBERS, TYPICAL.
- INTERIOR DIMENSIONS ARE MEASURED TO FACE OF STUD, UNLESS NOTED OTHERWISE
- A CERTIFICATE OF OCCUPANCY SHALL BE REQUIRED PRIOR TO OCCUPANCY OF THE BUILDING
- FIRE EXTINGUISHER TYPES, SIZES, AND LOCATIONS ARE TO BE VERIFIED WITH THE LOCAL FIRE AUTHORITY. TYPE K FIRE EXTINGUISHERS PROVIDED NEAR THE KITCHEN EQUIPMENT AND IN THE PATH OF TRAVEL. REFERENCE G1.4 SHEET FOR ADDITIONAL NOTES
- FIELD MEASURE LOCATIONS IN WHICH NEW CONSTRUCTION MEETS EXISTING CONDITIONS TO ENSURE FINISHES ARE FLUSH WITH ADJACENT TO REMAIN
- NO TOXIC OR COMBUSTIBLE MATERIALS WITH A SMOKE DEVELOPMENT RATING OF GREATER THAN 50 ARE PERMITTED WITHIN WALLS.
- G.C. SHALL PROVIDE 5/8" WATER RESISTANT "GREENBOARD" AT NON-RATED WALLS WITHIN WET AREAS
- ALL DOORS ARE 4" OFF ADJACENT WALL, UNLESS NOTED OTHERWISE
- G.C. TO VERIFY THAT ALL EXISTING TENANT DEMISING WALLS ARE COMPLETE TO DECK AND SEALED TO COMPLETE RATED ASSEMBLY AS REQUIRED BY GOVERNING CODES.
- G.C. TO PROVIDE ALL NECESSARY SIGNS FOR TOILETS, STAIRS, ELEVATORS (WHERE APPLICABLE). SEE SHEET G1.3 FOR REFERENCE.

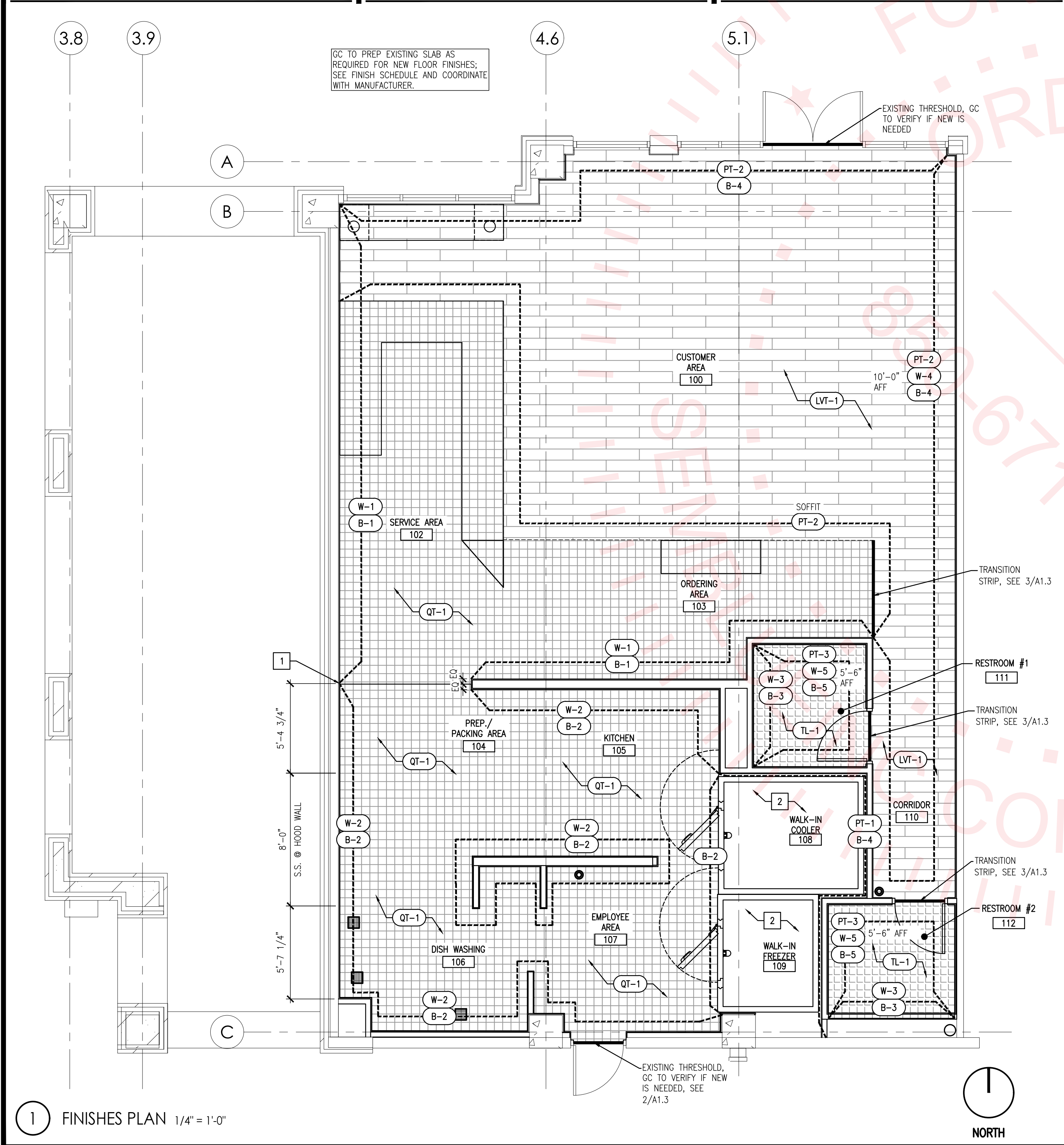
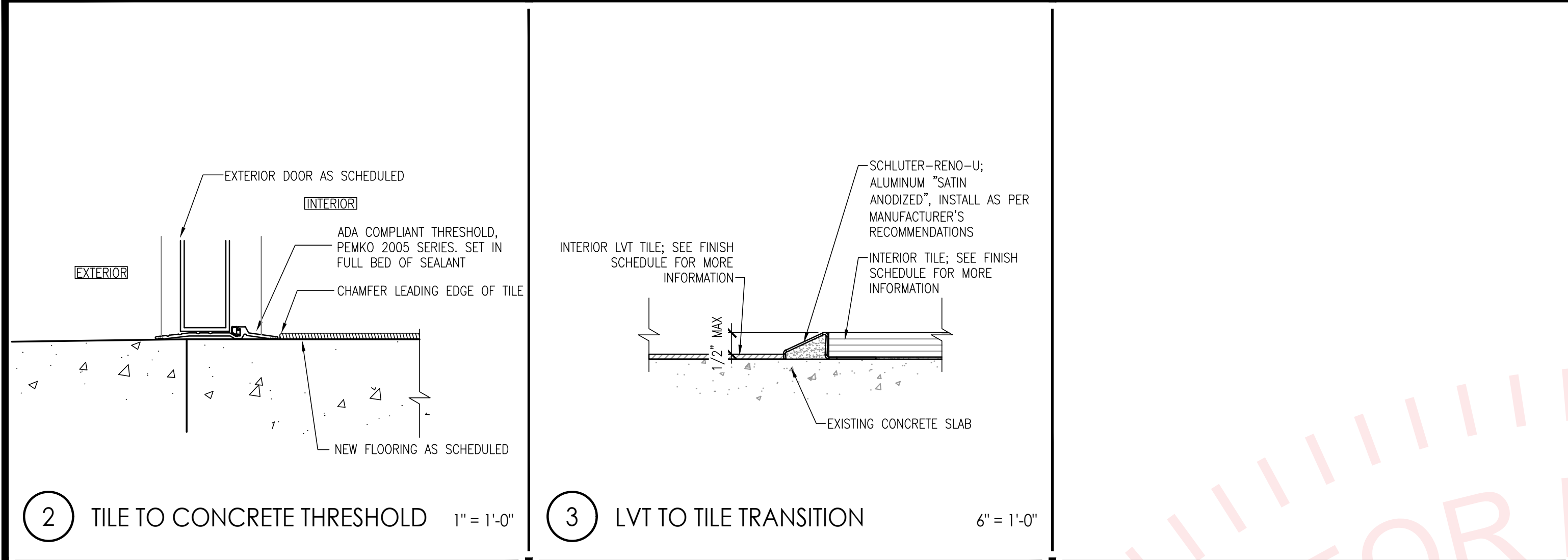
KEYED NOTES

- GC TO INSTALL ORDERING AREA COUNTER - REFERENCE MILLWORK DETAILS.
- GC TO INSTALL SERVICE AREA COUNTER - REFERENCE MILLWORK DETAILS.
- GC TO INSTALL COFFEE COUNTER - REFERENCE MILLWORK DETAILS.
- INSTALL ROOM SIGNAGE AT RESTROOM. SEE ADA STANDARDS DETAIL 11/G1.3.
- GALVALUME FLASHING BETWEEN COOLER AND WALL - REFERENCE DETAIL 4/A5.2.
- EXISTING STOREFRONT GLAZING SYSTEM AND ENTRY DOOR TO REMAIN. LANDLORD TO CONFIRM THAT ALL GLAZING COMPLIES WITH ALL CODE SAFETY REQUIREMENTS.
- ELECTRICAL PANEL; SEE ELECTRICAL DRAWINGS.
- WALK-IN COOLER AND FREEZER DETAILS AND SPECIFICATIONS BY OTHERS UNDER SEPARATE SUBMITTAL. WALK-INS TO HAVE A 2" AIR GAP BETWEEN COOLER WALLS AND WALL PARTITIONS.
- EXIST DEMISING WALL TO BE MIN 1 HR. FIRE RATED AS REQUIRED BY LOCAL CODES. GC TO MAINTAIN RATING AT ALL TIMES DURING DEMO/CONSTRUCTION.
- INSTALL WALL BASE AT BOTTOM OF COOLER WALLS - REFERENCE 5/A5.2.
- INSTALL FULL HEIGHT STAINLESS STEEL SCHLUTER CORNER PROTECTORS. SET ABOVE BASE. APPLY WITH CONSTRUCTION ADHESIVE AND SCREWS. REFER TO DETAIL 1/A5.2 OR SIMILAR. (HALF PARTITIONS AT MOP SINK TO HAVE CAP GUARDS TO 4'-0")
- NEW WALL FINISH TO ALIGN WITH EXISTING WALL.
- TRANSITION BETWEEN WALL FINISHES. SEE SHEET A1.3 FOR LOCATION.
- S.S. WALL PANEL @ HOOD WALL LOCATION.

INTERPLAN
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AR99238
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& CO.
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A1.2
CONSTRUCTION FLOOR
PLAN
CHECKED: DC DRAWN: WC



KEY NOTES

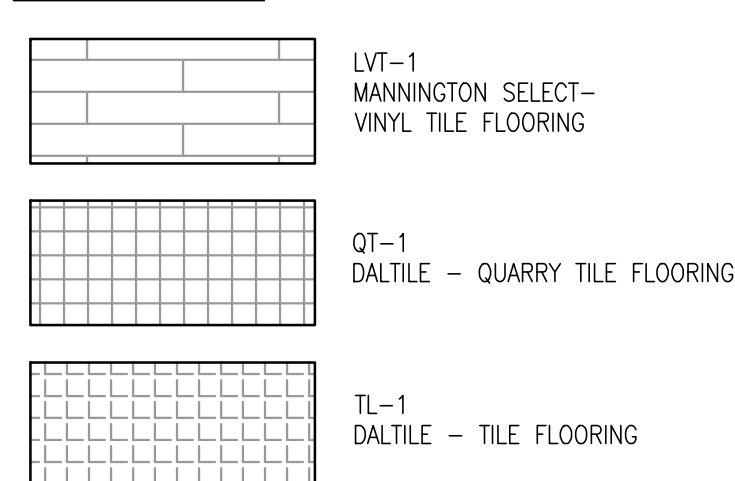


1. LOCATION OF WALL TYPE TRANSITION- COORDINATE WITH CONSTRUCTION PLAN A1.2.
2. COOLER WALL FINISHES PER MANUFACTURER.

LEGEND

FINISH SYMBOLS

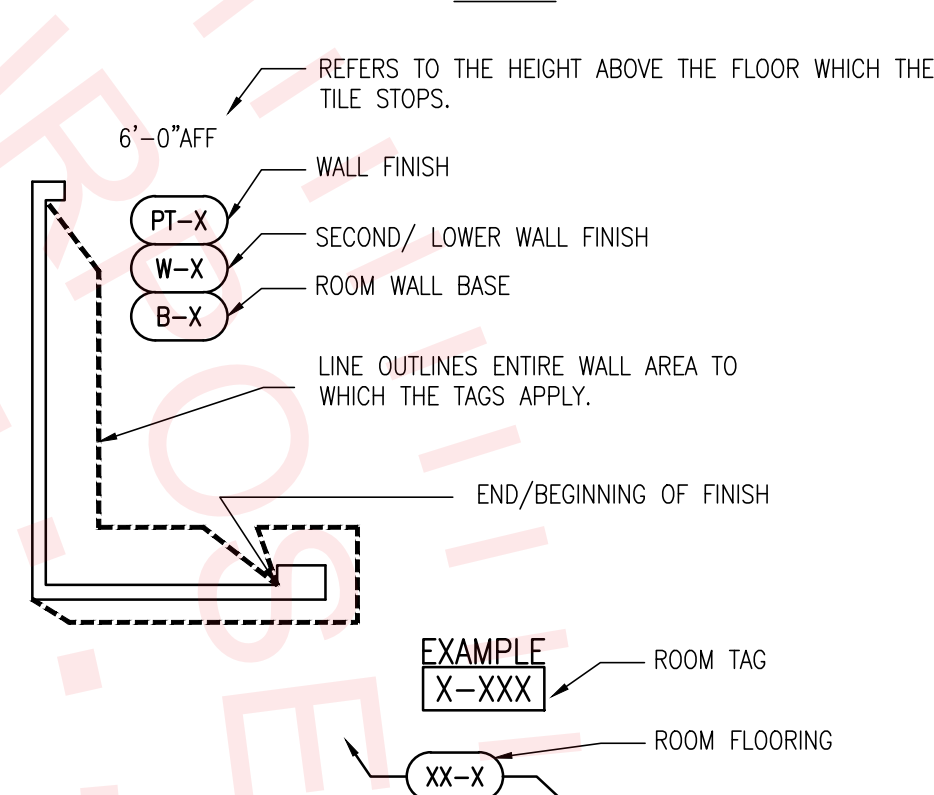
GENERAL ROOM FINISH



SYMBOL LEGEND:

FINISH TAGS MATCH NUMBER ON MATERIAL & FINISH LEGEND:

EXAMPLE:



GENERAL NOTES

- NOTIFY THE ARCHITECT/PROJECT MANAGER IMMEDIATELY FOR CLARIFICATION IF THERE ARE ANY DISCREPANCIES BETWEEN THIS SHEET AND THE FINISH SCHEDULE.
- REFER TO WALL SECTIONS AND ELEVATIONS FOR ADDITIONAL INFORMATION AND REQUIREMENTS.
- INTERIOR WALL AND CEILING FINISH MATERIALS SHALL BE INSTALLED IN ACCORDANCE WITH ALL APPLICABLE CODES.
- FINISHES SHALL MEET FLAME SPREAD AND SMOKE DEVELOPMENT RATINGS FOR THEIR USE AS REQUIRED BY GOVERNING CODES.
- TAPE, BED, FLOAT, AND SAND GWB. JOINTS. FINISH TO A LITE ROLLED TEXTURE AT ALL PAINTED AREAS. PAINT (1) COAT OF PRIMER AND (2) COATS OF COLOR.
- G.C. SHALL CAULK JOINT BETWEEN CEILING GRID AND FRP WALL PANELS IN ALL KITCHEN AREAS.
- G.C. SHALL PREP AND MAKE SMOOTH EXISTING CONCRETE SLAB PRIOR TO NEW FLOOR FINISH INSTALLATION.
- HOLLOW METAL DOORS AND FRAMES SHALL BE PRIMED AND PAINTED (UNLESS NOTED OTHERWISE).
- CHANGES IN LEVEL UP TO 1/4" BETWEEN DIFFERENT FLOOR FINISHES CAN BE MADE VERTICAL AND WITHOUT EDGE TREATMENT. CHANGES IN LEVEL OF 1/4" - 1/2" SHALL BE MADE WITH A BEVELED TRANSITION OR OTHER MATERIAL WITH A SLOPE NO GREATER THAN 1:2. CHANGES IN LEVEL GREATER THAN 1/2" SHALL BE ACCOMPLISHED BY MEANS OF A RAMP. VERIFY FINISH OF TRANSITION STRIP MATERIAL WITH OWNER.
- G.C. SHALL PLACE ORDERS FOR ALL FINISHES, MATERIALS, EQUIPMENT, ETC. AT THE START OF THE PROJECT. SUBSTITUTIONS ARE NOT ACCEPTED FOR ANY ITEMS, UNLESS NOTED OTHERWISE. OWNER MUST BE INFORMED OF LEAD TIME PROBLEMS WITHIN THE FIRST TWO WEEKS OF PROJECT.
- G.C. SHALL PROVIDE SERVICES FOR FINAL CLEAN UP, PRIOR TO TURNING STORE OVER, TO INCLUDE MOPPING AND BROOM CLEAN UP.
- NEW / EXISTING FLOOR FINISHES NOT TO EXCEED CHAMFERED 1/2" MAX. VERTICAL OFFSET.
- NO TOXIC OR COMBUSTIBLE MATERIALS TO BE USED ABOVE FINISH CEILING LINE.
- SEE PLUMBING DRAWINGS FOR DIMENSIONS AND LOCATIONS OF ALL FLOOR DRAINS AND FLOOR SINKS.
- COORDINATE FINISH FLOORING WITH WALK-IN COOLER/FREEZER MANUFACTURER.
- BOH FRP/CORNER GUARD AREAS: FRP W/ M370 TOP EDGE AND M350 INSIDE CORNERS. USE SEAM JOINTS BETWEEN SECTIONS. USE FULL HEIGHT, 2-1/2" WIDE, STAINLESS STEEL CORNER GUARDS AT ALL OUTSIDE CORNERS. GUARDS SECURED WITH CONSTRUCTION ADHESIVE NO VISIBLE FASTENERS. SEE DETAIL 1/A5.2

FINISH SCHEDULE

TAG	DESCRIPTION	MANUFACTURER	SPECIFICATION/COLOR	SIZE	FINISH	NOTES
FLOORS						
LVT-1	LUXURY VINYL TILE W/ MICRO-BEVELED EDGES	MANNINGTON COMMERCIAL	MANNINGTON SELECT/STONE/FIERA CARBON GRAY MSC222	9 X 18 X .098"	-	CUSTOMER AREA
QT-1	QUARRY TILE FLOOR	DALTILE	0042-ARID GRAY	6 X 6 X 1/2"	-	PREP./ PACKING AREA; KITCHEN; DISH WASHING; EMPLOYEE AREA GROUT: LATICRETE - COLOR RAVEN
TL-1	TILE FLOOR	DALTILE	WHITE	2 X 2	-	0.5 SCOF MIN.
WALLS						
W-1	WALL TILE	DALTILE	0190 ARCTIC WHITE GLOSSY	3 X 6 X 3/8	-	GROUT: LATICRETE - COLOR RAVEN
W-2	FRP APPLIED WALL FINISH	CRANE COMPOSITES	GLASBORD- WHITE EMBOSSED FINISH	4 X 8	-	INSTALL WITH MATCHING KEMLIGHT MOLDINGS AT ALL SEAMS, CORNERS, AND EXPOSED EDGES
W-3	WALL TILE	DALTILE	D617 - ARCTIC WHITE	2 X 2	-	GROUT: LATICRETE - COLOR RAVEN
W-4	RECLAIMED WOOD	STIKWOOD	WEATHERED WOOD WHITE	4'-0X5'X3/16"	-	SEE ELEVATIONS; WOOD TRIM AT 5'-0" IN RESTROOMS
W-5	WALL TILE	DALTILE	COLOR WHEEL CLASSIC: GALAXY 1469	6 X 6 X 5/16"	GLOSSY	BULLNOSE TO MATCH A\$469, GROUT: LATICRETE- COLOR RAVEN
PT-1	PAINT FINISH	BENJAMIN MOORE	ATRIUM WHITE OC-145	-	EGG SHELL	
PT-2	PAINT FINISH	BENJAMIN MOORE	DEEP ROYAL 2016-10	-	EGG SHELL	ALIGN WITH TOPS OF WINDOWS
PT-3	PAINT FINISH	BENJAMIN MOORE	ECLIPSE 2132-40	-	EGG SHELL	
BASE						
B-1	SANITARY COVE BASE	DALTILE	MATCH WALL	6 X 6 SHADE NUMBER: S3619T	-	INSTALL MATCHING INTERIOR AND EXTERIOR CORNER AS REQUIRED
B-2	SANI WALL BASE	CRANE COMPOSITES	MATCH ADJACENT FRP COLOR	-	-	SANI-COVER RISER AT TILE FLOORS; SANI-COVE BASE AT LVT AND CONCRETE
B-3	BUILD UP COVE BASE	DALTILE	MB5B ARCTIC WHITE	2 X 2	-	GROUT: LATICRETE COLOR RAVEN
B-4	VINYL COVE BASE	JOHNSONITE	TOE TRADITIONAL VINYL-BLACK	2.5"	-	
B-5	TILE COVE BASE	DALTILE	COLOR WHEEL CLASSIC: GALAXY A3601	6 X 6	-	MATCH W-5 TILE
MILLWORK (REFER TO SECTION A5.3)						
M-1	COUNTER TOP	WILSONART	KASTORIA Q4071 QUARTZ	-	-	GC TO PROVIDE SAMPLE FOR EQUIVALENT APPROVAL TO OWNER
M-2	LAMINATE	LAMINART	2449-T TUXEDO TEXTURED FINISH	-	-	GC TO PROVIDE SAMPLE FOR EQUIVALENT APPROVAL TO OWNER

SEAL:

THIS DOCUMENT IS NOT
FOR REGULATORY
APPROVAL, PERMITTING,
OR CONSTRUCTION.

NO DATE REMARKS
REVISIONS

**BAGELS
& CO.**

BAGEL & CO
TALLAHASSEE

799 W. GAINES STREET
SUITE 121
TALLAHASSEE, FL 32310

PROJECT NO: 2023.0719
DATE: 04/12/2024

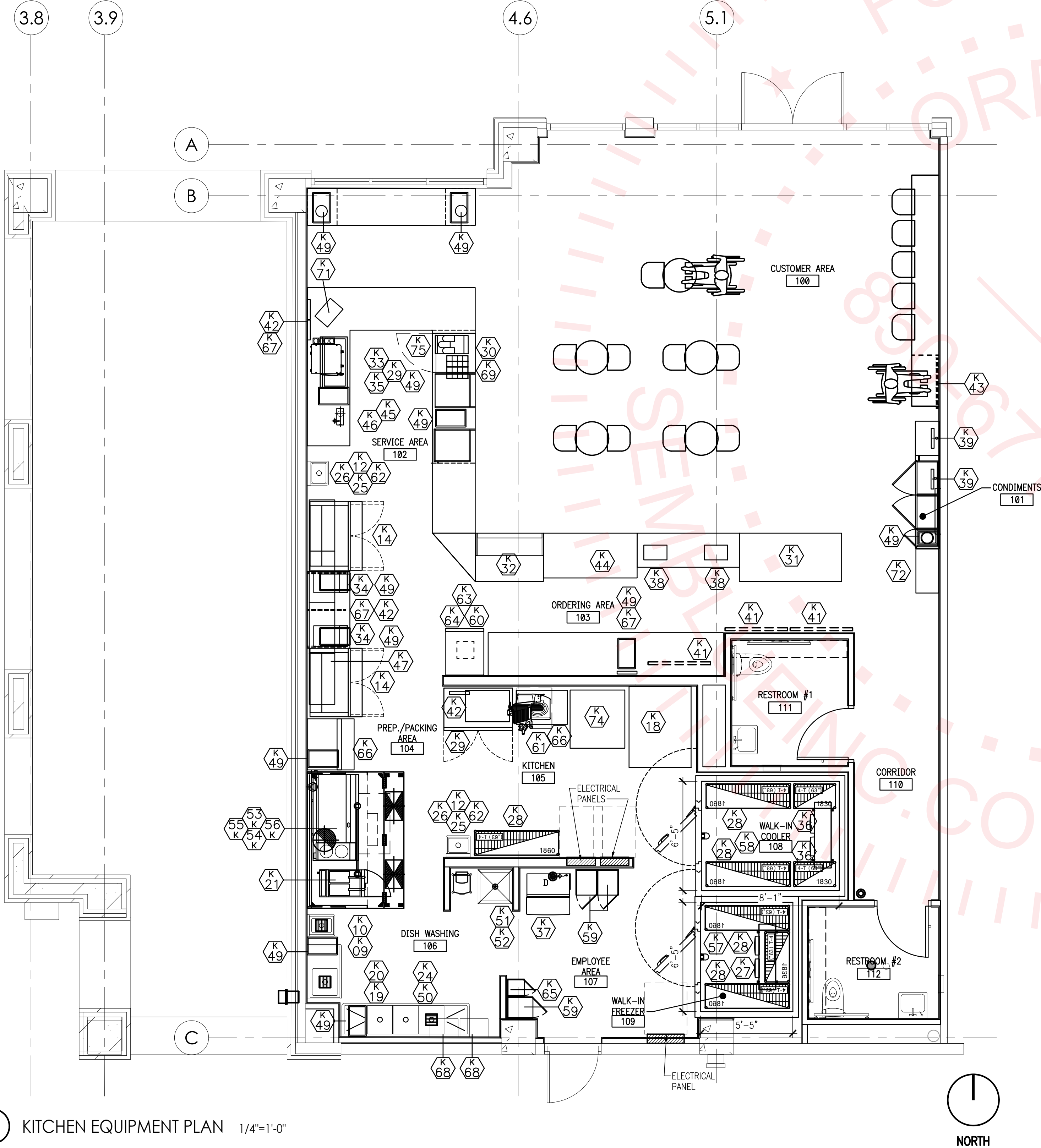
A1.3
WALL AND FLOOR FINISH
PLAN

CHECKED: DC DRAWN: EC/WC

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FIXTURE / EQUIPMENT NOTES

- ALL ITEMS LISTED UNLESS NOTE OTHERWISE (U.N.O) ARE TENANT PROVIDED, CONTRACTOR INSTALLED. G.C. TO VERIFY ALL ITEMS AND/OR SERVICES TO BE PROVIDED BY TENANT.
- MILLWORK ITEMS TO BE ORDERED THROUGH EQUIPMENT PROVIDER, REFERENCE FIXTURE PLAN AND MILLWORK DETAILS.
- EQUIPMENT SUPPLIER TO PROVIDE AND INSTALL COOLANT LINES. EQUIPMENT SUPPLIER TO INSTALL WALK-IN UNIT(S). G.C. AND EQUIPMENT SUPPLIER TO COORDINATE SCHEDULE AND ROOF PENETRATIONS.
- WALK-IN UNIT AND CONDENSATE LINES INSTALLED BY EQUIPMENT SUPPLIER.
- G.C. TO PROVIDE INSULATION AT ALL WALK-IN UNIT PENETRATIONS.
- ALL KITCHEN SHELVEING PROVIDED BY G.C. AND INSTALLED BY G.C.
- DECOR ITEMS, MILLWORK PACKAGE, AND EQUIPMENT TO BE INSTALLED BY G.C. U.N.O.
- EQUIPMENT FLOOR PLAN TAKES PRECEDENCE OVER EQUIPMENT SCHEDULE FOR LAYOUT, SIZES, AND CONFIGURATION. TENANT TO CONFIRM EQUIPMENT ORDER WITH SUPPLIER. EQUIPMENT SUPPLIER TO PROVIDE AND INSTALL BACK FOR WALK-IN CONDENSING UNIT(S). G.C. TO PROVIDE ROOF PENETRATION(S). COORDINATE LOCATIONS WITH EQUIPMENT SUPPLIER.

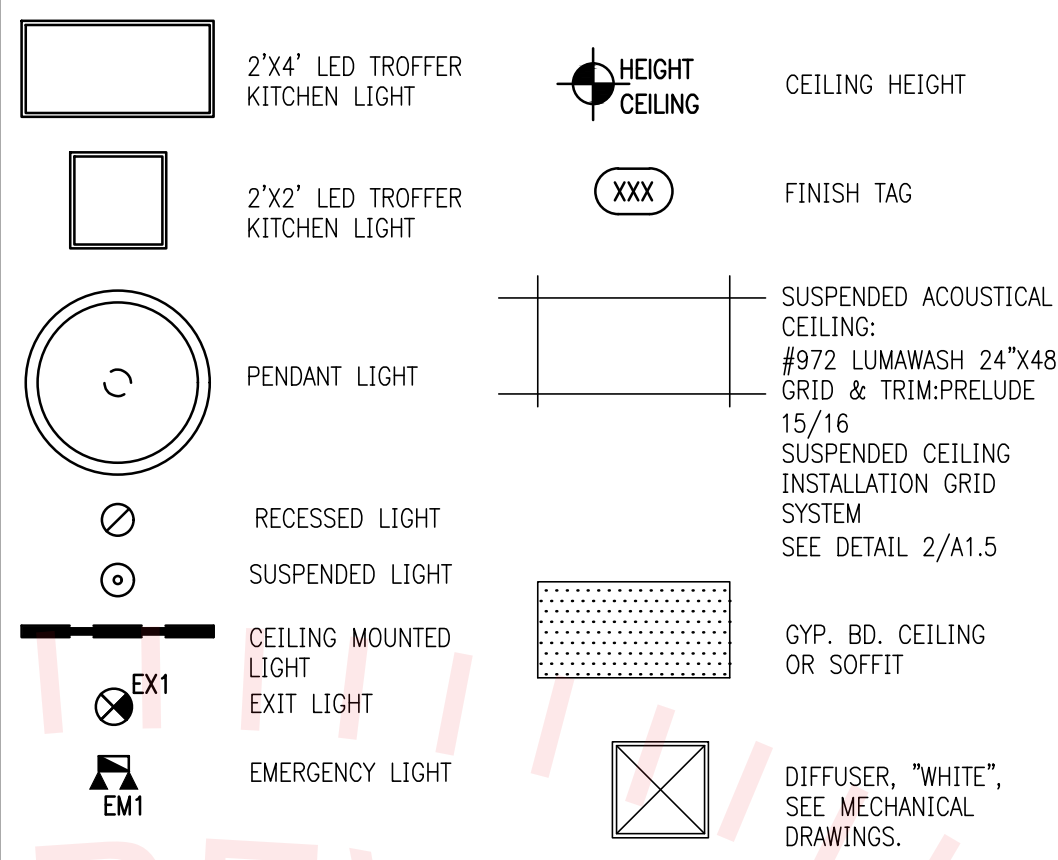


KITCHEN EQUIPMENT SCHEDULE

ITEM NO	EQUIPMENT CATEGORY	QTY.	HOT WATER SIZE (IN)	COLD WATER SIZE (IN)	DIRECT DRAIN SIZE (IN)	INDIRECT DRAIN SIZE (IN)	GAS SIZE (IN)	MBTUH	HP	KW	AMPS	PLUG	DIRECT	VOLTS	PHASE	NEMA	MANUFACTURER	MODEL NUMBER	REMARKS
09	8" CENTER SINK FAUCET	1	1/2	1/2	-	-	-	-	-	-	-	-	-	-	-	-	KROWNE	12-808L	-
10	FREE STANDING SINK	1	-	-	-	3 1/2	-	-	-	-	-	-	-	-	-	-	SAPPHIRE	SMS1014	DUMP & HAND SINK + DISPOSER, PROVIDE ST. STL. WINGS
11	NOT USED	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
12	SOAP DISPENSER	2	-	-	-	-	-	-	-	-	-	-	-	-	-	-	BOBRICK	B-26607	SURFACE MOUNTED LIQUID SOAP DISPENSER 500 ML
14	4' SANDWICH PREP. TABLE	2	-	-	-	-	-	-	1/7	-	2.3	X	-	115/60/1	-	5-15P	ATOSA	MSF8302GR	-
18	RACK OVEN	1	-	1/2	1	-	3/4	140	-	-	88.4	-	-	-	-	-	EMPIRE BAKERY EQUIPMENT	LFR-FOX-15E	-
19	PREP SINK	1	-	-	-	3 1/2	-	-	-	-	-	-	-	-	-	-	SAPPHIRE	SMS-1818R	-
20	8" CENTER SINK FAUCET	1	1/2	1/2	-	-	-	-	-	-	-	-	-	-	-	-	KROWNE	12-808L	-
21	GAS FRYER	1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	ASBER	AEF-50-S	-
24	8" 3 COMPARTMENT SINK FAUCET	1	1/2	1/2	-	-	-	-	-	-	-	-	-	-	-	-	KROWNE	17-108WL	ROYAL SERIES
25	HAND SINK	2	-	-	2	-	-	-	-	-	-	-	-	-	-	-	GREEN WORLD	TSS-1-H	PROVIDE ST. STL. WINGS AT SIDES AND BACK
26	4" CENTER HAND SINK FAUCET	2	1/2	1/2	-	-	-	-	-	-	-	-	-	-	-	-	GREEN WORLD	CD-804L	-
27	WIRE SHELVEING	1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	WINCO	VEXS-1836	-
28	WIRE SHELVEING	5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	WINCO	VEXS-1860	-
29	36" ICE BIN	2	-	-	-	-	-	-	-	-	-	-	-	-	-	-	NBR EQUIPMENT	IB-3618	-
30	ESPRESSO MACHINE	1	-	-	-	-	-	-	-	-	15	X	-	-	-	-	EVERSYS SWISS ESPRESSO SYSTEMS	CAMEO C'2M	-
31	GRAB & GO CASE	1	-	-	-	-	-	-	1/2	-	15	X	-	115	-	-	HOWARD-MCCRAY	SC-0040E-4L-B-LED	-
32	CREAM CHEESE CASE	1	-	-	-	-	-	-	1/4	-	5	X	-	120	-	5-15P	OUEDA USA	RDCH 6	-
33	COFFEE MACHINE	1	-	-	-	-	-	-	6000	25	X	-	-	120/240	1	-	BUNN	ICB TWIN SH PE	-
34	TOASTER	2	-	-	-	-	-	-	3/4	1800	-	X	-	240	-	5-15P	CHEF-BUILT	CCT-500	-
35	COFFEE GRINDER- DRIP	1	-	-	-	-	-	-	-	-	11	X	-	120	1	5-15P	BUNN	G3 HD RED - 22100.0001	-
36	WIRE SHELVEING	2	-	-	-	-	-	-	-	-	-	-	-	-	-	-	WINCO	VEXS-1830	ORDER INDIVIDUAL COMPONENTS IF SET IS NOT AVAILABLE
37	ICE MAKER	1	-	-	-	-	-	4.2	-	7.1	8.4	X	-	115/60/1	1	-	HOSHIZAKI	KM-301BAJ	-
38	POINT OF SALE DEVICE	2	-	-	-	-	-	3/4	450	14.5	X	-	-	110	1	5-15P	TOAST	-	PROVIDE CAT6 LAN CABLE
39	SELF ORDERING STATION	2	-	-	-	-	-	3/4	450	14.5	X	-	-	110	1	5-15P	TOAST	-	TOUCH SCREEN - PROVIDE CAT6 LAN CABLE
41	MENU DISPLAY	3	-	-	-	-	-	-	-	-	-	-	-	110	-	-	-	-	FLAT PANEL LED SCREEN - PROVIDE CAT6 LAN CABLE
42	KDS	3	-	-	-	-	-	-	-	-	-	-	-	110	-	-	-	TOAST	TOUCH SCREEN - PROVIDE CAT6 LAN CABLE & WALL MOUNT
43	NEON SIGN	1	-	-	-	-	-	-	-	-	-	X	-	-	-	-	-	-	REFER TO ELEVATIONS FOR LOCATION ON WALL
44	BAGEL DISPLAY	1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	ALL SOUTHERN FABRICATORS	EBAGEL-DISP 45786	-
45	DROP-IN WATER STATION	1	-	1/2	1/2	-	-	-	-	-	-	-	-	-	-	-	KROWNE	WS-1	-
46	6" STAINLESS STEEL SURFACE MOUNT GLASS RINSER	1	1/2	-	3 1/2	-	-	-	-	-	-	-	-	-	-	-	MICRO MATIC USA	MM-5821	-
47	BAGEL BINS	4	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
49	GARBAGE CANS	11	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
50	3 COMPARTMENT SINK	1	-	-	-	3 1/2	-	-	-	-	-	-	-	-	-	-	SAPPHIRE	SMS-31818D	-
51	MOP SINK	1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	ACORN ENGINEERING	1630	-
52	MOP SINK FAUCET	1	1/2	1/2	-	-	-	-	-	-	-	-	-	-	-	-	KROWNE	16-127	-
53	5' CHEF BASE	1	-	-	-	-	-	1/7	-	2.3	X	-	-	115/60/1	-	-	ATOSA	MGF8452GR	-
54	GAS GRIDDLE	1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	COOKRITE BY ATOSA	ATTG-48	-
55	GAS COOKTOP	1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	COOKRITE BY ATOSA	ACHP-2	-
56	96" HOOD	1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	CAPTIVEAIRE	5424 ND-2-PSP-F	SEE MECHANICAL DRAWINGS
57	WALK-IN FREEZER	1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
58	WALK-IN COOLER	1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
59	EMPLOYEE LOCKER - 15"x18" DOUBLE STACK	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-	BUNN	-	-
60	REACH-IN FREEZER	1	-	-	-	-	-	1/2	-	6.3	-	-	-	115/60/1	-	-	ATOSA USA	MBF8501GR	REFRIGERANT: R290
61	MANUAL MEAT SLICER	1	-	-	-	-	-	1/2	-	3	-	-	-	115/60/1	-	5-15	GLOBE	G12	-
62	PAPER TOWEL DISPENSER	2	-	-	-	-	-	-	-	-	-	-	-	-	-	-	BOBRICK	B-4262	SURFACE MOUNTED PAPER TOWEL DISPENSER
63	WIRELESS ROUTER	1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	SEE ELECTRICAL SYSTEMS SHEET E3.0 FOR DETAILS
64	MODEM	1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	SEE ELECTRICAL SYSTEMS SHEET E3.0 FOR DETAILS
65	LOCKERS	1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
66	STAINLESS STEEL PREP TABLE	2	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	30x24 WITH BOTTOM SHELF
67	TICKET PRINTER	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-	EPSON	-	PROVIDE WALL MOUNT
68	TANKLESS WATER HEATER	2	-	-	-	-	-	-	-	-	-	-	-	-	-	-	RHEEM	RTGH-CM95DVLN	NATURAL GAS COMMERCIAL CONDENSING TANKLESS WATER HEATER
69	SYRUP BOTTLE RACKS	1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
71	TEA BAG DISPLAY	1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
72	CHIP MERCHANDISING	1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
74	PROOFER	1	1/2	-	3/4	-	-	-	5	20	-	-	-	208	3	-	EMPIRE BAKERY EQUIPMENT	LRP1-40	-
75	UNDERCOUNTER REFRIGERATOR	1	-	-	-	-	-	1/7	-	2.3	X	-	-	115/60/1	-	5-15P	ATOSA	MGF8401GRL	REFRIGERANT: R290

Light Fixture Schedule									
Type	Description	Manufacturer	Lamp	Watts	Volts	Remarks	Location		
A AE A2 A2E	A 2x2 LAY-IN LED FIXTURE AE- 2x2 SAME AS ABOVE WITH EMERGENCY BATTERY PACK-REFER TO ELECTRICAL DRWGS A2- 2x4 LAY-IN LED FIXTURE A2E- 2x4 SAME AS ABOVE WITH EMERGENCY BATTER PACK- REFER TO ELECTRICAL DRWGS	KENALL-SIMPLESEALX CSEDO SERIES- TUNABLE WHITE LED	LED	67	120/277	4H HOUSING -24-GA CRS FH (FOOD ZONE)	KITCHEN		
B BE	LARGE CIRCULAR PENDANT FIXTURES 46" DIA. BE SAME AS ABOVE WITH EMERGENCY BATTER PACK- REFER TO ELECTRICAL DRWGS	EUREKA CALDERA 4246-46	LED- COLOR TEMP 4000K	-	120	PROVIDE DIMMING- BLACK FINISH	SEATING AREA		
C	RECESSED FIXTURE	HALO COMMERCIAL PR6	LED- COLOR TEMP	-	120/277	PROVIDE MOUNTING BARS FOR GRID CEILING	RESTROOMS, SERVICE AREA		
D DE	4 FT LINEAR LED FIXTURE DE- EMERGENCY BATTERY BACK UP	LUMINOSO LLS SERIES LED LINEAR FIXTURE	LED 4000K	-	120/277		SERVICE AREA		
E	LED TAPE LIGHT	COMMERCIAL ELECTRIC LED TAPE LIGHT	COOL WHITE	VARIES	120		UNDER CABINET LIGHTS		
F FE	PENDANT POINT LIGHT FE-SAME AS ABOVE WITH EMERGENCY BATTER PACK- REFER TO ELECTRICAL DRWGS	EUREKA SCOUT 4048-18	LED- NARROW BEAM ANGLE 24" COLOR TEMP 4000K	-	120	PROVIDE DIMMING- BLACK FINISH CLEAR GLASS LENS BLACK CABLE ROLIND CANOPY MOUNTING BLACK FINISH REFER TO ELECTRICAL	CORRIDOR		
ER1	EXTERIOR EMERGENCY LIGHTING						FRONT AND REAR EXTERIOR		
EX1	EMERGENCY EXIT LIGHT					REFER TO ELECTRICAL	THROUGHOUT		
EM	EGRESS LIGHTING					REFER TO ELECTRICAL	THROUGHOUT		
	WALL MOUNTED SPEAKER	SONOS	-	-	-	COORDINATE WITH RECEPTACLE LOCATIONS, REFER TO	THROUGHOUT		
	CEILING OR WALL MOUNTED SECURITY CAMERA					PROVIDE LOW VOLTAGE WIRING, REFER TO ELECTRICAL	THROUGHOUT		

LEGEND

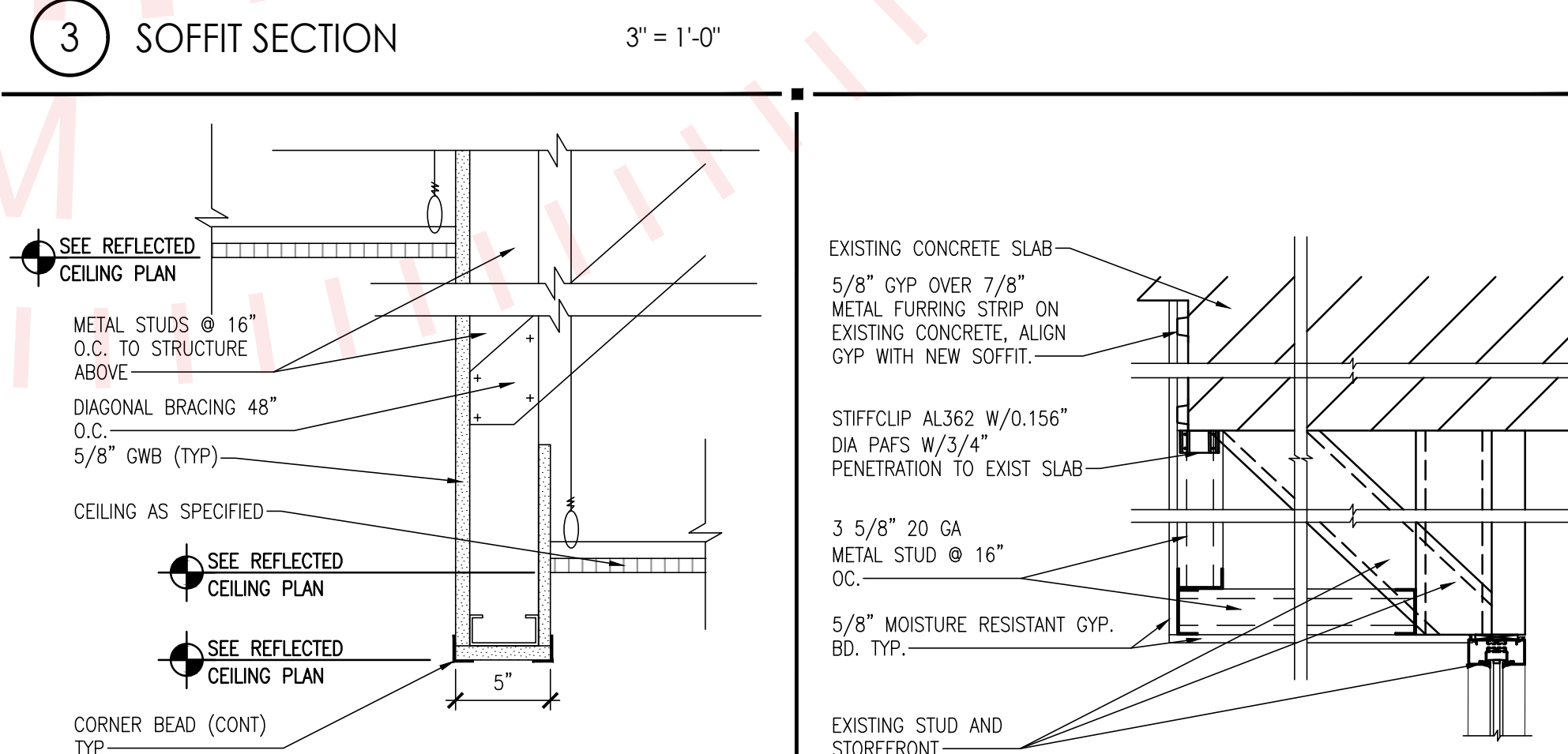
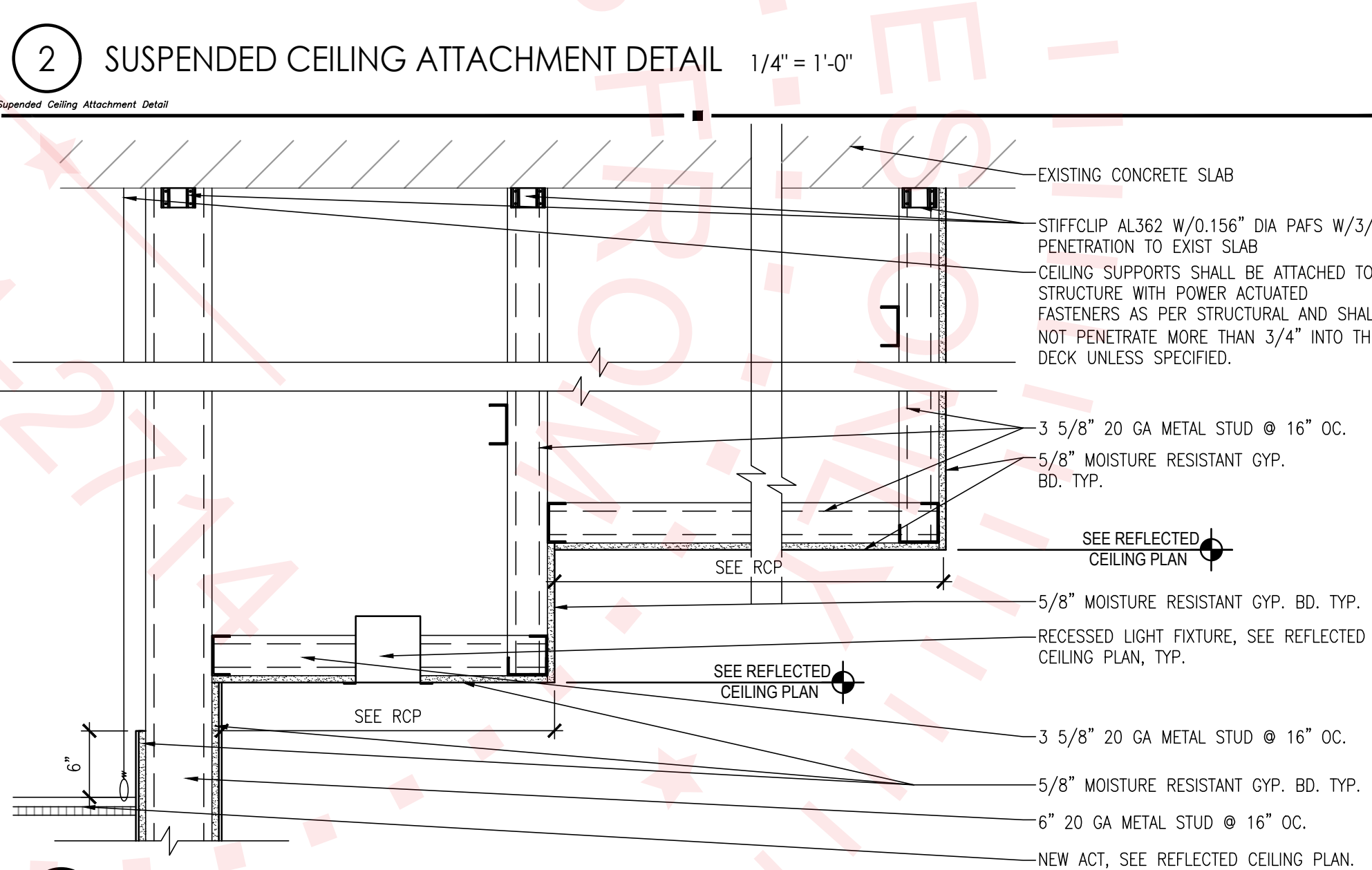
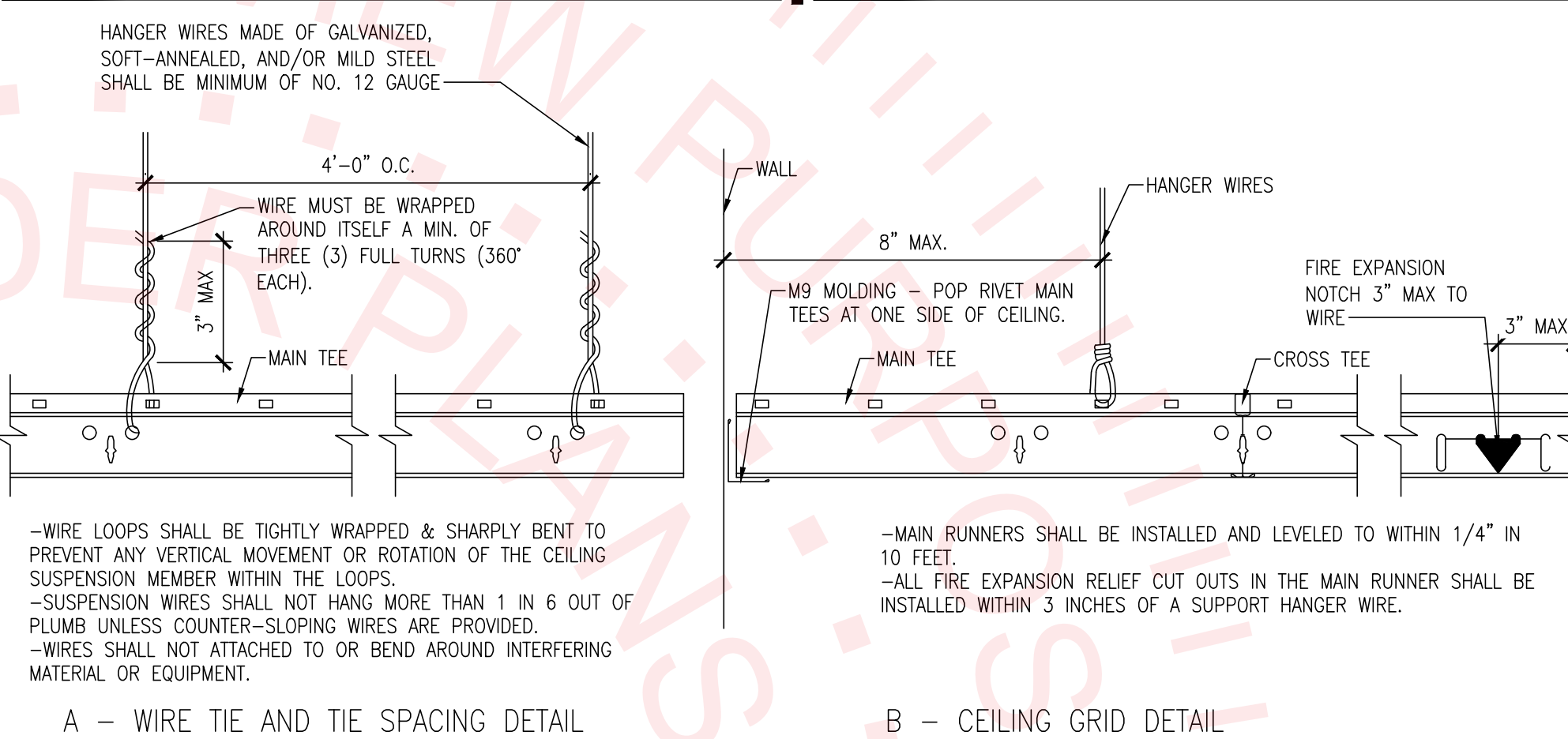
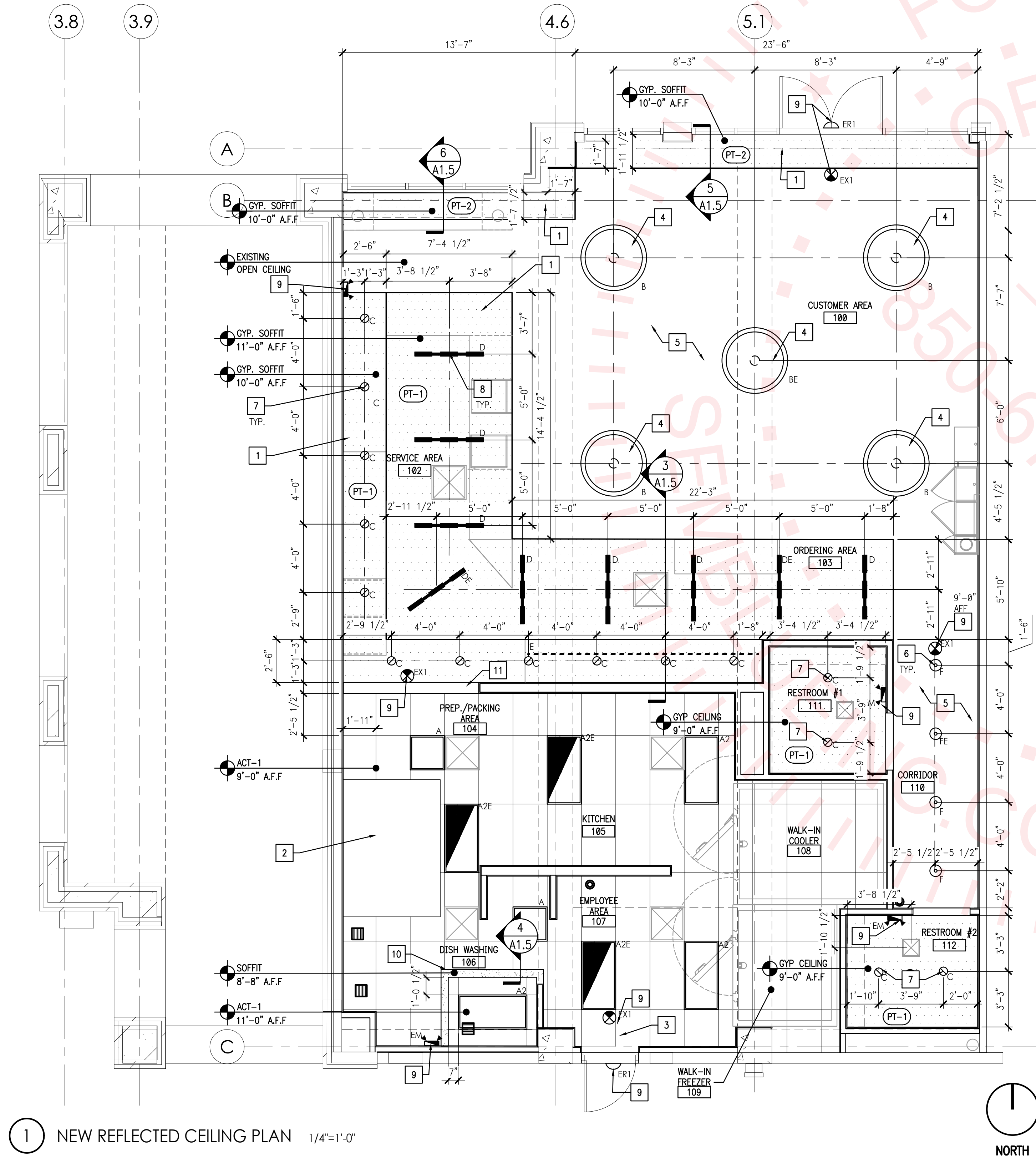


KEYED NOTES

- SOFFIT/ BULKHEAD. REFER TO A.F.F. TAGS.
- KITCHEN HOOD, REFER TO MECHANICAL DRAWINGS.
- AIR CURTAIN, SEE MECHANICAL DRWGS.
- PENDANT LIGHT ABOVE SEATING AREA. BOTTOM AT 10'-0" AFF.
- EXPOSED CEILING AREA TO BE PAINTED (PT-2), INCLUDING DUCTWORK, CONDUITS, ETC. U.N.O. REFER TO INTERIOR ELEVATIONS AND FINISH SCHEDULE.
- SUSPENDED PENDANT LIGHTING ABOVE CORRIDOR. BOTTOM AT 8'-6" AFF.
- RECESSED LIGHTING IN GYP CEILING/SOFFIT, SEE ELECTRICAL.
- SURFACE MOUNTED LIGHTING, SEE ELECTRICAL.
- NEW EMERGENCY EXIT, SEE ELECTRICAL.
- HEADER TO BE PAINTED PT-1.
- HEADER WITH STAINLESS STEEL WRAP, 8'-6" AFF.

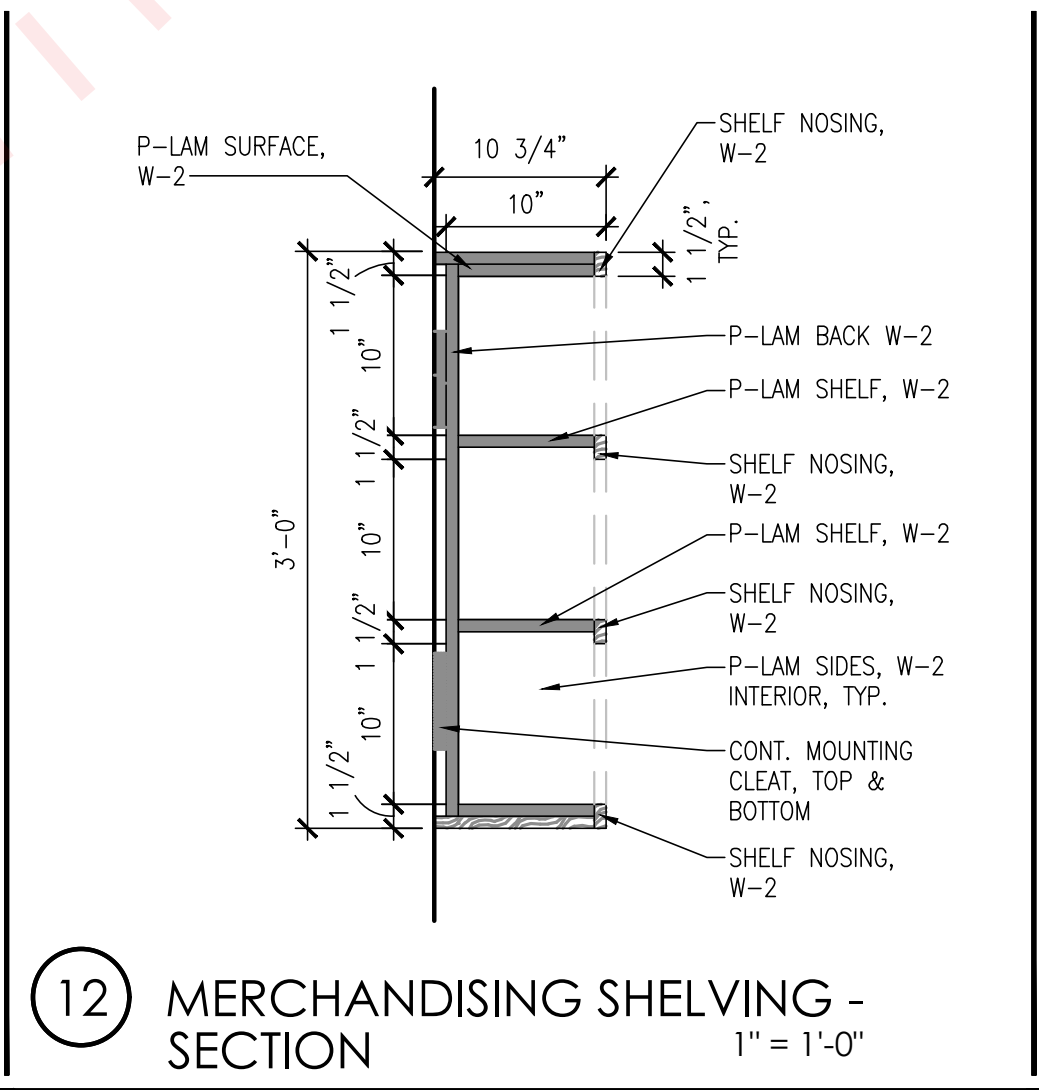
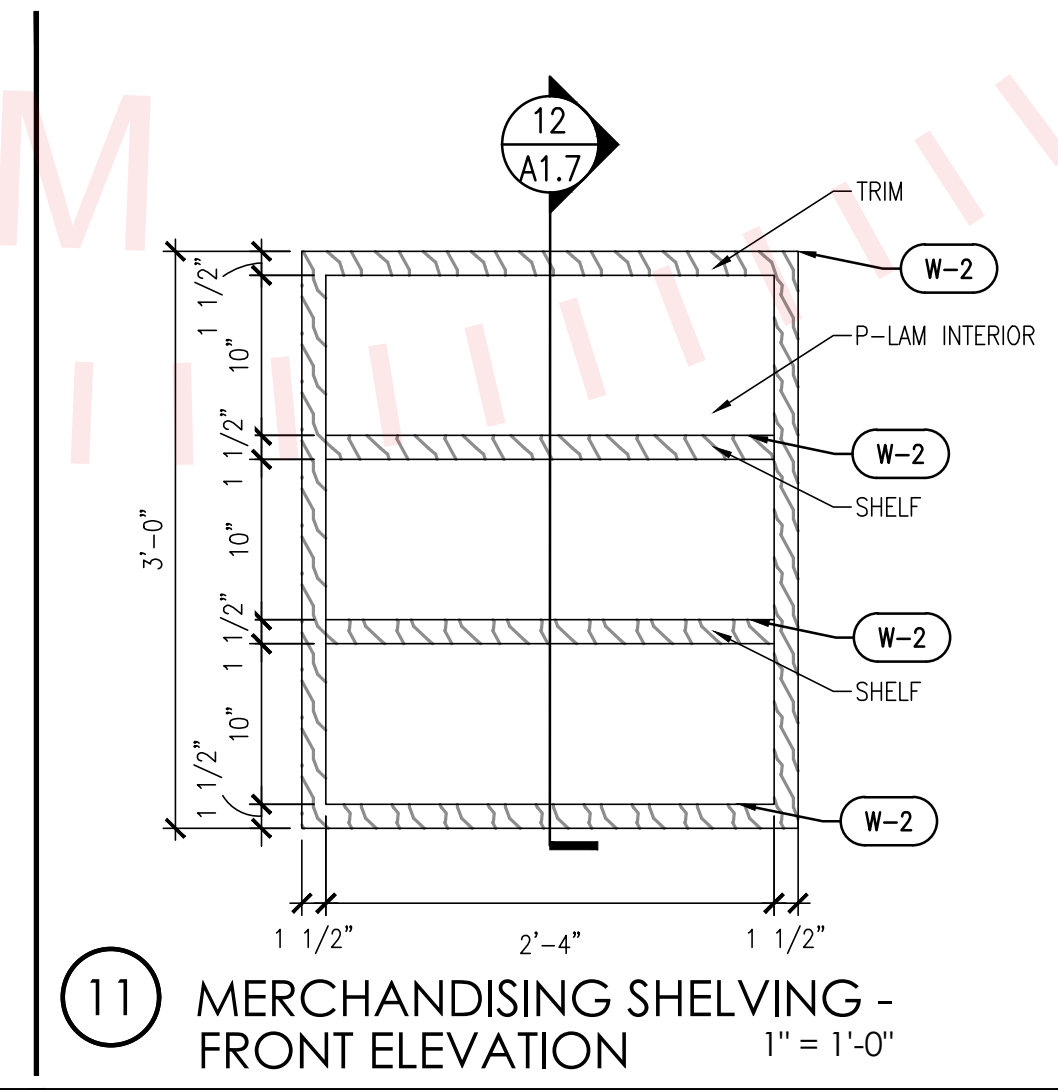
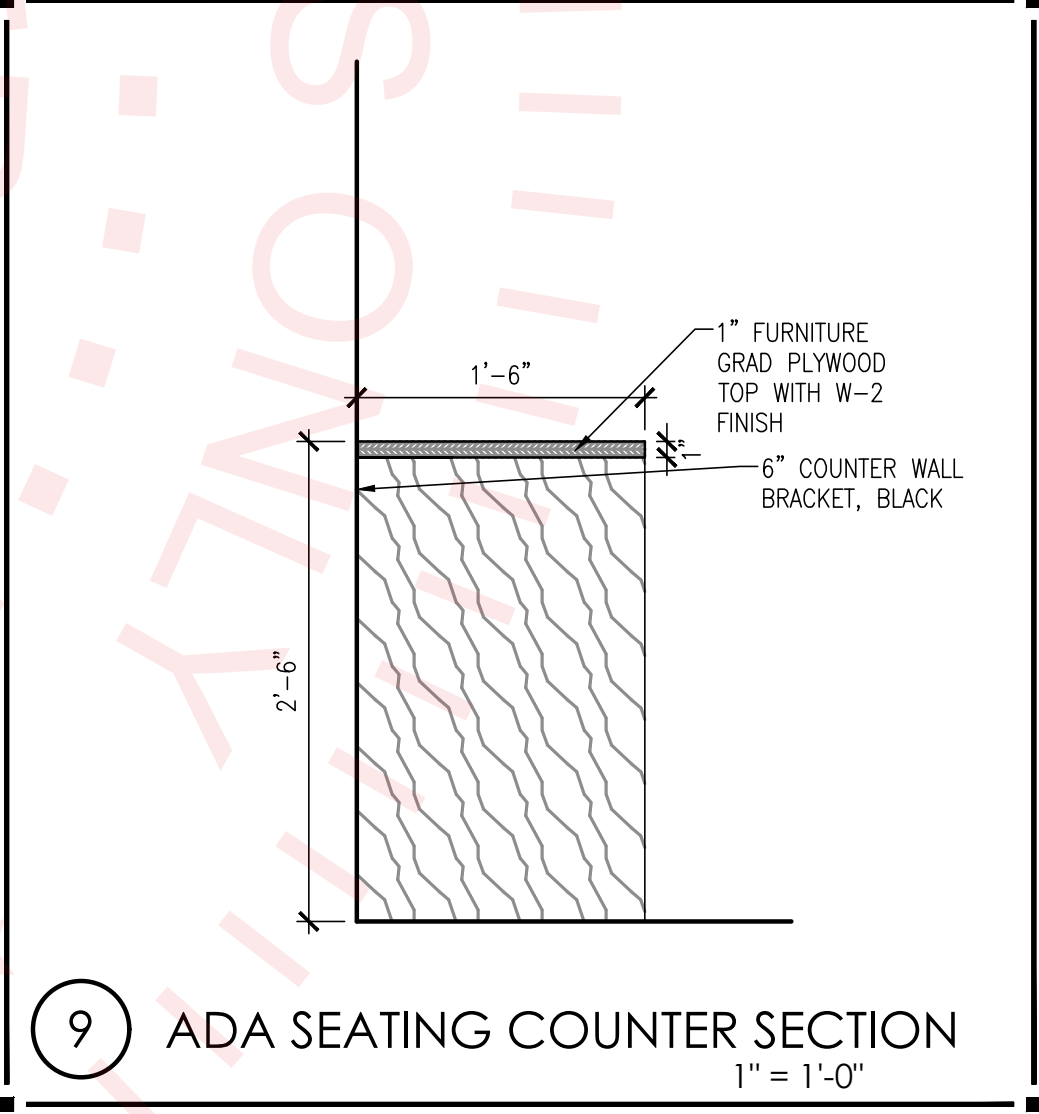
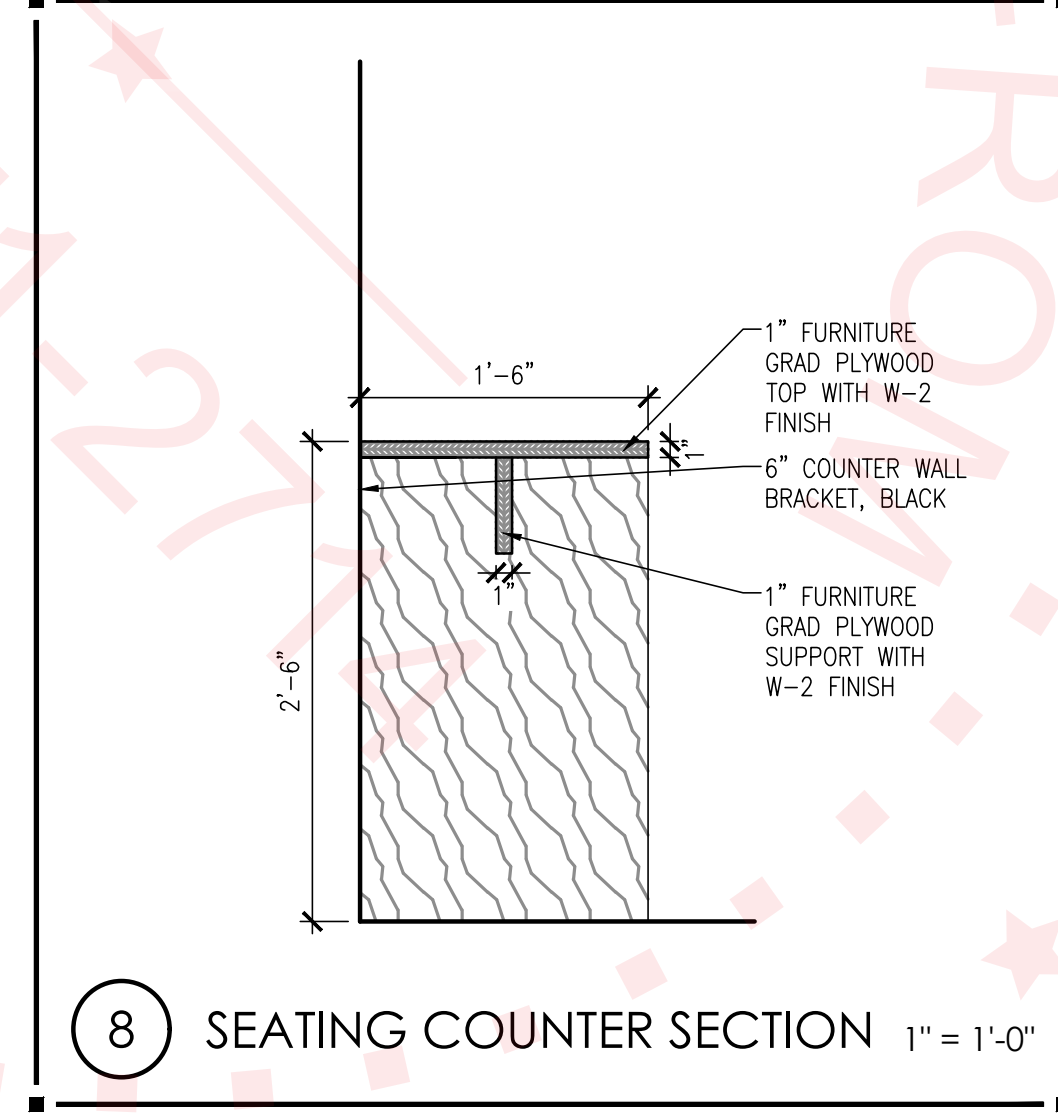
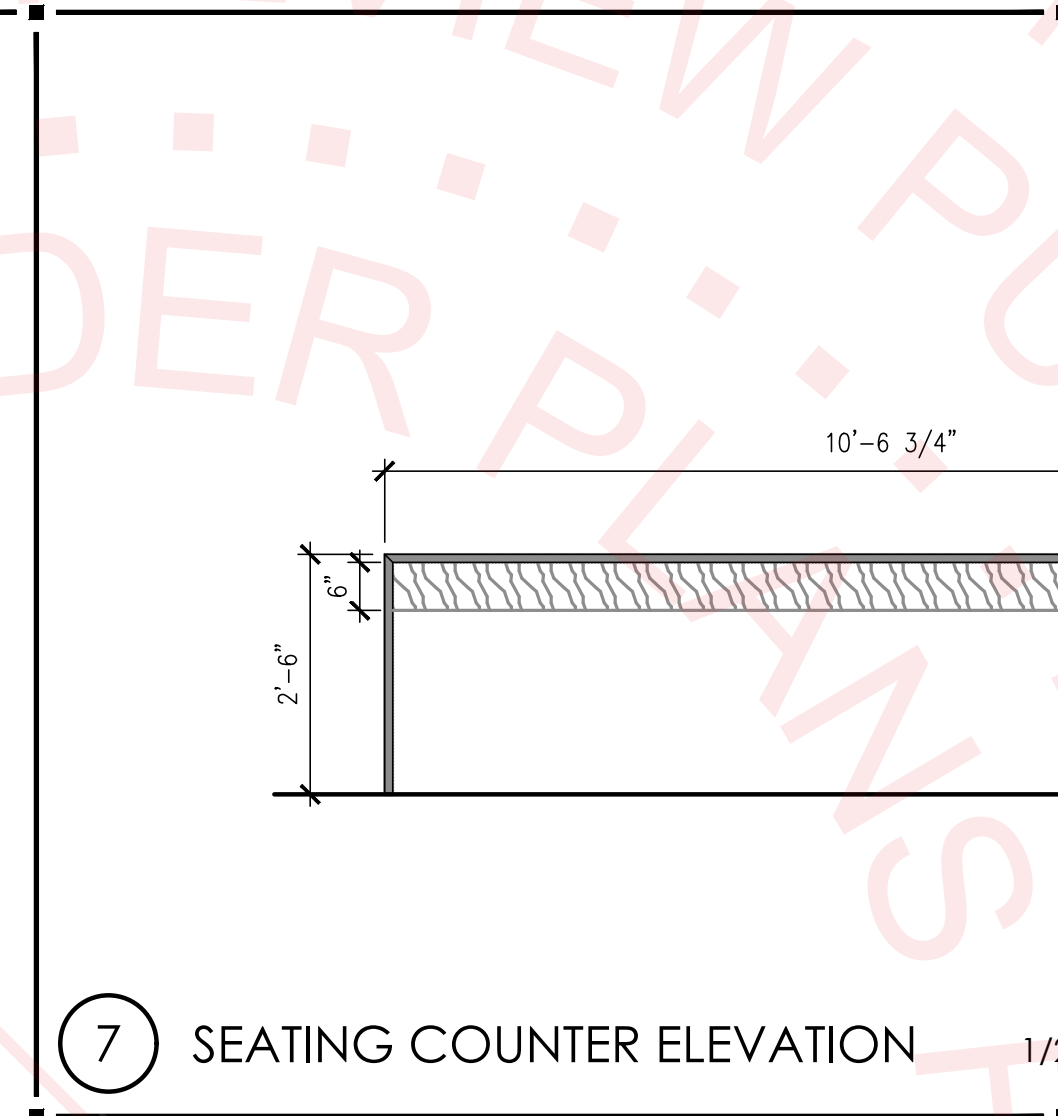
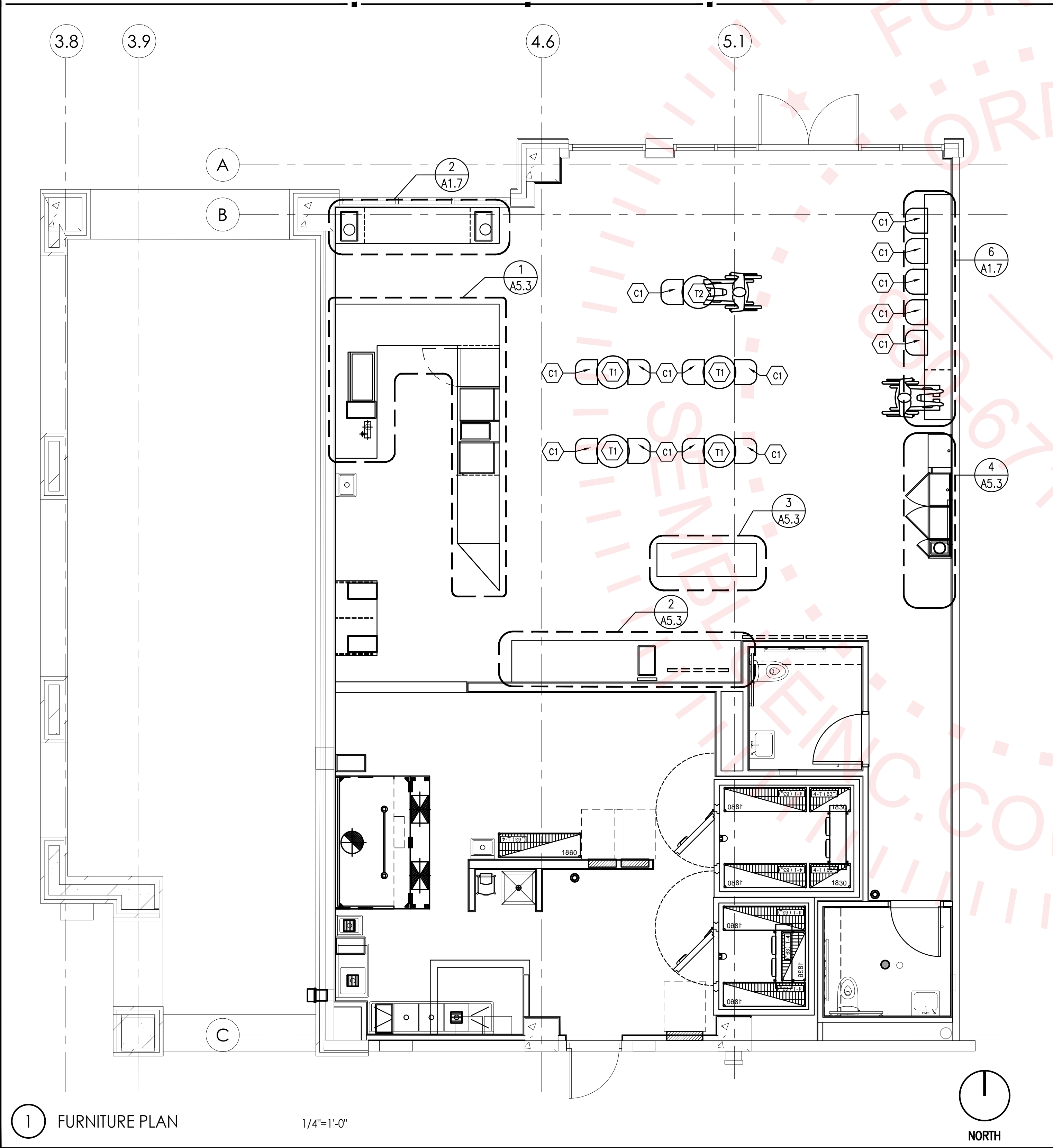
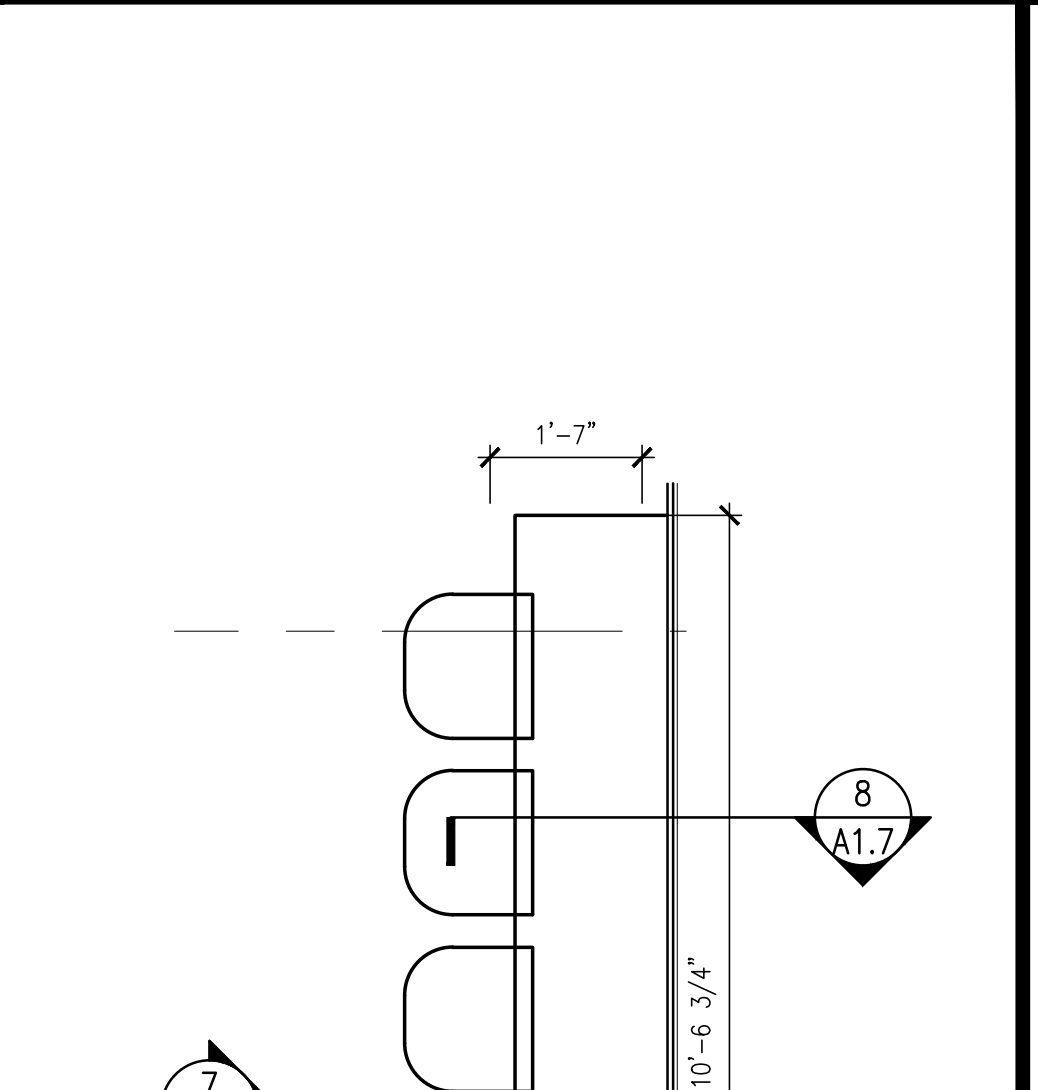
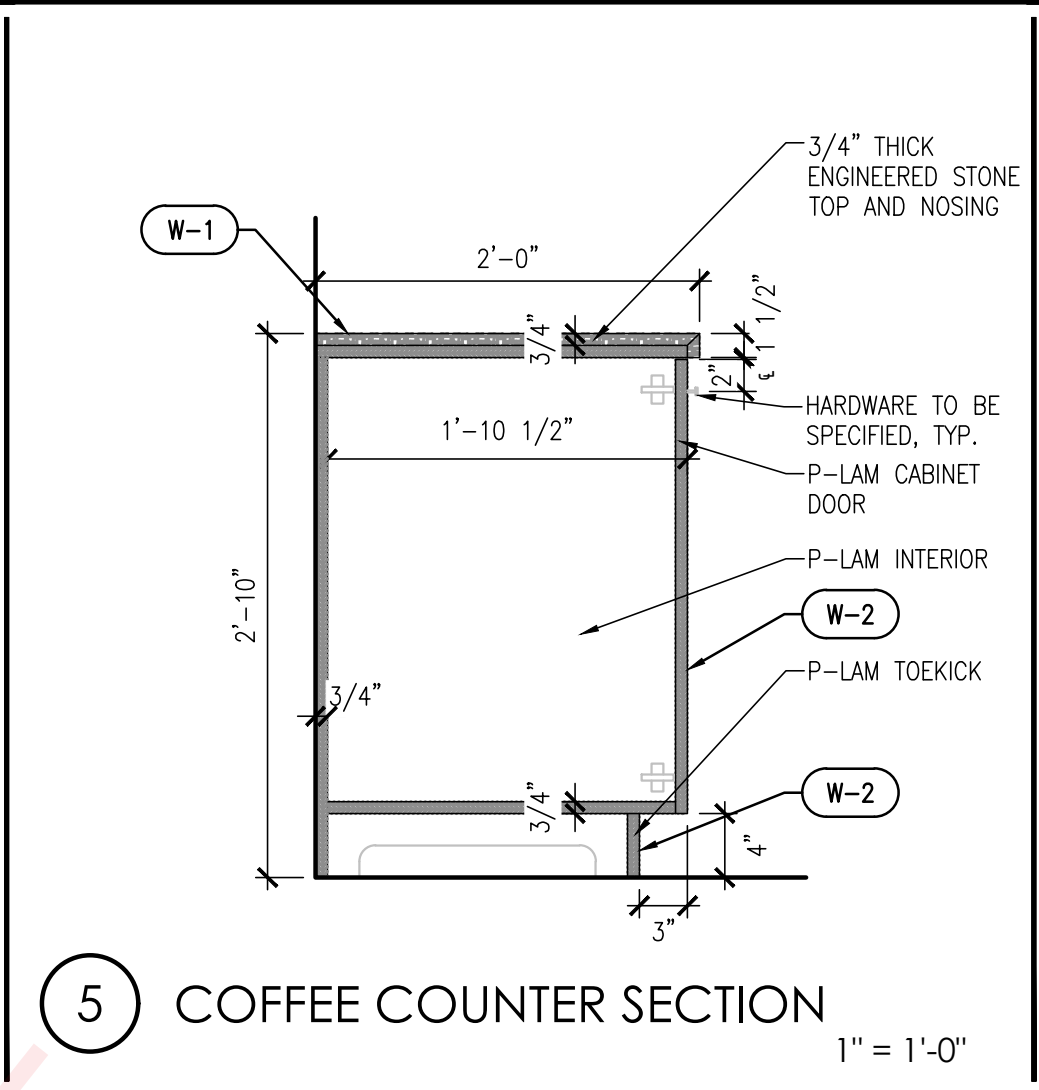
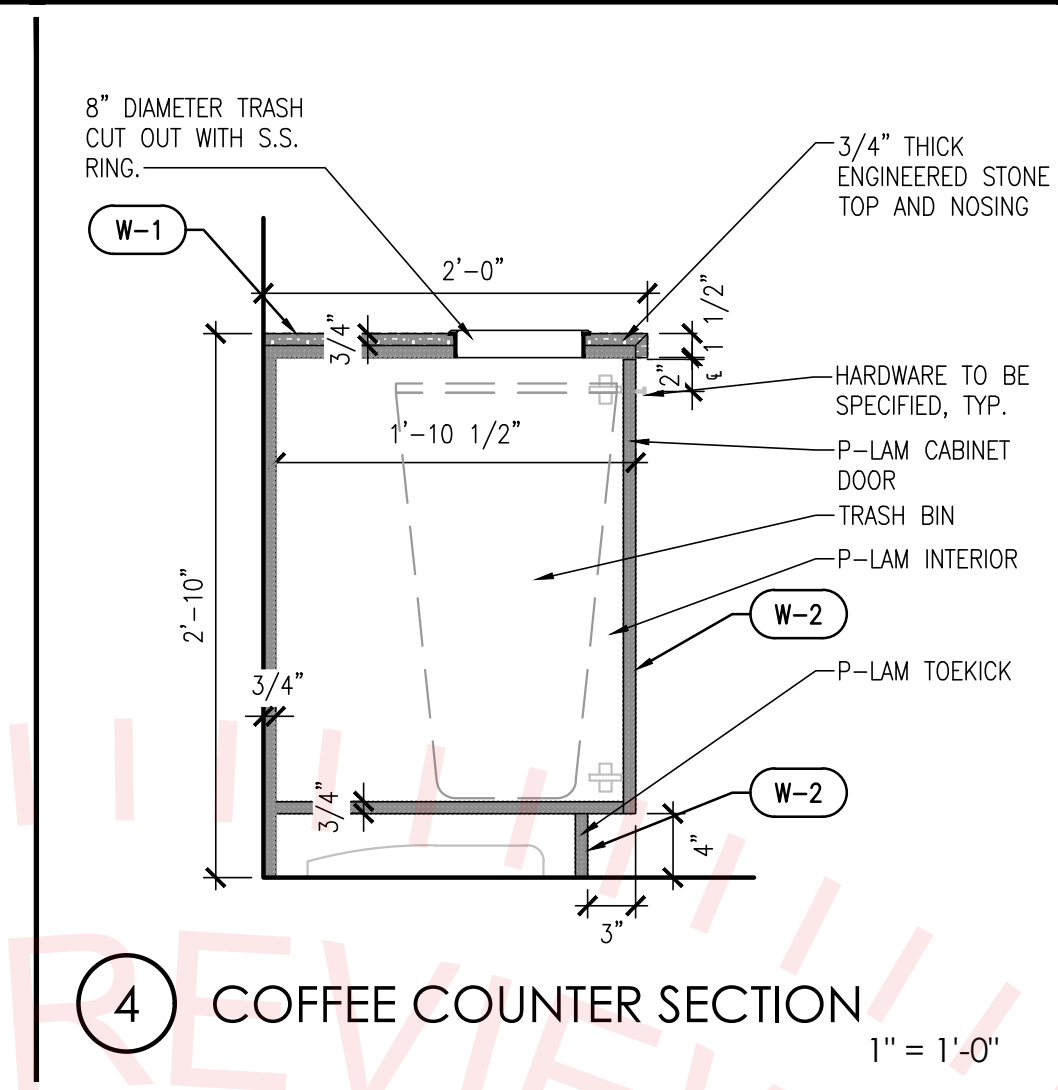
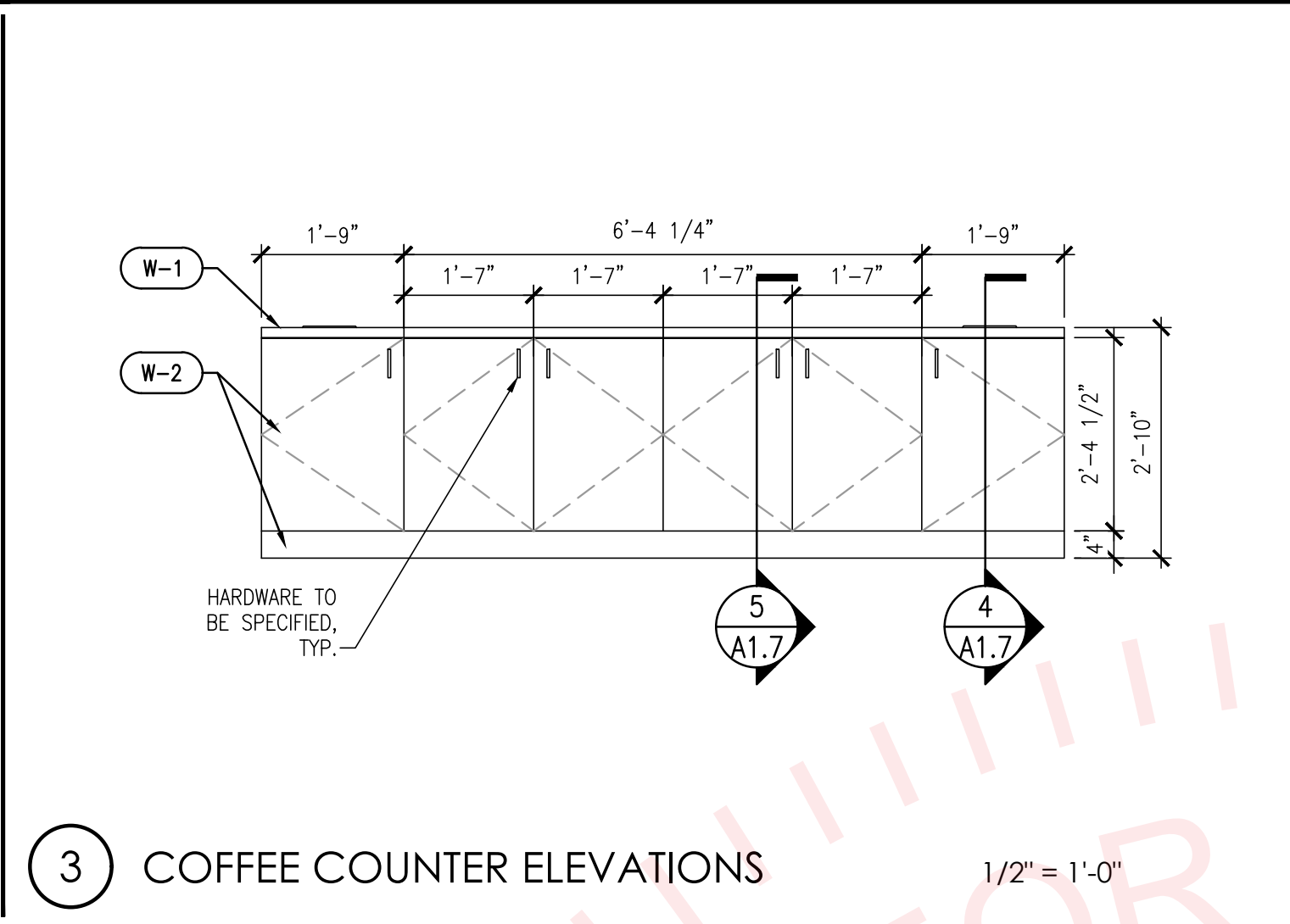
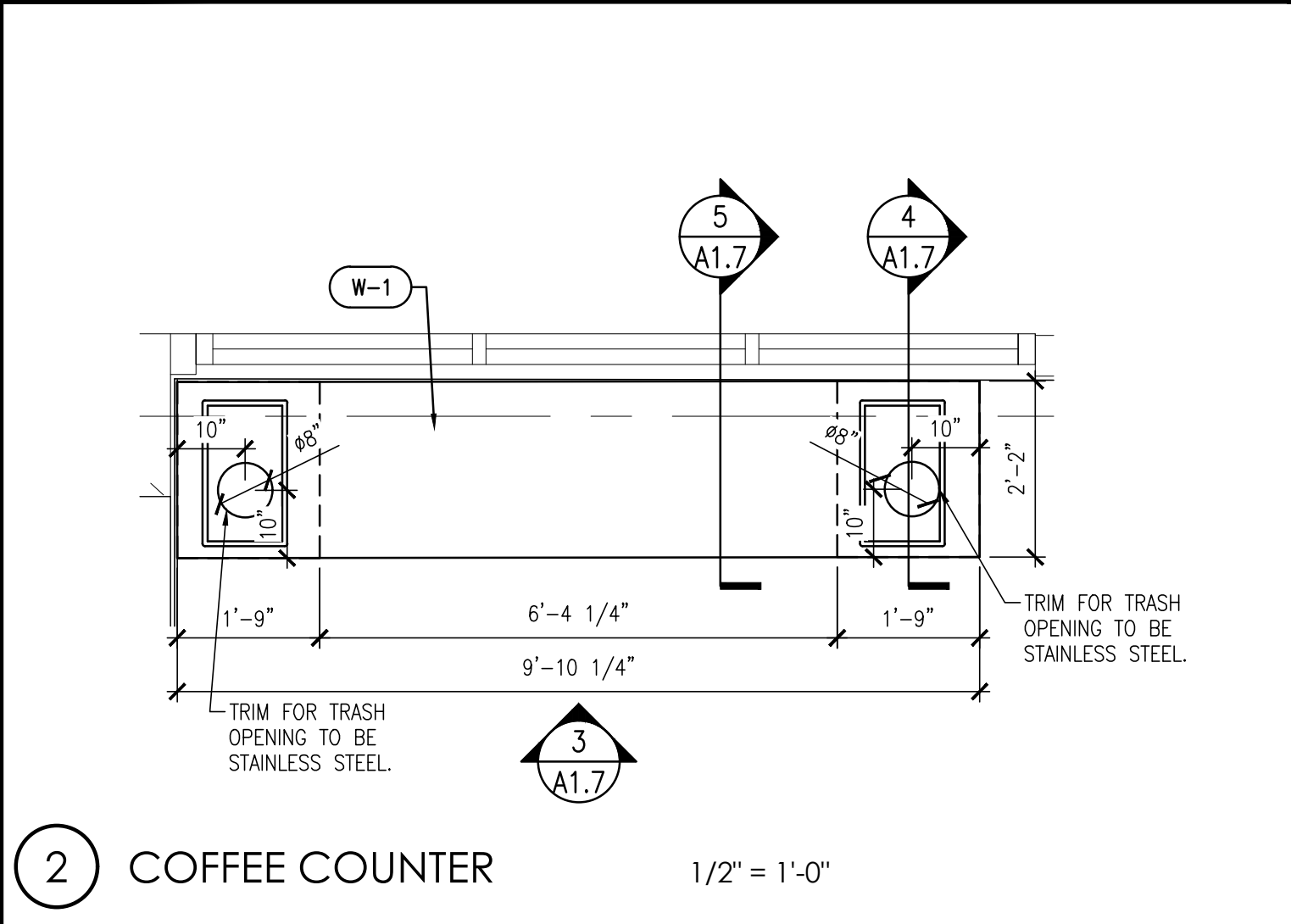
GENERAL NOTES

- ALL CEILING HEIGHTS INDICATED ARE TO BE FROM FINISHED FLOOR U.N.O.
- GENERAL CONTRACTOR SHALL CONFIRM ALL CEILING HEIGHTS, COORDINATE WITH EXISTING CONDITIONS. NOTIFY OWNER/ARCHITECT IF SPECIFIED HEIGHTS CANNOT BE ACHIEVED.
- ALL DIMENSIONS ARE TAKEN FROM END OF TRACK, CENTERLINE OF FIXTURE, OR FINISH FACE OF SURFACE U.N.O.
- LOCATE ALL TRACK AND RECESSED FIXTURES @ CENTERLINE OF TILE OR OPENING U.N.O.
- VERIFY LIGHT FIXTURES WITH ELECTRICAL DRAWING
- COORDINATE WITH ELECTRICAL PLAN FOR SWITCHING
- COORDINATE WITH MECHANICAL PLAN FOR LOCATION OF DIFFUSERS
- MOUNTED EQUIPMENT AND DUCTWORK SHALL BE MOUNTED TO PROVIDE CLEARANCE FOR CEILING MOUNTED LIGHT FIXTURES AND EQUIPMENT
- PROVIDE REQUIRED ACCESS PANELS IN CEILING FOR MECHANICAL, ELECTRICAL, AND PLUMBING SYSTEMS. LOCATION AND TYPE OF ACCESS PANEL SHALL BE VERIFIED WITH ARCHITECT PRIOR TO CONSTRUCTION
- PENETRATIONS AND OPENINGS IN CEILING FOR FIXTURES OR MECHANICAL SYSTEM PATHS AND REGISTERS SHALL BE PRECISELY CUT FOR INSTALLED ITEMS WITH EDGES CONCEALED BY MOUNTING TRIM OR COVER. INSTALLATION OF TRIM AND COVERS SHALL BE FLUSH TO SURFACE
- PENETRATION OF PARTITIONS FOR MECHANICAL SYSTEMS SHALL BE HERMETICALLY SEALED. PENETRATIONS WITHIN RATED PARTITION ASSEMBLIES SHALL PROVIDE FOR REQUIRED DAMPERING DEVICES, SLEEVES, AND FIREPROOFING TO MAINTAIN APPROVED RATING
- FIXTURES SHALL BE INSTALLED. EDGES OF CEILING CUTS SHALL BE CONCEALED BY MOUNTING TRIM OR COVER. INSTALLATION OF TRIM RINGS AND COVERS SHALL BE FLUSH WITH CEILING SURFACE
- LIGHT FIXTURES SHALL BE CENTERED IN CEILING TILE UNLESS OTHERWISE NOTED
- COORDINATE NEW CEILING/ TILE/ GRID WITH LEGEND.
- CONTRACTOR SHALL REMOVE DEBRIS OR MATERIAL FROM ABOVE CEILING PRIOR TO FINAL CLEANING. GENERAL CONTRACTOR IS RESPONSIBLE FOR DAMAGES CAUSED BY DEBRIS OR UNSECURED MATERIAL ABOVE FINISHED CEILING.
- MINIMUM CEILING HEIGHT ALONG ACCESSIBLE ROUTE SHALL BE 90" AFF (PER ADA GUIDELINES).



LIGHTING NOTES

- ALL FIXTURES TO BE PROVIDED BY, RECEIVED, STORED AND INSTALLED BY ELECTRICAL CONTRACTOR.
- ELECTRICAL CONTRACTOR SHALL SUPPLY 10% ADDITIONAL SPARE, OF ALL LAMP TYPES, TO BE LEFT AT JOB SITE AT END OF JOB.
- GC TO COORDINATE COLOR OF ALL GENERAL, EMERGENCY, AND DECORATIVE LIGHTING WITH OWNER PRIOR TO ORDERING.



FURNITURE COUNT AND SPEC			
TAG	TYPE	QUANTITY	MODEL #
T1	TABLE	4	LANCASTER #42724RDVE22D
T2	TABLE	1	WITH ADA COMPLIANT BASE- SUPPLIED BY OWNER TO MATCH LANCASTER #42724RDVE22D
C1	CHAIR	14	LANCASTER #164CMCAFEBLK

FURNITURE NOTES

- ACCESSIBLE DINING SURFACES SHALL COMPLY WITH ICC/ANSI A117.1-2017 AND WITH CLEAR FLOOR SPACE AS INDICATED ON G1.3.
- ALL FINISHES TO COMPLY WITH 2023 FBC CHAPTER 8 AND FBC TABLE 804, 806, AND 807.
- FURNITURE PROVIDED BY OWNER AND SHOWN FOR REPRESENTATION ONLY.

INTERPLAN
INTERPLAN LLC
AR99238
CA 8660

ARCHITECTURE
ENGINEERING
PERMITTING

220 E. CENTRAL PKWY, STE 4000
ALTAMONTE SPRINGS, FL 32701
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SEAL:

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NO DATE REMARKS
REVISIONS

BAGELS
CO.

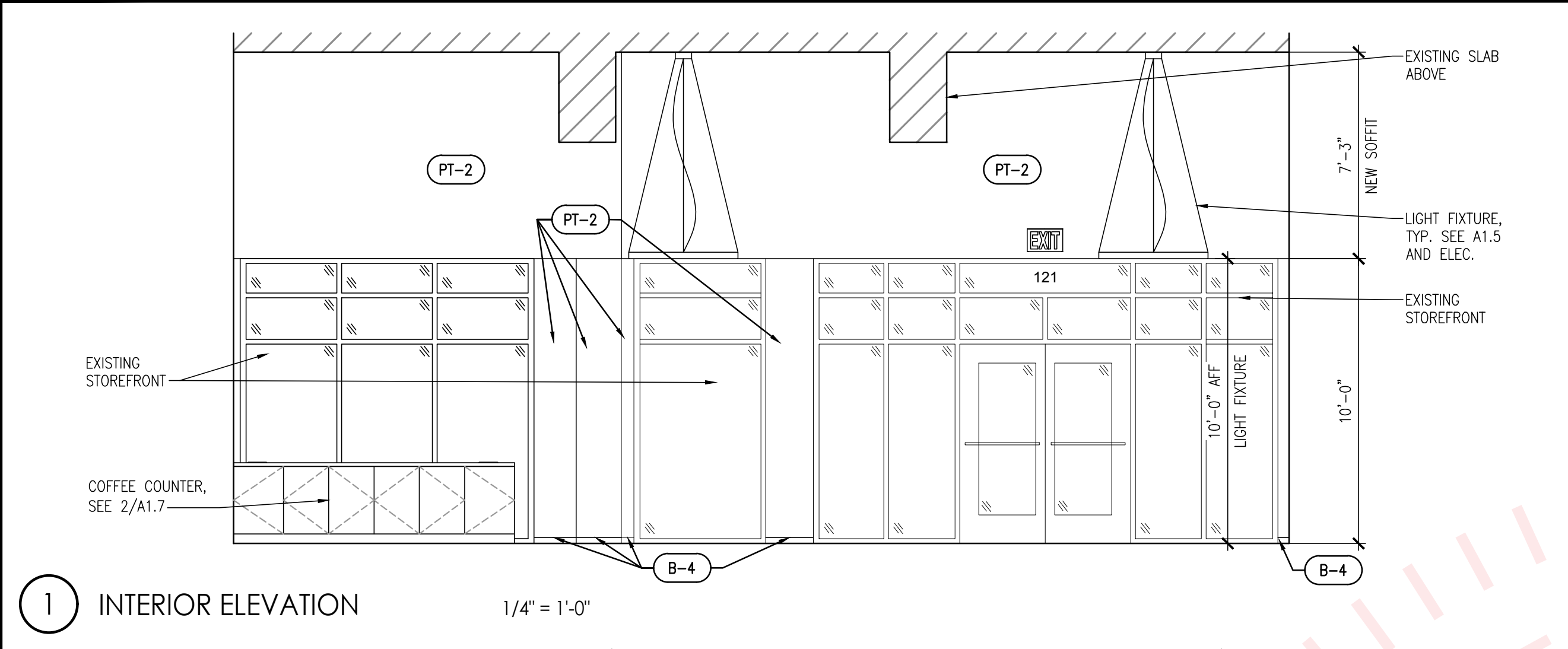
BAGEL & CO
TALLAHASSEE
799 W. GAINES STREET
SUITE 121
TALLAHASSEE, FL 32310

PROJECT NO: 2023.0719
DATE: 04/12/2024

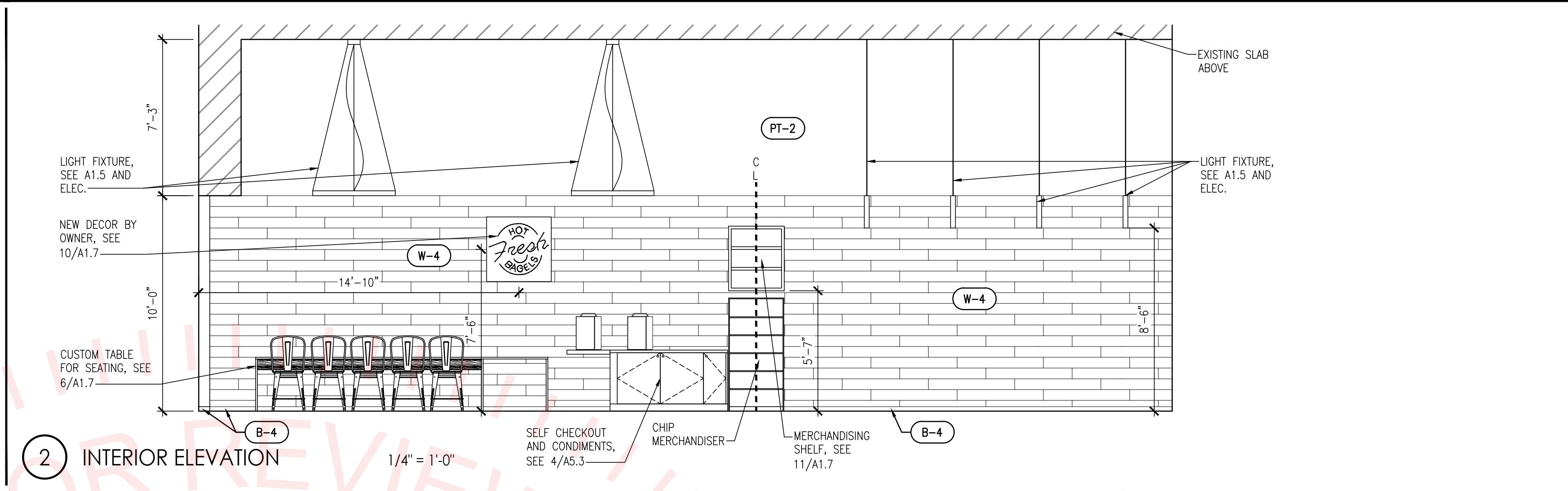
A1.7
FURNITURE FIXTURE PLAN &
SCHEDULE

CHECKED: DC DRAWN: MT

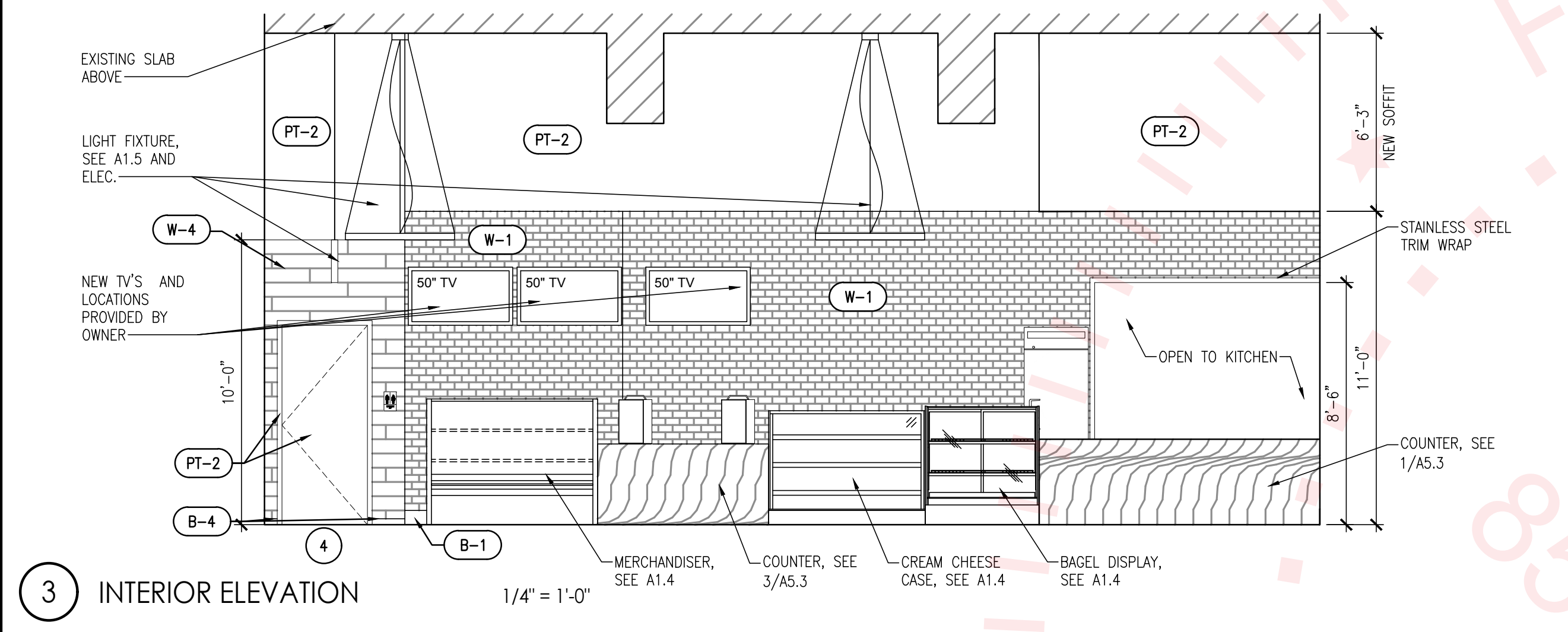
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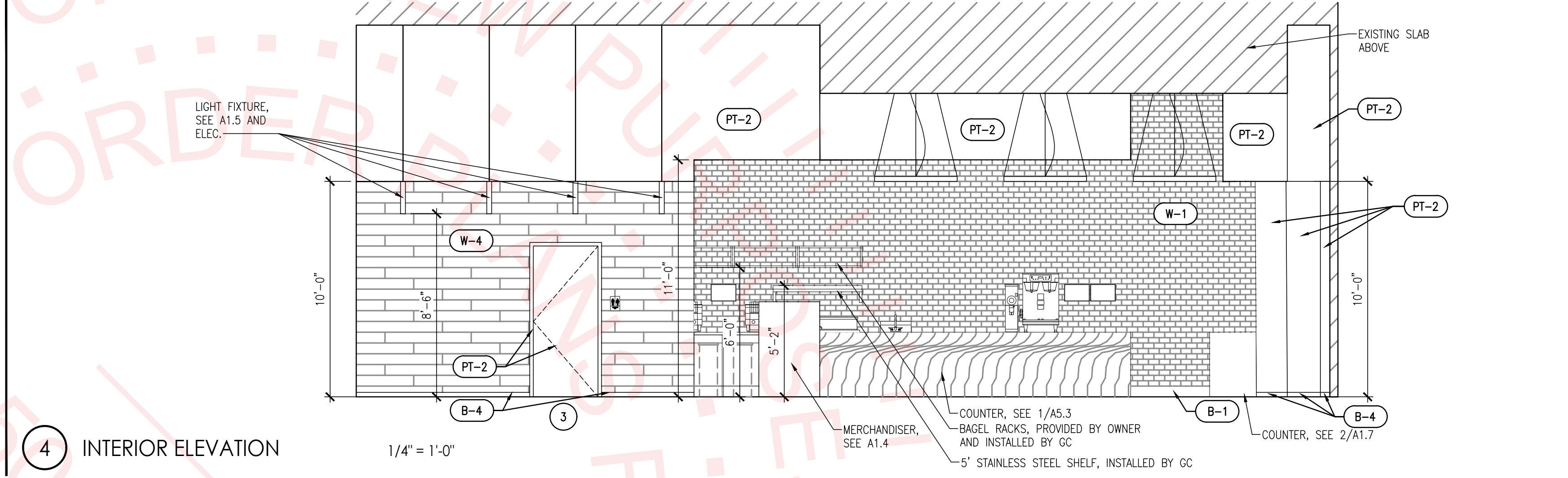
1 INTERIOR ELEVATION



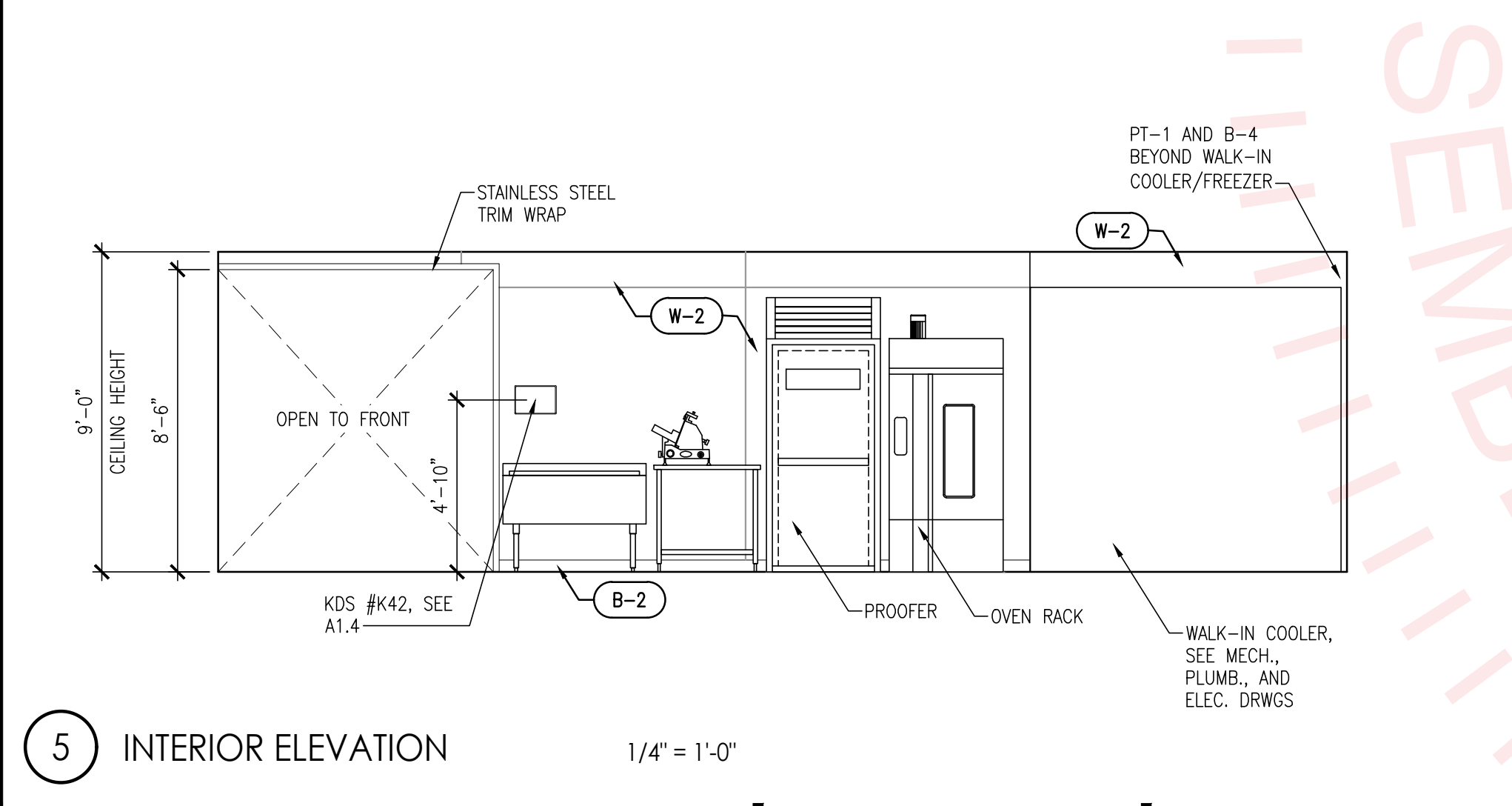
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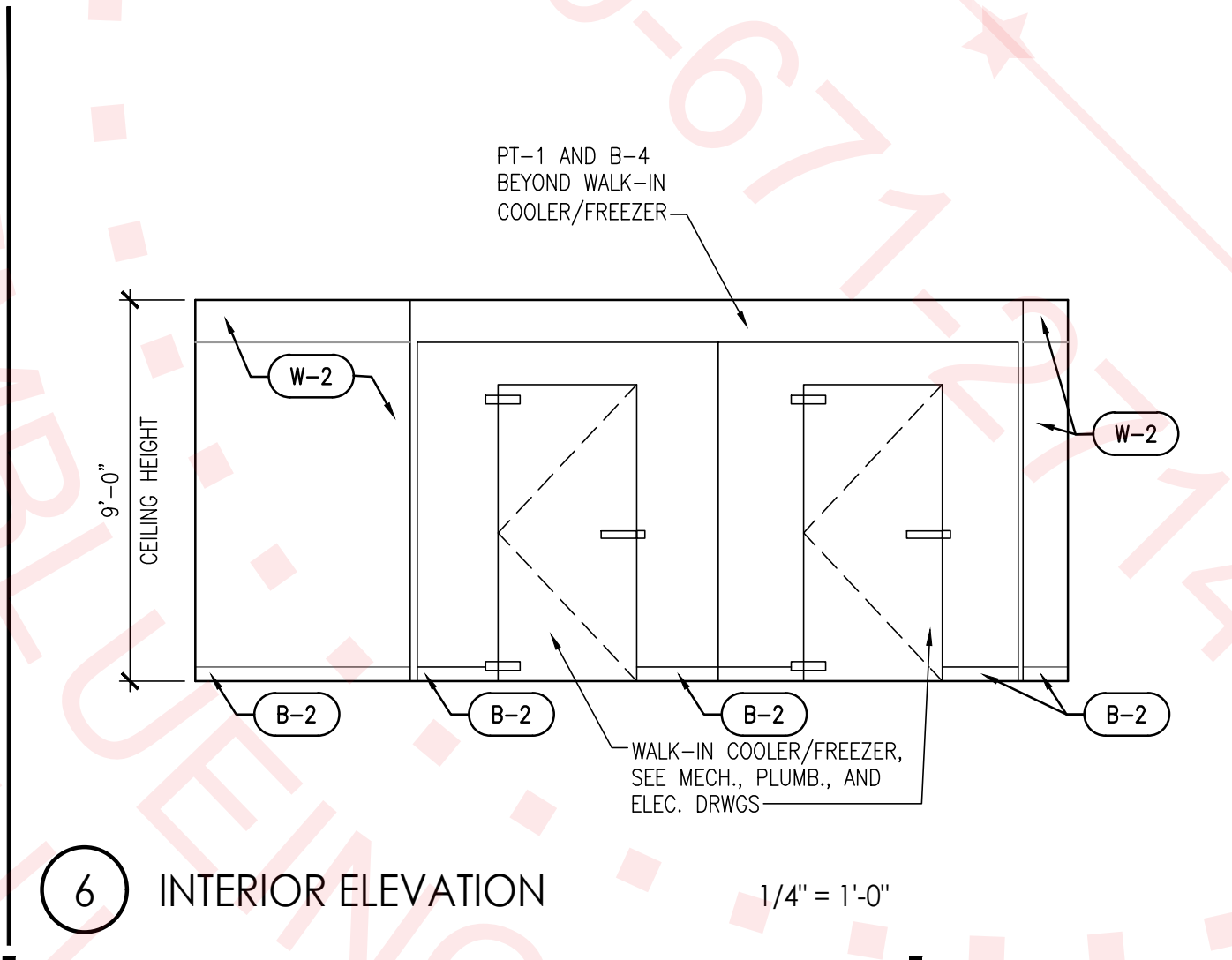
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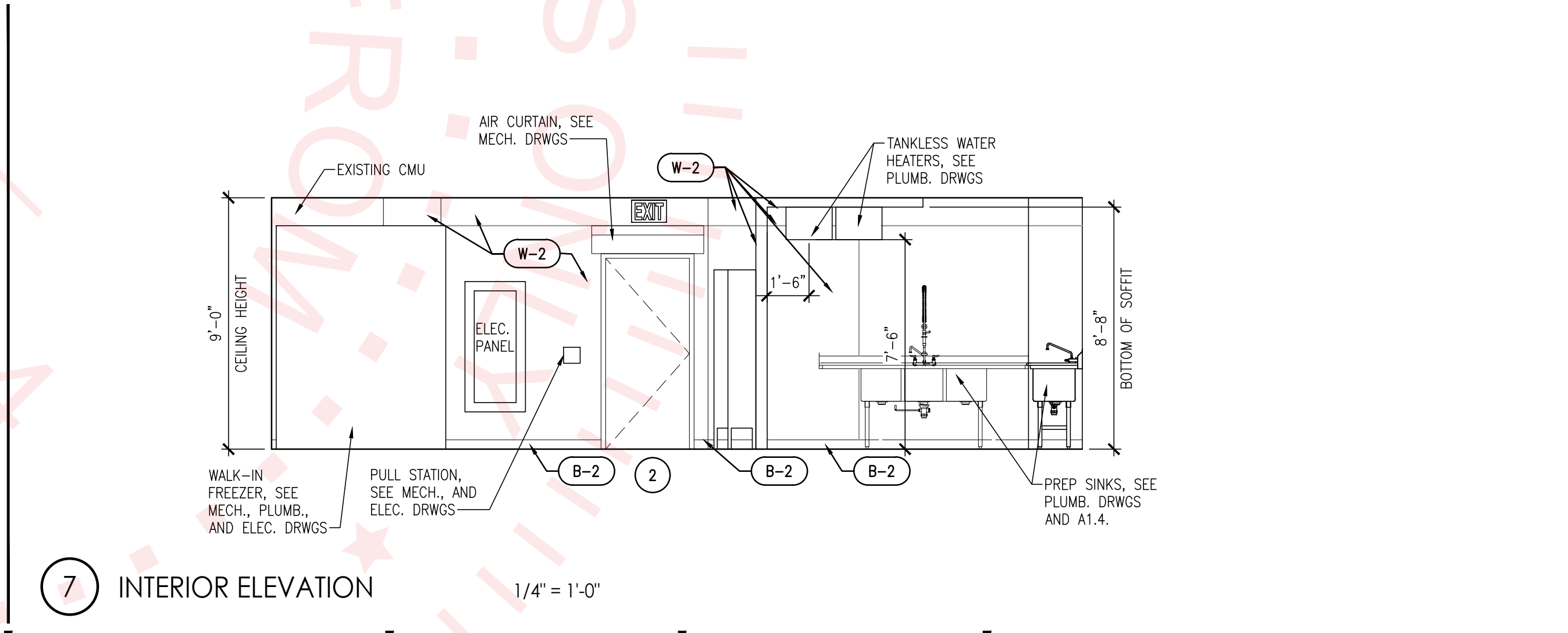
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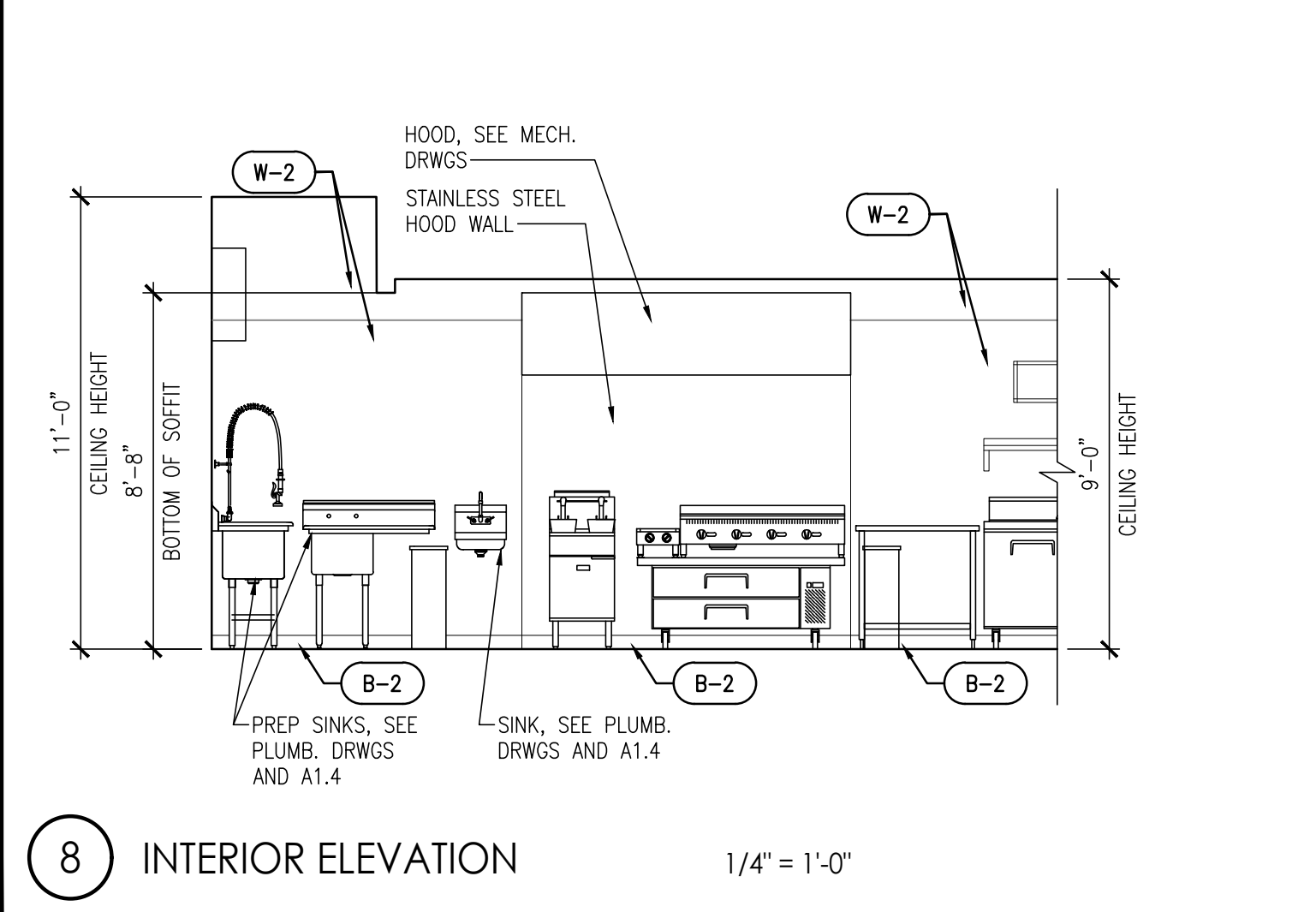
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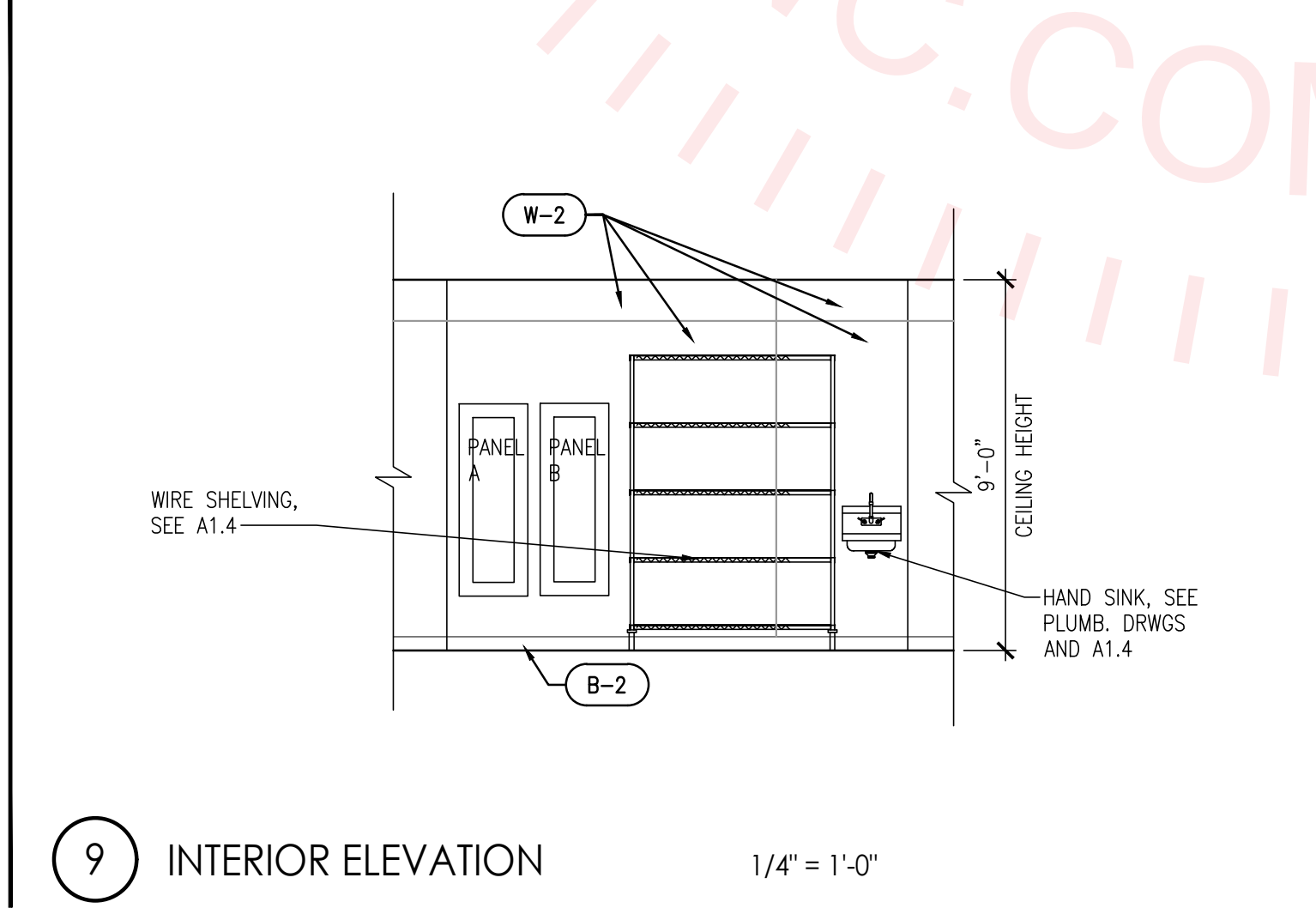
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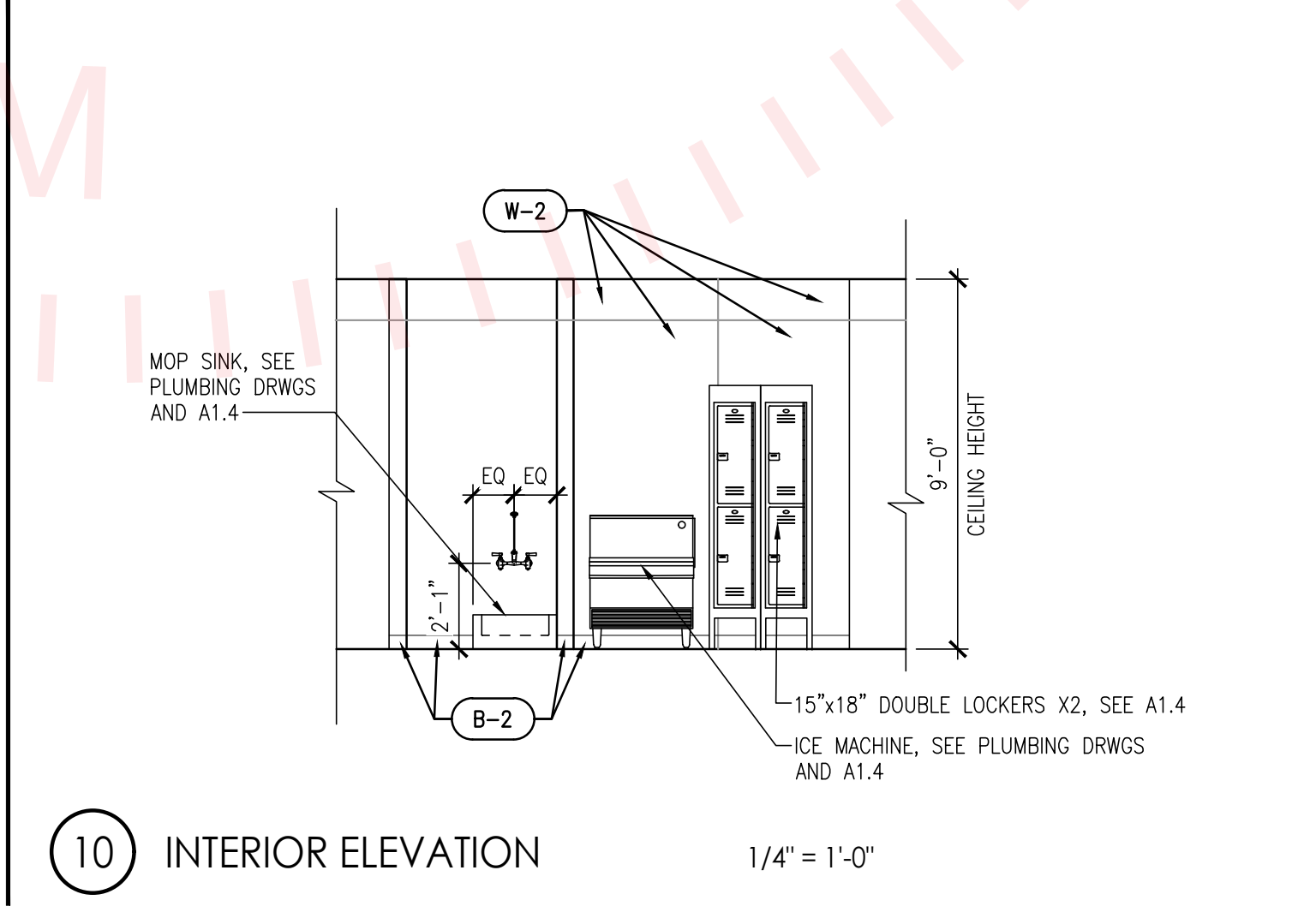
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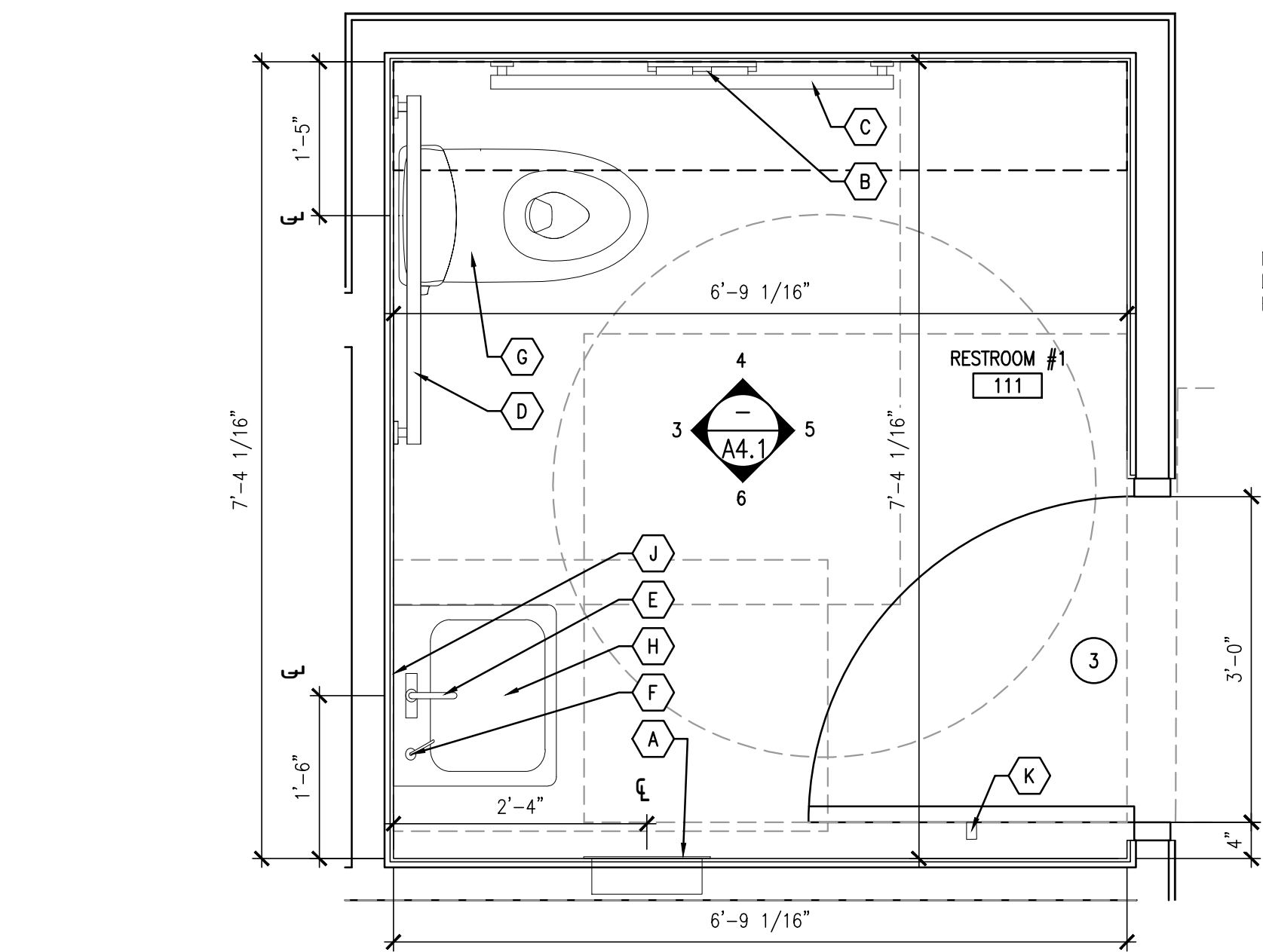
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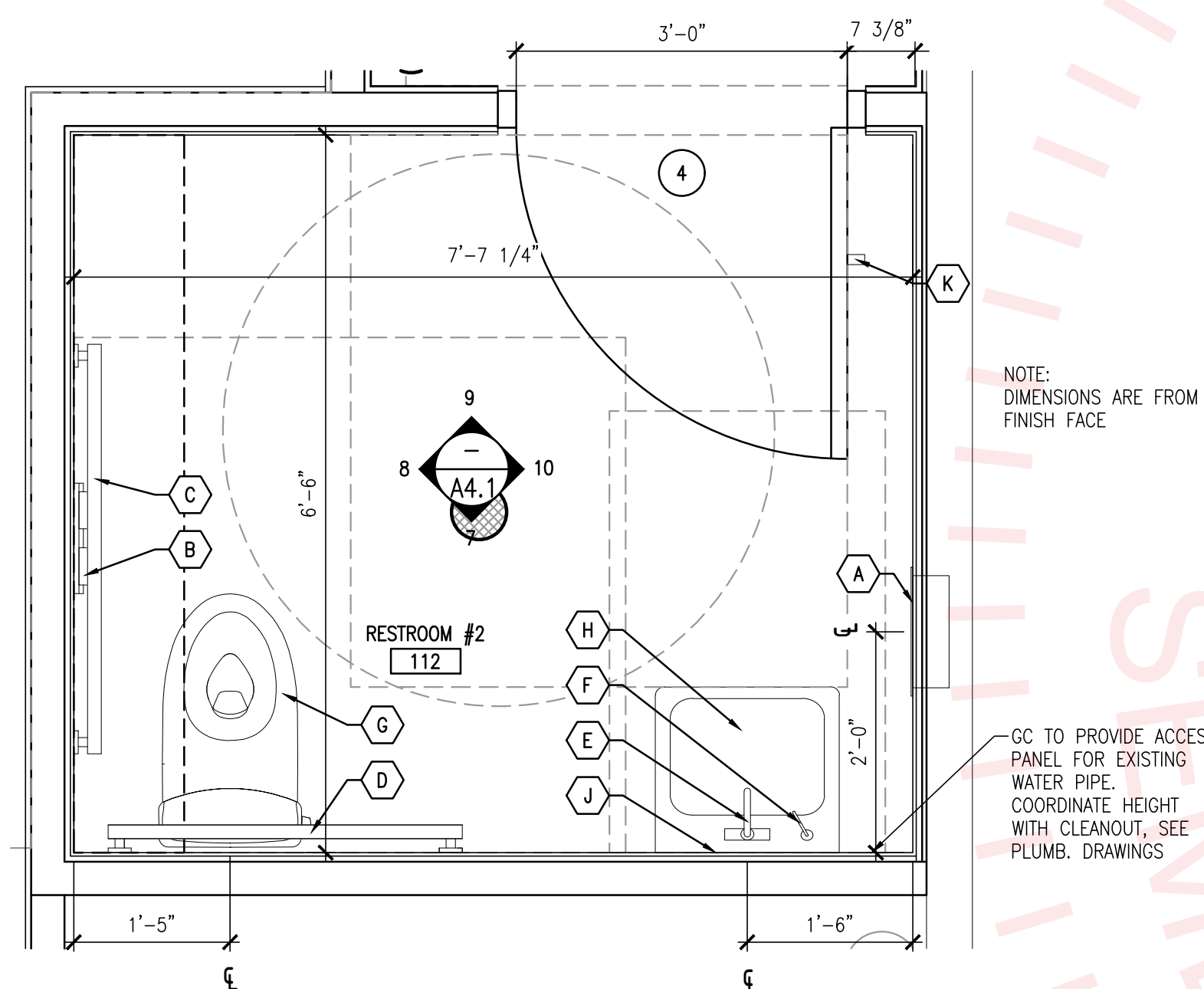
9 INTERIOR ELEVATION



10 INTERIOR ELEVATION



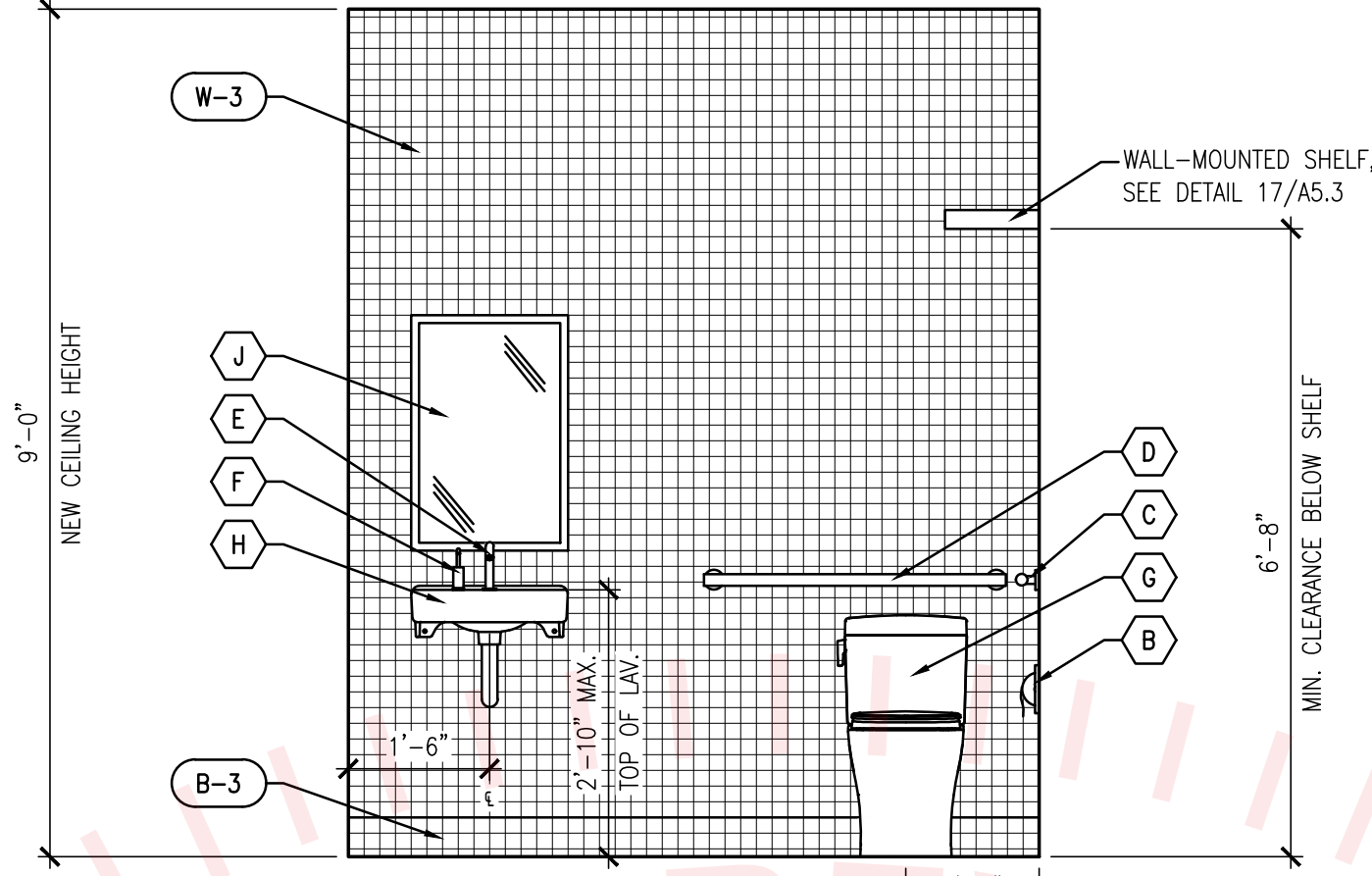
1 RESTROOM #1 ENLARGED PLAN 3/4"=1'-0"



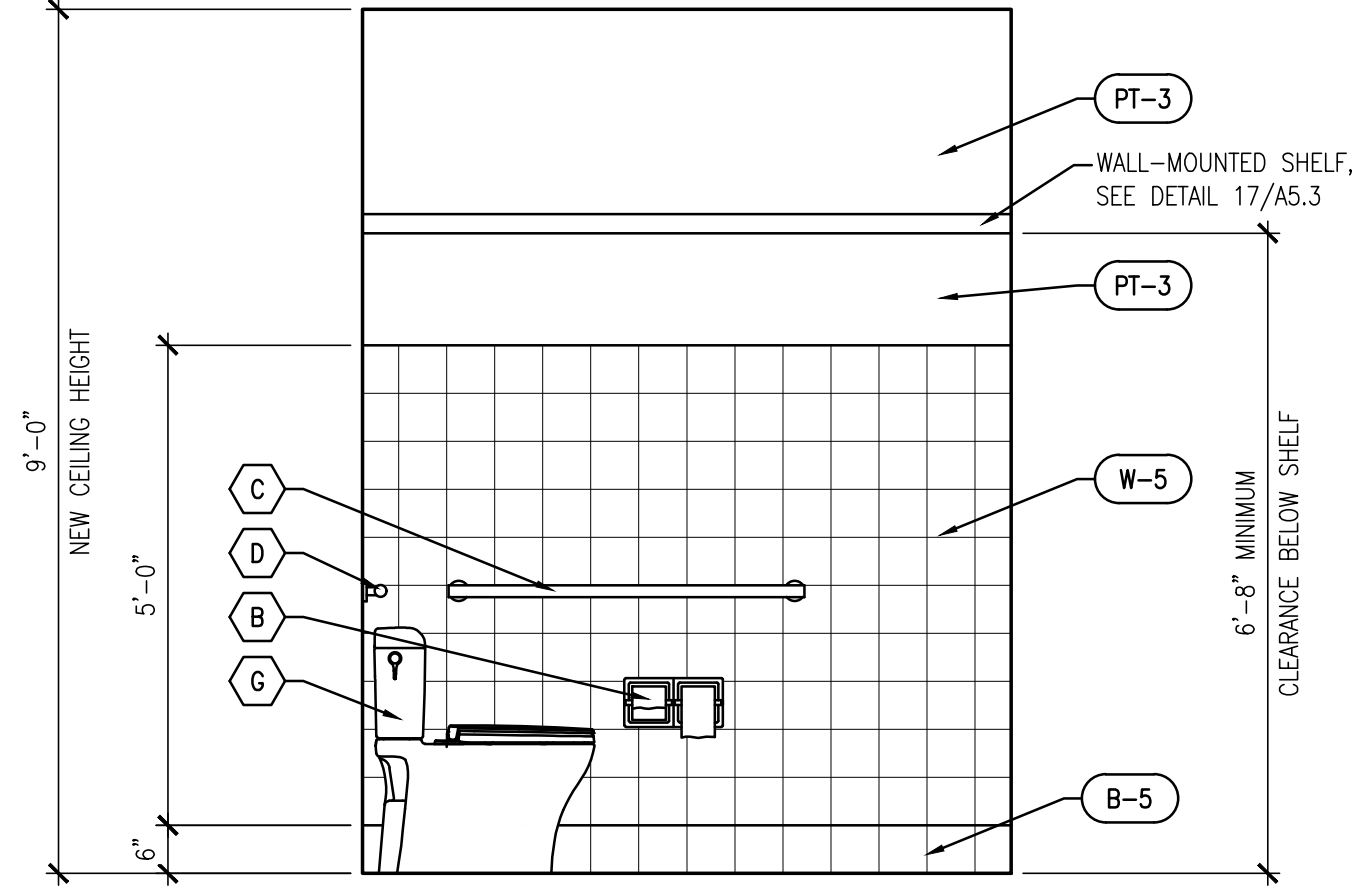
2 RESTROOM #2 ENLARGED PLAN 3/4"=1'-0"

RESTROOM ACCESSORY SCHEDULE				
NO.	DESCRIPTION	MANUFACTURER	MODEL NUMBER	LOCATION
A	PAPER TOWEL DISPENSER/WASTE RECEPTACLE	BOBRICK	B-38032	6 INCHES A.F.F.
B	TOILET PAPER DISPENSER	BOBRICK	B-2888	SURFACE MOUNTED 15" A.F.F. MINIMUM TO BOTTOM OF DISPENSER
C	42" GRAB BAR	BOBRICK	B-9806X42	WALL-MOUNTED 36" A.F.F. TO TOP OF GRAB BAR
D	36" GRAB BAR	BOBRICK	B-9806X36	WALL-MOUNTED 36" A.F.F. TO TOP OF GRAB BAR
E	AUTOMATIC FAUCET	TOTO	TEL115I2	MOUNTED TO LAVATORY
F	LIQUID SOAP DISPENSER	TOTO	TES100AA#CP	MOUNTED TO LAVATORY
G	WATER CLOSET	TOTO	MS436124CEMF GN	REFERENCE ACCESSIBILITY CODE FOR COMPLIANT WATER CLOSET INSTALLATION
H	WALL MOUNTED LAVATORY	KOHLER	K-2084-L	REFERENCE ACCESSIBILITY CODE FOR COMPLIANT LAVATORY INSTALLATION
J	24" x 36" FRAMED MIRROR	BOBRICK	B-165 2436	WALL MOUNTED 40" A.F.F. MAXIMUM TO BOTTOM OF REFLECTIVE SURFACE
K	COAT HOOK	BOBRICK	B-9541	FINO COLLECTION SURFACE-MOUNTED WITH BUMPER, REFERENCE G1.3 FOR INSTALLATION

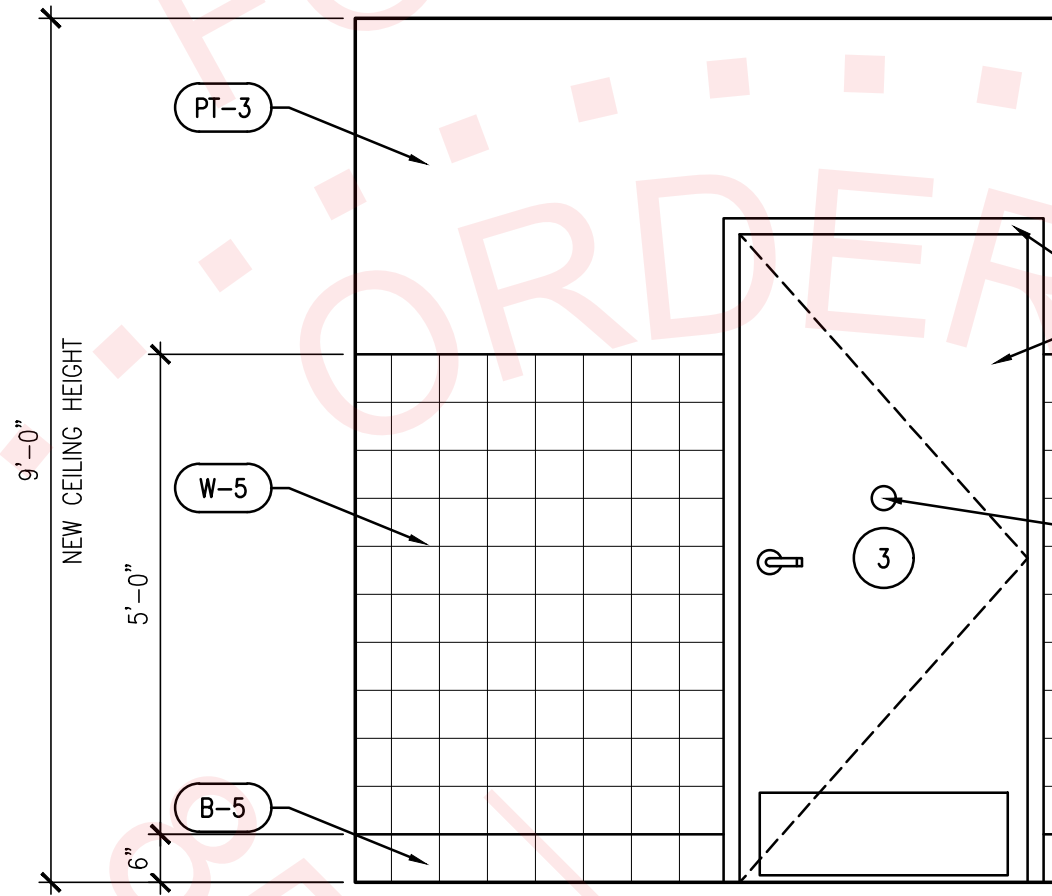
NOTE:
1. ELEVATIONS SHOW OUTLINE OF ITEMS ONLY. FOR ROUGH WALL OPENING DIMENSIONS AND PLACEMENT OF REQUIRED ELECTRICAL AND PLUMBING CONNECTIONS, SEE CURRENT TECHNICAL DATA SHEET AVAILABLE FOR EACH MODEL LISTED IN TOILET ROOM ACCESSORY SCHEDULE.
2. DIMENSIONS INDICATE AVERAGE MOUNTING HEIGHT FROM TOP OF ROUGH WALL OPENING (AT RECESSED ITEMS) OR TOP OF UNIT (AT SURFACE MOUNTED ITEMS) TO FINISHED FLOOR FOR CONDITIONS GENERALLY ENCOUNTERED IN PUBLIC WASHROOMS. MOUNTING HEIGHTS ARE IN COMPLIANCE WITH FLORIDA ACCESSIBILITY CODE.
3. ALTERNATE ACCESSORY MODELS CAN BE USED ONLY WITH OWNER APPROVAL.
4. ALL DIMENSIONS ON THIS SHEET ARE TO FINISHED SURFACE.



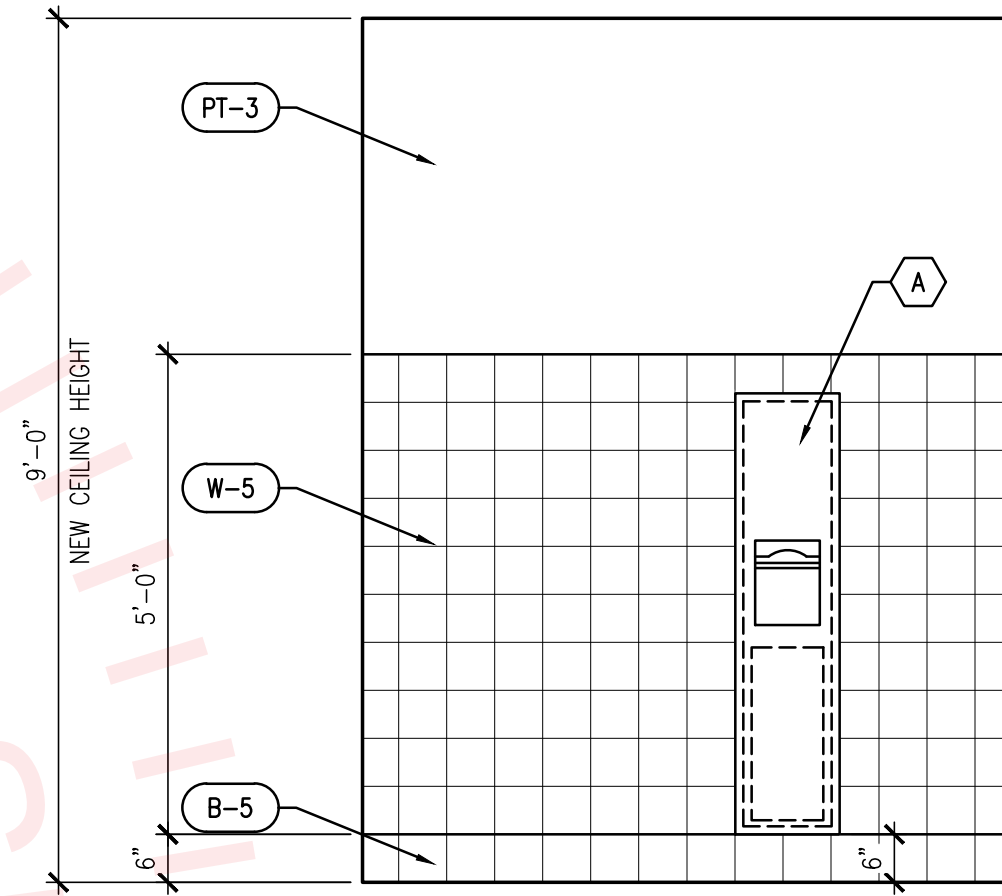
3 RESTROOM #1 ELEVATION 1/2"=1'-0"



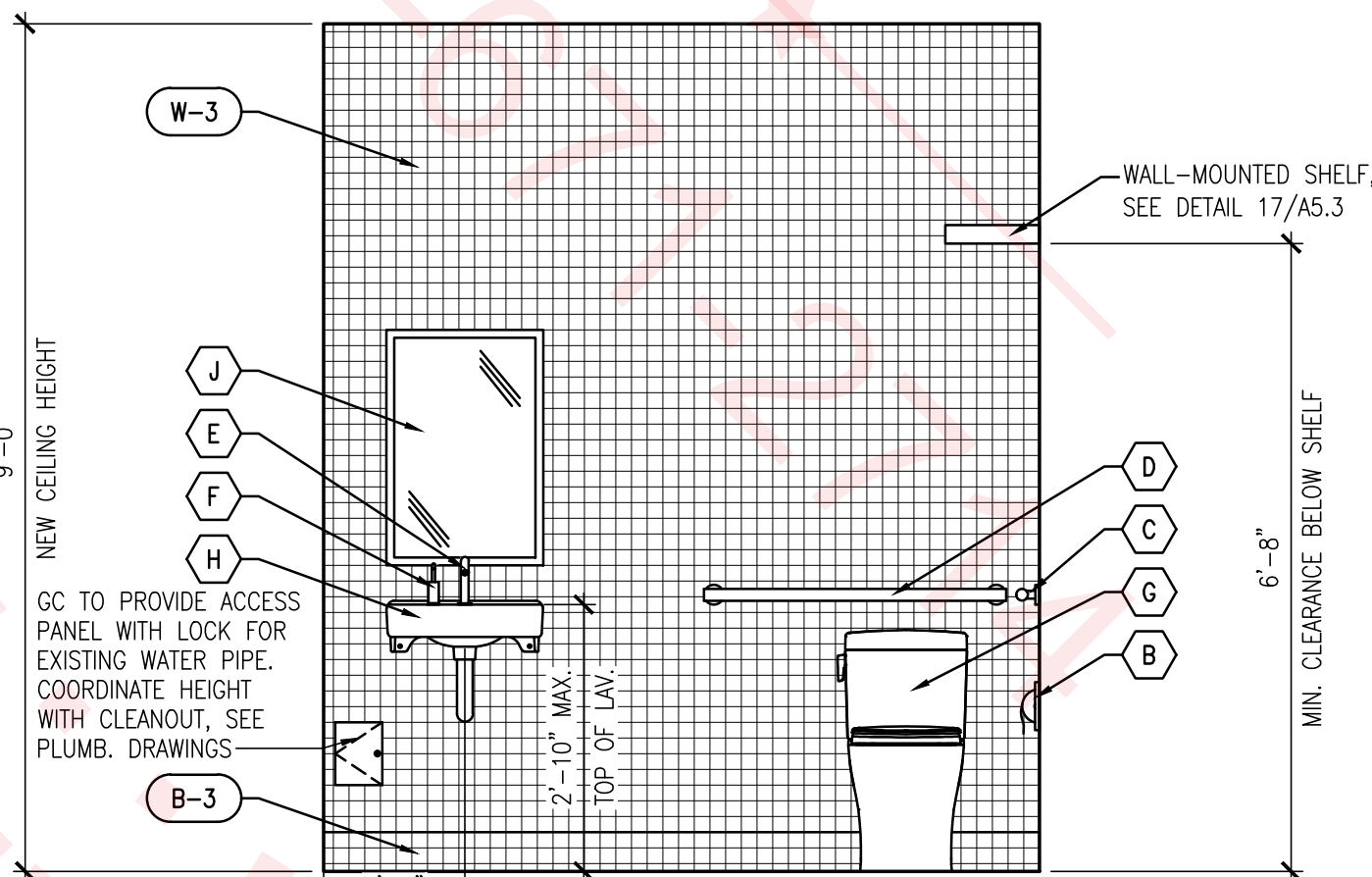
4 RESTROOM #1 ELEVATION 1/2"=1'-0"



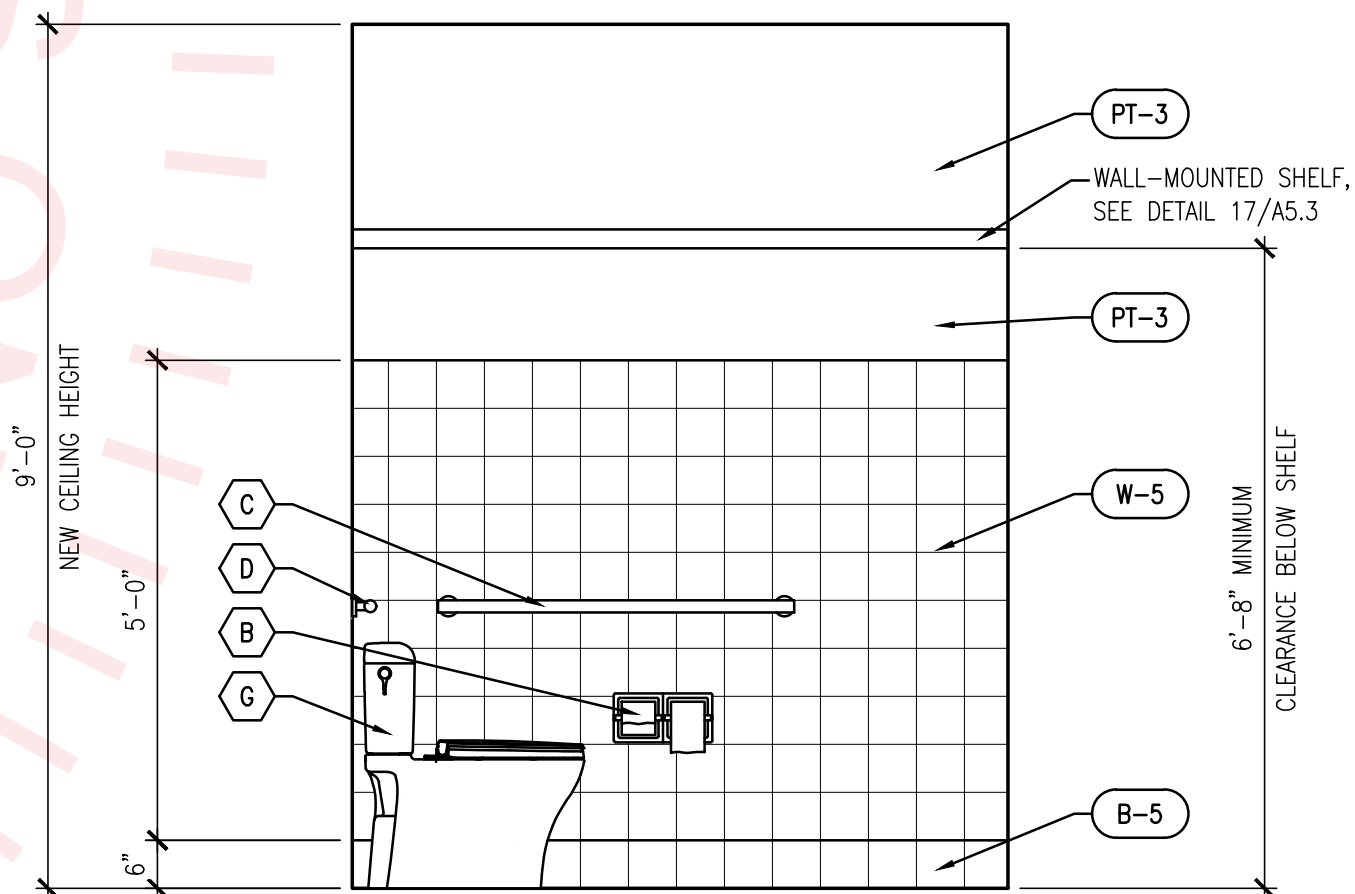
5 RESTROOM #1 ELEVATION 1/2"=1'-0"



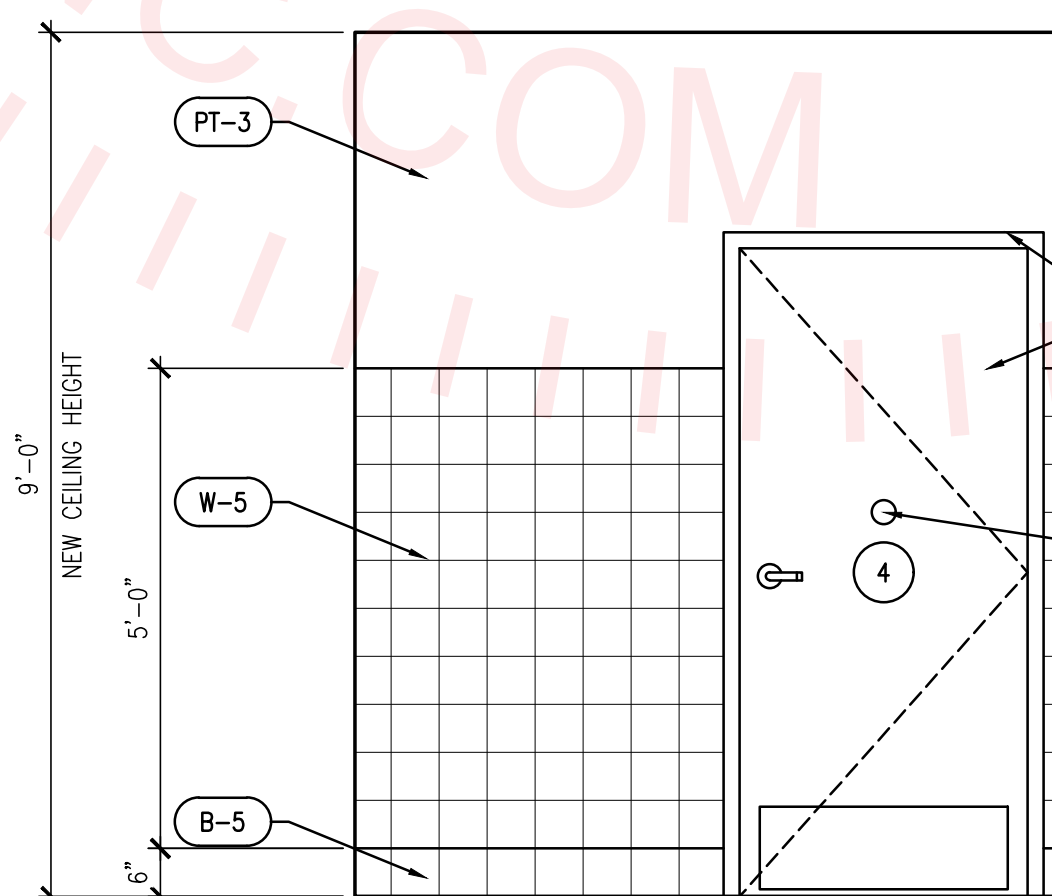
6 RESTROOM #1 ELEVATION 1/2"=1'-0"



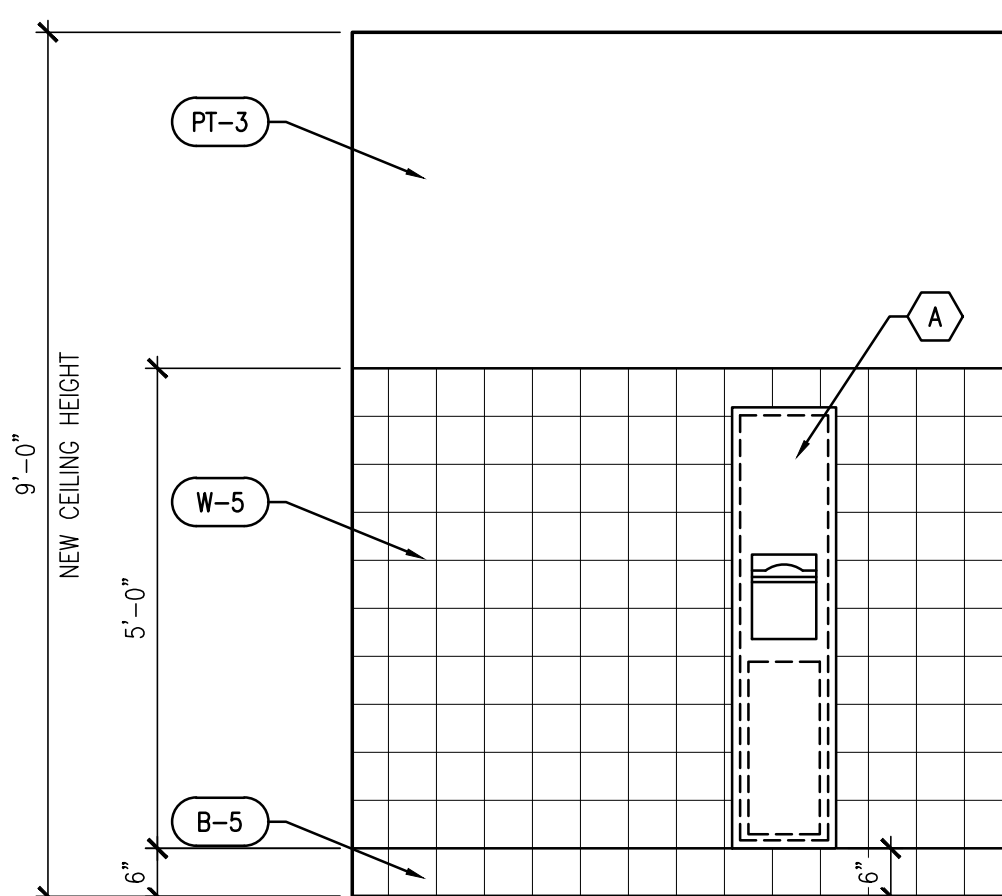
7 RESTROOM #2 ELEVATION 1/2"=1'-0"



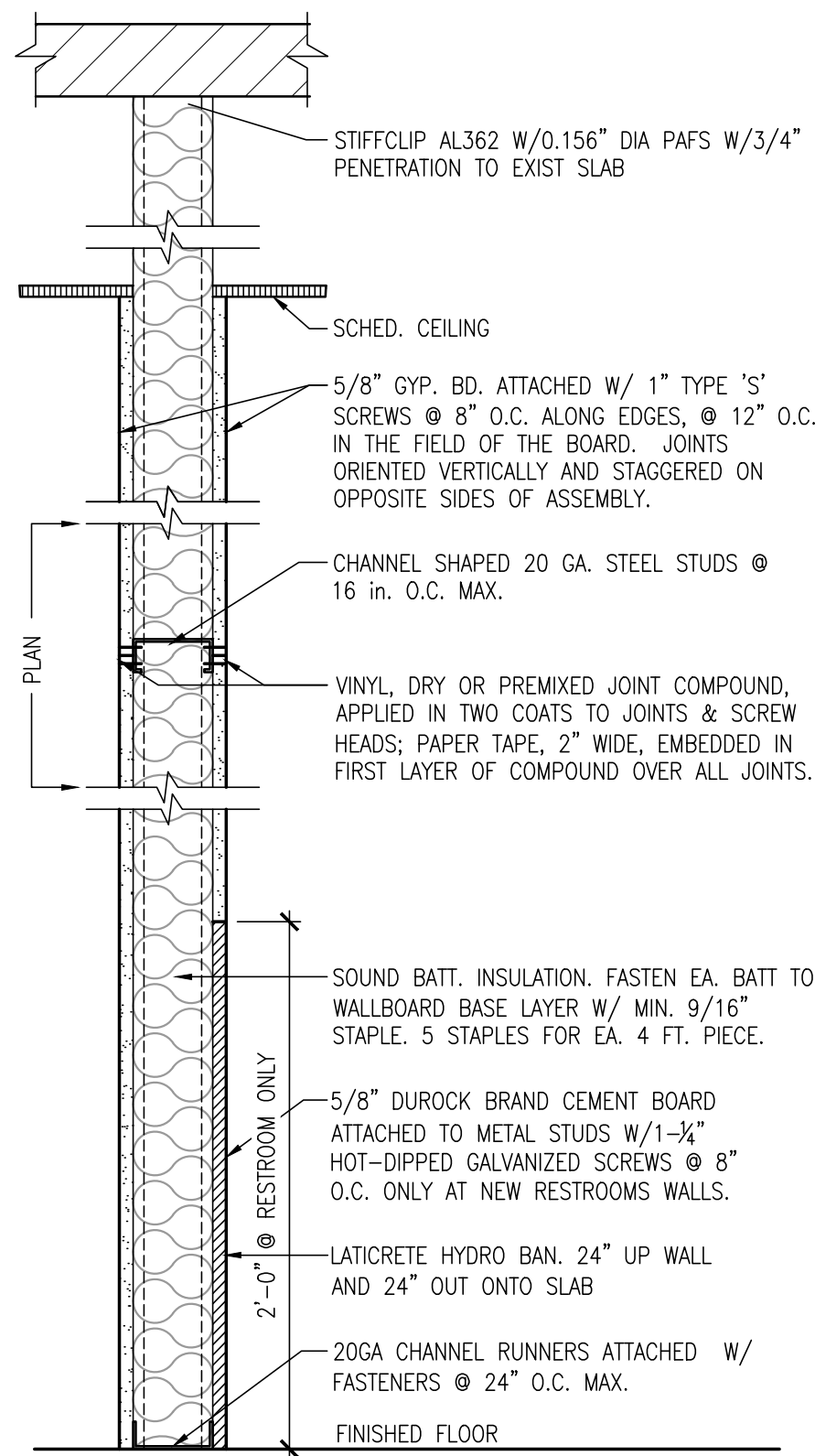
8 RESTROOM #2 ELEVATION 1/2"=1'-0"



9 RESTROOM #2 ELEVATION 1/2"=1'-0"

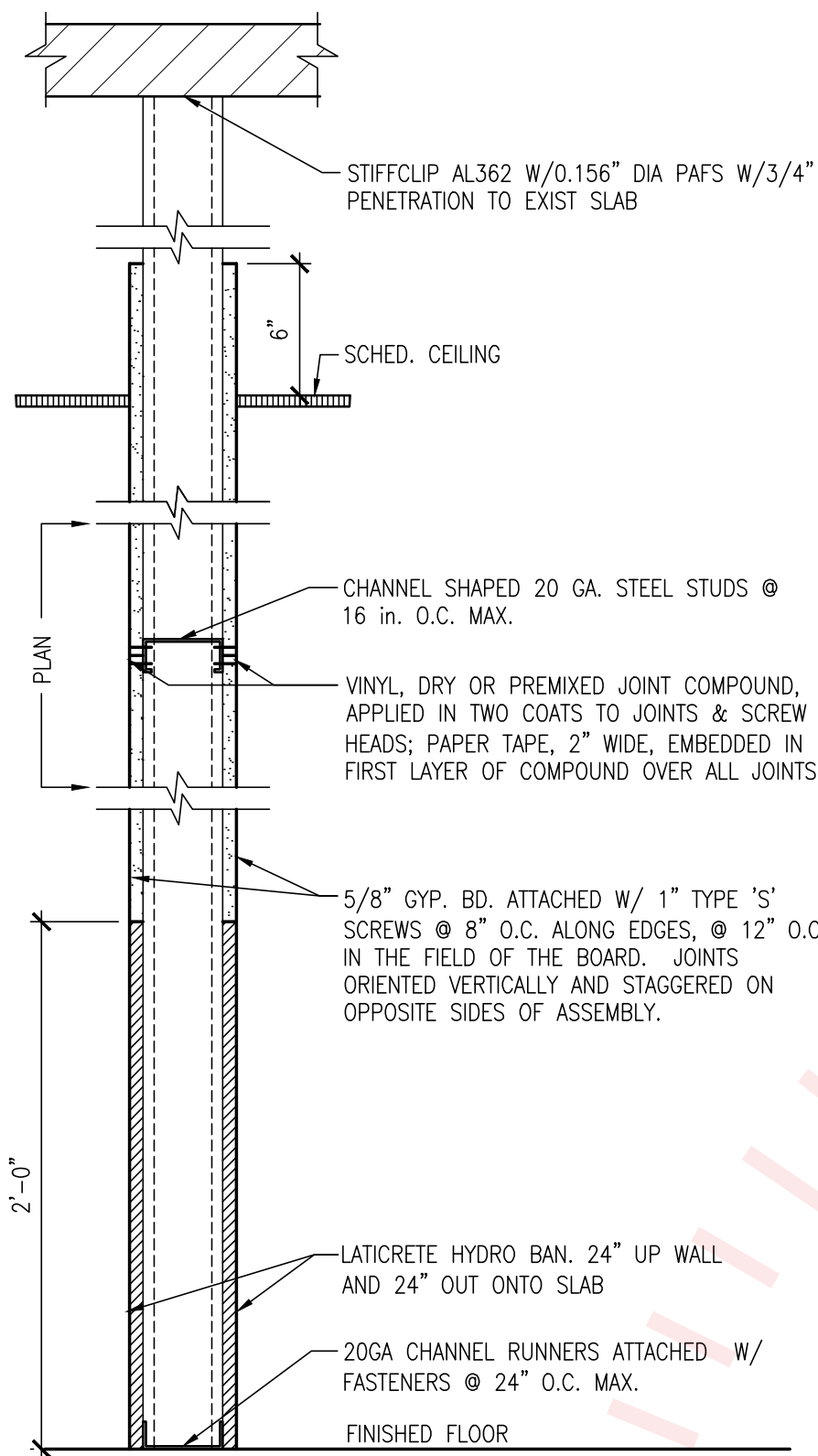


10 RESTROOM #2 ELEVATION 1/2"=1'-0"



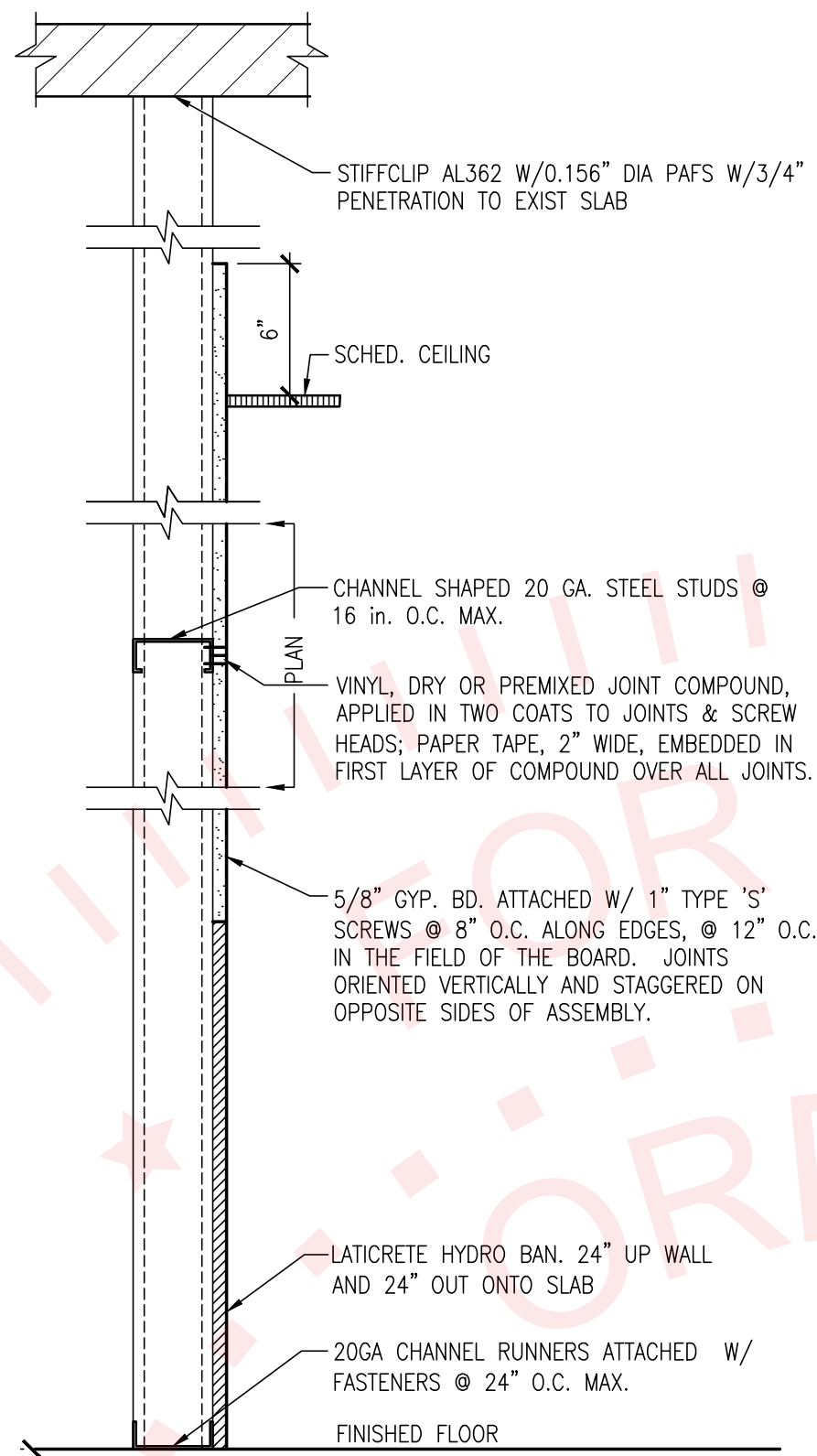
- A1** USE 3 5/8" MTL STUDS
- A2** SAME AS "A1" WHERE TILE WILL BE INSTALLED, G.C. TO PROVIDE TILE BACKER BOARD
- A3** USE 6" MTL STUDS
- A4** SAME AS "A3" WHERE TILE WILL BE INSTALLED, G.C. TO PROVIDE TILE BACKER BOARD
- A5** SAME AS "A1" WHERE TILE WILL BE INSTALLED, G.C. TO PROVIDE TILE BACKER BOARD AND GWB TO STRUCTURE ABOVE

A INTERIOR WALL TYPE- RESTROOM TO CORRIDOR WALL NTS



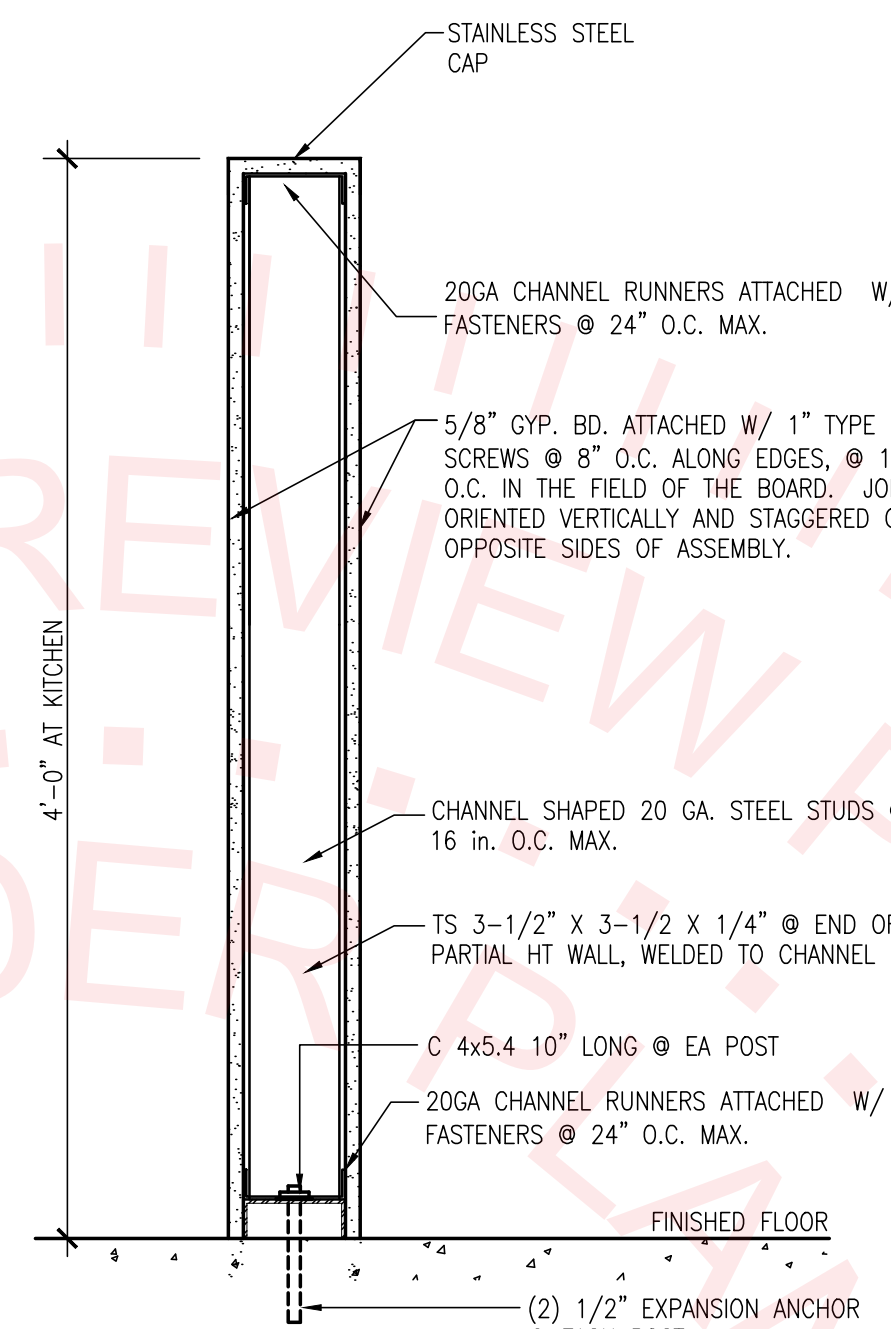
- B1** USE 3 5/8" MTL STUDS
- B2** SAME AS "B3" WHERE TILE WILL BE INSTALLED, G.C. TO PROVIDE TILE BACKER BOARD
- B3** USE 6" MTL STUDS
- B4** SAME AS "B1" WITHOUT CONCRETE BOARD ON BOTH SIDES -GYP BOARD TO FINISH FLOOR

B INTERIOR WALL TYPE- KITCHEN TO KITCHEN WALL NTS



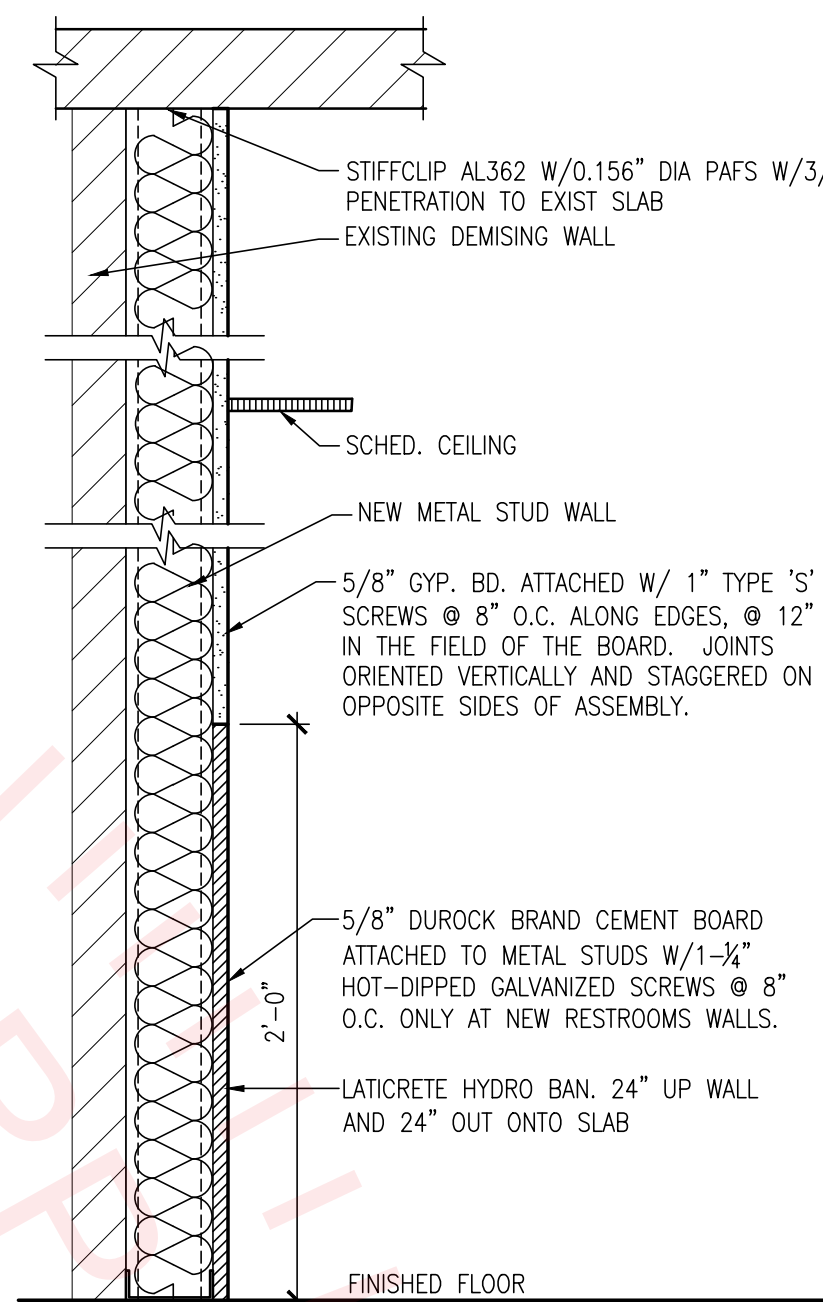
- C1** USE 3 5/8" MTL STUDS
- C2** SAME AS "C1" WHERE TILE WILL BE INSTALLED, G.C. TO PROVIDE TILE BACKER BOARD
- C3** USE 6" MTL STUDS
- C4** SAME AS "C1" WITHOUT CONCRETE BOARD - GYP BOARD TO FINISH FLOOR

C INTERIOR WALL TYPE- AT VOID SPACE NTS



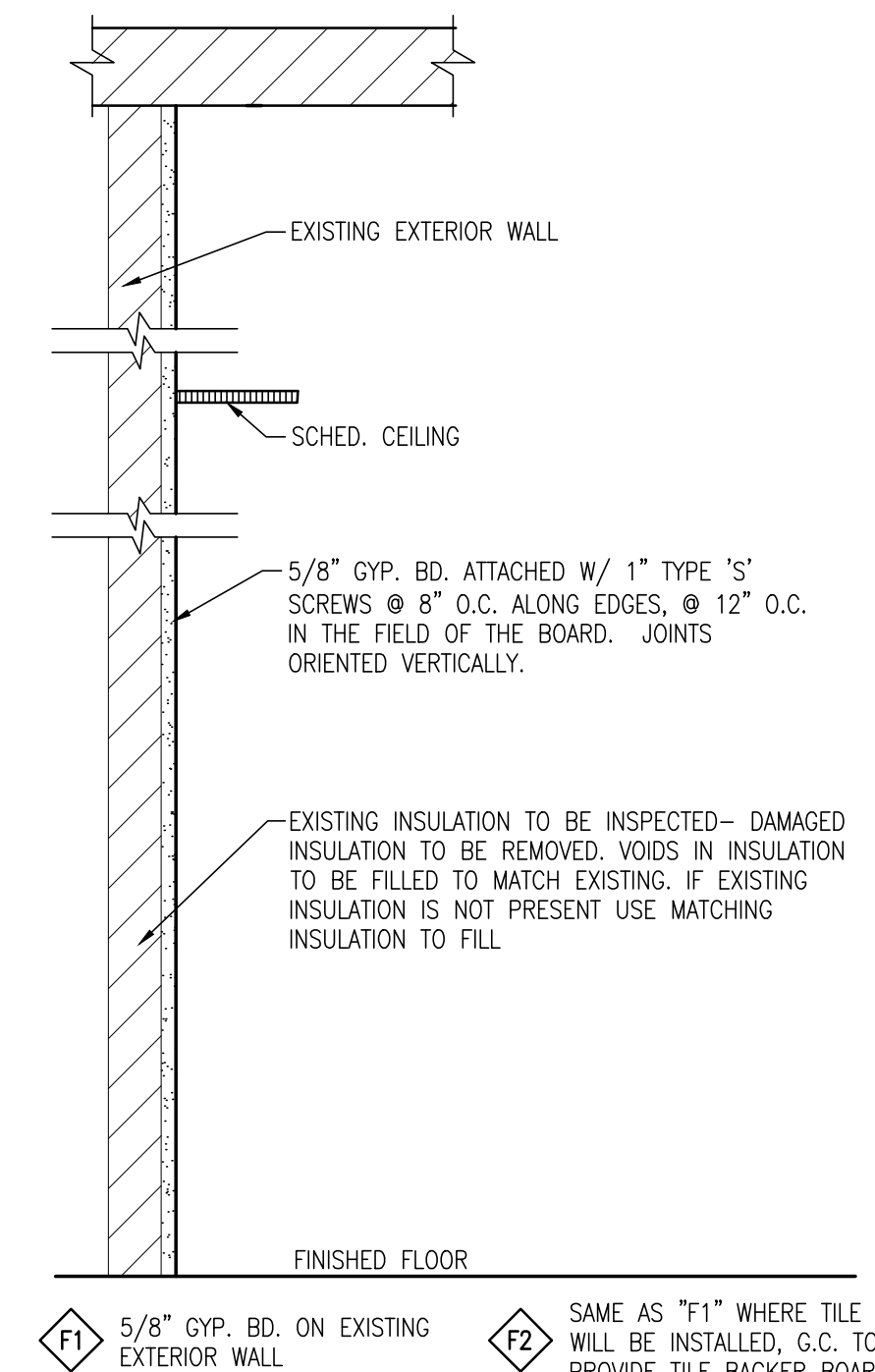
- D1** USE 3 5/8" MTL STUDS

D PARTIAL HEIGHT WALL TYPE- KITCHEN WALL NTS



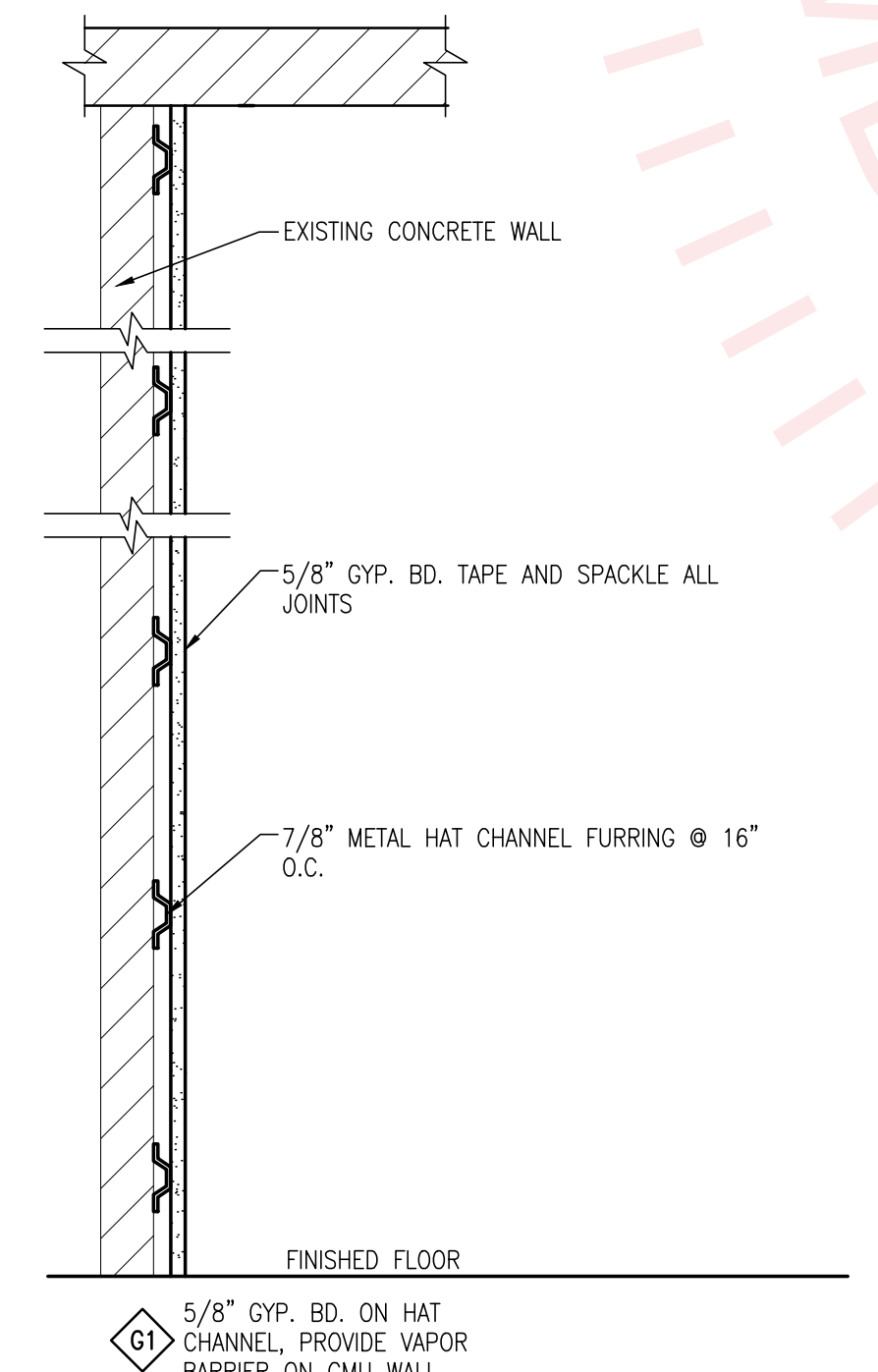
- E1** USE 3 5/8" MTL STUDS
- E2** SAME AS "E1" WHERE TILE WILL BE INSTALLED, G.C. TO PROVIDE TILE BACKER BOARD

E EXTERIOR WALL TYPE- EXISTING WALL NTS



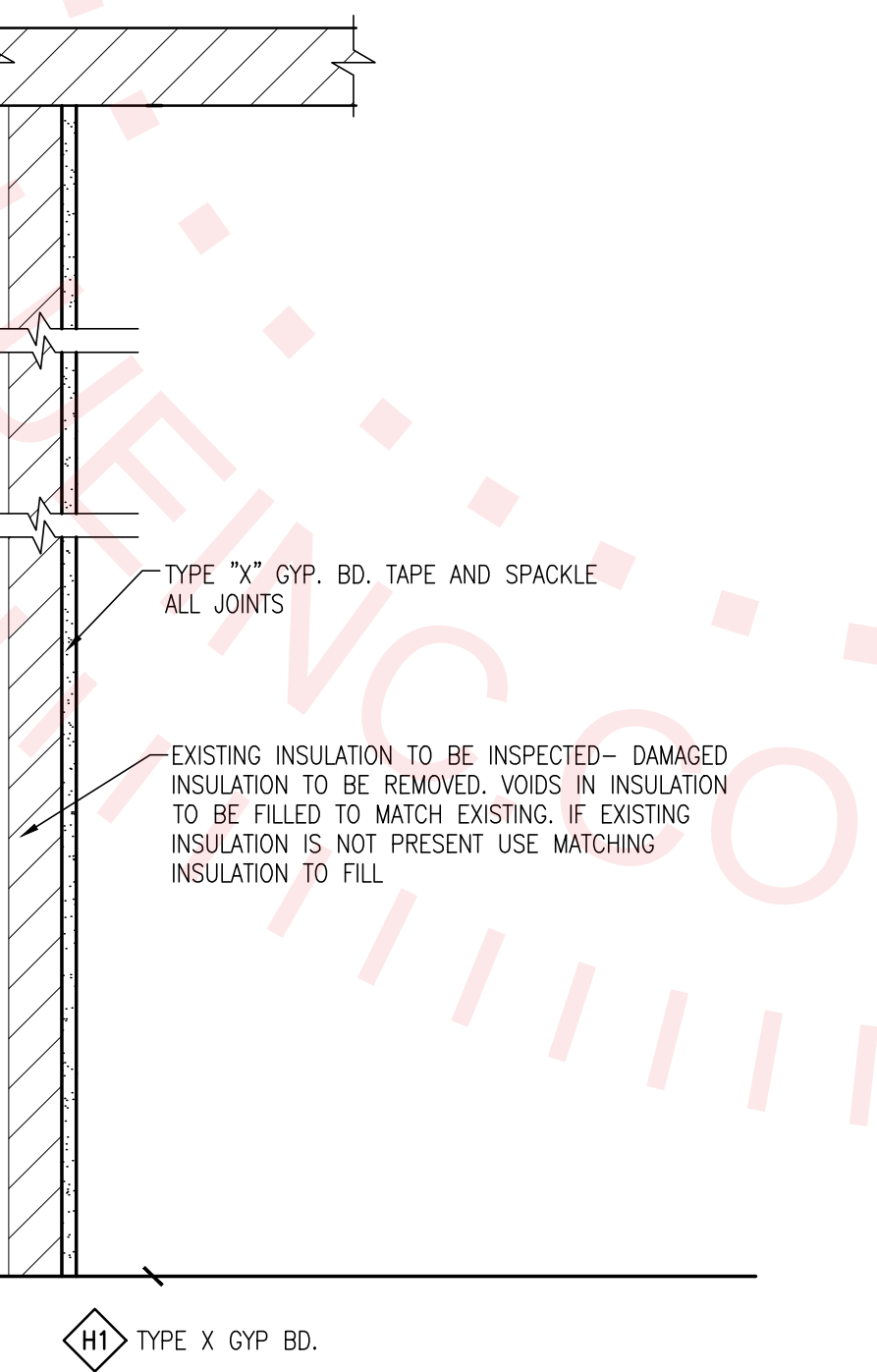
- F1** 5/8" GYP. BD. ON EXISTING EXTERIOR WALL
- F2** SAME AS "F1" WHERE TILE WILL BE INSTALLED, G.C. TO PROVIDE TILE BACKER BOARD

F EXTERIOR WALL TYPE- EXISTING WALL NTS



- G1** 5/8" GYP. BD. ON HAT CHANNEL, PROVIDE VAPOR BARRIER ON CMU WALL

G EXTERIOR WALL TYPE- EXISTING CONCRETE NTS

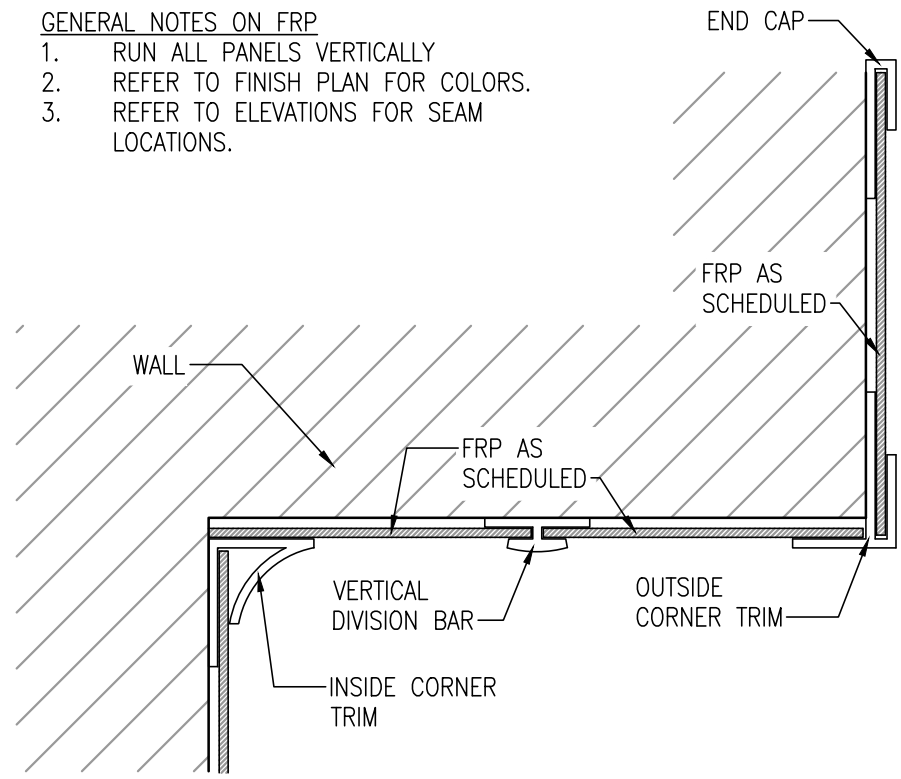


- H1** TYPE X GYP BD.

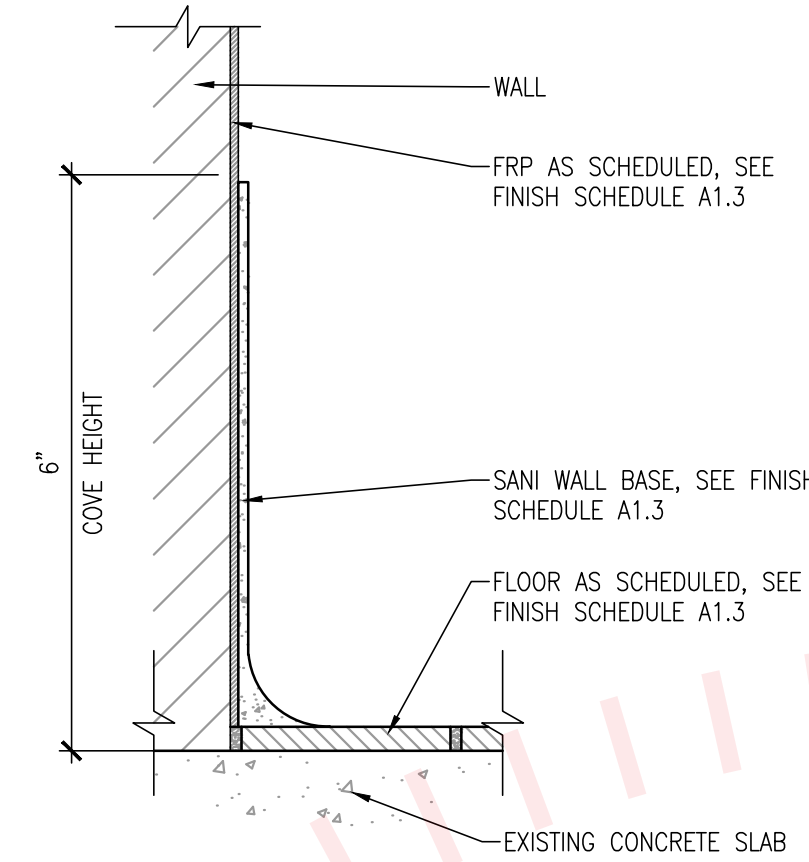
H INTERIOR WALL TYPE- EXISTING DEMISING WALL-UL U419 NTS

PARTITION NOTES:

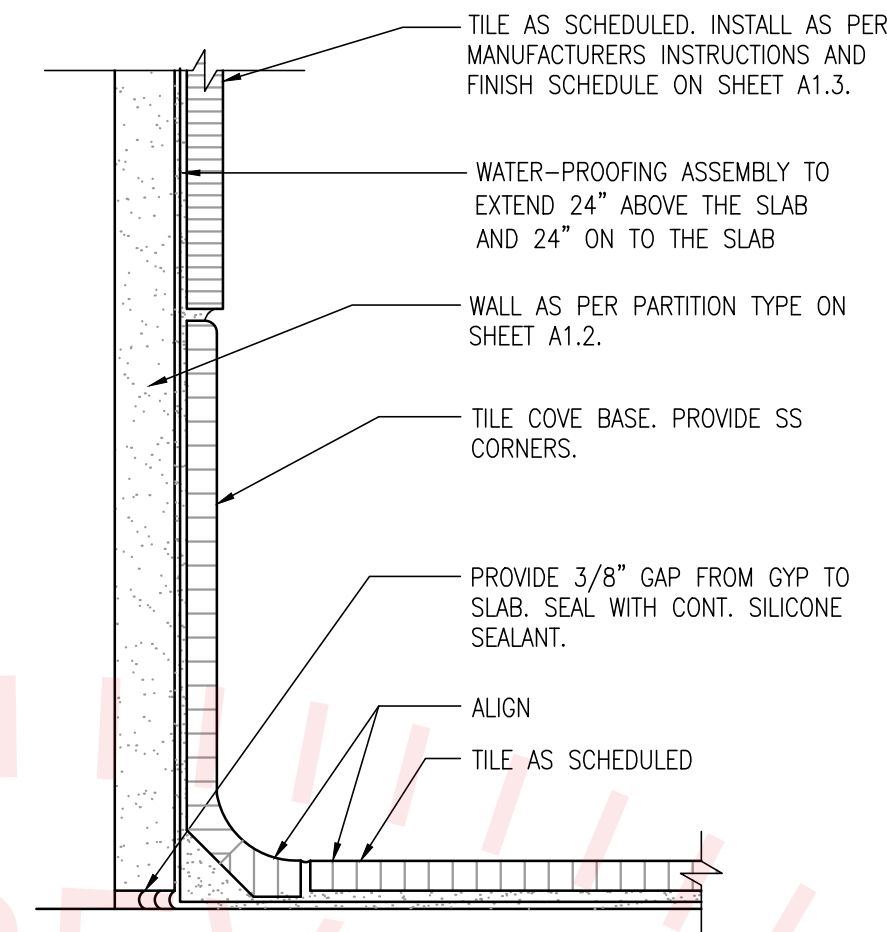
- GYPSUM BOARD WALL CONSTRUCTION THAT IS EXPOSED TO WETNESS OR HIGH HUMIDITY SHALL BE WATER RESISTANT. (SPECIFICALLY IN RESTROOM AND KITCHEN LOCATIONS).
- GENERAL CONTRACTOR TO BRACE TOPS OF FULL HEIGHT PARTITION TO STRUCTURE ABOVE.
- INTERIOR WALLS AND PARTITIONS THAT EXCEED 6 FEET IN HEIGHT INCLUDING THEIR FINISH MATERIALS, SHALL HAVE ADEQUATE STRENGTH TO RESIST THE LOADS TO WHICH THEY ARE SUBJECTED, BUT NOT LESS THAN HORIZONTAL LOAD OF 5PSF.
- USE TWO-STUD CORNERS WHERE POSSIBLE.
- LUMBER AND BLOCKING IN CONTACT WITH MASONRY AND CONCRETE SHALL BE PRESSURED TREATED.
- PROVIDE WOOD BLOCKING IN PARTITIONS AT ALL LOCATIONS WHERE WORK SURFACE, SHELVING, BRACKETS, DISPLAYS, GRAB BARS, HANDRAILS AND/OR EQUIPMENT WILL BE MOUNTED OR ATTACHED TO THE FACE OF WALL FOR STRUCTURAL STABILITY. REFERENCE FLOOR PLANS FOR LOCATIONS OF SUCH EQUIPMENT. ALL CONCEALED WOOD SHALL BE FIRE RETARDANT LUMBER AND INSTALLED WITH THE LABEL FACING OUT.
- GC TO VERIFY STUD SIZE AND SPACING.
- ALL DRYWALL SHALL BE PATCHED/REPAIRED TO A SMOOTH AND LEVEL SURFACE AND FREE OF VISIBLE IMPERFECTIONS.
- GENERAL CONTRACTOR TO PROVIDE WATERPROOFING ON TENANT SIDE OF DEMISING WALLS. 1/2" DENSIELD TO BE USED FOR WATERPROOFING DETAIL WHERE FINISH WILL BE TILE AND IN ALL WET AREAS.
- AT EXISTING WALLS WHERE INSULATION IS REQUIRED, VERIFY/INSTALL R-11.
- IN KITCHEN AREAS PROVIDED PLYWOOD IN LIEU GYP BOARD FROM 4'-0"-8'-0" AFF. (FOR WALL SHELVING), PLYOOD TO BE FIRE-RETARDANT WHERE REQUIRED BY CODE.
- TYPE X GYPSUM BOARD USED FOR ALL FIRE RATED WALL TYPES.



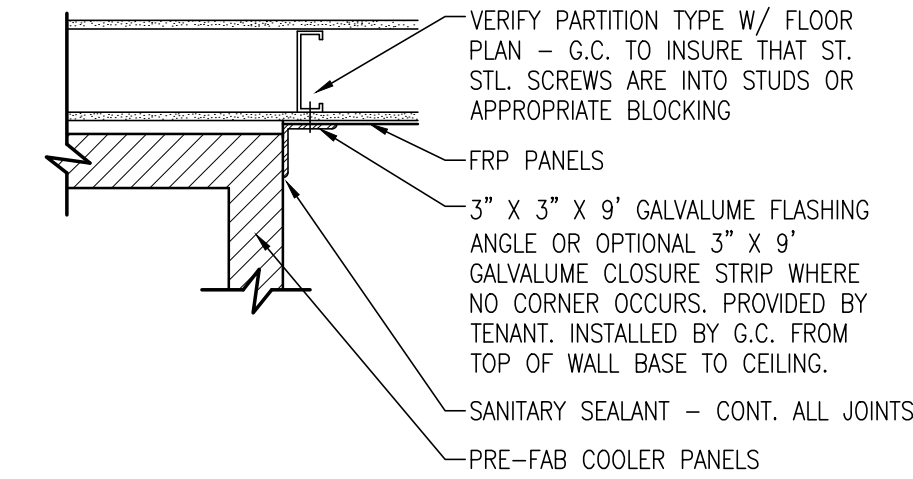
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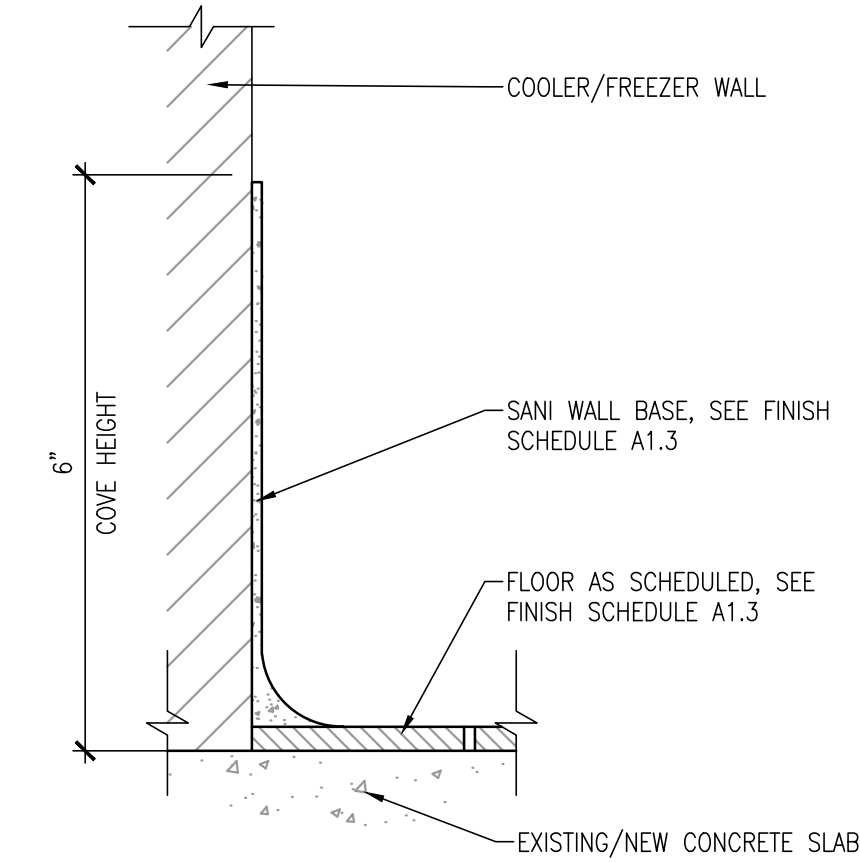
2 FRP DETAILS SECTION NTS



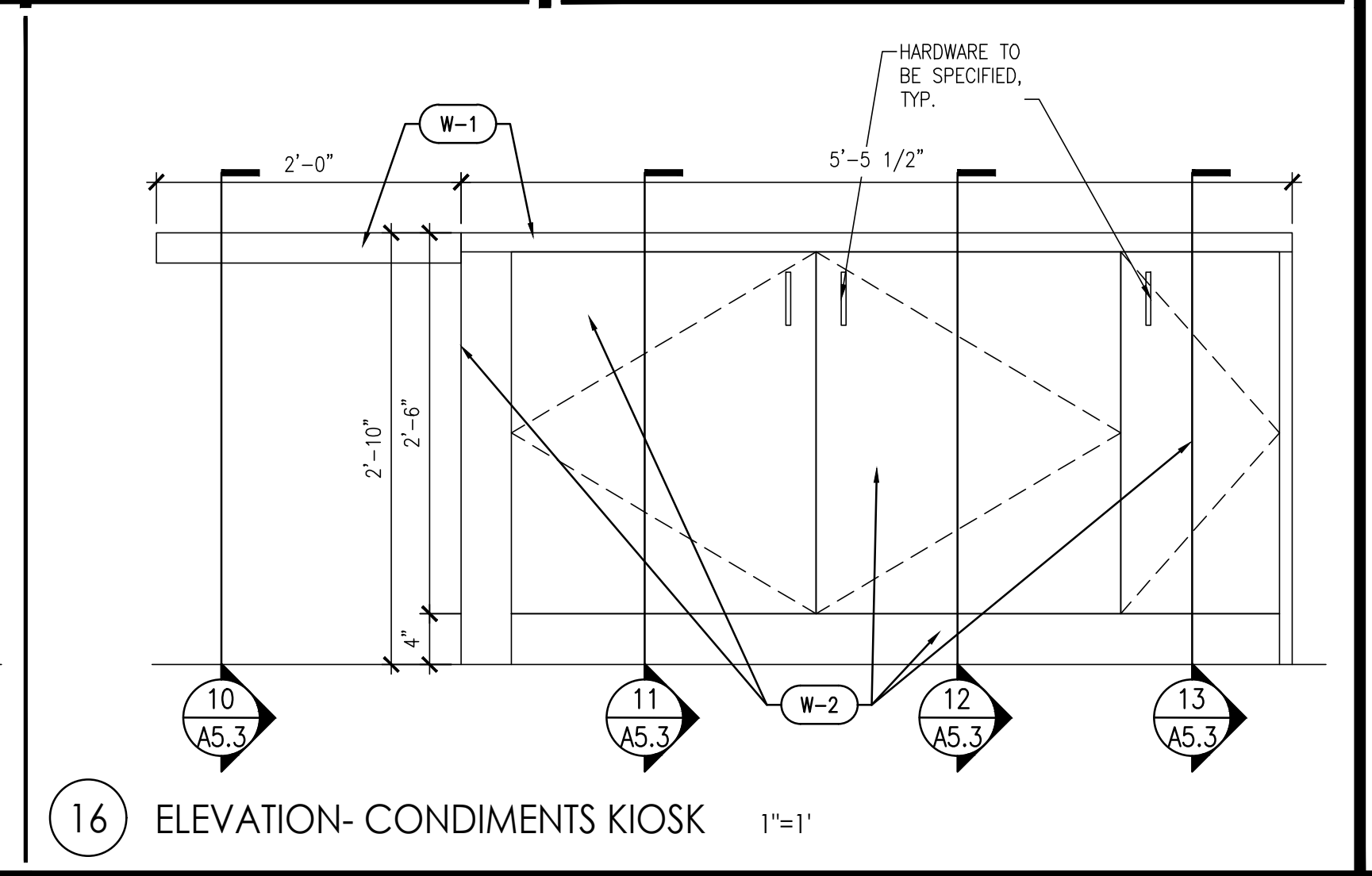
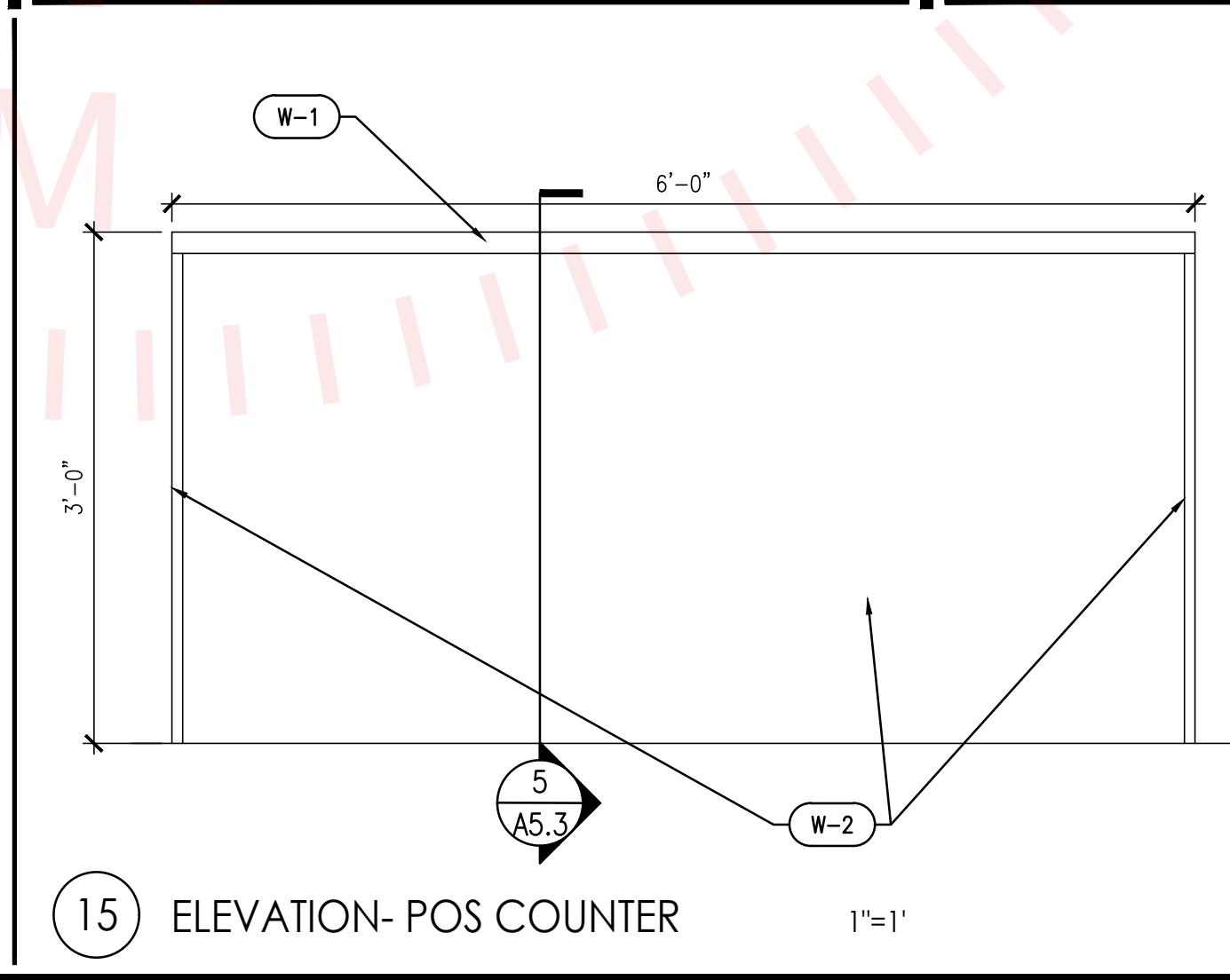
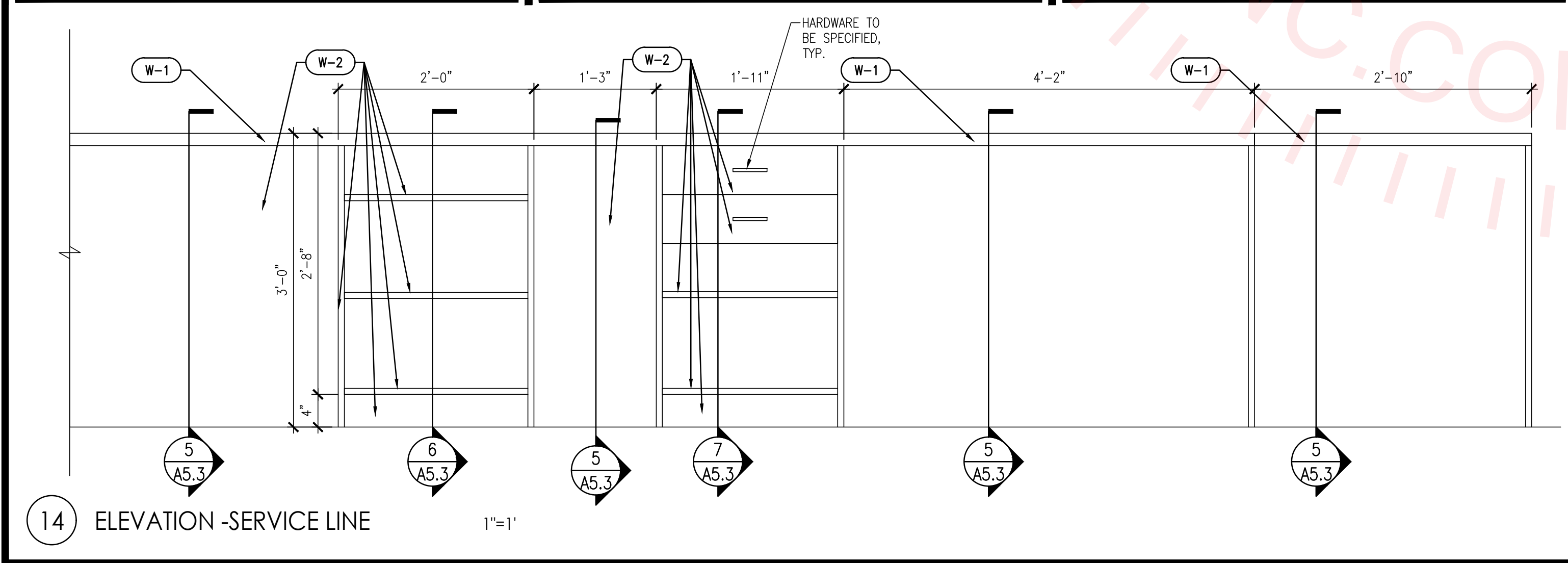
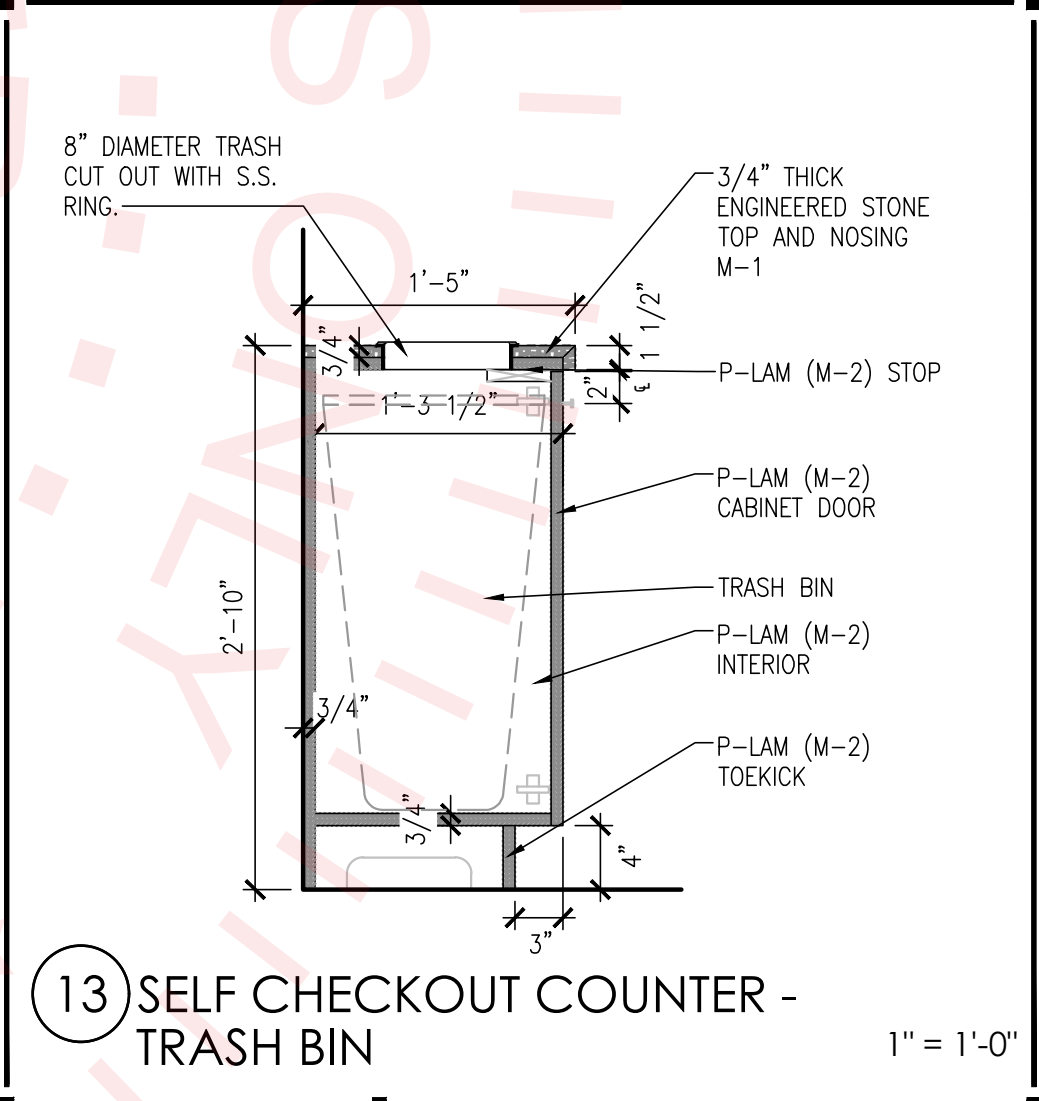
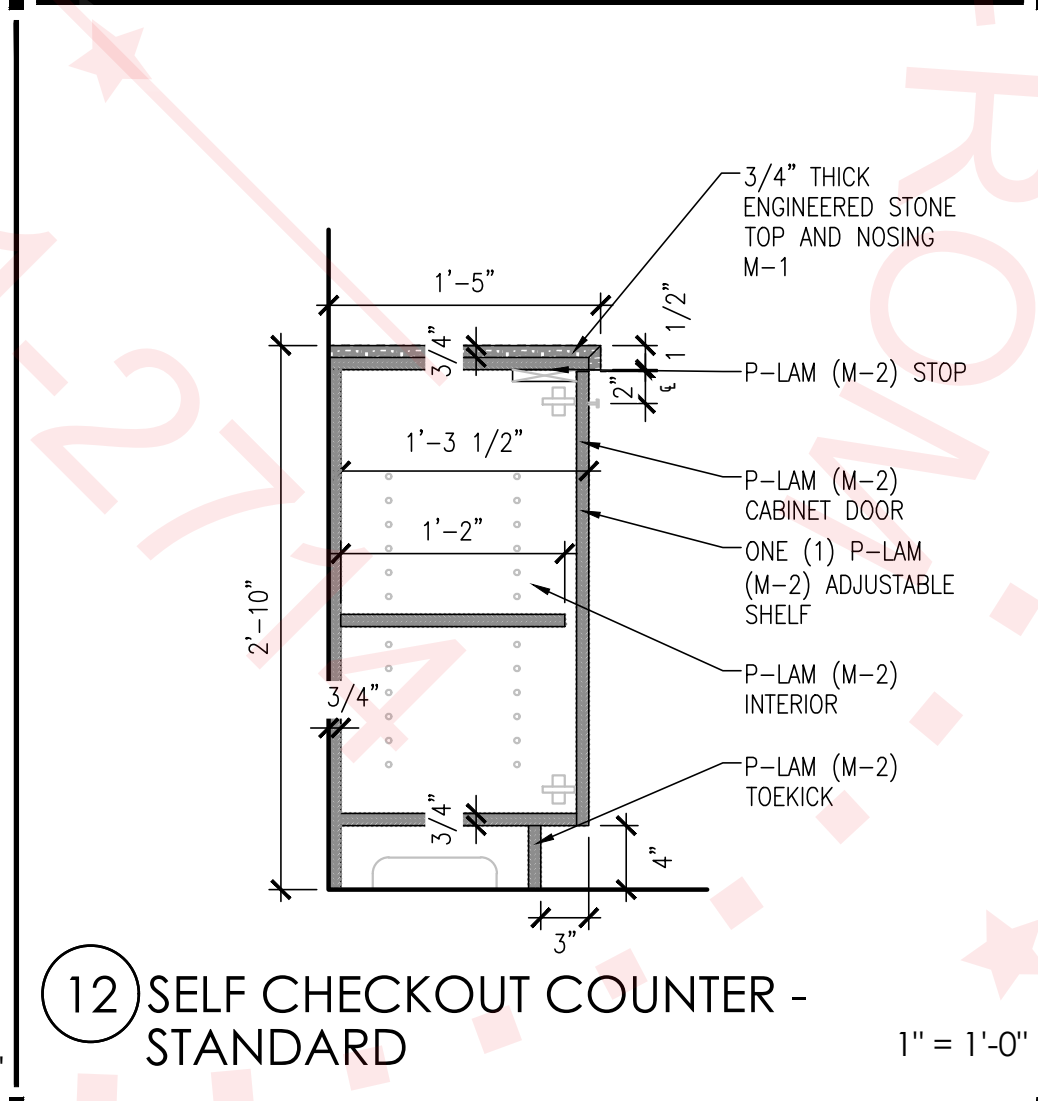
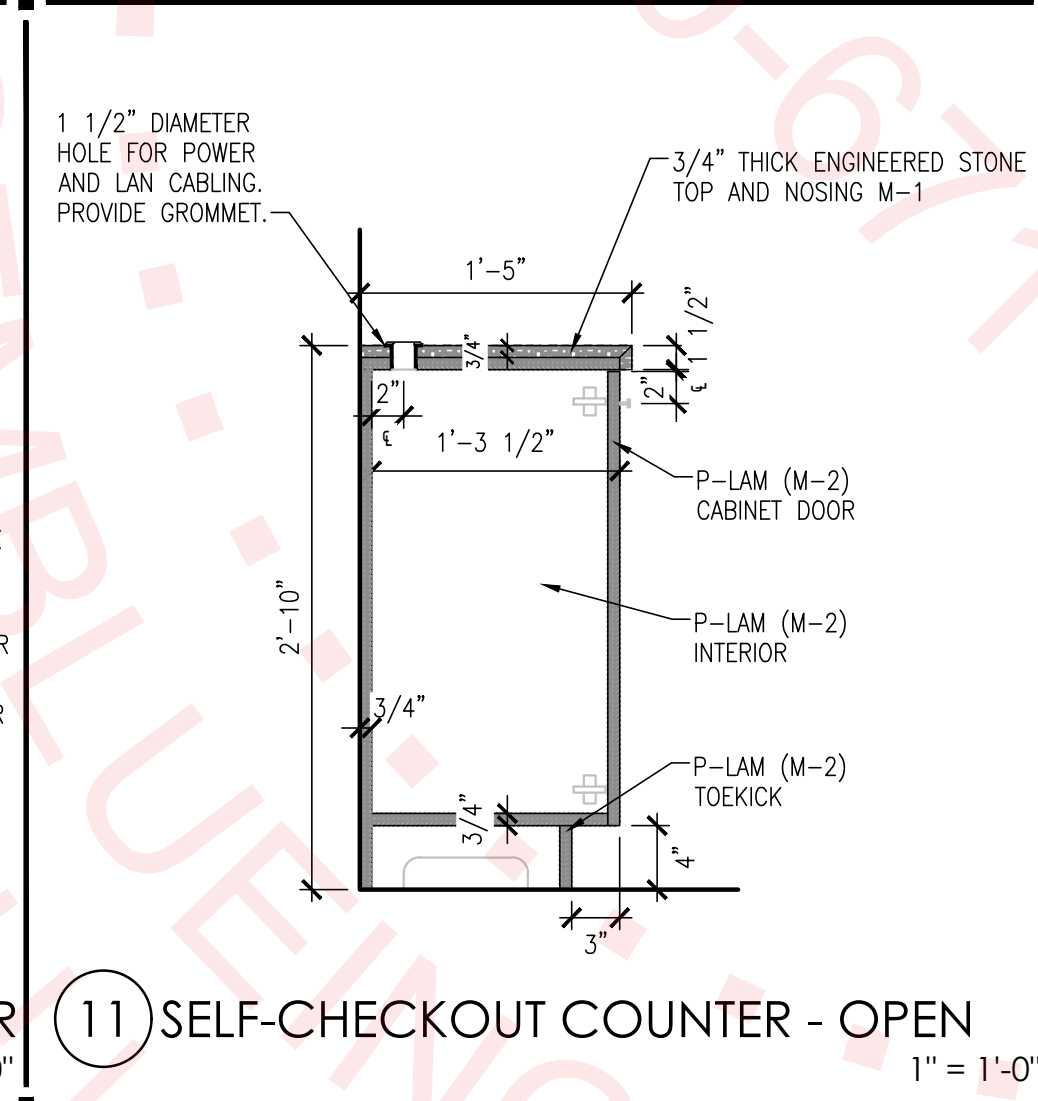
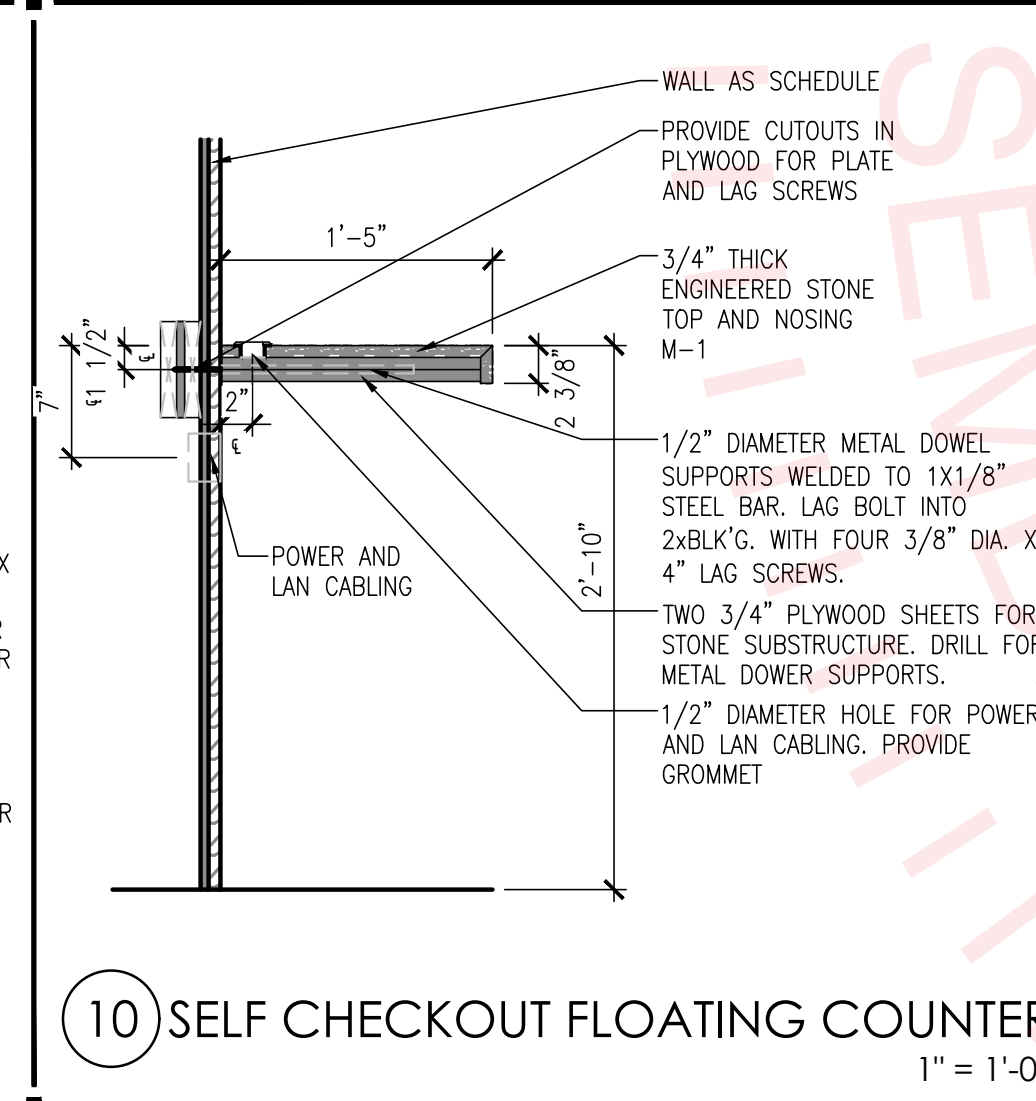
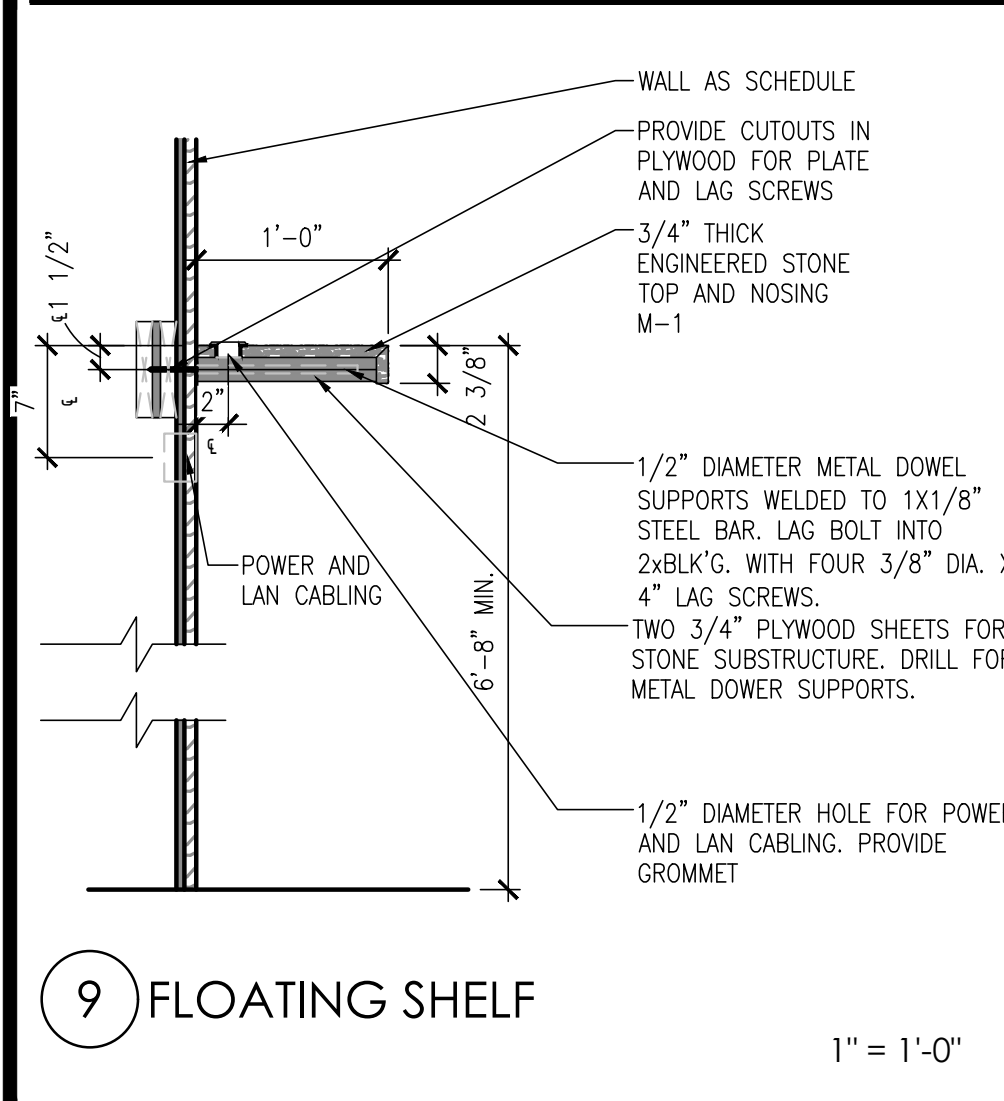
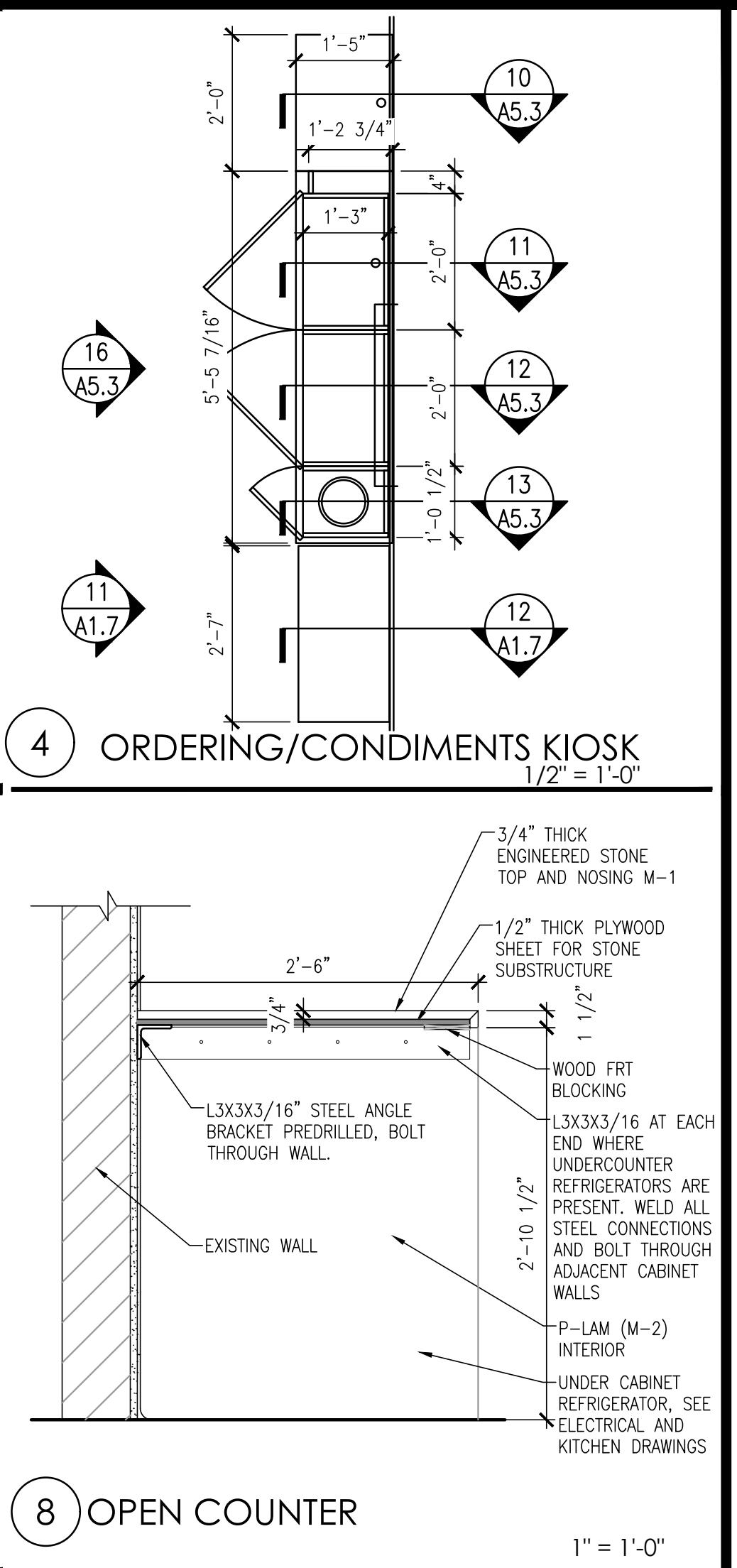
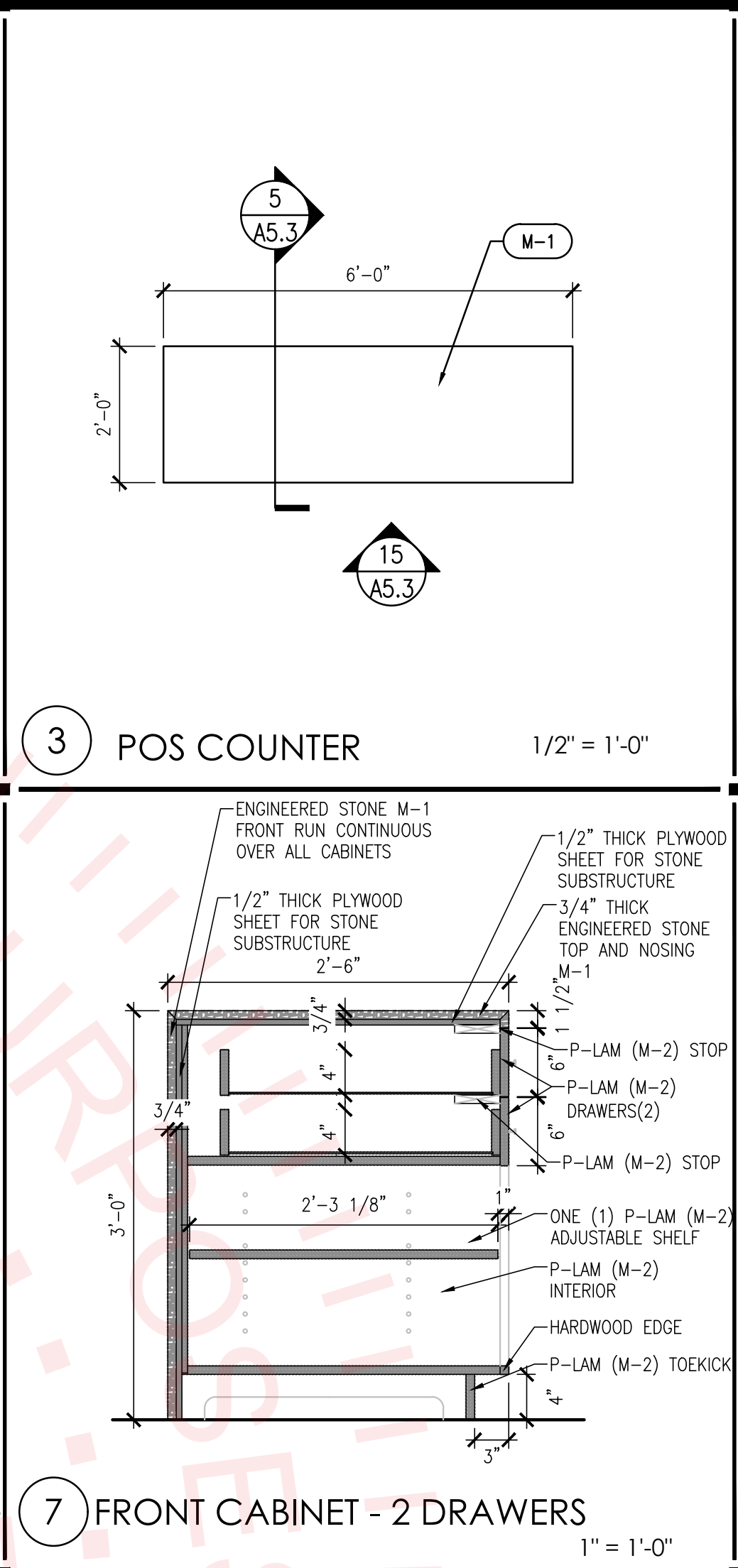
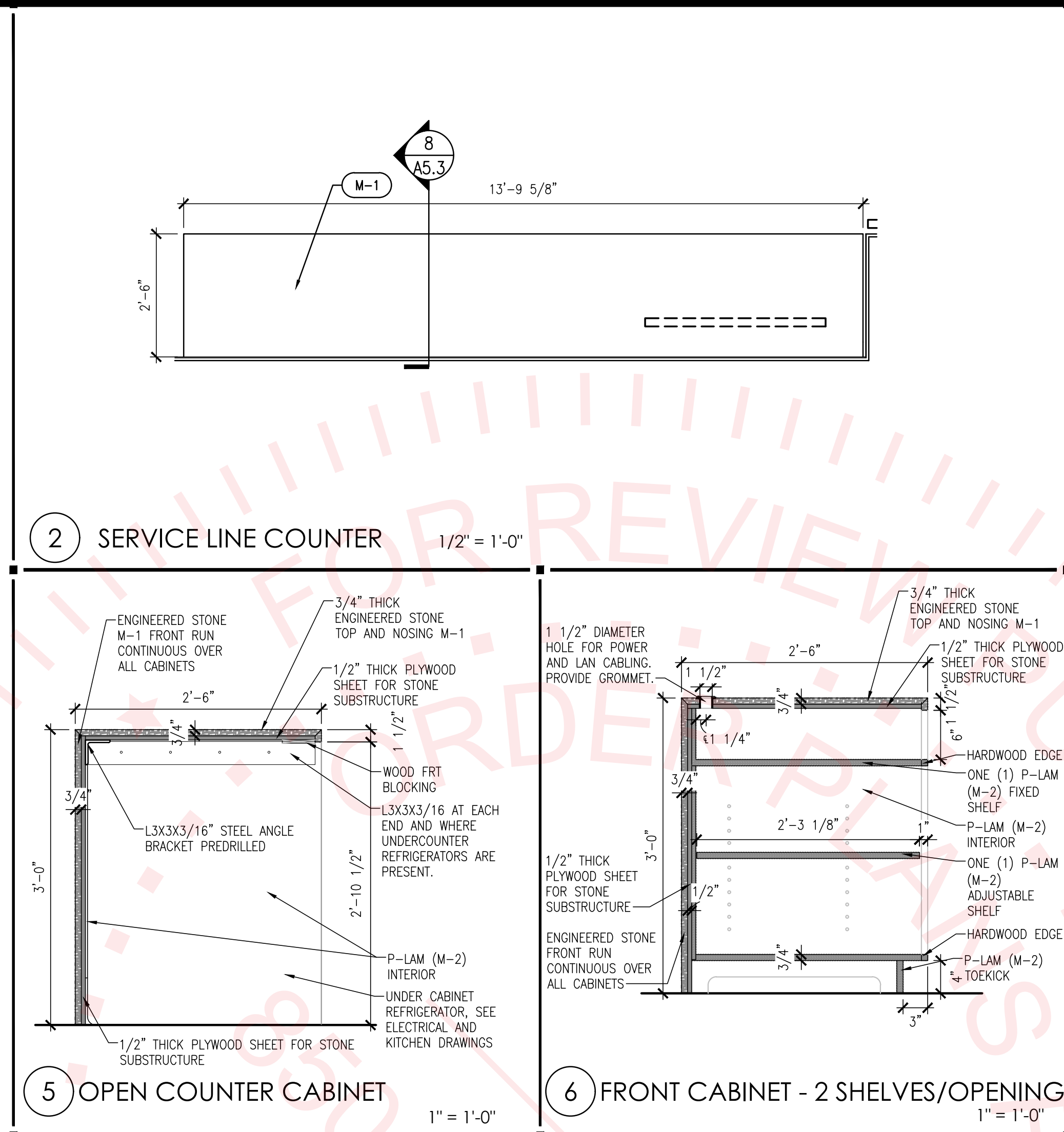
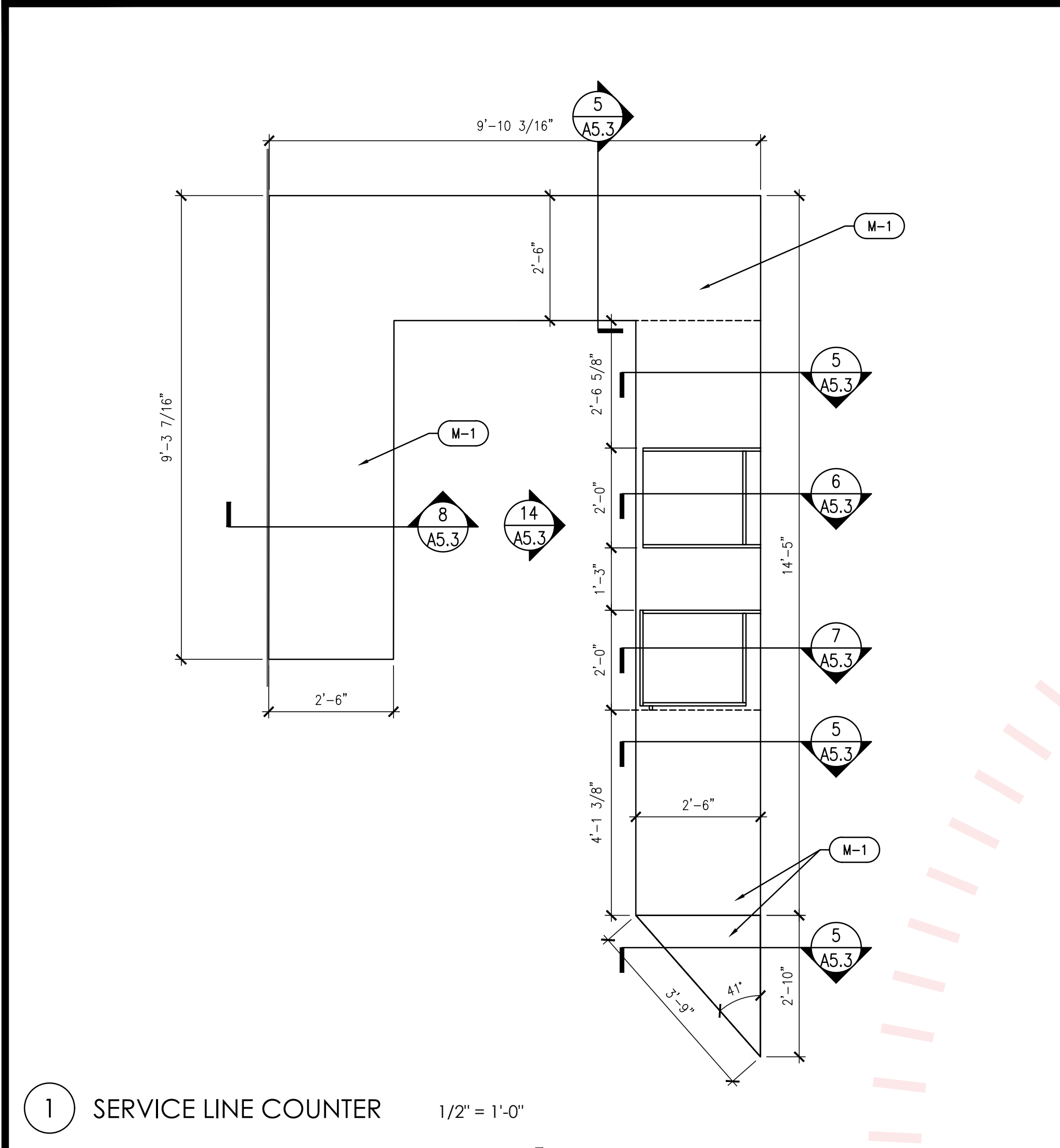
3 TILE COVE BASE 6" = 1'-0"



4 COOLER WALL CLOSURE DETAIL NTS



5 COOLER/FREEZER DETAILS SECTION NTS



DOOR SCHEDULE AND HARDWARE								
NO.	SIZE	DOOR TYPE	FRAME	HARDWARE	STATUS	LOCATION	FINISH	NOTES
1	6'-0" X 6'-8"	D1	EXISTING	EXISTING	EXISTING	CUSTOMER AREA	EXISTING-CLEAN	*
2	3'-0" X 7'-0"	D3	EXISTING	EXISTING	EXISTING	REAR EXIT-KITCHEN	EXISTING-CLEAN	PROVIDE ALARM SYSTEM
3	3'-0" X 7'-0"	D2	2/A6.1	HM-01	NEW	RESTROOM #1	PAINT PT-2	
4	3'-0" X 7'-0"	D2	2/6.1	HM-01	NEW	RESTROOM #2	PAINT PT-2	
* EXISTING DOOR & FRAME - VERIFY HARDWARE, PROVIDE ITEMS LISTED IF NOT EXISTING								

HARDWARE TYPES

HM-01
*DEADLATCH DEADLOCKS LATCHBOLT
*INSIDE LEVER RETRACTS LATCHBOLT (AND DEADBOLT SIMULTANEOUSLY) AT ALL TIMES
*KEY OUTSIDE RETRACTS LATCHBOLT
*WHEN DEADBOLT THROWN, OUTSIDE LEVER OPERABLE ONLY WITH EMERGENCY OR MASTER CODE
*THUMB TURN RETRACTS AND PROJECTS DEADBOLT
*CYLINDER OVERRIDE AND DEADBOLT

HARDWARE SPEC

MANUFACTURER: SARGENT
HARDWARE TYPE: MORTISE
SERIES: 7800
BACKSET: 2 3/4"
LEVER STYLE: W
INTERIOR ROSE: 0 ROSE
EXTERIOR ESCUTCHEON: LE2
THUMBTURN: T2 WITH 130KC
BACKPLATE
STRIKE PLATE: CURVED LIP STRIKE
W/ BOX STRIKE
FINISH: ANSI 630

HINGE MANUFACTURER: STANLEY
HINGE SIZE: 4 1/2"
HINGE FINISH ANSI 630

- REMARKS:
- PROVIDE DOOR AND HARDWARE SHOP DRAWINGS / SUBMITTAL FOR ARCHITECT REVIEW AND APPROVAL PRIOR TO ORDERING.
 - COORDINATE ALL HARDWARE WITH DOORS SHOWN ON PLANS.
 - FOR DOORS 1 3/4" THICK STANDARD.
 - LOOKSETS SHALL BE STANDARD MORTISE TYPE MANUFACTURED BY SARGENT MANUFACTURING COMPANY, U.O.N.
 - SPECIFY HAND ON ORDER.
 - EXTERIOR ESCUTCHEON FOR EXTERIOR DOORS.
 - THUMBTURN FOR HARDWARE WITH DEADLATCH.
 - LOOKS SHALL HAVE A ONE YEAR LIMITED WARRANTY.
 - ALL DOORS TO HAVE DOOR STOPS, U.O.N.
 - ALL MISCELLANEOUS HARDWARE FINISHES TO MATCH MORTISE FINISH AS SPECIFIED.

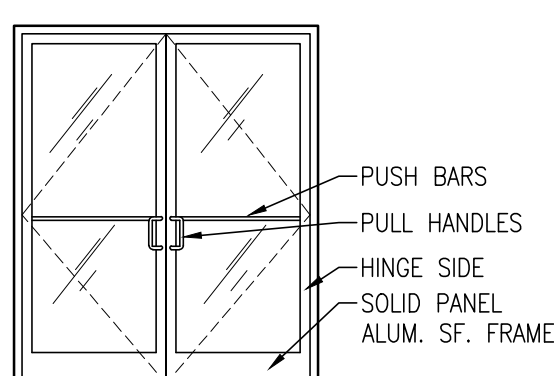
DOOR HARDWARE GENERAL NOTES:

- CONTRACTOR SHALL EXAMINE DRAWINGS AND SPECIFICATIONS AND PROVIDE PROPER HARDWARE FOR ALL DOOR OPENINGS SHOWN, WHETHER LISTED OR NOT AND IMMEDIATELY BRING OMISSIONS TO THE ATTENTION OF THE ARCHITECT PRIOR TO FINALIZING BID; OTHERWISE LIST WILL BE CONSIDERED COMPLETE. NO EXTRAS WILL BE ALLOWED.
- ALL HARDWARE TO BE INSTALLED IN ACCORDANCE WITH MANUFACTURER'S INSTRUCTIONS AND RECOMMENDATIONS.
- COMPLY WITH RECOMMENDATIONS OF HEIGHTS FOR ITEMS NOT SPECIFICALLY INDICATED IN DRAWINGS.
- CONTRACTOR TO FACTORY OR SHOP PREPARE ALL MORTISES, CUTOOTS, & HOLES IN DOORS FOR SPECIFIED HARDWARE.
- FIT HARDWARE PRIOR TO PAINTING, THEN REMOVE PRIOR TO PAINTING DOORS AND FRAMES; REINSTALL AFTER PAINTING IS COMPLETE.
- CONFORM WITH ADA REQUIREMENTS TO ENSURE COMPLIANCE WITH FEDERAL AND LOCAL REQUIREMENTS FOR ACCESSIBILITY.
- PRIOR TO SUBSTANTIAL COMPLETION CONTRACTOR SHALL ADJUST CLOSERS, LOCKS, AND CRITIC OPERATIONAL HARDWARE TO ENSURE PROPER FUNCTION OF ALL OPENINGS.
- ALL LOCKSETS TO BE HEAVY DUTY CYLINDER HARDWARE U.O.N.
- CONTRACTOR TO PROVIDE 4 SETS OF KEYS TO ALL LOCKED DOORS IN THE PROJECT.
- HARDWARE TO BE COORDINATED WITH DOOR SCHEDULE.
- PRIOR TO ORDERING ANY AND ALL HARDWARE, GENERAL CONTRACTOR TO PROVIDE SHOP DRAWINGS AND SUBMITTALS TO ARCHITECT FOR REVIEW AND APPROVAL THAT PROVIDE THE FOLLOWING INFORMATION FOR EACH OPENING IN THE PROJECT.
 - IDENTIFY EACH PIECE OF HARDWARE WITH OPENING NUMBER AND MANUFACTURER'S NAME AND CATALOGUE NUMBER.
 - INDICATE ALL HARDWARE LOCATIONS AND MOUNTING HEIGHTS.

METAL DOORS AND FRAMES GENERAL NOTES:

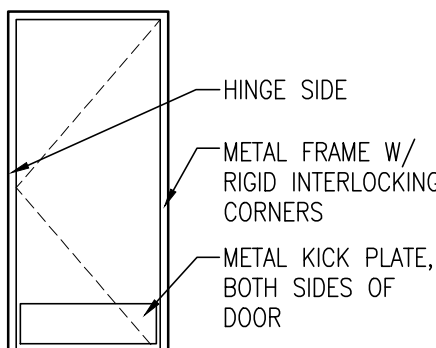
- METAL DOORS, FRAMES, SIDELIGHTS AND BORROWED LIGHTS SHALL CONFORM TO ALL RELEVANT NATIONAL REFERENCES STANDARDS INCLUDING BUT NOT LIMITED TO ASTM, ANSI, DHI, SDI, UL, AND NFPA.
- DETAILED METAL DOOR AND FRAME SCHEDULE SHALL BE SUBMITTED TO THE ARCHITECT FOR REVIEW AND APPROVAL. FABRICATION, ORDERING, OR PURCHASING OF MATERIAL SHALL NOT COMMENCE UNTIL SUCH APPROVAL HAS BEEN RECEIVED.
- METAL DOOR AND FRAME SUBMITTAL SHALL BE COORDINATED WITH DOOR OPENING/SWING DIRECTIONS AND WITH HARDWARE SCHEDULE AS SHOWN IN DRAWINGS.
- FIRE RATED DOORS AND FRAMES AS INDICATED IN SCHEDULE SHALL BE LABELED BY UL, WH, OR OTHER AGENCY ACCEPTABLE TO THE JURISDICTION.
- ALL DOOR AND FRAME PRODUCTS SHALL BE SUPPLIED FROM A SINGLE MANUFACTURER WHO IS A MEMBER OF THE STEEL DOOR INSTITUTE.
- PROVIDE UNITS OF GALVANIZED STEEL WHERE INDICATED ON THE DOOR SCHEDULE.
- PROVIDE LOUVERS AND VISION LIGHTS WHERE INDICATED IN THE DOOR SCHEDULE AND ON THE DRAWINGS.
- INSTALL FRAMES PLUMB, LEVEL, RIGID, AND IN TRUE ALIGNMENT IN ACCORDANCE WITH ANSI A250.11.
- INSPECT AND INSTALL FIRE RATED DOORS AND FRAMES IN ACCORDANCE WITH NFPA 80.
- DOOR SHALL BE INSTALLED AND FASTENED TO MAINTAIN ALIGNMENT WITH FRAMES TO ACHIEVE MAXIMUM OPERATIONAL EFFECTIVENESS AND APPEARANCE. DOORS SHALL BE ADJUSTED TO MAINTAIN PERIMETER CLEARANCES. SHIMMING SHALL BE PERFORMED BY THE INSTALLER AS NEEDED TO ASSURE THE PROPER CLEARANCES ARE ACHIEVED.
- ALL HOLLOW METAL (H.M.) DOORS TO BE 1 3/4" THICK U.O.N.
- ALL H.M. DOORS TO BE 18GA STEEL MIN.
- 3 HINGES TYPICAL ON DOORS NO TALLER THAN 7'-6".
- DOORS OVER 7'-6" TO 10'-0" REQUIRE 4 HINGES.
- WHERE MORTISE HARDWARE SPECIFIED PROVIDE MORTISE LOCK REINFORCEMENT.
- WHERE CLOSER SPECIFIED PROVIDE REINFORCEMENT AT TOP OF DOOR FOR CLOSER.
- WHERE RIM HARDWARE SPECIFIED PROVIDE REINFORCEMENT BOX FOR APPLICATION OF HARDWARE.
- WHERE HEAD AND SILL BOLTS SPECIFIED PROVIDE REINFORCEMENT AT HARDWARE MOUNTING LOCATIONS.
- FOR DOUBLE DOORS REFER TO PLANS TO DETERMINE ACTIVE AND INACTIVE LEAVES.
- ALL CUTOUTS FOR SCHEDULE HARDWARE SHALL BE FACTORY PREPARED.
- COORDINATE ANCHOR SYSTEMS WITH WALLS TYPES.
- WHERE FIRE RATED DOORS ASSEMBLIES IN FIRE RATED PARTITIONS ARE REQUIRED, FIRE RATING OF FRAME TO MATCH FIRE RATING OF DOOR.

DOOR TYPES



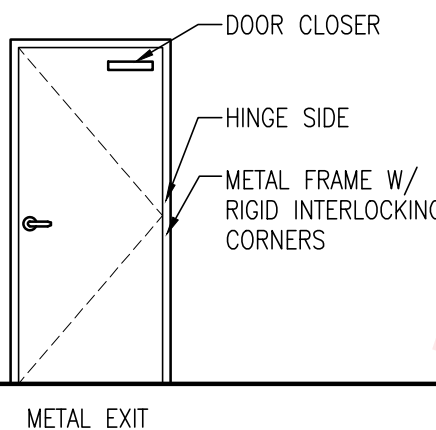
GLASS / ALUM.
ENTRY DOOR

D1 FOR REFERENCE ONLY EXISTING DOOR TYPE TO BE VERIFIED IN FIELD.



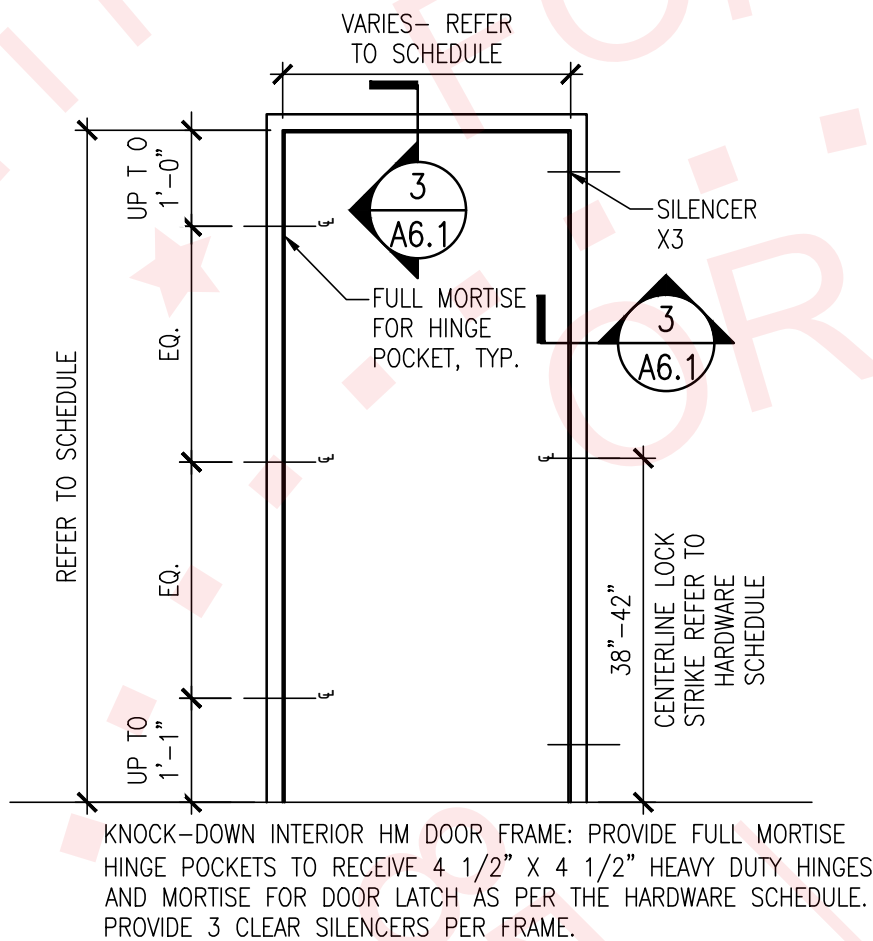
HOLLOW METAL TYPE "F" LEVEL ONE
STANDARD DOOR. PREP FOR
HARDWARE AS SPECIFIED.

D2 SINGLE
DOOR



METAL EXIT

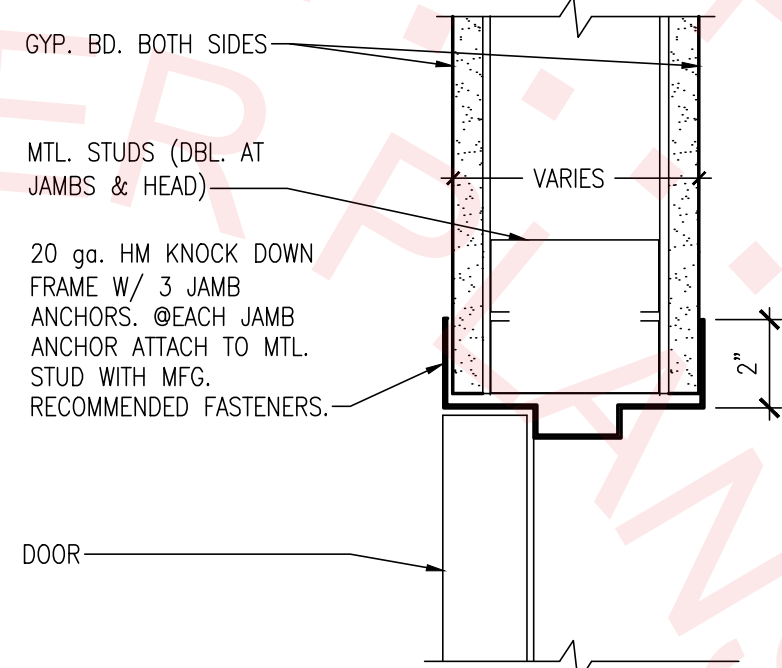
D3 FOR REFERENCE ONLY EXISTING DOOR TYPE TO BE VERIFIED IN FIELD.



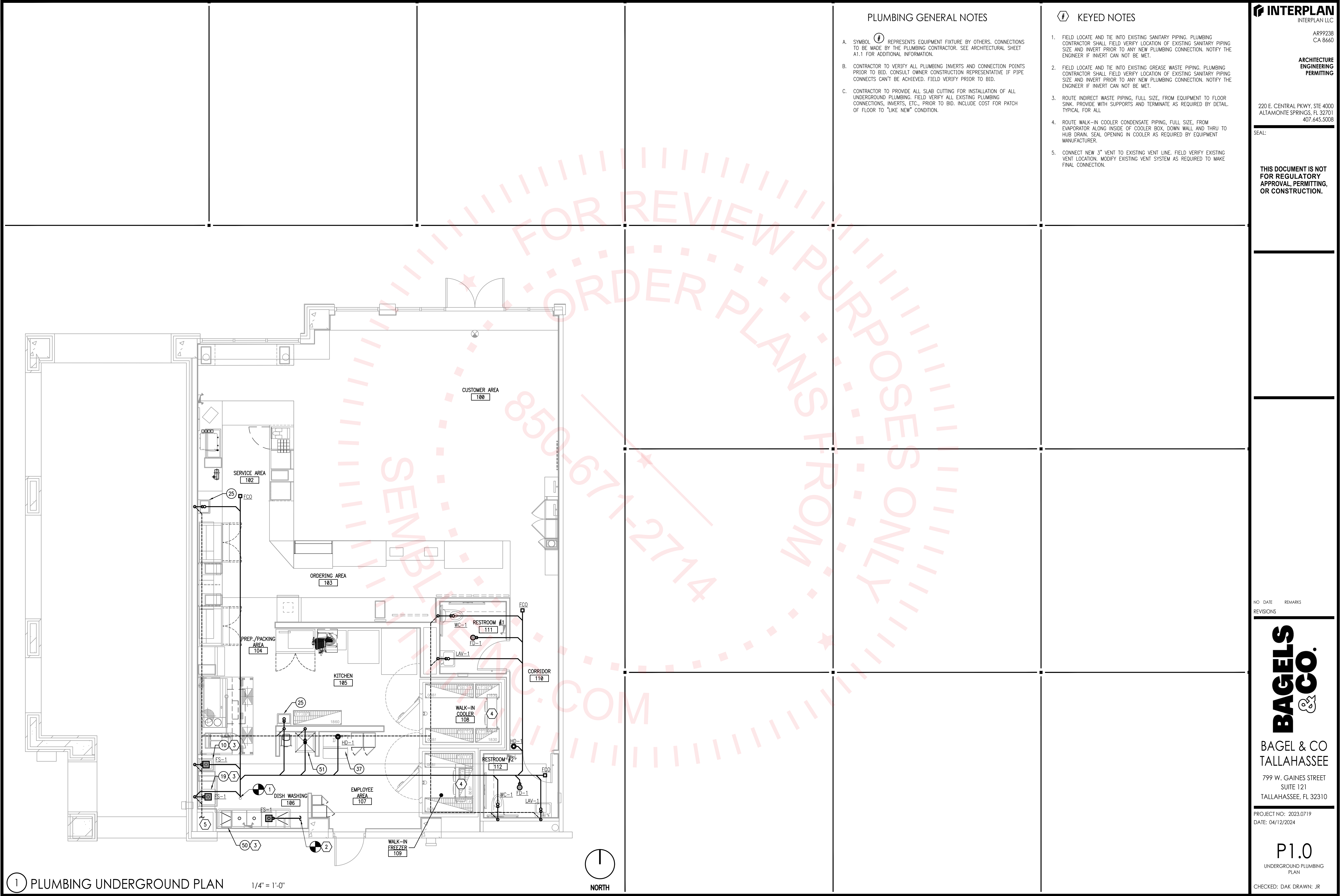
KNOCK-DOWN INTERIOR HM DOOR FRAME: PROVIDE FULL MORTISE
HINGE POCKETS TO RECEIVE 4 1/2" X 4 1/2" HEAVY DUTY HINGES
AND MORTISE FOR DOOR LATCH AS PER THE HARDWARE SCHEDULE.
PROVIDE 3 CLEAR SILENCERS PER FRAME.

2 FRAME

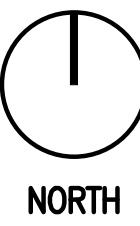
NTS



3 DOOR HEADER/JAMB DETAIL 3" = 1'-0"



1 PLUMBING UNDERGROUND PLAN 1/4" = 1'-0"



PLUMBING GENERAL NOTES

- A. SYMBOL  REPRESENTS EQUIPMENT FIXTURE BY OTHERS. CONNECTIONS TO BE MADE BY THE PLUMBING CONTRACTOR. SEE ARCHITECTURAL SHEET A1.1 FOR ADDITIONAL INFORMATION.
- B. CONTRACTOR TO VERIFY ALL PLUMBING INVERTS AND CONNECTION POINTS PRIOR TO BID. CONSULT OWNER CONSTRUCTION REPRESENTATIVE IF PIPE CONNECTS CAN'T BE ACHIEVED. FIELD VERIFY PRIOR TO BID.
- C. CONTRACTOR TO PROVIDE ALL SLAB CUTTING FOR INSTALLATION OF ALL UNDERGROUND PLUMBING. FIELD VERIFY ALL EXISTING PLUMBING CONNECTIONS, INVERTS, ETC., PRIOR TO BID. INCLUDE COST FOR PATCH OF FLOOR TO "LIKE NEW" CONDITION.

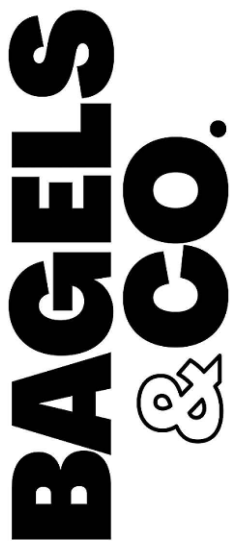
 KEYED NOTES

1. FIELD LOCATE AND TIE INTO EXISTING SANITARY PIPING. PLUMBING CONTRACTOR SHALL FIELD VERIFY LOCATION OF EXISTING SANITARY PIPING SIZE AND INVERT PRIOR TO ANY NEW PLUMBING CONNECTION. NOTIFY THE ENGINEER IF INVERT CAN NOT BE MET.
2. FIELD LOCATE AND TIE INTO EXISTING GREASE WASTE PIPING. PLUMBING CONTRACTOR SHALL FIELD VERIFY LOCATION OF EXISTING SANITARY PIPING SIZE AND INVERT PRIOR TO ANY NEW PLUMBING CONNECTION. NOTIFY THE ENGINEER IF INVERT CAN NOT BE MET.
3. ROUTE INDIRECT WASTE PIPING, FULL SIZE, FROM EQUIPMENT TO FLOOR SINK. PROVIDE WITH SUPPORTS AND TERMINATE AS REQUIRED BY DETAIL. TYPICAL FOR ALL.
4. ROUTE WALK-IN COOLER CONDENSATE PIPING, FULL SIZE, FROM EVAPORATOR ALONG INSIDE OF COOLER BOX, DOWN WALL AND THRU TO HUB DRAIN. SEAL OPENING IN COOLER AS REQUIRED BY EQUIPMENT MANUFACTURER.
5. CONNECT NEW 3" VENT TO EXISTING VENT LINE. FIELD VERIFY EXISTING VENT LOCATION. MODIFY EXISTING VENT SYSTEM AS REQUIRED TO MAKE FINAL CONNECTION.

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NO DATE REMARKS
REVISIONS



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TALLAHASSEE

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TALLAHASSEE, FL 32310

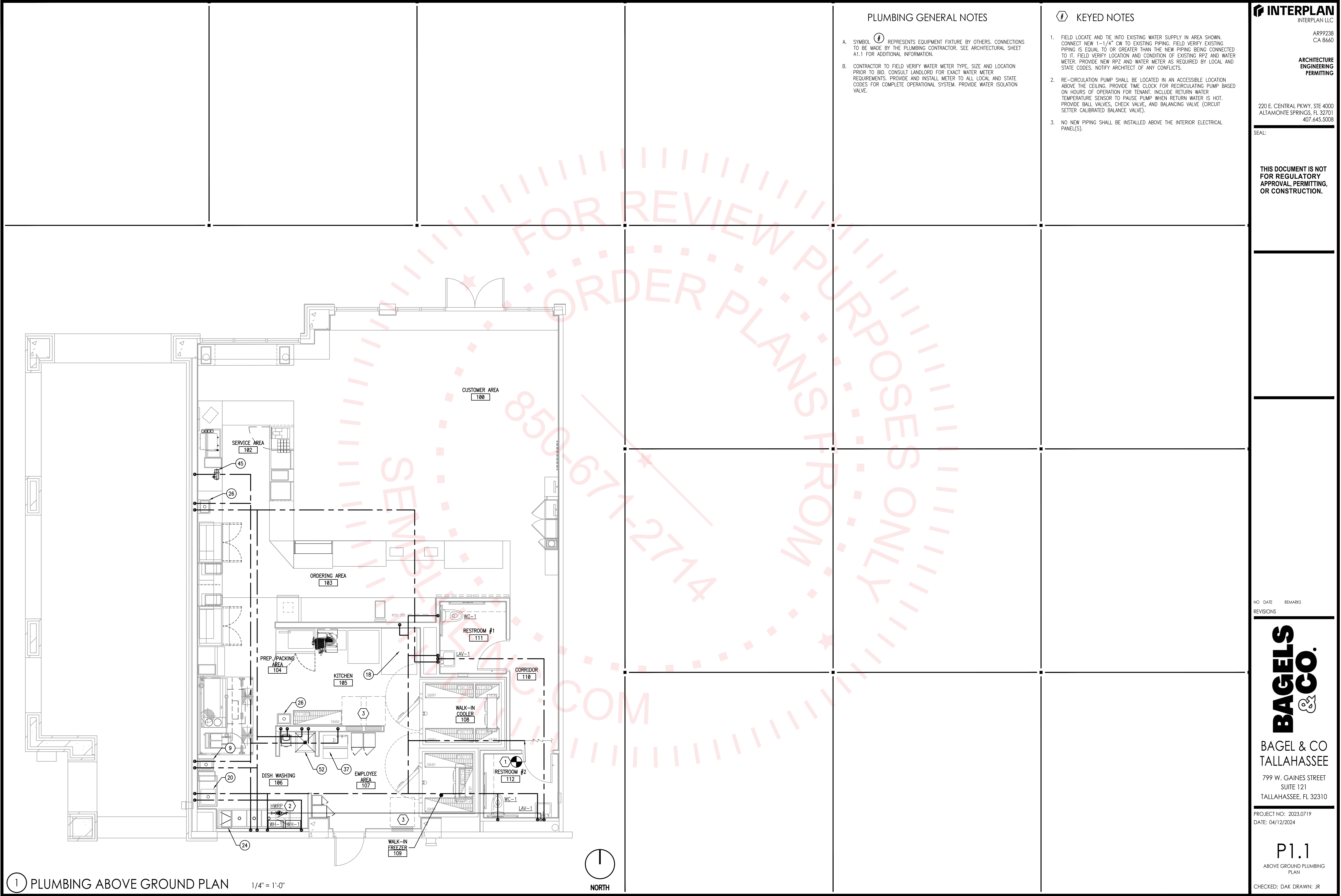
PROJECT NO: 2023.0719
DATE: 04/12/2024

P1.0

UNDERGROUND PLUMBING
PLAN

CHECKED: DAK DRAWN: JR

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1 PLUMBING ABOVE GROUND PLAN 1/4" = 1'-0"

PLUMBING GENERAL NOTES

- A. SYMBOL  REPRESENTS EQUIPMENT FIXTURE BY OTHERS. CONNECTIONS TO BE MADE BY THE PLUMBING CONTRACTOR. SEE ARCHITECTURAL SHEET A1.1 FOR ADDITIONAL INFORMATION.
- B. CONTRACTOR TO FIELD VERIFY WATER METER TYPE, SIZE AND LOCATION PRIOR TO BID. CONSULT LANDLORD FOR EXACT WATER METER REQUIREMENTS. PROVIDE AND INSTALL METER TO ALL LOCAL AND STATE CODES FOR COMPLETE OPERATIONAL SYSTEM. PROVIDE WATER ISOLATION VALVE.

KEYED NOTES

1. FIELD LOCATE AND TIE INTO EXISTING WATER SUPPLY IN AREA SHOWN. CONNECT NEW 1-1/4" CW TO EXISTING PIPING. FIELD VERIFY EXISTING PIPING IS EQUAL TO OR GREATER THAN THE NEW PIPING BEING CONNECTED TO IT. FIELD VERIFY LOCATION AND CONDITION OF EXISTING RPZ AND WATER METER. PROVIDE NEW RPZ AND WATER METER AS REQUIRED BY LOCAL AND STATE CODES. NOTIFY ARCHITECT OF ANY CONFLICTS.
2. RE-CIRCULATION PUMP SHALL BE LOCATED IN AN ACCESSIBLE LOCATION ABOVE THE CEILING. PROVIDE TIME CLOCK FOR RECIRCULATING PUMP BASED ON HOURS OF OPERATION FOR TENANT. INCLUDE RETURN WATER TEMPERATURE SENSOR TO PAUSE PUMP WHEN RETURN WATER IS HOT. PROVIDE BALL VALVES, CHECK VALVE, AND BALANCING VALVE (CIRCUIT SETTER CALIBRATED BALANCE VALVE).
3. NO NEW PIPING SHALL BE INSTALLED ABOVE THE INTERIOR ELECTRICAL PANEL(S).

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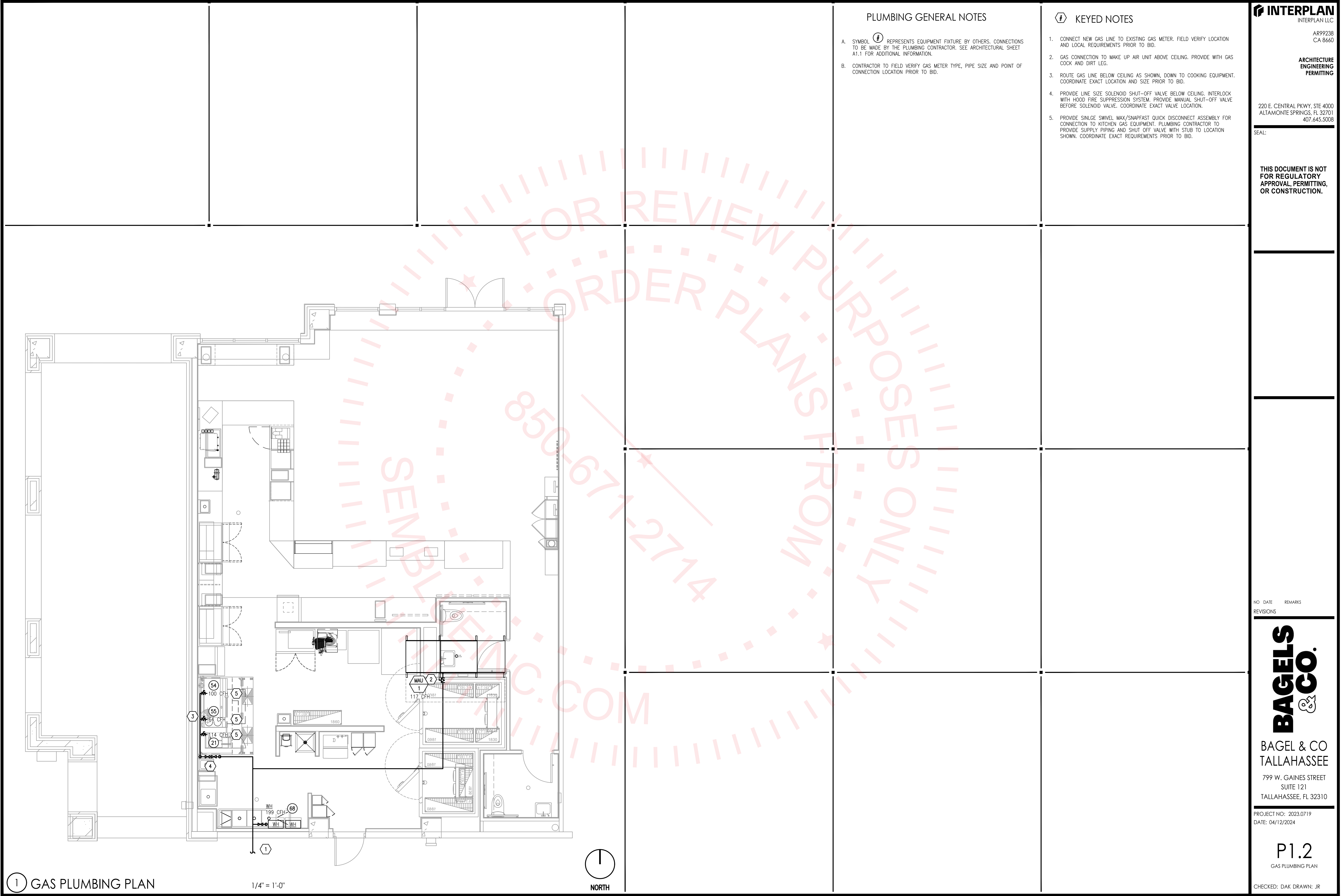
NO	DATE	REMARKS

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PROJECT NO: 2023.0719
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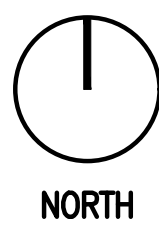
P1.1
ABOVE GROUND PLUMBING
PLAN
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1 GAS PLUMBING PLAN

1/4" = 1'-0"



PLUMBING GENERAL NOTES

- A. SYMBOL  REPRESENTS EQUIPMENT FIXTURE BY OTHERS. CONNECTIONS TO BE MADE BY THE PLUMBING CONTRACTOR. SEE ARCHITECTURAL SHEET A1.1 FOR ADDITIONAL INFORMATION.
- B. CONTRACTOR TO FIELD VERIFY GAS METER TYPE, PIPE SIZE AND POINT OF CONNECTION LOCATION PRIOR TO BID.

 KEYED NOTES

1. CONNECT NEW GAS LINE TO EXISTING GAS METER. FIELD VERIFY LOCATION AND LOCAL REQUIREMENTS PRIOR TO BID.
2. GAS CONNECTION TO MAKE UP AIR UNIT ABOVE CEILING. PROVIDE WITH GAS COCK AND DIRT LEG.
3. ROUTE GAS LINE BELOW CEILING AS SHOWN, DOWN TO COOKING EQUIPMENT. COORDINATE EXACT LOCATION AND SIZE PRIOR TO BID.
4. PROVIDE LINE SIZE SOLENOID SHUT-OFF VALVE BELOW CEILING. INTERLOCK WITH HOOD FIRE SUPPRESSION SYSTEM. PROVIDE MANUAL SHUT-OFF VALVE BEFORE SOLENOID VALVE. COORDINATE EXACT VALVE LOCATION.
5. PROVIDE SINGLE SWIVEL MAX/SNAPFAST QUICK DISCONNECT ASSEMBLY FOR CONNECTION TO KITCHEN GAS EQUIPMENT. PLUMBING CONTRACTOR TO PROVIDE SUPPLY PIPING AND SHUT OFF VALVE WITH STUB TO LOCATION SHOWN. COORDINATE EXACT REQUIREMENTS PRIOR TO BID.

INTERPLAN
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GAS PLUMBING PLAN

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TANKLESS WATER HEATER SIZING CALCULATION					
FIXTURE	FIXTURE DESIGNATION	QTY.	GPM		TOTAL
MOP SINK	K-52	1	2		2.0
HAND SINK	K-26	2	0.5		1.0
PRE-RINSE	K-9	1	2		2.0
3 COMP SINK	K-24	1	1.75		1.8
1 COMP SINK	K-20	1	1.75		1.8
LAVATORY	LAV-1	2	0.5		1.0
				TOTAL	9.5
	GPM	USAGE FACTOR			
	9.5	1.0			9.5
			TOTAL DEMAND GPM		10
			GPM PROVIDED		10.8

PIPING MATERIALS					
SERVICE	SIZE	PIPE	FITTINGS	UNIONS & FLANGES	JOINTS
WASTE AND VENT (ABOVE GRADE)	ALL	CAST IRON PIPE HUBLESS, ASTM A 888, CISPI 301, PVC	-	-	-
WASTE AND VENT (BELOW GRADE)	ALL	CAST IRON PIPE HUBLESS, ASTM A 888, CISPI 301, PVC	-	-	-
DOMESTIC COLD WATER AND HOT WATER INSIDE BUILDING	ALL	COPPER PIPING TYPE "L"	-	-	-
DOMESTIC COLD WATER AND HOT WATER INSIDE BUILDING (BELOW GRADE)	ALL	CROSS LINKED PEX PIPING	NO FITTINGS ALLOWED BELOW GRADE	NO FITTINGS ALLOWED BELOW GRADE	NO FITTINGS ALLOWED BELOW GRADE
GAS PIPING (ABOVE GRADE)	-	BLACK STEEL SCHEDULE 40 ASTM A 120 LAP	#150 BLACK MALLEABLE IRON SCREWED ANSI B16.3 BUSHINGS PROHIBITED	-	THREADED UP TO 2" WELDED OVER 4", CONFORM WITH LOCAL CODES
CONDENSATE WALK-IN REFRIGERATION AND FREEZER EVAPORATORS	ALL	SCHEDULE 40 CPVC	SCHEDULE 40 CPVC	SCHEDULE 40 CPVC	SCHEDULE 40 CPVC
EXPOSED WATER PIPING FROM COMPARTMENT SINK	ALL	COPPER "M"	-	-	

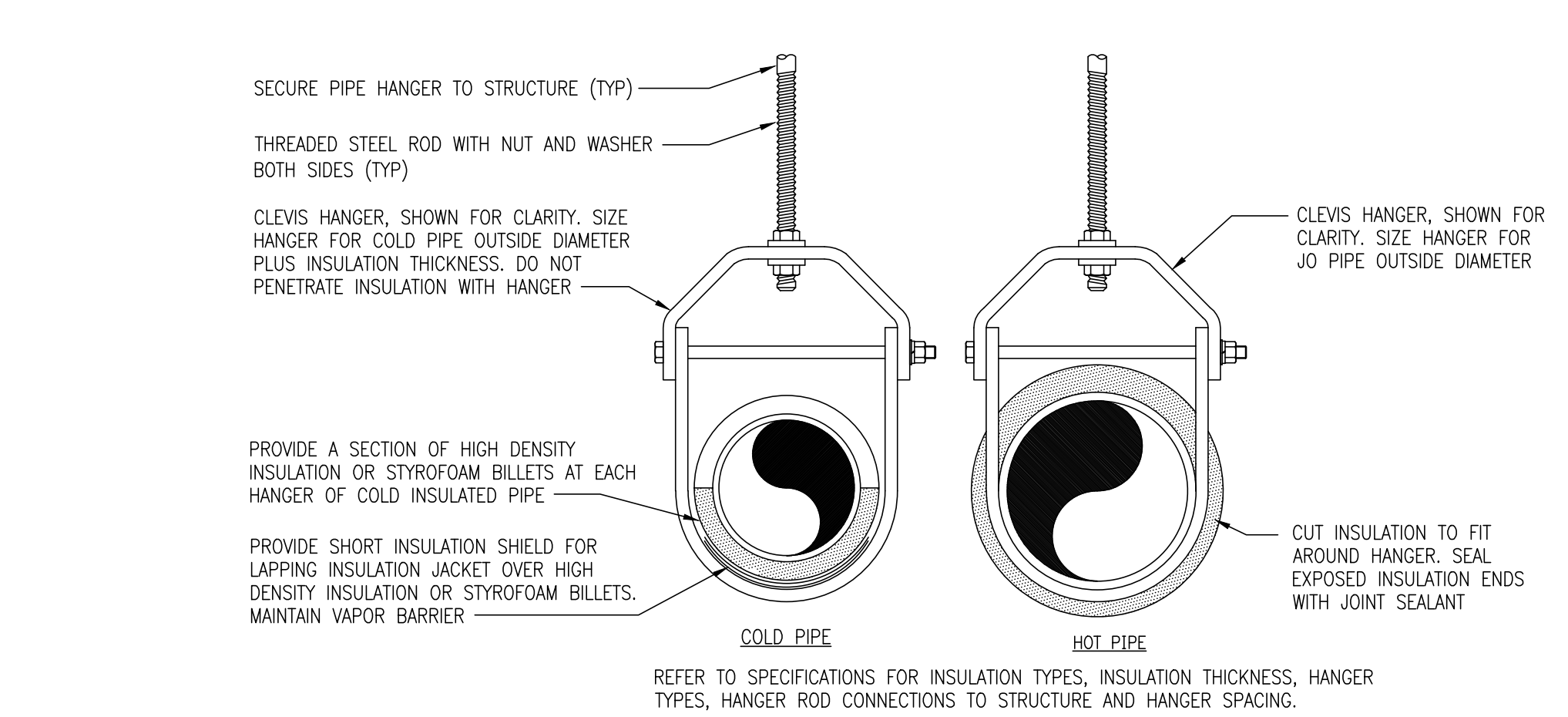
PLUMBING FIXTURE SCHEDULE							
MARK	FIXTURE TYPE	DESCRIPTION	FIXTURE CONNECTIONS				MANUFACTURER: MODEL NUMBER
			CW	HW	W	V	
WC-1	ADA WATER CLOSET	WATER CLOSET SHALL BE ELONGATED STYLE, 16.5" HIGH,1.6 GALLON FLUSH,SIPHON JET BOTTOM OUTLET.	3/4"	-	3"	2"	SLOAN BOWL: ST8029 TANK: ST8013A
LAV-1	ADA LAVATORY	WALL MOUNTED, 4" CENTERS, ADA COMPLIANT, 0.5 GPM, VOLUME AND HOT WATER LIMIT STOP. PROVIDE WITH CONCEALED ARM CARRIER MOUNT, ADA P-PIPE COVERING AND MIXING VALVE. COORDINATE WALL SUPPORT REQUIREMENTS WITH GENERAL TRADES CONTRACTOR.	1/2"	1/2"	2"	1-1/2"	SLOAN 3873055 FAUCET: CHICAGO FAUCET 420-E2805-717AB
FD-1	FLOOR DRAIN	CAST IRON BODY, MEMBRANE CLAMP, ADJUSTABLE COLLAR, 6" DIAMETER "TYPE B" POLISHED NICKEL BRONZE STRAINER, 1/2" TRAP SEAL UNLESS RESTRICTED BY LOCAL CODES THEN PROVIDE TRAP PRIMER CONNECTION AND ASSOCIATED PIPING.	-	-	3"	2"	MIFAB F1103-6-1-C-7
FS-1	FLOOR SINK	12"x12"x8" DEEP CAST IRON BODY WITH WHITE ACID RESISTING PORCELAIN ENAMEL INTERIOR, ANCHOR FLANGE WITH SEEPAGE HOLES AND CLAMP COLLAR, NICKEL BRONZE FRAME WITH 1/2 GRATE.	-	-	PER PLAN 3" MINIMUM	1-1/2"	MIFAB FS1733-FL-C-1-150
HD-1	HUB DRAIN	HUB DRAIN	-	-	3"	1-1/2"	MIFAB F1103-DD-50
TMV-1	MIXING VALVE	THERMOSTATIC MIXING VALVE. MOUNT VALVE EXPOSED BELOW FIXTURE. PROVIDE WITH CHROME FINISH. SET TEMPERATURE TO 110°F	-	-	-	-	LEONARD 170A-LF-BP-CP
TP-1	TRAP PRIMER	WATER SAVER TRAP PRIMER. SEE DETAIL	-	-	-	-	MIFAB MI-750
RP-1	RECIRCULATING PUMP	POTABLE HOT WATER RECIRCULATING PUMP: 1/25 HP,115V 1 PH, 0-15 GPM FLOW 0-15 FT. HEAD RANGE. PROVIDE WITH 24/7 TIME CLOCK AND AQUASAT CONTROL.	-	1/2"	-	-	TACO 008-IQSF6
FCQ	FLOOR CLEANOUT	SIZE PER PIPING LATERAL, 4" MAX. ADJUSTABLE, COATED CAST IRON BODY, ROUND SCORIATED NICKEL BRONZE TOP, BRONZE PLUG.	-	-	PER PLAN	-	MIFAB C1102-R-1
WCQ	WALL CLEANOUT	RAISED HEADPLUG POLISH S.S. COVER & VANDAL PROOF SCREW	-	-	PER PLAN	-	MIFAB C1462-RD?-6
WH-1 and WH-2	WATER HEATER	MULTI-UNIT COMMERCIAL CONDENSING INDOOR TANKLESS WATER HEATERS, 8.7 GPM @ 85 °F RISE, 10.7 GPM @ 70 °F RISE (2 x 199,900 BTU =399,800 BTU TOTAL SYSTEM)	(2)-3/4"	(2)-3/4"	-	-	RHEEM RTGH-CM95DVLN NATURAL GAS WATER HEATER (2)
ET-1	EXPANSION TANK	SIZE PER WATER HEATER CAPACITY	-	-	-	-	PROFLO: PFXTS1
AAV	AIR ADMITTANCE VALVE	2" AIR ADMITTANCE VALVE	-	-	-	2"	STUDOR
HB-1	HOSE BIBB	FREEZELESS, ANTI-SIPHON ASSE 1019 LISTED.	3/4"	-	-	-	WOODFORD MODEL 17
CONTACT SPIRIT GROUP JARED JOHNSON, JJOHNSON@SPIRITGROUPINC.COM							
NOTES: 1. WATER HEATER FURNISHED AND INSTALLED BY CONTRACTOR. 2. THERMAL EXPANSION TANK FURNISHED AND INSTALLED BY CONTRACTOR. 3. SET DISCHARGE TEMPERATURE OF WATER HEATER TO OPERATE AT 140 DEG F. 4. TEMPERATURE AND PRESSURE RELIEF VALVE FURNISHED AND INSTALLED BY CONTRACTOR. 5. REFER TO DETAIL FOR ADDITIONAL COMPONENTS AND CONFIGURATION. 6. PROVIDE MANUFACTURER'S VENT TERMINATION KIT FOR COMBUSTION AIR AND DISCHARGE OF FLUE GASES. 7. ALL TEMPERATURE AND PRESSURE RELIEF VALVES SHALL BE PIPED FULL SIZE TO INDIRECT WASTE SUCH AS THE NEAREST FLOOR DRAIN.							

GAS CONNECTION SCHEDULE	
EQUIPMENT	GAS LOAD
WH-1	199,000 BTUH
WH-2	199,000 BTUH
MAU-1	117,000 BTUH
GRIDDLE	100,000 BTUH
FRYER	114,000 BTUH
RANGE	64,000 BTUH
TOTAL CONNECTED LOAD	793,000 BTUH
REMARKS: 1. EQUIVALENT TO 793 CFH 2. INITIAL SUPPLY PRESSURE LESS THAN 2 PSI AND 0.5" WATER COLUMN PRESSURE DROP 3. DEVELOPED LENGTH IS ±250'-0"	

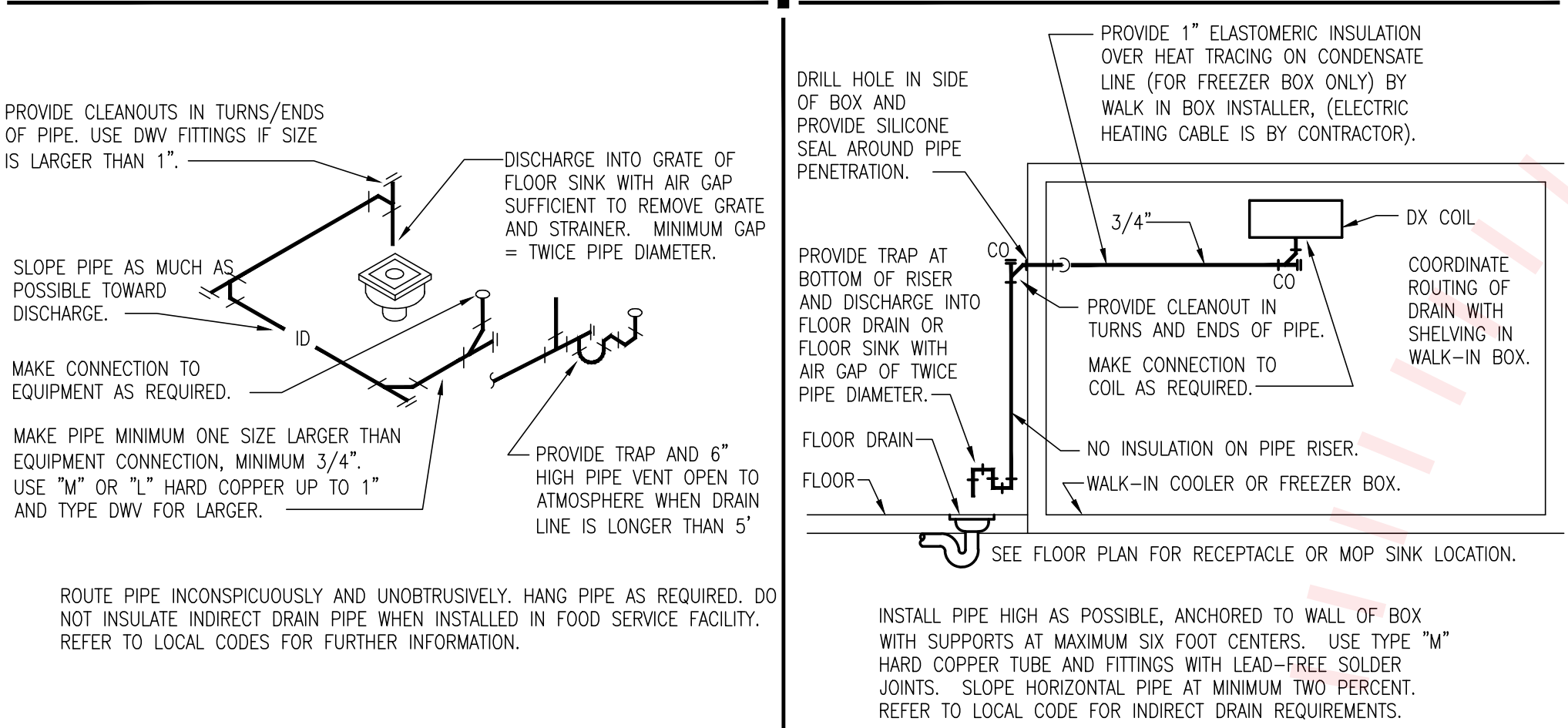
BACKFLOW DEVICE SCHEDULE					
TAG	SERIES	SIZE	TYPE	USAGE	APPROVAL
BFP-2	009QT	2"	REDUCED PRESSURE BACKFLOW PREVENTER	DOMESTIC WATER CONNECTION	ASSE 1015, AWWA 1013
BFP-3	SD-3	3/8"	DUAL CHECK VALVE W/ ATMOSPHERIC VENT	CARBONATED BEVERAGE	ASSE 1022
BFP-4	9D	1/2"	BACKFLOW PREVENTER W/ ATMOSPHERIC VENT	TEA MAKER, COFFEE MACHINE, JUICE DISPENSER, ETC	ASSE 1012
BFP-5	007QTS	1/2"	DUAL CHECK VALVE	NON-CARBONATED BEVERAGE, SOFT SERVE ICE CREAM, WATER FILTER, HUMIDIFIER, EYE WASH, ETC	ASSE 1015
BFP-6	8	3/4"	VACUUM BREAKER	WALL HYDRANT CONNECT VACUUM BREAKER	ASSE 1011
BFP-7	N9-CD	3/4"	DUAL CHECK VACUUM BREAKER	HOSE BIBB	ASSE 1052
BFP-8	008PCOT	1/2"	SPILL RESISTANT VACUUM BREAKER	SOAP DISPENSER, SPECIALTY SINK, CLEANING EQUIPMENT, DISHWASHER	ASSE 1056
NOTES: CONTRACTOR TO VERIFY EXACT REQUIREMENTS OF ALL REQUIRED BACKFLOW DEVICES AND FIXTURES WITH AUTHORITIES HAVING JURISDICTION PRIOR TO BID.					

DOMESTIC WATER SUPPLY FIXTURE UNITS								
TYPE OF FIXTURE	TYPE OF SUPPLY CONTROL	QUANTITY	HW (EACH)	HW (TOTAL)	CW (EACH)	CW (TOTAL)	TOTAL (EACH)	TOTAL (F.U.'S) (TOTAL)
LAVATORY	FAUCET	2	1.5	3.00	1.5	3.00	2.0	4.00
HAND SINK	FAUCET	2	1.5	3.00	1.5	3.00	2.0	4.00
MOP BASIN	FAUCET	1	2.25	2.25	2.25	2.25	3.0	3.00
PRE RINSE	FAUCET	1	1.5	1.50	1.5	1.50	2.0	2.00
1-COMPARTMENT SINK FAUCET	FAUCET	1	3.0	3.00	3.0	3.00	4.0	4.00
3-COMPARTMENT SINK FAUCET	FAUCET	1	3.0	3.00	3.0	3.00	4.0	4.00
ADA WATER CLOSET (TANK)	TANK	2	5.0	10.00	5	10.00	5.0	10.00
						HW	CW	TOTAL (FIXTURE UNITS) (TOTAL)
SERVICE TOTAL FIXTURE UNITS:						25.75	25.75	31.00
SERVICE TOTAL GPM:						21.5 GPM	21.5 GPM	23.3 GPM
SERVICE SIZE:						1-1/4"		

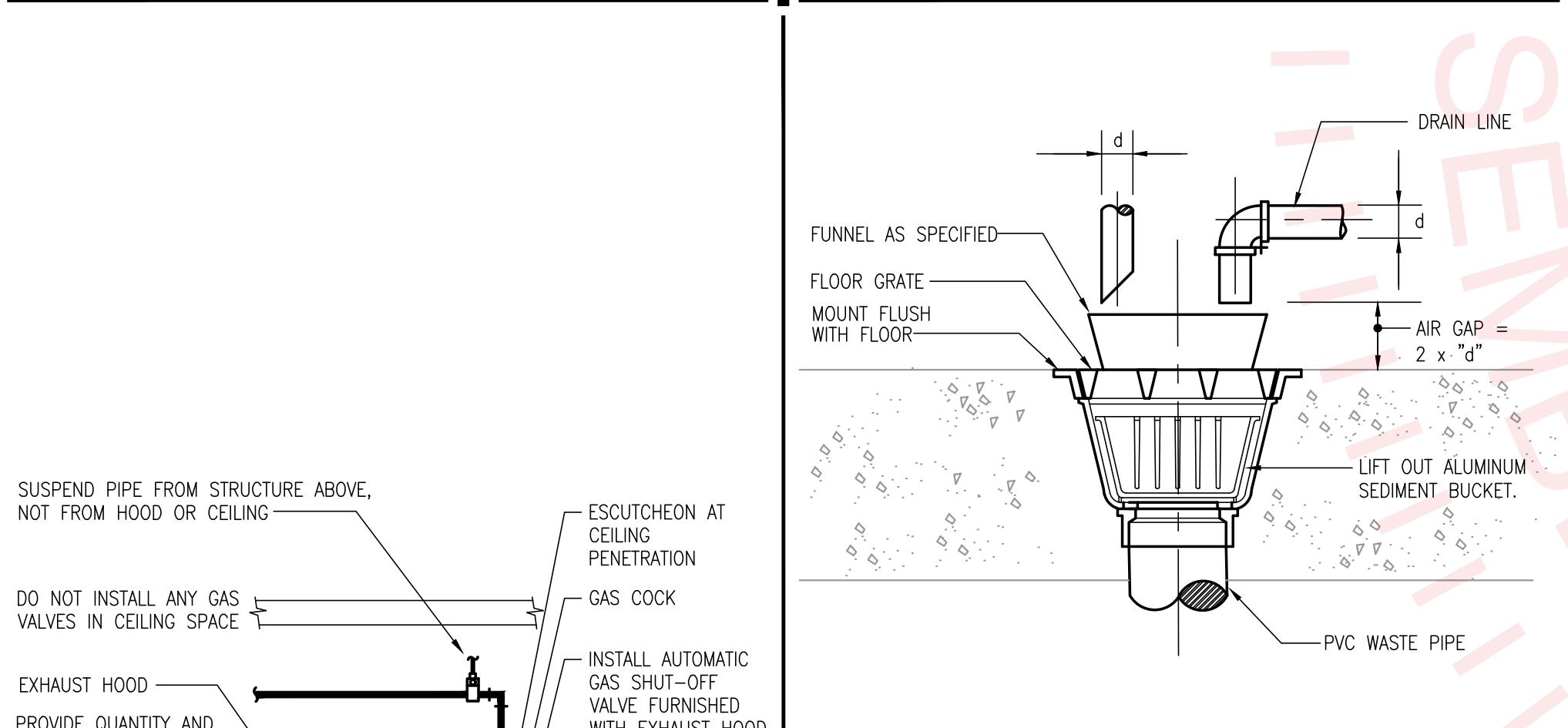
DRAINAGE FIXTURE UNITS SCHEDULE					
TYPE OF FIXTURE	QTY.	FIX. UNITS (EACH)	MINIMUM FIX. TRAP AND DRAIN SIZE	TOTAL FIX. UNITS	NOTES
FLOOR SINK	3	5.0	3"	15.0	X
HUB DRAIN	2	2.0	3"	4.0	X
FLOOR DRAIN	2	2.0	3"	4.0	X
LAVATORY	2	1.0	1-1/4"	2.0	X
HAND SINK	2	1.0	1-1/4"	2.0	X
MOP SINK	1	2.0	3"	2.0	X
PRE RINSE	1	2.0	2"	2.0	X
1-COMP SINK	1	2.0	1-1/2"	2.0	X
3-COMP SINK	1	3.0	1-1/2"	3.0	X
ADA WATER CLOSET	2	4.0	-	8.0	X
TOTAL DRAINAGE FIXTURE UNITS:				44.0	



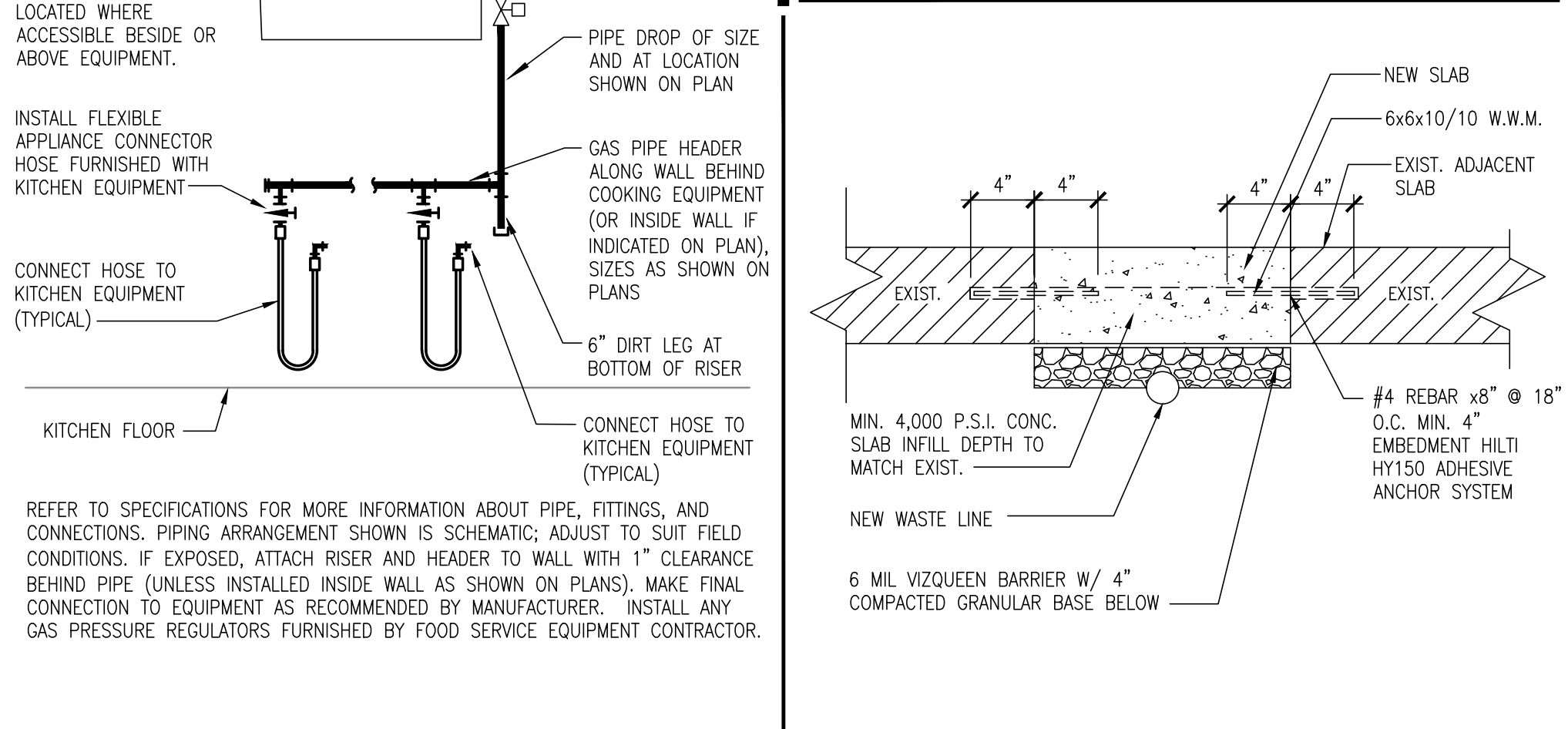
1 INSULATED PIPE HANGER DETAIL
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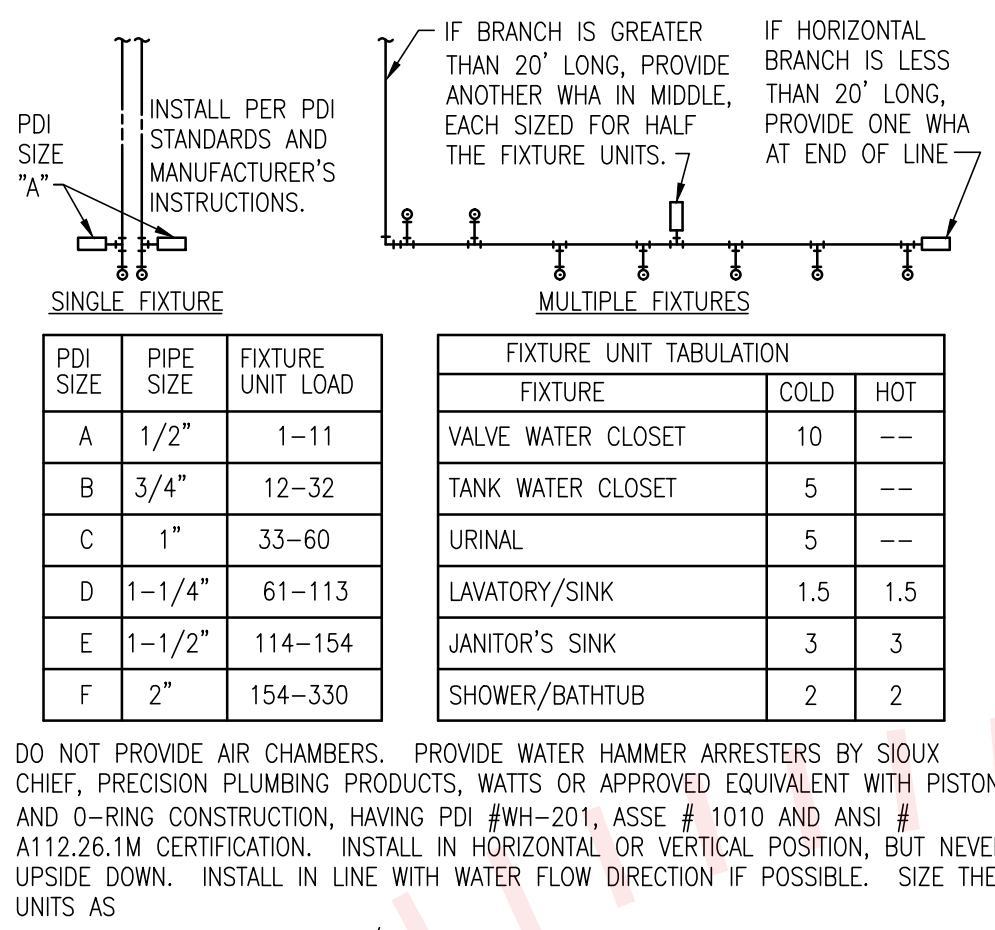
6 INDIRECT DRAIN DETAIL
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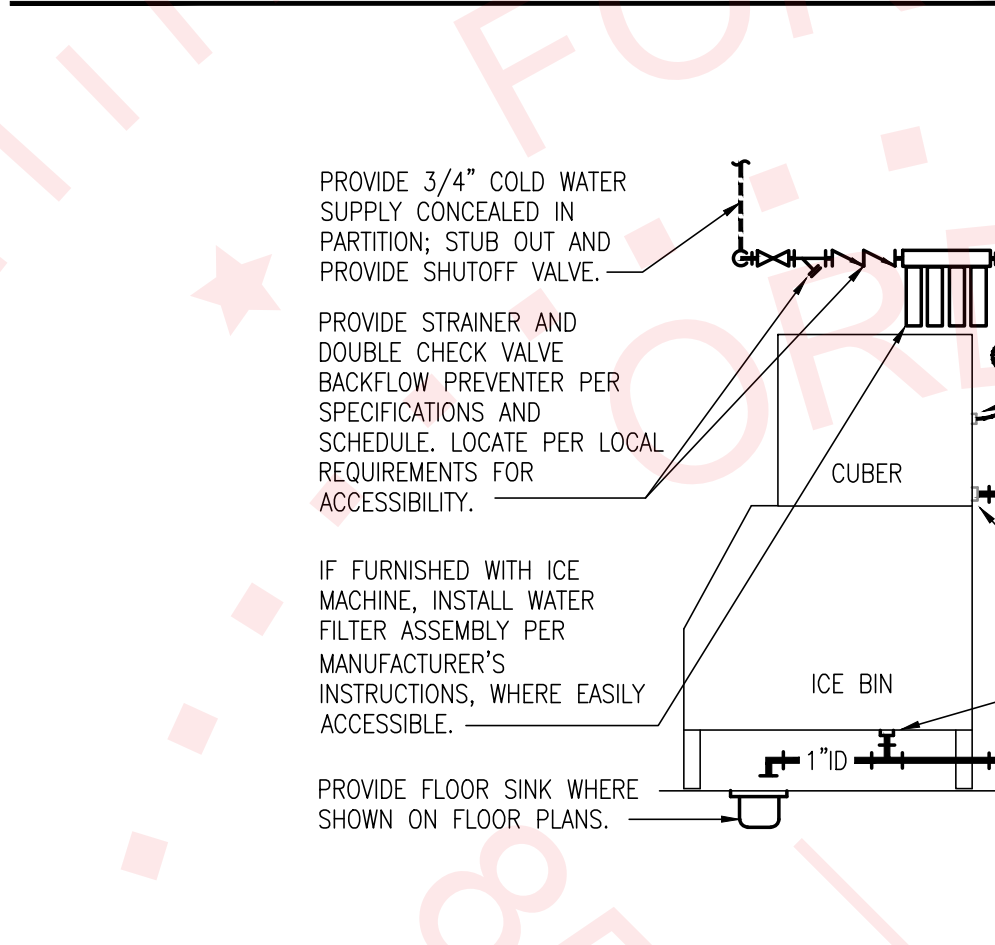
7 WALK-IN CONDENSATE DRAIN DETAIL
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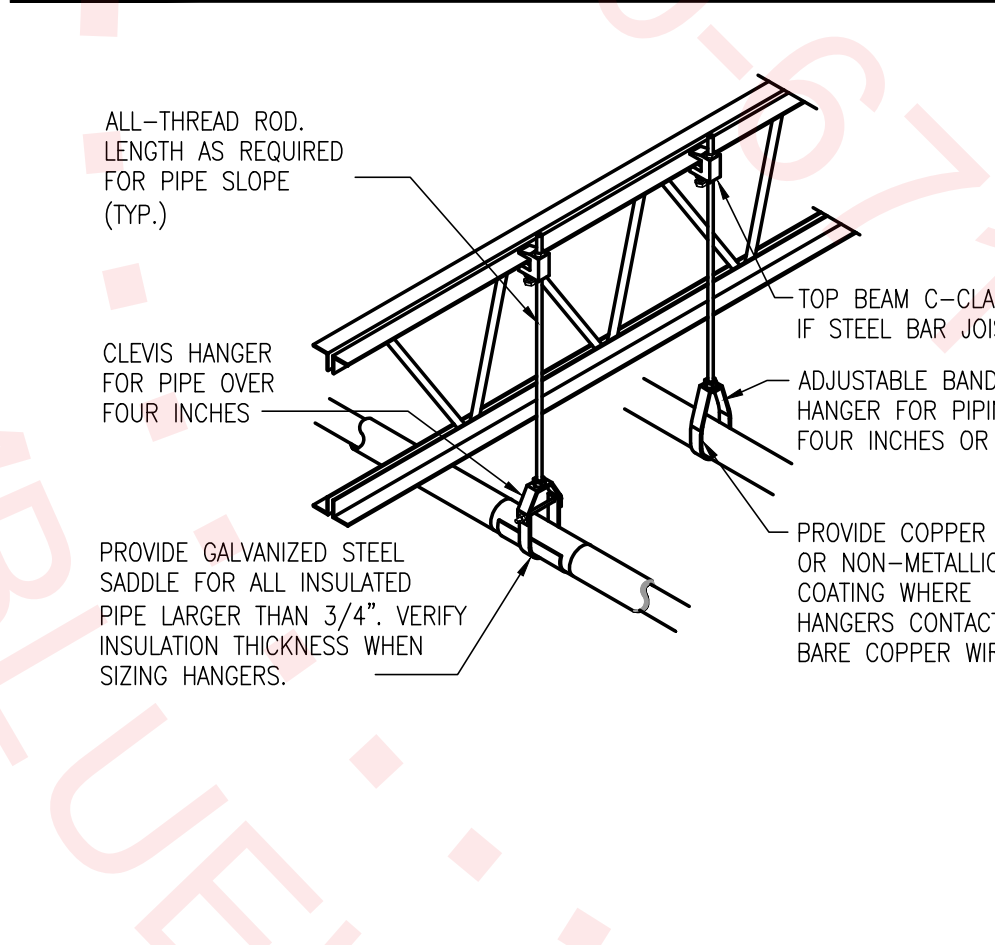
14 COOKING APPLIANCE GAS PIPE DETAIL
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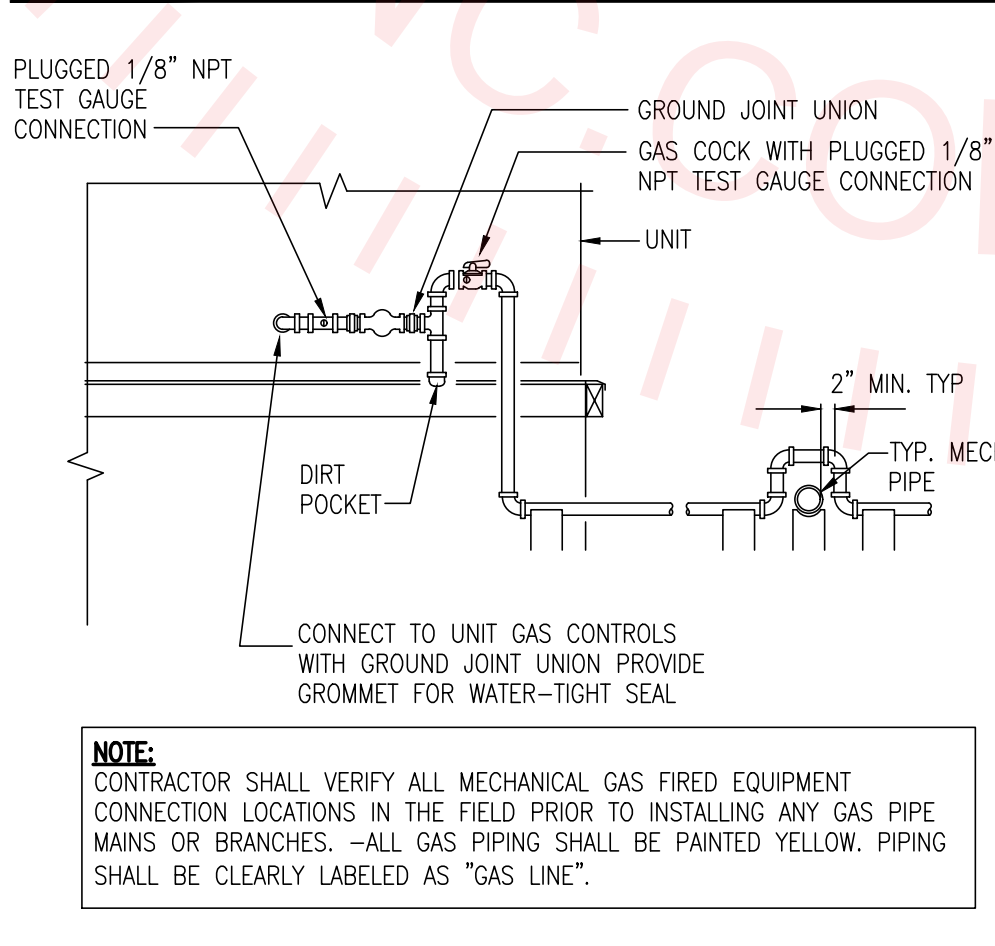
2 WATER HAMMER ARRESTORS DETAIL
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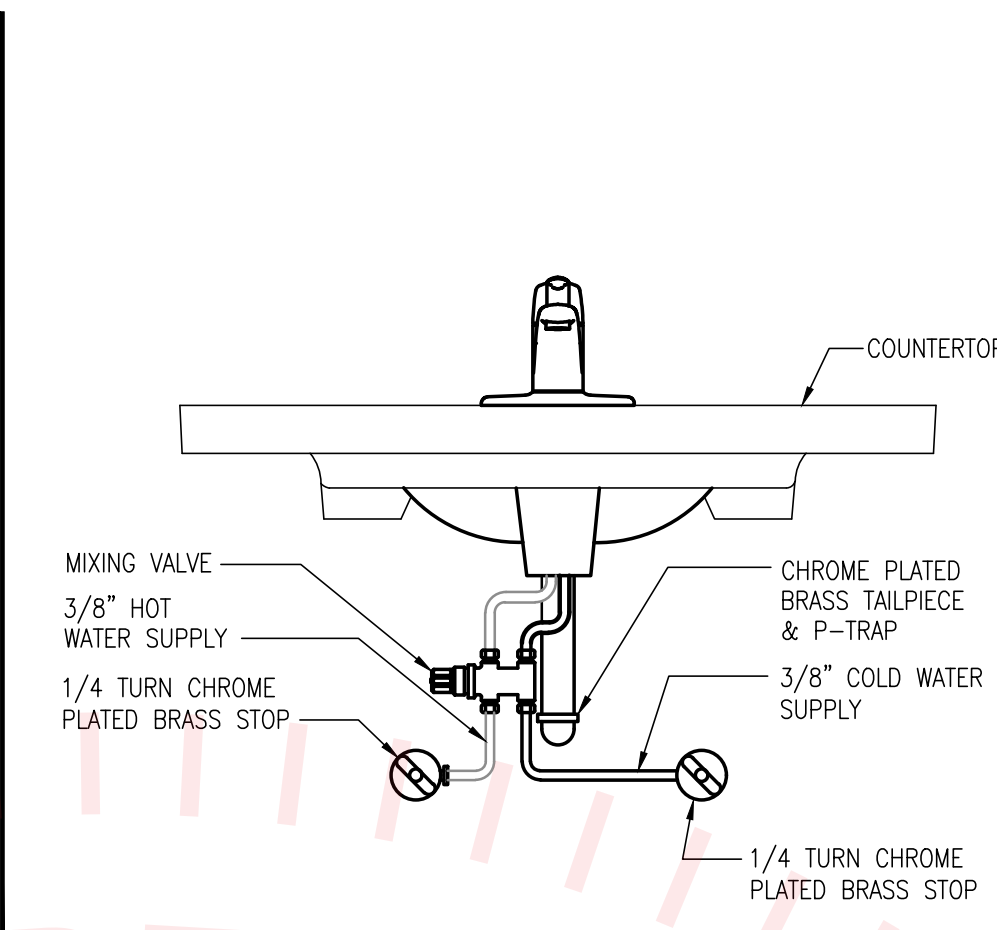
8 ICE MACHINE CONNECTION DETAIL
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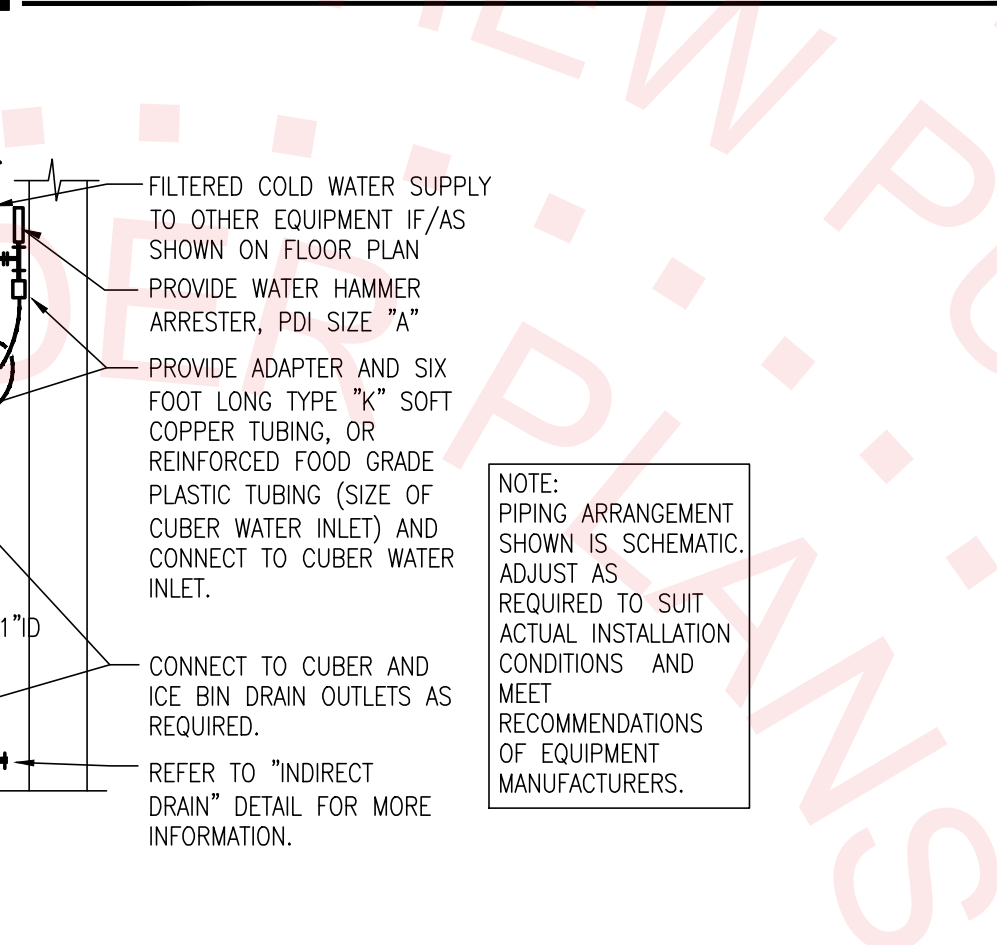
12 PIPE HANGER DETAIL
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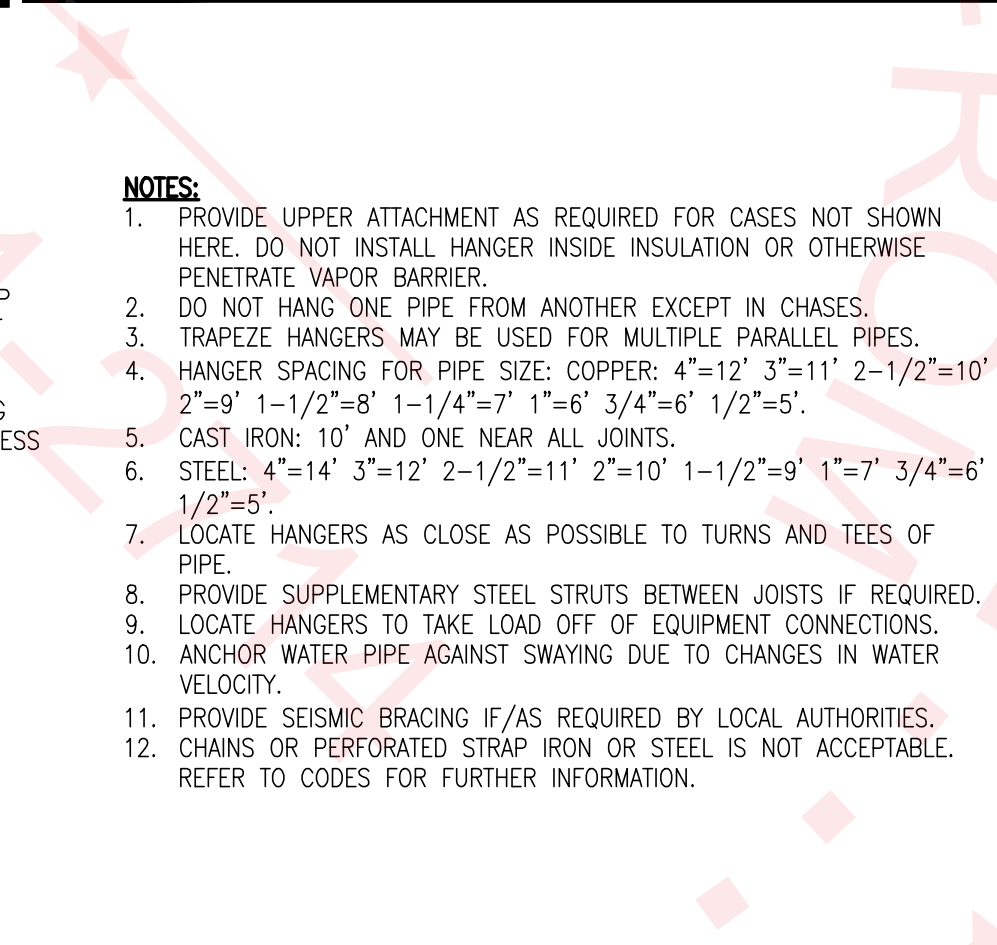
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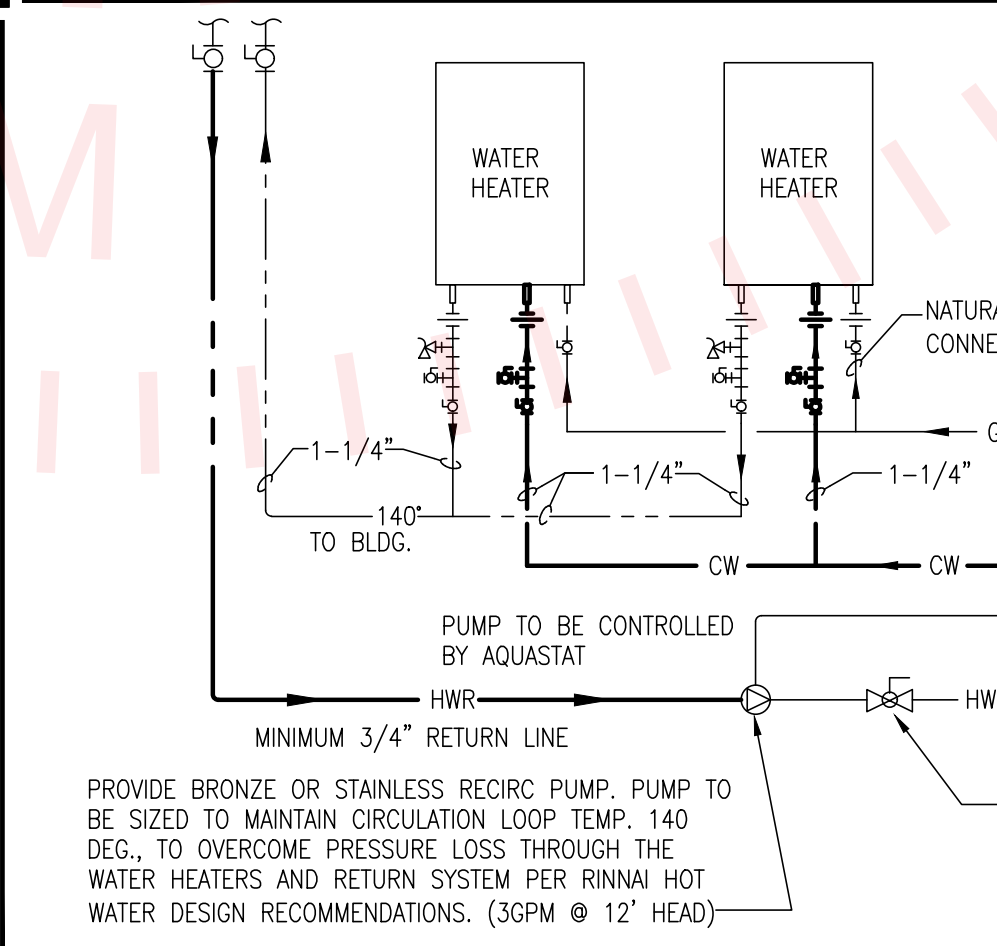
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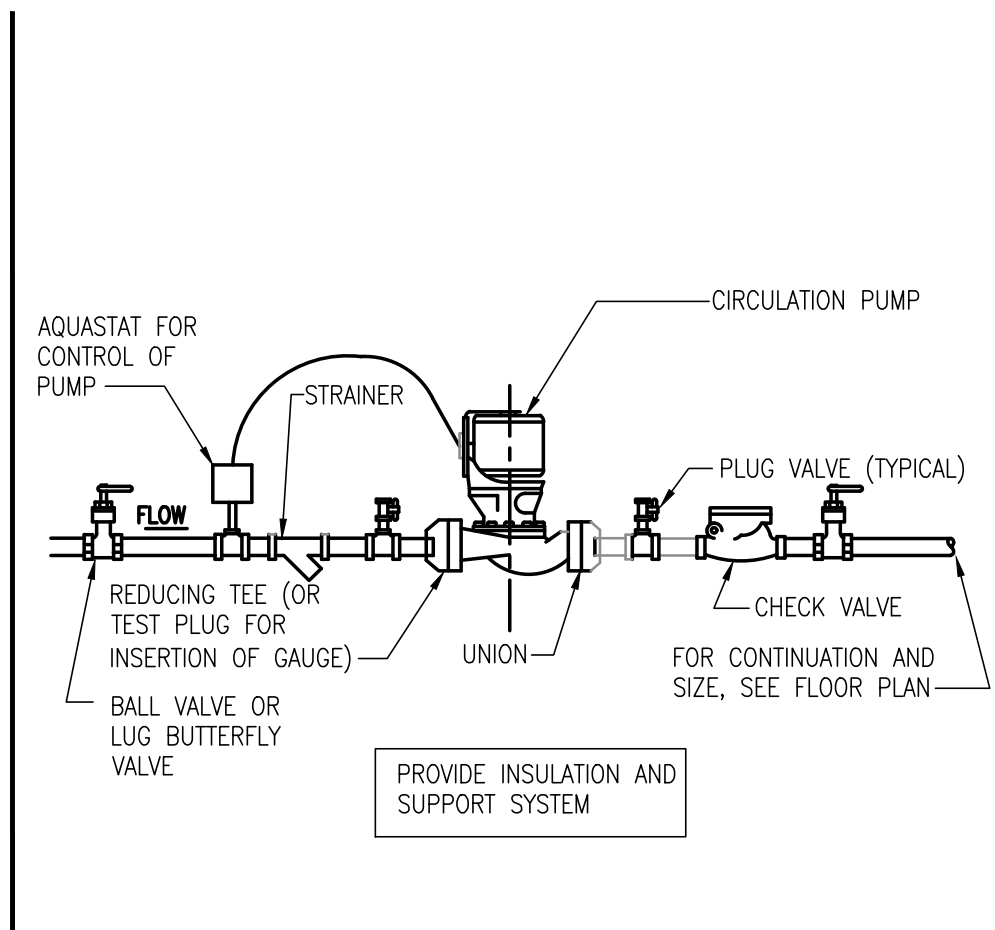
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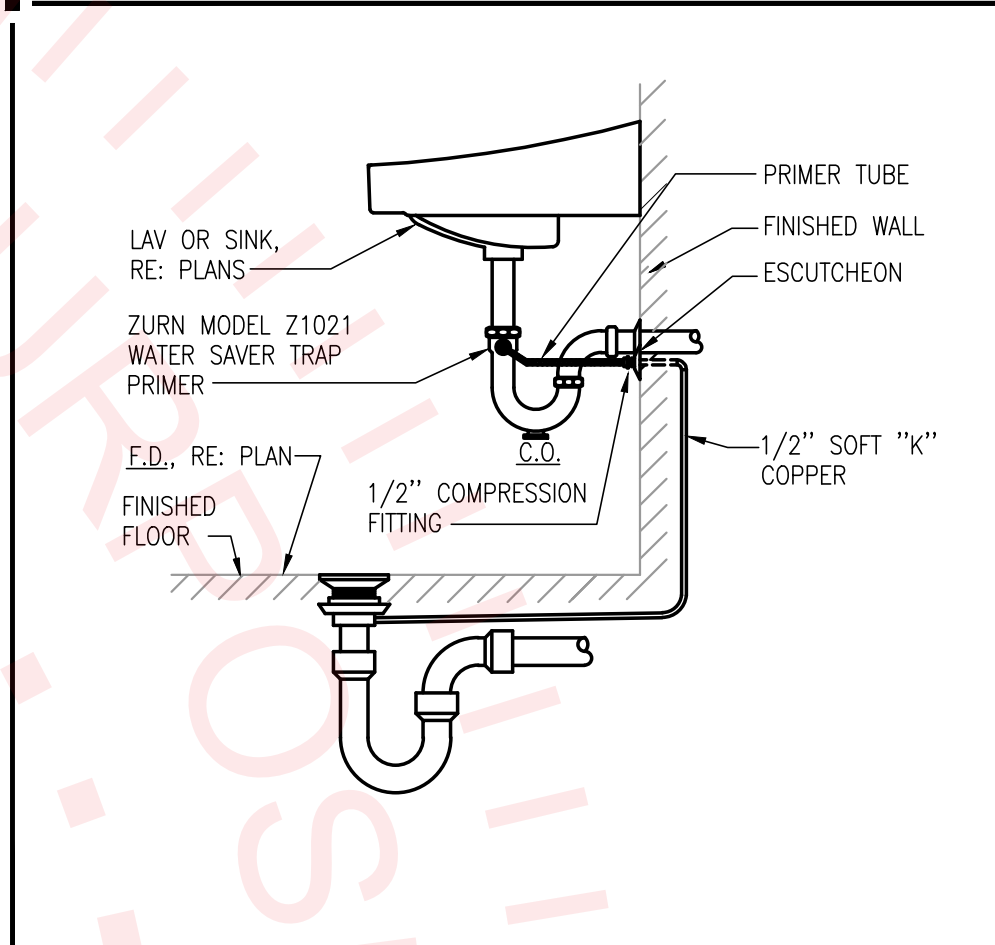
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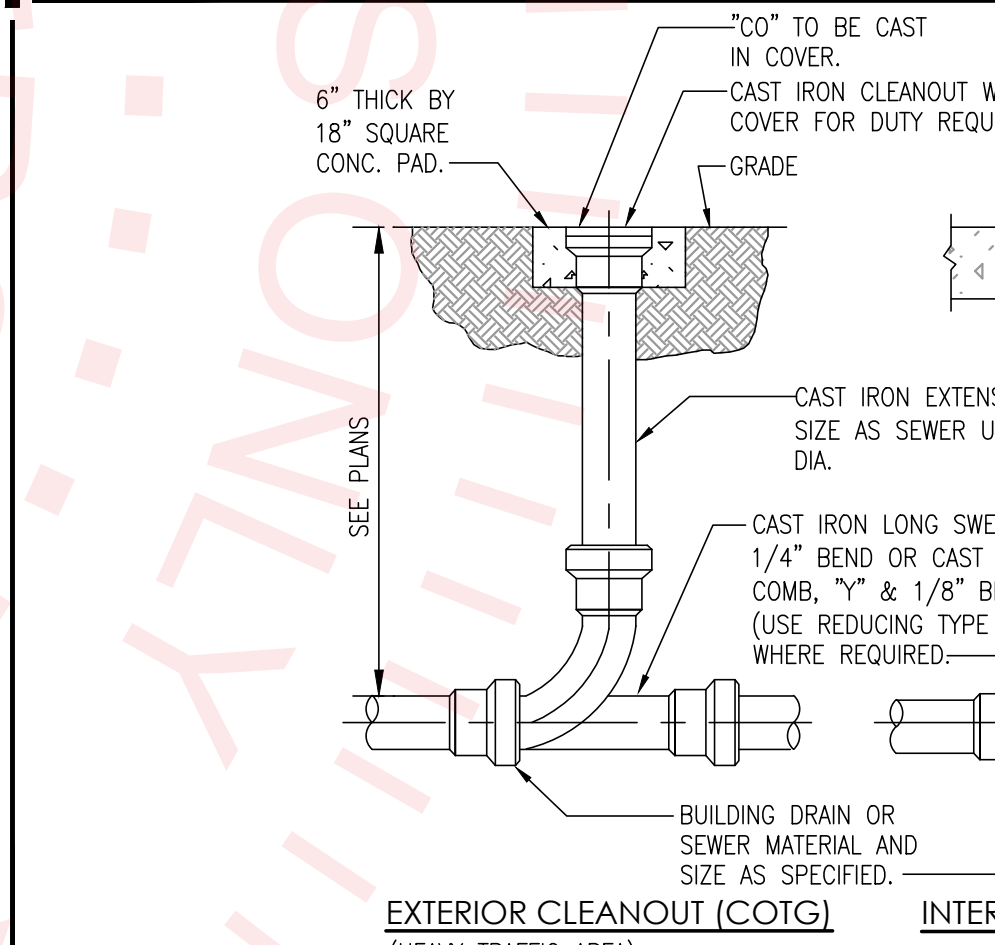
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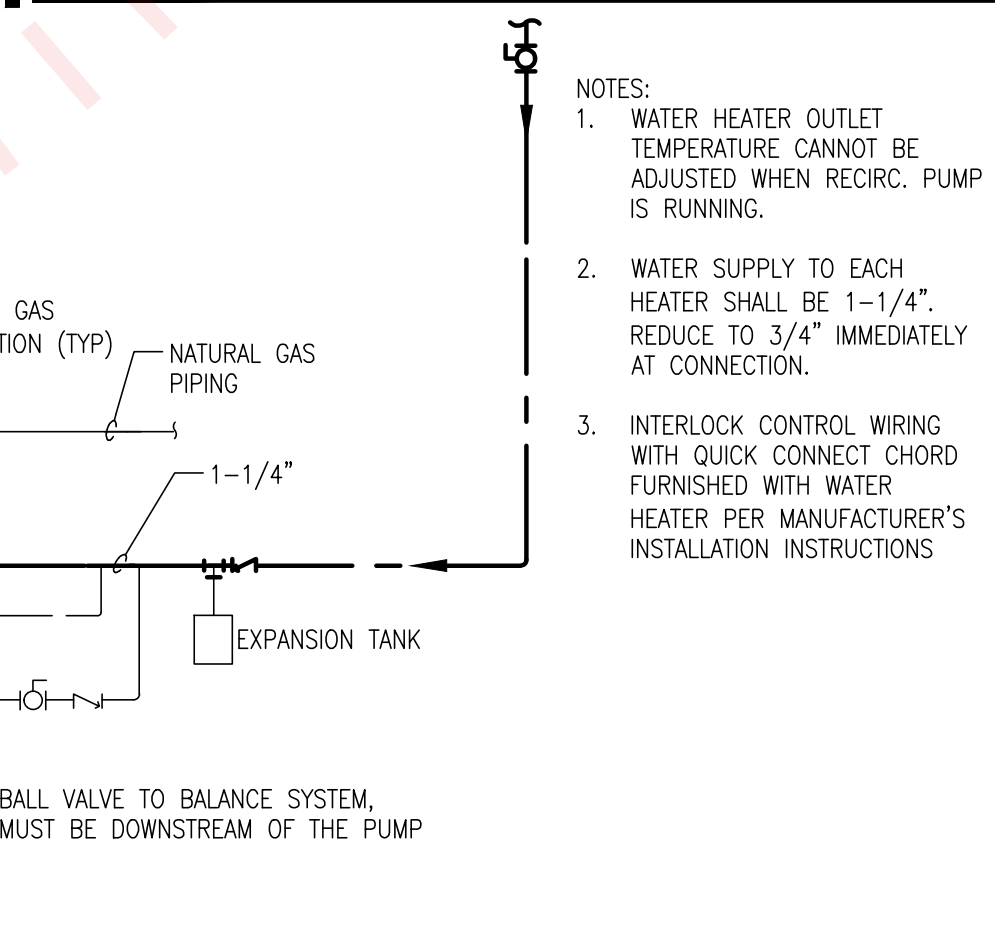
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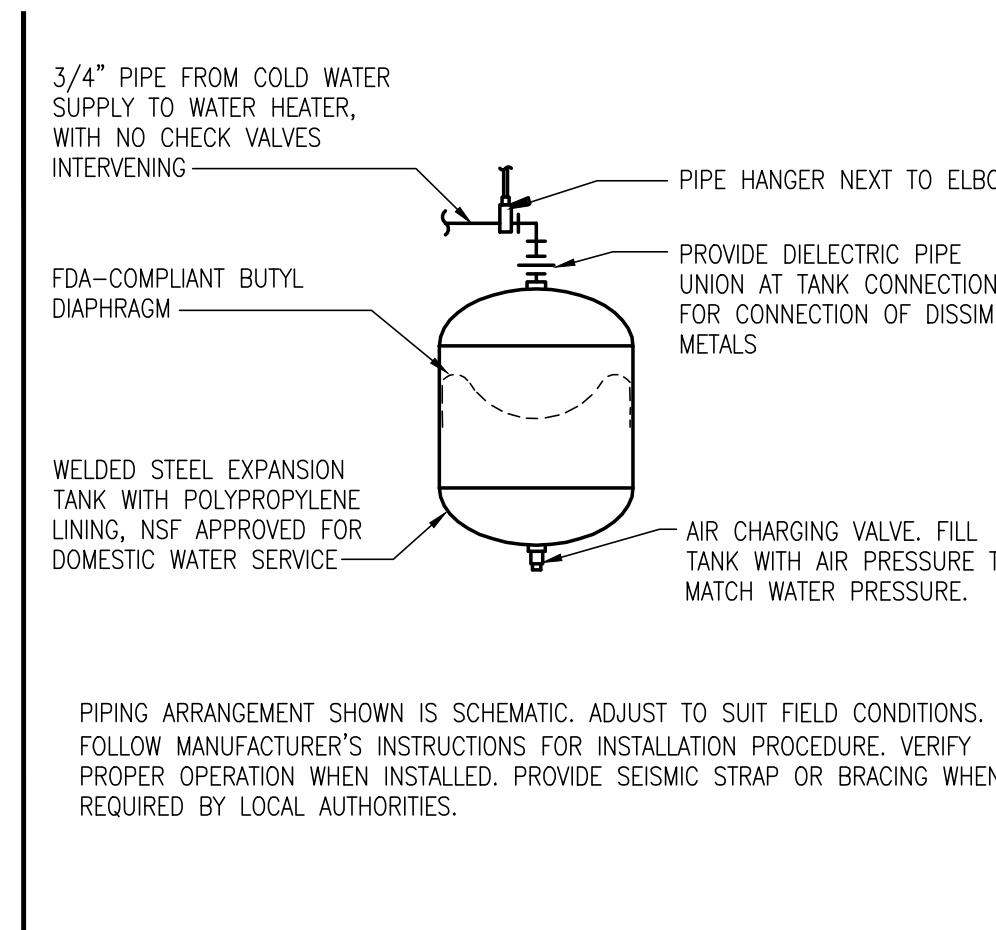
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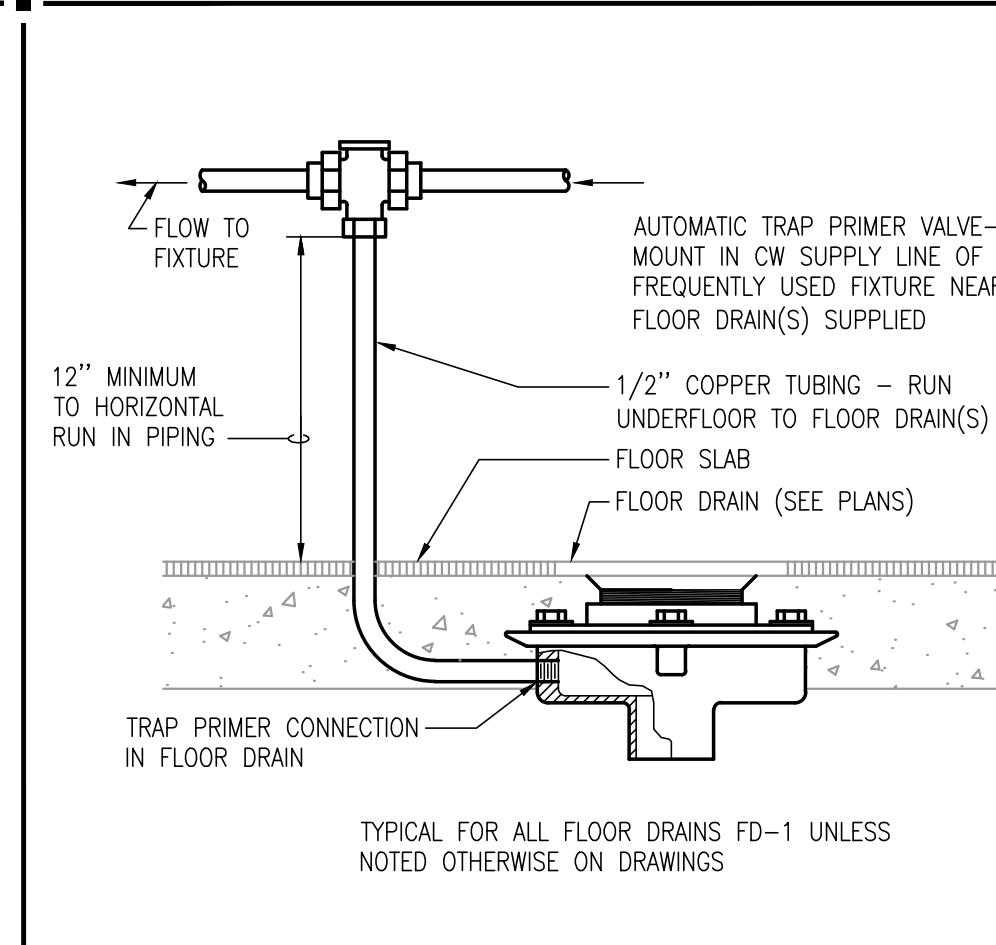
13 CLEANOUT DETAIL
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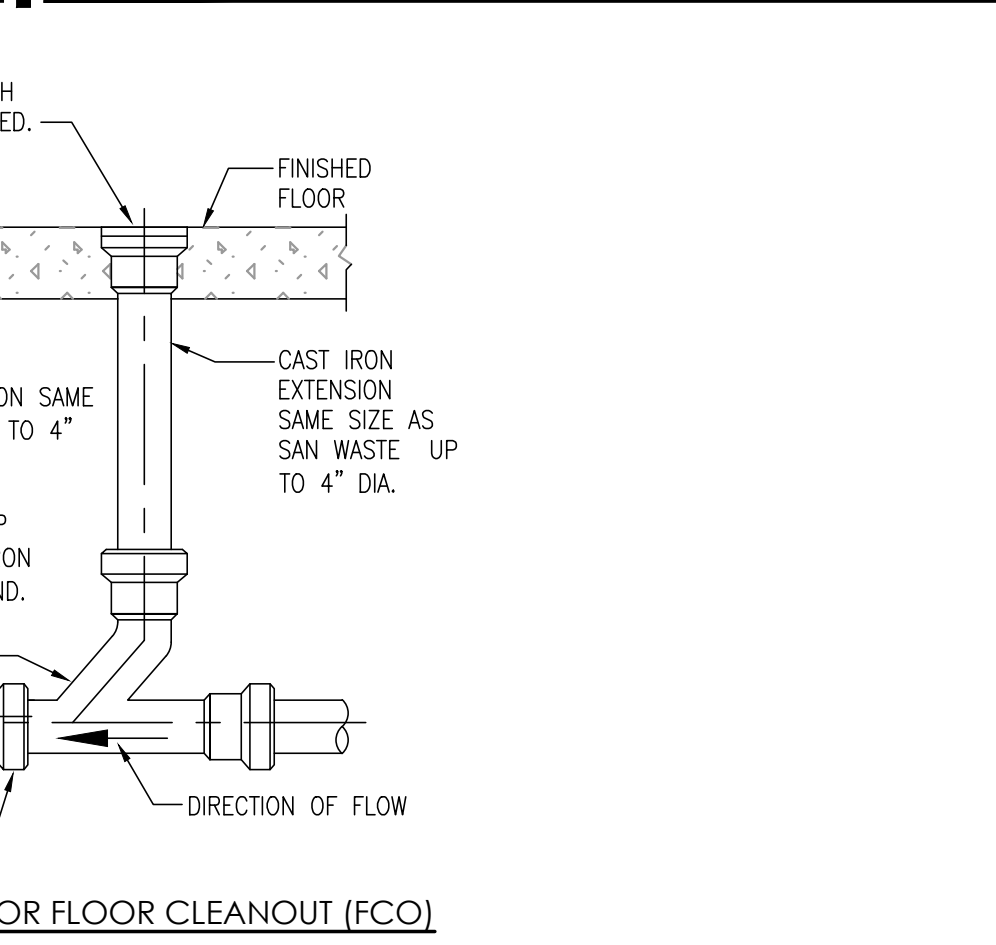
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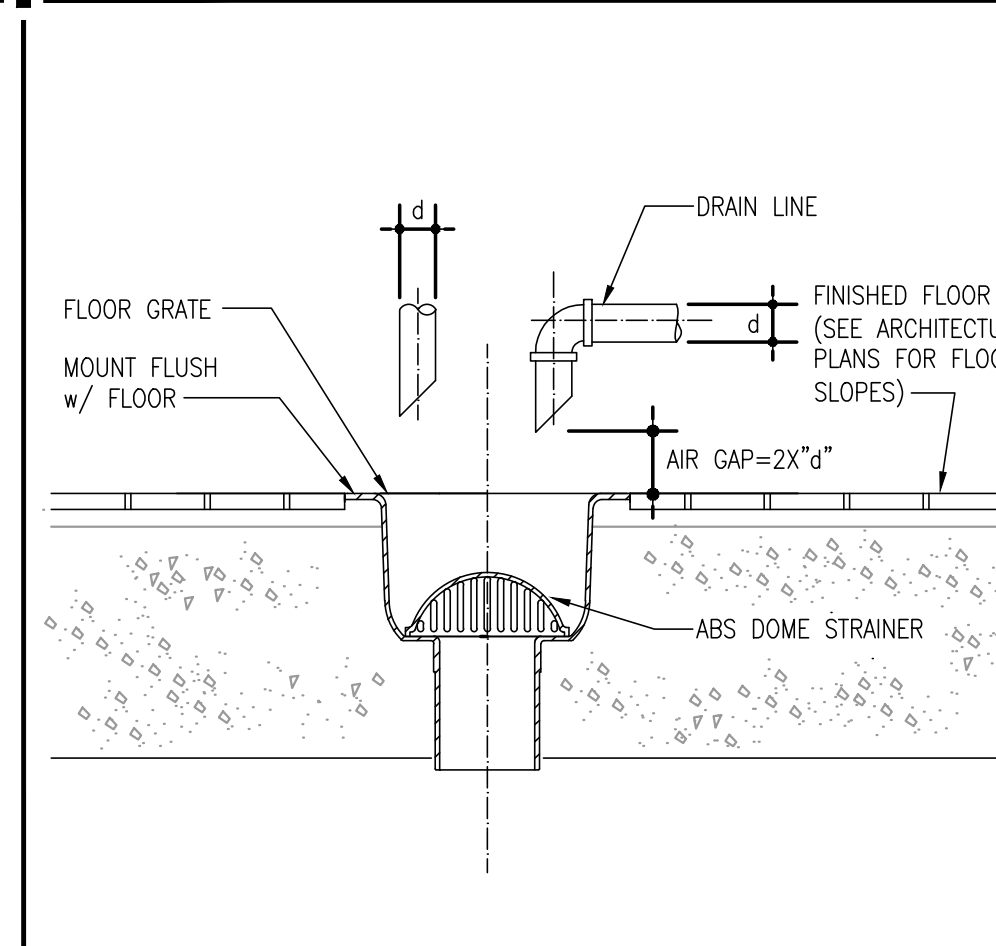
5 SMALL EXPANSION TANK DETAIL
NTS



10 TRAP PRIMER DETAIL
NTS



13 CLEANOUT DETAIL
NTS



18 FLOOR SINK DETAIL
NTS

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**BAGELS
& CO.**

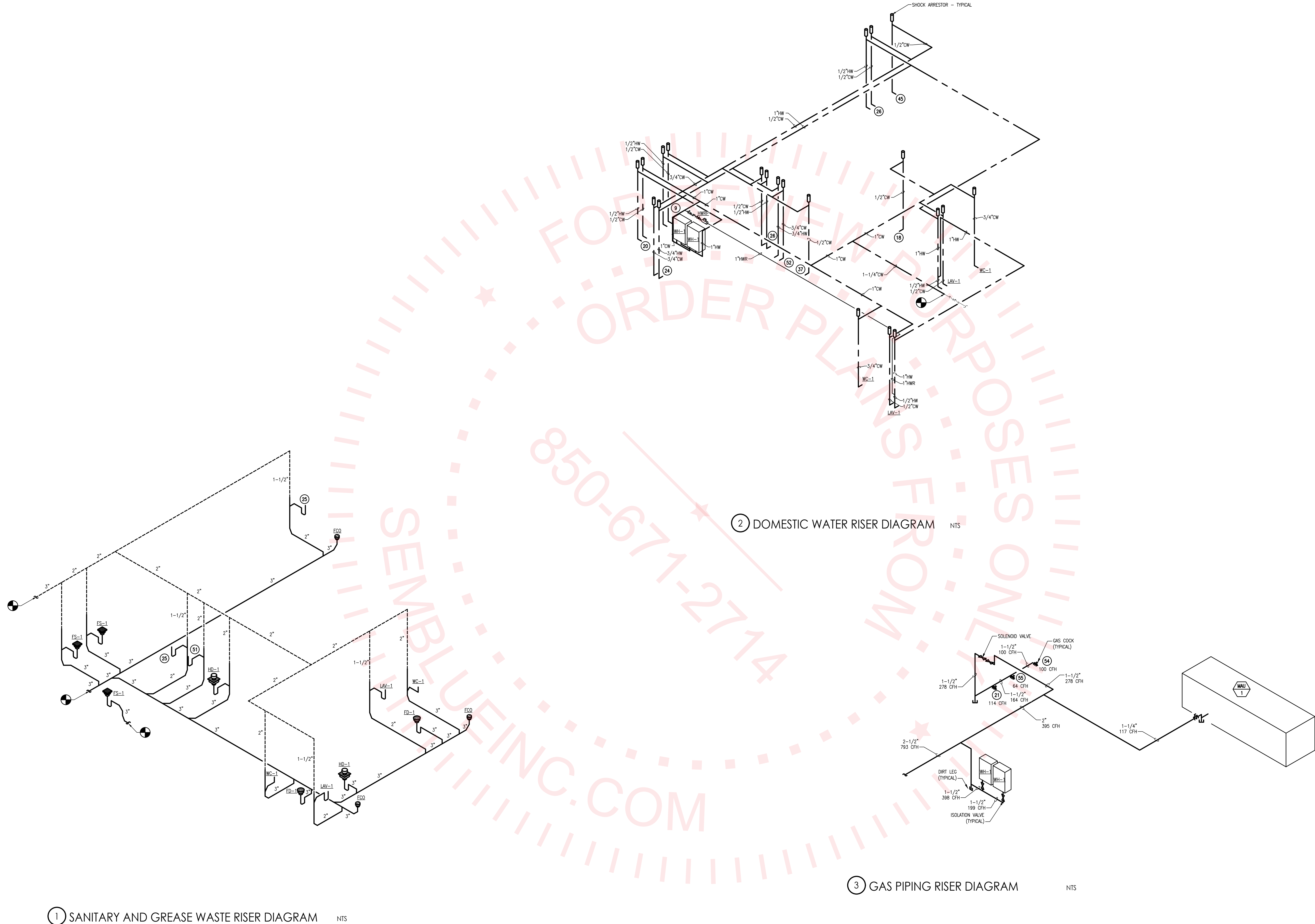
BAGEL & CO
TALLAHASSEE

799 W. GAINES STREET
SUITE 121
TALLAHASSEE, FL 32310

PROJECT NO: 2023.0719
DATE: 04/12/2024

P6.0
PLUMBING RISERS

CHECKED: DAK DRAWN: JR



PLUMBING SPECIFICATION

1. GENERAL INFORMATION

1.1. PROVIDE ALL SUPPLIES, MATERIAL, LABOR, EQUIPMENT, AND APPURTENANCES NECESSARY FOR COMPLETE INSTALLATION AND FULL OPERATION OF ALL PLUMBING AND PLUMBING RELATED WORK, INDICATED HEREINAFTER ON DRAWINGS AND SPECIFICATIONS, FOR A SAFE AND FULLY OPERATIONAL SYSTEM.

1.1.1. THE INSTALLED SYSTEM SHALL BE COMPLETE IN EVERY WAY AND FUNCTIONING ACCORDING TO THE DESIGN INTENT, WHETHER OR NOT ALL SUCH MATERIALS AND APPURTENANCES ARE SHOWN ON THE DRAWINGS OR DESCRIBED IN THE SPECIFICATIONS.

1.2. PERFORM ALL OPERATIONS INCLUDING EXCAVATION & BACKFILLING, SHORING, CUTTING, CHANNELING & CHASING, DE-WATERING, ETC. NECESSARY FOR INSTALLATION OF FULLY OPERATIONAL SYSTEM, WHETHER OR NOT SHOWN ON THE DRAWINGS.

1.3. DEFINITION OF TERMS

1.3.1. FURNISH – SUPPLY AND DELIVER TO THE PROJECT SITE, READY FOR UNLOADING, UNPACKING, ASSEMBLY, INSTALLATION, AND SIMILAR OPERATIONS.

1.3.2. INSTALL – OPERATIONS AT THE PROJECT SITE INCLUDING THE ACTUAL UNLOADING, UNPACKING, ASSEMBLY, ERECTION, PLACING, ANCHORING, APPLYING, WORKING TO DIMENSION, FINISHING, CURING, PROTECTING, CLEANING, AND SIMILAR OPERATIONS.

1.3.3. PROVIDE – FURNISH AND INSTALL, COMPLETE AND READY FOR THE INTENDED USE.

1.3.4. U.N.O. – UNLESS NOTED OTHERWISE.

1.3.5. M.S.D.S. – MATERIAL SAFETY DATA SHEET

1.3.6. CONTRACTOR – APPEARANCE ON DRAWINGS OR IN SPECIFICATIONS FOR PLUMBING WORK SHALL REFER TO PLUMBING SUB-CONTRACTOR.

1.3.7. COORDINATE WITH THE ELECTRICAL CONTRACTOR TO RELOCATE – DISCONNECT ELECTRICAL FEEDER, MAKE SAFE (INCLUDING LOCK OUT/TAG OUT), STORE AND PROTECT DEVICE, REINSTALL, REWORK AND EXTEND CONDUIT & WIRE TO NEW LOCATION, RE-ENERGIZE AND TEST.

1.3.8. EQUAL AND EQUIVALENT – TO MEAN OF THE SAME QUALITY, SIZE, NUMBER, VALUE, DEGREE, INTENSITY AND THE ITEMS ARE SIMILAR IN ALL RESPECTS.

1.3.8.1. THE FINAL DECISION OF ACCEPTANCE OF THESE ITEMS WILL BE MADE BY THE ENGINEER.

1.3.8.2. IT SHALL BE UNDERSTOOD THAT FOR ANY SPECIFIED ITEM ON THE DRAWINGS AND/OR IN THE SPECIFICATION, THIS TERM SHALL APPLY.

1.4. ALL WORK SHALL BE PERFORMED UNDER THE PERSONAL SUPERVISION OF A PROJECT SUPERINTENDENT ON-SITE. MAINTAIN A COMPLETE SET OF DRAWINGS AND SPECIFICATIONS ON SITE AT ALL TIMES DURING THE PROJECT.

1.5. ALL MATERIALS AND EQUIPMENT SHALL BE NEW AND SUITABLE FOR THE CONDITIONS AND DUTIES IMPOSED ON THEM AFTER INSTALLATION.

1.5.1. ALL MATERIALS, APPARATUS AND EQUIPMENT SHALL BEAR THE SEAL OF UNDERWRITERS LABORATORIES INC. (UL), OR A SIMILAR CREDIBLE TESTING AGENCY, LABEL WHERE REGULARLY SUPPLIED.

1.5.2. CERTAIN MANUFACTURERS OF MATERIAL AND EQUIPMENT ARE SPECIFIED AND PLANS ARE DETAILED ACCORDING TO THIS MATERIAL. CONTRACTOR SHALL BASE BID ON FURNISHING AND INSTALLING THIS MAKE OF MATERIAL AND EQUIPMENT.

1.5.3. WHERE MORE THAN ONE MAKE OF MATERIAL OR EQUIPMENT IS SPECIFIED, CONTRACTOR SHALL STATE IN BID WHICH MAKE THEY PROPOSE TO FURNISH. SHOP DRAWINGS SHALL BE SUBMITTED ON MATERIAL AND EQUIPMENT TO BE FURNISHED BY CONTRACTOR FOR ENGINEERS APPROVAL.

1.6. THIS APPROVAL TO BE OBTAINED PRIOR TO SHIPMENT OF EQUIPMENT.

1.7. ALL MATERIALS SHALL BE FABRICATED AND INSTALLED IN A NEAT AND WORKMANLIKE MANNER.

1.7.1. THE OWNER AND ENGINEER SHALL DETERMINE WHETHER WORKMANSHIP IS ACCEPTABLE. NO ALLOWANCES WILL BE MADE FOR REWORK OR DELAY DUE TO POOR WORKMANSHIP, COORDINATION DIFFICULTIES, OR INTERFERENCES BETWEEN INVOLVED TRADES.

1.7.2. PERFORM ALL WORK NECESSARY TO PREPARE THE STRUCTURE FOR THE INSTALLATION OF THE WORK. ALL HOLES, OPENINGS AND DAMAGED MATERIALS CREATED DURING CONSTRUCTION SHALL BE REPAIRED AND FINISHED BY EXPERIENCED WORKMEN.

1.7.3. COORDINATE AND SCHEDULE THE WORK WITH THE OWNER TO MINIMIZE DISRUPTIONS TO THE NORMAL OPERATIONS AT THE BUILDING.

1.7.3.1. IF AFTER HOURS WORK IS REQUIRED ON THIS PROJECT INCLUDE IN THE CONTRACT PRICE THE COST OF AFTER-HOURS WORK AND TEMPORARY PROVISIONS TO MINIMIZE DOWN TIME AND TO MAINTAIN FACILITY IN OPERATING CONDITION. COORDINATE WITH THE OWNER TO DETERMINE THE EXTENT OF THESE REQUIREMENTS PRIOR TO BID.

1.8. RELATED WORK SPECIFIED ELSEWHERE:

1.8.1. ALL DIVISION 1 REQUIREMENT, AND ALL TERMS AND CONDITIONS OF CONTRACT.

1.8.2. REFER TO ELECTRICAL SPECIFICATION FOR ELECTRICAL WORK TO BE DONE IN CONJUNCTION WITH THE PLUMBING WORK. CONTRACTOR IS RESPONSIBLE FOR ALL CONDUIT, WIRING, JUNCTION BOXES, ETC., REQUIRED FOR PLUMBING CONTROLS, UNLESS SPECIFICALLY NOTED OTHERWISE.

1.8.3. ALL PLUMBING EQUIPMENT AND WIRING PROVIDED UNDER DIVISION 22 SHALL COMPLY WITH THE ELECTRICAL SYSTEM CHARACTERISTICS INDICATED ON THE PLUMBING DRAWINGS AND SPECIFIED DIVISION 26.

1.8.4. ELECTRIC CONTROLS, CONTACTORS, STARTERS, PILOT LIGHTS, PUSH BUTTONS, ETC., SHALL BE PROVIDED COMPLETE AS PART OF THE MOTOR, HEATER OR OTHER EQUIPMENT WHICH IT OPERATES. ALL ELECTRICAL COMPONENTS SHALL BE IN CONFORMANCE WITH THE REQUIREMENTS OF THE NATIONAL ELECTRICAL CODE AND DIVISION 26.

1.9. WHERE EQUIPMENT SPECIFICATIONS INDICATE THAT A FACTORY-AUTHORIZED SERVICE ENGINEER OR TECHNICIAN SHALL OBSERVE INSTALLATION, TEST & ADJUST, OR START-UP OF EQUIPMENT, ETC., SUCH SERVICES WILL BE CONTRACTED BY OWNER AS PART OF THE EQUIPMENT PURCHASE.

1.9.1. CONTRACTOR SHALL ARRANGE FOR, SCHEDULE, AND COORDINATE SUCH FIELD SERVICES AS WORK INCLUDED IN THE CONTRACT.

1.9.2. CONTRACTOR SHALL FURNISH ALL LABOR AND MATERIALS NECESSARY TO SUPPORT ALL SUCH FACTORY REPRESENTATIVE'S FIELD SERVICES.

1.10. REGULARLY DURING EACH WORKING DAY, REMOVE REFUSE AND DEBRIS ACCUMULATING FROM PLUMBING CONSTRUCTION AND LEAVE AREA CLEAN AT END OF THE WORK DAY.

1.10.1. PRIOR TO ACCEPTANCE OF THIS WORK, LEAVE THE PREMISES "BROOM CLEAN" INSOFAR AS AFFECTED BY PLUMBING WORK.

1.10.2. CLEAN THE INTERIOR OF EACH PLUMBING COMPONENT OF DIRT AND CONSTRUCTION DUST INCLUDING BUT NOT LIMITED TO CONTROLLER OR CABINETS.

1.10.3. EXPOSED FINISHED MATERIALS AND EQUIPMENT SHALL BE CAREFULLY CLEANED AND

WIPED TO REMOVE GREASE, SMUDGES, FINGERPRINTS, DUST AND OTHER SPOTS AND LEFT SMOOTH AND CLEAN.

1.10.4. CLEAN THE EXTERIOR OF PLUMBING COMPONENTS PRIOR TO ACCEPTANCE OF WORK.

1.10.5. FOR ALL MATERIALS AND DEVICES REMOVED, THE CONTRACTOR SHALL DISPOSE OFF-SITE IN AN APPROVED MANNER. PROVIDE WRITTEN DOCUMENTATION FOR DISPOSAL OF ALL ITEMS.

1.11. A COMMISSIONING PLAN SHALL BE DEVELOPED BY A LICENSED DESIGN PROFESSIONAL, OR APPROVED AGENCY AND SHALL INCLUDE THE FOLLOWING ITEMS:

1.11.1. A NARRATIVE DESCRIPTION OF THE ACTIVITIES THAT WILL BE ACCOMPLISHED DURING EACH PHASE OF COMMISSIONING, INCLUDING THE PERSONNEL INTENDED TO ACCOMPLISH EACH OF THE ACTIVITIES.

1.11.2. A LISTING OF THE SPECIFIC EQUIPMENT, APPLANCES OR SYSTEMS TO BE TESTED AND A DESCRIPTION OF THE TESTS TO BE PERFORMED.

1.11.3. FUNCTIONS TO BE TESTED INCLUDING, BUT NOT LIMITED TO, PUMP AND FAN START UP, CALIBRATIONS AND CONTROLS.

1.11.4. NOTIFY THE ARCHITECT/ENGINEER/OWNER/ UNDERWRITER PRIOR TO PERFORMING THE TEST WHO, AT THEIR OPTION, MAY DESIRE TO WITNESS THE TEST. TEST SHALL BE SATISFACTORY TO THE ARCHITECT/ENGINEER/OWNER/UNDERWRITER OR AUTHORITIES HAVING JURISDICTION.

1.11.5. ABOVE AND BELOW GROUND PIPING: PERFORM A HYDROSTATIC TEST OF THE PIPING SYSTEM IN ACCORDANCE WITH THE CURRENT CODE REQUIREMENTS.

1.11.5.1 IF TEST PROVES UNSATISFACTORY, LEAKS SHALL BE REPAIRED AND A NEW TEST SHALL BE PERFORMED.

1.11.6. AFTER TESTING HAS BEEN COMPLETED, ALL VALVES SHALL BE LEFT IN THEIR RESPECTIVE NORMAL POSITIONS, AND THE OWNER AND HIS DULY APPOINTED REPRESENTATIVE SHALL BE INSTRUCTED IN THE CARE, OPERATION AND MAINTENANCE OF THE ENTIRE SYSTEM.

1.11.7. A REPORT OF TEST PROCEDURES AND RESULTS IDENTIFIED AS "FINAL COMMISSIONING REPORT" SHALL BE DELIVERED TO THE BUILDING OWNER OR OWNER'S AUTHORIZED AGENT. THE REPORT SHALL INCLUDE THE FOLLOWING:

1.11.7.1. RESULTS OF FUNCTIONAL PERFORMANCE TESTS.

1.11.7.2. DISPOSITION OF DEFICIENCIES FOUND DURING TESTING, INCLUDING DETAILS OF CORRECTIVE MEASURES USED OR PROPOSED.

1.11.7.3. FUNCTIONAL PERFORMANCE TEST PROCEDURES USED DURING THE COMMISSIONING PROCESS INCLUDING MEASURABLE CRITERIA FOR TEST ACCEPTANCE, PROVIDED HEREIN FOR REPEATABILITY.

1.12. PROVIDE ALL LABOR, INSTRUMENTS, AND OTHER SERVICES REQUIRED FOR COMPLETE AND SATISFACTORY TEST AND ADJUSTMENT OF PLUMBING SYSTEMS AND RELATED WORK.

1.12.1. CHECK ALL MOTORS AND ROTATING EQUIPMENT FOR PROPER ROTATION.

1.12.2. IMMEDIATELY REMEDIATE ALL EQUIPMENT PROVIDED UNDER THIS DIVISION THAT TESTS PROVE TO BE DEFECTIVE OR OPERATING IMPROPERLY AS A PART OF THIS CONTRACT.

1.13. CONTRACTOR AND VENDOR SHALL INSTRUCT THE OWNER'S TECHNICAL PERSONNEL ON ALL OWNER FURNISHED EQUIPMENT IN ACCORDANCE WITH SPECIFICATIONS.

1.13.1. INCLUDE IN THE CONTRACT PRICE A MINIMUM OF (2) HOURS OF OWNER TRAINING FOR THE NEW CONTROLS.

1.13.2. TRAIN BUILDING OWNER'S PERSONNEL DURING NORMAL WORKING HOURS ON START-UP & SHUT-DOWN PROCEDURES, TROUBLESHOOTING PROCEDURES, SERVICING & PREVENTATIVE MAINTENANCE SCHEDULE & PROCEDURES, AND OVERVIEW OF EQUIPMENT & CONTROLS IN THE EVENT OF SYSTEM FAILURE. REVIEW WITH THE OWNER'S PERSONNEL, THE DATA CONTAINED IN THE OPERATING AND MAINTENANCE MANUALS. SCHEDULE TRAINING WITH OWNER, PROVIDE AT LEAST 7-DAYS PRIOR NOTICE TO ARCHITECT/ENGINEER.

1.13.3. CONTRACTOR SHALL PROVIDE OWNER WITH SUFFICIENT SETS OF OPERATIONS AND MAINTENANCE MANUALS OF CONTRACTOR FURNISHED EQUIPMENT FOR INCLUSION INTO THE OWNER'S OPERATIONS AND MAINTENANCE MANUALS AS REQUIRED BY THE SPECIFICATIONS.

1.13.4. PROJECT WILL NOT BE COMPLETE UNTIL ACCURATE O/M MANUALS ARE DELIVERED.

1.13.5. O/M MANUALS SHALL INCLUDE CATALOG TECHNICAL DATA, RECOMMENDED SERVICE PROCEDURES, RECOMMENDED SERVICE INTERVALS, CALIBRATION INFORMATION, FACTORY TRAINING MANUALS, MAGNETIC MEDIA FOR SOFTWARE PROVIDED, AND RECOMMENDED SPARE PARTS.

1.13.5.1. PROVIDE (2) NEATLY BOUND (WITH TABBED SECTIONS) COPIES OF MAINTENANCE BOOKS, INSTRUCTION BOOKS, AND PARTS LIST PERTAINING TO ALL EQUIPMENT FURNISHED. SUBMIT TO THE OWNERS REPRESENTATIVE FOR APPROVAL. FINAL PAYMENT WILL NOT BE MADE UNTIL MAINTENANCE AND INSTRUCTION MANUALS ARE DELIVERED TO THE OWNERS REPRESENTATIVE.

1.13.5.1.1. ORGANIZE OPERATING AND MAINTENANCE DATA INTO SUITABLE SETS OF MANAGEABLE SIZE.

1.13.5.1.2. BIND PROPERLY INDEXED DATA IN INDIVIDUAL, HEAVY-DUTY, 2-INCH, 3-RING VINYL-COVERED BINDERS WITH POCKET FOLDERS FOR FOLDED SHEET INFORMATION.

1.13.5.2. MARK APPROPRIATE IDENTIFICATION ON FRONT AND SPINE OF EACH BINDER. INCLUDE THE FOLLOWING TYPES OF INFORMATION:

1.13.5.2.1. SPARE PARTS LIST

1.13.5.2.2. COPIES OF GUARANTEES AND WARRANTIES OF ALL EQUIPMENT AND SYSTEMS.

1.13.5.2.3. WIRING DIAGRAMS

1.13.5.2.4. INSPECTION REPORTS & APPROVALS

1.13.5.2.5. SUBMITTAL DATA STATING EQUIPMENT SIZE AND SELECTED OPTIONS FOR EACH PIECE OF EQUIPMENT REQUIRING MAINTENANCE.

1.13.5.2.6. MANUFACTURER'S OPERATION MANUALS AND MAINTENANCE MANUALS FOR EACH PIECE OF EQUIPMENT REQUIRING MAINTENANCE, EXCEPT EQUIPMENT NOT FURNISHED AS PART OF THE PROJECT. REQUIRED ROUTINE MAINTENANCE ACTIONS SHALL BE CLEARLY IDENTIFIED.

1.13.5.2.7. CONTRACTOR'S NAME, ADDRESS, AND TELEPHONE NUMBERS.

1.13.5.2.8. NAME AND ADDRESS AND 24-HOUR PHONE NUMBER OF THE PERSONNEL RESPONSIBLE FOR SERVICING DURING THE WARRANTY PERIOD.

1.13.5.2.9. SUBMITTAL DATA INDICATING ALL SELECTED OPTIONS FOR EACH PIECE OF EQUIPMENT

AND CONTROLS.

1.13.5.2.10. OPERATION AND MAINTENANCE MANUALS FOR EACH PIECE OF EQUIPMENT. REQUIRED ROUTINE MAINTENANCE ACTIONS AND CLEANING RECOMMENDATIONS SHALL BE CLEARLY IDENTIFIED.

1.13.5.2.11. A SCHEDULE FOR INSPECTING AND RECALIBRATING ALL CONTROLS.

1.13.5.2.12. A NARRATIVE OF HOW EACH SYSTEM IS INTENDED TO OPERATE, INCLUDING RECOMMENDED SET POINTS.

1.13.5.2.13. ALL MANUFACTURERS' DATA APPLICABLE TO THE INSTALLED EQUIPMENT SUCH AS INSTALLATION INSTRUCTIONS, MAINTENANCE INSTRUCTIONS, OPERATING INSTRUCTIONS, ETC.

1.13.5.2.14. CONTROL DIAGRAMS INCLUDING AN EXPLANATION OF THE CONTROL SEQUENCE OF EACH SYSTEM INCLUDING NORMAL STARTING, OPERATION, SHUTDOWN, POWER FAIL RE-START, AND WINTER SHUTDOWN.

1.13.5.2.15. SCHEDULE OF MAINTENANCE AND INSPECTION THAT SHOULD BE PERFORMED BY OWNER, AND THOSE WHICH SHOULD BE PROVIDED BY AN OUTSIDE SERVICE.

1.13.5.2.16. COPY OF COMPLETED FIELD ACCEPTANCE TEST PROCEDURE, LOGS, AND ALL DOCUMENTATION ASSOCIATED WITH THE TEST.

1.14. GUARANTEE ALL PLUMBING SYSTEM EQUIPMENT, MATERIALS, AND WORKMANSHIP TO BE FREE FROM DEFECTS FOR A PERIOD OF ONE (1) YEAR FROM THE DATE OF FINAL ACCEPTANCE AND PROPERLY CORRECT LATENT DEFECTS ARISING DURING THIS PERIOD UPON NOTIFICATION BY THE OWNER'S REPRESENTATIVE WITHOUT ADDITIONAL COMPENSATION AND TO THE SATISFACTION OF THE ENGINEER AND OWNER'S REPRESENTATIVE.

1.15. ALL EQUIPMENT, ETC., SHALL BE NEW UNLESS OTHERWISE NOTED, AND AS SPECIFIED FREE OF DEFECTS. ALL PLUMBING EQUIPMENT SHALL BE U.L. OR E.T.L. LISTED.

1.16. PROVIDE THUMB DRIVE TO THE OWNER REPRESENTATIVE WITH ALL THE ABOVE LISTED MATERIAL SEPERATED INTO DIRECTORIES TO MATCH CATALOGED INFORMATION.

2. CODES & PERMITS

2.1. ENTIRE INSTALLATION (INCLUDING EQUIPMENT, DEVICES, AND WIRING) SHALL BE IN ACCORDANCE WITH ALL APPLICABLE PROVISIONS OF THE NATIONAL ELECTRICAL CODE (N.E.C.), NATIONAL FIRE PROTECTION ASSOCIATION (NFPA 13, NFPA 70 & NFPA 101), OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (O.S.H.A.), CURRENT BUILDING CODE, CURRENT ENERGY CONSERVATION CODE (I.E.C.C.), AND ALL LAWS & ORDINANCES APPLICABLE TO WORK AT THIS SITE. IN ADDITION, INSTALLATION SHALL MEET APPROVAL OF LOCAL INSPECTION AUTHORITY HAVING JURISDICTION. REFER TO COVER SHEET FOR LIST OF CURRENT APPLICABLE CODE EDITIONS.

2.2. SECURE AND PAY ALL FEES ASSOCIATED WITH ALL PERMITS AND LICENSES REQUIRED FOR EXECUTION OF THE CONTRACT. ARRANGE FOR ALL INSPECTIONS REQUIRED BY CITY, COUNTY, STATE AND OTHER AUTHORITIES HAVING JURISDICTION, AND DELIVER CERTIFICATES OF APPROVAL TO THE ARCHITECT.

2.3. A CERTIFICATE OF APPROVAL FOR WORK FROM INSPECTION AUTHORITY SHALL BE GIVEN TO THE OWNER BEFORE FINAL ACCEPTANCE WILL BE GIVEN BY OWNERS REPRESENTATIVE.

2.4. THE CODE REQUIREMENTS ARE STRICTLY A MINIMUM AND SHALL BE MET WITHOUT INCURRING ADDITIONS TO THE CONTRACT. WHERE REQUIREMENTS OF THE DRAWINGS OR SPECIFICATIONS EXCEED THE CODE REQUIREMENTS, THE WORK SHALL BE PROVIDED IN ACCORDANCE WITH THESE DRAWINGS OR SPECIFICATIONS. IN THE EVENT OF CONFLICT OR AMBIGUITY BETWEEN THE VARIOUS CODES, THE MOST STRINGENT REQUIREMENTS SHALL GOVERN.

2.5. REFER TO ARCHITECTURAL COVERSHEET FOR APPLICABLE CODE REFERENCES FOR THIS PROJECT.

3. SAFETY

3.1. THE CONTRACTOR SHALL MAINTAIN A SAFE WORK ENVIRONMENT AT ALL TIMES.

3.1.1. COMPLY WITH ALL O.S.H.A., N.I.O.S.H., D.O.T., STATE & LOCAL REQUIREMENTS REGARDING SAFE HANDLING, STORING, TRANSPORTING, AND DISPENSING OF CHEMICALS.

3.1.2. MAINTAIN AND DISPLAY M.S.D.S. INFORMATION FOR ALL CHEMICAL PRODUCTS.

3.1.3. PROVIDE ALL NECESSARY MEANS TO MAINTAIN SAFE WORKING CONDITIONS, INCLUDING VENTILATION FANS, FIRE EXTINGUISHERS, EYE PROTECTION, RESPIRATORS, PROTECTIVE CLOTHING, VENTILATION, ETC.

3.1.4. ALL EQUIPMENT AND MATERIALS USED TO IMPLEMENT THE WORK SHALL BE USED IN STRICT ACCORDANCE WITH THE MANUFACTURER'S WRITTEN INSTRUCTIONS, INCLUDING ALL RECOMMENDED SAFETY PRECAUTIONS.

3.1.5. MAINTAIN A PROPER FIRE WATCH FOR ALL OPERATIONS WHERE SPARKS, FLAMES, OR OTHER SOURCES OF FIRE ARE PRODUCED.

3.1.6. FOR ALL MATERIALS CONTAINING SOLVENTS, MAINTAIN THE RECOMMENDED VENTILATION OF THE AREA TO PREVENT THE ACCUMULATION OF VAPORS WHICH POSE A HEALTH OR FIRE HAZARD.

4. INTENT OF DRAWINGS AND SPECIFICATIONS

4.1. THE IMPLIED AND STATED INTENT OF THE DRAWINGS & SPECIFICATIONS ARE TO ESTABLISH MINIMUM ACCEPTABLE STANDARDS FOR MATERIALS, EQUIPMENT, WORKMANSHIP, AND TO PROVIDE OPERABLE SYSTEMS THAT ARE COMPLETE IN EVERY RESPECT.

4.2. ENGINEERING DRAWINGS ARE DIAGRAMMATIC, INTENDED TO SHOW GENERAL ARRANGEMENT AND SIZES OF SYSTEM COMPONENTS, AND SHALL NOT BE SCALED.

4.2.1. ARCHITECTURAL AND STRUCTURAL DRAWINGS SHALL GOVERN SPACE CONSTRAINTS, DIMENSIONS AND FINISHES.

4.2.2. ALL OFFSETS AND FITTINGS THAT SHALL BE NECESSARY TO ACCOMPLISH A FINISHED INSTALLATION SHALL BE PROVIDED AT NO ADDITIONAL COST OR INCREASE THE CONTRACT.

4.2.3. WORK INTENDED, BUT HAVING MINOR DETAILS OBVIOUSLY OMITTED, SHALL BE PROVIDED COMPLETE AS A REQUIREMENT OF THIS CONTRACT.

4.2.4. LOCATIONS OF EQUIPMENT INDICATED ON PLANS SHALL BE FOLLOWED AS CLOSELY AS POSSIBLE TO THE PLANS SUBJECT TO BUILDING CONSTRUCTION AND INTERFERENCES WITH OTHER TRADES.

4.2.5. MAINTAIN MINIMUM SERVICE CLEARANCE AS REQUIRED BY THE EQUIPMENT MANUFACTURER AND N.E.C.

4.3. PROVIDE THE OWNER A COMPLETE SET OF RECORD DRAWINGS AT THE END OF THE PROJECT. PROJECT WILL NOT BE COMPLETE UNTIL ACCURATE RECORD DRAWINGS ARE DELIVERED.

4.4. THE RECORD DRAWINGS SHALL BE MACHINE DRAFTED, AND SHALL BE PROVIDED BOTH

ON REPRODUCIBLE VELLUMS AND MAGNETIC MEDIA CAD FILES TO THE ARCHITECT WHICH REFLECT ALL CHANGES, DEVIATIONS AND REVISIONS MADE TO THE ORIGINAL DESIGN DOCUMENTS. LOCATIONS OF ALL UNDERGROUND PIPING AND UTILITIES SHALL BE CLEARLY SHOWN AND DIMENSIONED FROM PERMANENT REFERENCE POINTS SUCH AS BUILDING COLUMN LINES.

4.5. DRAWING FORMAT FOR THIS PROJECT SHALL BE AUTOCAD 2010.

4.6. ALL ITEMS MOUNTED IN OR BELOW THE CEILING, AND ALL ITEMS PENETRATING THE CEILING, SHALL BE COORDINATED WITH THE ARCHITECTURAL REFLECTED CEILING PLANS. IF ANY ITEMS ARE NOT SHOWN ON THESE PLANS, OR ANY ITEMS NEED TO BE RELOCATED FOR COORDINATION PURPOSES, PREPARE A REFLECTED CEILING PLAN SUBMIT IT TO THE ARCHITECT FOR APPROVAL.

5. EXISTING CONDITIONS

5.1. ATTENTION IS CALLED TO THE FACT THAT THE WORK IS TO BE PERFORMED WITHIN AN EXISTING, OPERATIONAL FACILITY.

5.2. THE FOLLOWING GENERAL PROVISIONS OF THE CONTRACT, INCLUDING THE GENERAL & SUPPLEMENTAL CONDITIONS AND GENERAL REQUIREMENTS, SHALL APPLY TO THE WORK IN THIS DRAWING AND SPECIFICATION SET.

5.2.1. VISIT THE SITE OF THE WORK AND BECOME THOROUGHLY FAMILIAR WITH ALL EXISTING CONDITIONS, AND THOROUGHLY REVIEW ALL DRAWINGS, SPECIFICATIONS AND ADDENDA PRIOR TO BIDDING ON THIS WORK. NO EXTRA PAYMENTS TO THE CONTRACT AMOUNT WILL BE ALLOWED FOR FAILURE TO COMPLY WITH THIS REQUIREMENT.

5.2.2. TAKE MEASUREMENTS AND BE RESPONSIBLE FOR EXACT SIZE AND LOCATIONS OF ALL OPENINGS REQUIRED FOR THE INSTALLATION OF WORK.

5.2.3. FIELD DIMENSIONS ARE REASONABLY ACCURATE AND SHOULD GOVERN IN SETTING OUT WORK.

5.2.4. WHERE DETAILED METHOD OF INSTALLATION IS NOT INDICATED OR WHERE VARIATIONS EXIST BETWEEN DESCRIBED WORK AND APPROVED PRACTICE, DIRECTION OF THE OWNERS REPRESENTATIVE ON JOB SITE SHALL BE FOLLOWED.

5.3. CONTRACTOR SHALL VERIFY PROJECT CONDITIONS TO ENSURE THAT THE WORK WILL FIT INTO THE STRUCTURE IN THE MANNER INTENDED ON THE DRAWINGS.

5.3.1. SHOULD ANY CONDITIONS EXIST THAT ARE CONTRARY TO THOSE SHOWN ON THE DRAWINGS, CONTRACTOR SHALL NOTIFY THE ENGINEER PRIOR TO FABRICATION OR PERFORMING ANY WORK IN THE AREA INVOLVING THE DIFFERENCES.

5.3.2. NOTIFICATION SHALL BE IN THE FORM OF A DRAWING OR SKETCH INDICATING FIELD MEASUREMENTS OR NOTES RELATING TO THE AREA.

5.4. CONNECT NEW WORK TO EXISTING WORK IN A NEAT AND WORKMANLIKE MANNER.

5.4.1. WHERE AN EXISTING STRUCTURE MUST BE CUT OR EXISTING UTILITIES INTERFERE, SUCH OBSTRUCTION SHALL BE BYPASSED, REMOVED, REPLACE OR RELOCATED, PATCH AND REPAIR.

5.4.2. WORK DISTURBED OR DAMAGED SHALL BE REPLACED OR REPAIR TO ITS PRIOR CONDITION.

5.5. PRIOR TO THE START OF ANY DEMOLITION OR CONSTRUCTION, CONTRACTOR SHALL REVIEW THE EXISTING SITE AND SECURE THE SERVICES OF A QUALIFIED, EPA CERTIFIED ASBESTOS ABATEMENT AGENCY IF NEEDED TO CHECK THE EXISTING INSULATION, ETC. FOR ASBESTOS. SHOULD ASBESTOS BE FOUND, DO NOT PROCEED WITH DEMOLITION OR CONSTRUCTION; NOTIFY THE ARCHITECT IN ANY CASE IN WRITING OF THE AGENCY'S FINDING.

5.6. FOR RENOVATION PROJECTS – PROVIDE ALL DEMOLITION, PATCHING, SAW CUTTING, EXCAVATION, TRENCHING, SHORING, COMPACTING, DE-WATERING, ETC. REQUIRED FOR THE PROJECT, WHETHER OR NOT SHOWN ON THE DRAWINGS.

5.6.1. INFORMATION WAS TAKEN FROM VARIOUS ARCHIVE DRAWINGS AND LIMITED FIELD OBSERVATION. FIELD VERIFICATION OF EXISTING CONDITIONS AND POINTS OF CONNECTIONS ARE REQUIRED.

5.7. EXISTING SYSTEMS TO REMAIN – WHERE EXISTING SYSTEMS ARE INDICATED TO REMAIN, THEY SHALL BE ASSUMED TO BE IN GOOD WORKING ORDER REQUIRING NO WORK UNLESS SPECIFICALLY NOTED.

5.7.1. IF DURING THE CONSTRUCTION DEFICIENCIES ARE NOTED, THEN BRING THESE TO THE ATTENTION OF THE OWNER AND SEEK DIRECTION.

6. SHOP DRAWINGS, SUBMITTALS, AND SUBSTITUTIONS

6.1. ENGINEER OF RECORD SHALL BE PROVIDED WITH SHOP DRAWINGS, COORDINATION DRAWINGS, AND MANUFACTURER'S DATA OF ANY CONTRACTOR FURNISHED MATERIALS AND EQUIPMENT, PRIOR TO PURCHASE AND/OR FABRICATION, AND SHALL VERIFY, BY STAMPING AND SIGNING THE DATA AND DRAWINGS BEFORE RETURNING THEM TO THE CONTRACTOR, THAT THE ITEMS FURNISHED BY THE CONTRACTOR FIT THE SPACES AND DIMENSIONS DESCRIBED IN AND CONFORM TO THE SPIRIT AND INTENT OF THE CONTRACT DOCUMENTS.

6.1.1. ENGINEER SHALL, WITHIN FIVE (5) WORKING DAYS OF RECEIPT OF SHOP DRAWINGS AND PRODUCT DATA, NOTIFY THE CONTRACTOR OF ANY DISCREPANCY OR INCOMPATIBILITY WITH THE CONTRACT DOCUMENTS, AND SHALL RETURN THE SHOP DRAWINGS TO THE CONTRACTOR APPROPRIATELY ANNOTATED.

6.1.2. REVIEW OF SUBMITTALS SHALL NOT BE CONSTRUED AS AUTHORIZING ANY DEVIATIONS FROM THE PLANS AND SPECIFICATIONS UNLESS SUCH DEVIATIONS ARE CLEARLY IDENTIFIED AND SEPARATELY SUBMITTED IN THE FORM OF A LETTER THAT IS ENCLOSED WITH THE SUBMITTALS.

6.1.3. SUBMIT SHOP DRAWING CUT SHEETS AND TECHNICAL DATA ON THE FOLLOWING ITEMS: EQUIPMENT, PIPING AND CONNECTION METHOD, SUPPORTS, SAFETY SWITCHES, AND ANY OTHER ITEM REQUIRED BY NOTES.

6.1.4. SUBMIT CATALOG INFORMATION, FACTORY ASSEMBLY DRAWINGS, FIELD INSTALLATION DRAWINGS AND CERTIFICATIONS AS REQUIRED FOR COMPLETE EXPLANATION AND DESCRIPTION OF ALL ITEMS OF EQUIPMENT. THE SUBMITTAL DATA SHALL PROVIDE AMPLE, UNQUESTIONABLE COMPLIANCE WITH THE CONTRACT DOCUMENTS.

6.1.5. SUBMIT PROPOSED CONTROL SYSTEM FOR REVIEW BY THE ENGINEER PRIOR TO EQUIPMENT PURCHASE OR FABRICATION. DO NOT PROCEED WITH THE WORK WITHOUT APPROVED SUBMITTALS.

6.1.6. WHERE QUALIFICATIONS AND/OR QUALITY ASSURANCE REQUIREMENTS ARE SPECIFIED, THE SUBMITTAL SHALL INCLUDE EVIDENCE THAT THE STATED REQUIREMENTS HAVE BEEN MET. INCLUDE QUALIFICATIONS AND CERTIFICATIONS OF PROPOSED TEST AND BALANCE SUBCONTRACTOR.

6.1.7. EQUIPMENT PERFORMANCE SHALL BE VERIFIED BY THE EQUIPMENT MANUFACTURER AS PART OF THE SUBMITTAL. PRIOR TO ORDERING, VERIFY EQUIPMENT VOLTAGE AND ELECTRICAL REQUIREMENTS OF THE EQUIPMENT WITH ELECTRICAL CONTRACTOR PRIOR TO ORDERING EQUIPMENT. SHOP DRAWINGS FOR EQUIPMENT REQUIRING ELECTRIC POWER OR CONTROL WIRING CONNECTIONS SHALL INCLUDE COMPLETE WIRING DIAGRAMS.

6.2. REFER TO THE CONDITIONS OF THE CONTRACT (GENERAL AND SUPPLEMENTARY WHEN INCLUDED AND OTHER DIVISIONS) REFERENCING SUBMITTALS FOR SHOP DRAWINGS,

PRODUCT DATA, AND SAMPLES FOR SUBMITTAL DEFINITIONS, REQUIREMENTS AND PROCEDURES.

6.3. BEFORE PREPARING SUBMITTALS, STUDY ALL CONTRACTS DRAWINGS AND SPECIFICATIONS IN DETAIL, OBTAIN MANUFACTURER'S RECOMMENDED INSTRUCTIONS, AND HAVE SUBMITTALS PREPARED BASED ON SPECIFIC EQUIPMENT AND MATERIAL PROPOSED FOR INSTALLATION.

6.3.1. AN OFFICER OF THE CONTRACTING FIRM SHALL SIGN ALL SHOP DRAWINGS (CERTIFYING CONFORMANCE WITH PLANS AND SPECIFICATIONS) BEFORE SUBMITTING TO THE ARCHITECT OR RELEASING TO THE FIELD.

6.4. THE SUBMITTAL PROCESS SHALL NOT BE UTILIZED AS AN AVENUE TO SUBSTITUTE AFTER THE EXECUTION OF THE CONTRACT.

6.4.1. SHOULD AN UNSPECIFIED OR UNEQUAL PRODUCT BE SUBMITTED, IT WILL BE REJECTED. IF A SECOND ATTEMPT AT SUBSTITUTION IS MADE DURING THE RE-SUBMITTAL OF THE SAME PRODUCT, THEN NO MORE REVIEWS OF THAT PRODUCT WILL BE PERFORMED WITHOUT DIRECT COMPENSATION TO THE ENGINEER BEING PAID FOR THE ADDITIONAL SERVICES REQUIRED FOR THE THIRD REVIEW AND ANY FURTHER REVIEWS.

6.5. IN ADDITION TO THE OTHER REQUIREMENTS IN PARAGRAPHS ABOVE, THE FOLLOWING APPLIES TO SUBMITTALS OF THIS DIVISION:

6.5.1. DO NOT INSTALL OR ORDER PLUMBING EQUIPMENT OR PROCEED WITH THE WORK UNTIL SUBMITTALS HAVE BEEN ACCEPTABLY REVIEWED BY THE OWNER'S REPRESENTATIVE AND STAMPED ACCORDINGLY. EQUIPMENT OR WORK WHICH IS ORDERED OR INSTALLED WITHOUT PRIOR APPROVED SUBMITTALS SHALL, AT THE ENGINEER'S DISCRETION, BE REMOVED AT NO COST TO THE OWNER. NO ALLOWANCES WILL BE MADE FOR REWORK OR DELAY DUE TO NEGLECT OF REQUIRED APPROVAL PROCESS.

6.5.1. MAKE ALL PLUMBING SUBMITTALS AT ONE TIME AND WITHIN FOURTEEN (14) CALENDAR DAYS OF OWNER'S "NOTICE TO PROCEED".

6.5.2. SUBMIT A MINIMUM OF SIX (6) COMPLETE SETS TO OWNER'S REPRESENTATIVE. ONE SET SHALL REMAIN WITH THE OWNER'S REPRESENTATIVE, ONE SET SHALL BE DELIVERED TO THE OWNER AND THE OTHERS WILL BE RETURNED TO DISTRIBUTE AS REQUIRED.

6.5.2. EACH SET SHALL BE BOUND AND INDEXED INTO GROUPS SUCH AS FIXTURES, PANELS, WIRING DEVICES, DISCONNECT SWITCHES, ETC.

6.5.3. MAKE EACH SET IDENTICAL.

6.5.4. ALL SUBMITTALS SHALL BE ONLY ORIGINALS, COPIES OF CUT SHEETS ARE NOT ACCEPTABLE.

6.5.5. SUBMITTAL DATA WILL BE PROVIDED AND REVIEWED IN PDF FORMAT ELECTRONICALLY VIA EMAIL.

6.5.6. SUBMITTAL WILL NOT BE ACCEPTED FOR REVIEW UNLESS THEY:

6.5.7.1. COMPLY WITH THE REQUIREMENTS OF DIVISION 1.

6.5.7.2. INCLUDE COMPLETE INFORMATION PERTAINING TO ALL APPURTENANCE AND ACCESSORIES.

6.5.7.3. ARE SUBMITTED AS COMPLETE PACKAGES WHICH PERTAIN TO ALL RELATED ITEMS IN DIVISION 22.

6.5.7.4. ARE PROPERLY MARKED WITH EQUIPMENT, SERVICE OR FUNCTION IDENTIFICATION AS RELATED TO THE PROJECT AND ARE MARKED WITH PERTINENT SPECIFICATION PARAGRAPH NUMBER.

6.5.8. IF ANY ITEM IN THE SUBMITTAL IS "NOT ACCEPTABLE" FOR ANY REASON, IT AUTOMATICALLY VOIDS THE ENTIRE SET, AND A RE-SUBMIT OF ALL IS REQUIRED TO OBTAIN ACCEPTANCE OF ALL ITEMS.

6.5.9. IN

7. INSTALLATION
- 7.1. PROVIDE ALL EQUIPMENT FOR THIS CONTRACT NEATLY AND WITH WORKMANSHIP AS DEFINED BY A.C.C.A. "STANDARD 5 PRACTICES FOR GOOD WORKMANSHIP IN MECHANICAL CONSTRUCTION", LEVEL AND PLUMB, AND SECURELY SUPPORTED.
- 7.1.1. THE ENTIRE INSTALLATION, AND MANNER OF INSTALLATION SHALL MEET THE COMPLETE SATISFACTION OF THE OWNER'S REPRESENTATIVE OR IT SHALL BE REMOVED AND REWORKED AS DIRECTED BY THE OWNER'S REPRESENTATIVE AS INCLUDED IN THE CONTRACT AMOUNT.
- 7.2. CONTRACTOR SHALL CONSULT FIRE PROTECTION, HVAC, ELECTRICAL AND STRUCTURAL PLANS (WHERE APPLICABLE) IN ALL INSTANCES BEFORE INSTALLING WORK SO THAT WORK WILL NOT INTERFERE WITH THOSE BRANCHES.
- 7.2.1. IN THE EVENT OF A CONFLICT, CONTRACTOR SHALL REPORT TO THE OWNERS REPRESENTATIVE AT ONCE AND DO NO FURTHER WORK UNTIL A SATISFACTORY ARRANGEMENT IS DECIDED UPON.
- 7.2.2. ANY WORK DONE, OR EQUIPMENT PLACED IN POSITION, BY CONTRACTOR THAT CREATES A CONFLICT IN VIOLATION HEREOF, SHALL BE READJUSTED TO THE SATISFACTION OF THE OWNER'S REPRESENTATIVE AT THE EXPENSE OF THE CONTRACTOR. THE DECISION OF THE OWNERS REPRESENTATIVE SHALL BE FINAL IN REGARD TO CHANGES DUE TO CONFLICTING CONDITIONS.
- 7.2.3. CONTRACTOR SHALL COMPLETE WORK OR ANY PART THEREOF AT SUCH TIME AS MAY BE DESIGNATED BY THE OWNER, SO THAT IT CAN BE USED FOR TEMPORARY OR PERMANENT USE AND SUCH USE OF THE SYSTEM SHALL NOT BE CONSTRUED AS AN ACCEPTANCE OF SAME BY OWNER.
- 7.2.4. VERIFY VOLTAGE AND ALL ELECTRICAL REQUIREMENTS OF PLUMBING EQUIPMENT AND SYSTEMS WITH DIV. 26 CONTRACTOR PRIOR TO ORDERING.
- 7.2.5. COORDINATION OF SPACE REQUIREMENTS WITH RESPECT TO DIVISION 22 SHALL BE PERFORMED SUCH THAT:
- 7.2.5.1.NO EQUIPMENT, PIPING OR DUCTWORK, OTHER THAN ELECTRICAL, SHALL BE INSTALLED WITHIN 42" OF SWITCHBOARDS OR PANELBOARDS.
- 7.3. TWO SETS OF PLUMBING DRAWINGS SHALL BE PROVIDED AS RECORD DRAWINGS WHICH SHALL BE SEPARATE, CLEAN, COPIES RESERVED FOR THE PURPOSE OF SHOWING A COMPLETE PICTURE OF THE WORK AS ACTUALLY INSTALLED.
- 7.3.1. DRAWINGS SHALL ALSO SERVE AS WORK PROGRESS REPORT SHEETS AND THE PLUMBING CONTRACTOR SHALL MAKE ANY NOTATIONS, NEAT AND LEGIBLE THEREON DAILY AS WORK PROCEEDS.
- 7.3.2. DRAWINGS SHALL BE AVAILABLE FOR INSPECTION AT ALL TIMES AND SHALL BE KEPT AT THE JOB AT A LOCATION DESIGNATED BY THE OWNERS REPRESENTATIVE.
- 7.3.3. AT THE COMPLETION OF THE WORK, THESE RECORD DRAWINGS SHALL BE SIGNED BY THE PLUMBING CONTRACTOR, DATED AND RETURNED TO THE OWNERS REPRESENTATIVE. FINAL PAYMENT OF CONTRACT WILL NOT BE MADE UNTIL RECEIPT AND REVIEW OF SAID DRAWINGS.
- 7.4. PROVIDE ALL ROOF, WALL AND FLOOR PENETRATIONS REQUIRED TO COMPLETE INSTALLATION AND REMOVAL OF WORK (MAINTAIN FIRE RATING OF EXISTING STRUCTURE).
- 7.4.1. ALL PENETRATIONS SHALL BE PATCHED AND FINISHED TO MATCH SURROUNDING SURFACES AND FINISHES.
- 7.4.2. ALL EQUIPMENT OR PIPE PENETRATIONS THROUGH WALL, ROOF AND FLOORS SHALL BE SLEEVED AND SEALED SO AS TO BE WATER AND AIR TIGHT.
- 7.4.3. ALL ROOF CUTS AND REPAIRS SHALL BE PERFORMED BY OWNER APPROVED ROOFING CONTRACTOR IN ORDER TO MAINTAIN ROOF WARRANTY.
- 7.5. CUTTING OF HOLES THROUGH CONCRETE AND MASONRY SHALL BE BY DIAMOND CORE CONCRETE SAW.
- 7.5.1. PNEUMATIC HAMMER, IMPACT ELECTRIC AND HAND OR MANUAL HAMMER TYPE DRILLS WILL NOT BE ALLOWED, EXCEPT AS PERMITTED BY THE ARCHITECT WHERE REQUIRED BY LIMITED WORKING SPACE.
- 7.5.2. LOCATE HOLES SUCH THAT THEY WILL NOT AFFECT STRUCTURAL SECTIONS SUCH AS RIBS OR BEAMS. HOLES SHALL BE LAID OUT WELL IN ADVANCE OF THE INSTALLATION. THESE LAYOUT LOCATIONS SHALL BE APPROVED BY THE ARCHITECT PRIOR TO DRILLING.
- 7.6. PROVIDE ALL EXCAVATION AND TRENCHING TO THE CORRECT ELEVATIONS, FOR THE INSTALLATION OF ALL PIPING, MANHOLES, CATCH BASINS AND FOUNDATIONS INCLUDED UNDER THIS DIVISION OF THE WORK.
- 7.7. PROVIDE ALL BACKFILL IN STRICT ACCORDANCE WITH THE EXCAVATION AND BACKFILL SECTION OF DIVISION 1 SPECIFICATIONS.
- 7.7.1. EXCAVATE PIPE TRENCH. HAND DIG IN ALL AREAS WHERE EXISTING UTILITIES EXIST. HAND TRIM EXCAVATION FOR ACCURATE PLACEMENT OF PIPE TO ELEVATIONS INDICATED.
- 7.7.1.1.PROVIDE NECESSARY SHEETING AND SHORING TO COMPLY WITH O.S.H.A. REGULATIONS FOR SAFETY IN THE TRENCH.
- 7.7.2. PLACE BEDDING MATERIAL AT TRENCH BOTTOM, LEVEL MATERIALS IN CONTINUOUS LAYER NOT EXCEEDING 6 INCHES COMPACTED DEPTH, COMPACT TO 95 PERCENT.
- 7.7.2.1.MAINTAIN OPTIMUM MOISTURE CONTENT OF BEDDING MATERIAL TO ATTAIN REQUIRED COMPACTION DENSITY.
- 7.7.3. BACKFILL MATERIAL SHALL BE CLEAN EARTH FILL COMPOSED OF SAND, SLIGHTLY SILTY SAND, SAND AND ROCK, CRUSHED ROCK, OR AN APPROVED COMBINATION THEREOF.
- 7.7.3.1.THE BACKFILL MATERIAL SHALL HAVE NO MORE THAN 12 PERCENT PASSING THE NUMBER 200 SIEVE UNLESS APPROVED BY THE OWNER AND AHJ.
- 7.7.3.2.WHEN TRENCHES ARE CUT IN PAVEMENTS OR AREAS TO BE PAVED, COMPACTION AS DETERMINED BY ASSHTO SPECIFICATION T-180 (MODIFIED), SHALL BE NOT LESS THAN 98 PERCENT OF MAXIMUM DENSITY. ALL OTHER LOCATIONS SHALL BE 95 PERCENT.
- 7.7.3.3.DENSITY TESTS FOR CONFORMANCE TO THE COMPACTION REQUIREMENTS SHALL BE MADE BY TESTING LABORATORY SELECTED BY THE OWNER AT THE EXPENSE OF THE CONTRACTOR. TESTS SHALL BE MADE EVERY 100 FEET OR LESS, MINIMUM TWO TESTS PER SITE. IF ANY TEST RESULTS ARE UNSATISFACTORY,
- 7.7.3.4.THE CONTRACTOR SHALL RE-EXCAVATE AND RE-COMPACT THE BACKFILL AT HIS EXPENSE UNTIL THE DESIRED COMPACTION IS OBTAINED.
- 7.7.4. PAVEMENT OR ROADWAY SURFACES CUT OR DAMAGED SHALL BE REPLACED IN EQUAL OR BETTER CONDITION THAN THE ORIGINAL, INCLUDING STABILIZATION, BASE COURSE, SURFACE COURSE, CURB AND GUTTER, OR OTHER APPURTENANCES.
- 7.7.4.1.OPEN CUTS THROUGH ROADWAY SECTIONS SHALL BE REPAIRED USING NON-SHRINKABLE FILL MATERIAL AS APPROVED BY THE OWNER AND AHJ. THE MATERIALS OF CONSTRUCTION AND METHOD OF INSTALLATION SHALL RECEIVE PRIOR APPROVAL FROM THE OWNER AND AHJ.
- 7.7.4.2.WHERE EXISTING PAVEMENT IS REMOVED, THE SURFACING SHALL BE MECHANICALLY SAW CUT PRIOR TO TRENCH EXCAVATION, LEAVING A UNIFORM AND STRAIGHT EDGE WITH MINIMUM DISTURBANCE TO THE REMAINING ADJACENT SURFACING.

- 7.7.4.3.THE WIDTH OF THE PAVEMENT REMOVAL SHALL BE THE MINIMAL NECESSARY TO ALLOW FOR INSTALLATION OF THE NEW UTILITIES.
- 7.7.4.4.IMMEDIATELY FOLLOWING BACK-FILLING AND COMPACTION, AN ASPHALTIC SURFACE SHALL BE APPLIED TO THE CUT AREAS. THIS SURFACING SHALL PROVIDE A SMOOTH TRAFFIC SURFACE WITH THE EXISTING ROADWAY.
- 7.7.4.5.THE PAYING WORK SHALL BE GUARANTEED FROM DEFECTS IN WORKMANSHIP, INCLUDING SETTLING, FOR TWO YEARS.
- 7.8. ALL WIRING SHALL BE IN CONDUIT. THE USE OF E.N.T., BX, NM, ETC. OR PRE-MANUFACTURED CABLE ASSEMBLIES OR ALUMINUM WIRE WILL NOT BE PERMITTED.
- 7.9. VERIFY FINAL LOCATIONS FOR ROUGH-INS WITH SHOP DRAWING SUBMITTALS, FIELD MEASUREMENTS, AND WITH REQUIREMENTS OF THE ACTUAL EQUIPMENT TO BE CONNECTED PRIOR TO ROUGH-IN.
- 7.9.1. REFER TO EQUIPMENT SPECIFICATIONS AND DIMENSIONS SPECIFIED FOR ROUGH-IN REQUIREMENTS.
- 7.10. PROVIDE ALL CUTTING AND PATCHING REQUIRED FOR INSTALLATION OF PLUMBING WORK.
- 7.10.1.PATCH EXISTING FINISHED AND DISTURBED NEW FINISHED SURFACES AND BUILDING COMPONENTS USING NEW MATERIALS MATCHING EXISTING MATERIALS & EXPERIENCED INSTALLERS. INSTALLERS' QUALIFICATIONS REFER TO THE MATERIALS AND METHODS REQUIRED OR THE SURFACE AND BUILDING COMPONENTS BEING PATCHED.
- 7.10.2.NO ADDITIONAL COMPENSATION WILL BE AUTHORIZED FOR CUTTING AND PATCHING WORK THAT IS NECESSITATED BY ILL- TIMED, DEFECTIVE OR NON-CONFORMING INSTALLATIONS.
- 7.11. PROVIDE NAMEPLATES ON ALL ITEMS OF PLUMBING EQUIPMENT THAT HAS A CONTROL PANEL AND ALL ITEMS WHERE A SUBMITTAL WAS MADE, EXCEPT FOR PIPING.
- 7.11.1.MAKE NAMEPLATES FROM WHITE ENGRAVING STOCK WITH BLACK LETTERS AND BLACK FOUR EDGE BEVEL.
- 7.11.2.WORDING SHALL SUITABLY DESCRIBE ITEMS SUCH AS PANEL ID, SOURCE OCPD AND VOLTAGE.
- 7.11.3.NAMEPLATES SHALL BE ATTACHED USING PROPER SIZE AND TYPE STAINLESS STEEL BOLTS, LOCK WASHERS AND NUTS. GLUE ON, TAPE ON, OR TAPE TYPE NAMEPLATES ARE NOT ACCEPTABLE FOR THE EQUIPMENT.
- 7.12. BEARINGS THAT REQUIRE LUBRICATION SHALL BE LUBRICATED IN STRICT ACCORDANCE WITH THE MANUFACTURER'S RECOMMENDATIONS.
- 7.13. ARRANGE FOR CHASES, SLOTS, AND OPENINGS IN OTHER BUILDING COMPONENTS DURING PROGRESS OF CONSTRUCTION, TO ALLOW FOR PLUMBING INSTALLATIONS.
- 7.14. COORDINATE THE INSTALLATION OF REQUIRED SUPPORTING DEVICES AND SLEEVES TO BE SET IN POURED-IN-PLACE CONCRETE AND OTHER STRUCTURAL COMPONENTS, AS THEY ARE CONSTRUCTED.
- 7.15. SEQUENCE, COORDINATE, AND INTEGRATE INSTALLATIONS OF PLUMBING MATERIALS AND EQUIPMENT FOR EFFICIENT FLOW OF THE WORK. GIVE PARTICULAR ATTENTION TO LARGE EQUIPMENT REQUIRING POSITIONING PRIOR TO CLOSING IN THE BUILDING.
- 7.16. THE EXACT MOUNTING HEIGHT OF DEVICES SHALL BE DETERMINED IN THE FIELD WITH RELATION TO ARCHITECTURAL DETAILS AND EQUIPMENT BEING SERVED.
- 7.16.1.IT SHALL BE THE RESPONSIBILITY OF CONTRACTOR TO COORDINATE OUTLET LOCATION WITH EQUIPMENT. OWNERS REPRESENTATIVE SHALL BE PERMITTED TO RELOCATE ANY OUTLET PRIOR TO INSTALLATION WITHIN A 15 FOOT LIMIT AT NO ADDITIONAL CHARGE IN CONTRACT PRICE.
- 7.16.2.ALL FASTENERS, HANGERS, AND METHODS OF HANGING EXPOSED WORK IN FINISHED AREAS SHALL BE SUBMITTED TO THE OWNERS REPRESENTATIVE FOR APPROVAL BEFORE INSTALLATION.
- 7.16.3.WHERE MOUNTING HEIGHTS ARE NOT DETAILED OR DIMENSIONED, INSTALL SYSTEMS, MATERIALS, AND EQUIPMENT TO PROVIDE THE MAXIMUM HEADROOM POSSIBLE.
- 7.17. INSTALL PLUMBING EQUIPMENT TO FACILITATE SERVICING, MAINTENANCE, AND REPAIR OR REPLACEMENT OF EQUIPMENT COMPONENTS. AS MUCH AS PRACTICAL, CONNECT EQUIPMENT FOR EASE OF DISCONNECTING, WITH MINIMUM OF INTERFERENCE WITH OTHER INSTALLATIONS.
- 7.18. INSTALL SYSTEMS, MATERIALS, AND EQUIPMENT GIVING RIGHT-OF-WAY PRIORITY TO SYSTEMS REQUIRED TO BE INSTALLED AT A SPECIFIED SLOPE.
8. OWNER FURNISHED EQUIPMENT:
- 8.1. THE OWNER WILL PURCHASE THE FOLLOWING MATERIALS:
- 8.2. THE DIV. 22 CONTRACTOR SHALL RECEIVE, INSPECT, UNLOAD, STORE AND PROTECT THE OWNER FURNISHED MATERIAL AND EQUIPMENT; AND, AT THE APPROPRIATE TIME, TRANSPORT, RIG, ASSEMBLE, ERECT, INSTALL, HOOK UP, CONNECT, TEST, ADJUST AND OTHERWISE PLACE INTO FULL OPERATION THE OWNER FURNISHED MATERIAL AND EQUIPMENT, AS THOUGH THE EQUIPMENT HAD BEEN PROVIDED THROUGH HIS/HER OWN PURCHASE ORDER.
- 8.3. ALL PLUMBING AUXILIARY DEVICES, FITTINGS, HARDWARE, FASTENERS, ANCHORS, SUPPORTS, ETC. NOT FURNISHED BY THE EQUIPMENT SUPPLIER, BUT REQUIRED TO MAKE THE OWNER FURNISHED EQUIPMENT FULLY OPERATIONAL, SHALL BE FURNISHED AND INSTALLED BY THE DIV. 22 CONTRACTOR AND INCLUDED IN THE DIV. 22 CONTRACTOR SCOPE OF WORK.
- 8.4. THE CONTRACTOR SHALL BE PROVIDED WITH SHOP DRAWINGS AND PRODUCT DATA OF THE OWNER FURNISHED MATERIALS AND EQUIPMENT, AND SHALL VERIFY, BY STAMPING AND SIGNING THE DATA AND DRAWINGS AND RETURNING THEM TO THE OWNER, THAT THE ITEMS FURNISHED BY THE OWNER, FIT THE SPACES AND DIMENSIONS DESCRIBED IN, AND CONFORM TO THE SPIRIT AND INTENT OF THE CONTRACT DOCUMENTS. THE CONTRACTOR SHALL, WITHIN FIVE (5) WORKING DAYS OF RECEIPT OF SHOP DRAWINGS AND PRODUCT DATA, NOTIFY THE OWNER OF ANY DISCREPANCY OR INCOMPATIBILITY WITH THE CONTRACT DOCUMENTS, AND SHALL RETURN THE SHOP DRAWINGS TO THE OWNER APPROPRIATELY ANNOTATED.
- 8.5. WHERE EQUIPMENT SPECIFICATIONS INDICATE THAT A FACTORY-AUTHORIZED SERVICE ENGINEER OR TECHNICIAN SHALL OBSERVE INSTALLATION; TEST AND ADJUST; OR OBSERVE OR DIRECT TESTING, ADJUSTMENT, OR START-UP OF EQUIPMENT, SUCH SERVICES WILL BE CONTRACTED BY THE OWNER AS PART OF THE EQUIPMENT PURCHASE. THE CONTRACTOR SHALL ARRANGE FOR, SCHEDULE, AND COORDINATE SUCH FIELD SERVICES AS WORK INCLUDED IN THE CONTRACT. THE CONTRACTOR SHALL FURNISH ALL LABOR AND MATERIALS NECESSARY TO SUPPORT ALL SUCH FACTORY REPRESENTATIVE'S FIELD SERVICES.
- 8.6. THE CONTRACTOR AND VENDOR SHALL INSTRUCT THE OWNER'S TECHNICAL PERSONNEL ON ALL OWNER FURNISHED EQUIPMENT IN ACCORDANCE WITH SPECIFICATIONS.
- 8.6.1. THE OWNER WILL PROVIDE THE CONTRACTOR WITH SUFFICIENT SETS OF OPERATIONS AND MAINTENANCE MANUALS OF OWNER FURNISHED EQUIPMENT FOR INCLUSION INTO THE CONTRACTOR'S OPERATIONS AND MAINTENANCE MANUALS AS REQUIRED BY THE SPECIFICATIONS.
9. RESPONSIBILITY FOR LAYOUT OF PLUMBING SYSTEMS.

- 9.1. FURNISH AND INSTALL COMPLETE PLUMBING SYSTEM AS INDICATED ON THE DESIGN DRAWINGS AND AS OUTLINED WITHIN THESE SPECIFICATIONS. WORK SHALL INCLUDE BUT NOT BE LIMITED TO SELECTIVE DEMOLITION, THE FABRICATION AND/OR INSTALLATION OF THE SCHEDULED PLUMBING FIXTURES AND RELATED PIPING. THE COMPLETED PLUMBING SYSTEM SHALL MEET ALL APPLICABLE CODES AND STANDARDS.
- 9.2. ALL SOLDER OR BRAZE CONNECTIONS TO THE POTABLE WATER SYSTEM SHALL BE MADE WITH LEAD-FREE SOLDER.
- 9.3. PLUMBING DRAWINGS INDICATE FIXTURE TYPES, BUT NOT QUANTITIES. REFER TO PLUMBING PLANS AND ARCHITECTURAL PLANS TO DETERMINE REQUIRED NUMBER OF FIXTURES.
- 9.4. REFER TO ARCHITECTURAL DRAWINGS FOR EXACT FIXTURE LOCATION AND FIXTURE HEIGHTS.
10. WATER SUPPLY SYSTEM, REFER TO TABLES LOCATED IN CONSTRUCTION DOCUMENTS FOR PIPING AND MATERIALS USED.
11. INSULATION
- 11.1. SHALL BE AS MANUFACTURED BY OWENS-CORNING, MANVILLE, PITTSBURGH CORNING, ARMSTRONG, OR APPROVED EQUAL. INSULATION SUNDRIES AND ADHESIVES SHALL BE AS MANUFACTURED BY BENJAMIN FOSTER, CHILDERS, VIMASCO, OR APPROVED EQUAL.
- 11.2. ALL INSULATION SHALL BE SUITABLE FOR INSTALLATION IN A RETURN AIR PLENUM AND SHALL MEET THE CURRENT ENERGY CODE REQUIREMENTS.
- 11.3. ALL PIPE INSULATION SHALL BE SELECTED TO FIT SNUGLY AGAINST THE PIPING WITH NO VOIDS.
- 11.4. CONTRACTOR SHALL BE RESPONSIBLE FOR FURNISHING AND INSTALLING ADEQUATE AND PROPER INSULATION AND MOISTURE-SEAL IN A MANNER THAT WILL PERMANENTLY PREVENT THE ACCUMULATION OF ANY OBJECTIONABLE MOISTURE ON THE EXTERIOR OF AIR-CONDITIONING UNITS, REFRIGERANT-PIPING, CONDENSATE DRAIN PIPING, AIR DUCTS OR OTHER PARTS OF THE SYSTEM.
- 11.5. CONTRACTOR SHALL CORRECT THE CAUSE OF ANY CONDENSATION AND FULLY REPAIR, WITHOUT COST TO THE OWNER, ANY DAMAGES TO BUILDING SURFACES, FURNISHINGS OR EQUIPMENT CAUSED BY CONDENSATION FROM THIS SYSTEM, FOR THE FULL PERIOD OF THE GUARANTEE.
12. BACKFLOW PREVENTERS.
- 12.1. PROVIDE BACKFLOW PREVENTERS WHERE SHOWN AND/OR WHERE REQUIRED. BACKFLOW PREVENTERS SHALL BE REDUCED PRESSURE PRINCIPLE TYPE U.N.O.
- 12.2. PROVIDE INDIRECT WASTE FITTING AT BACKFLOW PREVENTER AND EXTEND WASTE PIPING FULL SIZE TO DRAIN.
- 12.3. THERE SHALL NOT BE A BYPASS ARRANGEMENT OF ANY KIND AROUND A BACKFLOW PREVENTER.
- 12.4. PROVIDE BACKFLOW PREVENTERS FOR POTABLE WATER CONNECTIONS TO NON-POTABLE PROCESSES, INCLUDING BUT NOT LIMITED TO:
- 12.4.1.COFFEE MACHINES.
- 12.4.2.OTHER SIMILAR APPLICATIONS.
13. VACUUM BREAKERS.
- 13.1. PROVIDE VACUUM BREAKERS WHERE SHOWN AND/OR WHERE REQUIRED.
- 13.2. FOR HOT WATER, PROVIDE BACKFLOW PREVENTER IN LIEU OF VACUUM BREAKER.
- 13.3. PROVIDE VACUUM BREAKER FOR ALL FIXTURES WITH THREADED ENDS, INCLUDING BUT NOT LIMITED TO:
- 13.3.1.HOSE BIBBS.
- 13.3.2.HYDRANTS.
- 13.3.3.OTHER HOSE-END FIXTURES CONNECTED TO POTABLE WATER.
13. SUPPLIES AND STOPS.
- 13.1. PROVIDE POLISHED CHROME PLATED BRASS SUPPLIES AND ANGLE STOPS FOR ALL FIXTURES. PLASTIC SUPPLIES AND STOPS ARE NOT ACCEPTABLE. PROVIDE SHUT OFF VALVES FOR WATER SERVICE TO ALL EQUIPMENT, UPSTREAM OF FIXTURE STOPS.
14. WATER HAMMER ARRESTORS.
- 14.1. PROVIDE WATER HAMMER ARRESTORS WHERE SHOWN, OR AT EACH GROUP OF FIXTURES, AT ENDS OF PRINCIPAL PIPING RUNS, AND AS REQUIRED.
- 14.2. AIR CHAMBERS ARE NOT ACCEPTABLE.
- 14.3. SIZE WATER HAMMER ARRESTORS SHALL BE BASED ON FIXTURE UNITS OR TWO FIXTURE UNITS PER GPM, WHICHEVER IS GREATER.
- 14.4. WATER HAMMER ARRESTORS SHALL COMPLY WITH PDI WH-201, GAS CHARGED PISTON TYPE, AS MANUFACTURED BY SIOUX CHIEF, ZURN, JOSAM, OR WADE.
15. CHLORINATION
- 15.1. BEFORE BEING PLACED IN SERVICE, ALL WATER DISTRIBUTION SYSTEMS SHALL BE FLUSHED CLEAN AND DISINFECTED. THE METHOD TO BE FOLLOWED SHALL BE THAT PRESCRIBED BY THE HEALTH AUTHORITY OR WATER PURVEYOR HAVING JURISDICTION OR, IN THE ABSENCE OF A PRESCRIBED METHOD, USE THE FOLLOWING PROCEDURE:
- 15.2. THE PIPE SYSTEM SHALL BE FLUSHED WITH CLEAN, POTABLE WATER UNTIL DIRTY WATER DOES NOT APPEAR AT THE OUTLETS. THE SYSTEM SHALL BE FILLED WITH A WATER / CHLORINE SOLUTION CONTAINING AT LEAST 50 PARTS PER MILLION OF CHLORINE, VALVED OFF, AND ALLOWED TO STAND FOR (24) HOURS. ALTERNATELY, A SOLUTION OF 200 PARTS PER MILLION AND A STANDING TIME OF (3) HOURS MAY BE USED. FLUSH THE SYSTEM WITH CLEAN POTABLE WATER UNTIL THE CHLORINE IS PURGED FROM THE SYSTEM. REPEAT THE PROCEDURE WHERE BACTERIAL CONTAMINATION REMAINS PRESENT IN THE SYSTEM.
16. CONNECTIONS TO MISCELLANEOUS EQUIPMENT.
- 16.1. ROUGH-IN AND CONNECT WATER, WASTE, AND VENT TO COMPLETE THE INSTALLATION OF EQUIPMENT LISTED ON THE DRAWINGS.
- 16.2. PLUMBING FIXTURES SHALL BE PROVIDED COMPLETE AS SHOWN ON THE DRAWINGS WITH ALL REQUIRED SUPPLY, WASTE, SOIL, AND VENT CONNECTIONS, TOGETHER WITH ALL FITTINGS, SUPPORTS, FASTENING DEVICES, COCKS, VALVES, AND TRAPS.
- 16.3. ALL FIXTURES SHALL HAVE STOP VALVES ON ALL WATER CONNECTIONS.
- 16.4. ALL EXPOSED METAL TRIM ON ALL FIXTURES SHALL BE POLISHED CHROMIUM PLATED.

- 16.5. ALL EXPOSED PIPES EXTENDING FROM WALL SHALL HAVE CHROMIUM PLATED BRASS ESCUTCHEON MOUNTED AGAINST WALL.
- 16.6. THE PLUMBING FIXTURES SHALL BE ROUGHED-IN ACCORDANCE WITH MANUFACTURER'S ROUGH-IN INFORMATION.
- 16.7. PROVISIONS FOR MOUNTING WALL FIXTURES SHALL BE MADE WHILE THE WALL IS BEING BUILT.
17. DRAINAGE SYSTEM, REFER TO TABLES LOCATED IN CONSTRUCTION DOCUMENTS FOR PIPING AND MATERIALS USED.
- 17.1. TRAP AND VENT ALL FIXTURES U.N.O.
- 17.2. EXTEND VENTS THROUGH THE BUILDING EXTERIOR AS REQUIRED BY CODE.
- 17.3. PROVIDE POLISHED CHROME PLATED CAST BRASS GROUND JOINT SWIVEL (ADJUSTABLE TYPE) P-TRAP WITH CLEAN OUT PLUG FOR ALL FIXTURES TRAPPED ABOVE GRADE. TRAP SHALL HAVE PROVISION FOR DISASSEMBLY TO ALLOW STRAIGHT-THROUGH CLEAN-OUT OPERATIONS.
- 17.4. WHERE INDIRECT CONNECTIONS ARE SHOWN OR REQUIRED, PROVIDE MANUFACTURER'S STANDARD AIR GAP FITTING OR OTHER PROVISION TO MAINTAIN AIR GAP EQUAL TO MINIMUM TWICE THE DIAMETER OF THE DRAINING PIPE, BUT NOT LESS THAN 2 INCHES.
- 17.5. DRAIN, WASTE AND VENT PIPING LOCATED IN FIRE RATED WALL ASSEMBLIES AND RETURN AIR PLENUMS SHALL BE SERVICE-WEIGHT CAST IRON WITH SPIGOT FITTINGS.
- 17.6. STORM DRAINAGE REFER TO TABLES LOCATED IN CONSTRUCTION DOCUMENTS FOR PIPING ANS MATERIALS USED.
- 17.7. BASE ELBOW AND FIRST THREE FEET OF THE BOTTOM OF THE STORM WATER RISER SHALL BE CAST IN CONCRETE AND ATTACHED TO THE BUILDING SLAB USING THREADED RODS. ALTERNATELY, USE ASPHALT COATED EXTRA HEAVY CAST IRON BASE ELBOW AND FIRST THREE FEET, THEN TRANSITION TO PVC.
18. INSULATION
- 18.1. SHALL BE AS MANUFACTURED BY OWENS-CORNING, MANVILLE, PITTSBURGH CORNING, ARMSTRONG, OR APPROVED EQUAL. INSULATION SUNDRIES AND ADHESIVES SHALL BE AS MANUFACTURED BY BENJAMIN FOSTER, CHILDERS, VIMASCO, OR APPROVED EQUAL.
- 18.2. ALL INSULATION SHALL BE SUITABLE FOR INSTALLATION IN A RETURN AIR PLENUM AND SHALL MEET THE CURRENT ENERGY CODE REQUIREMENTS.
- 18.3. ALL PIPE INSULATION SHALL BE SELECTED TO FIT SNUGLY AGAINST THE PIPING WITH NO VOIDS.
- 18.4. CONTRACTOR SHALL BE RESPONSIBLE FOR FURNISHING AND INSTALLING ADEQUATE AND PROPER INSULATION AND MOISTURE-SEAL IN A MANNER THAT WILL PERMANENTLY PREVENT THE ACCUMULATION OF ANY OBJECTIONABLE MOISTURE ON THE EXTERIOR OF AIR-CONDITIONING UNITS, REFRIGERANT-PIPING, CONDENSATE DRAIN PIPING, AIR DUCTS OR OTHER PARTS OF THE SYSTEM.
- 18.5. CONTRACTOR SHALL CORRECT THE CAUSE OF ANY CONDENSATION AND FULLY REPAIR, WITHOUT COST TO THE OWNER, ANY DAMAGES TO BUILDING SURFACES, FURNISHINGS OR EQUIPMENT CAUSED BY CONDENSATION FROM THIS SYSTEM, FOR THE FULL PERIOD OF THE GUARANTEE.
19. CLEAN OUTS
- 19.1. PROVIDE REQUIRED CLEANOUTS IN ALL DRAINAGE PIPING, WHETHER OR NOT SHOWN ON THE DRAWINGS. INCLUDE CLEAN-OUTS FOR THE FOLLOWING SYSTEMS:
- 19.1.1.SANITARY WASTE.
- 19.1.2.STORM DRAIN.
- 19.1.3.CONDENSATE DRAIN (A/C CLEAR WATER WASTE).
- 19.1.4.OTHER GRAVITY DRAINAGE SYSTEMS.
- 19.2. PROVIDE CLEANOUTS WHERE INDICATED ON DRAWINGS, AND WHERE REQUIRED, AT CHANGES IN DIRECTION, AT BASE OF DOWNSPOUTS AND AT BASE OF SOIL AND WASTE STACKS.
- 19.3. REQUIRED CLEANOUT SPACING.
- 19.4. MINIMUM 100 FOOT INTERVALS FOR ALL INDOOR AND OUTDOOR PIPING, SIZES 6 INCH AND SMALLER.
- 19.5. WHERE ONE OR MORE CHANGE OF DIRECTION GREATER THAN 45-DEGREES OCCURS WITHIN 40-FOOT OF DEVELOPED LENGTH OF PIPE THE FIRST CLEANOUT SHALL SERVE AS THE CLEANOUT FOR ALL CHANGES OF DIRECTION WITHIN 40-FOOT DOWNSTREAM FOR ALL INDOOR PIPING, ALL SIZES.
- 19.6. PROVIDE TWO-WAY CLEANOUT AT EACH SANITARY PIPING POINT OF BUILDING EXIT.
- 19.7. FOR INSTALLATION OF EXTERIOR CLEAN OUTS, CONSTRUCT MIN 12 IN. X 12 IN. X 4 IN. DEEP SQUARE CONCRETE CROWN WITH CHAMFERED EDGES, WITH TOP OF CROWN (2) IN. ABOVE GRADE.
- 19.8. CLEANOUTS WHICH WILL BE SUBJECTED TO VEHICULAR TRAFFIC LOADING AND ALL EXTERIOR CLEANOUTS SHALL BE PROTECTED WITH A TRAFFIC-DUTY SLEEVE, SEPARATE FROM THE CLEANOUT FERRULE, SUCH THAT THE VEHICLE LOAD SHALL NOT BEAR UPON THE CLEANOUT OR THE PIPING SYSTEM.
20. ABOVE AND UNDERGROUND PIPE SLOPE:
- 20.1. 2-1/2 INCH DIAMETER AND LESS SHALL BE INSTALLED WITH A FALL OF NOT LESS THAN 1/4 INCH PER FOOT.
- 20.2. 3 INCH DIAMETER OR LARGER SHALL BE INSTALLED WITH A FALL NOT LESS THAN 1/8 INCH PER FOOT.
- 20.3. A/C CONDENSATE DRAIN PIPING SHALL SLOPE AT MINIMUM 1/8 INCH PER FOOT, REGARDLESS OF SIZE.
- 20.4. GREASE WASTE SHALL SLOPE AT MINIMUM 1/8 INCH PER FOOT REGARDLESS OF SIZE.
21. FIXTURES REFER TO TABLES LOCATED IN CONSTRUCTION DOCUMENTS FOR MATERIALS USED.
- 21.1. FURNISH AS SCHEDULED ON THE DRAWINGS AND AS DESCRIBED IN THESE SPECIFICATIONS.
- 21.2. GENERAL REQUIREMENTS.
- 21.3. EXPOSED METAL TRIM ON ALL FIXTURES SHALL BE POLISHED CHROME PLATED.
- 21.4. HOSE BIBBS, HYDRANTS, AND FAUCETS WITH HOSE-END CONNECTIONS SHALL HAVE INTEGRAL VACUUM BREAKER.

- 21.5. PROVIDE FLOOR MOUNTED CARRIER FOR ALL WALL-HUNG FIXTURES, SUITABLE FOR SAFELY SUPPORTING A 250 LB. LOAD AT FIXTURE OUTSIDE EDGE.
- 21.6. FLOOR DRAINS AND FLOOR SINKS SHALL HAVE TRAP PRIMER CONNECTION, U.N.O. PROVIDE TRAP PRIMER VALVE AND ASSOCIATED U.G. COPPER TUBING TO EACH FLOOR DRAIN.
- 21.7. PROVIDE ADA APPROVED PADDED SEAMLESS WHITE SAFETY COVERS FOR WASTE AND SUPPLY PIPING OF HANDICAP FIXTURES.
- 21.8. PROVIDE OFFSET TAILPIECE FOR LAVATORIES.
- 21.9. PROVIDE ANTI-SCALD PRESSURE BALANCED TEMPERING VALVE FOR LAVATORIES AND SHOWER, UNLESS THE FIXTURE IS SERVED WITH TEMPERED (MIXED) SUPPLY WATER.
22. EXECUTION.
- 22.1. PLUMBING FIXTURES SHALL BE ROUGHED-IN ACCORDANCE WITH MANUFACTURERS ROUGH-IN INFORMATION. PROVISIONS FROM MOUNTING WALL FIXTURES SHALL BE MADE WHILE THE WALL IS BEING BUILT.
- 22.2. ALL EXPOSED METAL TRIM ON ALL FIXTURES SHALL BE POLISHED CHROME PLATED.
- 22.3. RUN PIPING CONCEALED WHERE POSSIBLE AND WITHIN WALLS FOR EQUIPMENT LOCATED AT WALLS.
- 22.4. PROVIDE ESCUTCHEONS AT ALL WALL PENETRATIONS, CHROME PLATED BRASS.
- 22.5. PROVIDE VENTS THROUGH ROOF WHERE SHOWN AND WHERE REQUIRED. COORDINATE WITH GENERAL CONTRACTOR FOR WATER PROOFING DETAILS, TO MAINTAIN WATERPROOF INTEGRITY OF ROOF SYSTEM.
- 22.6. INSTALLATION OF FLOOR DRAINS, CLEAN OUTS, ETC. IN EXISTING SLAB:
- 22.7. SAW CUT MIN. 18 IN. X 18 IN. SQUARE OPENING SO THAT ANCHOR FLANGES WILL BE SUPPORTED BY NEW CONCRETE. NEW CONCRETE TO FILL IN THE PATCH BY GENERAL CONTRACTOR.

HVAC GENERAL NOTES

1.

ALL DRAWINGS ARE CONCEPTUAL AND SCHEMATIC AND ARE INTENDED FOR USE AS A DESIGN/BUILD GUIDELINE. THE CONTRACTORS ARE RESPONSIBLE FOR VERIFYING ALL FIELD CONDITIONS AND ADJUSTING OR MODIFYING THE SPECIFIC ELEMENTS OF THEIR WORK AS REQUIRED TO MEET THE DESIGN INTENT. THE CONTRACTORS ARE RESPONSIBLE FOR THE FOLLOWING:

A.

COORDINATION WITH OTHER TRADES.

B.

PROVIDING ADDITIONAL DRAWINGS, CALCULATIONS AND OTHER DOCUMENTATION REQUIRED FOR THE BUILDING DEPARTMENT. THE MECHANICAL CONTRACTOR SHALL DOCUMENT THE INSTALLATION AND PROVIDE ALL TESTS REQUIRED TO SUBSTANTIATE CODE COMPLIANCE AS REQUIRED BY THE BUILDING DEPARTMENT AND LOCAL INSPECTOR. CONTRACTOR SHALL SUBMIT FINAL AS-BUILT DRAWINGS TO BUILDING DEPARTMENT FOR RECORD AT COMPLETION.
2.

MECHANICAL PLANS ARE DIAGRAMMATIC IN NATURE, NOT SHOWING EVERY ITEM IN EXACT LOCATION OR DETAIL. MEASUREMENTS AND LOCATIONS MUST BE FIELD VERIFIED AND COORDINATED WITH ARCHITECTURAL, HVAC, FIRE PROTECTION, STRUCTURAL, ELECTRICAL AND OTHER BUILDING DRAWINGS.
3.

THE CONTRACTOR SHALL BE RESPONSIBLE FOR COORDINATION OF ALL TRADES, LANDLORD REQUIREMENTS, CEILING HEIGHTS AND EXISTING STRUCTURAL CONDITIONS PRIOR TO FABRICATION OF ANY DUCTWORK OR ORDERING OF ANY EQUIPMENT.
4.

CONTRACTOR MUST FIELD VERIFY EXACT CONDITION PRIOR TO ANY DEMOLITION. COORDINATE EXISTING FIELD INSTALLATION WITH CAPTIVEAIRE PRIOR TO CONSTRUCTION. CONSULT ENGINEER AFTER CEILING DEMO IF INSTALLATION VARIES FROM DRAWING INTENT. VERIFY EXISTING DUCTWORK SYSTEM PRIOR TO BID.
5.

MECHANICAL CONTRACTOR TO FIELD VERIFY WITH STRUCTURE ALL DUCT ROUTING PRIOR TO FABRICATION.
6.

CONTRACTOR TO INCLUDE IN BID ALL COSTS TO MAKE FIELD COORDINATION AND ADJUSTMENT TO DUCTWORK FOR FIT INTO EXISTING STRUCTURE. CONTRACTOR SHALL VERIFY AND FIELD COORDINATE FINAL LOCATION OF MECHANICAL EQUIPMENT.
7.

CONTRACTOR SHALL SECURE AND PAY FOR ALL REQUIRED PERMITS AND INSPECTIONS AND PERFORM ALL TESTS CALLED FOR OR REQUIRED AS A PART OF HIS WORK. FURNISHED APPROVED CERTIFICATE OF FINAL INSPECTION, AND TURN OVER TO OWNER AT COMPLETION OF PROJECT.
8.

THE MECHANICAL CONTRACTOR SHALL BE RESPONSIBLE FOR ADMINISTERING ALL WARRANTIES ON EQUIPMENT WHICH HE INSTALLS. THIS INCLUDES ALL CONDENSERS, REFRIGERANT PIPES, AND OTHER ITEMS FURNISHED BY OTHERS AS WELL AS THOSE FURNISHED BY HIM.
9.

GARANTEE ALL HVAC WORK AND EQUIPMENT FOR A PERIOD OF ONE YEAR FROM DATE OF STORE OPENING. DURING THE ONE YEAR PERIOD, SERVICE SHALL BE PROVIDED TWENTY-FOUR HOURS PER DAY, SEVEN DAYS PER WEEK AND SHALL INCLUDE ALL SERVICE CALLS, PARTS, LABOR AND MAINTENANCE AT NO ADDITIONAL COST TO THE OWNER OR TENANT. ALL FILTERS SHALL BE CHANGED ONCE PER MONTH. IN ADDITION, ALL COMPRESSORS SHALL CARRY A FIVE YEAR WARRANTY FROM THE MANUFACTURER.
10.

ALL INSTALLATION OF THE MECHANICAL EQUIPMENT SHALL COMPLY WITH THE MANUFACTURER'S SPECIFICATION AND CLEARANCE REQUIREMENTS.
11.

ALL HVAC WORK SHALL BE IN ACCORDANCE WITH NFPA 90A, 90B, 96, 54 AND NEC 101. LIFE SAFETY CODE.
12.

INSTALLATION SHALL COMPLY WITH ALL LOCAL, STATE AND NATIONAL CODES, AND WITH LATEST ASHRAE PUBLICATIONS. WORK SHALL BE NEAT AND WORKMANSHIP SHALL BE ACCEPTABLE TO BUILDING STANDARDS.
13.

ALL WORK SHALL BE IN STRICT ACCORDANCE WITH STATE AND LOCAL CODES AND ORDINANCES AND THE NATIONAL ELECTRIC CODE.
14.

THE CONTRACTOR SHALL COOPERATE WITH OTHER TRADES. FURNISH WORKMANSHIP OF THE HIGHEST GRADE, BE RESPONSIBLE FOR ALL LAYOUT, POSITIONING AND FIELD MEASUREMENTS FOR A NEAT, ACCURATE AND COMPLETE HVAC INSTALLATION.
15.

FURNISH ALL LABOR, MATERIALS, TOOLS, INCIDENTALS AND DETAILS NECESSARY TO PROVIDE A COMPLETE HEATING, VENTILATING, AIR CONDITIONING SYSTEM. INCLUDE ANY LABOR AND MATERIAL NOT SPECIFICALLY MENTIONED, BUT NECESSARY TO PROVIDE A COMPLETE AND OPERATING SYSTEM. ALL WORK SHALL BE INSTALLED IN A PROFESSIONAL MANNER AND SHALL MEET ALL THE REQUIREMENTS OF THE STATE BUILDING CODE, CITY BUILDING CODE, SAFETY AND HEALTH CODES, NFPA CODES AND ALL OTHER APPLICABLE CODES AND REQUIREMENTS. ALL COSTS FOR SAID REQUIREMENTS SHALL BE INCLUDED IN THIS CONTRACTORS BID PRICE.
16.

DURING THE BIDDING PERIOD, EACH CONTRACTOR SHALL VISIT THE SITE TO DETERMINE CONDITIONS AFFECTING THE WORK. BIDS SHALL SERVE AS EVIDENCE OF KNOWLEDGE OF EXISTING CONDITIONS AND ANY MODIFICATIONS WHICH ARE REQUIRED TO MEET THE INTENT OF THE DRAWINGS AND SPECIFICATIONS. FAILURE TO VISIT THE SITE DOES NOT RELIEVE THE CONTRACTOR OF RESPONSIBILITY IN PERFORMANCE OF WORK REQUIRED CONDITIONS IN EVIDENCE THEREBY SHALL NOT BE JUSTIFICATION FOR ADDITIONAL COMPENSATION.
17.

THE CONTRACTOR SHALL PROVIDE SHOP DRAWINGS PRIOR TO PURCHASING OR INSTALLING EQUIPMENT AND SYSTEMS INDICATED ON THE CONTRACT DOCUMENTS. PRIOR TO THE SUBMITTAL THE CONTRACTOR SHALL VERIFY THAT ADEQUATE SPACE EXIST FOR THE SUBMITTED EQUIPMENT. SHOP DRAWINGS MUST BE REVIEWED BY THE ENGINEER AND ARCHITECT.
18.

ALL THE BARE METAL SURFACES SHALL BE PRIMED AND PAINTED TO PREVENT ANY RUST, INCLUDING, BUT NOT LIMITED TO, ANGLE FRAMING, UNIT SUPPORTS, INCLUDING HARDWARE, ETC. ANY PAINTING OF DUCTWORK SHALL BE VERIFIED WITH ARCHITECT.
19.

CONTRACTOR TO PROVIDE TENANT WITH AS-BUILT DRAWINGS OF ALL CHANGES OR MODIFICATIONS MADE IN THE FIELD, TO THE ORIGINAL SET OF CONSTRUCTION DOCUMENTS, FOR TURN-OVER TO THE ARCHITECT/ENGINEER UPON COMPLETION OF THE PROJECT. PROVIDE ALL EQUIPMENT SHOP DRAWINGS, INFORMATION ON CONTROL DEVICES, CONTROL WIRING DIAGRAMS AND OTHER PERTINENT INFORMATION AT COMPLETION OF PROJECT.
20.

IT IS THE CONTRACTOR'S RESPONSIBILITY TO ENSURE THAT THE MECHANICAL EQUIPMENT COMPONENTS ARE INSTALLED AT LOCATIONS AND ELEVATIONS WHICH MAKE THEM READILY ACCESSIBLE FOR ROUTINE MAINTENANCE WITHOUT REQUIRING ANY EXTRAORDINARY MEASURES.
21.

FIELD VERIFY THE EXACT LOCATION OF ALL EQUIPMENT WITH ARCHITECT/OWNER PRIOR TO INSTALLATION. INFORM OWNER OF ANY EQUIPMENT ITEMS THAT REQUIRE RELOCATION.
22.

CONTRACTOR SHALL VERIFY THAT ALL EQUIPMENT, AS SHOWN ON THESE DRAWINGS, WILL NOT CONFLICT WITH ANY DRAINS, VENTS, MECH. PIPING OF ANY KIND, ELECTRICAL, ETC.
23.

PROVIDE VIBRATION ISOLATION DEVICES AND FLEXIBLE CONNECTIONS TO ALL MOVING MACHINERY.
24.

CONTRACTOR IS RESPONSIBLE FOR COORDINATING BOX-OUT LOCATIONS FOR ALL DRYWALL MOUNTED AIR DEVICES WITH GENERAL CONTRACTOR AND CEILING FRAMING. CONTRACTOR SHALL COORDINATE ALL DUCT AND DIFFUSER LOCATIONS WITH LIGHTING LAYOUTS AS REQUIRED.
25.

THE CONTRACTOR SHALL ENGAGE AN INDEPENDENT AIR BALANCING AGENCY SUBSEQUENT TO THE APPROVAL OF THE OWNERS REPRESENTATIVE. THE T&B AGENCY CAN ONLY ACT AS HIS OWN REPORTING AGENCY IF SUITABLE INSTRUMENTS HEREINAFTER REQUIRED ARE DEMONSTRATED TO BE PART OF HIS NORMAL PROCEDURE TO THE SATISFACTION OF THE OWNERS REPRESENTATIVE. THE T&B AGENCY SHALL BE AABC OR NEBB CERTIFIED. CONTRACTOR SHALL PROVIDE LANDLORD WITH WATER AND AIR BALANCE REPORT.
26.

IT SHALL BE THE RESPONSIBILITY OF THIS T&B AGENCY TO PROVIDE THE LOCAL BUILDING DEPARTMENT AND OWNER WITH PROPER TEST & BALANCE DATA ON AABC OR NEBB FORMS.
27.

BUILDING AIR SYSTEMS SHALL BE BALANCED PER DATA INCLUDED ON THE DRAWINGS TO ACHIEVE RELATIVE AIR VOLUMES AS INDICATED ON THE DRAWINGS AND SCHEDULED HEREIN. REFER TO AIR FLOW DIAGRAM DETAIL.
28.

ALL EQUIPMENT FURNISHED SHALL FIT THE SPACE AVAILABLE WITH CONNECTIONS IN THE REQUIRED LOCATIONS AND WITH ADEQUATE SPACE FOR OPERATING AND SERVICING. THE DRAWINGS ARE GENERALLY DIAGRAMMATIC AND INDICATE THE INTENT OF THE INSTALLATION WHILE THE SPECIFICATIONS AND EQUIPMENT LIST DENOTE THE TYPE AND QUALITY OF MATERIAL AND WORKMANSHIP TO BE USED. THE DRAWINGS SHALL NOT BE SOLD FOR EXACT LOCATION. WHEN A CONFLICT EXISTS BETWEEN THE DRAWINGS AND THE SPECIFICATIONS, THE HIGHER AND/OR MORE COSTLY STANDARD WILL APPLY. THE CONTRACTOR SHALL PROMPTLY NOTIFY THE ENGINEER WHOSE DECISION SHALL BE FINAL. NO ALLOWANCE WILL BE MADE SUBSEQUENTLY IN THIS REGARD ON BEHALF OF THE CONTRACTOR AFTER AWARD OF THE CONTRACT.
29.

INSTALLATION OF EQUIPMENT SHALL COMPLY WITH EQUIPMENT MANUFACTURER'S INSTALLATION AND CLEARANCE REQUIREMENTS TO ALLOW FOR INSPECTION, SERVICE, REPAIR OR REPLACEMENT.
30.

CONTRACTOR IS RESPONSIBLE FOR COORDINATING DIFFUSERS/REGISTERS WITH ARCHITECTURAL CEILING TYPES.
31.

CONTRACTOR SHALL PERFORM A COMPLETE CERTIFIED TEST AND BALANCE OF EACH MECHANICAL SYSTEM INCLUSIVE OF EXHAUST FANS, SPLIT SYSTEM AIR HANDLING UNITS, ROOFTOP UNITS, VARIABLE AIR VOLUME BOXES, FAN COIL UNITS AND CEILING DEVICES. SEE SPECIFICATIONS FOR DETAILS AND PROVIDE THREE (3) WRITTEN COPIES OF THE COMPLETED TEST AND BALANCE REPORT TO THE OWNER FOR REVIEW.
32.

CONTRACTOR SHALL COMPLY WITH ALL LANDLORD'S TENANT CRITERIA. COORDINATE AND SCHEDULE ALL WORK WITH LANDLORD'S FIELD REPRESENTATIVE.
33.

AN INDEPENDENT TEST AND BALANCE REPORT FOR THE COOKING AREA IS REQUIRED BEFORE FINAL APPROVAL OF THE VENTILATION SYSTEM PER CURRENT MECHANICAL CODE.
34.

CAPTURE AND CONTAINMENT TEST WILL ALSO BE REQUIRED BEFORE FINAL APPROVAL OF THE VENTILATION SYSTEM PER CURRENT MECHANICAL CODE. PERFORM THIS TEST UNDER WITNESS OF THE BUILDING OFFICIAL. THIS WILL BE A VISUAL VERIFICATION OF THE PATH OF SMOKE AND VAPORS WITH THE COOKING EQUIPMENT UNDER THE HOOD AT OPERATING TEMPERATURE.
35.

GREASE DUCT TEST SHALL BE PERFORMED WITH THE PRESENCE OF A CODE OFFICIAL BEFORE APPROVAL OF GREASE DUCT SYSTEM. A LIGHT OR AN APPROVED EQUIVALENT TEST METHOD SHALL BE PERFORMED TO DETERMINE THAT ALL WELDED AND BRAZED JOINTS ARE AIR TIGHT, PER CURRENT CODE.
36.

ALL DUCTWORK SHALL MAINTAIN SYSTEM PRESSURE. THE AIR DISTRIBUTION COMPONENTS SHALL BE SEALED IN ACCORDANCE WITH SMACNA REQUIREMENTS. TWO INCH PRESSURE CLASS.
37.

ALL DUCTWORK SHALL HAVE AN APPROVED CLOSURE SYSTEM BASED ON LOCAL CODE REQUIREMENTS. ALL SEAMS SHALL BE PRESSURE TESTED BASED ON DUCT PRESSURE RATING.
38.

ALL NEW SUPPLY AND RETURN DUCTWORK SHALL BE GALVANIZED STEEL, EXTERNALLY INSULATED WITH 2 INCH THICK, 1 POUND DENSITY FIBERGLASS DUCTWRAP. INSTALLED R VALUE SHALL BE A MINIMUM OF 6. SEAL ALL JOINTS AND SEAMS PRIOR TO ADDING DUCTWRAP. SEAL WRAP WITH GLASS FABRIC AND MASTIC MEETING UL 181.
39.

RESTROOM EXHAUST DUCT SHALL BE MINIMUM 26 GA. GALVANIZED STEEL. PROVIDE RAIN CAP WITH INTEGRAL BIRD SCREEN, BACKDRAFT DAMPER, AND ACCESSORIES AS REQUIRED INCLUDING SPEED CONTROLLER CONCEALED AND ACCESSIBLE. EXHAUST FAN SHALL MAINTAIN 10' CLEARANCE FOR ANY OUTSIDE AIR INTAKE. ALL ROOF WORK SHALL BE DONE BY LANDLORD'S ROOFING CONTRACTOR AT TENANT'S EXPENSE UNLESS NOTED OTHERWISE.
40.

WHEN DUCTWORK IS INSTALLED IN ATTIC OR IN A NON CONDITIONED SPACE, EXTERNALLY INSULATE ALL SUPPLY AND RETURN AIR DUCTWORK IN UNCONDITIONED SPACES WITH JOHNS MANVILLE FIBERGLASS MICROLITE FIBER GLASS DUCT WRAP WITH VAPOR BARRIER AND AN AVERAGE INSTALLED "R" VALUE OF 8 OR PER CURRENT MECHANICAL CODE. WRAP INSULATION SHALL MEET NFPA 90A AND 90B, AND TO HAVE A FLAME SPREAD INDEX NOT MORE THAN 25 AND SMOKE DENSITY INDEX NOT TO EXCEED 50 WHEN TESTED AS A COMPLETE INSTALLATION. INSULATION MATERIAL SHALL BE RESISTANT TO MICROBIAL GROWTH.
41.

DUCT DIMENSIONS SHOWN ARE INSIDE NET DIMENSIONS, ADD TO SHEET METAL SIZE FOR INSULATION THICKNESS. HOLD DUCTWORK TIGHT TO UNDERSIDE OF STRUCTURE UNLESS OTHERWISE NOTED OR REQUIRED BY FIELD CONDITIONS. IT IS REQUIRED TO COORDINATE EXACT MOUNTING HEIGHT IN FIELD WITH SITE INVESTIGATION. SUPPLY, RETURN, OUTSIDE AIR AND RELIEF AIR DUCTS SHALL BE SHEET METAL AND BE EXTERNALLY INSULATED WITH OWENS CORNING TYPE 150 2" THICK, FOIL FACED FLEXIBLE FIBROUS GLASS BLANKET INSULATION WITH A MIN R-6.4 VALUE. EQUAL IS APPROVED. INSULATION WRAP SHALL BE SEALED WITH FAB AND MASTIC.
42.

DUCT SIZES MUST BE VERIFIED FOR CLEARANCES AT THE JOB SITE PRIOR TO FABRICATION. DIMENSIONS MAY BE CHANGED TO ACCOMMODATE CONSTRUCTION AS LONG AS EFFECTIVE CROSS-SECTIONAL AREA IS MAINTAINED. DUCT TRANSITIONS SHALL BE CONSTRUCTED WITH A SLOPE OF "1" TO "4". ALL DEVIATIONS FROM ORIGINAL CONTRACT DRAWINGS SHALL BE REVIEWED BY ENGINEER DURING THE SHOP DRAWING PROCESS.
43.

DUCT SIZES SERVING DIFFUSERS AND GRILLES ARE SAME SIZE AS DIFFUSER OR GRILLE NECK UNLESS NOTED OTHERWISE. TOTAL LENGTH OF FLEXIBLE DUCT RUN SHALL NOT EXCEED 5'-0" LONG AND SHALL MEET INSTALLATION AND MATERIAL REQUIREMENTS OF LOCAL CODES. EXTEND SHEETMETAL DUCT WITHIN 5'-0" OF THE AIR DEVICE FOR COMPLIANCE. FLEXIBLE DUCTWORK IS NOT PERMITTED FOR USE IN ANY PARTS OF THE RETURN OR EXHAUST AIR SYSTEMS. FLEXIBLE DUCTWORK SHALL NOT PASS THROUGH FIRE RATED CONSTRUCTION.
44.

INSTALL FLEX FULLY EXTENDED, DO NOT INSTALL IN THE COMPRESSED STATE OR USE EXCESS LENGTHS (USING MORE THAN NECESSARY TO GET FROM POINT A TO POINT B, AND LEAVING THE EXCESS CURLING AROUND). THIS WILL NOTICEABLY INCREASE FRICTION LOSSES". AVOID BENDING DUCTS ACROSS SHARP CORNERS OR INCIDENTAL CONTACT WITH METAL FIXTURES, PIPES OR CONDUITS. RADIUS AT CENTERLINE SHALL BE NO LESS THAN ONE DUCT DIAMETER. FLEX DUCT SUPPORT AT NO MORE THAN FOUR FOOT INTERVALS AND CLARIFIES THAT CONNECTIONS TO A MAIN DUCT OR AIR OUTLET ARE SUPPORTS. HANGER OR SADDLE MATERIAL IN CONTACT WITH THE FLEXIBLE DUCT
45.

ALL FLEXIBLE DUCTS SHALL BE SUPPORTED EVERY 4'-0" WITH 2" WIDE GALVANIZED STEEL BANDS. MINIMUM ONE PER EACH SECTION OF FLEXIBLE DUCT. MAXIMUM LENGTH OF FLEX DUCT SHALL BE 5'-0" LONG AND SHALL MEET INSTALLATION AND MATERIAL REQUIREMENTS OF LOCAL CODES.
46.

NO FLEXIBLE DUCTS SHALL PASS THROUGH FIRE WALLS, OR BE CONNECTED TO ANY METAL DUCT WITH-IN 5'-0" FROM EITHER SIDE OF THE FIREWALL.
47.

MAXIMUM LENGTH OF FLEX DUCTWORK SHALL BE 5 FEET LONG AND SHALL MEET INSTALLATION AND MATERIAL REQUIREMENTS OF LOCAL CODES.
48.

ALL DUCTWORK BEHIND RETURN AIR DEVICE PLENUMS SHALL BE PAINTED FLAT BLACK.
49.

ALL SUPPLY DUCT BENDS FROM THE VERTICAL TO HORIZONTAL AND ANGLED TURNS OF DUCTWORK SHALL HAVE TURNING VANES INSTALLED.
50.

PROVIDE SMOOTH TRANSITIONS AT EQUIPMENT AND AIR DEVICES TO MATCH CONNECTION SIZES. DUCTWORK SHALL BE SHEET METAL FABRICATED IN ACCORDANCE WITH ASHRAE GUIDE AND SMACNA MANUAL LATEST EDITIONS.
51.

IF NEW DUCTWORK IS REQUIRED, SIZE DUCTWORK AT 0.08" FRICTION PER 100 FT OF DUCT. FLEX SHALL BE SIZED 0.05" PER 100FT.
52.

PROVIDE ALL NECESSARY TRANSITIONS AND OFFSETS IN SUPPLY AND RETURN AIR DUCTWORK TO AVOID EXISTING STRUCTURE, WATER, GAS, SPRINKLER PIPING, OTHER DUCTWORK, OTHER TRADES, ETC. DUCTWORK SHALL BE INSTALLED AS HIGH AS CONDITIONS WILL ALLOW.
53.

WHERE CEILING SPACE IS NOT SUFFICIENT TO PERMIT TOP CONNECTION TO CEILING DIFFUSER WITH PROPER BEND RADIUS, FOR FLEXIBLE DUCTS, CONTRACTOR SHALL FABRICATE AND/OR PROVIDE AN ADAPTER BOX FOR DIFFUSER TO PERMIT SIDE CONNECTION TO FLEXIBLE DUCT.
54.

THE EQUIPMENT SHALL BE LOCATED TO ALLOW FOR EASY ACCESS FOR SERVICING, ADJUSTING OR MAINTENANCE AND SPACE FOR REMOVAL OF INTERNAL ASSEMBLIES. PROVIDE MINIMUM CLEARANCES FOR ALL EQUIPMENT PER THE MANUFACTURERS RECOMMENDATIONS.
55.

PROVIDE ACCESS TO ALL COMPONENTS REQUIRING PERIODIC INSPECTION AND SERVICE THAT ARE LOCATED WITHIN THE SPACE OR REQUIRE ACCESS THROUGH THE SPACE. LABEL ACCESS DOORS AND PANELS OR CEILING TILES UTILIZED FOR ACCESS WITH THE NAME OF THE HIDDEN COMPONENT(S). DEMONSTRATE ACCESS TO ALL HIDDEN COMPONENTS FOR THE FIELD REPRESENTATIVE PRIOR TO OCCUPANCY.
56.

ALL ACCESS DOORS REQUIRED IN GENERAL CONSTRUCTION ARE TO BE PROVIDED AND INSTALLED BY THE GENERAL CONTRACTOR. IT IS THE RESPONSIBILITY OF THE HVAC CONTRACTOR TO IDENTIFY SIZE, TYPE AND LOCATION OF SUCH DOORS FOR PROPER ACCESS TO ALL CONCEALED HVAC EQUIPMENT, VALVES AND OTHER RELATED EQUIPMENT. THE HVAC CONTRACTOR SHALL IDENTIFY THESE REQUIREMENTS ON A COORDINATED SHOP DRAWING PRIOR TO SYSTEM FABRICATION AND INSTALLATION.
57.

CONTRACTOR SHALL FURNISH AND INSTALL A COMPLETE TEMPERATURE CONTROL SYSTEM TO INCLUDE: PANELS, MODULES, RELAYS, WIRING, THERMOSTATS, SENSORS, DAMPERS, ACTUATORS AND ALL MISCELLANEOUS ITEMS AS REQUIRED TO FULFILL THE DESIGN INTENT AS INDICATED ON THE PLANS AND IN THE CODED NOTES. THERMOSTATS AND SENSORS SHALL BE LOCATED GENERALLY AS SHOWN BUT THEIR EXACT LOCATION SHALL BE FIELD COORDINATED TO AVOID INTERFERENCE WITH WALL MOUNTED WORK.
58.

PROVIDE ALL CONTROL EQUIPMENT, MOTOR STARTERS, RELAYS, LINE VOLTAGE CONTROLS, TRANSFORMERS, LOW VOLTAGE CONTROLS, AND DEVICES NECESSARY FOR THE COMPLETE OPERATION OF THE HEATING AND AIR CONDITIONING AND VENTILATING SYSTEM.
59.

ALL LOW VOLTAGE WIRING AND CONDUIT REQUIRED FOR MECHANICAL EQUIPMENT SHALL BE FURNISHED AND INSTALLED BY MECHANICAL CONTRACTOR.
60.

WIRE SMOKE DETECTORS TO BUILDING FIRE ALARM CONTROL PANEL, IF EXISTING, OR ELSE FURNISH AND INSTALL A COMPATIBLE REMOTE AUDIBLE/VISUAL ALARM DEVICE. FIELD VERIFY EXACT REQUIREMENTS. PER SIGNAL FROM SMOKE DETECTOR, UNIT SHALL SHUT DOWN AND ALL DAMPER SHALL CLOSE. CONTRACTOR SHALL TEXT SYSTEM, PRIOR TO TURNOVER, TO INSURE PROPER FUNCTION.
61.

SEE ELECTRICAL SPECIFICATIONS FOR SMOKE DETECTOR WIRING.
62.

IN THE EVENT OF FAN SHUT DOWN, ALL DUCT MOUNTED DETECTORS SHALL REMAIN IN OPERATION.
63.

PROVIDE ALL FANS AND ROOFTOP UNITS WITH RELAYS TO SHUT DOWN WHEN FIRE ALARM IS INITIATED. COORDINATE LOCATION WITH THE ELECTRICAL CONTRACTOR FOR THE FIRE ALARM WIRING.
64.

MATERIALS EXPOSED WITHIN PLENUMS SHALL BE NONCOMBUSTIBLE OR SHALL HAVE A FLAME SPREAD INDEX OF NOT MORE THAN 25 AND A SMOKE-DEVELOPED INDEX OF NOT MORE THAN 50 WHEN TESTED IN ACCORDANCE WITH ASTM E 84.
65.

PROVIDE VOLUME BALANCING DAMPER AT ALL NINETY-DEGREE DUCT TAKE-OFFS. THIS ALSO APPLIES TO TAKE-OFFS TO DIFFUSERS OR REGISTERS LOCATED DIRECTLY UNDER DUCTS.
66.

PROVIDE ELBOWS OR TEES WITH TURNING VANES FOR ALL CHANGES OF DUCT DIRECTION. PROVIDE SPLITTER DAMPERS WITH LOCKING QUADRANTS IN ALL TEES.
67.

PROVIDE MANUAL BALANCING DAMPERS AS REQUIRED TO PROPERLY BALANCE EACH INDIVIDUAL AIR DISTRIBUTION SYSTEM. THE FOLLOWING MINIMUM STANDARDS SHALL GOVERN THE LOCATION OF BALANCING DAMPERS: ALL SUPPLY, RETURN, AND EXHAUST MAIN BRANCHES FROM TRUNKS, EACH SPLIT, AND ALL SUB-BRANCHES FROM MAINS SHALL INCORPORATE BALANCING DAMPERS.
68.

ALL BALANCING DAMPER MUST BE ACCESSIBLE, CONTRACTOR MUST FIELD COORDINATE PRIOR TO CONSTRUCTION AND BID. PROVIDE REMOTE CABLE CONTROLS IF CONFLICT PERSIST. COORDINATE WITH AIR BALANCING CONTRACTOR. (TYPICAL ALL).
69.

ALL ROOFTOP EQUIPMENT TO BE SET LEVEL AND PLUMB.
70.

DUCT-TO-EXHAUST FAN CONNECTIONS SHALL BE FLANGED AND GASKETED AND BOLTED TO THE INLET OF THE FAN FOR SIDE-INLET UTILITY FANS; AND SHALL BE FLANGED, GASKETED AND BOLTED TO THE INLET AND OUTLET OF THE FAN FOR IN-LINE FANS. GASKET AND SEALING MATERIALS SHALL BE RATED FOR CONTINUOUS DUTY AT A TEMPERATURE OF NOT LESS THAN 1500°F (816°C).
71.

REFRIGERANT PIPE SIZING AND ROUTING DESIGN SHALL BE THE RESPONSIBILITY OF THE CONTRACTOR. CONTRACTOR SHALL TAKE INTO ACCOUNT LENGTH OF RUN AND FOR IN-LINE PIPES, GASKET AND PIPING TO ENSURE ALL LOSSES ARE ACCOUNTED FOR.
72.

ALL CEILING MOUNTED EQUIPMENT MUST BE SUPPORTED DIRECTLY FROM BUILDING STRUCTURE WITH COMBINATION SPRING AND NEOPRENE-IN-SHEAR HANGERS AND ROD. PROVIDE SUPPLEMENTARY
- STEEL AS REQUIRED TO ADEQUATELY SUPPORT THE LOAD.
73.

MECHANICAL EQUIPMENT, APPLIANCES AND SUPPORTS THAT ARE EXPOSED TO WIND SHALL BE DESIGNED AND INSTALLED TO RESIST THE WIND PRESSURES ON THE EQUIPMENT AND THE SUPPORTS AS DETERMINED IN ACCORDANCE WITH THE BUILDING CODE. ROOF MOUNTED MECHANICAL UNITS AND SUPPORTS SHALL BE SECURED TO THE STRUCTURE. THE USE OF WOOD "SLEEPERS" SHALL NOT BE PERMITTED.
74.

THE BUILDING OR STRUCTURE SHALL NOT BE WEAKENED BY THE INSTALLATION OF MECHANICAL SYSTEMS. THE CONTRACTOR SHALL VERIFY WITH STRUCTURAL THAT WEIGHT OF FAN SUPPORTED ON THE BOTTOM CORD OF THE ROOF TRUSSES CAN SUPPORT THE LOADS OF THE CEILING FANS.
75.

MECHANICAL CONTRACTOR TO FIELD VERIFY EXACT LOCATION OF ALL ROOF TOP EQUIPMENT PRIOR TO CUTTING ROOF OPENINGS. ALL OPENINGS SHOULD FALL BETWEEN JOIST.
76.

IDENTIFY ALL ROOF MOUNTED EQUIPMENT WITH STORE NAME AND UNIT NUMBER USING PERMANENT WEATHER PROOF 2" HIGH DIE-CUT LETTERS.
- Sheet List Table
- | Sheet Number | Sheet Title |
|--------------|---------------------------|
| M0.0 | GENERAL NOTES AND LEGEND |
| M1.0 | MECHANICAL FLOOR PLAN |
| M2.0 | MECHANICAL ROOF PLAN |
| M4.0 | MECHANICAL SCHEDULES |
| M5.0 | MECHANICAL DETAILS |
| M6.0 | HOOD DRAWING |
| M6.1 | HOOD DRAWING |
| M6.2 | HOOD DRAWING |
| M6.3 | HOOD DRAWING |
| M6.4 | HOOD DRAWING |
| M8.0 | MECHANICAL SPECIFICATIONS |
| M8.1 | MECHANICAL SPECIFICATIONS |
| M8.2 | MECHANICAL SPECIFICATIONS |
- HVAC ABBREVIATIONS
- | | |
|---------|--------------------------------|
| A | AMPS |
| AFF | ABOVE FINISHED FLOOR |
| AMB | AMBIENT |
| BTU | BRITISH THERMAL UNIT |
| CFM | CUBIC FEET PER MINUTE |
| C.O. | CLEAN OUT |
| DIA | DIAMETER |
| (E) | EXISTING |
| EAT | ENTERING AIR TEMPERATURE |
| EDB | ENTERING DRY BULB TEMPERATURE |
| EXH | EXHAUST FAN |
| (E)RTU | EXISTING ROOF TOP UNIT FAN |
| ESP | EXTERNAL STATIC PRESSURE |
| EWB | ENTERING WET BULB TEMPERATURE |
| 'F | DEGREE FAHRENHEIT |
| FLA | FULL LOAD AMPS |
| PFM | FEET PER MINUTE |
| FT | FOOT OR FEET |
| HP | HORSEPOWER |
| kW | KILOWATT |
| HZ | HERTZ |
| IN | INCH |
| LAT | LEAVING AIR TEMPERATURE |
| LDB | LEAVING DRY BULB TEMPERATURE |
| LRA | LOCKED ROTOR AMPS |
| LWB | LEAVING WET BULB TEMPERATURE |
| MAX | MAXIMUM |
| MBH | 1,000 BTU'S PER HOUR |
| MCA | MAXIMUM CIRCUIT AMPACITY |
| MIN | MINIMUM |
| MOCP | MAXIMUM OVERCURRENT PROTECTION |
| (N) | NEW |
| N T S | NOT TO SCALE |
| OA | OUTSIDE AIR |
| PH | PHASE |
| RLA | RUNNING LOAD AMPS |
| RPM | REVOLUTIONS PER MINUTE |
| RTU | ROOF TOP UNIT |
| SA | SUPPLY AIR |
| SD | SMOKE DETECTOR |
| SHC | SENSIBLE HEAT CAPACITY |
| SQ. FT. | SQUARE FEET |
| T | THERMOSTAT |
| TA | TRANSFER AIR |
| UC | UNDERCUT |
- HOOD & GREASE EXHAUST DUCT NOTES
1.

EXHAUST HOODS SHALL BE CONSTRUCTED OF 16 GAUGE GALVANIZED OR 18 GA. STAINLESS STEEL WITH ALL EXTERNAL SEAMS AND JOINTS CONTINUOUSLY WELDED 100% LIQUID TIGHT. EXHAUST HOODS SHALL MEET OR EXCEED THE REQUIREMENTS OF NFPA 96, ALL LOCAL CODES AND SHALL BEAR THE NSF SEAL OF APPROVAL. SEE CAPTIVEAIRE HOOD DRAWING FOR INFORMATION.

2.

ALL LIGHTS USED IN THE HOODS SHALL BE U.L. LISTED FOR CANOPY HOOD USE AND OF THE INCANDESCENT TYPE AND SHALL BE WIRED TO COME ON THRU A SWITCH LOCATED ON THE HOOD FACE.

3.

THE EXHAUST HOODS SHALL HAVE ALL STAINLESS STEEL BAFFLE FILTERS AND SHALL HAVE A FIRE ACTUATED DAMPER IN THE MAKE-UP AIR COLLAR.

4.

THE EXHAUST HOODS SHALL HAVE PREPARED AUTOMATIC U.L. FIRE EXTINGUISHING SYSTEMS FOR PROTECTION OF THE EXHAUST PLENUM & DUCT AND COOKING SURFACES. FIRE CONTROL CABINETS SHALL ALSO BE PROVIDED AS SHOWN AND SHALL HAVE MICRO SWITCHES FURNISHED AS REQUIRED FOR EQUIPMENT SHUT OFF. THERE SHALL BE A MANUAL PULL STATION NEAR THE EXIT DOOR AND MINIMUM OF 10'-0" FROM THE HOOD. (MUST BE FLUSH MOUNTED, CONDUIT RUN IN THE WALL).

5.

THE SUPPLY FAN SWITCHES, 40 VA TRANSFORMERS, SUPPLY & EXHAUST FAN STARTERS, THERMAL OVERLOADS AND MECHANICAL GAS VALVE SHALL BE FURNISHED BY THE HOOD MANUFACTURER, AND SUPPLY FAN SWITCHES SHALL BE MOUNTED ON THE HOOD FACES. THE 40 VA TRANSFORMERS AND FAN STARTERS SHALL BE MOUNTED IN THE FIRE CONTROL CABINETS AND THE MECHANICAL GAS VALVE SHALL BE INSTALLED AS SHOWN ON THE PLUMBING DRAWINGS.

6.

ALL EXHAUST COLLARS AND EXHAUST DUCTWORK ARE SIZED TO MAINTAIN BETWEEN 1500 AND 2000 FPM EXHAUST AIR VELOCITY. ALL GREASE EXHAUST DUCTWORK SHALL BE CONSTRUCTED AND INSTALLED IN ACCORDANCE WITH NFPA-96. GREASE EXHAUST DUCTWORK SHALL HAVE ALL SEAMS, JOINTS AND PENETRATIONS CONTINUOUSLY WELDED LIQUID TIGHT.

7.

ALL HORIZONTAL RUNS OF GREASE EXHAUST DUCT SHALL SLOPE BACK TOWARD THE HOOD AT A SLOPE OF 1/4" PER FOOT IF NOT LISTED. FOLLOW MANUFACTURER LISTING AND INSTALLATION REQUIREMENTS IF LISTED. GREASE DUCT, PROVIDE A RESIDUE TRAP AT THE BASE OF EACH VERTICAL RISER.

8.

PROVIDE U.L. LISTED CLEANOUTS IN GREASE EXHAUST DUCTWORK AT A MINIMUM OF 10'-0" INTERVALS, AT EACH CHANGE OF DIRECTION AND AT EACH RESIDUE TRAP.

9.

THE DISCHARGE OF THE GREASE EXHAUST FANS SHALL BE A MINIMUM OF 10'-0" FROM ANY OUTSIDE AIR INTAKE.

10.

ALL GREASE EXHAUST DUCTWORK SHALL HAVE STANDARD OR RADIUS ELBOWS.

11.

THIS CONTRACTOR SHALL BE RESPONSIBLE FOR ALL WARRANTIES.

A.)

STAINLESS STEEL HOODS AS SPECIFIED ABOVE WITH FIRE PROTECTION SYSTEMS, CONTROLS, STARTERS, FIRE CONTROL CABINETS CLOSURE STRIP AND MECH. GAS VALVE.

B.)

ALL HOODS, HOOD SUPPLY AND EXHAUST FANS WITH ROOF CURBS. PROVIDE ALL HOOD EXHAUST FANS WITH GREASE GUARDS SUPPLIED BY CAPTIVEAIRE.

C.)

WIRING BETWEEN HOODS, FANS AND FIRE SYSTEM - INSTALLED BY GC.

13.

KITCHEN HOOD TEST AND BALANCE REPORT SHALL BE SUBMITTED TO BUILDING DEPARTMENT PRIOR TO FINAL INSPECTION.

14.

UPON ACTIVATION OF ANY FIRE EXTINGUISHING SYSTEM FOR A COOKING OPERATION, ALL SOURCES OF FUEL AND ELECTRIC POWER THAT PRODUCE HEAT TO ALL EQUIPMENT REQUIRING PROTECTION BY THAT SYSTEM SHALL AUTOMATICALLY SHUT OFF. ACTIVATION OF THE AUTOMATIC FIRE EXTINGUISHING SYSTEM MUST IMMEDIATELY SHUT OFF GAS AND ELECTRIC SUPPLY TO ALL APPLIANCES UNDER THE PROTECTED HOOD. THE PLUMBING CONTRACTOR SHALL PROVIDE A MASTER SOLENOID VALVE IN GAS LINE TO DISCONNECT ALL GAS APPLIANCES. MANUAL GAS AND ELECTRIC RESETS ARE REQUIRED.

15.

ALL REMOTE MANUAL OPERATING DEVICES SHALL BE IDENTIFIED AS THE "HAZARD PROTECTED" PROVIDE PLAQUE AND SIGN AS REQUIRED BY LOCAL JURISDICTION.

16.

ONE PLENUM NOZZLE SHALL BE PROVIDE FOR EVERY 10 FEET OF HOOD. REFER TO HOOD DRAWINGS FOR EXACT LOCATION.

17.

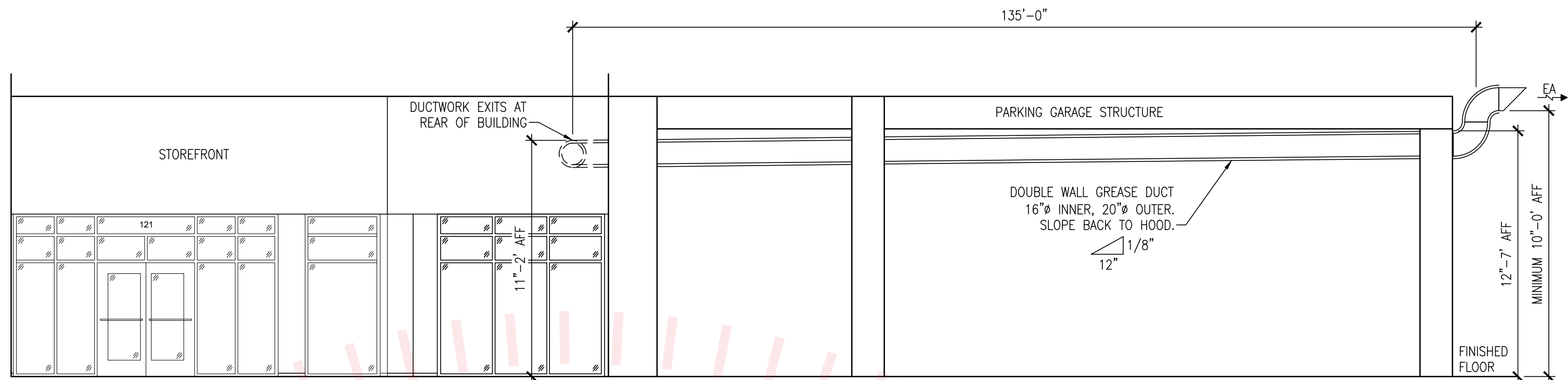
THE HOOD INSTALLING CONTRACTOR SHALL PROVIDE THE LATEST SYSTEM MANUAL AS PROVIDED BY THE MANUFACTURER TO VERIFY THE SYSTEM INSTALLATION.

18.

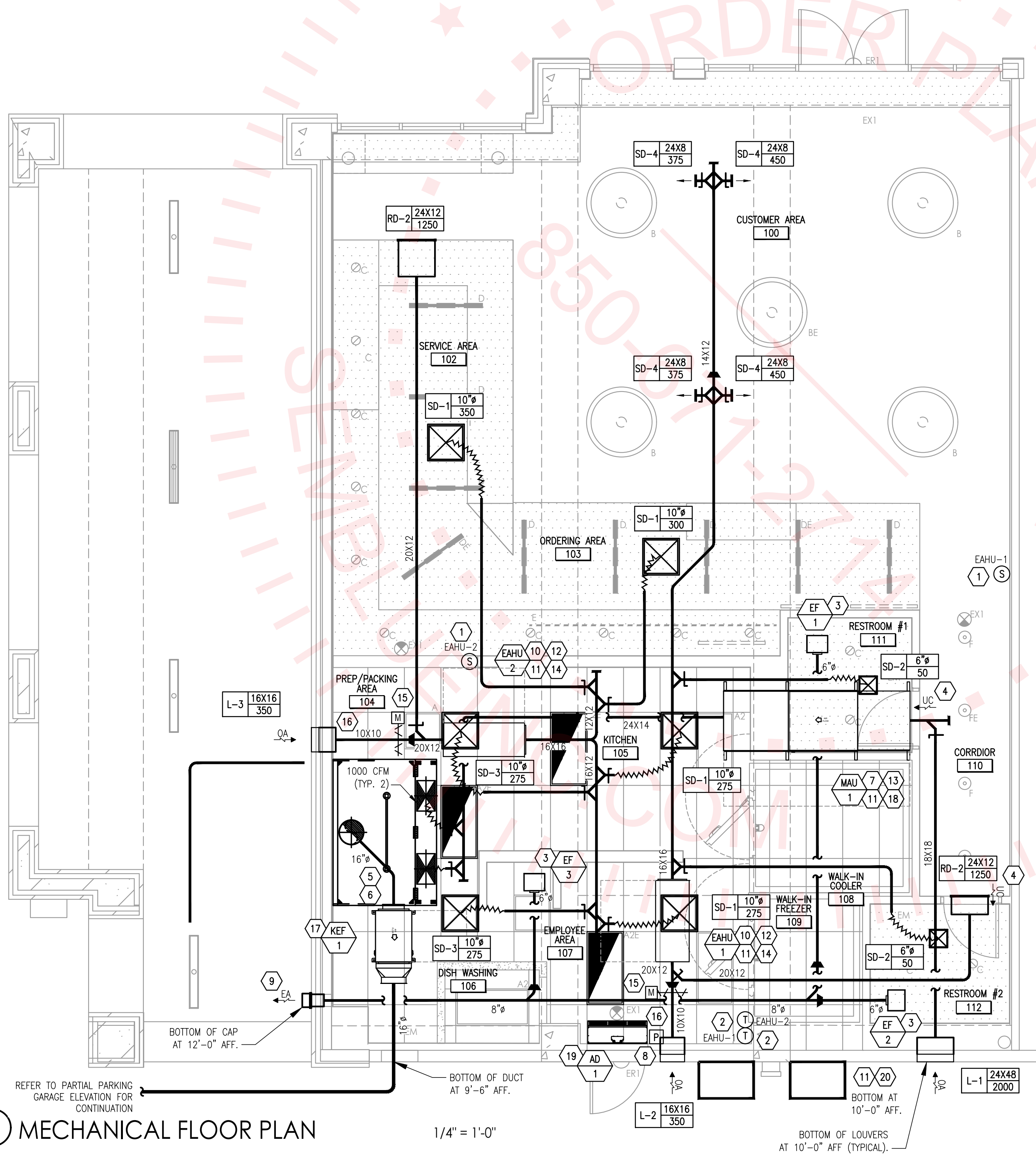
A PERFORMANCE TEST SHALL BE CONDUCTED UPON COMPLETION AND BEFORE FINAL APPROVAL OF THE INSTALLATION OF A VENTILATION SYSTEM SERVING COMMERCIAL COOKING APPLIANCES. THE TEST SHALL VERIFY THE RATE OF EXHAUST AIRFLOW REQUIRED BY CODE, MAKEUP AIR FLOW REQUIRED CODE AND PROPER OPERATION AS REQUIRED BY LOCAL CODE. THE PERMIT HOLDER SHALL THE NECESSARY TEST EQUIPMENT AND DEVICES REQUIRED.

19.

THE PERMIT HOLDER SHALL VERIFY CAPTURE AND CONTAINMENT PERFORMANCE OF THE EXHAUST SYSTEM. THIS FIELD TEST SHALL BE CONDUCTED WITH ALL APPLIANCES UNDER THE HOOD AT OPERATING TEMPERATURES, WITH ALL SOURCES OF OUTDOOR AIR PROVIDING MAKEUP AIR FOR THE HOOD OPERATING AND WITH ALL SOURCES OF RECIRCULATED AIR PROVIDING CONDITIONING FOR THE SPACE IN WHICH THE HOOD IS LOCATED OPERATING. CAPTURE AND CONTAINMENT SHALL BE VERIFIED VISUALLY BY OBSERVING SMOKE OR STEAM PRODUCED BY ACTUAL OR SIMULATED COOKING, SUCH AS WITH SMOKE CANDLES, SMOKE PUFFERS AND SIMILAR MEANS.
- NOTE:
- EACH HOOD SHALL BEAR THE FOLLOWING APPROVALS: NSF #1362, SBCCI #8469, U.L. CLASSIFICATION #9166, NFPA #90A, 90B, 96-101
- HVAC SYMBOLS LEGEND
-
-
- FIELD VERIFY ALL CONDITIONS
- NOTE:
- AS NOTED IN THE SPECIFICATIONS, ALL WIRING LAYOUTS, PIPING LAYOUTS AND DUCT LAYOUTS ARE SCHEMATIC. EXACT LOCATIONS SHALL BE DETERMINED BY THE CONSTRUCTOR OF THE BUILDING AND SHALL BE VERIFIED AND COORDINATED IN THE FIELD. EACH TRADE CONTRACTOR SHALL VERIFY WITH THE GENERAL CONTRACTOR THAT HE HAS THOROUGHLY REVIEWED AND COORDINATED ALL LOCATIONS AND ROUTINGS WITH ALL OTHER TRADES PRIOR TO FABRICATION OF CONDUITS, DUCTS, OR PIPING, AND START OF INSTALLATION OF SAME (INCLUDING SPRINKLER PIPING WHEN PRESENT ON JOB). ANY INSTALLATION OR CONSTRUCTION CONFLICTS WHICH OCCUR IN THE FIELD SHALL BE RESOLVED BY THE TRADE CONTRACTOR TO THE SATISFACTION OF THE OWNER, ARCHITECT AND AT NO EXPENSE TO THE OWNER, ARCHITECT AND/OR GENERAL CONTRACTOR.
- THE CONTRACTOR SHALL CONTACT THE ARCHITECT, ENGINEER OR OWNER PRIOR TO BIDDING FOR INTERPRETATIONS AND CLARIFICATIONS OF THE DESIGN AND INCLUDE HIS BID ALL COSTS TO MEET THE DESIGN INTENT. CLARIFICATIONS MADE BY THE ARCHITECT, ENGINEER OR OWNER AFTER BIDDING WILL BE FINAL AND SHALL BE IMPLEMENTED AT CONTRACTORS COST.
- BIDDING CONTRACTORS SHALL HAVE A WORKING KNOWLEDGE OF LOCAL CODES AND ORDINANCES AND SHALL INCLUDE IN THEIR BIDS THE COSTS FOR ALL WORK INSTALLED IN STRICT ACCORDANCE WITH GOVERNING CODES, THE PLANS AND SPECIFICATIONS NOTWITHSTANDING. THE CONTRACTOR SHALL ALERT ARCHITECT, ENGINEER OR OWNER OF ANY APPARENT DISCREPANCIES BETWEEN GOVERNING CODES AND DESIGN INTENT.
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- PROJECT NO: 20230719
DATE: 04/12/2024
- M0.0
GENERAL NOTES AND
LEGEND
- CHECKED: DAK DRAWN: RAN



② PARTIAL PARKING GARAGE ELEVATION NOT TO SCALE



① MECHANICAL FLOOR PLAN

1/4" = 1'-0"

KEYED NOTES

- PROVIDE REMOTE TEMPERATURE SENSORS ON WALL AS SHOWN AT 66" AFF. COORDINATE EXACT LOCATION WITH TENANT CONSTRUCTION MANAGER AND DO NOT MOUNT ON EXTERIOR WALL. THE ENTIRE CONTROL SYSTEM SHALL BE PROVIDED COMPLETE IN EVERY RESPECT BY THE MECHANICAL CONTRACTOR. REMOTE SENSORS TO BE CONNECTED TO 7-DAY PROGRAMMABLE THERMOSTAT. WIRE TO HVAC EQUIPMENT PER MANUFACTURER'S PRINTED INSTRUCTIONS.
- CONTRACTOR SHALL PROVIDE A 7-DAY DIGITAL PROGRAMMABLE THERMOSTAT IN MOUNTED 48" AFF TO TOP OF DEVICE. MOUNT THERMOSTAT AT 46" IF ADA REQUIREMENTS APPLY. COORDINATE EXACT LOCATION WITH CONSTRUCTION MANAGER. THE ENTIRE CONTROL SYSTEM SHALL BE PROVIDED COMPLETE IN EVERY RESPECT BY THE MECHANICAL CONTRACTOR. TYPICAL OF (3).
- PROVIDE A CABINET STYLE EXHAUST FAN. INTERLOCK EXHAUST FAN PER THE FAN SCHEDULE. ROUTE DUCTWORK TO EXTERIOR WALL AS SHOWN.
- UNDERCUT DOOR 1" FOR AIR PASSAGE.
- DURAVENT LISTED, DOUBLE WALL GREASE EXHAUST DUCT ROUTED FROM HOOD THROUGH EXTERIOR WALL. PROVIDE WITH FIRE RATED PENETRATION. DUCTWORK TO RUN TO END OF GARAGE AND DISCHARGE MINIMUM 10' AWAY FROM OUTSIDE AIR INTAKES AND OPENINGS INTO THE BUILDING. REFER TO MECHANICAL DETAILS FOR ADDITIONAL INFORMATION. INSTALL TIGHT TO STRUCTURE. FIELD VERIFY ROUTING PRIOR TO BID AND FABRICATION. SLOPE BACK TOWARDS HOOD AT MINIMUM 1/8" PER FOOT PER MANUFACTURER REQUIREMENTS. SEE HOOD DRAWING FOR HOOD OPENING SIZE.
- GREASE DUCT CLEANOUTS LOCATED ON HORIZONTAL SECTIONS OF DUCTS SHALL BE SPACED NOT MORE THAN 20 FEET APART. THE CLEANOUTS SHALL BE LOCATED ON THE SIDE OF THE DUCT WITH THE OPENING NOT LESS THAN 1.5 INCHES ABOVE THE BOTTOM OF THE DUCT, AND NOT LESS THAN 1 INCH BELOW THE TOP OF THE DUCT. MINIMUM OPENING SHALL 12/12 INCHES. CONTRACTOR SHALL VERIFY ALL REQUIREMENTS TO MEET LOCAL CODES PRIOR TO BID AND INSTALLATION.
- MAKE-UP AIR DUCT ROUTED FROM INTAKE LOUVER TO MAKE-UP AIR UNIT. ROUTE DUCTWORK TO HOOD(S) MAKE-UP AIR OPENINGS AND TRANSITION AS REQUIRED. PROVIDE BALANCING DAMPER IN MAKEUP AIR DUCT IN ADDITION TO DAMPER AT HOOD. PLACE INTO DUCTWORK WHERE DAMPER CAN BE ACCESSED FOR BALANCING. FIELD VERIFY ROUTING PRIOR TO BID AND FABRICATION.
- A MANUAL ACTUATION DEVICE (PULL STATION) SHALL BE LOCATED AT OR NEAR A MEANS OF EGRESS FROM THE COOKING AREA A MINIMUM OF 10 FEET (3048 MM) AND A MAXIMUM OF 20 FEET (6096 MM) FROM THE KITCHEN EXHAUST SYSTEM. THE MANUAL ACTUATION DEVICE SHALL BE INSTALLED NOT MORE THAN 48 INCHES (1200 MM) NOR LESS THAN 42 INCHES (1067 MM) ABOVE THE FLOOR AND SHALL CLEARLY IDENTIFY THE HAZARD PROTECTED. THE MANUAL ACTUATION SHALL REQUIRE A MAXIMUM FORCE OF 40 POUNDS (178 N) AND A MAXIMUM MOVEMENT OF 14 INCHES (356 MM) TO ACTUATE THE FIRE SUPPRESSION SYSTEM.
- PROVIDE BROAN MODEL 643 EXHAUST VENT OR EQUIVALENT WITH BACKDRAFT DAMPER. VENT SHALL MAINTAIN 10" CLEARANCE FOR ANY OUTSIDE AIR INTAKE. MUST MEET LOCAL CODE REQUIREMENTS. FIELD VERIFY ALL ROUTING AND REQUIREMENTS PRIOR TO BID. SEAL ALL PENETRATIONS WEATHER TIGHT.
- AIR HANDLING UNIT TO BE RELOCATED TO LOCATION SHOWN ON PLANS. ENERGIZE UNIT AND CONFIRM PROPER OPERATION BEFORE COMMENCEMENT OF WORK. NOTIFY OWNER OF ANY DISCREPANCIES OR ISSUES. UNIT TO BE SUPPORTED FROM STRUCTURE ABOVE. INSTALL PER MANUFACTURER'S RECOMMENDATIONS AND PROVIDE WITH VIBRATION ISOLATORS. REFER TO MECHANICAL SCHEDULE.
- MAINTAIN MINIMUM SERVICE CLEARANCES AS REQUIRED BY THE EQUIPMENT MANUFACTURER'S RECOMMENDATIONS.
- EXISTING CONDENSATE LINES TO BE MODIFIED PER NEW LOCATION OF AIR HANDLING UNITS.
- PROVIDE NEW CONDENSATE LINE FROM MAKE UP AIR UNIT AND TIE INTO EXISTING CONDENSATE LINE.
- EXISTING REFRIGERANT LINES TO BE MODIFIED PER NEW LOCATION OF AIR HANDLING UNITS. REFRIGERANT LINES SHALL BE INSTALLED AND SIZED PER MANUFACTURER'S RECOMMENDATIONS. REFER TO ROOF PLAN.
- ROUTE RETURN DUCT FROM GRILLE TO UNIT AS SHOWN. MAKE FINAL CONNECTION TO AIR HANDLING UNIT AS REQUIRED.
- ROUTE OUTSIDE AIR DUCT FROM LOUVER TO PLENUM AS SHOWN ON PLAN. PROVIDE WITH MOTORIZED DAMPER. OUTSIDE AIR DAMPER SHALL OPEN WHEN UNIT IS ENERGIZED. REFER TO SEQUENCE OF OPERATION FOR ADDITIONAL INFORMATION.
- INLINE GREASE EXHAUST FAN SUPPORTED FROM STRUCTURE ABOVE. INSTALL PER MANUFACTURER'S RECOMMENDATIONS AND PROVIDE WITH VIBRATION ISOLATORS. REFER TO MECHANICAL SCHEDULE.
- MAKE UP AIR UNIT HUNG ABOVE CEILING, TIGHT TO STRUCTURE. UNIT TO BE SUPPORTED FROM UNDER STRUCTURE. PROVIDE WITH VIBRATION ISOLATORS.
- AIR CURTAIN WALL MOUNTED DIRECTLY ABOVE THE BACK DOOR, ON THE INSIDE OF THE BUILDING, FLUSH TO THE WALL.
- APPROXIMATE LOCATION OF REFRIGERATION EQUIPMENT CONDENSING UNIT. COORDINATE WITH FOOD SERVICE, ARCHITECTURAL, AND ELECTRICAL DRAWINGS. PROVIDE SUPPORT BRACKET AND MOUNT TO WALL, INSTALL PER MANUFACTURER'S REQUIREMENTS. COORDINATE EXACT LOCATION PRIOR TO BID. REFER TO MECHANICAL DETAILS FOR MORE INFORMATION.

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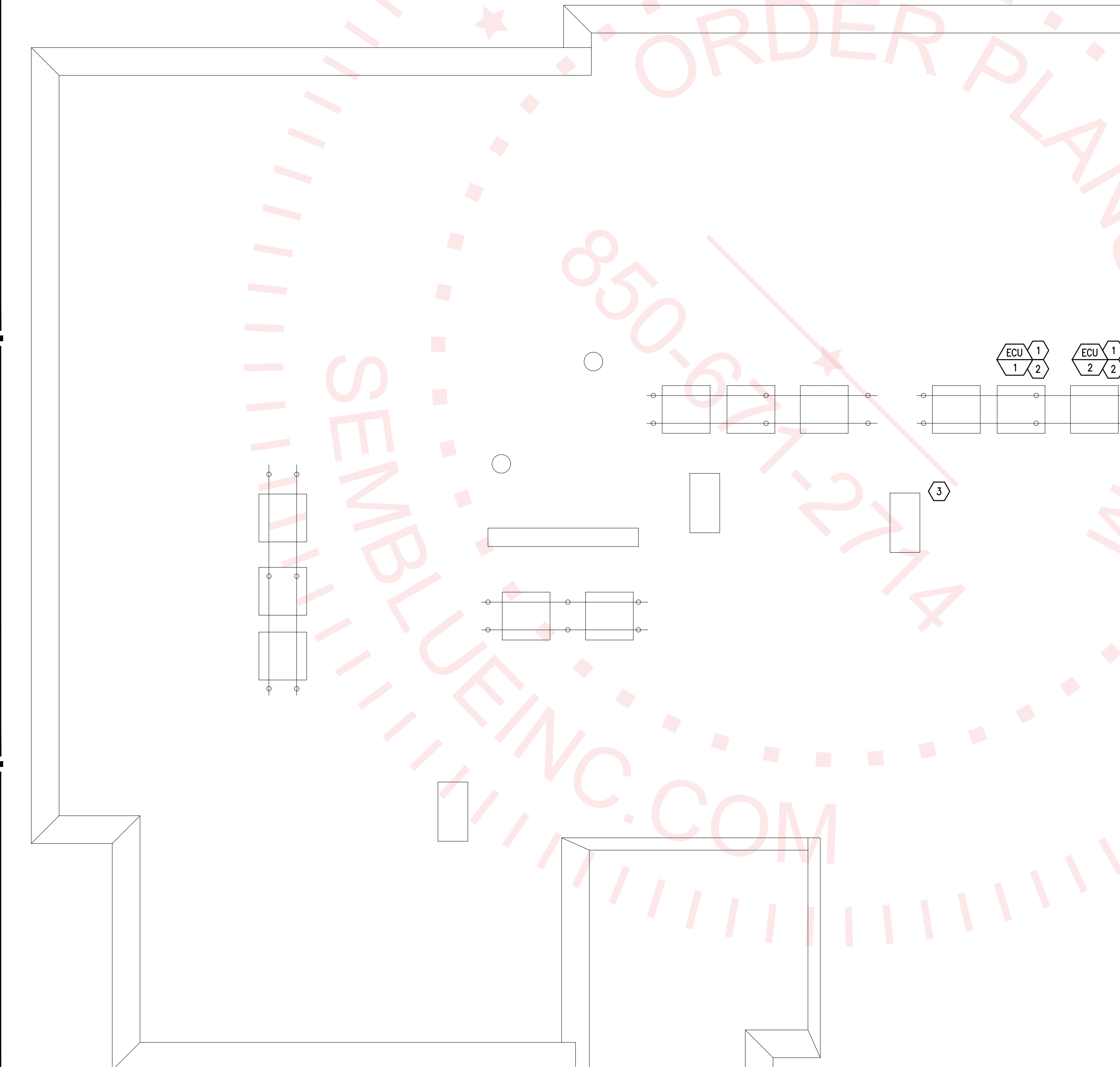
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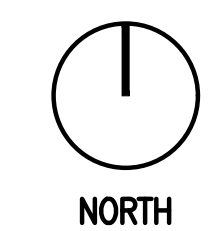
M1.0
MECHANICAL FLOOR PLAN

CHECKED: DAK DRAWN: RAN



① MECHANICAL ROOF PLAN

1/4" = 1'-0"



KEYED NOTES

- APPROXIMATE LOCATION OF EXISTING SPLIT SYSTEM CONDENSING UNITS FOR AIR HANDLING UNITS IN THE SPACE. FIELD VERIFY AND COORDINATE EXACT LOCATION.
- CONTRACTOR SHALL ENERGIZE CONDENSING UNITS PRIOR TO COMMENCEMENT OF WORK TO CONFIRM PROPER OPERATION. NOTIFY OWNER OF ANY DISCREPANCIES OR ISSUES.
- APPROXIMATE LOCATION OF EXISTING REFRIGERANT CHASES. CONTRACTOR SHALL MODIFY EXISTING REFRIGERANT LINES TO THE AIR HANDLING UNITS IN THE SPACE AS REQUIRED PER NEW WORK. REFRIGERANT LINES SHALL BE SIZED PER MANUFACTURER'S RECOMMENDATIONS.

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THERMOSTAT SCHEDULE					
MARK	SERVICE LOCATION	OCCUPIED		UNOCCUPIED	
		COOLING	HEATING	COOLING	HEATING
EAHU-1	DINING/ RESTROOMS	74	68	78	60
EAHU-2	KITCHEN	74	68	78	60
NOTES: 1. CONTRACTOR SHALL COORDINATE EXACT OPERATIONAL TIMES WITH OWNER/MANAGER PRIOR TO PROGRAMMING.					

FAN SCHEDULE												
TAG	MANUFACTURER	MODEL	AREA SERVED	LOCATION	CFM	ESP (IN)	MOTOR (HP)	FAN (RPM)	FLA	VOLTS/ PHASE	WEIGHTS (LBS)	NOTES
KEF-1	CAPTIVEAIRE	SIF15DD	KITCHEN	ABOVE CEILING	2400	1.500	2.00	1835	5.7	208V/3PH	250	A-D
MAU-1	CAPTIVEAIRE	A1-D.250-20D	KITCHEN	ABOVE CEILING	2000	0.750	1.50	1086	6.6	208V/3PH	565	A,C,D-E
EF-1	COOK	GC-148	RESTROOMS	CEILING	70	0.250	48W	1300	.42	115V/1PH	20	F,G
EF-2	COOK	GC-148	RESTROOMS	CEILING	70	0.250	48W	1300	.42	115V/1PH	20	F,G
EF-3	COOK	GC-148	MOP SINK	CEILING	50	0.250	48W	1200	.42	115V/1PH	20	F,H
NOTES: A. EXHAUST FANS, MAKE-UP AIR UNIT, AND ACCESSORIES TO BE FURNISHED THRU CAPTIVEAIRE, INSTALLED BY MECHANICAL CONTRACTOR. B. PROVIDE WITH HANGING SPRING ISOLATORS, AND UNI-STRUT. C. PROVIDE WITH FACTORY DISCONNECT SWITCH. D. PROVIDE WITH HOOD ON/OFF TOGGLE SWITCH, INTERLOCK TO CONTROL PANEL. E. PROVIDE WITH INDOOR HANGING OPTION, HANGING SPRING ISOLATORS, AND UNI-STRUT. F. MECHANICAL CONTRACTOR TO PROVIDE EXHAUST FANS WITH WIRED SPEED CONTROLLER CONCEALED AND IN AN ACCESSIBLE LOCATION. G. INTERLOCK WITH RESTROOM LIGHTS. H. FAN TO RUN CONTINUOUSLY.												

AIR DISTRIBUTION SCHEDULE										
TAG	SERVICE	MANUFACTURER & MODEL NO.	NECK SIZE	FACE SIZE	FRAME TYPE	PATTERN	DAMPER	MATERIAL	FINISH	NOTES
SD-1	SUPPLY	TITUS / TMS	SEE PLAN	24"X24"	LAY-IN	SEE PLAN	YES	ALUMINUM	WHITE	1-7
SD-2	SUPPLY	TITUS / TMS	SEE PLAN	12"X12"	LAY-IN	SEE PLAN	YES	ALUMINUM	WHITE	1-7
SD-3	SUPPLY	TITUS / PSS	SEE PLAN	24"X24"	LAY IN	SEE PLAN	YES	ALUMINUM	WHITE	1-7
SD-4	SUPPLY	TITUS / 250	SEE PLAN	SEE PLAN	SIDEWALL	SEE PLAN	YES	ALUMINUM	WHITE	1-2,4-5
RD-1	RETURN	TITUS / 350RL	SEE PLAN	SEE PLAN	SIDEWALL	SEE PLAN	YES	ALUMINUM	WHITE	2,4,6
NOTES: 1. PROVIDE WITH OPPOSED BLADE DAMPER AT AIR DEVICE. 2. MAX NC LEVEL 30. 3. PROVIDE SQUARE TO ROUND NECK ADAPTOR. 4. SEE ARCHITECTURAL DRAWINGS FOR COLOR AND FINISH. 5. PROVIDE 4-WAY AIR THROW PATTERN UNLESS OTHERWISE NOTED OR INDICATED. 6. PROVIDE INSULATED BACK ON ALL AIR DEVICES MIN. R6 MIN. 7. ALL BALANCING DAMPERS MUST BE ACCESSIBLE. CONTRACTOR TO FIELD COORDINATE PRIOR TO INSTALLATION.										

LOUVER SCHEDULE										
TAG	SERVICE	MANUFACTURER & MODEL NO.	WIDTH (INCH)	HEIGHT (INCH)	AIR FLOW (CFM)	FREE AREA (%)	DAMPER	MATERIAL	FINISH	NOTES
L-1	INTAKE	RUSKIN / EME520DD	24	48	SEE PLAN	41	NO	ALUM.	PER ARCH	1
L-2	INTAKE	RUSKIN / EME520DD	16	16	SEE PLAN	33	NO	ALUM.	PER ARCH	1
L-3	INTAKE	RUSKIN / EME520DD	16	16	SEE PLAN	33	NO	ALUM.	PER ARCH	1
NOTES: 1. BASED ON RUSKIN OR APPROVED EQUAL.										

OUTSIDE AIR CALCULATIONS						
ROOM NUMBER & NAME	AREA SQ. FT.	NO. OF PEOPLE	REQ'D OA CFM/PERSON	REQ'D OA CFM/ SQ FT	ZONE EFFECTIVENESS	TOTAL OSA
CUSTOMER AREA 100	715	16	7.5	0.18	0.8	310.88
SERVICE AREA 102	226	2	7.5	0.18	0.8	69.60
ORDERING AREA 103	166	2	7.5	0.18	0.8	56.10
PREP/PACKING AREA 104/KITCHEN 105/DISH WASHING 106/EMPLOYEE AREA 107	477	10	7.5	0.12	0.8	165.30
RESTROOMS 111 & 112	104	0	0	0	0	0.00
TOTALS	REQUIRED OUTSIDE AIR					602
	PROVIDED OUTSIDE AIR					700
NOTES: 1. ESTIMATED MAXIMUM OCCUPANCY AND REQUIRED OUTSIDE AIR BASED ON CURRENT EDITION IMC.						

AIR DOOR CURTAIN SCHEDULE								
TAG	MANUFACTURER/ MODEL	LOCATION	NOZZLE VELOCITY (FPM)	VOLTS/PHASE	AMPS	MOCp	WEIGHT (LBS)	NOTES
AD-1	BERNER/ CHD10-1036AA	BACK DOOR	2062	120/1PH/60HZ	3	15	50	A-C
NOTES: A. MOUNTED ABOVE BACK DOOR. B. INCLUDES FACTORY MOUNTED ON/OFF SWITCH. C. PROVIDE WITH DOOR SWITCH, INSTALLED BY MECHANICAL CONTRACTOR TO ENGAGE AIR CURTAIN WHEN THE DOOR IS OPENED AND SHUTS OFF WHEN THE DOOR IS CLOSED.								

CONTROL SEQUENCE OF OPERATIONS: SPLIT STSTEM

ELECTRIC HEATING/ELECTRIC COOLING SPLIT SYSTEM SEQUENCE OF OPERATION

DAY CYCLE - COOLING

- SUPPLY AIR FAN SHALL RUN CONTINUOUSLY.
- OUTSIDE AIR DAMPER SHALL BE IN MINIMUM POSITION.
- THERMOSTAT SHALL CYCLE CONDENSING COMPRESSOR(S) TO MAINTAIN ROOM SET TEMPERATURE.

DAY CYCLE - HEATING

- SUPPLY AIR FAN SHALL RUN CONTINUOUSLY.
- OUTSIDE AIR DAMPER SHALL BE IN MINIMUM POSITION.
- THERMOSTAT SHALL CYCLE ELECTRIC HEATER TO MAINTAIN ROOM SET TEMPERATURE.
- WHEN SET TEMPERATURE IS REACHED DAY CYCLE SHALL COMMENCE.

MORNING WARM-UP

- SUPPLY AIR FAN SHALL RUN CONTINUOUSLY.
- OUTSIDE AIR DAMPER SHALL BE IN CLOSED POSITION.
- THERMOSTAT SHALL CYCLE ELECTRIC HEATER TO ACHIEVE ROOM SET TEMPERATURE.
- WHEN SET TEMPERATURE IS REACHED DAY CYCLE SHALL COMMENCE.

NIGHT SETBACK

- OUTSIDE AIR DAMPER SHALL BE IN CLOSED POSITION.
- RETURN AIR DAMPER WILL BE IN 100% OPEN POSITION.
- THERMOSTAT SHALL CYCLE EITHER COOLING OR HEATING AND SUPPLY AIR FAN TO MAINTAIN ROOM SET TEMPERATURE.

SMOKE DETECTOR

- WHEN SMOKE DETECTOR IS ACTIVATED SUPPLY AND RETURN AIR FAN SHALL SHUTDOWN.
- FIRE ALARM SHALL BE SIGNALLED.
- SUPPLY AIR FAN SHALL BE MANUALLY RESET.

SPLIT SYSTEM SCHEDULE (ELECTRIC HEATING/ELECTRIC COOLING)		
MARK	EAHU-1	EAHU-2
MANUFACTURER	BRYANT	BRYANT
MODEL NUMBER	FB4C060	FB4C060
NOMINAL TONNAGE	5	5
AREA SERVED	DINING	KITCHEN
CONFIGURATION	HORIZONTAL	HORIZONTAL
SUPPLY AIR (CFM)	1750	1750
OUTDOOR AIR (CFM)	350	350
OUTDOOR AIR PERCENTAGE	20%	20%
ENTERING AIR TEMP. DB. / WB. °F	-	-
UNIT DIMENSIONS W X H X D (IN)	53-7/16" X 21-1/8" X 22-1/16"	53-7/16" X 21-1/8" X 22-1/16"
UNIT WEIGHT (LB)	175	175
TOTAL COOLING CAPACITY (BTU/H)	60	60
SENSIBLE COOLING CAPACITY (BTU/H)	40	40
TOTAL HEATING CAPACITY (KW)	11.3	11.3
HEATING DELTA T	28	28
VOLTS / PHASE	208V/1PH/60HZ	208V/1PH/60HZ
INDOOR UNIT MCA	76.3	76.3
INDOOR UNIT MOCp	80	80
CONDENSER	ECU-1	ECU-1
EER/IEER	-	-
MANUFACTURER	BRYANT	BRYANT
MODEL NUMBER	113APAD60	113APAD60
NOMINAL TONNAGE	5	5
LOCATION	ROOF	ROOF
UNIT DIMENSIONS W X H X D (IN)	31-3/16" X 28-15/16" X 31-3/16"	31-3/16" X 28-15/16" X 31-3/16"
UNIT WEIGHT (LB)	230	230
VOLTS / PHASE	208V/3PH/60HZ	208V/3PH/60HZ
OUTDOOR UNIT MCA	21.4	21.4
OUTDOOR UNIT MOCp	30	30
UNIT OPTIONS:	A-F	A-F
INCLUDED SYSTEM OPTIONS AHU/CU A. UNITS ARE EXISTING TO REMAIN AND HAVE NOT BEEN ENERGIZED IN 7+ YEARS. B. ENERGIZE UNITS BEFORE COMMENCEMENT OF WORK TO CONFIRM PROPER OPERATION. NOTIFY OWNER OF ANY DISCREPANCIES OR ISSUES. C. CLEAN AND REFURBISH TO LIKE NEW CONDITION. D. BALANCE AS REQUIRED PER NEW WORK. E. PROVIDE UNITS WITH SPRING ISOLATORS FOR CEILING MOUNTING. F. CONDENSING UNITS ARE POWERED SEPARATELY.		
AHU/CU NOTES 1. PROVIDE CONDENSATE DRAIN "P" TRAP MINIMUM 4" DEEP, OR TWICE THE TOTAL STATIC PRESSURE, WHICHEVER IS GREATER. 2. COORDINATE MANUFACTURER'S RECOMMENDED CLEARANCES AROUND UNIT. 3. ALTERNATE MANUFACTURES AND MODELS WILL BE REVIEW. THERE MAY BE ARCHITECTURAL STRUCTURAL, AND ELECTRICAL CHANGES RESULTING FROM THE ALTERNATES. THE COST OF IMPLEMENTING AND ENGINEERING THESE CHANGES SHALL BE BORNE BY THE MECHANICAL SUBCONTRACTOR.		
BUILDING DEPARTMENT NOTE: 1. INTERLOCK FIRE SYSTEM WITH MAKE UP AIR UNIT AND KITCHEN AIR HANDLING UNITS.		
* CONTRACTOR SHALL VERIFY EXACT ELECTRICAL CONNECTIONS, WIRE SIZES, BREAKERS, DISCONNECT, ETC. PRIOR TO ORDERING AND BID.		

BUILDING AIR BALANCE SCHEDULE		
MARK	OUTSIDE AIR	EXHAUST
KEF-1		-2400
EF-1		-70
EF-2		-70
EF-3		-50
EAHU-1	+350	
EAHU-2	+350	
MAU-1	+2000	
TOTAL:	2700	-2590
BUILDING POSITIVE AIR BALANCE	110	



9 SIDEWALL RETURN GRILLE DETAIL
NTS



FOR QUESTIONS, CALL THE
Central Florida Office
REGION 66
PHONE: (407) 682 - 0317
EMAIL: reg66@capliveair.com

PATENT NUMBERS
AC-PSP (UNITED STATES) - US PATENT 7963830 B2.
AC-PSP WALL (CANADA) - CA PATENT 2820509.
AC-PSP ISLAND (CANADA) - CA PATENT 2520330.

HOOD INFORMATION - JOB#6622389

HOOD NO	TAG	MODEL	MANUFACTURER	LENGTH	MAX COOKING TEMP	TYPE	APPLIANCE DUTY	DESIGN CFM/FT	TOTAL EXH CFM	EXHAUST PLENUM RISER(S)							TOTAL SUPPLY CFM	HOOD CONSTRUCTION	HOOD CONFIG	
										WIDTH	LENG	HEIGHT	DIA	CFM	VEL	SP			END TO END	ROW
1		5424 ND-2-PSP-F	CAPTIVEAIRE	8' 0"	450 DEG	I	MEDIUM	200	2400			4'	16"	1600	1718	-0.785"	2000	430 SS WHERE EXPOSED	ALONE	ALONE

HOOD INFORMATION

HOOD NO	TAG	FILTER(S)				LIGHT(S)				UTILITY CABINET(S)				FIRE SYSTEM	HOOD SYSTEM	HANGING PIPING WEIGHT
		TYPE	QTY	HEIGHT	LENGTH	EFFICIENCY @ 7 MICRONS	QTY	TYPE	WIRE GUARD	LOCATION	SIZE	FIRE SYSTEM	SIZE	ELECTRICAL MODEL #	SWITCHES QUANTITY	
1		CAPTRATE SOLID FILTER	5	20"	16"	85% SEE FILTER SPEC	2	RECESSED ROUND	NO						YES	541 LBS

HOOD OPTIONS

HOOD NO	TAG	OPTION									
		FIELD WRAPPER	18.00"	HIGH	FRONT, LEFT, RIGHT.						
1		BACKSPLASH	122.00"	HIGH X	132.00"	LONG	430 SS	VERTICAL.			
		RISER SENSOR	INSTALL	6IN	PLEN.						
		RIGHT VERTICAL END PANEL	27"	TOP WIDTH,	21"	BOTTOM WIDTH,	80"	HIGH	INSULATED	430 SS.	
		LEFT VERTICAL END PANEL	27"	TOP WIDTH,	21"	BOTTOM WIDTH,	80"	HIGH	INSULATED	430 SS.	

PERFORATED SUPPLY PLENUM(S)

HOOD NO	TAG	POS	LENGTH	WIDTH	HEIGHT	TYPE	RISER(S)				
							WIDTH	LENG	DIA	CFM	SP
1		Front	96'	14'	6'	MUA	12"	20"		1000	0.219"
						MUA	12"	20"		1000	0.219"

WALL-MOUNT UTILITY CABINET

HOOD NO	LOCATION	SIZE	TYPE	FIRE SYSTEM	ELECTRICAL	SWITCHES	WEIGHT
					SIZE	MODEL #	QUANTITY
1	WALL MNT	12"x54"x24"	TANK FS		40/4.0	SC-311110MA	1 LIGHT 1 FAN

SYSTEM DESIGN VERIFICATION (SDV)

IF ORDERED, CAS SERVICE WILL PERFORM A SYSTEM DESIGN VERIFICATION (SDV) ONCE ALL EQUIPMENT HAS HAD A COMPLETE START UP PER THE OPERATION AND INSTALLATION MANUAL. TYPICALLY, THE SDV WILL BE PERFORMED AFTER ALL INSPECTIONS ARE COMPLETE.

ANY FIELD RELATED DISCREPANCIES THAT ARE DISCOVERED DURING THE SDV WILL BE BROUGHT TO THE ATTENTION OF THE GENERAL CONTRACTOR AND CORRESPONDING TRADES ON SITE. THESE ISSUES WILL BE DOCUMENTED AND FORWARDED TO THE APPROPRIATE SALES OFFICE. IF CAS SERVICE HAS TO RESOLVE A DISCREPANCY THAT IS A FIELD ISSUE, THE GENERAL CONTRACTOR WILL BE NOTIFIED AND BILLED FOR THE WORK. SHOULD A RETURN TRIP BE REQUIRED DUE TO ANY FIELD RELATED DISCREPANCY THAT CANNOT BE RESOLVED DURING THE SDV, THERE WILL BE ADDITIONAL TRIP CHARGES.

DURING THE SDV, CAS SERVICE WILL ADDRESS ANY DISCREPANCY THAT IS THE FAULT OF THE MANUFACTURER. SHOULD A RETURN TRIP BE REQUIRED, THE GENERAL CONTRACTOR AND APPROPRIATE SALES OFFICE WILL BE NOTIFIED. THERE WILL BE NO ADDITIONAL CHARGES FOR MANUFACTURER DISCREPANCIES.

SPECIFICATION: CAPTRATE® GREASE-STOP® SOLID FILTER

THE CAPTRATE GREASE-STOP SOLID FILTER IS A SINGLE-STAGE FILTER FEATURING A UNIQUE S-Baffle DESIGN IN CONJUNCTION WITH A SLOTTED REAR Baffle DESIGN, TO DELIVER EXCEPTIONAL FILTRATION EFFICIENCY.

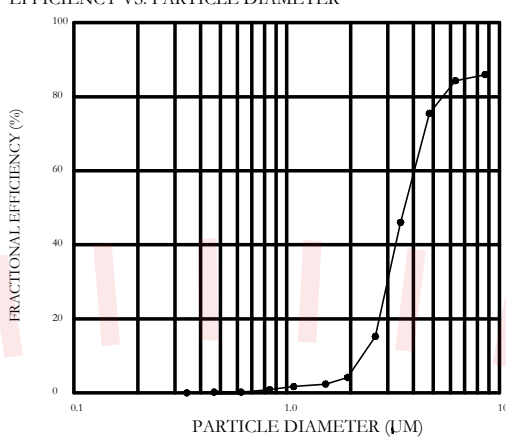
FILTER IS STAINLESS STEEL CONSTRUCTION, AND SIZED TO FIT INTO STANDARD 2-INCH DEEP HOOD CHANNEL(S).

UNITS SHALL INCLUDE STAINLESS STEEL HANDLES AND A FASTENING DEVICE TO SECURE THE TWO COMPONENTS WHEN ASSEMBLED.

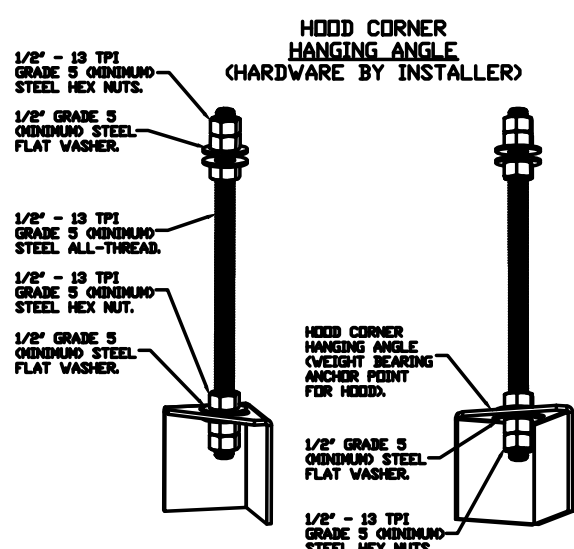
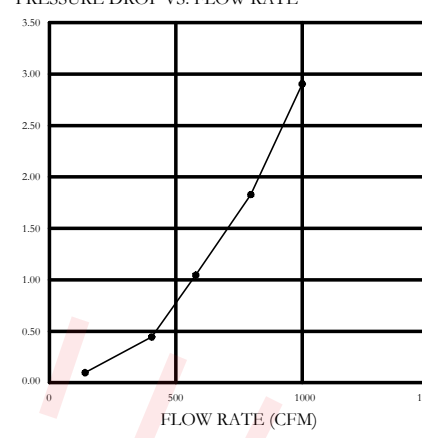
GREASE EXTRACTION EFFICIENCY PERFORMANCE SHALL REMOVE AT LEAST 75% OF GREASE PARTICLES FIVE MICRONS IN SIZE, AND 85% GREASE PARTICLES SEVEN MICRONS IN SIZE AND LARGER, WITH A CORRESPONDING PRESSURE DROP NOT TO EXCEED 1.0 INCHES OF WATER GAUGE.

THE CAPTRATE GREASE-STOP SOLID WAS TESTED TO ASTM STANDARD ASTM F2519-05. MANUFACTURER APPROVED FOR USE IN SOLID FUEL APPLICATIONS AS A SPARK ARRESTER.

EFFICIENCY VS. PARTICLE DIAMETER

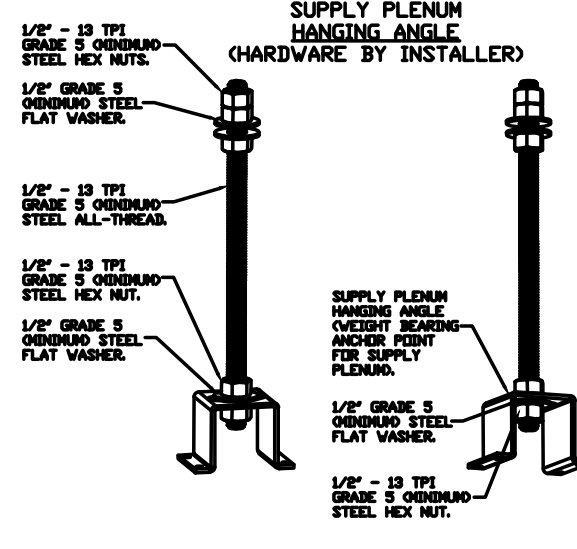


CAPTRATE FILTERS ARE BUILT IN COMPLIANCE WITH:
NFPA #96.
NSF STANDARD #2.
UL STANDARD #1046.
INT. MECH. CODE (IMC).
ULC-S649.



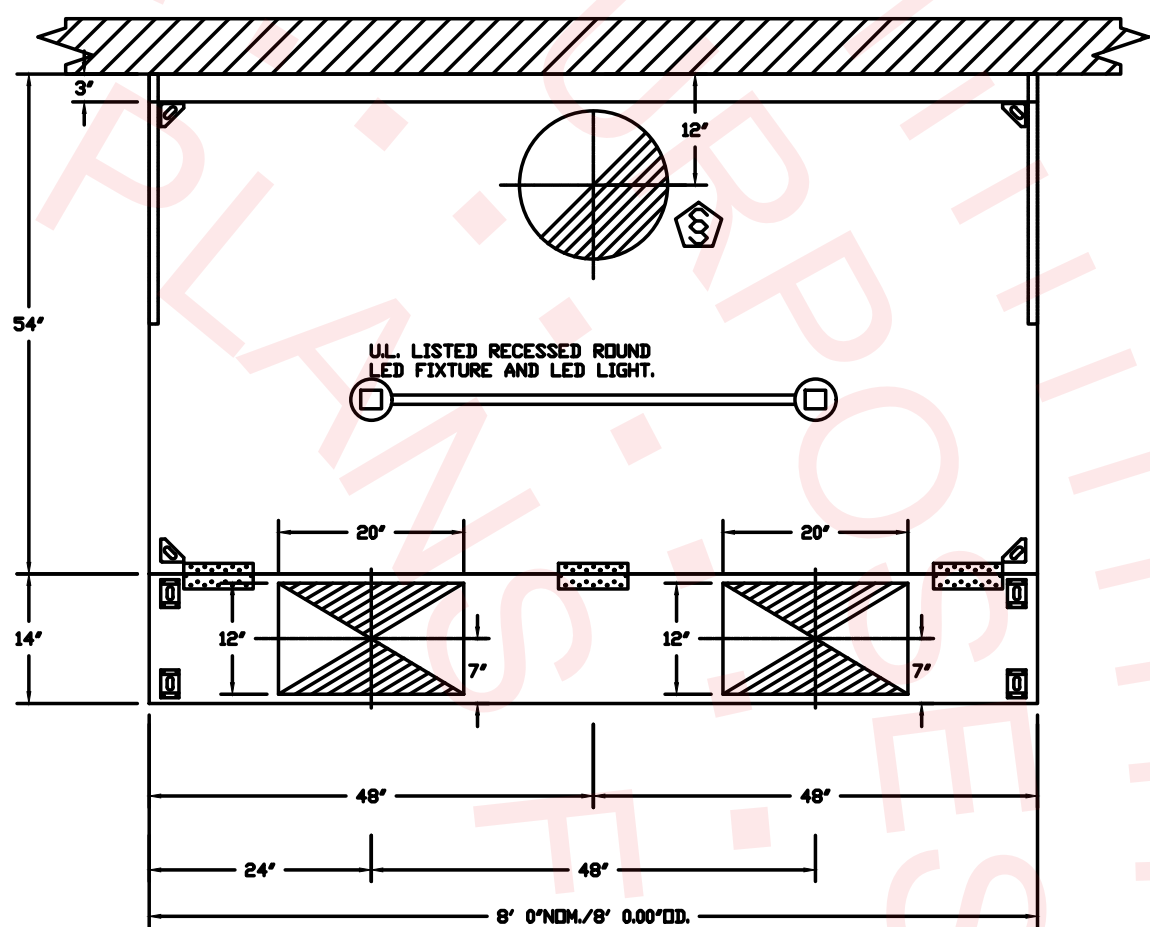
ASSEMBLY INSTRUCTIONS

HANGING ANGLE MUST BE SUPPORTED WITH 1/2" - 13 TPI GRADE 5 (MINIMUM) ALL-THREAD, SANDWICH HANGING ANGLES AND CEILING ANCHOR POINTS WITH 1/2" GRADE 5 (MINIMUM) STEEL FLAT WASHERS AND 1/2" - 13 TPI GRADE 5 (MINIMUM) HEX NUTS AS SHOWN. MUST USE DOUBLED HEX NUT CONFIGURATION ABOVE CEILING ANCHORS. SINGLE HEX NUT BENEATH HANGING ANGLES AND ABOVE CEILING ANCHORS. MAINTAIN 1/4" OF EXPOSED THREADS BENEATH BOTTOM HEX NUT. TORQUE ALL HEX NUTS TO 57 FT-LBS.

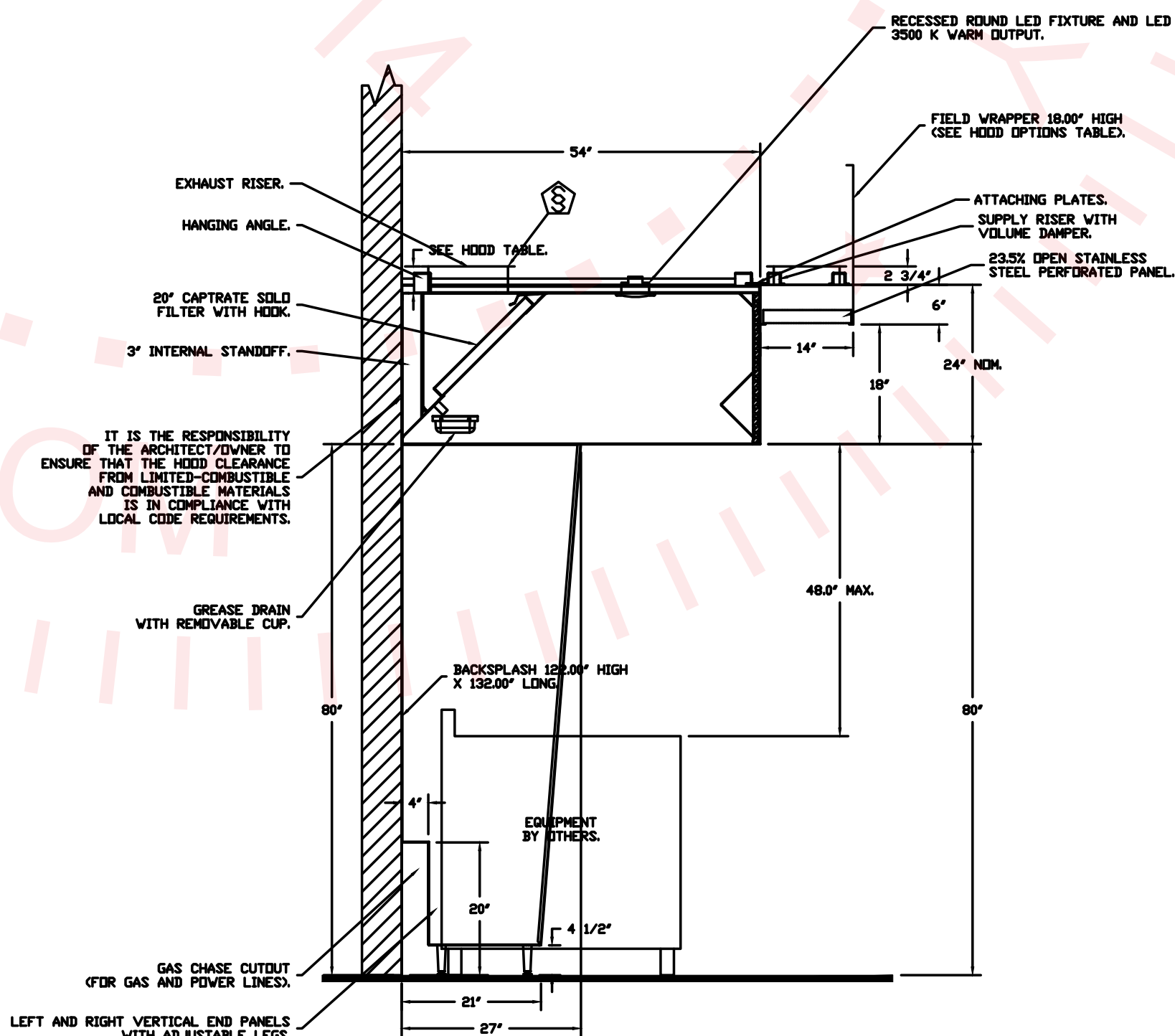


ASSEMBLY INSTRUCTIONS

HANGING ANGLE MUST BE SUPPORTED WITH 1/2" - 13 TPI GRADE 5 (MINIMUM) ALL-THREAD, SANDWICH HANGING ANGLES AND CEILING ANCHOR POINTS WITH 1/2" GRADE 5 (MINIMUM) STEEL FLAT WASHERS AND 1/2" - 13 TPI GRADE 5 (MINIMUM) HEX NUTS AS SHOWN. MUST USE DOUBLED HEX NUT CONFIGURATION ABOVE CEILING ANCHORS. SINGLE HEX NUT BENEATH HANGING ANGLE IS ACCEPTABLE FOR PSP HANGING ANGLES. MAINTAIN 1/4" OF EXPOSED THREADS BENEATH BOTTOM HEX NUT. TORQUE ALL HEX NUTS TO 57 FT-LBS.



PLAN VIEW - HOOD #1
8' 0.00" LONG 5424ND-2-PSP-F

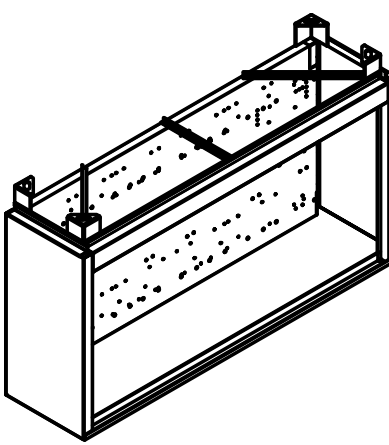
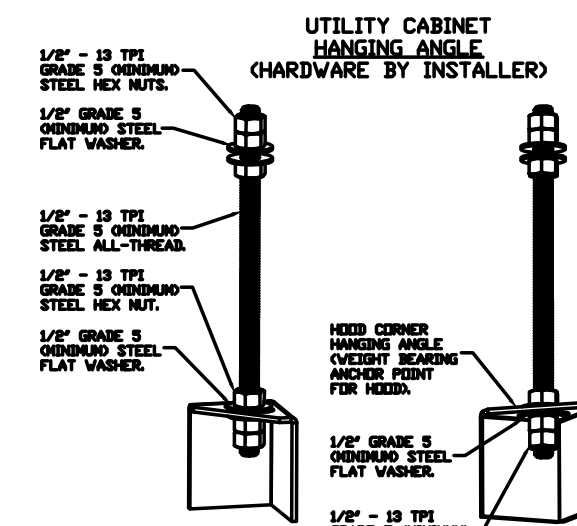


SECTION VIEW - MODEL 5424ND-2-PSP-F
HOOD - #1

WALL-MOUNT UTILITY CABINET
ASSEMBLY INSTRUCTIONS

HANGING ANGLE MUST BE SUPPORTED WITH 1/2" - 13 TPI GRADE 5 (MINIMUM) ALL-THREAD, SANDWICH HANGING ANGLES AND CEILING ANCHOR POINTS WITH 1/2" GRADE 5 (MINIMUM) STEEL FLAT WASHERS AND 1/2" - 13 TPI GRADE 5 (MINIMUM) HEX NUTS AS SHOWN. MUST USE DOUBLED HEX NUT CONFIGURATION BENEATH UTILITY CABINET HANGING ANGLES AND ABOVE CEILING ANCHORS. MAINTAIN 1/4" OF EXPOSED THREADS BENEATH BOTTOM HEX NUT. TORQUE ALL HEX NUTS TO 57 FT-LBS.

CABINET TO BE HUNG BY HOOD INSTALLER. SEE UTILITY CABINET SCHEDULE FOR CABINET SIZE.



Bagel & Co - Tallahassee, FL (Gas)

799 W. Gaines Street, Suite 121
Tallahassee, FL, 32310

DATE: 2/15/2024

DWG.#:
6622389

DRAWN BY: N.J.L.

SCALE:
3/4" = 1'-0"

MASTER DRAWING

SHEET NO.
1

SEAL:

THIS DOCUMENT IS NOT
FOR REGULATORY
APPROVAL, PERMITTING,
OR CONSTRUCTION.

FOR
INFORMATION
ONLY

NO DATE REMARKS
REVISIONS

BAGELS
BAGEL & CO.

BAGEL & CO
TALLAHASSEE

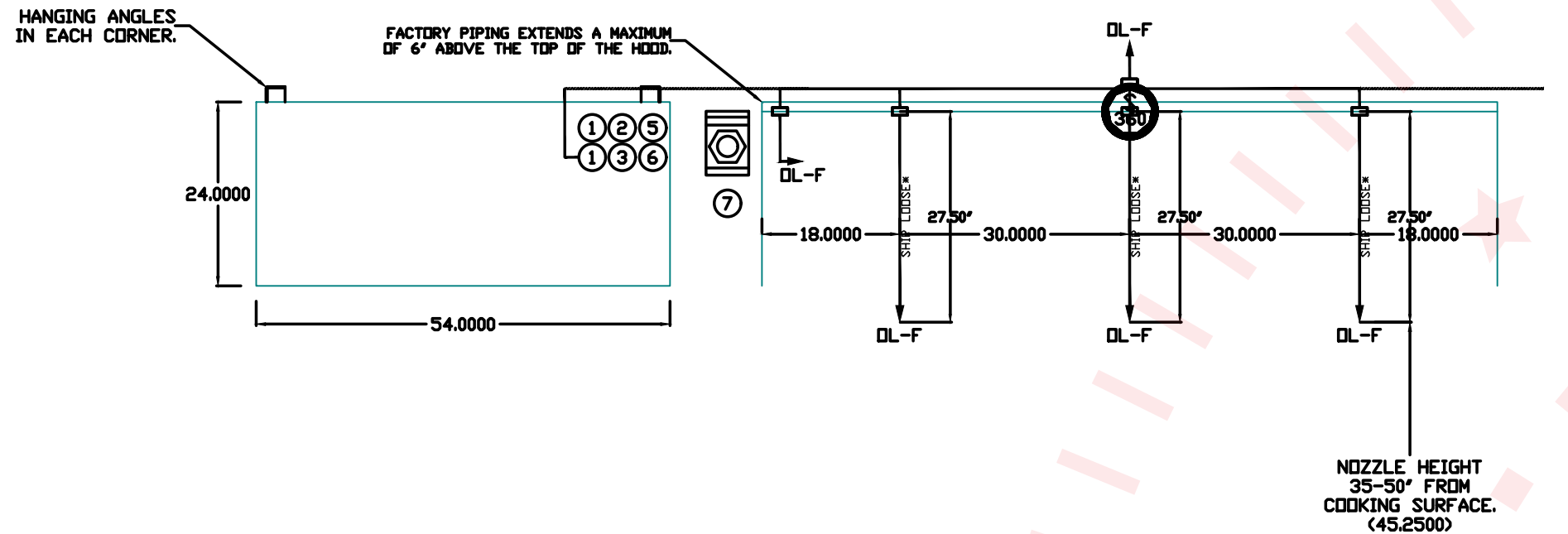
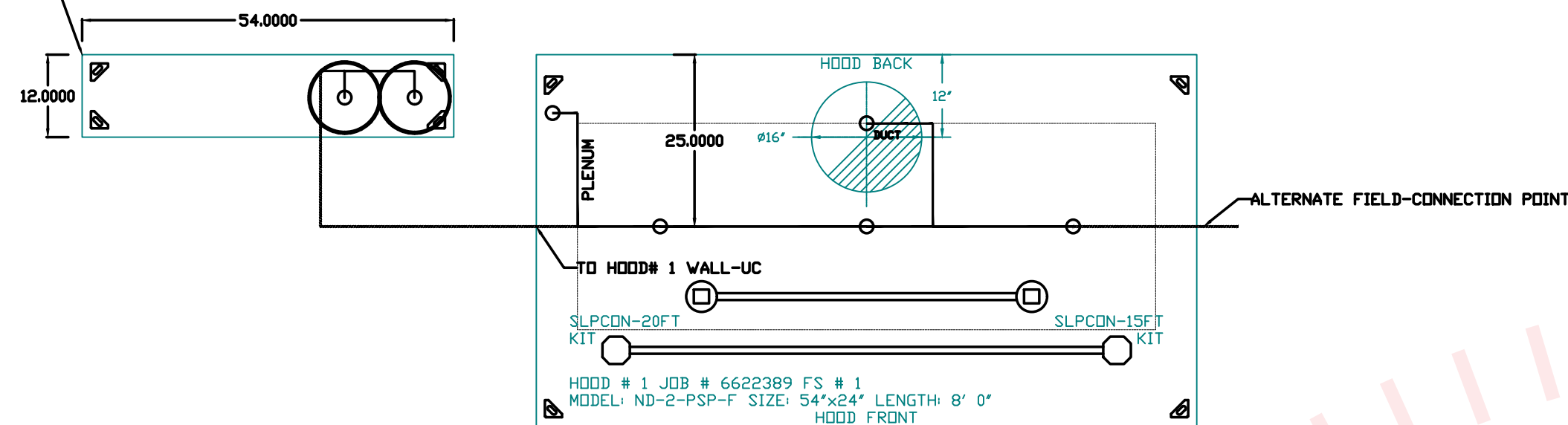
799 W. GAINES STREET
SUITE 121
TALLAHASSEE, FL 32310

PROJECT NO: 2023.0719
DATE: 04/12/2024

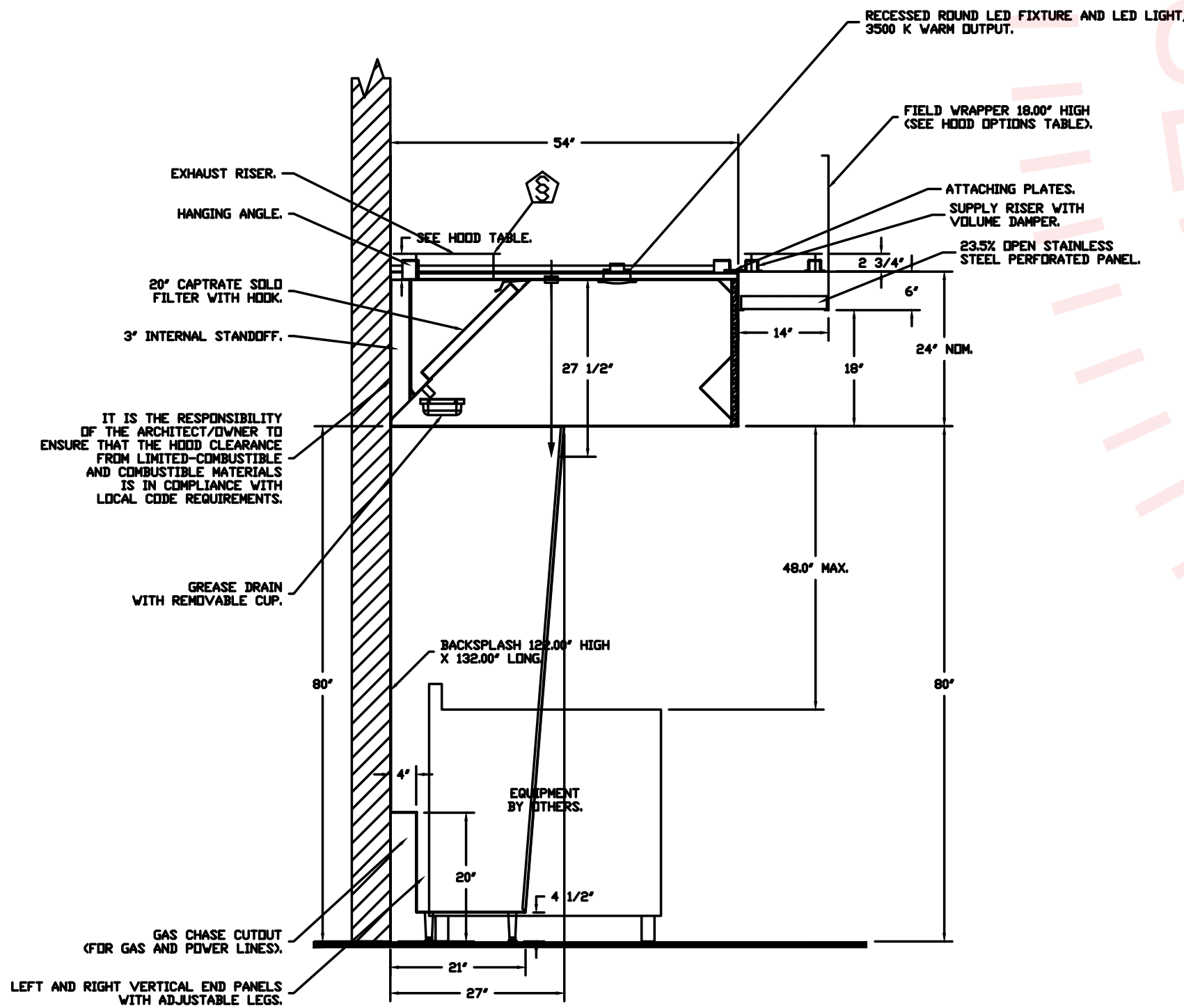
M6.0
HOOD DRAWING

CHECKED: DAK DRAWN: RAN

- SYSTEM REQUIRES A MINIMUM OF 7 FT OF EQUIVALENT PIPE LENGTH BETWEEN TANK AND NEAREST APPLIANCE NOZZLE FOR MOST APPLIANCES. EACH 90 DEGREE ELBOW ADDS 1.5 FT OF EQUIVALENT LENGTH. SEE MANUAL FOR DETAILS.



TANK OVERLAPPING PROTECTED - 30" HIGH PROXIMITY 84.00" L X 33.00" D



SECTION VIEW - MODEL 5424ND-2-PSP-F HOOD - #1

- NOTES
- FIELD PIPE DROPS AS SHOWN
 - PIPING, ELBOWS, TEES, AND NOZZLES SUPPLIED BY GAS.
 - FIELD INSTALLED DROP: FACTORY WILL PROVIDE QTY 2 60IN LONG PIECES OF CHROME PLATED PIPING SHIPPED LOOSE TO BE FIELD-INSTALLED.
 - SHIP LOOSE DROP: FACTORY WILL PROVIDE THE EXACT CHROME PIPE LENGTH NEEDED SHIPPED LOOSE TO BE FIELD-INSTALLED.
 - RELOCATE NOZZLES IF FLOW PATTERN IS BLOCKED BY SHELVEING, SALAMANDERS, ETC.
 - OVERLAPPING COVERAGE SHALL NOT BE USED ON ANY APPLIANCE WITH AN OBSTRUCTION.
 - IF APPLICABLE, EXTENDED PRE-PIPED DROPS ARE SHIPPED LOOSE.
 - FACTORY PIPING EXTENDS A MAXIMUM OF 6' ABOVE THE TOP OF THE HOOD.

- APPLIANCE DIMENSIONS LISTED REPRESENT THE COOKING SURFACE SIZE, NOT THE OVERALL APPLIANCE SIZE.
- THIS FIRE SYSTEM COMPLIES WITH U.L. 300 REQUIREMENTS.

- DL-F NOZZLE PART NUMBER REPLACES 3070-3/8H-10-SS

JOB #: 6622389.
JOB NAME: BAGEL & CO - TALLAHASSEE, FL (GAS).

SYSTEM SIZE: TANK-SP-2-WC. DESIGN FP: 23. MAXIMUM FP: 40.
HOOD # 1 8' 0.00' LONG X 54' WIDE X 24' HIGH.

RISE # 1 SIZE: 16" DIA.
HOOD # 1 METAL BLOW-OFF CAPS INCLUDED.

- HEAVY-DUTY APPLIANCES (RATED 600°F) WILL REQUIRE AN ADDITIONAL DOWNSTREAM FIRESTAT IN THE EVENT THAT THE DUCTWORK CONTAINS ANY HORIZONTAL RUNS OVER 25 FT IN LENGTH
- MEDIUM TO LIGHT-DUTY APPLIANCES (RATED 450°F) WILL NOT REQUIRE ANY ADDITIONAL DOWNSTREAM DETECTION.

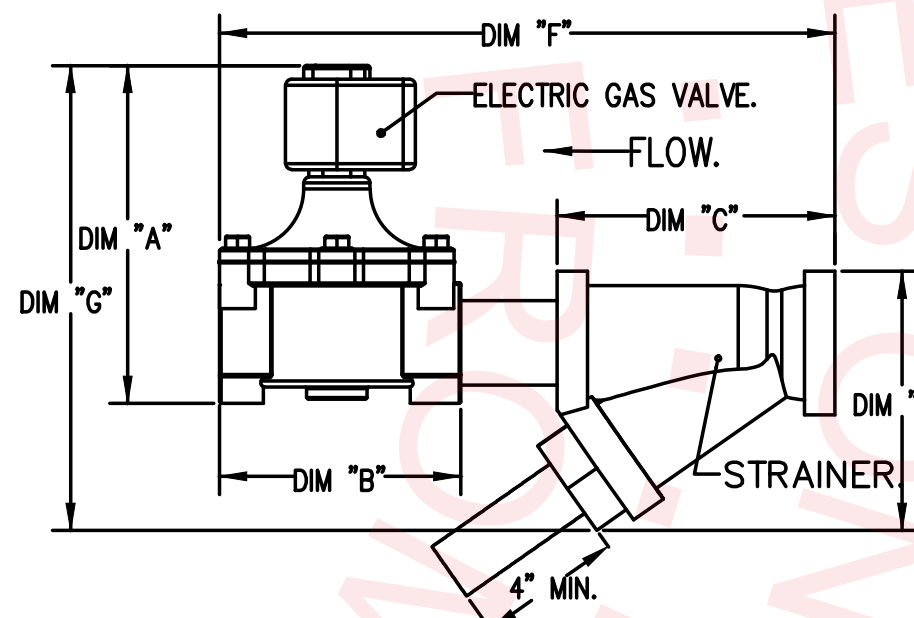
LEGEND - FIRE CABINET TANK SYSTEM

- 4 GALLON TANK.
- PRIMARY ACTUATOR RELEASE.
- SECONDARY ACTUATOR RELEASE.
- PRESSURE SUPERVISION SWITCH.
- PRIMARY HOSE ASSEMBLY.
- SECONDARY HOSE ASSEMBLY.
- REMOTE MANUAL ACTUATION DEVICE.

GAS VALVES AND STRAINERS									
GAS VALVE SIZING					GAS VALVE DIMENSIONS				
TYPE	SIZE	VOLTAGE	MIN. INLET PRESSURE	MAX. INLET PRESSURE	FLOW AT 1 IN.W.C. DROP NATURAL GAS	FLOW AT 1 IN.W.C. DROP PROPANE	DIM "A"	DIM "B"	DIM "C"
ELECTRICAL	2"	120 VAC	0 PSIG (0 IN.W.C.)	5 PSIG (138 IN.W.C.)	2,040,000 BTU/HR	1,000,040 BTU/HR	7-5/8"	6-3/8"	7-1/4"

ALL GAS VALVES/STRAINERS
PROPER CLEARANCE MUST BE PROVIDED IN ORDER TO SERVICE THE STRAINERS A MINIMUM OF 4" CLEARANCE DISTANCE MUST BE PROVIDED AT THE BASE OF THE STRAINER. CUSTOMER MUST VERIFY BTU CONSUMPTION AS WELL AS PRESSURE RATING SPECIFIC GRAVITY OF NATURAL GAS = 0.64, SPECIFIC GRAVITY OF LP = 1.52.

CALCULATIONS
TO CALCULATE GAS FLOW FOR OTHER THAN 1 IN.W.C. PRESSURE DROP
NEW BTU/HR = (BTU/HR AT 1 IN.W.C. PRESSURE DROP) X NEW PRESSURE DROP^{3/4}
TO CALCULATE GAS FLOW FOR OTHER THAN 0.64 SPECIFIC GRAVITY
NEW BTU/HR = (BTU/HR AT 0.64) X (0.64 / NEW SPECIFIC GRAVITY)^{3/4}.



FIRE SYSTEM INFORMATION - JOB#6622389

FIRE SYSTEM NO.	TAG	TYPE	SIZE	MAX FP	DESIGN FP	INSTALLATION	
1		TANK FS	4.0/4.0	40	23	SYSTEM	LOCATION ON HOOD
1		TANK FS	4.0/4.0	40	23	WALL UTILITY CABINET LEFT	N/A

GAS VALVE(S)

FIRE SYSTEM NO.	TAG	TYPE	SIZE	SUPPLIED BY
1		SC ELECTRICAL	2,000	CAPTIVEAIRE SYSTEMS

FIRE SYSTEM PARTS LIST KEY

FIRE SYSTEM NO.	TAG	KEY NUMBER - PART DESCRIPTION	QTY BY FACTORY	QTY BY DIST
1		0 - 0 - TANK FIRE SUPPRESSION POST-DISCHARGE PROCEDURE UTILITY CABINET LABEL SHEET.	1	0
		0 - 0 - TANK FIRE SUPPRESSION MAINTENANCE GUIDE UTILITY CABINET LABEL SHEET.	1	0
		0 - 0 - 12-F28021-32144-DT-360 DUCT FIRE THERMOSTAT WITH 12 FOOT WIRE LEADS. NO. CLOSE ON TEMP RISE AT 360°F.	1	0
		0 - 0 - 4429K153 1/2" MALE NPT TO 1/2" FEMALE NPT ELBOW, BRASS.	2	0
		0 - 0 - 4429K422 1/2" X 1/4" BRASS REDUCING BUSHING.	1	0
		0 - 0 - 79525 1/2" 90 PRO-PRESS ELBOW WITH 1/2" NPT FEMALE CONNECTION, VIEGA.	1	0
		0 - 0 - 79580 1/2" X 1/2" PRO-PRESS TEE X 1/2" NPT FEMALE CONNECTION, VIEGA.	2	0
		0 - 0 - 87-120042-001 SECONDARY ACTUATOR VALVE (SVA) - SINGLE ACTUATOR, REQUIRES PRIMARY RELEASE ACTUATOR, TANK FIRE SUPPRESSION.	1	0
		0 - 0 - 87-120045-001 HOSE, SECONDARY ACTUATOR HOSE, 7.5' BRAIDED STAINLESS STEEL, TANK FIRE SUPPRESSION.	1	0
		0 - 0 - 87-300001-001 TANK - PRESSURIZED TANK USED FOR TANK FIRE SUPPRESSION.	2	0
		0 - 0 - 87-300030-001 PRIMARY ACTUATOR KIT (PAK) - ACTUATOR AND RELEASE SOLENOID ASSEMBLY, ONE NEEDED PER FIRE SYSTEM, SUPERVISED, TANK FIRE SUPPRESSION.	1	0
		0 - 0 - 87-300152-001 HARDWARE, SVA BOLTS, TANK FIRE SUPPRESSION.	8	0
		0 - 0 - 986944115 HARDWARE, DATANKLOCK LOCKING BRACKET SQUARE NUTS 5/16" ZINC, TANK FIRE SUPPRESSION.	4	0
		0 - 0 - A0034332 JUNCTION BOX FOR MANUAL PULL STATION, 1.5" DEEP BACK BOX, RED COLOR.	1	0
		0 - 0 - A31484 1/4" NPT SCHRADER VALVE AND CAP, J.B. INDUSTRIES. 1/4" FLARE X 1/4" NPT HALF UNION. USED ON TANK SERVICE PORT.	1	0
		0 - 0 - DATANKLOCK DISCHARGE ADAPTER TANK LOCKING PLATE FOR FIRE SYSTEM TANK INSTALLATION IN UTILITY CABINETS, TANK FIRE SUPPRESSION.	2	0
		0 - 0 - SLPICDN-15FT SUPERVISED LOOP CONNECTION KIT. CONTAINS THE PARTS NEEDED TO CONNECT THE SUPERVISED LOOP BETWEEN END TO END HOODS WITH LESS THAN A 14" GAP OR BACK TO BACK HOODS. KIT CONTAINS 17 FEET OF BLACK MG WIRE, 17 FEET OF TAN MG WIRE, 15 FEET OF FLEXIBLE CONDUIT, AND TWO 7/8" CONNECTORS.	1	0
		0 - 0 - SLPICDN-20FT SUPERVISED LOOP CONNECTION KIT. CONTAINS THE PARTS NEEDED TO CONNECT THE SUPERVISED LOOP BETWEEN HOODS WITH UP TO 19" GAP. KIT CONTAINS 22 FEET OF BLACK MG WIRE, 22 FEET OF TAN MG WIRE, 20 FEET OF FLEXIBLE CONDUIT, AND TWO 7/8" CONNECTORS.	1	0
		0 - 0 - TANK STRAP TANK STRAP - USED FOR TANK FIRE SUPPRESSION.	6	0
		0 - 0 - TFS-15-TANKBRACKET TANK BRACKET FOR FIRE SYSTEM TANK INSTALLATION IN UTILITY CABINETS, TANK FIRE SUPPRESSION.	2	0
		0 - 0 - VK-283952-000 DISCHARGE ADAPTER, TANK FIRE SUPPRESSION.	2	0
		34 - 34 - A0034331 24VDC SINGLE ACTION MANUAL ACTUATION DEVICE (PUSH/PULL STATION) WITH PROTECTIVE COVER, ONE (1) NORMALLY OPEN CONTACT. RED COLOR.	1	0

REVISIONS

NO.	DESCRIPTION	DATE
1		
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Central Florida Office

2901 W. SF 434, Suite 101, Longwood, FL 32779 PHONE: (407) 982-0317 FAX: 9192275976 EMAIL: reg66@capwheie.com

Bagel & Co - Tallahassee, FL (Gas)
799 W. Gaines Street, Suite 121
Tallahassee, FL, 32310

DATE: 2/15/2024
DWG.#: 6622389
DRAWN BY: NJL
SCALE: 3/4" = 1'-0"
MASTER DRAWING

SHEET NO. 2

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NO DATE REMARKS
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Exhaust Motor

Model GP0024 • 2.000 HP • 3 Phs • 208 V • 60Hz • 5.7 FLA • TEFC, Premium (E-Plus3) Eff.

Exhaust Performance

Volume:	2400	RPM:	1835	TS:	7566ft/min
SP:	1.5" w.g.	BHP:			1.125
Altitude:	69	Ambient Temp:			70°F

Exhaust Installation Information

Unit	3 phs 208 V 60Hz
Voltage:	
Unit Main Input:	7.2 Amps MCA •15 Amps MDP •208 V •14 AWG Wire Min.
SCCR:	10

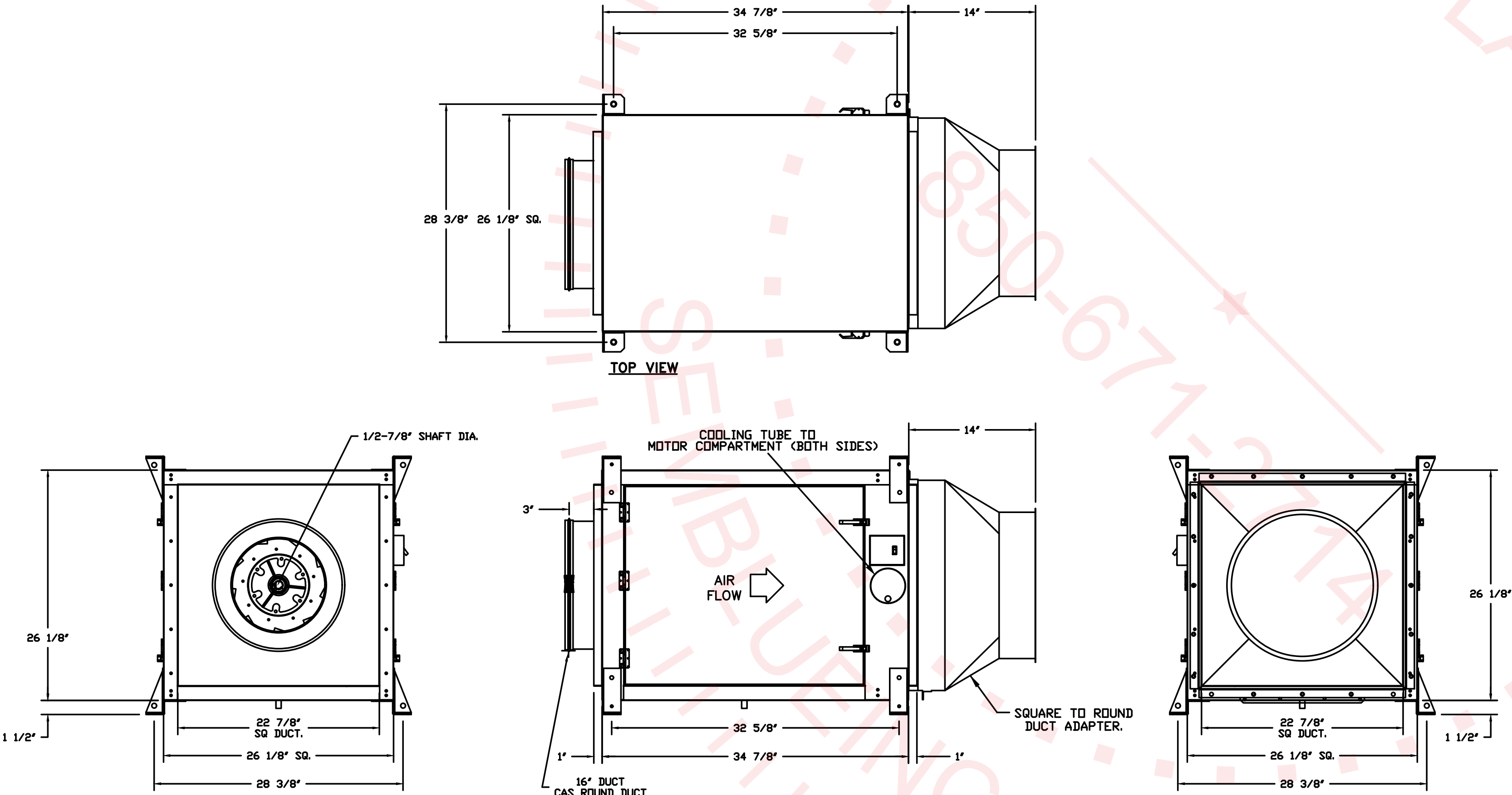
SIF15DD Sound (1835 RPM) at 5 ft.							
Octaves:							
1	2	3	4	5	6	7	8
77.3	73.7	78.4	75.6	74.5	74.4	71.5	64.6
LWA:	80.5		Sones:	17.2		DBA:	69

Configured Options

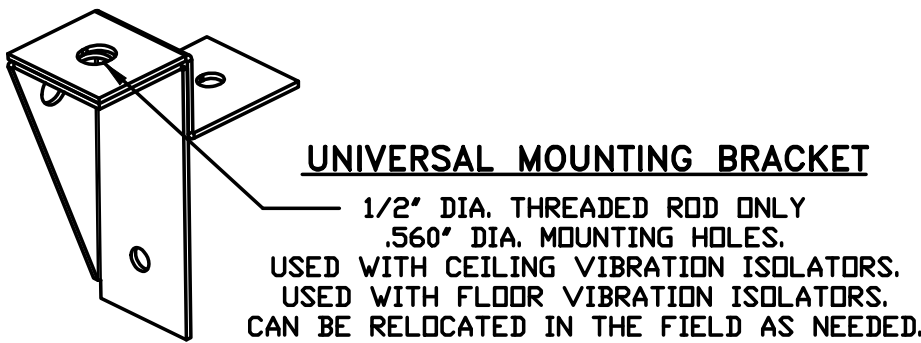
- VAV (Variable-Air-Volume) Wiring Package for Commercial Fans. Manual Speed Control Variable Frequency Drive Included
- Exhaust Variable Frequency Drive - 2 HP Max., 200/240 V, Single or Three Phase Input, 7.0 A Max., NEMA 1 Enclosure PART NEEDS PROGRAMMING
- Hanging Vibration Isolators for Square Inline Fans. Contains four HSA125 Hanging Isolators.
- SIF - Horizontal Overhead Pre-Installed Mounts (Used on sizes 11-36).
- SIF 15. Standard 16" Duct Inlet Connection.
- Single Wall Duct Adapter - Square To Round - 23" X 23" To 16" - 14" Tall - Stainless Steel. Standard Part. SIF 15 Discharge Adapter

FEATURES:

- DIRECT DRIVE CONSTRUCTION (NO BELTS/PULLEYS).
- TWO ACCESS DOORS FOR EASY ACCESS.
- BACKWARD INCLINED NON-OVERLOADING WHEELS.
- UL762 LISTING.
- AMCA AIR & SOUND CERTIFIED.
- THERMAL OVERLOAD PROTECTION (SINGLE PHASE).
- NEMA 3R SAFETY DISCONNECT SWITCH.
- DRAIN MUST BE CONNECTED TO OPTIONAL GREASE BOX OR PIPED TO GREASE RESERVOIR (RECOMMENDED).



- NOTES:
- DWXXFRISER USED WHEN CONNECTING TO STANDARD DUCT.
 - DWXXFRISERS ARE INSTALLED ON THE INTAKE AND/OR DISCHARGE SIDES.



REVISIONS

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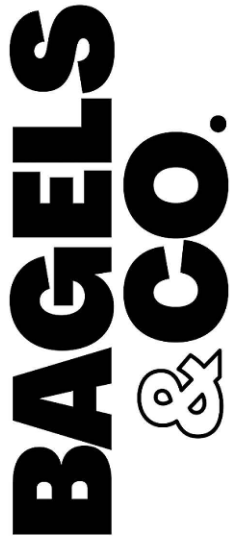
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BAGEL & CO
TALLAHASSEE

799 W. GAINES STREET
SUITE 121
TALLAHASSEE, FL 32310

PROJECT NO: 2023.0719
DATE: 04/12/2024

M6.2
HOOD DRAWING

CHECKED: DAK DRAWN: RAN

Supply Motor

Model DTP1/56 • 1.500 HP, 3 Phs, 208 V, 60 Hz, 6.6 FLA, ODP, Premium (E-Plus3) Eff.

Burner

Min Output BTU:	18,000 BTU/Hr
Max Output BTU:	275,000 BTU/Hr
Size:	6" long
Gas Type:	Natural

Supply Performance

Volume:	2000 CFM	RPM:	1086
Tip Speed:	5686 ft/min	BHP	0.668
Static Pressure:	0.796 " w.g. (0.750"Ext+0.037"Int+0.009"Opt)		
Altitude:	69 ft	Entering Dry Bulk Temp:	29 °F

Supply Installation Information

Gas Pressure:	7 in. w.c. - 14 in. w.c.
Insurance Requirements:	No Insurance Requirement (ANSI)
Unit Voltage:	3 phs 208 V 60Hz
Unit Main Input:	9.5 Amps MCA •15 Amps MDP •208 V •14 AWG Wire Min.
SCCR:	10

Heating Schedule

Altitude:	69 ft
Winter Outdoor Air Dry Bulb Temp:	29°F
Temp Rise:	51°F
Output BTU:	107692
Input BTU:	117057
Based Off:	BTUs Actual Air Density

Configured Options

- Profile Plate Configuration for size 2 Direct Fired Unit for low cfm applications.
- Gas Pressure Gauge, 0-35", 2.5" Diameter, 1/4" Thread Size
- Gas Pressure Gauge, -5 to +15 Inches Wc., 2.5" Diameter, 1/4" Thread Size
- Butterfly Mod Valve Option for Mod Size 2 (1" Mod Valve).
- Ship Loose Gas Strainer. To be installed upstream of unit connection. 1" Connection
- CASLink building monitoring system communications module. Requires internet & field wired ethernet connection or 3G cellular service. Includes Rev 3 Comm Module, RJ45 to modbus converter, 3 FT cat5 cable, and 1 FT of shielded twisted pair.
- Motorized Back Draft Damper 22.75" X 24" for Size 2 Standard & Modular Heater Units w/Extended Shaft, Standard Galvanized Construction, 3/4" Rear Flange, Low Leakage, LF120S Actuator Included
- VAV (Variable-Air-Volume) Wiring Package for Commercial Fans.
 - Manual Speed Control Variable Frequency Drive Included
- Supply Variable Frequency Drive - 2 HP Max., 200/240 V, Single or Three Phase Input, 7.0 A Max., NEMA 1 Enclosure, with 2RJ-45 FOR MODBUS
- VFD factory mounted and wired in unit control vestibule.
- Indoor Hanging Cradle For The Size 2 Direct Fired Unit. 2 HSA125 Hanging Isolators per Uni-Strut Included.
- "Insulation" for V-bank Intake Option.

Construction Features

- Housing constructed of heavy duty G90 galvanized steel
- Vibration isolation
- Ball bearing motors
- Heavy duty, pre-lubricated bearings rated for 200,000 hours of operation
- Service doors on both sides
- Horizontal & down discharge
- Large intake area ensures low pressure drop across unit
- Spring loaded profile plates automatically adjust for any airflow - no manual setting required!
- Weatherproof safety disconnect switch
- Modular design provides design flexibility
- Fully insulated casing

Blower

- 20 Inch Mixed Flow Plenum Fan for size 2 MOD. Hub specified to match motor shaft.

Temperature Control

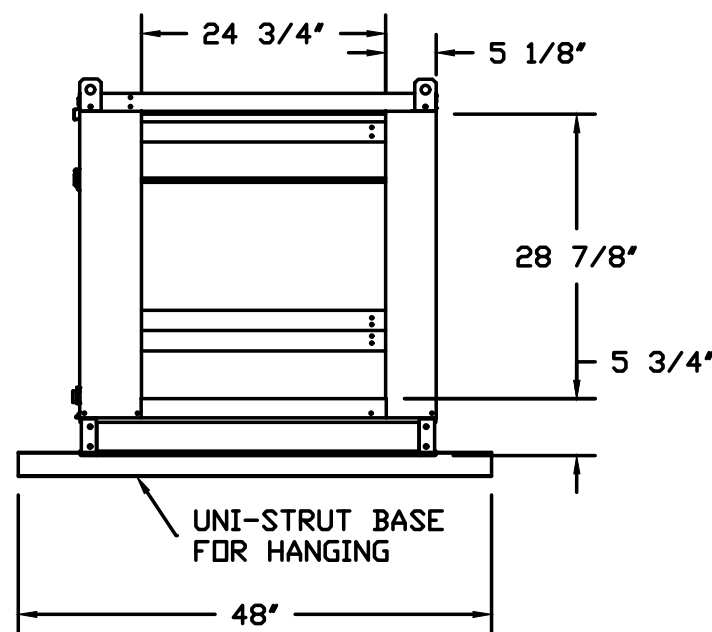
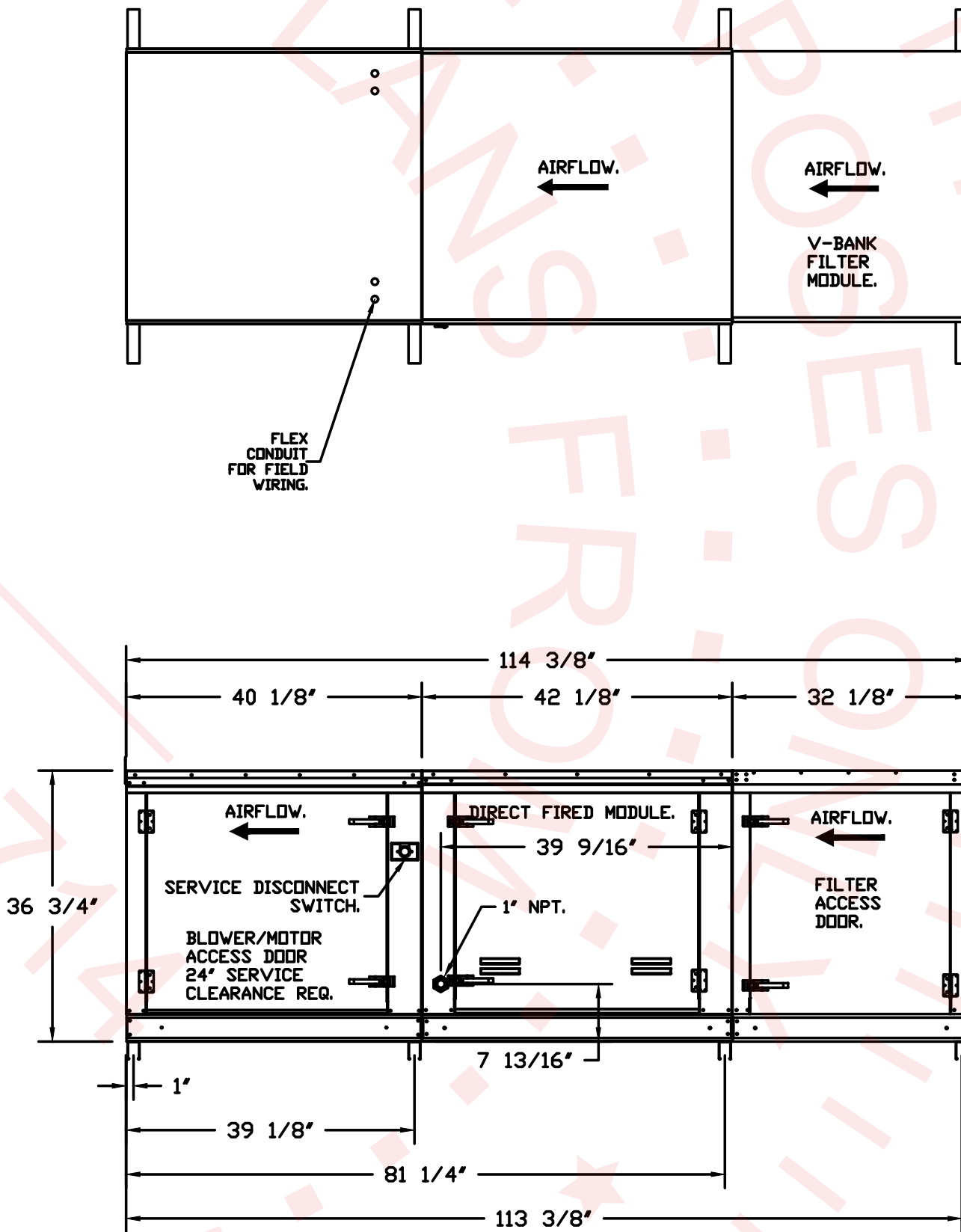
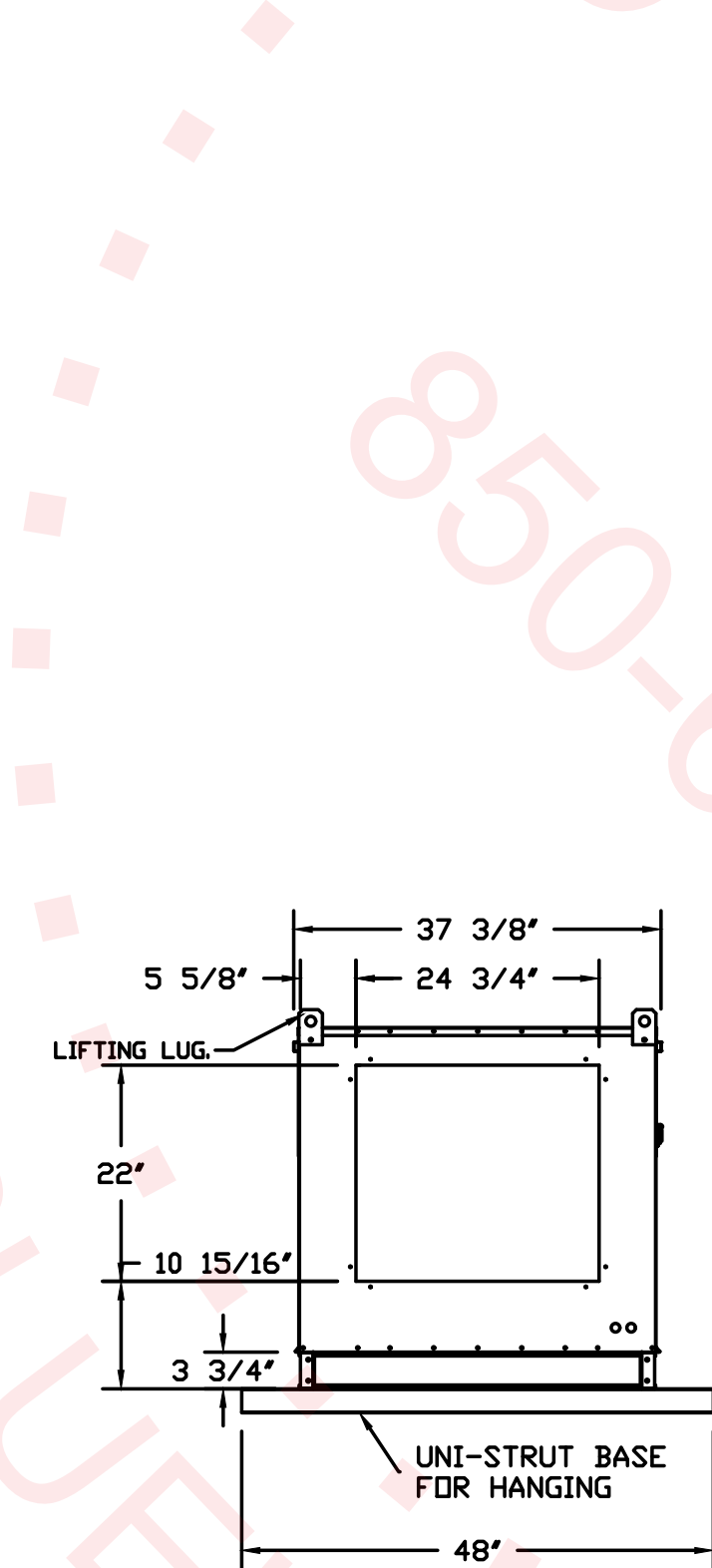
- 0-150°F Discharge Temp Control • Field Wired On/Off Start Command • Heating Activation Based On Intake Set Point. Use with MUA

Intake

- V-Bank for Size # 2 Modular Heater with TA filters.
For indoor installation. MERV 8 V-Bank Filters.

Filters

- 8x Filter throw away 16" x 20" x 2". MERV 8, (DP40)



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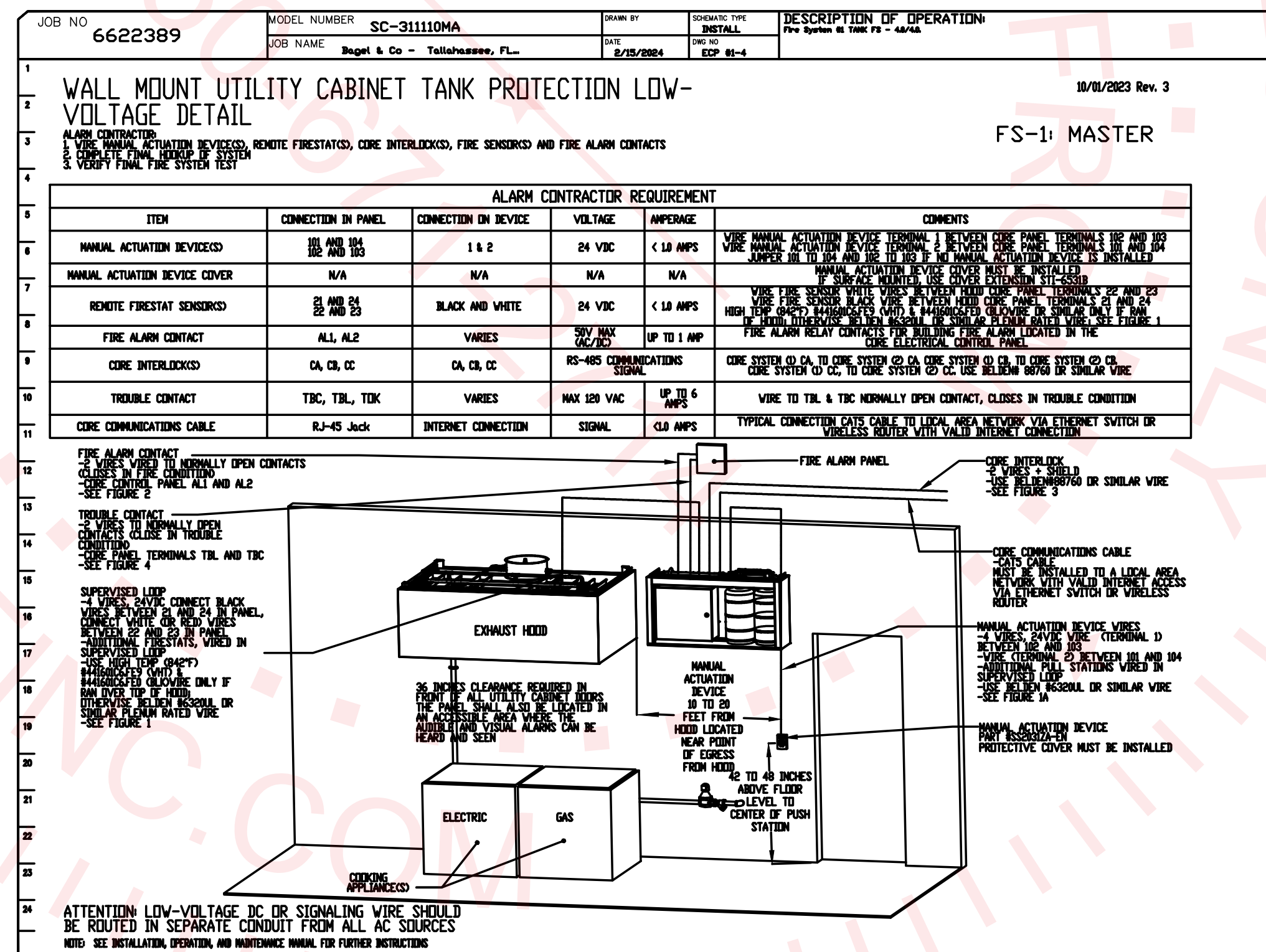
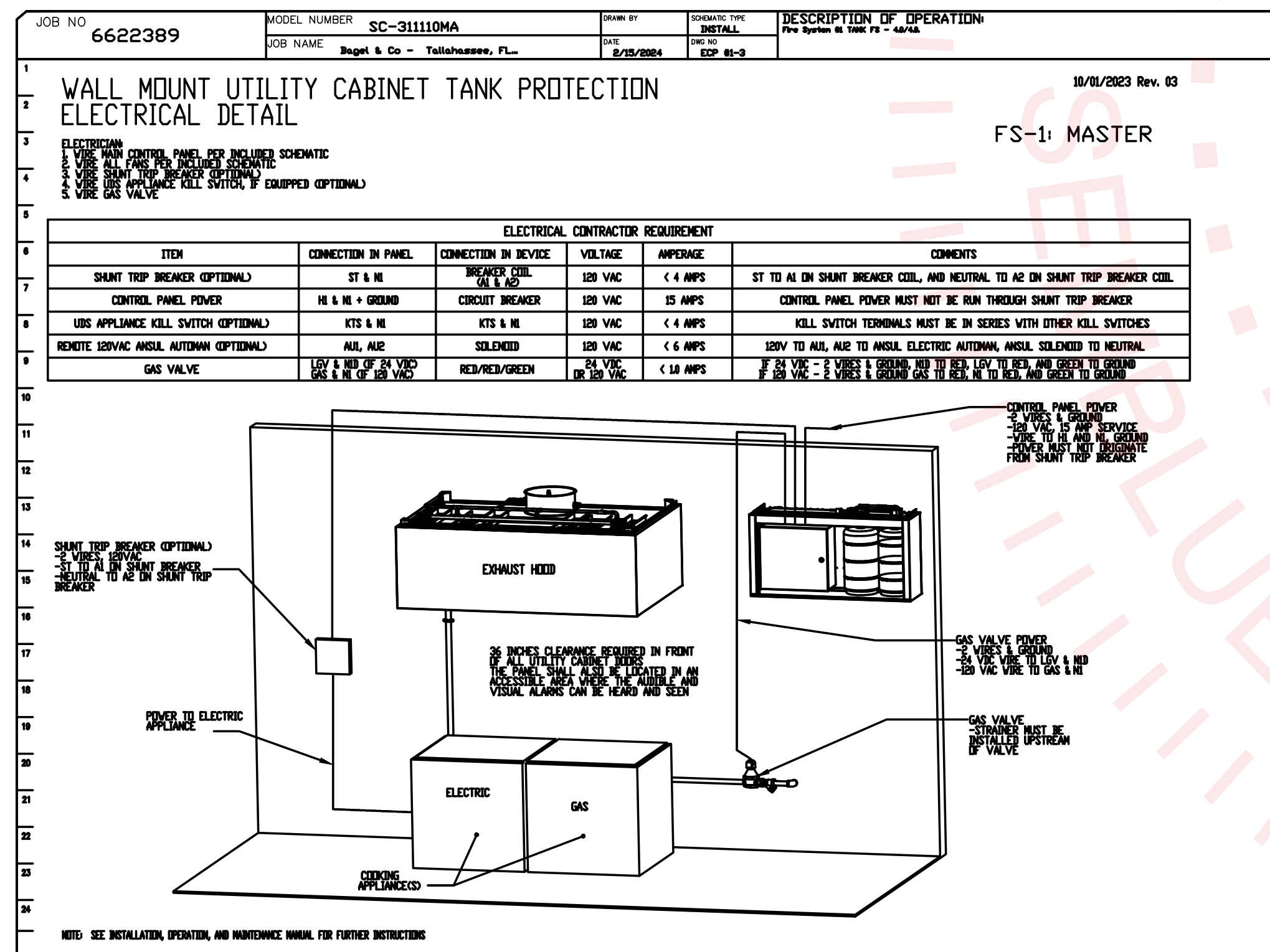
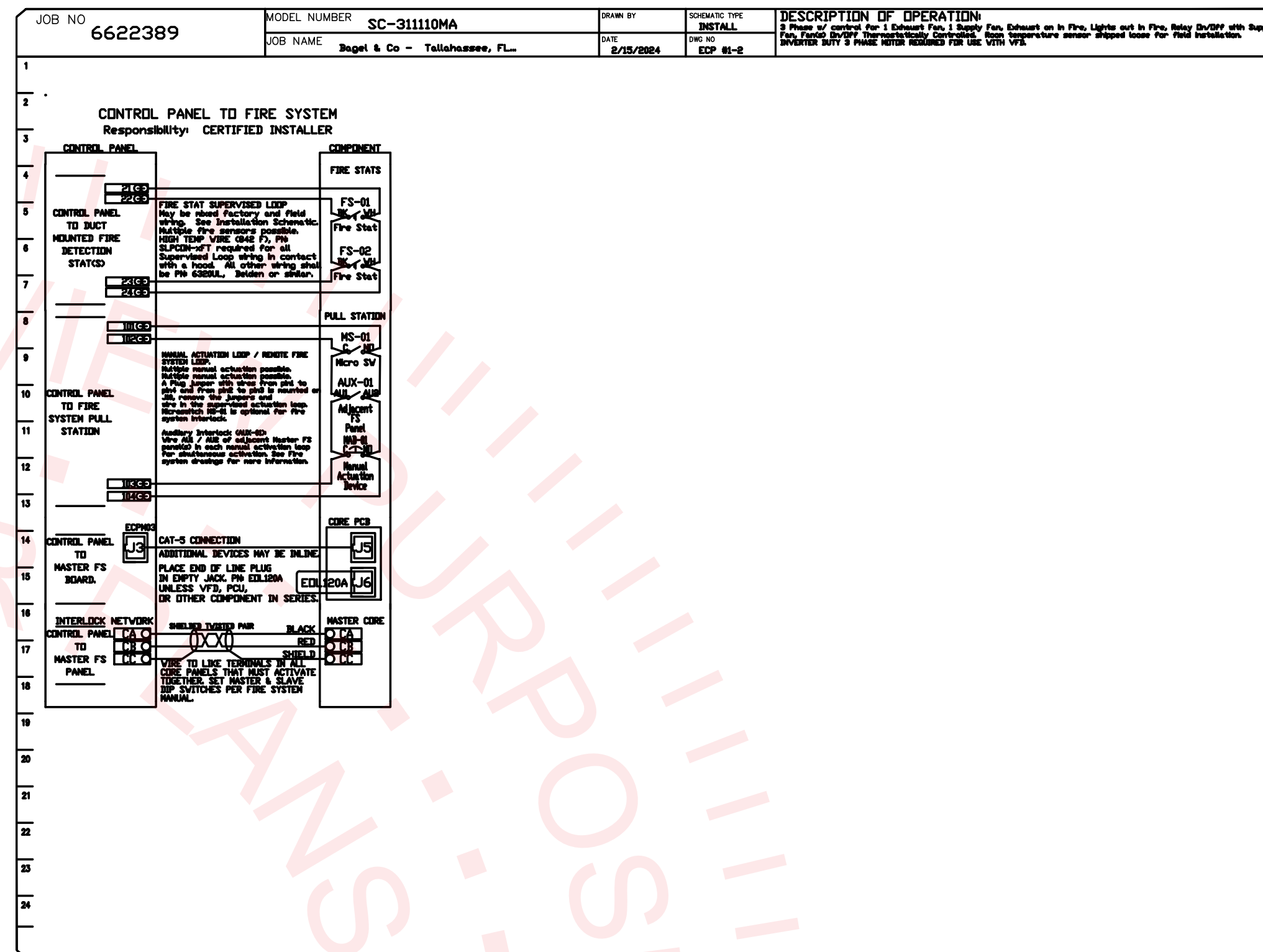
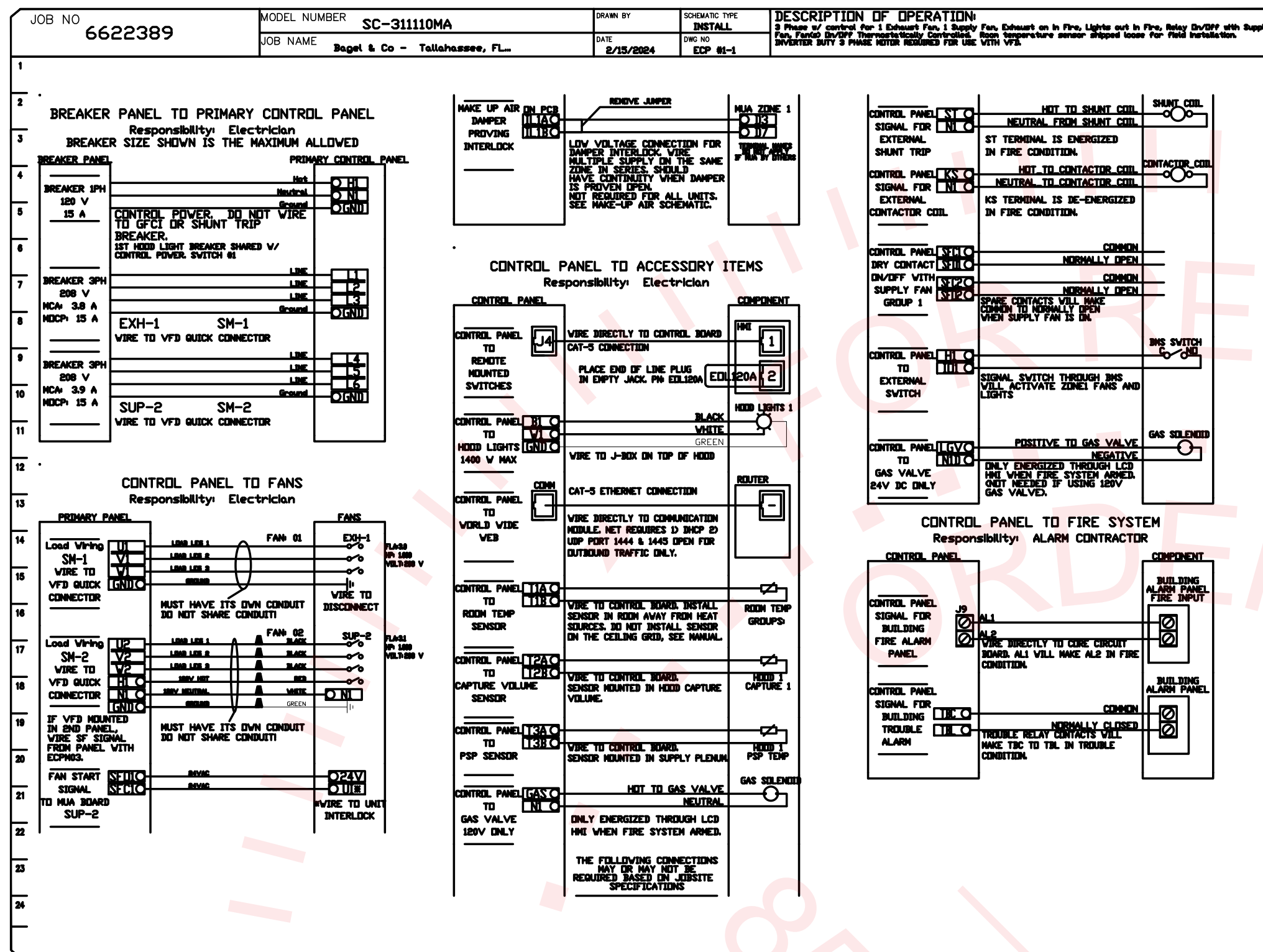
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NO	TAG	PACKAGE #	LOCATION	SWITCHES		OPTION	FANS CONTROLLED				
				LOCATION	QUANTITY		TYPE	φ	HP	VOLTS	FLA
1		SC-311110MA	WALL UTILITY CABINET RIGHT	FACE MOUNT RIGHT SIDE OF HOOD	1 LIGHT	SMART CONTROLS THERMOSTATIC CONTROL W/ RELAY ON/OFF WITH SUPPLY	EXHAUST	3	1,000	208	3.0
				HOOD # 1	1 FAN		SUPPLY	3	1,000	208	3.1



- Hood control panel to support communications to cloud-based Building Management System.
- Hood Control Panel to allow cloud-based Building Management System to monitor real time parameters outlined as MONITOR in the points list.
- Hood Control Panel to allow cloud-based Building Management System to control parameters outlined as CONTROL in the points list.
- Hood Control Panel to allow cloud-based Building Management System to implement SYSTEM ECONOMIZER control strategies for fully integrated Building Management.

RTV Pinholes	Pass/Fail	RTV Pinholes	Pass/Fail
Room Temperature	ASMT0705	Room Temperature(s)	ASMT0705
Dual Temperature(s)	ASMT0705	Dual Temperature(s)	ASMT0705
RTU Dinohang Temperature	ASMT0705	RTU Dinohang Temperature	ASMT0705
Milohun RTU Dinohang Temperature	ASMT0705	Milohun RTU Dinohang Temperature	ASMT0705
Fun Speed	ASMT0705	Controlled Fun(s)	ASMT0705
Fun Jangpung	ASMT0705	Fun Pin(s)	ASMT0705
Fun Power	ASMT0705	Fun Shobo	ASMT0705
YFD Pin(s)	ASMT0705	PCU Pin(s)	ASMT0705
Controlled Fun(s)	ASMT0705	PCU Filter Chip Percentage	ASMT0705
Fun Pin(s)	ASMT0705	Fire Condition	ASMT0705
Fun Shobo	ASMT0705	COSE Fire System	ASMT0705
PCU Pin(s)	ASMT0705	Building Pressure	ASMT0705
PCU Filter Chip Percentage	ASMT0705	Fun Pin(s)	ASMT0705 & CONTROL
Fire Condition	ASMT0705	Light(s) Button(s)	ASMT0705 & CONTROL
COSE Fire System	ASMT0705	Yash Button	ASMT0705 & CONTROL
Building Pressure	ASMT0705		
Prep Time Button	ASMT0705 & CONTROL		
Fun(s) Button	ASMT0705 & CONTROL		
Light(s) Button	ASMT0705 & CONTROL		
Yash Button	ASMT0705 & CONTROL		



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SHEET NO.

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TALLAHASSEE

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PROJECT NO: 2023.0719
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M6.4

HOOD DRAWING

CHECKED: DAK DRAWN: RAN

MECHANICAL SPECIFICATION

1. GENERAL INFORMATION
- 1.1. PROVIDE ALL SUPPLIES, MATERIAL, LABOR, EQUIPMENT, AND APPURTENANCES NECESSARY FOR COMPLETE INSTALLATION AND FULL OPERATION OF ALL MECHANICAL AND MECHANICAL RELATED WORK, INDICATED HEREINAFTER ON DRAWINGS AND SPECIFICATIONS, FOR A SAFE AND FULLY OPERATIONAL SYSTEM.
- 1.1.1. THE INSTALLED SYSTEM SHALL BE COMPLETE IN EVERY WAY AND FUNCTIONING ACCORDING TO THE DESIGN INTENT, WHETHER OR NOT ALL SUCH MATERIALS AND APPURTENANCES ARE SHOWN ON THE DRAWINGS OR DESCRIBED IN THE SPECIFICATIONS.
- 1.2. PERFORM ALL OPERATIONS INCLUDING EXCAVATION & BACKFILLING, SHORING, CUTTING, CHANNELING & CHASING, DE-WATERING, ETC. NECESSARY FOR INSTALLATION OF FULLY OPERATIONAL SYSTEM, WHETHER OR NOT SHOWN ON THE DRAWINGS.
- 1.3. DEFINITION OF TERMS
- 1.3.1. FURNISH – SUPPLY AND DELIVER TO THE PROJECT SITE, READY FOR UNLOADING, UNPACKING, ASSEMBLY, INSTALLATION, AND SIMILAR OPERATIONS.
- 1.3.2. INSTALL – OPERATIONS AT THE PROJECT SITE INCLUDING THE ACTUAL UNLOADING, UNPACKING, ASSEMBLY, ERECTION, PLACING, ANCHORING, APPLYING, WORKING TO DIMENSION, FINISHING, CURING, PROTECTING, CLEANING, AND SIMILAR OPERATIONS.
- 1.3.3. PROVIDE – FURNISH AND INSTALL, COMPLETE AND READY FOR THE INTENDED USE.
- 1.3.4. U.N.O. – UNLESS NOTED OTHERWISE.
- 1.3.5. M.S.D.S. – MATERIAL SAFETY DATA SHEET
- 1.3.6. CONTRACTOR – APPEARANCE ON DRAWINGS OR IN SPECIFICATIONS FOR MECHANICAL WORK SHALL REFER TO MECHANICAL SUB-CONTRACTOR.
- 1.3.7. COORDINATE WITH THE ELECTRICAL CONTRACTOR TO RELOCATE – DISCONNECT ELECTRICAL FEEDER, MAKE SAFE (INCLUDING LOCK OUT/TAG OUT), STORE AND PROTECT DEVICE, REINSTALL, REWORK AND EXTEND CONDUIT & WIRE TO NEW LOCATION, RE-ENERGIZE AND TEST.
- 1.3.8. EQUAL AND EQUIVALENT – TO MEAN OF THE SAME QUALITY, SIZE, NUMBER, VALUE AND THE ITEMS ARE SIMILAR IN ALL RESPECTS.
- 1.3.8.1. THE FINAL DECISION OF ACCEPTANCE OF THESE ITEMS WILL BE MADE BY THE ENGINEER.
- 1.3.8.2. IT SHALL BE UNDERSTOOD THAT FOR ANY SPECIFIED ITEM ON THE DRAWINGS AND/OR IN THE SPECIFICATION, THIS TERM SHALL APPLY.
- 1.4. ALL WORK SHALL BE PERFORMED UNDER THE PERSONAL SUPERVISION OF A PROJECT SUPERINTENDENT ON-SITE. MAINTAIN A COMPLETE SET OF DRAWINGS AND SPECIFICATIONS ON SITE AT ALL TIMES DURING THE PROJECT.
- 1.5. ALL MATERIALS AND EQUIPMENT SHALL BE NEW AND SUITABLE FOR THE CONDITIONS AND DUTIES IMPOSED ON THEM AFTER INSTALLATION.
- 1.5.1. ALL MATERIALS, APPARATUS AND EQUIPMENT SHALL BEAR THE SEAL OF UNDERWRITERS LABORATORIES INC. (UL), OR A SIMILAR CREDIBLE TESTING AGENCY, LABEL WHERE REGULARLY SUPPLIED.
- 1.5.2. CERTAIN MANUFACTURERS OF MATERIAL AND EQUIPMENT ARE SPECIFIED AND PLANS ARE DETAILED ACCORDING TO THIS MATERIAL. CONTRACTOR SHALL BASE BID ON FURNISHING AND INSTALLING THIS MAKE OF MATERIAL AND EQUIPMENT.
- 1.5.3. WHERE MORE THAN ONE MAKE OF MATERIAL OR EQUIPMENT IS SPECIFIED, CONTRACTOR SHALL STATE IN BID WHICH MAKE THEY PROPOSE TO FURNISH. SHOP DRAWINGS SHALL BE SUBMITTED ON MATERIAL AND EQUIPMENT TO BE FURNISHED BY CONTRACTOR FOR ENGINEERS APPROVAL.
- 1.6. THIS APPROVAL TO BE OBTAINED PRIOR TO SHIPMENT OF EQUIPMENT.
- 1.7. ALL MATERIALS SHALL BE FABRICATED AND INSTALLED IN A NEAT AND WORKMANLIKE MANNER.
- 1.7.1. THE OWNER AND ENGINEER SHALL DETERMINE WHETHER WORKMANSHIP IS ACCEPTABLE. NO ALLOWANCES WILL BE MADE FOR REWORK OR DELAY DUE TO POOR WORKMANSHIP, COORDINATION DIFFICULTIES, OR INTERFERENCES BETWEEN INVOLVED TRADES.
- 1.7.2. PERFORM ALL WORK NECESSARY TO PREPARE THE STRUCTURE FOR THE INSTALLATION OF THE WORK. ALL HOLES, OPENINGS AND DAMAGED MATERIALS CREATED DURING CONSTRUCTION SHALL BE REPAIRED AND FINISHED BY EXPERIENCED WORKMEN.
- 1.7.3. COORDINATE AND SCHEDULE THE WORK WITH THE OWNER TO MINIMIZE DISRUPTIONS TO THE NORMAL OPERATIONS AT THE BUILDING.
- 1.7.3.1. IF AFTER HOURS WORK IS REQUIRED ON THIS PROJECT INCLUDE IN THE CONTRACT PRICE THE COST OF AFTER-HOURS WORK AND TEMPORARY PROVISIONS TO MINIMIZE DOWN TIME AND TO MAINTAIN FACILITY IN OPERATING CONDITION. COORDINATE WITH THE OWNER TO DETERMINE THE EXTENT OF THESE REQUIREMENTS PRIOR TO BID.
- 1.8. RELATED WORK SPECIFIED ELSEWHERE:
- 1.8.1. ALL DIVISION 1 REQUIREMENT, AND ALL TERMS AND CONDITIONS OF CONTRACT.
- 1.8.2. REFER TO ELECTRICAL SPECIFICATION FOR ELECTRICAL WORK TO BE DONE IN CONJUNCTION WITH THE MECHANICAL WORK. CONTRACTOR IS RESPONSIBLE FOR ALL CONDUIT, WIRING, JUNCTION BOXES, ETC., REQUIRED FOR HVAC CONTROLS, UNLESS SPECIFICALLY NOTED OTHERWISE.
- 1.8.3. ALL MECHANICAL EQUIPMENT AND WIRING PROVIDED UNDER DIVISION 23 SHALL COMPLY WITH THE ELECTRICAL SYSTEM CHARACTERISTICS INDICATED ON THE MECHANICAL DRAWINGS AND SPECIFIED DIVISION 26.
- 1.8.4. ELECTRIC CONTROLS, CONTACTORS, STARTERS, PILOT LIGHTS, PUSH BUTTONS, ETC., SHALL BE PROVIDED COMPLETE AS PART OF THE MOTOR, HEATER OR OTHER EQUIPMENT WHICH IT OPERATES. ALL ELECTRICAL COMPONENTS SHALL BE IN CONFORMANCE WITH THE REQUIREMENTS OF THE NATIONAL ELECTRICAL CODE AND DIVISION 26.
- 1.9. WHERE EQUIPMENT SPECIFICATIONS INDICATE THAT A FACTORY-AUTHORIZED SERVICE ENGINEER OR TECHNICIAN SHALL OBSERVE INSTALLATION, TEST & ADJUST, OR START-UP OF EQUIPMENT, ETC., SUCH SERVICES WILL BE CONTRACTED BY OWNER AS PART OF THE EQUIPMENT PURCHASE.
- 1.9.1. CONTRACTOR SHALL ARRANGE FOR, SCHEDULE, AND COORDINATE SUCH FIELD SERVICES AS WORK INCLUDED IN THE CONTRACT.
- 1.9.2. CONTRACTOR SHALL FURNISH ALL LABOR AND MATERIALS NECESSARY TO SUPPORT ALL SUCH FACTORY REPRESENTATIVE'S FIELD SERVICES.
- 1.10. REGULARLY DURING EACH WORKING DAY, REMOVE REFUSE AND DEBRIS ACCUMULATING FROM MECHANICAL CONSTRUCTION AND LEAVE AREA CLEAN AT END OF THE WORK DAY.
- 1.10.1. PRIOR TO ACCEPTANCE OF THIS WORK, LEAVE THE PREMISES "BROOM CLEAN" INSOFAR AS AFFECTED BY MECHANICAL WORK.
- 1.10.2. CLEAN THE INTERIOR OF EACH MECHANICAL COMPONENT OF DIRT AND CONSTRUCTION DUST INCLUDING BUT NOT LIMITED TO AIR HANDLER, CONDENSING UNITS, RTUS AND CONTROLLERS BEFORE ENERGIZING.

- 1.10.3. EXPOSED FINISHED MATERIALS AND EQUIPMENT SHALL BE CAREFULLY CLEANED AND WIPED TO REMOVE GREASE, SMUDGES, FINGERPRINTS, DUST AND OTHER SPOTS AND LEFT SMOOTH AND CLEAN.
- 1.10.4. CLEAN THE EXTERIOR OF MECHANICAL COMPONENTS PRIOR TO ACCEPTANCE OF WORK.
- 1.10.5. FOR ALL MATERIALS AND DEVICES REMOVED, THE CONTRACTOR SHALL DISPOSE OFF-SITE IN AN APPROVED MANNER. PROVIDE WRITTEN DOCUMENTATION FOR DISPOSAL OF ALL ITEMS.
- 1.11. A COMMISSIONING PLAN SHALL BE DEVELOPED BY A LICENSED DESIGN PROFESSIONAL, OR APPROVED AGENCY AND SHALL INCLUDE THE FOLLOWING ITEMS:
- 1.11.1. A NARRATIVE DESCRIPTION OF THE ACTIVITIES THAT WILL BE ACCOMPLISHED DURING EACH PHASE OF COMMISSIONING, INCLUDING THE PERSONNEL INTENDED TO ACCOMPLISH EACH OF THE ACTIVITIES.
- 1.11.2. A LISTING OF THE SPECIFIC EQUIPMENT, APPLIANCES OR SYSTEMS TO BE TESTED AND A DESCRIPTION OF THE TESTS TO BE PERFORMED.
- 1.11.3. FUNCTIONS TO BE TESTED INCLUDING, BUT NOT LIMITED TO, EQUIPMENT COMMISSIONING, CALIBRATIONS AND ECONOMIZER CONTROLS AND ADJUSTING BAROMETRIC RELIEF DAMPER.
- 1.11.4. CONDITIONS UNDER WHICH THE TEST WILL BE PERFORMED. TESTING SHALL AFFIRM WINTER AND SUMMER DESIGN CONDITIONS AND FULL OUTSIDE AIR CONDITIONS.
- 1.11.5. MEASURABLE CRITERIA FOR PERFORMANCE.
- 1.11.5.1. A REPORT OF TEST PROCEDURES AND RESULTS IDENTIFIED AS "FINAL COMMISSIONING REPORT" SHALL BE DELIVERED TO THE BUILDING OWNER OR OWNER'S AUTHORIZED AGENT. THE REPORT SHALL INCLUDE THE FOLLOWING:
- 1.11.5.2. RESULTS OF FUNCTIONAL PERFORMANCE TESTS.
- 1.11.5.3. DISPOSITION OF DEFICIENCIES FOUND DURING TESTING, INCLUDING DETAILS OF CORRECTIVE MEASURES USED OR PROPOSED.
- 1.11.5.4. FUNCTIONAL PERFORMANCE TEST PROCEDURES USED DURING THE COMMISSIONING PROCESS INCLUDING MEASURABLE CRITERIA FOR TEST ACCEPTANCE, PROVIDED HEREIN FOR REPEATABILITY.
- 1.11.5.5. EXCEPTION: DEFERRED TESTS THAT CANNOT BE PERFORMED AT THE TIME OF REPORT PREPARATION DUE TO CLIMATIC CONDITIONS.
- 1.12. PROVIDE ALL LABOR, INSTRUMENTS, AND OTHER SERVICES REQUIRED FOR COMPLETE AND SATISFACTORY TEST AND ADJUSTMENT OF MECHANICAL SYSTEMS AND RELATED WORK.
- 1.12.1. CHECK ALL MOTORS AND ROTATING EQUIPMENT FOR PROPER ROTATION.
- 1.12.2. CHECK ALL ELECTRICAL AND CONTROL WIRING, INTERLOCKS, ETC., RELATED TO MECHANICAL EQUIPMENT TO DETERMINE THAT ALL WIRING IS CORRECT.
- 1.12.3. IMMEDIATELY REMEDIATE ALL EQUIPMENT PROVIDED UNDER THIS DIVISION THAT TESTS PROVE TO BE DEFECTIVE OR OPERATING IMPROPERLY AS A PART OF THIS CONTRACT.
- 1.13. CONTRACTOR AND VENDOR SHALL INSTRUCT THE OWNER'S TECHNICAL PERSONNEL ON ALL OWNER FURNISHED EQUIPMENT IN ACCORDANCE WITH SPECIFICATIONS.
- 1.13.1. INCLUDE IN THE CONTRACT PRICE A MINIMUM OF (2) HOURS OF OWNER TRAINING FOR THE NEW CONTROLS.
- 1.13.2. TRAIN BUILDING OWNER'S PERSONNEL DURING NORMAL WORKING HOURS ON START-UP & SHUT-DOWN PROCEDURES, TROUBLESHOOTING PROCEDURES, SERVICING & PREVENTATIVE MAINTENANCE SCHEDULE & PROCEDURES, AND OVERRIDE OF EQUIPMENT & CONTROLS IN THE EVENT OF SYSTEM FAILURE. REVIEW WITH THE OWNER'S PERSONNEL, THE DATA CONTAINED IN THE OPERATING AND MAINTENANCE MANUALS. SCHEDULE TRAINING WITH OWNER, PROVIDE AT LEAST 7-DAYS PRIOR NOTICE TO ARCHITECT/ENGINEER.
- 1.13.3. CONTRACTOR SHALL PROVIDE OWNER WITH SUFFICIENT SETS OF OPERATIONS AND MAINTENANCE MANUALS OF CONTRACTOR FURNISHED EQUIPMENT FOR INCLUSION INTO THE OWNER'S OPERATIONS AND MAINTENANCE MANUALS AS REQUIRED BY THE SPECIFICATIONS.
- 1.13.4. PROJECT WILL NOT BE COMPLETE UNTIL ACCURATE O/M MANUALS ARE DELIVERED.
- 1.13.5. O/M MANUALS SHALL INCLUDE CATALOG TECHNICAL DATA, RECOMMENDED SERVICE PROCEDURES, RECOMMENDED SERVICE INTERVALS, CALIBRATION INFORMATION, FACTORY TRAINING MANUALS, MAGNETIC MEDIA FOR SOFTWARE PROVIDED, AND RECOMMENDED SPARE PARTS.
- 1.13.5.1. PROVIDE (2) NEATLY BOUND (WITH TABBED SECTIONS) COPIES OF MAINTENANCE BOOKS, INSTRUCTION BOOKS, AND PARTS LIST PERTAINING TO ALL EQUIPMENT FURNISHED. SUBMIT TO THE OWNERS REPRESENTATIVE FOR APPROVAL. FINAL PAYMENT WILL NOT BE MADE UNTIL MAINTENANCE AND INSTRUCTION MANUALS ARE DELIVERED TO THE OWNERS REPRESENTATIVE.
- 1.13.5.1.1.ORGANIZE OPERATING AND MAINTENANCE DATA INTO SUITABLE SETS OF MANAGEABLE SIZE.
- 1.13.5.1.2.BIND PROPERLY INDEXED DATA IN INDIVIDUAL, HEAVY-DUTY, 2-INCH, 3-RING VINYL-COVERED BINDERS WITH POCKET FOLDERS FOR FOLDED SHEET INFORMATION.
- 1.13.5.2. MARK APPROPRIATE IDENTIFICATION ON FRONT AND SPINE OF EACH BINDER. INCLUDE THE FOLLOWING TYPES OF INFORMATION:
- 1.13.5.2.1.SPARE PARTS LIST
- 1.13.5.2.2.COPIES OF GUARANTEES AND WARRANTIES OF ALL EQUIPMENT AND SYSTEMS.
- 1.13.5.2.3.WIRING DIAGRAMS
- 1.13.5.2.4.INSPECTION REPORTS & APPROVALS
- 1.13.5.2.5.SUBMITTAL DATA STATING EQUIPMENT SIZE AND SELECTED OPTIONS FOR EACH PIECE OF EQUIPMENT REQUIRING MAINTENANCE.
- 1.13.5.2.6.MANUFACTURER'S OPERATION MANUALS AND MAINTENANCE MANUALS FOR EACH PIECE OF EQUIPMENT REQUIRING MAINTENANCE, EXCEPT EQUIPMENT NOT FURNISHED AS PART OF THE PROJECT. REQUIRED ROUTINE MAINTENANCE ACTIONS SHALL BE CLEARLY IDENTIFIED.
- 1.13.5.2.7.CONTRACTOR'S NAME, ADDRESS, AND TELEPHONE NUMBERS.
- 1.13.5.2.8.NAME AND ADDRESS AND 24-HOUR PHONE NUMBER OF THE PERSONNEL RESPONSIBLE FOR SERVICING DURING THE WARRANTY PERIOD.
- 1.13.5.2.9.SUBMITTAL DATA INDICATING ALL SELECTED OPTIONS FOR EACH PIECE OF EQUIPMENT AND CONTROLS.

- 1.13.5.2.10. OPERATION AND MAINTENANCE MANUALS FOR EACH PIECE OF EQUIPMENT. REQUIRED ROUTINE MAINTENANCE ACTIONS, CLEANING RECOMMENDATIONS SHALL BE CLEARLY IDENTIFIED.
- 1.13.5.2.11. A SCHEDULE FOR INSPECTING AND RECALIBRATING ALL CONTROLS.
- 1.13.5.2.12. A NARRATIVE OF HOW EACH SYSTEM IS INTENDED TO OPERATE, INCLUDING RECOMMENDED SET POINTS.
- 1.13.5.2.13. ALL MANUFACTURERS' DATA APPLICABLE TO THE INSTALLED EQUIPMENT SUCH AS INSTALLATION INSTRUCTIONS, MAINTENANCE INSTRUCTIONS, OPERATING INSTRUCTIONS, ETC.
- 1.13.5.2.14. CONTROL DIAGRAMS INCLUDING AN EXPLANATION OF THE CONTROL SEQUENCE OF EACH SYSTEM INCLUDING NORMAL STARTING, OPERATION, SHUTDOWN, POWER FAIL RE-START, AND WINTER SHUTDOWN.
- 1.13.5.2.15. SCHEDULE OF MAINTENANCE AND INSPECTION THAT SHOULD BE PERFORMED BY OWNER, AND THOSE WHICH SHOULD BE PROVIDED BY AN OUTSIDE SERVICE.
- 1.13.5.2.16. COPY OF COMPLETED FIELD ACCEPTANCE TEST PROCEDURE, LOGS, AND ALL DOCUMENTATION ASSOCIATED WITH THE TEST.
- 1.14. GUARANTEE ALL MECHANICAL SYSTEM EQUIPMENT, MATERIALS, AND WORKMANSHIP TO BE FREE FROM DEFECTS FOR A PERIOD OF ONE (1) YEAR FROM THE DATE OF FINAL ACCEPTANCE AND PROPERLY CORRECT LATENT DEFECTS ARISING DURING THIS PERIOD UPON NOTIFICATION BY THE OWNER'S REPRESENTATIVE WITHOUT ADDITIONAL COMPENSATION AND TO THE SATISFACTION OF THE ENGINEER AND OWNER'S REPRESENTATIVE.
- 1.15. ALL EQUIPMENT, ETC., SHALL BE NEW UNLESS OTHERWISE NOTED, AND AS SPECIFIED FREE OF DEFECTS. ALL MECHANICAL EQUIPMENT SHALL BE U.L. OR E.T.L. LISTED.
- 1.16. PROVIDE THUMB DRIVE TO THE OWNER REPRESENTATIVE WITH ALL THE ABOVE LISTED MATERIAL SEPARATED INTO DIRECTORIES TO MATCH CATALOGED INFORMATION.
2. CODES & PERMITS
- 2.1. ENTIRE INSTALLATION (INCLUDING EQUIPMENT, DEVICES, AND WIRING) SHALL BE IN ACCORDANCE WITH ALL APPLICABLE PROVISIONS OF THE NATIONAL ELECTRICAL CODE (N.E.C.), NATIONAL FIRE PROTECTION ASSOCIATION (NFPA 13, NFPA 70 & NFPA 101), OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (O.S.H.A.), CURRENT BUILDING CODE, CURRENT ENERGY CONSERVATION CODE (I.E.C.C.), AND ALL LAWS & ORDINANCES APPLICABLE TO WORK AT THIS SITE. IN ADDITION, INSTALLATION SHALL MEET APPROVAL OF LOCAL INSPECTION AUTHORITY HAVING JURISDICTION. REFER TO COVER SHEET FOR LIST OF CURRENT APPLICABLE CODE EDITIONS.
- 2.2. SECURE AND PAY ALL FEES ASSOCIATED WITH ALL PERMITS AND LICENSES REQUIRED FOR EXECUTION OF THE CONTRACT. ARRANGE FOR ALL INSPECTIONS REQUIRED BY CITY, COUNTY, STATE AND OTHER AUTHORITIES HAVING JURISDICTION, AND DELIVER CERTIFICATES OF APPROVAL TO THE ARCHITECT.
- 2.3. A CERTIFICATE OF APPROVAL FOR WORK FROM INSPECTION AUTHORITY SHALL BE GIVEN TO THE OWNER BEFORE FINAL ACCEPTANCE WILL BE GIVEN BY OWNERS REPRESENTATIVE.
- 2.4. THE CODE REQUIREMENTS ARE STRICTLY A MINIMUM AND SHALL BE MET WITHOUT INCLUDING ADDITIONS TO THE CONTRACT. WHERE REQUIREMENTS OF THE DRAWINGS OR SPECIFICATIONS EXCEED THE CODE REQUIREMENTS, THE WORK SHALL BE PROVIDED IN ACCORDANCE WITH THESE DRAWINGS OR SPECIFICATIONS. IN THE EVENT OF CONFLICT OR AMBIGUITY BETWEEN THE VARIOUS CODES, THE MOST STRINGENT REQUIREMENTS SHALL GOVERN.
- 2.5. REFER TO ARCHITECTURAL COVERSHEET FOR APPLICABLE CODES REFERENCES FOR THIS PROJECT.
3. SAFETY
- 3.1. THE CONTRACTOR SHALL MAINTAIN A SAFE WORK ENVIRONMENT AT ALL TIMES.
- 3.1.1. COMPLY WITH ALL O.S.H.A., N.I.O.S.H., D.O.T., STATE & LOCAL REQUIREMENTS REGARDING SAFE HANDLING, STORING, TRANSPORTING, AND DISPENSING OF CHEMICALS.
- 3.1.2. MAINTAIN AND DISPLAY M.S.D.S. INFORMATION FOR ALL CHEMICAL PRODUCTS.
- 3.1.3. PROVIDE ALL NECESSARY MEANS TO MAINTAIN SAFE WORKING CONDITIONS, INCLUDING VENTILATION FANS, FIRE EXTINGUISHERS, EYE PROTECTION, RESPIRATORS, PROTECTIVE CLOTHING, VENTILATION, ETC.
- 3.1.4. ALL EQUIPMENT AND MATERIALS USED TO IMPLEMENT THE WORK SHALL BE USED IN STRICT ACCORDANCE WITH THE MANUFACTURER'S WRITTEN INSTRUCTIONS, INCLUDING ALL RECOMMENDED SAFETY PRECAUTIONS.
- 3.1.5. MAINTAIN A PROPER FIRE WATCH FOR ALL OPERATIONS WHERE SPARKS, FLAMES, OR OTHER SOURCES OF FIRE ARE PRODUCED.
- 3.1.6. FOR ALL MATERIALS CONTAINING SOLVENTS, MAINTAIN THE RECOMMENDED VENTILATION OF THE AREA TO PREVENT THE ACCUMULATION OF VAPORS WHICH POSE A HEALTH OR FIRE HAZARD.
4. INTENT OF DRAWINGS AND SPECIFICATIONS
- 4.1. THE IMPLIED AND STATED INTENT OF THE DRAWINGS & SPECIFICATIONS ARE TO ESTABLISH MINIMUM ACCEPTABLE STANDARDS FOR MATERIALS, EQUIPMENT, WORKMANSHIP, AND TO PROVIDE OPERABLE SYSTEMS THAT ARE COMPLETE IN EVERY RESPECT.
- 4.2. ENGINEERING DRAWINGS ARE DIAGRAMMATIC, INTENDED TO SHOW GENERAL ARRANGEMENT AND SIZES OF SYSTEM COMPONENTS, AND SHALL NOT BE SCALED.
- 4.2.1. ARCHITECTURAL AND STRUCTURAL DRAWINGS SHALL GOVERN SPACE CONSTRAINTS, DIMENSIONS AND FINISHES.
- 4.2.2. ALL OFFSETS AND FITTINGS THAT SHALL BE NECESSARY TO ACCOMPLISH A FINISHED INSTALLATION SHALL BE PROVIDED AT NO ADDITIONAL COST OR INCREASE THE CONTRACT.
- 4.2.3. WORK INTENDED, BUT HAVING MINOR DETAILS OBVIOUSLY OMITTED, SHALL BE PROVIDED COMPLETE AS A REQUIREMENT OF THIS CONTRACT.
- 4.2.4. LOCATIONS OF EQUIPMENT INDICATED ON PLANS SHALL BE FOLLOWED AS CLOSELY AS POSSIBLE TO THE PLANS SUBJECT TO BUILDING CONSTRUCTION AND INTERFERENCES WITH OTHER TRADES.
- 4.2.5. MAINTAIN MINIMUM SERVICE CLEARANCE AS REQUIRED BY THE EQUIPMENT MANUFACTURER AND N.E.C.
- 4.3. PROVIDE THE OWNER A COMPLETE SET OF RECORD DRAWINGS AT THE END OF THE PROJECT. PROJECT WILL NOT BE COMPLETE UNTIL ACCURATE RECORD DRAWINGS ARE DELIVERED.
- 4.4. THE RECORD DRAWINGS SHALL BE MACHINE DRAFTED, AND SHALL BE PROVIDED BOTH ON REPRODUCIBLE VELLUMS AND MAGNETIC MEDIA CAD FILES TO THE ARCHITECT WHICH REFLECT ALL CHANGES, DEVIATIONS AND REVISIONS MADE TO THE ORIGINAL DESIGN DOCUMENTS. LOCATIONS OF ALL UNDERGROUND PIPING AND UTILITIES SHALL BE CLEARLY SHOWN AND DIMENSIONED FROM PERMANENT REFERENCE POINTS SUCH AS BUILDING

COLUMN LINES.

- 4.5. DRAWING FORMAT FOR THIS PROJECT SHALL BE AUTOCAD 2010.
- 4.6. ALL ITEMS MOUNTED IN OR BELOW THE CEILING, AND ALL ITEMS PENETRATING THE CEILING, SHALL BE COORDINATED WITH THE ARCHITECTURAL REFLECTED CEILING PLANS. IF ANY ITEMS ARE NOT SHOWN ON THESE PLANS, OR ANY ITEMS NEED TO BE RELOCATED FOR COORDINATION PURPOSES, PREPARE A REFLECTED CEILING PLAN SUBMIT IT TO THE ARCHITECT FOR APPROVAL.
5. EXISTING CONDITIONS
- 5.1. ATTENTION IS CALLED TO THE FACT THAT THE WORK IS TO BE PERFORMED WITHIN AN EXISTING, OPERATIONAL FACILITY.
- 5.2. THE FOLLOWING GENERAL PROVISIONS OF THE CONTRACT, INCLUDING THE GENERAL & SUPPLEMENTAL CONDITIONS AND GENERAL REQUIREMENTS, SHALL APPLY TO THE WORK IN THIS DRAWING AND SPECIFICATION SET.
- 5.2.1. VISIT THE SITE OF THE WORK AND BECOME THOROUGHLY FAMILIAR WITH ALL EXISTING CONDITIONS, AND THOROUGHLY REVIEW ALL DRAWINGS, SPECIFICATIONS AND ADDENDA PRIOR TO BIDDING ON THIS WORK. NO EXTRA PAYMENTS TO THE CONTRACT AMOUNT WILL BE ALLOWED FOR FAILURE TO COMPLY WITH THIS REQUIREMENT.
- 5.2.2. TAKE MEASUREMENTS AND BE RESPONSIBLE FOR EXACT SIZE AND LOCATIONS OF ALL OPENINGS REQUIRED FOR THE INSTALLATION OF WORK.
- 5.2.3. FIELD DIMENSIONS ARE REASONABLY ACCURATE AND SHOULD GOVERN IN SETTING OUT WORK.
- 5.2.4. WHERE DETAILED METHOD OF INSTALLATION IS NOT INDICATED OR WHERE VARIATIONS EXIST BETWEEN DESCRIBED WORK AND APPROVED PRACTICE, DIRECTION OF THE OWNERS REPRESENTATIVE ON JOB SITE SHALL BE FOLLOWED.
- 5.3. CONTRACTOR SHALL VERIFY PROJECT CONDITIONS TO ENSURE THAT THE WORK WILL FIT INTO THE STRUCTURE IN THE MANNER INTENDED ON THE DRAWINGS.
- 5.3.1. SHOULD ANY CONDITIONS EXIST THAT ARE CONTRARY TO THOSE SHOWN ON THE DRAWINGS, CONTRACTOR SHALL NOTIFY THE ENGINEER PRIOR TO FABRICATION OR PERFORMING ANY WORK IN THE AREA INVOLVING THE DIFFERENCES.
- 5.3.2. NOTIFICATION SHALL BE IN THE FORM OF A DRAWING OR SKETCH INDICATING FIELD MEASUREMENTS OR NOTES RELATING TO THE AREA.
- 5.4. CONNECT NEW WORK TO EXISTING WORK IN A NEAT AND WORKMANLIKE MANNER.
- 5.4.1. WHERE AN EXISTING STRUCTURE MUST BE CUT OR EXISTING UTILITIES INTERFERE, SUCH OBSTRUCTION SHALL BE BYPASSED, REMOVED, REPLACE OR RELOCATED, PATCH AND REPAIR.
- 5.4.2. WORK DISTURBED OR DAMAGED SHALL BE REPLACED OR REPAIR TO ITS PRIOR CONDITION.
- 5.5. PRIOR TO THE START OF ANY DEMOLITION OR CONSTRUCTION, CONTRACTOR SHALL REVIEW THE EXISTING SITE AND SECURE THE SERVICES OF A QUALIFIED, EPA CERTIFIED ASBESTOS ABATEMENT AGENCY IF NEEDED TO CHECK THE EXISTING INSULATION, ETC. FOR ASBESTOS. SHOULD ASBESTOS BE FOUND, DO NOT PROCEED WITH DEMOLITION OR CONSTRUCTION; NOTIFY THE ARCHITECT IN ANY CASE IN WRITING OF THE AGENCY'S FINDING.
- 5.6. FOR RENOVATION PROJECTS – PROVIDE ALL DEMOLITION, PATCHING, SAW CUTTING, EXCAVATION, TRENCHING, SHORING, COMPACTING, DE-WATERING, ETC. REQUIRED FOR THE PROJECT, WHETHER OR NOT SHOWN ON THE DRAWINGS.
- 5.6.1. INFORMATION WAS TAKEN FROM VARIOUS ARCHIVE DRAWINGS AND LIMITED FIELD OBSERVATION. FIELD VERIFICATION OF EXISTING CONDITIONS AND POINTS OF CONNECTIONS ARE REQUIRED.
- 5.7. EXISTING SYSTEMS TO REMAIN – WHERE EXISTING SYSTEMS ARE INDICATED TO REMAIN, THEY SHALL BE ASSUMED TO BE IN GOOD WORKING ORDER REQUIRING NO WORK UNLESS SPECIFICALLY NOTED.
- 5.7.1. IF DURING THE CONSTRUCTION DEFICIENCIES ARE NOTED, THEN BRING THESE TO THE ATTENTION OF THE OWNER AND SEEK DIRECTION.
6. SHOP DRAWINGS, SUBMITTALS, AND SUBSTITUTIONS
- 6.1. ENGINEER OF RECORD SHALL BE PROVIDED WITH SHOP DRAWINGS, COORDINATION DRAWINGS, AND MANUFACTURER'S DATA OF ANY CONTRACTOR FURNISHED MATERIALS AND EQUIPMENT, PRIOR TO PURCHASE AND/OR FABRICATION, AND SHALL VERIFY, BY STAMPING AND SIGNING THE DATA AND DRAWINGS BEFORE RETURNING THEM TO THE CONTRACTOR, THAT THE ITEMS FURNISHED BY THE CONTRACTOR FIT THE SPACES AND DIMENSIONS DESCRIBED IN AND CONFORM TO THE SPIRIT AND INTENT OF THE CONTRACT DOCUMENTS.
- 6.1.1. ENGINEER SHALL, WITHIN FIVE (5) WORKING DAYS OF RECEIPT OF SHOP DRAWINGS AND PRODUCT DATA, NOTIFY THE CONTRACTOR OF ANY DISCREPANCY OR INCOMPATIBILITY WITH THE CONTRACT DOCUMENTS, AND SHALL RETURN THE SHOP DRAWINGS TO THE CONTRACTOR APPROPRIATELY ANNOTATED.
- 6.1.2. REVIEW OF SUBMITTALS SHALL NOT BE CONSTRUED AS AUTHORIZING ANY DEVIATIONS FROM THE PLANS AND SPECIFICATIONS UNLESS SUCH DEVIATIONS ARE CLEARLY IDENTIFIED AND SEPARATELY SUBMITTED IN THE FORM OF A LETTER THAT IS ENCLOSED WITH THE SUBMITTALS.
- 6.1.3. SUBMIT SHOP DRAWING CUT SHEETS AND TECHNICAL DATA ON THE FOLLOWING ITEMS: EQUIPMENT, DUCTWORK, PIPING, AIR DEVICES AND MOTOR STARTERS, TRANSFORMERS, AND ANY OTHER ITEM REQUIRED BY NOTES.
- 6.1.4. SUBMIT CATALOG INFORMATION, FACTORY ASSEMBLY DRAWINGS, FIELD INSTALLATION DRAWINGS AND CERTIFICATIONS AS REQUIRED FOR COMPLETE EXPLANATION AND DESCRIPTION OF ALL ITEMS OF EQUIPMENT. THE SUBMITTAL DATA SHALL PROVIDE AMPLE, UNQUESTIONABLE COMPLIANCE WITH THE CONTRACT DOCUMENTS.
- 6.1.5. SUBMIT PROPOSED CONTROL SYSTEM FOR REVIEW BY THE ENGINEER PRIOR TO EQUIPMENT PURCHASE OR FABRICATION. DO NOT PROCEED WITH THE WORK WITHOUT APPROVED SUBMITTALS.
- 6.1.6. WHERE QUALIFICATIONS AND/OR QUALITY ASSURANCE REQUIREMENTS ARE SPECIFIED, THE SUBMITTAL SHALL INCLUDE EVIDENCE THAT THE STATED REQUIREMENTS HAVE BEEN MET. INCLUDE QUALIFICATIONS AND CERTIFICATIONS OF PROPOSED TEST AND BALANCE SUBCONTRACTOR.
- 6.1.7. EQUIPMENT PERFORMANCE SHALL BE VERIFIED BY THE EQUIPMENT MANUFACTURER AS PART OF THE SUBMITTAL. PRIOR TO ORDERING, VERIFY EQUIPMENT VOLTAGE AND ELECTRICAL REQUIREMENTS OF THE EQUIPMENT WITH ELECTRICAL CONTRACTOR PRIOR TO ORDERING EQUIPMENT. SHOP DRAWINGS FOR EQUIPMENT REQUIRING ELECTRIC POWER OR CONTROL WIRING CONNECTIONS SHALL INCLUDE COMPLETE WIRING DIAGRAMS.
- 6.2. REFER TO THE CONDITIONS OF THE CONTRACT (GENERAL AND SUPPLEMENTARY WHEN INCLUDED AND OTHER DIVISIONS) REFERENCING SUBMITTALS FOR SHOP DRAWINGS, PRODUCT DATA, AND SAMPLES FOR SUBMITTAL DEFINITIONS, REQUIREMENTS AND PROCEDURES.
- 6.3. BEFORE PREPARING SUBMITTALS, STUDY ALL CONTRACTS DRAWINGS AND SPECIFICATIONS

- IN DETAIL, OBTAIN MANUFACTURER'S RECOMMENDED INSTRUCTIONS, AND HAVE SUBMITTALS PREPARED BASED ON SPECIFIC EQUIPMENT AND MATERIAL PROPOSED FOR INSTALLATION.
- 6.3.1. AN OFFICER OF THE CONTRACTING FIRM SHALL SIGN ALL SHOP DRAWINGS (CERTIFYING CONFORMANCE WITH PLANS AND SPECIFICATIONS) BEFORE SUBMITTING TO THE ARCHITECT OR RELEASING TO THE FIELD.
- 6.4. THE SUBMITTAL PROCESS SHALL NOT BE UTILIZED AS AN AVENUE TO SUBSTITUTE AFTER THE EXECUTION OF THE CONTRACT.
- 6.4.1. SHOULD AN UNSPECIFIED OR UNEQUAL PRODUCT BE SUBMITTED, IT WILL BE REJECTED. IF A SECOND ATTEMPT AT SUBSTITUTION IS MADE DURING THE RE-SUBMITTAL OF THE SAME PRODUCT, THEN NO MORE REVIEWS OF THAT PRODUCT WILL BE PERFORMED WITHOUT DIRECT COMPENSATION TO THE ENGINEER BEING PAID FOR THE ADDITIONAL SERVICES REQUIRED FOR THE THIRD REVIEW AND ANY FURTHER REVIEWS.
- 6.5. IN ADDITION TO THE OTHER REQUIREMENTS IN PARAGRAPHS ABOVE, THE FOLLOWING APPLIES TO SUBMITTALS OF THIS DIVISION:
- 6.5.1. DO NOT INSTALL OR ORDER MECHANICAL EQUIPMENT OR PROCEED WITH THE WORK UNTIL SUBMITTALS HAVE BEEN ACCEPTABLY REVIEWED BY THE OWNER'S REPRESENTATIVE AND STAMPED ACCORDINGLY. EQUIPMENT OR WORK WHICH IS ORDERED OR INSTALLED WITHOUT PRIOR APPROVED SUBMITTALS SHALL, AT THE ENGINEER'S DISCRETION, BE REMOVED AT NO COST TO THE OWNER. NO ALLOWANCES WILL BE MADE FOR REWORK OR DELAY DUE TO NEGLECT OF REQUIRED APPROVAL PROCESS.
- 6.5.1. MAKE ALL MECHANICAL SUBMITTALS AT ONE TIME AND WITHIN FOURTEEN (14) CALENDAR DAYS OF OWNER'S "NOTICE TO PROCEED".
- 6.5.2. SUBMIT A MINIMUM OF SIX (6) COMPLETE SETS TO OWNER'S REPRESENTATIVE. ONE SET SHALL REMAIN WITH THE OWNER'S REPRESENTATIVE, ONE SET SHALL BE DELIVERED TO THE OWNER AND THE OTHERS WILL BE RETURNED TO DISTRIBUTE AS REQUIRED.
- 6.5.2. EACH SET SHALL BE BOUND AND INDEXED INTO GROUPS SUCH AS FIXTURES, PANELS, WRING DEVICES, DISCONNECT SWITCHES, ETC.
- 6.5.3. MAKE EACH SET IDENTICAL.
- 6.5.4. ALL SUBMITTALS SHALL BE ONLY ORIGINALS, COPIES OF CUT SHEETS ARE NOT ACCEPTABLE.
- 6.5.5. SUBMITTAL DATA WILL BE PROVIDED AND REVIEWED IN PDF FORMAT ELECTRONICALLY VIA EMAIL.
- 6.5.6. SUBMITTAL WILL NOT BE ACCEPTED FOR REVIEW UNLESS THEY:
- 6.5.7.1. COMPLY WITH THE REQUIREMENTS OF DIVISION 1.
- 6.5.7.2. INCLUDE COMPLETE INFORMATION PERTAINING TO ALL APPURTENANCE AND ACCESSORIES.
- 6.5.7.3. ARE SUBMITTED AS COMPLETE PACKAGES WHICH PERTAIN TO ALL RELATED ITEMS IN DIVISION 23.
- 6.5.7.4. ARE PROPERLY MARKED WITH EQUIPMENT, SERVICE OR FUNCTION IDENTIFICATION AS RELATED TO THE PROJECT AND ARE MARKED WITH PERTINENT SPECIFICATION PARAGRAPH NUMBER.
- 6.5.8. IF ANY ITEM IN THE SUBMITTAL IS "NOT ACCEPTABLE" FOR ANY REASON, IT AUTOMATICALLY VOIDS THE ENTIRE SET, AND A RE-SUBMIT OF ALL IS REQUIRED TO OBTAIN ACCEPTANCE OF ALL ITEMS.
- 6.5.9. IN CASE OF DISCREPANCIES BETWEEN SETS OF SUBMITTALS, THE SET RETAINED BY THE OWNER'S REPRESENTATIVE SHALL HAVE PRECEDENCE.
- 6.6. EQUIPMENT AND DESIGN OF SYSTEMS INDICATED ON THE DESIGN DRAWINGS AND WITHIN THESE SPECIFICATIONS SHALL BE CONSIDERED AS SPECIFIED STANDARD OF QUALITY.
- 6.6.1. NO SUBSTITUTIONS SHALL BE MADE WITHOUT PRIOR WRITTEN APPROVAL OF THE ENGINEER.
- 6.6.2. ALL COSTS ARISING FROM A SUBSTITUTION SHALL BE THE RESPONSIBILITY OF THE CONTRACTOR MAKING THE SUBSTITUTION, INCLUDING VERIFICATION OF FIT AND ACCESS, FIELD-INSTALLED ACCESSORIES, SUPPORTS, ELECTRICAL REQUIREMENTS, AND REVISIONS TO DOCUMENTS (DESIGN COSTS).
- 6.7. WHERE MATERIAL OR EQUIPMENT IS IDENTIFIED BY PROPRIETARY NAME, MODEL NUMBER AND/OR MANUFACTURER, FURNISH THE NAMED ITEM OR EQUAL THEREOF. SUBSTITUTED ITEMS SHALL BE EQUAL OR BETTER IN QUALITY AND PERFORMANCE AND MUST BE SUITABLE FOR THE AVAILABLE SPACE, REQUIRED ARRANGEMENT AND APPLICATION.
- 6.7.1. MANUFACTURER'S NAME, SERIES AND MODEL NUMBERS, AS NOTED OR SPECIFIED, ARE FOR THE PURPOSE OF DESCRIBING TYPE, CAPACITY, AND QUALITY OF EQUIPMENT, MATERIALS AND PRODUCTS TO BE USED.
- 6.7.2. UNLESS "OR EQUAL" IS SPECIFICALLY STATED, BIDS SHALL BE BASED ONLY ON THE SPECIFIED BASIS OF DESIGN MANUFACTURER.
- 6.7.3. THE LISTING OF A PARTICULAR MANUFACTURER AS AN "EQUAL" OR "ACCEPTABLE" SUBSTITUTE MANUFACTURER SHALL NOT BE MISCONSTRUED AS APPROVING NOR ALLOWING THE SUBSTITUTION OF THAT MANUFACTURER'S STANDARD PRODUCT IN PLACE OF THE BASIS DESIGN.
- 6.7.4. NO CONSIDERATION WILL BE GIVEN TO A PRODUCT WHICH WOULD REQUIRE DIMENSIONAL, SPATIAL OR AESTHETIC CHANGES TO THE PROJECT. "ACCEPTABLE SUBSTITUTE" AND "EQUAL" MANUFACTURERS SHALL ONLY BID THOSE PRODUCTS WHICH EXACTLY MATCH THE SIZE AND OTHER CHARACTERISTIC OF THE SPECIFIED BASIS OF DESIGN.
- 6.7.5. ANY CHANGES TO OTHER DISCIPLINES AND TRADES OF WORK REQUIRED BY AN "OR EQUAL" OR "SUBSTITUTE" PRODUCT SHALL BE DULY CONSIDERED AND PRICED ACCORDINGLY PRIOR TO BIDDING OR PRICING.
- 6.7.6. THE DECISION AS TO WHETHER OR NOT A PROPOSED SUBSTITUTE OR "EQUAL" PRODUCT IS ACTUALLY EQUAL TO THAT SPECIFIED SHALL REST SOLELY WITH THE ENGINEER.
- 6.7.7. REQUEST TO PROVIDE "EQUAL" PRODUCTS IN LIEU OF THOSE SPECIFIED SHALL BE SUBMITTED TO THE ARCHITECT IN WRITING AT LEAST TEN (10) DAYS PRIOR TO FINAL PRICING AND EXECUTION OF THE CONTRACT. NO CONSIDERATION WILL BE GIVEN TO SUBSTITUTE PRODUCTS AFTER FINAL PRICING AND EXECUTION OF THE CONTRACT.
- 6.7.8. ANY "OR EQUAL" PRODUCT OR PROPOSED PRODUCTION SUBSTATION WHICH WILL CAUSE A CHANGE IN THE APPEARANCE, DIMENSIONS OR DESIGN OF ANY PART OF THE BUILDING, IF STRUCTURE, MECHANICAL SYSTEM OR ANY OTHER ENGINEERED SYSTEM SHALL BE ACCOMPANIED BY A SCALED DRAWING AND WRITTEN DESCRIPTION OF THE REQUIRED CHANGES(S) FOR APPROVAL BY THE ARCHITECT.
- 6.7.8.1. IF DEEMED NECESSARY BY ARCHITECT, DESIGN CHANGES SHALL BE SIGNED AND SEAL BY A REGISTERED PROFESSIONAL ENGINEER, CURRENTLY LICENSED IN THIS STATE.



AR9238
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ARCHITECTURE
ENGINEERING
PERMITTING

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SEAL:

THIS DOCUMENT IS NOT
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OR CONSTRUCTION.

NO DATE REMARKS
REVISIONS



BAGEL & CO
TALLAHASSEE

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PROJECT NO: 2023.0719
DATE: 04/12/2024

M8.0
MECHANICAL
SPECIFICATIONS

CHECKED: DAK DRAWN: RAN

	7. INSTALLATION 7.1. PROVIDE ALL EQUIPMENT FOR THIS CONTRACT NEATLY AND WITH WORKMANSHIP AS DEFINED BY A.C.C.A. "STANDARD 5 PRACTICES FOR GOOD WORKMANSHIP IN MECHANICAL CONSTRUCTION", LEVEL AND PLUMB, AND SECURELY SUPPORTED. 7.1.1. THE ENTIRE INSTALLATION, AND MANNER OF INSTALLATION SHALL MEET THE COMPLETE SATISFACTION OF THE OWNER'S REPRESENTATIVE OR IT SHALL BE REMOVED AND REWORKED AS DIRECTED BY THE OWNER'S REPRESENTATIVE AS INCLUDED IN THE CONTRACT AMOUNT. 7.2. CONTRACTOR SHALL CONSULT PLUMBING, FIRE PROTECTION, ELECTRICAL, AND STRUCTURAL PLANS (WHERE APPLICABLE) IN ALL INSTANCES BEFORE INSTALLING WORK SO THAT WORK WILL NOT INTERFERE WITH THOSE BRANCHES. 7.2.1. IN THE EVENT OF A CONFLICT, CONTRACTOR SHALL REPORT TO THE OWNERS REPRESENTATIVE AT ONCE AND DO NO FURTHER WORK UNTIL A SATISFACTORY ARRANGEMENT IS DECIDED UPON. 7.2.2. ANY WORK DONE, OR EQUIPMENT PLACED IN POSITION, BY CONTRACTOR THAT CREATES A CONFLICT IN VIOLATION HEREOF, SHALL BE READJUSTED TO THE SATISFACTION OF THE OWNER'S REPRESENTATIVE AT THE EXPENSE OF THE CONTRACTOR. THE DECISION OF THE OWNERS REPRESENTATIVE SHALL BE FINAL IN REGARD TO CHANGES DUE TO CONFLICTING CONDITIONS. 7.2.3. CONTRACTOR SHALL COMPLETE WORK OR ANY PART THEREOF AT SUCH TIME AS MAY BE DESIGNATED BY THE OWNER, SO THAT IT CAN BE USED FOR TEMPORARY OR PERMANENT USE AND SUCH USE OF THE SYSTEM SHALL NOT BE CONSTRUED AS AN ACCEPTANCE OF SAME BY OWNER. 7.2.4. VERIFY VOLTAGE AND ALL ELECTRICAL REQUIREMENTS OF MECHANICAL EQUIPMENT AND SYSTEMS WITH DIV. 26 CONTRACTOR PRIOR TO ORDERING. 7.2.5. COORDINATION OF SPACE REQUIREMENTS WITH RESPECT TO DIVISION 23 SHALL BE PERFORMED SUCH THAT: 7.2.5.1. NO EQUIPMENT, PIPING OR DUCTWORK, OTHER THAN ELECTRICAL, SHALL BE INSTALLED WITHIN 42" OF SWITCHBOARDS OR PANELBOARDS. 7.3. TWO SETS OF MECHANICAL DRAWINGS SHALL BE PROVIDED AS RECORD DRAWINGS WHICH SHALL BE SEPARATE, CLEAN, COPIES RESERVED FOR THE PURPOSE OF SHOWING A COMPLETE PICTURE OF THE WORK AS ACTUALLY INSTALLED. 7.3.1. DRAWINGS SHALL ALSO SERVE AS WORK PROGRESS REPORT SHEETS AND THE MECHANICAL CONTRACTOR SHALL MAKE ANY NOTATIONS, NEAT AND LEGIBLE THEREON DAILY AS WORK PROCEEDS. 7.3.2. DRAWINGS SHALL BE AVAILABLE FOR INSPECTION AT ALL TIMES AND SHALL BE KEPT AT THE JOB AT A LOCATION DESIGNATED BY THE OWNERS REPRESENTATIVE. 7.3.3. AT THE COMPLETION OF THE WORK, THESE RECORD DRAWINGS SHALL BE SIGNED BY THE MECHANICAL CONTRACTOR, DATED AND RETURNED TO THE OWNERS REPRESENTATIVE. FINAL PAYMENT OF CONTRACT WILL NOT BE MADE UNTIL RECEIPT AND REVIEW OF SAID DRAWINGS. 7.4. PROVIDE ALL ROOF, WALL AND FLOOR PENETRATIONS REQUIRED TO COMPLETE INSTALLATION AND REMOVAL OF WORK (MAINTAIN FIRE RATING OF EXISTING STRUCTURE). 7.4.1. ALL PENETRATIONS SHALL BE PATCHED AND FINISHED TO MATCH SURROUNDING SURFACES AND FINISHES. 7.4.2. ALL EQUIPMENT OR PIPE PENETRATIONS THROUGH WALL, ROOF AND FLOORS SHALL BE SLEEVED AND SEALED SO AS TO BE WATER AND AIR TIGHT. 7.4.3. ALL ROOF CUTS AND REPAIRS SHALL BE PERFORMED BY OWNER APPROVED ROOFING CONTRACTOR IN ORDER TO MAINTAIN ROOF WARRANTY. 7.5. CUTTING OF HOLES THROUGH CONCRETE AND MASONRY SHALL BE BY DIAMOND CORE CONCRETE SAW. 7.5.1. PNEUMATIC HAMMER, IMPACT ELECTRIC AND HAND OR MANUAL HAMMER TYPE DRILLS WILL NOT BE ALLOWED, EXCEPT AS PERMITTED BY THE ARCHITECT WHERE REQUIRED BY LIMITED WORKING SPACE. 7.5.2. LOCATE HOLES SUCH THAT THEY WILL NOT AFFECT STRUCTURAL SECTIONS SUCH AS RISBS OR BEAMS. HOLES SHALL BE LAID OUT WELL IN ADVANCE OF THE INSTALLATION. THESE LAYOUT LOCATIONS SHALL BE APPROVED BY THE ARCHITECT PRIOR TO DRILLING. 7.6. ALL WIRING SHALL BE IN CONDUIT. THE USE OF E.N.T., BX, NM, ETC. OR PRE-MANUFACTURED CABLE ASSEMBLIES OR ALUMINUM WIRE WILL NOT BE PERMITTED. 7.7. VERIFY FINAL LOCATIONS FOR ROUGH-INS WITH SHOP DRAWING SUBMITTALS, FIELD MEASUREMENTS, AND WITH REQUIREMENTS OF THE ACTUAL EQUIPMENT TO BE CONNECTED PRIOR TO ROUGH-IN. 7.7.1. REFER TO EQUIPMENT SPECIFICATIONS AND DIMENSIONS SPECIFIED FOR ROUGH-IN REQUIREMENTS. 7.8. PROVIDE ALL CUTTING AND PATCHING REQUIRED FOR INSTALLATION OF MECHANICAL WORK. 7.8.1. PATCH EXISTING FINISHED AND DISTURBED NEW FINISHED SURFACES AND BUILDING COMPONENTS USING NEW MATERIALS MATCHING EXISTING MATERIALS & EXPERIENCED INSTALLERS. INSTALLERS' QUALIFICATIONS REFER TO THE MATERIALS AND METHODS REQUIRED OR THE SURFACE AND BUILDING COMPONENTS BEING PATCHED. 7.8.2. NO ADDITIONAL COMPENSATION WILL BE AUTHORIZED FOR CUTTING AND PATCHING WORK THAT IS NECESSITATED BY ILL- TIMED, DEFECTIVE OR NON-CONFORMING INSTALLATIONS. 7.9. PROVIDE NAMEPLATES ON ALL ITEMS OF MECHANICAL EQUIPMENT THAT HAS A CONTROL PANEL AND ALL ITEMS WHERE A SUBMITTAL WAS MADE, EXCEPT FOR PIPING. 7.9.1. MAKE NAMEPLATES FROM WHITE ENGRAVING STOCK WITH BLACK LETTERS AND BLACK FOUR EDGE BEVEL. 7.9.2. WORDING SHALL SUITABLY DESCRIBE ITEMS SUCH AS PANEL ID, SOURCE OCPD AND VOLTAGE. 7.9.3. NAMEPLATES SHALL BE ATTACHED USING PROPER SIZE AND TYPE STAINLESS STEEL BOLTS, LOCK WASHERS AND NUTS. GLUE ON, TAPE ON, OR TAPE TYPE NAMEPLATES ARE NOT ACCEPTABLE FOR THE EQUIPMENT. 7.10. BEARINGS THAT REQUIRE LUBRICATION SHALL BE LUBRICATED IN STRICT ACCORDANCE WITH THE MANUFACTURER'S RECOMMENDATIONS. 7.11. ALL COILS SHALL BE THOROUGHLY CLEANED AND COMBED PRIOR TO FINAL INSPECTION. 7.12. ARRANGE FOR CHASES, SLOTS, AND OPENINGS IN OTHER BUILDING COMPONENTS DURING PROGRESS OF CONSTRUCTION, TO ALLOW FOR MECHANICAL INSTALLATIONS. 7.13. COORDINATE THE INSTALLATION OF REQUIRED SUPPORTING DEVICES AND SLEEVES TO BE SET IN POURED-IN-PLACE CONCRETE AND OTHER STRUCTURAL COMPONENTS, AS THEY	ARE CONSTRUCTED. 7.14. SEQUENCE, COORDINATE, AND INTEGRATE INSTALLATIONS OF MECHANICAL MATERIALS AND EQUIPMENT FOR EFFICIENT FLOW OF THE WORK. GIVE PARTICULAR ATTENTION TO LARGE EQUIPMENT REQUIRING POSITIONING PRIOR TO CLOSING IN THE BUILDING. 7.15. THE EXACT MOUNTING HEIGHT OF DEVICES SHALL BE DETERMINED IN THE FIELD WITH RELATION TO ARCHITECTURAL DETAILS AND EQUIPMENT BEING SERVED. 7.15.1. IT SHALL BE THE RESPONSIBILITY OF CONTRACTOR TO COORDINATE OUTLET LOCATION WITH EQUIPMENT. OWNERS REPRESENTATIVE SHALL BE PERMITTED TO RELOCATE ANY OUTLET PRIOR TO INSTALLATION WITHIN A 15 FOOT LIMIT AT NO ADDITIONAL CHARGE IN CONTRACT PRICE. 7.15.2. ALL FASTENERS, HANGERS, AND METHODS OF HANGING EXPOSED WORK IN FINISHED AREAS SHALL BE SUBMITTED TO THE OWNERS REPRESENTATIVE FOR APPROVAL BEFORE INSTALLATION. 7.15.3. WHERE MOUNTING HEIGHTS ARE NOT DETAILED OR DIMENSIONED, INSTALL SYSTEMS, MATERIALS, AND EQUIPMENT TO PROVIDE THE MAXIMUM HEADROOM POSSIBLE. 7.16. INSTALL MECHANICAL EQUIPMENT TO FACILITATE SERVICING, MAINTENANCE, AND REPAIR OR REPLACEMENT OF EQUIPMENT COMPONENTS. AS MUCH AS PRACTICAL, CONNECT EQUIPMENT FOR EASE OF DISCONNECTING, WITH MINIMUM OF INTERFERENCE WITH OTHER INSTALLATIONS. 7.17. INSTALL SYSTEMS, MATERIALS, AND EQUIPMENT GIVING RIGHT-OF-WAY PRIORITY TO SYSTEMS REQUIRED TO BE INSTALLED AT A SPECIFIED SLOPE. 8. OWNER FURNISHED EQUIPMENT: 8.1. THE OWNER WILL PURCHASE THE FOLLOWING MATERIALS: NONE CURRENTLY LISTED. 9. SPACE PRIORITY: 9.1. ENSURE OPTIMUM USE OF AVAILABLE SPACE FOR MATERIALS AND EQUIPMENT INSTALLED ABOVE CEILING. ALLOCATE SPACE IN THE ORDER OF PRIORITY AS LISTED BELOW EXCEPT AS OTHERWISE DETAILED. ITEMS ARE LISTED IN THE ORDER OF PRIORITY, WITH ITEMS OF EQUAL IMPORTANCE LISTED UNDER A SINGLE PRIORITY NUMBER. 9.1.1. GRAVITY FLOW PIPING SYSTEMS 9.1.2. VENT PIPING SYSTEM 9.1.3. RECESSED LIGHTING FIXTURES 9.1.4. CONCEALED HVAC TERMINAL AND EQUIPMENT 9.1.5. AIR DUCT SYSTEMS 9.1.6. SPRINKLER PIPING SYSTEMS 9.1.7. PRESSURIZED PIPING SYSTEMS 9.1.8. ELECTRICAL CONDUIT, WIRING, CONTROL AIR TUBING 10. ORDER OF SPACE PRIORITY DOES NOT DICTATE INSTALLATION SEQUENCE. INSTALLATION SEQUENCE SHALL BE AS REQUIRED TO INSTALL ALL AFFECTED TRADES. 11. THE WORK OF THIS DIVISION 23 SHALL NOT OBSTRUCT ACCESS FOR INSTALLATION, OPERATION AND MAINTENANCE OF THE WORK OF ANY OTHER DIVISION. 12. ALL MAJOR ITEMS OF EQUIPMENT SHALL BE ARRANGED SO AS TO PROVIDE A MINIMUM 28" CLEAR AISLE SPACE. ADDITIONAL SPACE SHALL BE PROVIDED BETWEEN AND AROUND EQUIPMENT FOR MAINTENANCE AND PROPER OPERATION AS SHOWN IN THE EQUIPMENT MANUFACTURE'S LITERATURE AND NEC REQUIREMENTS. 13. SLEEVES, SEALS AND ESCUTCHEONS: 13.1. SLEEVES SHALL BE PROVIDED THROUGH ALL PIPE PENETRATION OF CONCRETE OR MASONRY WALLS, ELEVATED FLOORS AND ROOFS, EXCEPT THOSE PLUMBING PIPING PENETRATION FOR FIXTURES, VENTS, ETC. 13.2. SLEEVES SHALL BE FABRICATED FROM SCHEDULE 40 STEEL PIPE THROUGH 10" AND STANDARD WALL STEEL PIPE FOR SLEEVE SIZES 1/2" AND LARGER. ALL SLEEVES PENETRATING EXTERIOR WALLS, UNDERGROUND WALLS, PIT OR VAULT WALLS SHALL BE PROVIDED WITH 3"x3/8" THICK WATERSTOP RING WELDED COMPLETELY TO THE MIDPOINT OF THE SLEEVE. 13.3. ALL SLEEVES PENETRATING EXTERIOR WALLS, UNDERGROUND WALLS, PIT OR VAULT WALLS AND ELEVATED FLOORS SHALL BE PACKED AND SEAL WATER TIGHT. 13.4. SLEEVES THROUGH ROOFS SHALL EXTEND ABOVE THE ROOF SURFACE AND BE FLASHED WATERTIGHT. 13.5. SLEEVES THROUGH WALLS SHALL BE CUT AND FINISHED FLUSH WITH EACH SURFACE OF THE WALL IN WHICH THEY ARE INSTALLED. 13.6. SLEEVES SHALL BE SIZED TO PROVIDE A MINIMUM OF 3/4" CLEARANCE BETWEEN THE INSIDE OF THE SLEEVE AND THE OUTSIDE FINISHED SURFACE OF THE PIPE PLUS ANY INSULATION SPECIFIED. 13.7. PROVIDE ROUND, CHROME-PLATED ESCUTCHEONS ON ALL EXPOSED PIPING PENETRATIONS PASSING THROUGH WALLS, FLOOR, PARTITIONS AND CEILING. 14. FIRE-STOPS: 14.1. WHERE DUCTWORK, PIPING, CONDUIT, ETC., PASS THROUGH FIRE PARTITIONS, FIRE WALLS AND FLOORS, A FIRE-STOP SHALL BE PROVIDED THAT WILL ENSURE AND EFFECTIVE BARRIER AGAINST THE SPREAD OF FIRE, SMOKE AND GASES. FIRE-STOP MATERIAL SHALL BE PACKED TIGHT AND COMPLETELY FILL GAPS BETWEEN THE DUCTWORK, PIPING, CONDUIT, ETC. AND THE PERIMETER OF THEIR ROUGH OPENINGS. 14.2. FIRE-STOPPING MATERIAL SHALL MAINTAIN ITS DIMENSION AND INTEGRITY WHILE PREVENTING THE PASSAGE OF FLAME, SMOKE AND GASES UNDER CONDITIONS OF INSTALLATION AND USE WHEN EXPOSED TO THE ASTM E119 TIME-TEMPERATURE CURVE FOR A TIME PERIOD EQUIVALENT TO THE RATING OF THE ASSEMBLY PENETRATED. FIRE-STOPPING MATERIAL SHALL BE NONCOMBUSTIBLE AS DEFINED BY ASTM E116; AND, FOR INSULATION MATERIALS, MELT POINT SHALL BE A MINIMUM OF 1700 DEGREES F. FOR 1-HOUR PROTECTION AND 1850 DEGREES F. FOR 2-HOUR PROTECTION. FIRE-STOPPING MATERIAL SHALL BE DOW-CORNING RTV FOAM OR 3M FIRE BARRIER PRODUCTS OR SOHIO CARBORUNDUM FYRE PUTTY. 14.3. COORDINATE WITH FIRE STOP VENDOR FOR PROPER APPLICATION FOR PVC PIPING. 15. GENERAL CONTRACTOR'S DUTIES: 15.1. THE GENERAL CONTRACTOR SHALL START-UP AND MAINTAIN THE HVAC SYSTEMS AND SHALL CONTINUE THE OPERATION OF THE HVAC SYSTEMS DURING EACH DAY OF TESTING AND BALANCING. START-UP AND OPERATION SHALL INCLUDE, AS A MINIMUM, THE FOLLOWING: 15.1.1. ALL EQUIPMENT OPERABLE AND IN SAFE CONDITION. 15.1.2. TEMPERATURE CONTROL SYSTEM COMPLETE.	15.1.3. PROPER THERMAL OVERLAP PROTECTION IN PLACE FOR ELECTRICAL EQUIPMENT. 15.1.4. DUCTWORK LEAKAGE RATES NOT EXCEEDING THOSE SPECIFIED AND ALL DUCT SYSTEM CLEAN OF DEBRIS. 15.1.5. AIR SIDE SYSTEMS SHALL HAVE: 15.1.5.1. CORRECT FAN ROTATION AND RPM. 15.1.5.2. COIL FINS CLEANED AND COMBED 15.1.5.3. FILTERS CLEAN AND IN PLACE. 15.1.5.4. ACCESS DOORS CLOSED. 15.1.5.5. ALL DAMPERS IN PLACE AND OPEN. 15.1.5.6. ALL GRILLES, REGISTERS AND DIFFUSERS INSTALLED. 15.2. PROVIDE SUFFICIENT TIME BEFORE FINAL COMPLETION DATE SO THAT TESTING AND BALANCING CAN BE ACCOMPLISHED. COORDINATE THE SUBMITTED T&B SCHEDULE. 15.3. PROVIDE IMMEDIATE LABOR AND TOOLS TO MAKE REQUIRED CORRECTIONS AND REPAIRS WITHOUT UNDUE DELAY. 15.4. THE GENERAL CONTRACTOR AND HIS SUBCONTRACTORS SHALL COOPERATE FULLY WITH THE AGENCY TO PROVIDE THE FOLLOWING: 15.4.1. ACCESS TO HVAC SYSTEM COMPONENTS. 15.4.2. THE RIGHT TO ADJUST THE SYSTEMS. 15.5. ANY CONDITIONS WHICH PREVENT A PROPER HVAC TEST AND BALANCE SHALL BE REPORTED BY THE AGENCY TO THE GENERAL CONTRACTOR AND ARCHITECT WITHIN 7 DAYS OF THEIR DISCOVERY. 15.6. IF IT IS DETERMINED BY THE AGENCY AND CONFIRMED BY THE ARCHITECT THAT DRIVE CHANGES OR ADDITIONAL BALANCING DAMPERS ARE REQUIRED, THE CONTRACTOR SHALL OBTAIN AND INSTALL ALL NECESSARY COMPONENTS. 15.7. THE AGENCY SHALL COOPERATE WITH THE ARCHITECT AND THE CONTRACTOR AND ALL HIS SUBCONTRACTORS TO PERFORM THE WORK IN SUCH A MANNER AS TO MEET THE JOB SCHEDULE. 15.8. THE AGENCY SHALL VERIFY THAT ALL SYSTEM COMPONENTS ARE IN PLACE AND IN PROPER WORKING ORDER PRIOR TO LEAVING THE PROJECT. 15.9. ALL REPORTED, RECORDED DATA SHALL REPRESENT TRUE MEASURED CONDITIONS. 16. INTENT: 16.1. IT IS THE INTENT OF THIS SECTION OF THE SPECIFICATIONS TO PROVIDE COMPLETE, OPERABLE, ADJUSTED AIR CONDITIONING SYSTEM AS SCHEDULED, AS SHOWN AND SPECIFIED WHICH ARE FREE OF EXCESSIVE NOISE AND VIBRATION. 16.2. AIR CONDITIONING EQUIPMENT BASIS OF DESIGN: 16.2.1. THE BASIS OF DESIGN IS AS NOTED ON PLANS. CONTRACTOR TO PROVIDE PRICE FOR SCHEDULED EQUIPMENT AND AN ALTERNATIVE PRICE FOR ANY PROPOSED SUBSTITUTIONS OR EQUALS BY OTHER MANUFACTURERS SHALL BE PROVED EQUAL IN ALL RESPECTS TO THE EQUIPMENT SPECIFIED AS THE BASIS OF DESIGN. 16.3. STARTUP: 16.3.1. PROVIDE THE SERVICES OF A FACTORY TRAINED AND QUALIFIED SERVICE TECHNICIAN EMPLOYED BY THE UNIT MANUFACTURER WHO SHALL INSPECT THE INSTALLATION INCLUDING EXTERNAL CONTROL INTERLOCK AND ELECTRICAL POWER CONNECTIONS; SUPERVISE LEAK TESTING, INITIAL OPERATION, CALIBRATION OF OPERATING AND SAFETY CONTROLS AND SUPERVISE ELECTRICAL TESTING INCLUDING INSULATION RESISTANCE OF MOTORS AND VOLTAGE BALANCE BETWEEN PHASES DURING STARTING AND RUNNING. 16.3.2. THIS SERVICE TECHNICIAN SHALL FORWARD A REPORT IN FOUR (4) COPIES TO THE OWNER WHEN THE UNIT IS IN SAFE AND PROPER OPERATING CONDITION. THIS REPORT SHALL INCLUDE ALL PRESSURE AND CONTROL SETTINGS, MEG READINGS, VOLTAGE READINGS PER PHASE DURING START AND RUN, AND SHALL LIST MINOR DISCREPANCIES TO BE CORRECTED THAT AFFECT SAFE AND RELIABLE OPERATION. ONE ADDITIONAL COPY OF THE REPORT SHALL BE LEFT IN THE UNIT CONTROL PANEL. ONE COPY OF BOUND INSTALLATION, OPERATION, MAINTENANCE SERVICE AND PARTS BROCHURES, INCLUDING APPLICABLE SERIAL NUMBERS, FULL UNIT DESCRIPTION AND PARTS ORDERING SOURCES, SHALL BE PLACED IN THE UNIT CONTROL PANEL AT THE TIME OF STARTUP; FOR (4) ADDITIONAL COPIES SHALL BE FORWARDED TO THE OWNER. 16.4. SMOKE DETECTORS: 16.4.1. SMOKE DETECTORS REQUIRED IN DUCTWORK SHALL BE IONIZATION TYPE AND LISTED BY UL OR FM FOR DUCT INSTALLATION. DUCT DETECTORS SHALL BE PROVIDED WITH AN APPROVED DUCT HOUSING, MOUNTED EXTERIOR TO THE DUCT, AND SHALL BE PROVIDED WITH PERFORATED SAMPLING TUBES EXTENDING ACROSS THE WIDTH OF THE DUCT. ALSO PROVIDE SMOKE DETECTORS WITH A REMOTE AUDIO/VISUAL LED LIGHT IN CEILING BELOW DETECTOR ALARM INDICATOR WITH KEY SWITCH FOR RESET. MOUNT 5' A.F.F. NEAR DUCT. ACTIVATION OF A DUCT DETECTOR SHALL CAUSE SHUTDOWN OF IT'S RESPECTIVE AIR CONDITIONING UNIT. CONTROL AND INTERLOCK WIRING SHALL RUN IN CONDUIT WHICH SHALL BE SIZED TO SUIT THE NUMBER, TYPE AND SIZE OF CONDUCTORS, SHALL BE PROVIDED BY THE CONTROL CONTRACTOR. CONTROL AND INTERLOCK WIRING SHALL BE SEPARATE FROM POWER WIRING OF DIVISION 26. ALL WIRING SHALL BE IN ACCORDANCE WITH DIVISION 26. DUCT SMOKE DETECTOR SHALL BE WIRED AND INTERLOCKED WITH BUILDING FIRE ALARM SYSTEM. DUCT SMOKE DETECTORS SHALL BE SUPPLIED BY THE UNIT MANUFACTURER, AND WIRED BY THE ELECTRICAL CONTRACTOR. 16.4.2. THE REMOTE TEST/RESET ANNUNCIATORS SHALL BE PROVIDED BY THE ELECTRICAL CONTRACTOR AND INSTALLED BY THE MECHANICAL CONTRACTOR. INSTALLATION BY MECHANICAL SHALL INCLUDE MOUNTING OF THE ANNUNCIATORS AND ALL WIRING FROM EACH DEVICE TO THE RTU. ELECTRICAL WILL PROVIDE A JUNCTION BOX IN THE WALL WITH 1/2" CONDUIT STUBBED UP ABOVE THE CEILING FOR EACH. 16.4.3. ACCEPTABLE SUBSTITUTE MANUFACTURERS: 16.4.4. ACCEPTABLE SUBSTITUTE MANUFACTURERS ARE TRANE, CARRIER, CAPTIVE AIRE, AAOH, OR EQUAL. 16.5. FAN EQUIPMENT BASIS OF DESIGN: 16.5.1. THE BASIS OF DESIGN IS AS NOTED ON PLANS. CONTRACTOR TO PROVIDE PRICE FOR SCHEDULED EQUIPMENT AND AN ALTERNATIVE PRICE FOR ANY PROPOSED SUBSTITUTIONS OR EQUALS BY OTHER MANUFACTURERS SHALL BE PROVED EQUAL IN ALL RESPECTS TO THE EQUIPMENT SPECIFIED AS THE BASIS OF DESIGN.	16.5.2. ACCEPTABLE SUBSTITUTE MANUFACTURERS: 16.5.3. ACCEPTABLE SUBSTITUTE MANUFACTURERS ARE COOK, GREENHECK, CAPTIVE AIRE, CARNES, ACME, PENN OR EQUAL. 16.6. LOVERS, AIR INTAKE AND OUTLET DEVICES. 16.6.1. LOUVERS SHALL BE FITTED WITH A 1/2" MESH 16 GAUGE ALUMINUM BIRD SCREEN IN AN ALUMINUM FRAME. 16.6.2. FINISH SHALL BE DURANAR KYNAR 500 IN A COLOR SELECTED BY THE ARCHITECT AT THE TIME OF SUBMITTAL REVIEW. 16.6.3. THE PERFORMANCE STANDARDS SHALL BE CERTIFIED BY THE MANUFACTURER IN ACCORDANCE WITH THE AMCA CERTIFIED RATINGS PROGRAM AND THE LOUVER SHALL CARRY THE AMCA SEAL. 16.6.4. LOUVER PERFORMANCE STANDARDS: 16.6.4.1. MAXIMUM STATIC PRESSURE DROP AT 900 FPM VELOCITY THROUGH THE FREE AREA -0.065" W.C. 16.6.4.2. NO WATER PENETRATION AT UP TO 1,000 FPM VELOCITY THROUGH THE FREE AREA. 16.6.4.3. MINIMUM FREE AREA IN RELATION TO GROSS OVERALL AREA - 33%. 16.6.5. RELIEF OR INTAKE HOOD PERFORMANCE STANDARDS: 16.6.5.1. RELIEF HOODS SHALL BE FITTED WITH A 1/2' X 1/2" GALVANIZED BIRD SCREEN AND BACKDRAFT DAMPER. 16.6.5.2. INTAKE HOODS SHALL BE FITTED WITH 1 INCH THICK CLEANABLE METAL MESH FILTERS. 16.6.5.3. ALL HOODS SHALL BE PROVIDED COMPLETE WITH 12" HIGH ROOF CURBS. ROOF CURBS SHALL BE OF ALUMINUM CONSTRUCTION, INSULATED, CANTED AND COMPLETE WITH WOOD NAILER STRIPS. INSULATION SHALL MEET NFPA 25/50 FLAME SPREAD/SMOKE DEVELOPED RATINGS. 16.6.5.4. ALL HOODS WITH A THROAT AREA OF 12 SQ. FT. OR LESS SHALL HAVE HINGED HOOKS. 16.6.5.5. PERFORMANCE STANDARDS. 16.6.5.5.1. MAXIMUM TOTAL PRESSURE DROP AT 600 FPM THROAT VELOCITY THROUGH THE FREE AREA. 16.6.5.5.2. INTAKE HOODS - 0.125" W.C. 16.6.5.5.3. RELIEF HOODS - 0.08" W.C. 16.6.6. AIR INTAKE OR OUTLET EQUIPMENT BASIS OF DESIGN: 16.6.7. THE BASIS OF DESIGN IS AS NOTED ON PLANS. CONTRACTOR TO PROVIDE PRICE FOR SCHEDULED EQUIPMENT AND AN ALTERNATIVE PRICE FOR ANY PROPOSED SUBSTITUTIONS OR EQUALS BY OTHER MANUFACTURERS SHALL BE PROVED EQUAL IN ALL RESPECTS TO THE EQUIPMENT SPECIFIED AS THE BASIS OF DESIGN. 16.6.8. ACCEPTABLE SUBSTITUTE MANUFACTURERS: 16.6.9. ACCEPTABLE SUBSTITUTE MANUFACTURERS ARE RUSKIN, GREENHECK, OR EQUAL. 16.7. DUCTWORK: 16.7.1. THE CONSTRUCTION OF THE DUCTWORK SYSTEMS AS SPECIFIED HEREIN AS SHOWN. THESE SYSTEMS INCLUDE, BUT ARE NOT LIMITED TO, THE FOLLOWING: 16.7.1.1. SUPPLY AIR DUCTWORK 16.7.1.2. RETURN, TRANSFER AND RELIEF AIR DUCTWORK 16.7.1.3. EXHAUST DUCTWORK 16.7.1.4. GREASE EXHAUST DUCTWORK (FROM KITCHEN HOODS) 16.7.1.5. DISHWASHER EXHAUST DUCTWORK 16.7.1.6. OUTSIDE AIR DUCTWORK 16.7.1.7. FLUES
			NO DATE REMARKS REVISIONS BAGELS & CO. TALLAHASSEE 799 W. GAINES STREET SUITE 121 TALLAHASSEE, FL 32310 PROJECT NO: 2023.0719 DATE: 04/12/2024 M8.1 MECHANICAL SPECIFICATIONS CHECKED: DAK DRAWN: RAN	

		17. INTENT 17.1. IT IS THE INTENT OF THIS SECTION OF THE SPECIFICATIONS TO PROVIDE A COMPLETE OPERABLE DUCT SYSTEM AS SHOWN AND SPECIFIED WHICH IS AIRTIGHT, FREE OF NOISE, VIBRATION AND SWEATING, AND FABRICATED SO AS TO FIT INTO THE SPACE ALLOTTED AND TO EXHIBIT A MINIMUM RESISTANCE TO AIRFLOW. 17.2. DESIGN AND CONSTRUCTION 17.2.1. DUCTWORK SHALL BE PROVIDED IN STRICT ACCORDANCE WITH THE FIRST LATEST EDITION OF THE SMACNA HVAC DUCT CONSTRUCTION STANDARDS – METAL AND FLEXIBLE, NFPA NO. 90A, 90B, 91 AND 96, AND UL 181. 17.2.2. DUCTWORK DIMENSIONS SHOWN ARE NET, CLEAR, INSIDE DIMENSIONS WITH NO ALLOWANCE SHOWN FOR DUCT LINER. ALL DUCTWORK SPECIFIED TO BE LINED SHALL BE 2" LARGER THAN SHOWN IN EACH DIMENSION TO COMPENSATE FOR THE LINER. DUCTWORK SHALL BE SQUARE, RECTANGULAR, ROUND, SPIRAL OR FLAT OVAL AS NOTED. CONVERSION OF DUCT SHAPES AND SIZES SHOWN SHALL BE ACCOMPLISHED WITHOUT INCREASING AIR VELOCITIES OR FRICTION LOSSES AND IS SUBJECT TO PRIOR APPROVAL BY ARCHITECT. 17.2.3. ELBOWS SHALL BE EITHER FULL RADIUS TYPE (INSIDE RADIUS EQUAL TO DUCT WIDTH), GIVE-GORE RADIUSED FLAT-OVAL TYPE OR, IN LOW PRESSURE SYSTEMS ONLY, MITERED WITH DOUBLE-THICKNESS TURNING VANES. 17.2.4. ABRUPT CHANGES IN DUCT SIZES AND SHAPES SHALL NOT BE PERMITTED. THE TOTAL ANGLE OF DIVERGING TRANSITIONS SHALL BE NOT MORE THAN 15 DEGREES; CONVERGING TRANSITIONS SHALL BE NOT MORE THAN 30 DEGREES UNLESS OTHERWISE NOTED OR REQUIRED DUE TO STRUCTURAL CONSTRAINTS. 17.2.5. OFFSETS, TRANSITIONS, RISES AND DROPS ARE NOT INDIVIDUALLY CALLED OUT ON THE DESIGN DRAWINGS. THEY SHALL BE PROVIDED AS REQUIRED TO FIT THE DUCTWORK INTO THE ALLOCATED SPACES. 17.2.6. TRANSITION RECTANGULAR DUCTWORK ON BOTTOM AND SIDES. MAINTAIN TOP OF DUCTWORK LEVEL AND AS HIGH AS POSSIBLE. 17.2.7. GREASE EXHAUST DUCTWORK JOINTS SHALL BE CONTINUOUSLY WELDED AND BE LIQUID TIGHT. 17.2.8. PROVIDE THE FOLLOWING TYPES OF DUCTWORK MATERIAL FOR THE SERVICES INDICATED: 17.2.8.1. TYPE OF MATERIAL– SERVICE 17.2.8.2. GALVANIZED SHEETMETAL– SUPPLY, RETURN, EXHAUST AND RELIEF OF COMFORT CONDITIONED AND OUTSIDE AIR. 17.2.8.3. BLACK STEEL– GREASE EXHAUST. 17.2.8.4. STAINLESS STEEL OR ALUMINUM– DISHWATER EXHAUST 17.3. ALL DUCT SHOULD BE HELD AS HIGH TO STRUCTURE OR ROOF DECK AS POSSIBLE AND STILL OPERATE PROPERLY MEETING ALL REQUIREMENTS OF THIS DIVISION. 17.4. PRODUCTS: 17.4.1. GALVANIZED SHEETMETAL 17.4.1.1. GALVANIZED SHEETMETAL SHALL BE LOCK-FORMING GRADE G90-ASTM A 525 HOT DIP GALVANIZED STEEL SHEETS. SHEETMETAL SHALL BE GALVANIZED ON EACH SIDE WITH NOT LESS THAN 1.25 OUNCES OF ZINC PER SQUARE FOOT. 17.4.2. SPIRAL DUCT 17.4.2.1. SPIRAL DUCT SHALL BE UTILIZED FOR ALL FLAT-OVAL AND ROUND DUCTWORK IN MEDIUM AND HIGH-PRESSURE SYSTEMS. 17.4.2.2. SPIRAL DUCT SHALL BE THE PRODUCT OF AIRFLOW LLC OR AN APPROVED EQUAL. 17.4.2.3. SPIRAL RIBBED DUCT THAT REDUCES THE METAL THICKNESS IS NOT ACCEPTABLE. 17.4.2.4. ALL SPIRAL DUCTWORK SHALL BE INTERNALLY LINED. 17.5. BLACK STEEL EXHAUST DUCTWORK 17.5.1. BLACK STEEL UTILIZED FOR GREASE EXHAUST DUCTWORK SHALL BE MINIMUM 0.058" THICK (16 GAUGE MSG) BLACK CARBON STEEL. 17.5.2. ACCESS PANELS SHALL BE SECURED TO THE DUCT IN A SIMILAR MANNER TO THE PRESCRIBED METHOD IN NFPA 96 FOR THE HOOD-TO-DUCT CONNECTION. 17.5.3. ALL SEAMS SHALL BE WELDED AND POLISHED. DUCTWORK SHALL SLOPE TOWARD EQUIPMENT AT A MINIMUM OF 1/4" PER FOOT. 17.6. STAINLESS STEEL 17.6.1. STAINLESS STEEL UTILIZED FOR DISHWATER EXHAUST DUCTWORK SHALL BE MINIMUM 0.045" THICK (18 GAUGE) STAINLESS STEEL WITH #4 FINISH. 17.6.2. ALL SEAMS SHALL BE WELDED AND POLISHED. DUCTWORK SHALL SLOPE TOWARD EQUIPMENT AT A MINIMUM OF 1/4" PER FOOT. 17.6.3. DUCTWORK ABOVE CEILING SHALL BE INSULATED. 17.7. ALUMINUM 17.7.1. ALUMINUM UTILIZED FOR DISHWATER EXHAUST DUCTWORK SHALL BE MINIMUM 0.027" THICK (20 GAUGE) TYPE 3003 ALUMINUM. 17.7.2. ALL SEAMS SHALL BE WELDED AND POLISHED. DUCTWORK SHALL SLOPE TOWARD EQUIPMENT AT A MINIMUM OF 1/4" PER FOOT. 17.7.3. DUCTWORK ABOVE CEILING SHALL BE INSULATED. 17.8. FLUES 17.8.1. ALL FLUES SHALL BE TYPE "B", DOUBLE-WALL, AS MANUFACTURED BY METALBESTOS OR AN APPROVED EQUAL. FLUES SHALL BE COMPLETE WITH STORM COLLARS, WEATHERPROOF CAPS AND ALL ACCESSORIES. 17.9. DAMPERS 17.9.1. MANUAL VOLUME DAMPERS:	17.9.1.1. SINGLE BLADE BUTTERFLY DAMPERS ARE ACCEPTABLE UP TO 12" ROUND OR 12" X 12" SQUARE. DAMPERS LARGER THAN THESE DIMENSIONS SHALL BE MULTI-BLADE TYPE. SINGLE BLADE DAMPERS SHALL BE CONSTRUCTED OF 16 GAUGE OR HEAVIER GALVANIZED SHEETMETAL. 17.9.1.2. NO MULTI-BLADE DAMPER BLADE SHALL EXCEED 8" IN WIDTH. ALL MULTIPLE BLADE DAMPERS SHALL BE CONSTRUCTED OF 16 GAUGE GALVANIZED STEEL OR HEAVIER. THE DAMPER FRAME SHALL BE 16 GAUGE OR HEAVIER. THE DAMPER ACTION SHALL BE OPPOSED-BLADE TYPE. 17.9.1.3. EACH BLADE SHALL PIVOT ON A 1/4" CADMIUM PLATED, COLD-ROLLED STEEL AXLE WHICH PIVOTS WITHIN SELF-LUBRICATING, IOUITE BRONZE BEARINGS. 17.9.1.4. THE TOP AND BOTTOM EDGES OF EACH RECTANGULAR DAMPER BLADE SHALL BE CRIMPED FOR STIFFNESS. 17.9.1.5. THE OPERATING ROD FOR ALL DAMPERS SHALL BE EXTENDED OUTSIDE THE DAMPER FRAME FOR ATTACHMENT OF AN OPERATOR. EACH OPERATOR SHALL HAVE A POSITION INDICATOR AND LOCKING QUADRANT. 17.9.1.6. ALL DAMPERS UTILIZED FOR INTRODUCTION OF OUTSIDE AIR SHALL HAVE FLEXIBLE, GASKETED EDGE AND END SEALS. THE LEAKAGE RATE SHALL BE LESS THAN 4 CFM PER SQ. FT. OF FACE AREA AGAINST A 1" W.G. DIFFERENTIAL PRESSURE, BASED ON A NOMINAL 48" X 48" DAMPER SIZE. 17.9.1.7. MANUAL VOLUME DAMPERS SHALL BE AS MANUFACTURED BY LOUVERS & DAMPERS, INC. OR AN APPROVED EQUAL. 17.10. CONTROL DAMPERS 17.10.1. CONTROL DAMPERS SHALL BE OF THE SAME CONSTRUCTION AS MANUAL VOLUME DAMPERS, EXCEPT THAT NO MANUAL OPERATOR AND QUADRANT IS REQUIRED. THE OPERATING ROD SHALL BE SUITABLE FOR OPERATION BY AUTOMATIC PNEUMATIC OR ELECTRIC OPERATOR. 17.11. FIRE DAMPERS 17.11.1. FIRE DAMPERS SHALL BE UL-LISTED AND LABELED FOR 1 1/2 HOURS AND SHALL BE PROVIDED WITH 160 DEGREES F. LINKS. DAMPERS INSTALLED WITHIN DUCTS SHALL BE TYPE B OR TYPE C WITH THE BLADES OUT OF THE AIRSTREAM. AREAS INDICATED SHALL BE NET, CLEAR, OPEN AREAS. 17.12. SMOKE DAMPERS 17.12.1. SMOKE DAMPERS SHALL BE UL-LISTED AS CLASS 1 LOW-LEAKAGE SMOKE DAMPERS AND SHALL BE PRODUCTS OF RUSKIN. 17.12.2. LOW-PRESSURE DUCT BRANCHES 17.12.2.1. SPLITTER DAMPERS SHALL BE PROVIDED AT ALL LOW-PRESSURE DUCTWORK BRANCHES. ALL LOW-PRESSURE DUCTWORK BRANCHES SHALL BE RADIUSED OR 45 DEGREE TAKE-OFFS; STRAIGHT TAPS ARE UNACCEPTABLE. THE LENGTH OF THE DAMPER BLADE SHALL BE THE SAME AS THE WIDTH OF THE WIDEST DUCT SECTION AT THE SPLIT, BUT IN NO CASE SHALL BLADE LENGTH BE LESS THAN 12". EACH OPERATOR ROD SHALL HAVE A LOCKING SWIVEL JOINT. 17.13. FLEXIBLE DUCT 17.13.1. FLEXIBLE DUCTWORK SHALL BE CLASS 1, UL 181 AIR DUCT AND MEET NFPA 90A AND 90B STANDARDS. 17.13.2. THE INTERNAL DUCT SURFACE SHALL BE ACOUSTICALLY RATED, BLACK CPE BONDED TO A COATED STEEL WIRE HELIX. THE EXTERNAL JACKET SHALL BE A FIBERGLASS, BI-DIRECTIONALLY REINFORCED, METALIZED VAPOR BARRIER WITH A STANDING, TRIPLE PLY SEAM. FIBERGLASS INSULATION SHALL BE PROVIDED BETWEEN THE DUCT SURFACE AND THE JACKET TO ACHIEVE A MAXIMUM THERMAL CONDUCTANCE OF 0.23 BTU/HR./SQ. FT./DEGREE F. AT 75 DEGREES F. MEAN. 17.13.3. FLEXIBLE DUCTWORK SHALL BE SUITABLE FOR 10" W.G. POSITIVE PRESSURE AND 1" W.G. NEGATIVE PRESSURE. 17.13.4. FLEXIBLE DUCT SHALL BE THERMAFLEX M-KE OR AN APPROVED EQUAL. 17.13.5. THE MAXIMUM ALLOWABLE INSTALLED LENGTH OF FLEXIBLE DUCTWORK SHALL BE AS FOLLOWS: 17.13.5.1.5'-0" ON LOW-PRESSURE SUPPLY AIR SYSTEMS LIMITED TO SHORT RUNOUTS AND END OF RUNS CONNECTED TO ROUND NECK SUPPLY DIFFUSERS AND REGISTERS. 17.13.5.2.2'-0" ON CONNECTIONS FROM ROUND NECK GRILLES TO SHEETMETAL DUCTWORK ON RETURN, EXHAUST AND TRANSFER DUCTWORK. 17.14. PROVIDE A SPIN-IN FITTING WITH INTEGRAL SCOOP AND VOLUME DAMPER AT ALL FLEXIBLE RUN-OUT CONNECTIONS IN LOW-PRESSURE SUPPLY AIR DUCTWORK ONLY. 17.15. FLEXIBLE CONNECTIONS 17.15.1. PROVIDE FLEXIBLE DUCT CONNECTIONS AT THE INLET AND OUTLET OF EACH BELT-DRIVEN FAN, INDOOR UNIT, FAN COIL UNIT, AIR HANDLING UNIT, ETC., AND AT ALL OTHER LOCATIONS INDICATED. FLEXIBLE CONNECTIONS SHALL BE FABRICATED FROM A GLASS FABRIC COATED ON BOTH SIDES WITH NEOPRENE. MINIMUM WEIGHT SHALL BE 30 OZ. PER SQ. YARD. 17.16. DUCT HARDWARE 17.16.1. DUCT HARDWARE SHALL BE AS MANUFACTURED BY YOUNG REGULATOR OR AN APPROVED EQUAL. 17.17. ACCESS DOORS 17.17.1. A DUCT ACCESS DOOR SHALL BE PROVIDED AT EACH FIRE DAMPER AND FIRE/SMOKE DAMPER. ACCESS DOORS SHALL BE DESIGNATED FOR 1.5 TIMES THE PRESSURE OF THE DUCT IN WHICH THEY ARE MOUNTED. ACCESS DOORS SHALL BE OF SUFFICIENT SIZE TO PROVIDE ACCESS TO THE DAMPERS FOR RESETTING THE BLADES AND REPLACING THE LINKS. ACCESS DOORS IN MEDIUM AND HIGH-PRESSURE DUCTWORK SHALL BE INSTALLED DOWNSTREAM OF FIRE DAMPERS AND SHALL BE IMPLOSION TYPE. WHERE ACCESS IS PROVIDED THROUGH GYPSUM BOARD WALLS OR CEILINGS, FURNISH ACCESS DOORS FOR INSTALLATION UNDER DIVISION 9. EACH DOOR SHALL MATCH THE FIRE-RATING OF THE WALL OR CEILING INDICATED. 17.18. DUCT LINER 17.18.1. DUCT LINER SHALL BE ONE INCH THICK, 1 1/2 LB. DENSITY (3 LB. DENSITY ON MEDIUM- AND HIGH-PRESSURE SUPPLY AIR SYSTEMS) FIBROUS GLASS WITH ONE FACE COATED WITH A BLACK FIRE RETARDANT COMPOUND. THE PERMANENT COMPOSITE FIRE AND SMOKE HAZARD RATING OF THE LINER SHALL BE STENCILED ON THE LINER FACE AND SHALL BE: 17.18.1.1. MAXIMUM FLAME SPREAD 25	17.18.1.2. MAXIMUM SMOKE DEVELOPED 50 17.19. DUCT INSTALLATION 17.19.1. DUCT INSTALLATION SHALL BE 2" THICK, MINIMUM 3/4 LB. DENSITY FIBERGLASS WITH AN FSKL 0.00035" THICK ALUMINUM FOIL JACKET, REINFORCED WITH FIBERGLASS SCRM. THERMAL CONDUCTIVITY SHALL BE A MAXIMUM OF R=6.0 AT 75 DEGREES F. MEAN TEMPERATURE, INSTALLED IN CONDITIONED SPACE, R=8.0 IN UNCONDITIONED SPACES. 17.19.2. INSULATION ADHESIVE SHALL BE BENJAMIN FOSTER 85-20. TAPE SHALL BE ALUMINUM FOIL AND SHALL BE SMACNA LISTED AND LABELED. 17.19.3. THE COMPOSITE NFPA 90A AND 90B, ASTM E84, UL RATING OF THE INSTALLED INSULATION SHALL NOT EXCEED 25/50. 17.20. INSTALLATION 17.20.1. DUCTWORK SHALL BE INSTALLED IN A STRICT ACCORDANCE WITH SMACNA, UL AND NFPA STANDARDS. 17.20.2. DUCT LINER SHALL BE PROVIDED FOR THE FOLLOWING MINIMUM DISTANCES OR THROUGH THE FIRST ELBOW(S) OR AS OTHERWISE INDICATED ON THE DRAWINGS, WHICHEVER IS GREATER, DOWNSTREAM OF EACH UNIT INDICATED BELOW: 17.20.2.1. PACKAGED ROOFTOP UNIT 5 FEET 17.20.2.2. AIR HANDLING UNIT 5 FEET 17.20.2.3. ALL EXPOSED SPIRAL DUCT SHALL HAVE DUCT LINER PROVIDED THROUGHOUT ALL SUPPLY AIR, RETURN AIR, TRANSFER AIR AND PLENUMS. 17.20.3. DUCT LINER SHALL BE CUT TO PROVIDE OVERLAPPED AND COMPRESSED LONGITUDINAL CORNER JOINTS. LINER SHALL BE INSTALLED WITH THE COATED SURFACE FACING THE AIR STREAM. DUCT LINER SHALL BE ADHERED TO THE DUCTWORK WITH A 100% COVERAGE OF THE SHEET METAL SURFACES USING A FIRE RETARDANT ADHESIVE APPLIED BY SPRAYING. COAT ALL EXPOSED LEADING EDGES AND ALL TRANSVERSE JOINTS WITH FIRE RETARDANT ADHESIVE. THE LINER SHALL BE ADDITIONALLY SECURED USING METAL PINS WELDED TO THE DUCT AND SPEED WASHERS. ALL LEADING EDGES SHALL BE SECURED WITH SHEET METAL AIRFOILS. 17.20.4. ALL SUPPLY AIR DUCTWORK WHICH IS NOT LINED SHALL BE INSULATED. NOTE THAT ALL EXPOSED SPIRAL DUCT SHALL BE INTERNALLY LINED. ALL OUTSIDE AIR DUCTWORK SHALL BE INSULATED. INSULATION SHALL BE CUT SLIGHTLY LONGER THAN CIRCUMFERENCE OF DUCT TO INSURE FULL THICKNESS AT CORNERS. ALL INSULATION SHALL BE APPLIED WITH EDGES TIGHTLY BANDED. INSULATION SHALL BE ADHERED TO DUCT WITH FIRE RESISTANT ADHESIVE. ADHESIVE SHALL BE APPLIED SO THAT INSULATION CONFORMS TO DUCT SURFACES UNIFORMLY AND FIRMLY. IN ADDITIONAL TO THE ADHESIVE, THE INSULATION SHALL BE ADDITIONALLY SECURED TO THE BOTTOM OF DUCTS 18" OR WIDER BY MEANS OF WELDED PINS AND SPEED CLIPS. THE PROTRUDING END OF THE PINS SHALL BE CUT OFF FLUSH AFTER THE SPEED CLIPS HAVE BEEN APPLIED. THE VAPOR-BARRIER FACING SHALL BE THOROUGHLY SEALED WITH TAPE WHERE THE PINS HAVE PIERCED THROUGH. ALL JOINTS SHALL BE SEALED WITH 2" WIDE SMACNA TAPE. ANY CUTS OR TEARS SHALL BE SEALED WITH SMACNA TAPE. 17.20.5. FLEXIBLE DUCTS UTILIZED IN THE LOW-PRESSURE DUCTWORK SYSTEMS SHALL BE INSTALLED WITHOUT KINKS OR BENDS WHICH ARE LESS THAN A CENTERLINE RADIUS EQUAL TO OR GREATER THAN TWICE THE DIAMETER OF THE FLEXIBLE DUCT BEING INSTALLED. 17.21. AIR DEVICES: 17.21.1. ALL AIR DISTRIBUTION DEVICES SHALL BE SELECTED IN ACCORDANCE WITH THE FOLLOWING MINIMUM CRITERIA UNLESS OTHERWISE NOTED BELOW OR ON THE DRAWINGS: 17.21.1.1. METHOD OF MOUNTING SHALL BE COMPATIBLE WITH THE CEILING, WALL OR DUCT SURFACE WHICH IT MOUNTS ON OR IN; I.E. LAY-IN, SURFACE MOUNTING, PLASTER FRAME, DUCT COLLAR, ETC. THE ARCHITECTURAL DRAWINGS SHALL BE REFERENCED TO DETERMINE THE MOUNTING METHOD FOR EACH DEVISE. ALL FLANGES ON SURFACE MOUNTED DEVICES SHALL BE PROVIDED WITH A GASKET. 17.21.1.2. FINISH OF ALL CEILING MOUNTED DEVICES SHALL BE SELECTED TO MATCH THE COLOR OF THE ADJACENT CEILING. FINISH OF ALL WALL MOUNTED DEVICES SHALL BE PRIMER WHICH IS COMPATIBLE WITH THE FINISH COATING SPECIFIED FOR THE ADJACENT WALL; FINISH COAT WILL BE APPLIED UNDER DIVISION 9. 17.21.2. REFER TO SCHEDULE FOR ADDITIONAL INFORMATION. 17.22. BASIS OF DESIGN: 17.22.1. THE BASIS OF DESIGN IS LISTED ON THE DRAWING IN THE AIR DISTRIBUTION SCHEDULE. ANY PROPOSED SUBSTITUTIONS SHALL BE PROVEN EQUAL IN ALL RESPECTS TO THE EQUIPMENT SPECIFIED IN THE BASIS OF DESIGN. ANY MODIFICATIONS TO DUCTWORK, CONTROLS, CEILINGS, BUILDING STRUCTURE, ETC., THAT RESULT FROM ANY SUBSTITUTION SHALL BE COORDINATED WITH ALL TRADES. THIS COORDINATION SHALL OCCUR BEFORE DELIVERY OF EQUIPMENT AND ANY MODIFICATIONS SHALL BE PERFORMER WITHOUT INCURRING ADDITIONS TO THE CONTRACT. 17.23. ACCEPTABLE MANUFACTURERS 17.23.1. ACCEPTABLE MANUFACTURERS ARE PRICE, TITUS, CARNES, METAL AIRE OR EQUAL; PROVIDED THAT THEIR UNITS, PERFORMANCE, APPEARANCE AND PHYSICAL CHARACTERISTICS ARE EQUAL IN ALL RESPECTS FOR THIS SPECIFIC PROJECT.

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REMARKS

REVISIONS

BAGELS
CO.

BAGEL & CO
TALLAHASSEE

799 W. GAINES STREET
SUITE 121
TALLAHASSEE, FL 32310

PROJECT NO: 2023.0719
DATE: 04/12/2024

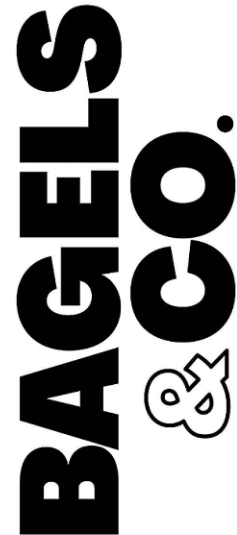
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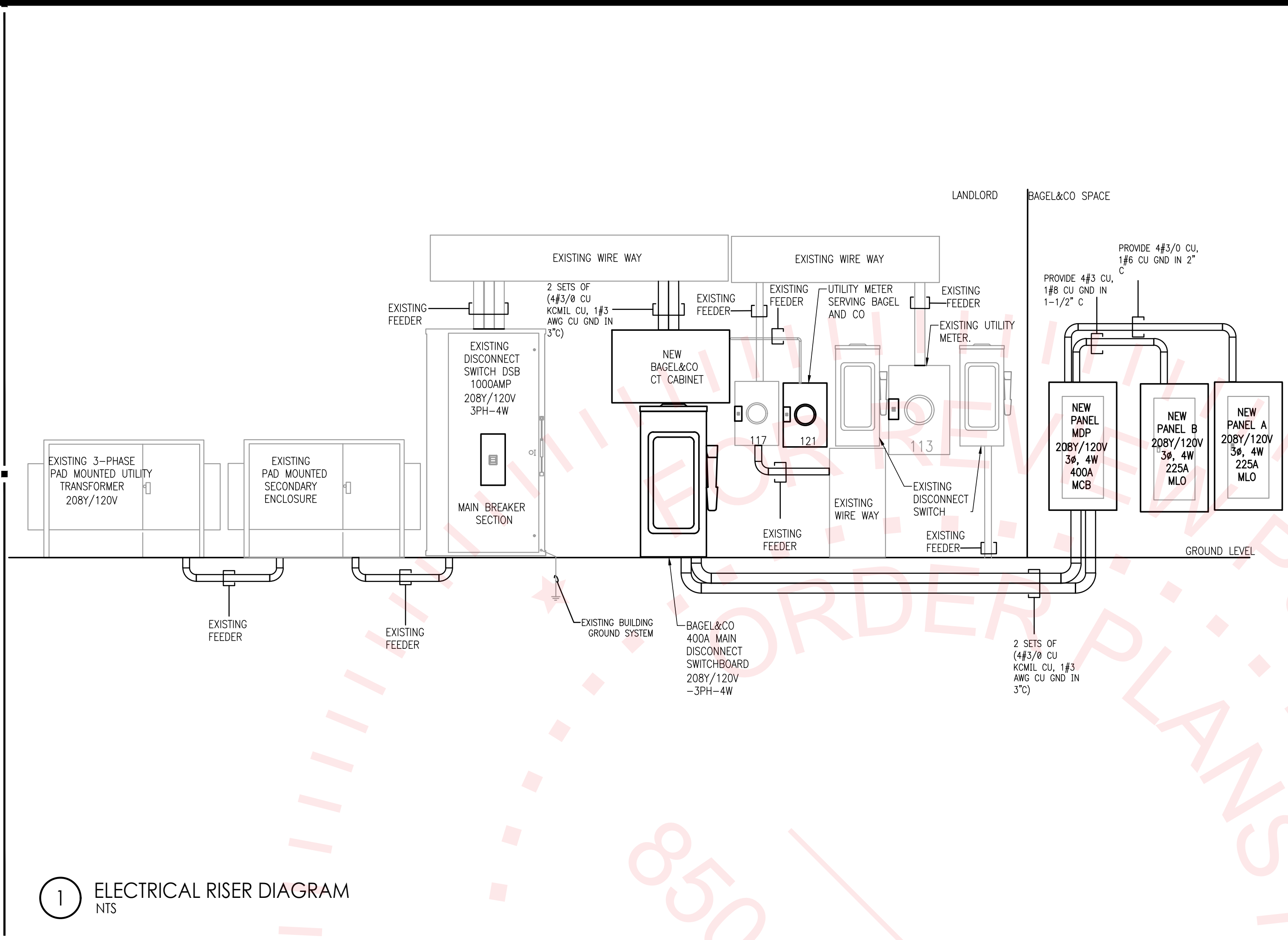
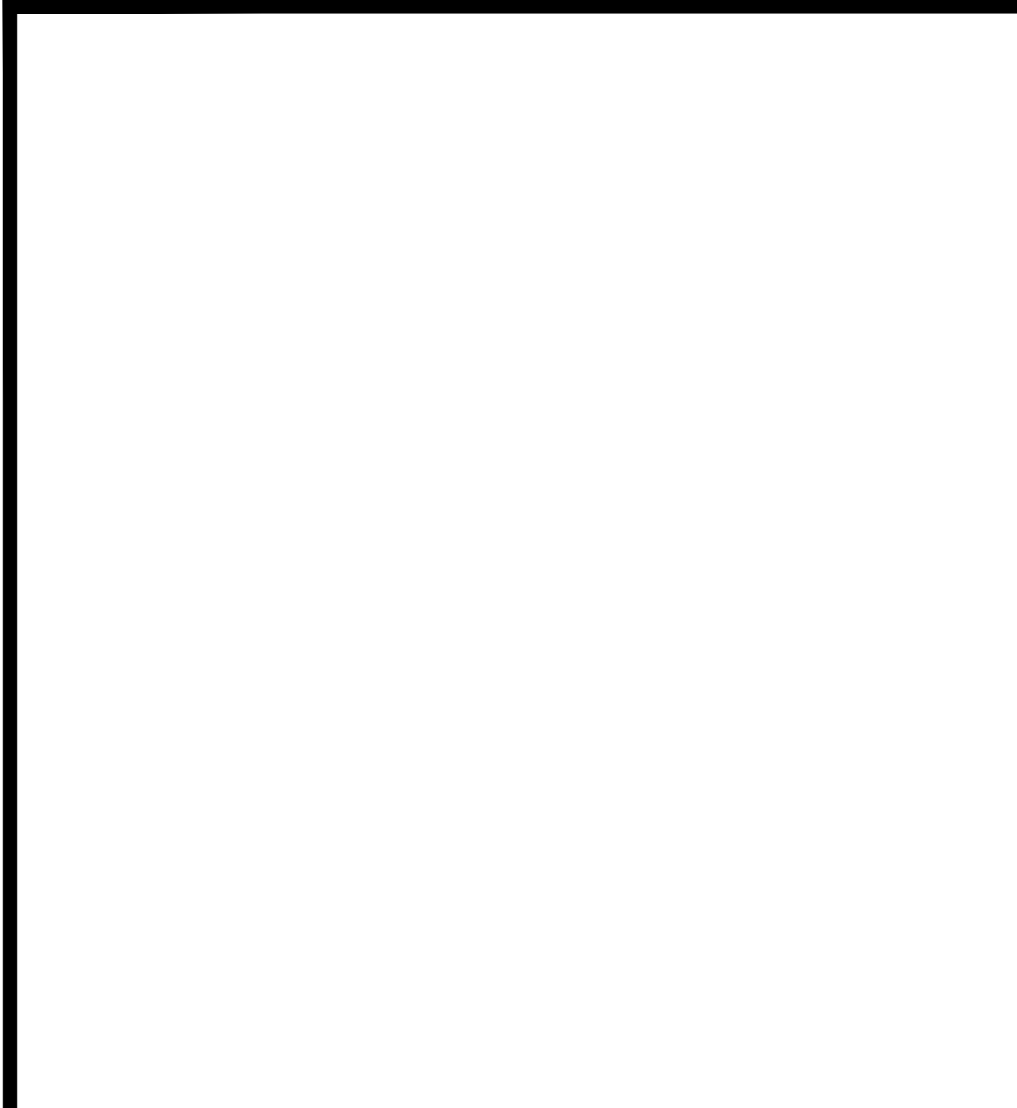
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1 ELECTRICAL RISER DIAGRAM
NTS

DSB RETAIL ELECTRICAL LOAD SUMMARY		
WEST RETAIL BUILDING 121 (BAGELS & CO.)	170.4	
WEST RETAIL BUILDING 117	57.2	
WEST RETAIL BUILDING 113	115.0	
TOTAL KVA	342.6	
TOTAL AMPS =	952.1	
EXISTING 1000 AMPS SERVICE SIZE PER INFORMATION AVAILABLE AT TIME OF DESIGN		

Sheet List Table	
Sheet Number	Sheet Title
E0.0	LEGEND SPECS AND RISER DIAGRAM
E1.0	ELECTRICAL PLAN--LIGHTING
E2.0	ELECTRICAL PLAN--POWER
E2.1	ELECTRICAL PLAN--HVAC POWER
E3.0	ELECTRICAL PLAN -- SYSTEMS
E4.0	ELECTRICAL DETAILS
E5.0	ELECTRICAL SCHEDULES
E6.0	ELECTRICAL PANEL SCHEDULES
E7.0	ENERGY CALCULATION -- LIGHTING
E8.0	ELECTRICAL SPECIFICATIONS (1 OF 2)
E8.1	ELECTRICAL SPECIFICATIONS (2 OF 2)

NOTE:
ALL EQUIPMENT LOCATED FOR METERING OR OTHER NEEDS EXTERIOR TO THE BUILDING OR WITHIN THE MAIN ELECTRICAL ROOM SHALL BE INSTALLED IN AN ORDERLY MANNER, STACKED AND KEPT TIGHT TO MINIMIZE REQUIRED WALL SPACE. ALL EQUIPMENT INSTALLED IN NON-ORGANIZED MANNER OR LAYOUT UN-ACCEPTABLE TO THE OWNER SHALL BE SUBJECT TO REMOVAL AND RE-INSTALLATION.

NOTE:
CONTRACTOR TO PROVIDE TYPED-WRITTEN PANEL SCHEDULES FOR ALL PANELS SHOWING ALL CIRCUIT LABELS. PROVIDE ENGRAVED NAME PLATES FOR ALL PANELS AS WELL AS ENGRAVED NAME PLATES WITH SUITE NUMBER AND TENANT NAME ON METER SOCKET/DISCONNECT.

NOTE:
PANEL A&B COMPONENTS INCLUDING OPD SHALL BE FULLY RATED FOR THE AVAILABLE FAULT CURRENT. PROVIDE ARC FLASH AND SHOCK HAZARD WARNING IDENTIFICATION PER NEC ARTICLE 110.16.

GENERAL NOTES

- ALL WORK SHOWN SHALL COMPLY WITH ALL NATIONAL, STATE AND LOCAL CODES, ORDINANCES, ETC.
- SEE LUMINAIRE SCHEDULE E5.0 FOR INFORMATION.
- EXIT SIGNS AND EMERGENCY LIGHTS SHALL HAVE THEIR OWN SELF-CONTAINED STANDBY BATTERY POWER SUPPLY. IF LOCAL CODE REQUIRES A DIRECT TAP BEFORE ANY CIRCUIT BREAKERS THEN INCORPORATE INTO THE FEEDER DIAGRAM.
- PROVIDE EMERGENCY LIGHTING TO MEET THE REQUIRED FOOT CANDLE LEVEL PER LOCAL CODE.
- LUMINAIRES INSTALLED IN CONTINUOUS ROWS SHALL BE GROUNDED WITH A CONDUCTOR ROUTED FROM LUMINAIRE TO LUMINAIRE, MOUNTED TO EACH WITH GROUNDING LUG OR SCREW. ALIGNING CLIPS ARE NOT ACCEPTABLE. LUMINAIRE GROUNDING SHALL BE INSTALLED IN COMPLIANCE WITH NATIONAL ELECTRICAL CODE, ARTICLE 410-21.
- CEILING LIGHTS ARE TO BE WIRED TO THE BAR JOIST MEMBERS AT THE DIAGONAL CORNERS.
- COORDINATE LIGHTING WITH SPRINKLER & MECHANICAL DRAWINGS.
- REFERENCE TO HEIGHT OF OUTLET OR RECEPTACLE SHALL BE MEASURED FROM FINISHED FLOOR TO CENTER OF OUTLET OR RECEPTACLE.
- ALL RECEPTACLES SHALL BE GROUNDING TYPE.
- ALL RECEPTACLES AND SWITCHES SHALL HAVE TWO (2) REVOLUTIONS OF ELECTRICAL TAPE (SCOTCH 33+) OVER ALL THE TERMINALS, TO PREVENT ACCIDENTAL CONTACT WITH THE JUNCTION BOX OR OUTLET BOX.
- CONDUIT SIZE TO BE PER NEC. PVC CONDUIT AND FITTINGS ARE ACCEPTABLE ONLY BELOW SUBBASE MATERIAL OF GROUND BEARING FLOOR SLABS WHERE SUCH USE IS ACCEPTABLE TO THE AUTHORITY HAVING JURISDICTION.
- CONDUIT SHALL BE ELECTRICAL METALLIC TUBING (EMT), RIGID STEEL (SIZE PER NEC), OR MC CABLE WHERE ALLOWED BY CODE. WHERE MC CABLE IS USED, PROPER SECUREMENT AND SUPPORT (AT INTERVALS NOT EXCEEDING 6 FEET) SHALL BE FOLLOWED PER NEC ART. 330.
- ELECTRICAL CONTRACTOR TO PROVIDE JUNCTION BOX AND CONDUIT STUB-UP ABOVE CEILING FOR ALL LOW VOLTAGE CABLING.
- DIMENSION IS FROM FINISHED WALL, SEE ARCHITECTURAL FOR WALL THICKNESS.
- FIELD COORDINATE EXACT LOCATION OF OUTLETS AS DETERMINED BY THE ACTUAL FURNITURE LAY OUT. VERIFY WITH FIXTURE PLAN.
- COORDINATE WITH OTHER DISCIPLINES FOR ELECTRICAL REQUIREMENTS OF EQUIPMENT NOT SHOWN ON DETAILS (i.e. ROOF-TOP UNITS, UNIT HEATERS, FANS, ETC.).
- EC SHALL VERIFY IN FIELD ALL KITCHEN EQUIPMENT CONNECTIONS AND REQUIREMENTS PRIOR TO ROUGH-IN AND INSTALLATION OF DEVICES. VERIFY ALL DEVICE LOCATIONS AND MOUNTING HEIGHTS WITH FOOD SERVICE CONTRACTOR. EC SHALL MAKE ADJUSTMENTS IN FIELD TO MATCH ACTUAL EQUIPMENT BEING INSTALLED, AS DIRECTED BY THE FOOD SERVICE CONTRACTOR.
- CONTRACTOR IS REQUIRED TO INSTALL STANDARD RECEPTACLES FOR CIRCUITS DEDICATED TO SPECIFIC EQUIPMENT. SUBJECT TO THE APPROVAL OF CODE ENFORCING AGENCY.
- ALL SINGLE-PHASE RECEPTACLES RATED 150 VOLTS TO GROUND OR LESS, 50 AMPERES OR LESS AND THREE-PHASE RECEPTACLES RATED 150 VOLTS TO GROUND OR LESS, 100 AMPERES OR LESS INSTALLED, INSTALLED IN THE LOCATIONS SPECIFIED IN 210.8(B)(1) THROUGH (B)(10) SHALL HAVE GROUND-FAULT CIRCUIT-INTERRUPTER PROTECTION FOR PERSONNEL.
- SERVICE CONDUCTORS ARE DESIGNED BASED ON THE SERVICE DISTANCE OF 100' TO MEET THE 2% VOLTAGE DROP PER NEC. IF SERVICE CONDUCTOR IS BETWEEN:
100' TO 150' -INCREASE THE CONDUCTOR BY ONE SIZE(S) LARGER THAN DESIGN
150' TO 200' -INCREASE THE CONDUCTOR BY TWO SIZE(S) LARGER THAN DESIGN
200' TO 250' -INCREASE THE CONDUCTOR BY THREE SIZE(S) LARGER THAN DESIGN
- BRANCH CONDUCTORS ARE DESIGNED BASED ON A DISTANCE OF 100' TO MEET THE 3% VOLTAGE DROP PER NEC. IF BRANCH CONDUCTOR IS BETWEEN:
100' TO 150' -INCREASE THE CONDUCTOR BY ONE SIZE(S) LARGER THAN DESIGN
150' TO 200' -INCREASE THE CONDUCTOR BY TWO SIZE(S) LARGER THAN DESIGN
200' TO 250' -INCREASE THE CONDUCTOR BY THREE SIZE(S) LARGER THAN DESIGN
- ELECTRICAL CONTRACTOR IS TO VERIFY WITH LOCAL AHJ FOR REQUIREMENT OF EXTERIOR EMERGENCY EGRESS LIGHTING AND ADJUST BID TO INCORPORATE AS NEEDED.
- A.I.C. OF PANELS AND SERVICE ENTRANCE EQUIPMENT IS BASED ON TYPICAL TRANSFORMER SPECIFICATIONS. CONTRACTOR SHALL FIELD VERIFY EXACT UTILITY A.I.C. RATING AND MAKE EQUIPMENT ADJUSTMENTS AS REQUIRED IN THE FIELD AT NO EXTRA COST TO OWNER. CONTRACTOR SHALL NOTIFY ENGINEER OF ANY FIELD CHANGES BASED ON UTILITY EQUIPMENT DATA.

LEGEND (NOT ALL SYMBOLS ARE USED WITHIN THIS SET OF DRAWINGS)

- LIGHTING**
- EXIT LIGHT (HATCHING DENOTES FACE)
 - SURFACE MOUNTED OR RECESSED EMERGENCY LIGHTING FIXTURE
 - SWITCH (P) PILOT LIGHT (R) ROTARY SWITCH
 - MOTOR RATED SWITCH
 - TOGGLE SWITCH (X) DENOTES CONTROL LABEL
 - OCCUPANCY SENSOR
 - PHOTOCELL OR DAYLIGHT SENSOR
 - TIMECLOCK
- POWER**
- SINGLE RECEPTACLE
 - DUPLEX RECEPTACLE
 - QUADRUPLUX RECEPTACLE
 - CEILING MOUNTED RECEPTACLE
 - LINE THRU CENTER OF RECEPTACLE DENOTES ABOVE COUNTER
 - CENTER SHADING IN RECEPTACLE DENOTES ISOLATED GROUND
 - TOP SHADING IN RECEPTACLE DENOTES GFCI PROTECTED
 - SPECIAL PURPOSE RECEPTACLE (AS NOTED)
 - TELEPHONE OUTLET
 - DATA OUTLET
 - VOICE/DATA COMBINATION OUTLET
 - JUNCTION BOX
 - SYMBOL IN SQUARE DENOTES FLOORMOUNTED
 - MOTOR, FAN, PUMP OR AIR CONDITIONING UNIT
 - PANELBOARD
 - FUSED DISCONNECT SWITCH, RATING AS NOTED.
 - NON-FUSED DISCONNECT SWITCH, RATING AS NOTED.
 - THERMOSTAT
 - REMOTE SENSOR/HUMIDISTAT COMBO
 - HUMIDITY SENSOR
- FIRE ALARM**
- FIRE ALARM PULL STATION NOM. 42" A.F.F. PER ADA
 - FIRE ALARM AUDIO/VISUAL STATION 80" A.F.F. PER ADA
 - FIRE ALARM VISUAL STATION 80" A.F.F. PER ADA
 - FIRE ALARM AUDIO STATION 80" A.F.F. PER ADA
 - REMOTE INDICATOR LIGHT/TEST SWITCH FOR DUCT SMOKE DETECTORS
 - DUCT SMOKE DETECTOR WITH REMOTE INDICATOR LIGHTS AND TEST SWITCH
 - SMOKE DETECTOR
 - FLOW SWITCH
 - TAMPER SWITCH
 - FACP
 - FIRE ALARM CONTROL PANEL
 - FIRE ALARM ANNUNCIATOR PANEL

ABBREVIATIONS

EC	ELECTRICAL CONTRACTOR
WP	WEATHERPROOF
AFF	ABOVE FINISHED FLOOR
UNO	UNLESS NOTED OTHERWISE
GFI	GROUND FAULT INTERRUPT
O.C	ON CENTER
ETR	EXISTING TO REMAIN

STANDARD MOUNTING HEIGHTS

A.F.F. (IN.)	DESCRIPTION
84"	AUDIBLE APPLIANCES
48"	ALARMS
48"	ANNUNCIATOR PANELS
84"	CLOCK OUTLETS (CENTERLINE)
48"	CONTROLS (CENTERLINE)
80"	EXIT SIGNS (WALL MOUNTED, BOTTOM)
60"	FIRE ALARM ANNUNCIATOR PANEL (DISPLAY)
120"	FIRE ALARM BELL (EXTERIOR)
60"	FIRE ALARM CONTROL PANEL/UNIT (DISPLAY)
36"	INTERCOM (AREA ONLY)
48"	INTERCOMS
72"	PANELS/PANELBOARDS (TOP)
48"	PULL STATIONS (TOP OF BOX)
144"	PHOTOCELLS
18"	RECEPTACLES (CENTERLINE)
24"	RECEPTACLES (EXTERIOR)
26"	RECEPTACLES (GARAGES)
48"	RECEPTACLES IN EQUIPMENT ROOMS
48"	REMOTE INDICATING LIGHT (EQUIPMENT ROOMS)
CEILING	REMOTE INDICATING LIGHT (FINISHED AREAS)
48"	SAFETY SWITCHES
48"	STARTERS
48"	SWITCHES (TOP OF BOX)
1048", 1036"	TELEPHONES (PUBLIC)
18"	TELEPHONE DATA OUTLETS (CENTERLINE)
6"	TELEPHONE TERMINAL BOARD (BOTTOM)
18"	TELEVISION OUTLETS
84"	VISIBLE APPLIANCES (CENTERLINE)

FIELD VERIFY ALL CONDITIONS

NOTE! ALL WIRING LAYOUTS, PIPING LAYOUTS AND DUCT LAYOUTS ARE SCHEMATIC. EXACT LOCATIONS SHALL BE DETERMINED BY THE CONSTRUCTION AND STRUCTURE OF THE BUILDING AND SHALL BE VERIFIED AND COORDINATED IN THE FIELD. EACH TRADE CONTRACTOR SHALL VERIFY WITH THE GENERAL CONTRACTOR THAT A THOROUGHLY REVIEWED AND COORDINATION OF ALL LOCATIONS AND ROUTINGS WITH ALL OTHER TRADES PRIOR TO FABRICATION OF CONDUITS, DUCTS, OR PIPING, AND START OF INSTALLATION OF SAME (INCLUDING SPRINKLER PIPING WHEN PRESENT ON JOB). ANY INSTALLATION OR CONSTRUCTION CONFLICTS WHICH OCCUR IN THE FIELD SHALL BE RESOLVED BY THE TRADE PROVIDER TO THE SATISFACTION OF THE OWNER AND/OR ARCHITECT AND AT NO EXPENSE TO THE OWNER, ARCHITECT AND/OR GENERAL CONTRACTOR.

THE CONTRACTOR SHALL CONTACT THE ARCHITECT, ENGINEER OR OWNER PRIOR TO BIDDING FOR INTERPRETATIONS AND CLARIFICATIONS OF THE DESIGN AND INCLUDE IN THE BID ALL COSTS TO MEET THE DESIGN INTENT. CLARIFICATIONS MADE BY THE ARCHITECT, ENGINEER OR OWNER AFTER BIDDING WILL BE FINAL AND SHALL BE IMPLEMENTED AT CONTRACTORS COST.

BIDDING CONTRACTORS SHALL HAVE A WORKING KNOWLEDGE OF ALL CODES, REGULATIONS, UTILITY REQUIREMENTS, LAWS AND ORDINANCES APPLICABLE TO THIS SITE AND SHALL INCLUDE IN THE BID THE COSTS FOR ALL WORK PROVIDED IN STRICT ACCORDANCE WITH THESE GOVERNING ITEMS, THE PLANS AND SPECIFICATIONS NOT WITHSTANDING. THE CONTRACTOR SHALL ALERT ARCHITECT, ENGINEER AND/OR OWNER OF ANY APPARENT DISCREPANCIES BETWEEN GOVERNING CODES AND DESIGN INTENT FOR DIRECTIONS.

Short-Circuit and Voltage Drop Calculations																														
FAULT POINT	BUS/FEEDER DESCRIPTION	PHASE	SOURCE POINT	SOURCE ISC (AMPS)	FEEDER				CONDUCTOR 'C' VALUE	BUSWAY 'C' VALUE	L-L VOLTS (E)	CIRCUIT LENGTH (L)	LOAD POWER FACTOR (PF)	CIRCUIT LOAD (A)	CONDUCTOR					F	M	FAULT CURRENT ISC	VOLTAGE DROP %VD	TOTAL V.D. %VD CUM	FAULT POINT (F#)					
					CONDUIT TYPE	MATERIAL CU OR AL	#	WIRE/BUS SIZE							RESISTANCE (R)	REACTANCE (X)	KVA	DEGREE RISE	Z							V SEC				
1	UTILITY SERVICE POINT			48041	AT THE SECONDARY OF THE UTILITY TRANSFORMER 208Y/120V - 3 PHASE																	SOURCE ISC + 6X MOTOR CONTRIBUTION =					48095			1
	MOTOR CONTRIBUTION			9																										
2	LANDLORD SERVICE DISCONNECT	3	1	48095	PVC	CU	2	SET(S) OF 3/0 KCMIL	13923		208	80	0.9	800	0.079	0.052					1.151	0.46	22,364	2.50	2.50	2				
3	BAGEL & CO SERVICE DISCONNECT	3	2	22364	M	CU	2	SET(S) OF 3/0 KCMIL	13923		208	30	0.9	284	0.079	0.052					0.201	0.83	18,626	0.33	2.83	3				
4	NEW PANEL MDP	3	3	18626	M	CU	2	SET(S) OF 3/0 KCMIL	13923		208	30	0.9	284	0.079	0.052					0.167	0.86	15,960	0.33	3.16	4				
5	NEW PANEL A	3	4	15960	M	CU	1	SET(S) OF 3/0 AWG	12844		208	30	0.9	170	0.079	0.052					0.310	0.76	12,179	0.40	3.56	5				
6	NEW PANEL B	3	5	12179	M	CU	1	SET(S) OF 3 AWG	7293		208	30	0.9	64	0.057	0.160					0.417	0.71	8,594	0.19	3.76	6				

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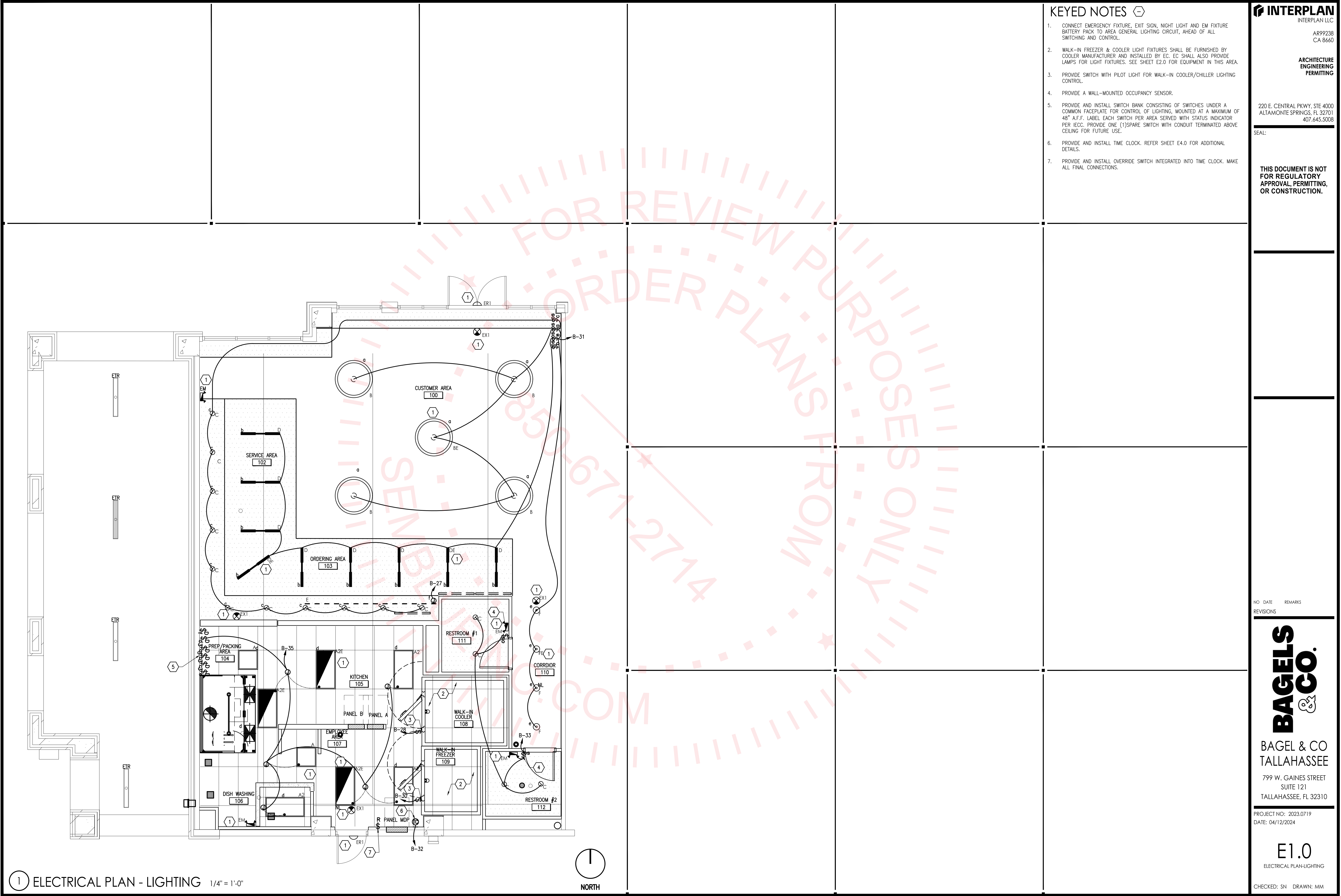
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SUITE 121
TALLAHASSEE, FL 32310

PROJECT NO: 2023.0719
DATE: 04/12/2024

E0.0

LEGEND SPECS AND RISER
DIAGRAM

CHECKED: SN DRAWN: MM



1 ELECTRICAL PLAN - LIGHTING 1/4" = 1'-0"

KEYED NOTES

1.

CONNECT EMERGENCY FIXTURE, EXIT SIGN, NIGHT LIGHT AND EM FIXTURE BATTERY PACK TO AREA GENERAL LIGHTING CIRCUIT, AHEAD OF ALL SWITCHING AND CONTROL.

2.

WALK-IN FREEZER & COOLER LIGHT FIXTURES SHALL BE FURNISHED BY COOLER MANUFACTURER AND INSTALLED BY EC. EC SHALL ALSO PROVIDE LAMPS FOR LIGHT FIXTURES. SEE SHEET E2.0 FOR EQUIPMENT IN THIS AREA.

3.

PROVIDE SWITCH WITH PILOT LIGHT FOR WALK-IN COOLER/CHILLER LIGHTING CONTROL.

4.

PROVIDE A WALL-MOUNTED OCCUPANCY SENSOR.

5.

PROVIDE AND INSTALL SWITCH BANK CONSISTING OF SWITCHES UNDER A COMMON FACEPLATE FOR CONTROL OF LIGHTING, MOUNTED AT A MAXIMUM OF 48" A.F.F. LABEL EACH SWITCH PER AREA SERVED WITH STATUS INDICATOR PER IECC. PROVIDE ONE (1)SPARE SWITCH WITH CONDUIT TERMINATED ABOVE CEILING FOR FUTURE USE.

6.

PROVIDE AND INSTALL TIME CLOCK. REFER SHEET E4.0 FOR ADDITIONAL DETAILS.

7.

PROVIDE AND INSTALL OVERRIDE SWITCH INTEGRATED INTO TIME CLOCK. MAKE ALL FINAL CONNECTIONS.

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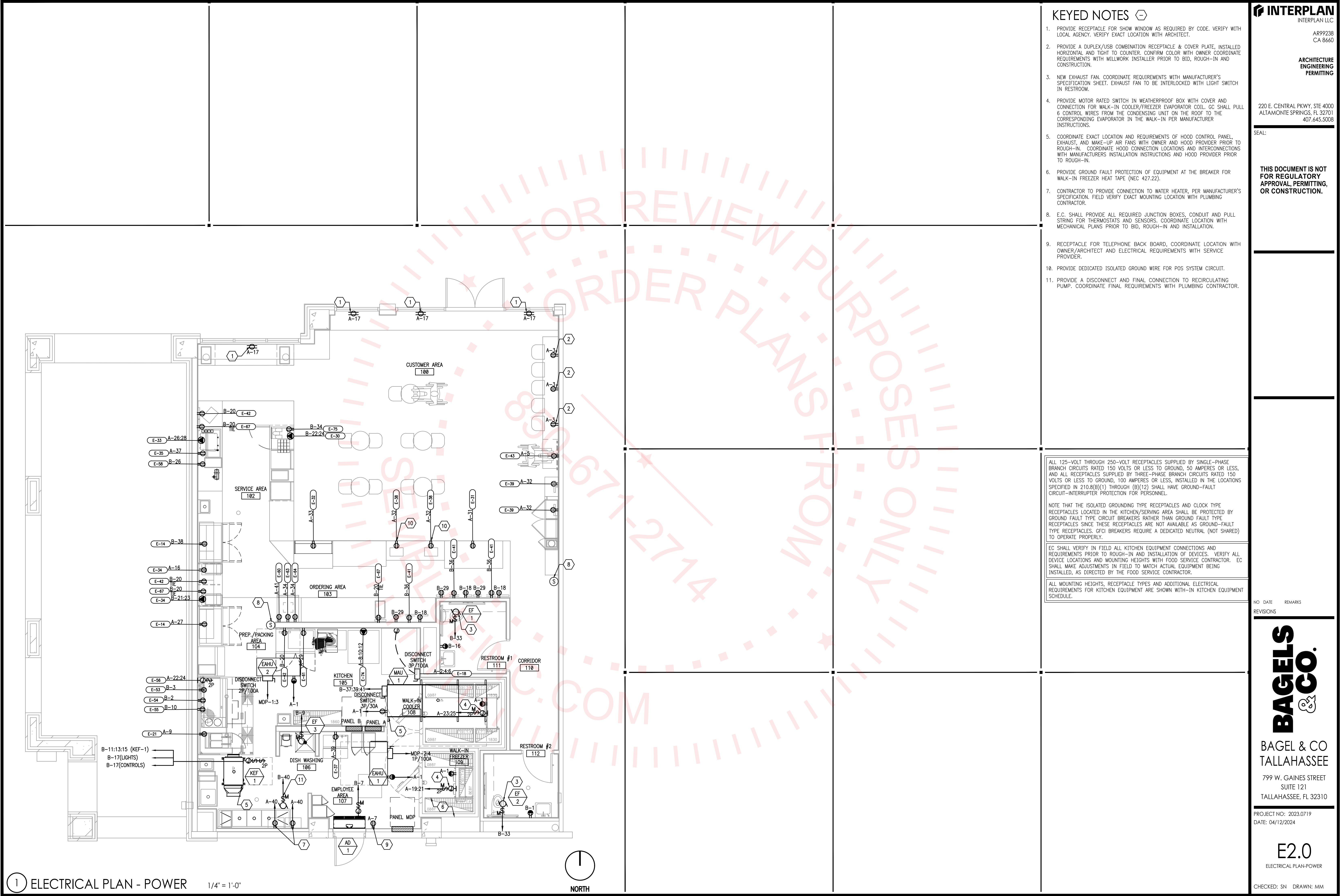
E1.0

ELECTRICAL PLAN-LIGHTING

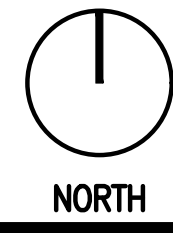
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1 ELECTRICAL PLAN - POWER 1/4" = 1'-0"



KEYED NOTES

1. PROVIDE RECEPTACLE FOR SHOW WINDOW AS REQUIRED BY CODE. VERIFY WITH LOCAL AGENCY. VERIFY EXACT LOCATION WITH ARCHITECT.

2. PROVIDE A DUPLEX/USB COMBINATION RECEPTACLE & COVER PLATE, INSTALLED HORIZONTAL AND TIGHT TO COUNTER. CONFIRM COLOR WITH OWNER COORDINATE REQUIREMENTS WITH MILLWORK INSTALLER PRIOR TO BID, ROUGH-IN AND CONSTRUCTION.

3. NEW EXHAUST FAN. COORDINATE REQUIREMENTS WITH MANUFACTURER'S SPECIFICATION SHEET. EXHAUST FAN TO BE INTERLOCKED WITH LIGHT SWITCH IN RESTROOM.

4. PROVIDE MOTOR RATED SWITCH IN WEATHERPROOF BOX WITH COVER AND CONNECTION FOR WALK-IN COOLER/FREEZER EVAPORATOR COIL. GC SHALL PULL 6 CONTROL WIRES FROM THE CONDENSING UNIT ON THE ROOF TO THE CORRESPONDING EVAPORATOR IN THE WALK-IN PER MANUFACTURER INSTRUCTIONS.

5. COORDINATE EXACT LOCATION AND REQUIREMENTS OF HOOD CONTROL PANEL, EXHAUST, AND MAKE-UP AIR FANS WITH OWNER AND HOOD PROVIDER PRIOR TO ROUGH-IN. COORDINATE HOOD CONNECTION LOCATIONS AND INTERCONNECTIONS WITH MANUFACTURERS INSTALLATION INSTRUCTIONS AND HOOD PROVIDER PRIOR TO ROUGH-IN.

6. PROVIDE GROUND FAULT PROTECTION OF EQUIPMENT AT THE BREAKER FOR WALK-IN FREEZER HEAT TAPE (NEC 427.22).

7. CONTRACTOR TO PROVIDE CONNECTION TO WATER HEATER, PER MANUFACTURER'S SPECIFICATION. FIELD VERIFY EXACT MOUNTING LOCATION WITH PLUMBING CONTRACTOR.

8. E.C. SHALL PROVIDE ALL REQUIRED JUNCTION BOXES, CONDUIT AND PULL STRING FOR THERMOSTATS AND SENSORS. COORDINATE LOCATION WITH MECHANICAL PLANS PRIOR TO BID, ROUGH-IN AND INSTALLATION.

9. RECEPTACLE FOR TELEPHONE BACK BOARD, COORDINATE LOCATION WITH OWNER/ARCHITECT AND ELECTRICAL REQUIREMENTS WITH SERVICE PROVIDER.

10. PROVIDE DEDICATED ISOLATED GROUND WIRE FOR POS SYSTEM CIRCUIT.

11. PROVIDE A DISCONNECT AND FINAL CONNECTION TO RECIRCULATING PUMP. COORDINATE FINAL REQUIREMENTS WITH PLUMBING CONTRACTOR.

ALL 125-VOLT THROUGH 250-VOLT RECEPTACLES SUPPLIED BY SINGLE-PHASE BRANCH CIRCUITS RATED 150 VOLTS OR LESS TO GROUND, 50 AMPERES OR LESS, AND ALL RECEPTACLES SUPPLIED BY THREE-PHASE BRANCH CIRCUITS RATED 150 VOLTS OR LESS TO GROUND, 100 AMPERES OR LESS, INSTALLED IN THE LOCATIONS SPECIFIED IN 210.8(B)(1) THROUGH (B)(12) SHALL HAVE GROUND-FAULT CIRCUIT-INTERRUPTER PROTECTION FOR PERSONNEL.

NOTE THAT THE ISOLATED GROUNDING TYPE RECEPTACLES AND CLOCK TYPE RECEPTACLES LOCATED IN THE KITCHEN/SERVING AREA SHALL BE PROTECTED BY GROUND FAULT TYPE CIRCUIT BREAKERS RATHER THAN GROUND FAULT TYPE RECEPTACLES SINCE THESE RECEPTACLES ARE NOT AVAILABLE AS GROUND-FAULT TYPE RECEPTACLES. GFCI BREAKERS REQUIRE A DEDICATED NEUTRAL (NOT SHARED) TO OPERATE PROPERLY.

EC SHALL VERIFY IN FIELD ALL KITCHEN EQUIPMENT CONNECTIONS AND REQUIREMENTS PRIOR TO ROUGH-IN AND INSTALLATION OF DEVICES. VERIFY ALL DEVICE LOCATIONS AND MOUNTING HEIGHTS WITH FOOD SERVICE CONTRACTOR. EC SHALL MAKE ADJUSTMENTS IN FIELD TO MATCH ACTUAL EQUIPMENT BEING INSTALLED, AS DIRECTED BY THE FOOD SERVICE CONTRACTOR.

ALL MOUNTING HEIGHTS, RECEPTACLE TYPES AND ADDITIONAL ELECTRICAL REQUIREMENTS FOR KITCHEN EQUIPMENT ARE SHOWN WITH-IN KITCHEN EQUIPMENT SCHEDULE.

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SUITE 121

TALLAHASSEE, FL 32310

PROJECT NO: 2023.0719

DATE: 04/12/2024

E2.0

ELECTRICAL PLAN-POWER

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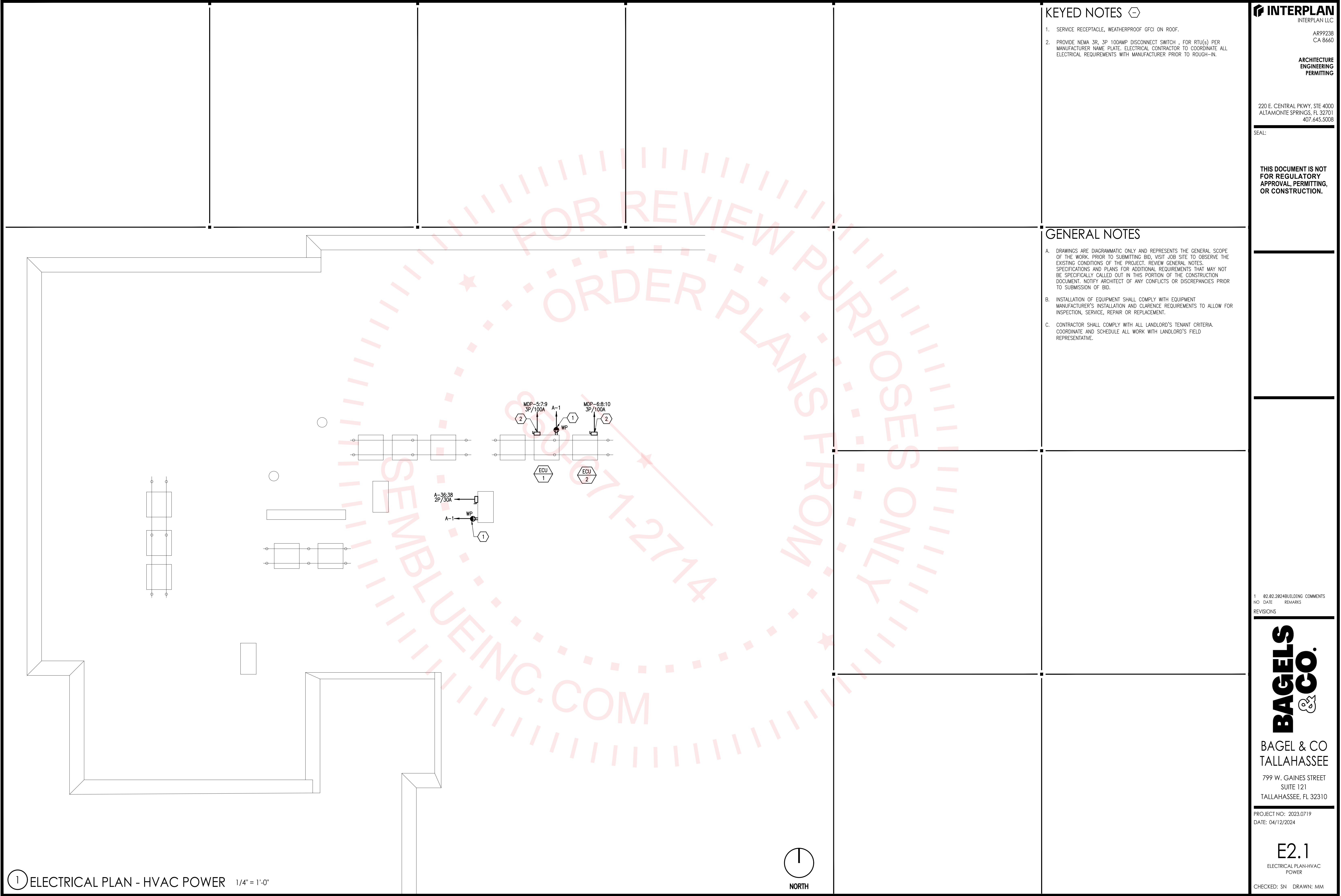
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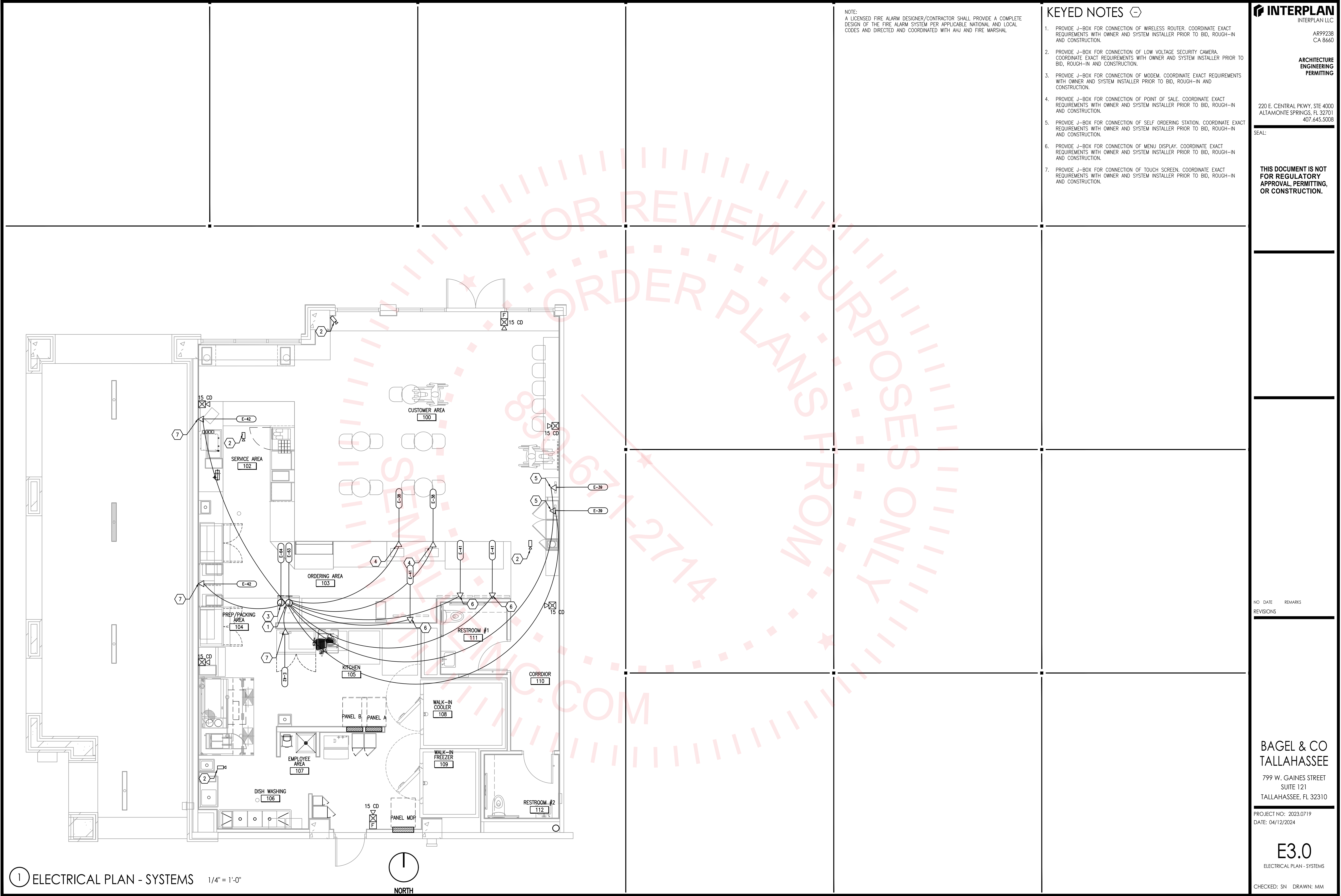
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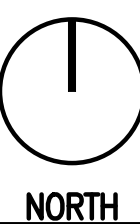
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1 ELECTRICAL PLAN - SYSTEMS 1/4" = 1'-0"



NOTE:
A LICENSED FIRE ALARM DESIGNER/CONTRACTOR SHALL PROVIDE A COMPLETE DESIGN OF THE FIRE ALARM SYSTEM PER APPLICABLE NATIONAL AND LOCAL CODES AND DIRECTED AND COORDINATED WITH AHJ AND FIRE MARSHAL.

KEYED NOTES

1. PROVIDE J-BOX FOR CONNECTION OF WIRELESS ROUTER. COORDINATE EXACT REQUIREMENTS WITH OWNER AND SYSTEM INSTALLER PRIOR TO BID, ROUGH-IN AND CONSTRUCTION.
2. PROVIDE J-BOX FOR CONNECTION OF LOW VOLTAGE SECURITY CAMERA. COORDINATE EXACT REQUIREMENTS WITH OWNER AND SYSTEM INSTALLER PRIOR TO BID, ROUGH-IN AND CONSTRUCTION.
3. PROVIDE J-BOX FOR CONNECTION OF MODEM. COORDINATE EXACT REQUIREMENTS WITH OWNER AND SYSTEM INSTALLER PRIOR TO BID, ROUGH-IN AND CONSTRUCTION.
4. PROVIDE J-BOX FOR CONNECTION OF POINT OF SALE. COORDINATE EXACT REQUIREMENTS WITH OWNER AND SYSTEM INSTALLER PRIOR TO BID, ROUGH-IN AND CONSTRUCTION.
5. PROVIDE J-BOX FOR CONNECTION OF SELF ORDERING STATION. COORDINATE EXACT REQUIREMENTS WITH OWNER AND SYSTEM INSTALLER PRIOR TO BID, ROUGH-IN AND CONSTRUCTION.
6. PROVIDE J-BOX FOR CONNECTION OF MENU DISPLAY. COORDINATE EXACT REQUIREMENTS WITH OWNER AND SYSTEM INSTALLER PRIOR TO BID, ROUGH-IN AND CONSTRUCTION.
7. PROVIDE J-BOX FOR CONNECTION OF TOUCH SCREEN. COORDINATE EXACT REQUIREMENTS WITH OWNER AND SYSTEM INSTALLER PRIOR TO BID, ROUGH-IN AND CONSTRUCTION.

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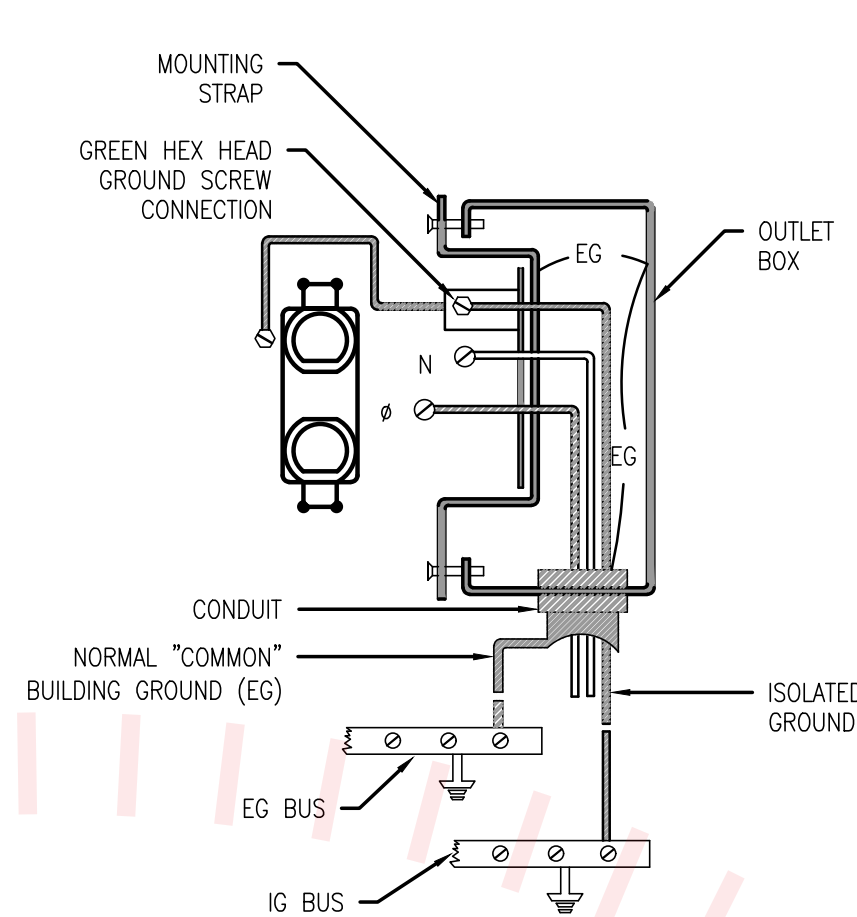
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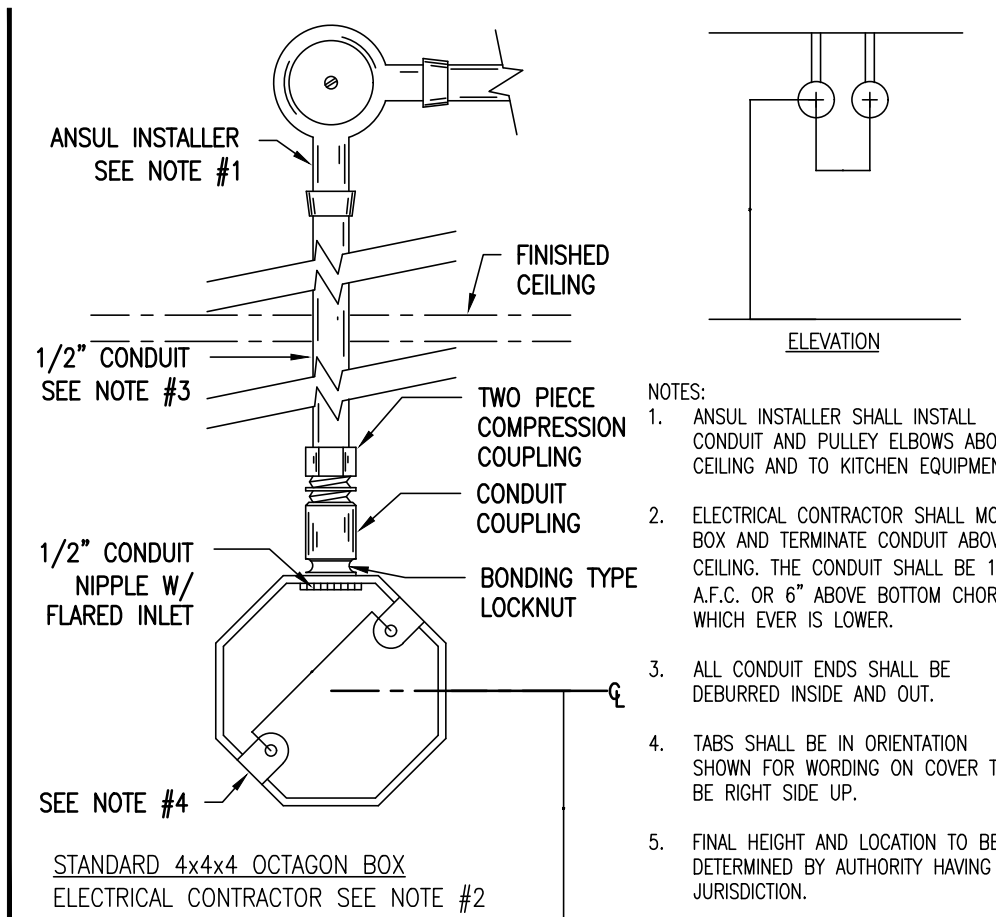
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ELECTRICAL PLAN - SYSTEMS

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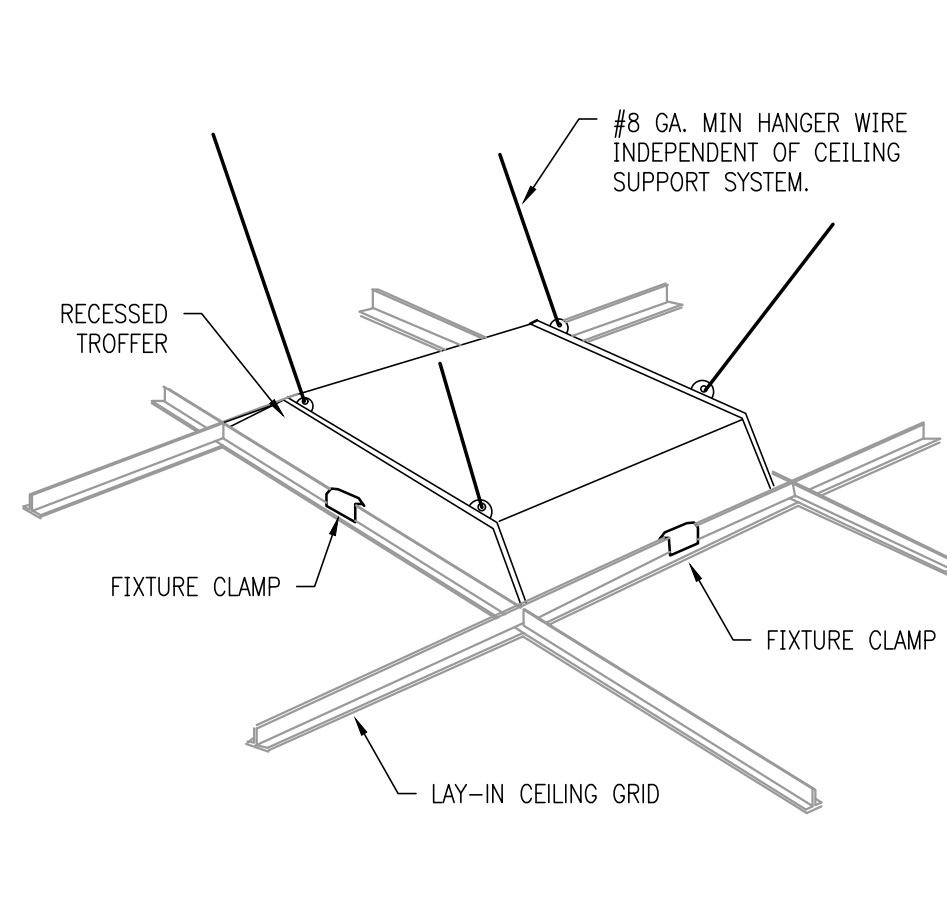
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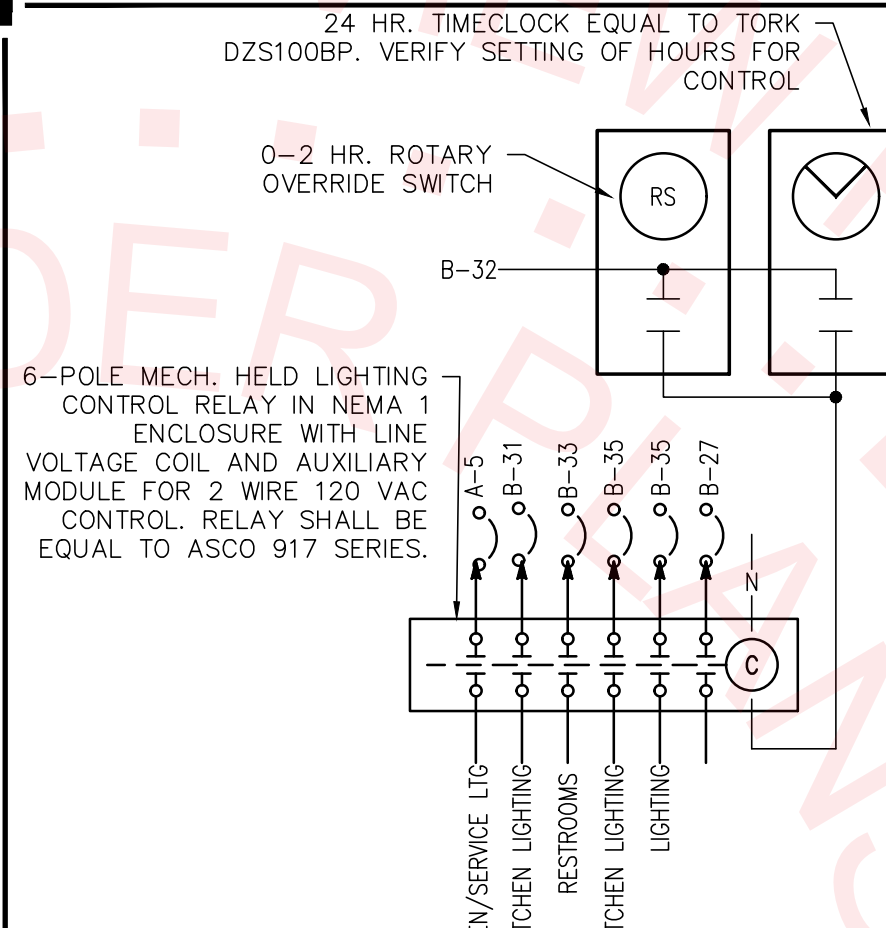
3 ISOLATED GROUND RECEPTACLE
NTS



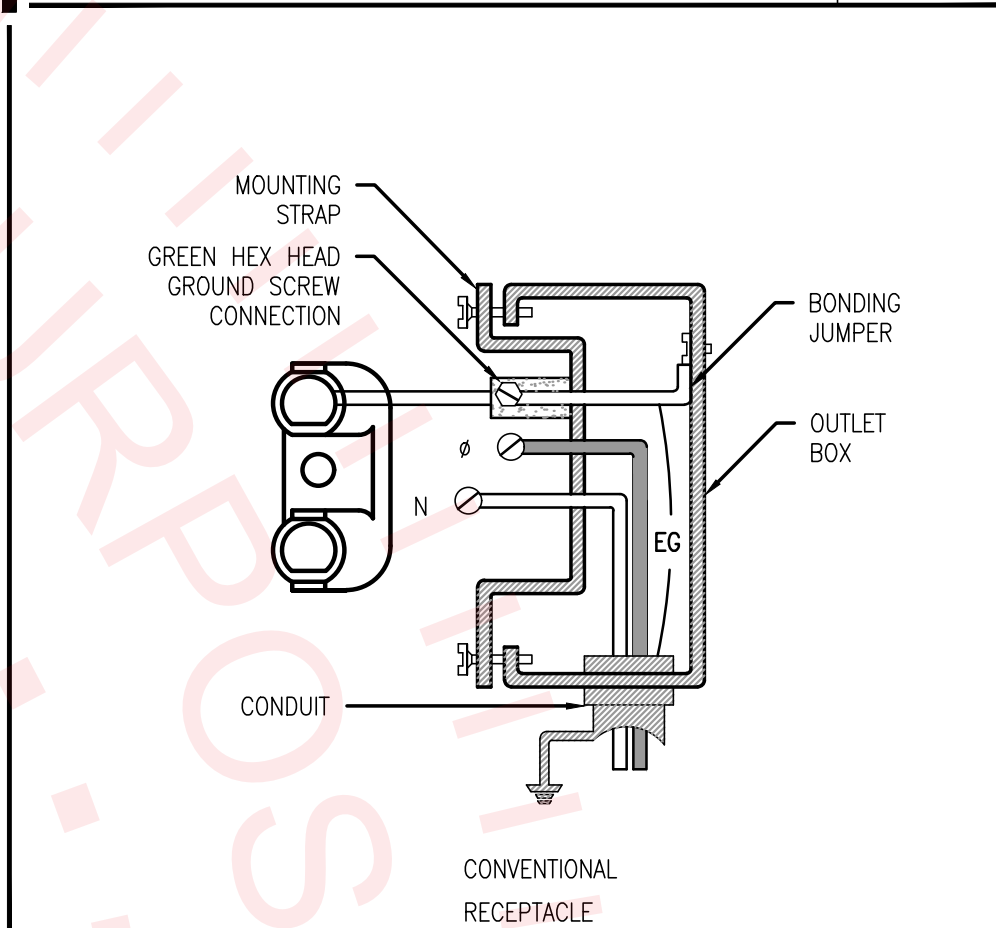
2 ANSUL PULL STATION
NTS



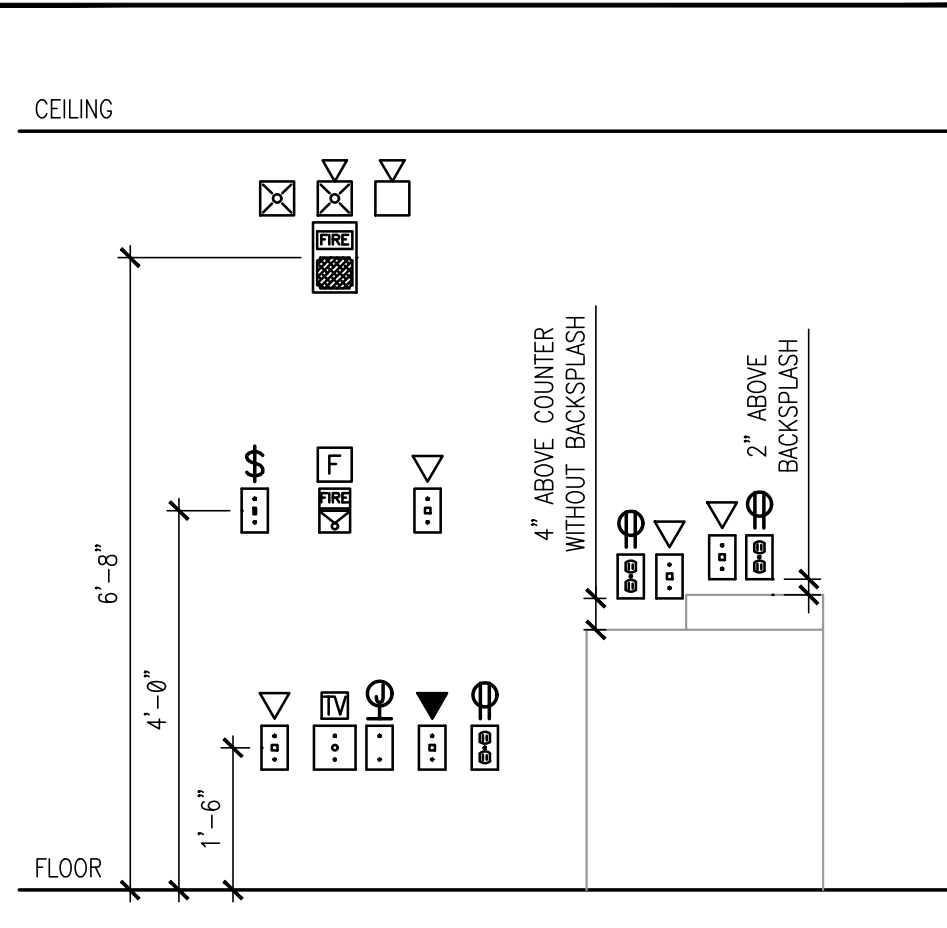
1 RECESSED FIXTURE MOUNTING
NTS



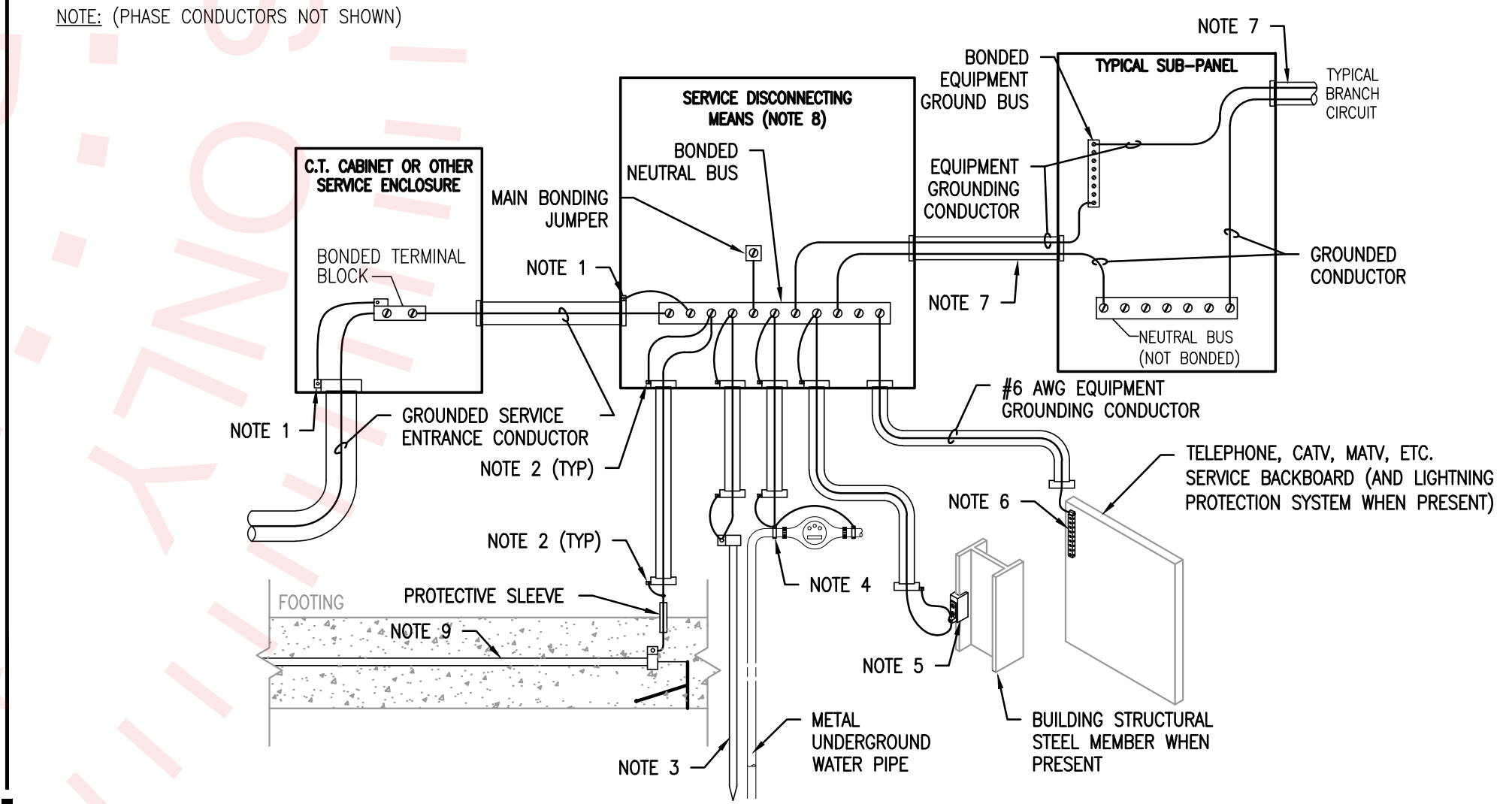
6 INTERIOR LIGHTING CONTROL DIAGRAM
NTS



5 CONVENTIONAL RECEPTACLE
NTS



4 TYPICAL MOUNTING HEIGHTS
NTS



- GROUNDING DETAIL NOTES:
1. ALL METAL CONDUITS ENCLOSING ANY SERVICE CONDUCTORS SHALL BE FITTED WITH A "BONDING BUSHING". SIZE THE JUMPER PER NEC.
 2. ALL METAL CONDUITS ENCLOSING ANY GROUNDING ELECTRODE CONDUCTOR SHALL BE FITTED WITH A "BONDING BUSHING" AT EACH END. SIZE THE JUMPER PER NEC.
 3. PROVIDE AT LEAST ONE SUPPLEMENTAL GROUNDING ELECTRODE PER NEC IN THE FORM OF A 10'-0" x 3/4" COPPER CLAD GROUND ROD INSTALLED PER CURRENT NEC REQUIREMENTS.
 4. CONNECT TO THE BUILDING'S METAL UNDERGROUND WATER PIPE WITHIN 5'-0" OF ITS ENTRANCE INTO THE BUILDING AND JUMPER ANY WATER METER PER NEC REQUIREMENTS.
 5. IF STRUCTURAL STEEL MEMBER IS AVAILABLE, BOND IT TO THE SERVICE USING A UL LISTED IRREVERSIBLE CLAMP OR WELDED LUG.
 6. PROVIDE AN EQUIPMENT GROUND BAR AND ATTACH IT TO THE PHONE BOARD.
 7. ALL BRANCH CIRCUIT AND FEEDER CONDUITS ARE TO HAVE AN INSULATED EQUIPMENT GROUNDING CONDUCTOR REGARDLESS OF THE CONDUIT MATERIAL.
 8. WHEN THE SERVICE CONSISTS OF MULTIPLE DISCONNECTING MEANS IN SEPARATE ENCLOSURES, CONNECT A TAP CONDUCTOR FROM THE MAIN GROUNDING ELECTRODE CONDUCTOR TO EACH DISCONNECTING MEANS. SIZE THIS TAP BASED ON THE LARGEST SERVICE DISCONNECT ENCLOSURE.
 9. PROVIDE A GROUNDING ELECTRODE ENCASED IN AT LEAST 2" OF CONCRETE AND LOCATED NEAR THE BOTTOM OF A CONCRETE FOUNDATION OR FOOTING THAT IS IN DIRECT CONTACT WITH EARTH. GROUNDING ELECTRODE SHALL CONSIST OF AT LEAST 20'-0" OF ONE OR MORE BARE OR ZINC GALVANIZED OR OTHER ELECTRICALLY CONDUCTIVE COATED STEEL REINFORCING BARS OR RODS OF NOT LESS THAN 1/2" IN DIAMETER OR AT LEAST 20'-0" OF #4 AWG BARE COPPER CONDUCTOR.

7 MAIN SERVICE ENTRANCE GROUND CONNECTIONS
N.T.S

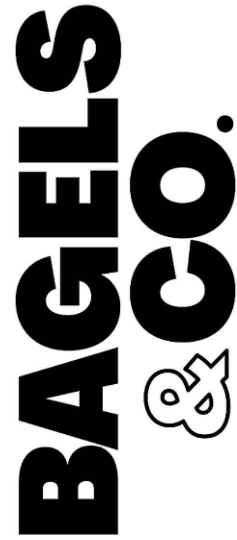
LIGHT FIXTURE SCHEDULE							
TYPE	MFG.	PART NO.	DESCRIPTION	VOLTAGE	LAMPS	FIXTURE WATTAGE	NOTES
A	KENALL	SIMPLESEALX CSED0 SERIES-22-67L	2X2 LAY-IN LED FIXTURE	120/277	LED	67	4H HOUSING -24-GA CRS FH (FOOD ZONE)
AE	KENALL	SIMPLESEALX CSED0 SERIES-22-67L	EMERGENCY BATTERY BACK UP	120/277	LED	67	4H HOUSING -24-GA CRS FH (FOOD ZONE)
A2	KENALL	SIMPLESEALX CSED0 SERIES-24-67L	2X4 LAY-IN LED FIXTURE	120/277	LED	67	4H HOUSING -24-GA CRS FH (FOOD ZONE)
A2E	KENALL	SIMPLESEALX CSED0 SERIES-24-67L	EMERGENCY BATTERY BACK UP	120/277	LED	67	4H HOUSING -24-GA CRS FH (FOOD ZONE)
B	EUREKA	CALDERA 4246-46	LARGE CIRCULAR PENDANT FIXTURES 46" DIA.	120	LED- COLOR TEMP 4000K	49.9	PROVIDE DIMMING- BLACK FINISH
BE	EUREKA	3981 EA	EMERGENCY BATTERY BACK UP	120	LED- COLOR TEMP 4000K	49.9	PROVIDE DIMMING- BLACK FINISH
C	HALO COMMERCIAL	PR6FS12	RECESSED FIXTURE	120/277	LED	13.4	PROVIDE MOUNTING BARS FOR GRID CEILING
CE	HALO COMMERCIAL		RECESSED FIXTURE	120/277	LED	13.4	PROVIDE MOUNTING BARS FOR GRID CEILING
D	LUMINOSO	LLS SERIES LED LINEAR FIXTURE	4 FT LINEAR LED FIXTURE	120	LED 4000K	35	
DE	LUMINOSO	LLS SERIES LED LINEAR FIXTURE	EMERGENCY BATTERY BACK UP	120	LED 4000K	35	
E	LED TAPE LIGHT	COMMERCIAL ELECTRIC LED TAPE LIGHT		120	COOL WHITE	36	UNDER CABINET LIGHTS
F	EUREKA	SCOUT 4048-18	PENDANT POINT LIGHT	120	LED- NARROW BEAM ANGLE 24" COLOR TEMP 4000K	9.5	PROVIDE DIMMING- BLACK FINISH CLEAR GLASS LENS BLACK CABLE ROUND CANOPY MOUNTING BLACK FINISH
FE	EUREKA	3981 EA	EMERGENCY BATTERY BACK UP	120	LED- NARROW BEAM ANGLE 24" COLOR TEMP 4000K	9.5	PROVIDE DIMMING- BLACK FINISH CLEAR GLASS LENS BLACK CABLE ROUND CANOPY MOUNTING BLACK FINISH
EM	COOPER SURE-LITES	CU2-LED	LED EMERGENCY LIGHT UNIT; WHITE HOUSING, NICKEL CADMIUM BATTERY		LED INCLUDED	2.9	
EX1	LITHONIA	ECC-R-M6	COMBINATION EMERGENCY EXIT SIGN W/ (2) LED EMERGENCY FLOOD LAMPS	-	LED INCLUDED	1.6	EXTERIOR INSTALLATION: MOUNT 12" ABOVE TOP OF DOOR; INTERIOR INSTALLATION: MOUNT 12" FROM CEILING; BATTERY PACK SHALL BE UNSWITCHED
ER1	LITHONIA	ER-GY-T-WP-SQ	WEATHERPROOF TWIN REMOTE LED SQUARES. NOTE: VERIFY EXACT FINISH COLOR WITH ARCHITECT AND OWNER PRIOR TO ORDERING.	MVOLT	LED INCLUDED	2	BATTERY PACK SHALL BE UNSWITCHED
ADDITIONAL NOTES: 1. EMERGENCY BALLASTS IN EMERGENCY LIGHTS, NIGHT LIGHTS, AND EXIT SIGNS SHALL BE WIRED AHEAD OF ALL LOCAL CONTROL (NOT SWITCHED)THE EMERGENCY BALLASTS SHALL PROVIDE A MINIMUM OF NINETY (90) MINUTES OF CODE REQUIRED EMERGENCY LIGHTING. EACH EMERGENCY BALLAST PROVIDED SHALL PRODUCE THE MAXIMAM LUEN OUTPUT AVAILABLE FOR THE LAN USED. EMERGENCY LIGHTING BALLASTS SHALL BE BODINE OR APPROVED EQUAL. 2. REFER TO ARCHITECTURAL REFLECTED CEILING PLAN AND/OR ROOM FINISH SCHEDULE TO DETERMINE PROPER TYPE OF LIGHT FIXTURE REQUIRED FOR THE CEILING CONSTRUCTION PRIOR TO ORDERING THE FIXTURES & PROVIDE FIXTURES THAT ARE COMPATIBLE WITH THE CEILING SYSTEM. 3. ALL FIXTURES ARE TO BE ORDERED BY E.C. 4. EXIT SIGNS COMPLYING WITH 101:5-10 SHALL DEFINE EXITS AND ACCESS TO EXITS WHERE THE EXIT IS NOT IMMEDIATELY APPARENT. 5. ALL LIGHT FIXTURES OVER FOOD SERVICE AREA SHALL HAVE LENS COVERS OR SHATTER PROOF BULBS.							

KITCHEN EQUIPMENT SCHEDULE															
ITEM	QTY.	DESCRIPTION	MANFACTURER	MODEL NUMBER	VOLTAGE	PHASE	KW	HP	AMPS	PLUG	DIRECT	NEMA	HEIGHT	NOTES	
E-14	2	4' SANDWICH PREP TABLE	ATOSA	MSF8302GR	115	1	-	1/7	2.3	X	-	5-15P	-	-	
E-18	1	RACK OVEN	EMPIRE BAKERY EQUIPMENT	LFR-FOX-15E	220	3	-	-	88.4	-	X	-	-	-	
E-21	1	GAS FRYER	ASBER	AEF-50-S											
E-30	1	ESPRESSO MACHINE	EVERSYS SWISS ESPRESSO SYSTEMS	CAMEO C'2M	220/240	1	5.6	-	26	-	-	-	-	-	
E-31	1	GRAB & CO CASE	HOWARD-MCORAY	SC-0D40E-4L-B-LED	115	-	-	1/2	15	X	-	5-20P	-	-	
E-32	1	CREAM CHEESE CASE	OJEDA USA	RDCH 6	120	-	-	1/4	5	X	-	5-15P	-	-	
E-33	1	COFFEE MACHINE	BUNN	ICB TWIN SH PE	120/240	1	6	-	25	X	-	-	-	-	
E-34	2	TOASTER	CHEF-BUILT	CCT-500	120	-	1.8	3/4	-	X	-	5-15P	-	-	
E-35	1	COFFE GRINDER - DRIP	BUNN	G3 HD RED-22100.0001	120	1	1.32	-	11	X	-	5-15P		-	
E-37	1	ICE MAKER	ATOSA	YR280-AP-161	115	1	0.704	-	7.2	X	-	-	-	-	
E-38	2	POINT OF SALE DEVICE	TOAST	-	110	1	0.45	3/4	14.5	X	-	5-15P	-	PROVIDE CAT6 LAN CABLE	
E-39	2	SELF ORDERING STATION	TOAST	-	110	1	0.45	3/4	14.5	X	-	5-15P	-	TOUCH SCREEN - PROVIDE CAT6 LAN CABLE	
E-41	3	MENU DISPLAY	-	-	110	-	0.18	-	-	-	-	-	-	FLAT PANEL LED SCREEN - PROVIDE CAT6 LAN CABLE	
E-42	3	KDS	TOAST	-	110	-	0.18	-	-	-	-	-	-	TOUCH SCREEN - PROVIDE CAT6 LAN CABLE & WALL MOUNT	
E-43		NEON SIGN						-		X				REFER TO ELEVATION FOR LOCATION ON WALL	
E-44	1	BAGEL DISPLAY	ALL SOUTHERN FABRICATIONS	EBAGEL-DISP 45786											
E-53	1	4' CHEF BASE	ATOSA	MGF8450GR	115	1	2.3	1/7		X		5-15P			
E-54	1	36" GAS GRIDDLE	COOK RITCH	ATTG-48	-	-	-	-	-	-	-	-	-	-	
E-55	1	15" GAS COOKTOP	COOK RITCH	ACHP-2	-	-	-	-	-	-	-	-	-	-	
E-56		96" HOOD													
E-60	1	REACH - IN FREEZER	ATOSA	MBF8501GR	115	1	-	-	6.3	X	-	5-15P	-	-	
E-61	1	MEAT SLICER	GLOBE	G12	115	1	-	1/2	3	X	-	5-15P	-	-	
E-63	1	WIRELESS ROUTER			120		0.6								
E-64	1	MODEM			120		0.6								
E-67	3	TICKET PRINTER	ESPON		120		0.6							PROVIDE WALL MOUNT	
E-74	1	PROOFER	EMPIRE BAKERY EQUIPMENT	LRP-1	208/240	3	-	-	20	-	-	15-50P	-	-	
E-75	1	UNDER COUNTER REFRIGERATOR	ATOSA	MGF-RGR	115	1			2.3			5-15P			

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OR CONSTRUCTION.

NO DATE REMARKS
REVISIONS



BAGEL & CO
TALLAHASSEE

799 W. GAINES STREET
SUITE 121
TALLAHASSEE, FL 32310

PROJECT NO: 2023.0719
DATE: 04/12/2024

E5.0

ELECTRICAL SCHEDULES

CHECKED: SN DRAWN: MM

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Note 1. Feeder size: SEE RISER DIAGRAM

NOTE:

CONTRACTOR TO PROVIDE TYPED PANEL SCHEDULES FOR ALL PANELS SHOWING ALL CIRCUIT LABELS. PROVIDE ENGRAVED NAME PLATES FOR ALL PANELS AS WELL AS ENGRAVED NAME PLATES WITH SUITE NUMBER AND TENANT NAME ON METER SOCKET/DISCONNECT.

NOTE:

PER NEC 210.4(B) ALL MULTI WIRE BRANCH CIRCUITS ARE TO BE PROVIDED WITH A DEVICE THAT WILL DISCONNECT POWER TO ALL UNGROUNDED CONDUCTORS SIMULTANEOUSLY AT THE POINT OF ORIGIN.

PANELBOARD NOTES

1. PROVIDE GFI RATED BREAKER(S) FOR CIRCUIT
2. PROVIDE HACR RATED BREAKER(S) FOR CIRCUIT
3. PROVIDE SHUNT TRIP BREAKER(S) FOR CIRCUIT
4. PROVIDE ISOLATED GROUND CONDUCTOR(S) FOR CIRCUIT
5. VIA CONTACTOR
6. PROVIDE GROUND FAULT PROTECTION FOR EQUIPMENT.
7. PROVIDE LOCKABLE BREAKER

NO	DATE	REMARKS
1	02.02.24	BUILDING COMMENTS

REVISIONS

REVISIONS

BAGELS & CO.

BAGEL & CO
TALLAHASSEE

799 W. GAINES STREET
SUITE 121
TALLAHASSEE, FL 32310

PROJECT NO: 2023.0719
DATE: 04/12/2024

E6.0

ELECTRICAL PANEL SCHEDULES

CHECKED: SN DRAWN: MM



COMcheck Software Version COMcheckWeb
Interior Lighting Compliance Certificate

Project Information

Energy Code: 2018 IECC
Project Title: 2023.0719 - Bagels and Co. - Tallahassee, FL
Project Type: Alteration

Construction Site: 799 W. GAINES STREET SUITE 121, TALLAHASSEE, Florida 32310
Owner/Agent: MATTHEW AILEY, BAGEL & CO, 1200 S PINE ISLAND RD, PLANTATION, Florida 33324, MAILEY@GEMROCK.COM
Designer/Contractor: INTERPLAN, LLC, 220 E. CENTRAL PKWY, STE 4000, ALTAMONTE SPRINGS, Florida 32701, 407.645.5008

Allowed Interior Lighting Power

A Area Category	B Floor Area (ft ²)	C Allowed Watts / ft ²	D Allowed Watts
1-RESTROOM 1 (Common Space Types:Restrooms)	50	0.85	42
2-RESTROOM 2 (Common Space Types:Restrooms)	52	0.85	44
3-KITCHEN (Common Space Types:Food Preparation)	929	1.06	985
4-CUSTOMER AREA (Common Space Types:Dining Area - General)	478	0.63	301
5-CORRIDOR (Common Space Types:Corridor/Transition <8 ft wide)	108	0.66	71
Total Allowed Watts = 1444			

Proposed Interior Lighting Power

Fixture ID : Description / Lamp / Wattage Per Lamp / Ballast	B Lamps/ Fixture	C # of Fixture	D Fixture Watt.	E (C X D)
RESTROOM 1 (Common Space Types: Restrooms, 50 sq.ft.) LED: C: Other:	1	2	13	27
RESTROOM 2 (Common Space Types: Restrooms, 52 sq.ft.) LED: C: Other:	1	2	13	27
KITCHEN (Common Space Types: Food Preparation, 929 sq.ft.) LED: A2: 2x4: Other: LED: A: 2x2: Other: LED: D: Other: LED: C: Other:	1 1 1 1	6 2 9 11	67 67 35 13	402 134 315 147
CUSTOMER AREA (Common Space Types: Dining Area - General, 478 sq.ft.) LED: B: Other:	1	5	50	250
CORRIDOR (Common Space Types: Corridor/Transition <8 ft wide, 108 sq.ft.) LED: F: Other:	1	4	10	38
Total Proposed Watts = 1340				

Project Title: 2023.0719 - Bagels and Co. - Tallahassee, FL
Data filename: Report date: 02/19/24
Page 1 of 6

Interior Lighting PASSES

Interior Lighting Compliance

Statement

Compliance Statement: The proposed interior lighting alteration project represented in this document is consistent with the building plans, specifications, and other calculations submitted with this permit application. The proposed interior lighting systems have been designed to meet the 2018 IECC requirements in COMcheck Version COMcheckWeb and to comply with any applicable mandatory requirements listed in the Inspection Checklist.

Hajar Mahran

Name - Title

Hajar Mahran

Signature

02/19/2024

Date

Project Title: 2023.0719 - Bagels and Co. - Tallahassee, FL
Data filename: Report date: 02/19/24
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Section # & Req ID	Rough-In Electrical Inspection	Complies?	Comments/Assumptions
C405.2.2 [EL22]†	Spaces required to have light-reduction controls have a manual control that allows the occupant to reduce the connected lighting load in a reasonably uniform illumination pattern >= 50 percent.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	Requirement will be met.
C405.2.1 [EL18]†	Occupancy sensors installed in classrooms/lecture/training rooms, conference/meeting/multipurpose rooms, copy/print rooms, lounges/breakrooms, enclosed offices, open plan office areas, restrooms, storage rooms, locker rooms, warehouse storage areas, and other spaces <= 300 sqft that are enclosed by floor-to-ceiling height partitions. Reference section language C405.2.1.2 for control function in warehouses and section C405.2.1.3 for open plan office spaces.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	Requirement will be met.
C405.2.1 [EL19]†	Occupancy sensors control function in warehouses: In warehouses, the lighting in aislesways and open areas is controlled with occupant sensors that automatically reduce lighting power by 50% or more when the areas are unoccupied. The occupant sensors control lighting in each aisleway independently and do not control lighting beyond the aisleway being controlled by the sensor.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	Exception: Requirement does not apply.
C405.2.1 [EL20]†	Occupant sensor control function in open plan office areas: Occupant sensor controls in open office spaces >= 300 sq.ft. have controls 1) configured so that general lighting can be controlled separately in control zones with floor areas <= 600 sq.ft. within the space, 2) automatically turn off general lighting in all control zones within 20 minutes after all occupants have left the space, 3) are configured so that general lighting power in each control zone is reduced by >= 80% of the full zone general lighting power within 20 minutes of all occupants leaving that control zone, and 4) are configured such that any daylight responsive control will activate space general lighting or control zone general lighting only when occupancy for the same area is detected.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	Exception: Requirement does not apply.
C405.2.2 [EL21]†	Each area not served by occupancy sensors (per C405.2.1) have time-switch controls and functions detailed in sections C405.2.2.1 and C405.2.2.2.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	Requirement will be met.

Additional Comments/Assumptions:

1 | High Impact (Tier 1) | 2 | Medium Impact (Tier 2) | 3 | Low Impact (Tier 3)
Project Title: 2023.0719 - Bagels and Co. - Tallahassee, FL
Data filename: Report date: 02/19/24
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COMcheck Software Version COMcheckWeb

Inspection Checklist

Energy Code: 2018 IECC

Requirements: 100.0% were addressed directly in the COMcheck software

Text in the "Comments/Assumptions" column is provided by the user in the COMcheck Requirements screen. For each requirement, the user certifies that a code requirement will be met and how that is documented, or that an exception is being claimed. Where compliance is itemized in a separate table, a reference to that table is provided.

Section # & Req ID	Plan Review	Complies?	Comments/Assumptions
C103.2 [PR4]†	Plans, specifications, and/or calculations provide all information with which compliance can be determined for the interior lighting and electrical systems and equipment and document where exceptions to the standard are claimed. Information provided should include interior lighting power calculations, wattage of bulbs and ballasts, transformers and control devices.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	Requirement will be met.

Additional Comments/Assumptions:

1 | High Impact (Tier 1) | 2 | Medium Impact (Tier 2) | 3 | Low Impact (Tier 3)
Project Title: 2023.0719 - Bagels and Co. - Tallahassee, FL
Data filename: Report date: 02/19/24
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Section # & Req ID	Rough-In Electrical Inspection	Complies?	Comments/Assumptions
C405.2.2 [EL22]†	Spaces required to have light-reduction controls have a manual control that allows the occupant to reduce the connected lighting load in a reasonably uniform illumination pattern >= 50 percent.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	Requirement will be met.
C405.2.1 [EL18]†	Occupancy sensors installed in classrooms/lecture/training rooms, conference/meeting/multipurpose rooms, copy/print rooms, lounges/breakrooms, enclosed offices, open plan office areas, restrooms, storage rooms, locker rooms, warehouse storage areas, and other spaces <= 300 sqft that are enclosed by floor-to-ceiling height partitions. Reference section language C405.2.1.2 for control function in warehouses and section C405.2.1.3 for open plan office spaces.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	Requirement will be met.
C405.2.1 [EL19]†	Occupancy sensors control function in warehouses: In warehouses, the lighting in aislesways and open areas is controlled with occupant sensors that automatically reduce lighting power by 50% or more when the areas are unoccupied. The occupant sensors control lighting in each aisleway independently and do not control lighting beyond the aisleway being controlled by the sensor.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	Exception: Requirement does not apply.
C405.2.1 [EL20]†	Occupant sensor control function in open plan office areas: Occupant sensor controls in open office spaces >= 300 sq.ft. have controls 1) configured so that general lighting can be controlled separately in control zones with floor areas <= 600 sq.ft. within the space, 2) automatically turn off general lighting in all control zones within 20 minutes after all occupants have left the space, 3) are configured so that general lighting power in each control zone is reduced by >= 80% of the full zone general lighting power within 20 minutes of all occupants leaving that control zone, and 4) are configured such that any daylight responsive control will activate space general lighting or control zone general lighting only when occupancy for the same area is detected.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	Exception: Requirement does not apply.
C405.2.2 [EL21]†	Each area not served by occupancy sensors (per C405.2.1) have time-switch controls and functions detailed in sections C405.2.2.1 and C405.2.2.2.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	Requirement will be met.

1 | High Impact (Tier 1) | 2 | Medium Impact (Tier 2) | 3 | Low Impact (Tier 3)
Project Title: 2023.0719 - Bagels and Co. - Tallahassee, FL
Data filename: Report date: 02/19/24
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Section # & Req ID	Final Inspection	Complies?	Comments/Assumptions
C303.3 C408.2.5 [F117]†	Furnished O&M instructions for systems and equipment to the building owner or designated representative.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	Requirement will be met.
C405.4.1 [F118]†	Interior installed lamp and fixture lighting power is consistent with what is shown on the approved lighting plans, demonstrating proposed watts are less than or equal to allowed watts.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	See the Interior Lighting fixture schedule for values.
C408.1.1 [F157]†	Building operations and maintenance documents will be provided to the owner. Documents will cover manufacturers' information, specifications, programming procedures and means of illustrating to owner how building, equipment and systems are intended to be installed, maintained, and operated.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	Requirement will be met.
C408.2.5 [F116]†	Furnished as-built drawings for electric power systems within 90 days of system acceptance.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	Requirement will be met.
C408.3 [F133]†	Lighting systems have been tested to ensure proper calibration, adjustment, programming, and operation.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	Requirement will be met.

Additional Comments/Assumptions:

1 | High Impact (Tier 1) | 2 | Medium Impact (Tier 2) | 3 | Low Impact (Tier 3)
Project Title: 2023.0719 - Bagels and Co. - Tallahassee, FL
Data filename: Report date: 02/19/24
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TALLAHASSEE

799 W. GAINES STREET
SUITE 121
TALLAHASSEE, FL 32310

PROJECT NO: 2023.0719
DATE: 04/12/2024

E7.0

ENERGY CALCULATION -
LIGHTING

CHECKED: SN DRAWN: RZ

1.1.	PROVIDE ALL SUPPLIES, MATERIAL, LABOR, EQUIPMENT, AND APPURTENANCES NECESSARY FOR COMPLETE INSTALLATION AND FULL OPERATION OF ALL ELECTRICAL AND ELECTRICAL RELATED WORK, INDICATED HEREINAFTER ON DRAWINGS AND SPECIFICATIONS, FOR A SAFE AND FULLY OPERATIONAL SYSTEM.	1.1.1.	THE INSTALLED SYSTEM SHALL BE COMPLETE IN EVERY WAY AND FUNCTIONING ACCORDING TO THE DESIGN INTENT, WHETHER OR NOT ALL SUCH MATERIALS AND APPURTENANCES ARE SHOWN ON THE DRAWINGS OR DESCRIBED IN THE SPECIFICATIONS.	1.2.	PERFORM ALL OPERATIONS INCLUDING EXCAVATION & BACKFILLING, SHORING, CUTTING, CHAMFING & CHASING, DE-WATERING, ETC. NECESSARY FOR INSTALLATION OF FULLY OPERATIONAL SYSTEM, WHETHER OR NOT SHOWN ON THE DRAWINGS.	1.3.	DEFINITION OF TERMS FURNISH – SUPPLY AND DELIVER TO THE PROJECT SITE, READY FOR UNLOADING, UNPACKING, ASSEMBLY, INSTALLATION, AND SIMILAR OPERATIONS. INSTALL – OPERATIONS AT THE PROJECT SITE INCLUDING THE ACTUAL UNLOADING, UNPACKING, ASSEMBLY, ERECTION, PLACING, ANCHORING, APPLYING, WORKING TO DIMENSION, FINISHING, CURING, PROTECTING, CLEANING, AND SIMILAR OPERATIONS. PROVIDE – FURNISH AND INSTALL, COMPLETE AND READY FOR THE INTENDED USE. U.N.O. – UNLESS NOTED OTHERWISE. M.S.D.S. – MATERIAL SAFETY DATA SHEET CONTRACTOR – APPEARANCE ON DRAWINGS OR IN SPECIFICATIONS FOR ELECTRICAL WORK SHALL REFER TO ELECTRICAL, SUB-CONTRACTOR, RELOCATE – RELOCATE ELECTRICAL EQUIPMENT MAKE SAFE (INCLUDING LOCK OUT/OUT), STORE AND PROTECT DEVICE, RE-INSTALL, RENOVAR AND EXTEND CONDUIT & WIRE TO NEW LOCATION, RE-ENERGIZE AND TEST. EQUAL AND EQUIVALENT – TO MEAN OF THE SAME QUALITY, SIZE, NUMBER, VALUE, DEGREE, INTENSITY AND THE ITEMS ARE SIMILAR IN ALL RESPECTS.	1.3.1.	THE FINAL DECISION OF ACCEPTANCE OF THESE ITEMS WILL BE MADE BY THE ENGINEER.	1.3.2.	IT SHALL BE UNDERSTOOD THAT FOR ANY SPECIFIED ITEM ON THE DRAWINGS AND/OR IN THE SPECIFICATION, THIS TERM SHALL APPLY. ALL WORK SHALL BE PERFORMED UNDER THE PERSONAL SUPERVISION OF A PROJECT SUPERINTENDENT ON-SITE. MAINTAIN A COMPLETE SET OF DRAWINGS AND SPECIFICATIONS ON SITE AT ALL TIMES DURING THE PROJECT.	1.3.3.	ALL MATERIALS AND EQUIPMENT SHALL BE NEW AND SUBSTITUTES FOR THE CONDITIONS AND TYPES INDICATED ON THEM AFTER INSTALLATION.	1.3.4.	ALL MATERIALS, APPARATUS AND EQUIPMENT SHALL BEAR THE SEAL OF UNDERWRITERS LABORATORIES INC. (UL), OR A SIMILAR CREDIBLE TESTING AGENCY, LABEL WHERE REGULARLY SUPPLIED.	1.3.5.	CERTAIN MANUFACTURERS OF MATERIAL AND EQUIPMENT ARE SPECIFIED AND PLANS ARE DETAILED ACCORDING TO THIS MATERIAL. CONTRACTOR SHALL BASE BID ON FURNISHING AND INSTALLING THIS MAKE OF MATERIAL AND EQUIPMENT.	1.3.6.	WHERE MORE THAN ONE MAKE OF MATERIAL OR EQUIPMENT IS SPECIFIED, CONTRACTOR SHALL STATE IN BID WHICH MAKE THEY PROPOSE TO FURNISH. SHOP DRAWINGS SHALL BE SUBMITTED ON MATERIAL AND EQUIPMENT TO BE FURNISHED BY CONTRACTOR FOR ENGINEERS APPROVAL.	1.3.7.	THIS APPROVAL TO BE OBTAINED PRIOR TO SHIPMENT OF EQUIPMENT MATERIALS.	1.3.8.	COORDINATE CONNECTION OF SECONDARY ELECTRICAL SYSTEMS WITH EXTERIOR UNDERGROUND AND OVERHEAD UTILITIES AND SERVICES. COMPLY WITH REQUIREMENTS OF GOVERNING REGULATIONS, FRANCHISED SERVICE COMPANIES, AND CONTROLLING AGENCIES. PROVIDE REQUIRED CONNECTION FOR EACH SERVICE.	1.3.9.	CONTRACTOR SHALL VERIFY TRANSFORMER LOCATION AND METERING SCHEME WITH LOCAL AGENCIES.	1.3.10.	ALL MATERIALS SHALL BE FABRICATED AND INSTALLED IN A NEAT AND WORKMANLIKE MANNER.	1.3.11.	THE OWNER AND ENGINEER SHALL DETERMINE WHETHER WORKMANSHIP IS ACCEPTABLE. NO ALLOWANCES WILL BE MADE FOR WORK OR DELAY DUE TO POOR WORKMANSHIP, COORDINATION DIFFICULTIES, OR INTERFERENCES BETWEEN INVOLVED TRADES.	1.3.12.	PERFORM ALL WORK NECESSARY TO PREPARE THE STRUCTURE FOR THE INSTALLATION OF THE WORK. ALL HOLES, OPENINGS AND DAMAGED MATERIALS CREATED DURING CONSTRUCTION SHALL BE REPAIRED AND FINISHED BY EXPERIENCED WORKMEN.	1.3.13.	COORDINATE AND SCHEDULE THE WORK WITH THE OWNER TO MINIMIZE DISRUPTIONS TO THE NORMAL OPERATIONS AT THE BUILDING.	1.3.14.	AFTER HOURS WORK IS REQUIRED ON THIS PROJECT.	1.3.15.	INCLUDE IN THE CONTRACT PRICE THE COST OF AFTER-HOURS WORK AND EQUIPMENT TO MINIMIZE DELAY TO THE OWNER AND TO MAINTAIN FACILITY IN OPERATING CONDITION, COORDINATE WITH THE OWNER TO DETERMINE THE EXTENT OF THESE REQUIREMENTS PRIOR TO BID.	1.3.16.	RELATED WORK SPECIFIED ELSEWHERE:	1.3.17.	ALL DIVISION 1 REQUIREMENTS, AND ALL TERMS AND CONDITIONS OF CONTRACT.	1.3.18.	REFER TO MECHANICAL SPECIFICATION FOR MECHANICAL WORK TO BE DONE IN CONJUNCTION WITH THE ELECTRICAL WORK. CONTRACTOR IS RESPONSIBLE FOR ALL CONDUIT, WIRING, JUNCTION BOXES, ETC., REQUIRED FOR HVAC CONTROLS, UNLESS SPECIFICALLY NOTED OTHERWISE.	1.3.19.	ALL ELECTRICAL EQUIPMENT AND WIRING PROVIDED UNDER DIVISION 23 SHALL COMPLY WITH THE ELECTRICAL SYSTEM CHARACTERISTICS INDICATED ON THE ELECTRICAL DRAWINGS AND SPECIFICATION DIVISION 26.	1.3.20.	ELECTRIC CONTROLS, CONTACTORS, STARTERS, PILOT LIGHTS, PUSH BUTTONS, ETC., SHALL BE PROVIDED COMPLETE AS PART OF THE MOTOR, HEATER OR OTHER EQUIPMENT WHICH IT OPERATES. ALL ELECTRICAL COMPONENTS SHALL BE IN CONFORMANCE WITH THE REQUIREMENTS OF THE NATIONAL ELECTRICAL CODE AND DIVISION 26.	1.3.21.	WHERE EQUIPMENT SPECIFICATIONS INDICATE THAT A FACTORY-AUTHORIZED SERVICE ENGINEER OR TECHNICIAN SHALL OBTAIN INSTALLATION, TEST & ADJUST, OR START-UP OF EQUIPMENT, ETC., SUCH SERVICES WILL BE CONTRACTED BY OWNER AS PART OF THE EQUIPMENT PURCHASE.	1.3.22.	CONTRACTOR SHALL ARRANGE FOR, SCHEDULE, AND COORDINATE SUCH FIELD SERVICES AS WORK INCLUDED IN THE CONTRACT.	1.3.23.	CONTRACTOR SHALL FURNISH ALL LABOR AND MATERIALS NECESSARY TO SUPPORT ALL SUCH FACTORY REPRESENTATIVE'S FIELD SERVICES.	1.3.24.	REGULARLY DURING EACH WORKING DAY, REMOVE REFUSE AND DEBRIS ACCUMULATING FROM ELECTRICAL CONSTRUCTION AND LEAVE AREA CLEAN AT END OF THE WORK DAY.	1.3.25.	PRIOR TO ACCEPTANCE OF THIS WORK, LEAVE THE PREMISES "PROTECT CLEAN" INsofar AS AFFECTED BY ELECTRICAL WORK.	1.3.26.	CLEAN ALL LIGHT FIXTURES, LAMPS AND LENSES PRIOR TO FINAL ACCEPTANCE.	1.3.27.	CLEAN THE INTERIOR OF EACH ELECTRICAL COMPONENT OF DIRT AND CONSTRUCTION DUST INCLUDING BUT NOT LIMITED TO PANEL BOARDS, CONTROLLERS AND SWITCHES BEFORE ENERGIZING.	1.3.28.	EXPOSED FINISHED MATERIALS AND EQUIPMENT SHALL BE CAREFULLY CLEANED AND WIPED TO REMOVE GREASE, SMUDGES, FINGERPRINTS, DUST AND OTHER SPOTS AND LEFT SMOOTH AND CLEAN.	1.3.29.	CLEAN THE EXTERIOR OF ELECTRICAL COMPONENTS PRIOR TO ACCEPTANCE OF WORK.	1.3.30.	FOR ALL MATERIALS AND DEVICES REMOVED, THE CONTRACTOR SHALL DISPOSE OFF-SITE IN AN APPROVED MANNER. PROVIDE WRITTEN DOCUMENTATION FOR DISPOSAL OF ALL ITEMS.	1.3.31.	PROVIDE ALL LABOR, INSTRUMENTS, AND OTHER SERVICES REQUIRED FOR COMPLETE AND SATISFACTORY TEST AND ADJUSTMENT OF ELECTRICAL SYSTEMS AND RELATED WORK.	1.3.32.	1.11.1. CHECK ALL MOTORS AND ROTATING EQUIPMENT FOR PROPER ROTATION. 1.11.2. TEST ALL FEEDERS WITH MEGGER PRIOR TO ENERGIZING TO ASSURE CODE RESISTANCE IS MET, AND (WITHOUT "SHORTS" OR "OPEN CIRCUITS") CHECK ALL FUSES AND OVERLOADS FOR PROPER SIZING. VERIFY FUSE LABELS ARE VISIBLE. 1.11.3. CHECK ALL ELECTRICAL POWER AND CONTROL WIRING, INTERLOCKS, ETC., RELATED TO MECHANICAL EQUIPMENT TO DETERMINE THAT ALL WIRING IS CORRECT. 1.11.5. IMMEDIATELY DEMANDATE ALL EQUIPMENT PROVIDED UNDER THIS DIVISION THAT TESTS PROVE TO BE DEFECTIVE OR OPERATING IMPROPERLY AS A PART OF THIS CONTRACT.	1.3.33.	1.1.1. GUARANTEE ALL ELECTRICAL SYSTEM EQUIPMENT, MATERIALS, AND WORKMANSHIP TO BE FREE FROM DEFECTS FOR A PERIOD OF ONE (1) YEAR FROM THE DATE OF FINAL ACCEPTANCE AND PROPERLY CORRECT LATENT DEFECTS ARISING DURING THIS PERIOD UPON NOTIFICATION BY THE OWNER'S REPRESENTATIVE WITHOUT ADDITIONAL COMPENSATION AND TO THE SATISFACTION OF THE ENGINEER AND OWNER'S REPRESENTATIVE. 1.1.3. ALL EQUIPMENT, ETC., SHALL BE NEW UNLESS OTHERWISE NOTED, AND AS SPECIFIED FREE OF DEFECTS. ALL ELECTRICAL EQUIPMENT SHALL BE UL OR E.T.L. LISTED.	1.3.34.	2. CODES & PERMITS 2.1. ENTIRE INSTALLATION (INCLUDING EQUIPMENT, DEVICES, AND WIRING) SHALL BE IN ACCORDANCE WITH ALL APPLICABLE PROVISIONS OF THE NATIONAL ELECTRICAL CODE (N.E.C.), NATIONAL FIRE PROTECTION ASSOCIATION (NFPA 70 & NFPA 101), OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (O.S.H.A.), INTERNATIONAL BUILDING CODE (I.B.C.), INTERNATIONAL ENERGY CONSERVATION CODE (E.E.C.), AND ALL LAWS & ORDINANCES APPLICABLE TO WORK AT THIS SITE. IN ADDITION, INSTALLATION SHALL MEET APPROVAL OF LOCAL INSPECTION AUTHORITY HAVING JURISDICTION. REFER TO COVER SHEET FOR LIST OF CURRENT APPLICABLE CODE EDITIONS. 2.2. SECURE AND PAY ALL FEES ASSOCIATED WITH ALL PERMITS AND LICENSES REQUIRED FOR EXECUTION OF THE CONTRACT. ARRANGE FOR ALL INSPECTIONS REQUIRED BY CITY, COUNTY, STATE AND OTHER AUTHORITIES HAVING JURISDICTION, AND DELIVER CERTIFICATES OF APPROVAL TO THE ARCHITECT. A CERTIFICATE OF APPROVAL FOR WORK FROM INSPECTION AUTHORITY SHALL BE GIVEN TO THE OWNER BEFORE FINAL ACCEPTANCE WILL BE GIVEN BY OWNERS REPRESENTATIVE. 2.4. THE CODE REQUIREMENTS ARE STRICTLY A MINIMUM AND SHALL BE MET WITHOUT INCURRING ADDITIONS TO THE CONTRACT. WHERE REQUIREMENTS OF THE DRAWINGS OR SPECIFICATIONS EXCEED THE CODE REQUIREMENTS, THE WORK SHALL BE PROVIDED IN ACCORDANCE WITH THESE DRAWINGS OR SPECIFICATIONS, IN THE EVENT OF CONFLICT OR AMBIGUITY BETWEEN THE VARIOUS CODES, THE MOST STRINGENT REQUIREMENTS SHALL GOVERN.	1.3.35.	3. SAFETY 3.1. THE CONTRACTOR SHALL MAINTAIN A SAFE WORK ENVIRONMENT AT ALL TIMES. 3.1.1. COMPLY WITH ALL O.S.H.A., N.I.O.S.H., D.O.T., STATE & LOCAL REQUIREMENTS REGARDING SAFE HANDLING, STORING, TRANSPORTING, AND DISPOSING OF CHEMICALS. 3.1.2. MAINTAIN AND DISPLAY M.S.D.S. INFORMATION FOR ALL CHEMICAL PRODUCTS. 3.1.3. PROVIDE ALL NECESSARY MEANS TO MAINTAIN SAFE WORKING CONDITIONS, INCLUDING VENTILATION FANS, FIRE EXTINGUISHERS, EYE PROTECTION, RESPIRATORS, PROTECTIVE CLOTHING, VENTILATION, ETC. 3.1.4. ALL EQUIPMENT AND MATERIALS USED TO IMPLEMENT THE WORK SHALL BE USED IN STRICT ACCORDANCE WITH THE MANUFACTURER'S WRITTEN INSTRUCTIONS, INCLUDING ALL RECOMMENDED SAFETY PRECAUTIONS. MAINTAIN A PROPER FIRE WATCH FOR ALL OPERATIONS WHERE SPARKS, FLAMES, OR OTHER SOURCES OF FIRE ARE PRODUCED. 3.1.5. FOR ALL MATERIALS CONTAINING SOLVENTS, MAINTAIN THE RECOMMENDED VENTILATION OF THE AREA TO PREVENT THE ACCUMULATION OF VAPORS WHICH POSE A HEALTH OR FIRE HAZARD.	1.3.36.	4. INTENT OF DRAWINGS AND SPECIFICATIONS 4.1. THE IMPLIED AND STATED INTENT OF THE DRAWINGS & SPECIFICATIONS ARE TO ESTABLISH MINIMUM ACCEPTABLE STANDARDS FOR MATERIALS, EQUIPMENT, WORKMANSHIP, AND TO PROVIDE OPERABLE SYSTEMS THAT ARE COMPLETE IN EVERY RESPECT. 4.2. ENGINEERING DRAWINGS ARE DIAGRAMMATIC, INTENDED TO SHOW GENERAL ARRANGEMENT AND SIZES OF SYSTEM COMPONENTS, AND SHALL NOT BE SCALED. 4.3.1. ARCHITECTURAL AND STRUCTURAL DRAWINGS SHALL GOVERN SPACE CONSTRAINTS, DIMENSIONS AND FINISHES. 4.3.2. ALL OFFSETS AND FITTINGS THAT SHALL BE NECESSARY TO ACCOMPLISH A FINISHED INSTALLATION SHALL BE PROVIDED AT NO ADDITIONAL COST OR INCREASE THE CONTRACT. 4.3.3. WORK INTENDED, BUT HAVING MINOR DETAILS OBVIOUSLY OMITTED, SHALL BE PROVIDED COMPLETE AS A REQUIREMENT OF THIS CONTRACT. 4.3.4. LOCATIONS OF EQUIPMENT INDICATED ON PLANS SHALL BE FOLLOWED AS CLOSELY AS POSSIBLE TO THE PLANS, SUBJECT TO BUILDING CONSTRUCTION AND INTERFERENCES WITH OTHER TRADES. 4.3.5. MAINTAIN MINIMUM SERVICE CLEARANCE AS REQUIRED BY THE EQUIPMENT MANUFACTURER AND N.E.C. 4.4. PROVIDE THE OWNER A COMPLETE SET OF RECORD DRAWINGS AT THE END OF THE PROJECT. PROJECT WILL NOT BE COMPLETE UNTIL ACCURATE RECORD DRAWINGS ARE DELIVERED. 4.5. THE RECORD DRAWINGS SHALL BE MACHINE DRAFTED, AND SHALL BE PROVIDED ON MAGNETIC MEDIA AND CD FILES TO THE ARCHITECT WHICH REFLECT ALL CHANGES, DEVIATIONS AND REVISIONS MADE TO THE ORIGINAL DESIGN DOCUMENTS.<
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DIVISION 26 – ELECTRICAL SPECIFICATIONS CONTINUED

9.14.7. GALVANIZED STEEL CLAMPS, 1-1/4" X 3/16" STOCK; 3/8" CROSS BOLT; FLANGE WIDTH 2". HEXAGON NUTS FOR 1/2" ROD SIZE.	10.11.1. 208Y/120V SYSTEMS: PHASE A – BLACK PHASE B – RED PHASE C – BLUE NEUTRAL – WHITE GROUND – GREEN	11.19.2. OUTLET MOUNTING HEIGHTS INDICATED ON THE DRAWINGS AT OUTLETS TAKE PRECEDENCE. REFER TO DRAWINGS FOR DETAILS OF OTHER EQUIPMENT MOUNTING HEIGHTS. MOUNTING HEIGHTS FOR FLUSH OUTLETS IN BLOCK WALLS MAY BE CHANGED FOR INSTALLATION. CONSULT OWNER'S REPRESENTATIVE IN FIELD PRIOR TO ANY SUCH INSTALLATION.	13.5.9. REFER TO ARCHITECTURAL FINISH SCHEDULES AND OWNER REPRESENTATIVE FOR ADDITIONAL INFORMATION.	BE OF SIZE, NUMBER OF POLES AND FUSED OR NON-FUSED, AS REQUIRED FOR JOINT CONDITIONS AND THE NATIONAL ELECTRICAL CODE.	16.14. ALL LAMPS OF THE SAME TYPE SHALL BE OF THE SAME MANUFACTURER AND CATALOG NUMBER.
9.14.8. GALVANIZED STEEL LEAD EXPANSION ANCHORS, 1/2".			13.6. FLOOR SERVICE FITTINGS	15.11.3. PROVIDE SWITCHES WITH ELECTRICAL CHARACTERISTICS INDICATED:	16.15. LED (LIGHT EMITTING DIODE) LIGHT FIXTURES
9.15. COORDINATE INSTALLATION OF ROOF CURBS, EQUIPMENT SUPPORTS, AND ROOF PENETRATIONS.			13.6.1. TYPE – MODULAR, FLUSH-TYPE, FLAP-TYPE, OR ABOVE-FLOOR, DUAL-SERVICE UNITS SUITABLE FOR WIRING METHOD USED.	15.11.3.1. SWITCHES SHALL BE EQUIPPED WITH FUSE CONTACTS AND JAWS WHICH INSURE POSITIVE FUSE AND JAW CONTACT BY MEANS OF REINFORCING SPRING CLIPS OR OTHER APPROVED MEANS.	16.15.1. PROVIDE FACTORY INSTALLED LED MODULES THAT ARE SPECIFICALLY DESIGNED FOR, AND MATCHED AND MATED TO, THE RESPECTIVE LUMINAIRE IN WHICH THEY ARE USED.
9.16. STEEL SLOTTED SUPPORT SYSTEMS COMPLY WITH MFMA-4, FACTORY-FABRICATED COMPONENTS FOR FIELD ASSEMBLY.			13.6.2. COMPARTMENTS – BARRIER SEPARATES POWER FROM VOICE AND DATA COMMUNICATION CABLEING.	15.11.3.2. HINGES SHALL BE NON-CURRENT CARRYING.	16.15.1.1. PROVIDE LED MODULES THAT CAN EASILY BE REPLACED IN THE FIELD AND ARE EASILY ACCESSIBLE FOR REPLACEMENT.
9.16.1. CONSTRUCT WITH 9116" DIA. HOLES, NOMINAL 2" O.C. ON TOP SURFACE, WITH STANDARD FACTORY FINISH, AND WITH THE ALL NECESSARY FITTINGS WHICH MATCH AND MATCH WITH CHANNEL.	10.12. AWG #10 AND SMALLER SHALL BE SOLID. AWG #8 AND LARGER SHALL BE STRANDED.	12. ELECTRICAL SYSTEM GROUNDING	13.6.3. GROUNDING CONDUCTOR TO THE WATER PIPING SYSTEM WHICH SHALL BE SIZED IN ACCORDANCE WITH THE NATIONAL ELECTRICAL CODE.	15.11.3.3. SWITCHES SHALL BE SO DESIGNED THAT THEY CAN BE LOCKED IN EITHER OPEN OR CLOSED POSITION.	16.15.1.2. PROVIDE COLOR TEMPERATURE AS INDICATED IN LUMINAIRE SCHEDULE.
9.16.2. PROVIDE METALLIC CONTINGS THAT ARE HOT-DIP GALVANIZED AFTER FABRICATION AND APPLIED ACCORDING TO MFMA-4.	10.13. MAINTAIN A UNIFORM ELEVATION FOR ALL CABLE RUNS WHEREVER POSSIBLE. ALL CABLES SHALL BE SUPPORTED/ANCHORED AT MAXIMUM 4 FOOT INTERVALS AND WITHIN 12" OF BOX OR OUTLET AND SHALL NOT SAG.	12.1. MAIN SERVICE GROUNDING SYSTEM SHALL CONSIST OF THREE BRANCHES.	13.6.4. POWER RECEPTACLE – NEMA WD 6 CONFIGURATION 5-20R, GRAY FINISH, UNLESS OTHERWISE INDICATED.	15.11.3.4. SWITCHES SHALL HAVE REJECTION CLIP PROVISIONS SO THAT ONLY CLASS RK-1 CURRENT LIMITING FUSES CAN BE INSTALLED.	16.15.2. PROVIDE FACTORY INSTALLED DRIVER(S) FOR THE LED SOURCE UTILIZED THAT ARE SPECIFICALLY COORDINATED TO THE LED SOURCE AND LUMINAIRE IN WHICH THEY ARE USED.
9.16.3. PROVIDE CHANNEL DIMENSIONS THAT ARE SELECTED FOR APPLICABLE LOAD CRITERIA.	10.14. INSTALL CABLES IN A MANNER THAT PREVENTS OVERHEATING.	12.1.1. GROUNDING CONDUCTOR TO THE WATER PIPING SYSTEM WHICH SHALL BE SIZED IN ACCORDANCE WITH THE NATIONAL ELECTRICAL CODE.	13.6.5. VOICE AND DATA COMMUNICATION OUTLET – BLANK COVER WITH BUSHED CABLE OPENING OR TWO MODULAR, KEYS, COLOR-CODED, RJ-45 CATEGORY 5E JACKS FOR UTP CABLE.	15.11.3.5. ALL SAFETY SWITCHES SHALL BE QUICK-MAKE, QUICK-BREAK, AND HAVE INTERLOCKING COVER WITH HANDLE THAT MAY EITHER BE FRONT OR SIDE OPERATING.	16.15.2.1. PROVIDE DRIVER(S) HAVING SPECIFIC OPERATING CHARACTERISTICS DEFINED IN THE LUMINAIRE SCHEDULE.
9.16.4. COMPLY WITH NECA 1 AND NECA 101 UNLESS REQUIREMENTS IN THIS OR OTHER SPECIFICATION SECTIONS ARE STRICTER.	10.15. CABLES SHALL BE FASTENED DIRECTLY TO THE STRUCTURE USING FACTORY CLAMPS/CLIPS SPECIFICALLY DESIGNED FOR THE RESPECTIVE CABLE.	12.2. GROUNDING CONDUCTOR SHALL BE BONDED AT BOTH ENDS TO THE CONDUIT IN WHICH IT IS INSTALLED.	13.7. PROVIDE GROUNDING (NEUTRAL) CONDUCTORS IN ALL WALL SWITCH, DIMMER, AND OTHER LIGHTING CONTROL OUTLET BOXES, EVEN IF NOT IMMEDIATELY UTILIZED.	15.11.3.6. SWITCH BLADE SHALL BE VISIBLE IN OFF POSITION WITH THE DOOR OPEN.	16.15.2.2. PROVIDE DRIVER(S) THAT CAN EASILY BE REPLACED IN THE FIELD AND ARE EASILY ACCESSIBLE FOR REPLACEMENT.
9.17. FOR SUPPORTING RIGID METAL:	10.16. WIRE CONNECTIONS	12.3. MAIN SERVICE GROUND TO THE WATER PIPING SYSTEM SHALL BE CONNECTED ON THE STREET SIDE OF THE WATER METER, OR ON A COLD WATER PIPE AS NEAR AS PRACTICABLE TO THE WATER SERVICE ENTRANCE TO THE BUILDING.	13.8. ALL DEVICE WALL PLATES SHALL BE STANDARD SIZE: "MIDWAY", "OVERSIZED" ("JUMBO") OR "EXTRA DEEP" WALL PLATES SHALL NOT BE ACCEPTABLE.	15.11.3.7. EQUIP WITH OPERATING HANDLE THAT IS AN INTEGRAL PART OF ENCLOSURE BASE AND WHOSE OPERATING POSITION IS CLEARLY INDICATED AND IS PADLOCKABLE IN THE OFF POSITION.	16.15.2.3. PROVIDE SPECIFICATION SHEET FOR THE SPECIFIC DRIVER AS PART OF THE LUMINAIRE SUBMITTAL.
9.17.1. RISER CLAMPS: GALVANIZED STEEL, WITH 2 BOLTS AND NUTS, AND 4" EARS.	10.16.1. FOR EACH ELECTRICAL CONNECTION INDICATED, PROVIDE COMPLETE ASSEMBLY OF MATERIALS, INCLUDING BUT NOT LIMITED TO, PRESSURE CONNECTORS, TERMINALS (LUGS), ELECTRICAL INSULATING TAPE, HEAT-SHRINKABLE INSULATING TUBING, CABLE TIES, SOLDERLESS WIRE-NUTS, AND OTHER ITEMS AND ACCESSORIES AS NEEDED TO COMPLETE SPLICES AND TERMINATIONS OF TYPES INDICATED.	12.4. BONDING JUMPERS SHALL BE PROVIDED WHERE REQUIRED BY THE NATIONAL ELECTRICAL CODE.	13.9. SCREW HEADS COLORED TO MATCH FINISH OF PLATES.	15.11.3.8. CONSTRUCT CURRENT CARRYING PARTS OF HIGH-CONDUCTIVITY CONCRETE AND SILVER-TUNGSTEN TYPE SWITCH CONTACTS.	16.15.3. PROVIDE TOTAL HARMONIC DISTORTION (THD) RATING OF LESS THAN 20 PERCENT.
9.17.2. CLEVIS HANGERS: GALVANIZED STEEL WITH 1/2" DIA. HOLE FOR ROUND STEEL ROD.	10.16.2. ALL FEEDER AND SUB-FEEDER WIRING CONNECTIONS SHALL BE MADE WITH COMPRESSION CONNECTIONS BY SQUARE D OR ACCEPTABLE EQUIVALENT.	12.4.1. BOND ALL STRUCTURAL STEEL OF THE BUILDING TO THE MAIN SERVICE GROUND BUS.	13.10. GROUND, PHASE AND NEUTRAL CONDUCTORS SHALL BE PIG-TAILED IN OUTLET BOXES, OR MULTI-OUTLET ASSEMBLY FOR RECEPTABLES, SO THAT GROUND AND ELECTRICAL SERVICE WILL NOT BE DISTURBED DUE TO OTHER RECEPTABLES ON THE SAME MULTI-WIRE CIRCUIT IF RECEPTABLE IS REMOVED.	15.11.3.9. PROVIDE NEUMA TYPE 1 FOR INTERIOR AND NEUMA TYPE 3R FOR EXTERIOR AND PROVIDE TYPE INDICATED FOR SPECIAL CASES.	16.15.3.1. PROVIDE FACTORY-INSTALLED INTEGRAL FILTERING SYSTEM TO ENSURE THD DOES NOT EXCEED 20 PERCENT REGARDLESS OF QUANTITIES AND/OR MIXES WITH OTHER MANUFACTURED LED SYSTEMS
9.17.3. TWO-HOLE CONDUIT STRAPS: GALVANIZED STEEL; 3/4" STRAP WIDTH; AND 2-1/8" BETWEEN CENTER OF SCREW HOLES.	10.16.3. ALL BRANCH WIRING CONNECTIONS SHALL BE 3M SCOTCH LOCK CONNECTORS OR ACCEPTABLE EQUIVALENT.	12.4.2. BOND THE NATURAL GAS SERVICE TO THE GROUND ELECTRODE SYSTEM.	14. PANEL BOARDS	15.11.3.10. PROVIDE FUSEABLE SWITCHES WITH FUSES REQUIRED:	
9.17.4. OFFSET CONDUIT CLAMPS: GALVANIZED STEEL.	10.16.4. WHERE CABLE CONNECTIONS REQUIRE INSULATION, SCOTCH #33, ELECTRICAL TAPE SHALL BE USED FOR WRAPPING.	12.5. CONTRACTOR SHALL PROVIDE A GROUNDING SYSTEM CONSISTING OF DRIVEN GROUND RODS WITH INTERCONNECTING CABLES.	14.1. PROVIDE PANELS MATCHING VOLTAGES, PHASE AND WIRES AS NOTED ON THE PANEL SCHEDULES.	15.11.3.10.1. FUSES SHALL BE ULL LISTED CLASS RK-1 CIRCUIT LIMITING TYPE. PROVIDE BUSSMAN "LOW PEAK" OR APPROVED EQUAL.	16.16. INCANDESCENT LIGHT FIXTURES
9.18. SUPPORT FOR CONDUCTORS IN VERTICAL CONDUIT: FACTORY-FABRICATED ASSEMBLY CONSISTING OF THREADED BODY AND INSULATING WEDGING PLUG OR PLUGS FOR NON-ARMORED ELECTRICAL CONDUCTORS OR CABLES IN RISER CONDUITS. PLUGS SHALL HAVE NUMBER, SIZE, AND SHAPE OF CONDUCTOR GRIPPING PIECES AS REQUIRED TO SUE INDIVIDUAL CONDUCTORS OR CABLES SUPPORTED BODY SHALL BE MALLEABLE IRON.	10.16.5. NO LESS THAN 8" LONG AT THE IR OUTLET FITTINGS TO FACILITATE INSTALLATION OF DEVICES OR LUMINAIRES.	12.5.1. GROUND RODS SHALL BE INSTALLED WITH TWO FEET OF COVER AND CABLES EXTERMICALLY WELDED.	14.2. AS A BASIS OF DESIGN, USE SQUARE-TYPE "NUT", "NUT" UNLESS OTHERWISE NOTED ON THE PANEL SCHEDULE OR AS A NOTE ASSOCIATED WITH THE PANEL.	15.11.3.11. PROVIDE CONTROL INTERLOCKS WHEN INDICATED.	16.16.1. PROVIDE INCANDESCENT LAMPS IN THE SIZES AND RATED AS INDICATED AND 130 VOLT RATED.
9.19. MOUNTING TO WOOD:	10.16.6. MAKE TERMINATIONS SO THERE IS NO BARE CONDUCTOR AT THE TERMINAL.	12.5.2. GROUND RODS SHALL BE 3/4" DIAMETER BY 10 FEET LONG COPPER CLAD STEEL, ONE PIECE, COPPERWELD #9450, OR APPROVED EQUAL.	14.3. ACCEPTABLE MANUFACTURERS	15.12. MOTOR STARTERS	16.17. MOUNTING
9.19.1. FASTEN WITH LAG SCREWS OR THROUGH-BOLTS.	10.17. GROUND AND BONDING CONDUCTORS FOR ELECTRICAL SYSTEMS	12.5.3. GROUND GRID CONDUCTORS SHALL BE #1/0 BARE DIRECT BURIED.	14.3.1. SQUARE D CO. (BOLT ON BREAKERS)	15.12.1. STARTERS AND CONTROL DEVICES FOR MECHANICAL EQUIPMENT SHALL BE PROVIDED UNDER DIVISION 26 – ELECTRICAL WHERE SHOWN ON THE DRAWINGS.	16.17.1. ALL SURFACE AND RECESSED CEILING LUMINAIRES INSTALLED ON GRID OR TILE CEILINGS SHALL BE INSTALLED TO AGREE WITH MODULE OF CEILING EITHER DISPLACING A TILE, OR UNIT ON CENTER OF TILE, OR CENTERED ON GRID LINES.
9.19.2. PROVIDE STANDARD GRADE, LIGHT-FRAMING-SIZE LUMBER OF ANY SPECIES, NUMBER 3 COMMON OR STANDARD GRADE BOARDS COMPLYING WITH WCLB OR ANPA RULES, OR NUMBER 3 BOARDS COMPLYING WITH SPB RULES.	10.17.1. ALL METALLIC CONDUIT, SURFACE RACEWAYS, WIREWAYS, SUPPORTS, CABINET AND EQUIPMENT SHALL BE GROUNDING.	12.5.4. GROUND SYSTEM SHALL BE SO CONSTRUCTED THAT THE RESISTANCE BETWEEN THE EQUIPMENT AND THE GROUND SHALL NOT EXCEED 25 OHMS.	14.3.2. SIEMENS (BOLT ON BREAKERS)	15.12.2. UNLESS SPECIFIED OTHERWISE UNDER OTHER DIVISION 23 SPECIFICATIONS, MECHANICAL DRAWINGS, OR INDIVIDUAL EQUIPMENT SECTIONS, MOTOR STARTERS SHALL CONFORM TO THE FOLLOWING REQUIREMENTS:	16.17.2. RECESSED INCANDESCENT AND FLUORESCENT FIXTURES IN CEILINGS MAY NOT BE SUPPORTED FROM THE SUSPENDED CEILING CONSTRUCTION BOX AND FIXTURE SUPPORTS SHALL BE FASTENED SECURELY TO CONCRETE SLAB OR BAR JOIST EXCEPT AS NOTED. WHERE FIXTURES ARE SURFACE MOUNTED, NEAT HOLES SHALL BE CUT IN THE HUNG CEILINGS AS REQUIRED FOR THE FIXTURE SUPPORTS. ALL SUPPORT HANGERS, CHANNELS, BOLTS, ETC., SHALL BE GALVANIZED OR GALV-KROM.
9.19.3. LUMBER SHALL BE PRESERVATIVE TREATED IN ACCORDANCE WITH ANPB LP-2, AND KILN DRIED TO A MOISTURE CONTENT OF NOT MORE THAN 19 PERCENT.	10.17.2. PROVIDE GROUNDING ELECTRODE CONDUCTORS FOR SERVICE ENTRANCES AND DERIVED SYSTEMS		14.3.3. EATON/CUTLER HAMMER (BOLT ON BREAKERS)	15.12.2.1. ALL PANEL BOARDS INTERIORS SHALL BE FACTORY ASSEMBLED COMPLETE WITH CIRCUIT BREAKERS AS SCHEDULED ON THE DRAWINGS. ALL CIRCUIT BREAKERS SHALL BE QUICK-MAKE AND SHALL BE TRIP INDICATING.	16.17.3. PROVIDE ADEQUATE SUPPORTS FOR ALL FIXTURES SEPARATE FROM THE SUSPENDED CEILING SYSTEM. CONTRACTOR SHALL FURNISH AND INSTALL ALL NECESSARY ACCESSORIES, AS REQUIRED, TO SUPPORT THE FIXTURES. PROVIDE A MINIMUM OF TWO (2) GALVANIZED STEEL #12 GAUGE HANGER WIRES (ALTERNATE CORNERS) ON ALL RECESSED FIXTURES. PROVIDE LUMINAIRES AND/OR LUMINAIRE OUTLET BOXES WITH HANGERS TO PROPERLY SUPPORT LUMINAIRE WEIGHT. ALL LUMINAIRES INSTALLED IN OR ON SUSPENDED CEILING SYSTEMS SHALL BE ANCHORED DIRECTLY TO THE BUILDING STRUCTURE SYSTEM ABOVE. SUCH ANCHORING SHALL BE INDEPENDENT OF THE CEILING SUPPORT SYSTEM.
9.19.4. PROVIDE MARINE GRADE PRODUCTS WHERE SUBJECT TO MOISTURE CONDITIONS.	10.17.3. PROVIDE ALL FEEDERS AND BRANCH CIRCUITS WITH INSULATED (GREEN COVERING) EQUIPMENT GROUNDING.		14.3.4. PROVIDE DEAD-FRONT, SAFETY CONSTRUCTED, FACTORY-ASSEMBLED CIRCUIT BREAKER TYPE PANEL BOARDS IN SIZES AND RATINGS AS INDICATED.	15.12.2.2. EACH STARTER FOR A THREE-PHASE MOTOR SHALL BE FURNISHED WITH THREE (3) OVERLOAD RELAYS SIZED FOR THE FULL LOAD RUNNING CURRENT OF THE MOTOR ACTUALLY PROVIDED.	16.17.4. ALL LUMINAIRES SHALL BE INSTALLED PLUMB AND SUPPORT SURFACE MOUNTED LUMINAIRES GREATER THAN 2 FEET IN LENGTH AT A POINT IN ADDITION TO THE OUTLET BOX LUMINAIRE STUD.
9.19.5. PROVIDE SIMPSON STRONG TIE (OR EQUAL) EXPANSION SCREW ANCHORS.	10.17.4. DO NOT SHARE NEUTRALS WHEN AMONGST MULTIPLE BRANCH CIRCUITS OR WITH MULTI-WIRE BRANCH CIRCUITS.		14.4.1. SUCH RATINGS SHALL BE ESTABLISHED BY TEST CONDUCTED IN ACCORDANCE WITH UNDERWRITERS LABORATORIES STANDARD UL 67. THE USE OF CONDUCTOR DIMENSIONS WILL NOT BE ACCEPTED IN LIEU OF ACTUAL HEAT TEST.	15.12.2.3. PROVIDE AN EXTERNAL "HAND-OFF-AUTO" SELECTOR SWITCH WITH RED "RUNNING" LIGHT.	16.17.5. INSTALL FLUSH MOUNTED FIXTURES TO ELIMINATE LIGHT LEAKAGE BETWEEN FRAME AND FINISH SURFACE.
9.19.6. CUT, FIT, AND PLACE WOOD RODS, NAILERS, BLOCKING, AND ANCHORAGE ACCURATELY IN LOCATION, ALIGNMENT, AND ELEVATION TO SUPPORT AND ANCHOR ELECTRICAL MATERIALS AND EQUIPMENT.	10.17.5. UNLESS SPECIFICALLY INDICATED OTHERWISE ON DRAWINGS, PROVIDE GROUNDING (NEUTRAL) CONDUCTORS THAT ARE AT LEAST COMPARABLY-SIZED WITH CORRESPONDING PHASE CONDUCTORS FOR ALL APPLICATIONS.		14.4.2. EACH BOLT-ON PANEL BOARD SHALL BE PROVIDED WITH AN EQUIPMENT GROUND BUS BONDED TO THE PANEL BACK BOX. IN ADDITION, PROVIDE 200K NEUTRAL BUS AND ISOLATED GROUND BUS WHERE INDICATED.	15.12.2.4. EACH PILOT LIGHT SHALL HAVE A LEGEND PLATE INDICATING REASON FOR SIGNAL.	16.17.6. ALL RECESSED LUMINAIRES SHALL BE EQUIPPED WITH NECESSARY PLASTER FRAMES AND SURFACE TRIM.
9.19.7. MAKE TIGHT CONNECTIONS BETWEEN MEMBERS. INSTALL FASTENERS WITHOUT SPLITTING WOOD MEMBERS. ATTACH TO SUBSTRATES AS REQUIRED TO SUPPORT APPLIED LOADS.	10.18. NORMAL SYSTEM POWER FEEDERS AND BRANCH CIRCUITS SHALL BE INSTALLED IN SEPARATE RACEWAYS FROM EMERGENCY SYSTEM POWER. ALL WIRING FOR DIFFERENT POWER VOLTAGES SHALL BE INSTALLED IN RACEWAY SYSTEMS SEPARATE FROM EACH OTHER. ALL WIRING FOR THE VARIOUS ELECTRICAL SYSTEMS SHALL BE INSTALLED IN RACEWAY SYSTEMS SEPARATE FROM EACH OTHER.		14.4.3. EQUIPMENT ENCLOSURES SHALL BE PROVIDED WITH A BUS STRUCTURE AND MAIN LUGS OR MAIN BREAKER SHALL HAVE CURRENT RATINGS AS SHOWN ON THE PANEL BOARD SCHEDULE.	15.12.2.5. ALL MOTOR STARTERS, PUSH BUTTONS AND PILOT LIGHTS SHALL BE OF THE SAME MANUFACTURER AS THE SWITCHBOARD.	16.17.7. ALL JUNCTION BOXES AND SERVICEABLE COMPONENTS FOR RECESSED LUMINAIRES SHALL BE READILY ACCESSIBLE FOR SERVICE OR REPLACEMENT FROM BELOW THE CEILING, WITHOUT REMOVING ANY CEILING COMPONENTS (OTHER THAN TILES).
9.19.8. MAKE TIGHT CONNECTIONS BETWEEN MEMBERS. INSTALL FASTENERS WITHOUT SPLITTING WOOD MEMBERS. ATTACH TO SUBSTRATES AS REQUIRED TO SUPPORT APPLIED LOADS.			14.4.4. BUS STRUCTURE AND MAIN LUGS OR MAIN BREAKER SHALL HAVE CURRENT RATINGS AS SHOWN ON THE PANEL BOARD SCHEDULE.	15.12.3. EACH OVERLOAD RELAY SHALL HAVE A NORMAL OPEN ALARM CONTACT WHICH WILL CLOSE ONLY WHEN ACTUATED BY AN OVERLOAD (NOT TO BE CONFUSED WITH N.A. OR N.C. AUXILIARY CONTACTS). THESE CONTACTS SHALL BE PROPERLY WIRED TO THEIR RESPECTIVE BLUE PILOT LIGHT PROVIDED ON THE STARTER FRONT COVER AND HAVING A "TRIPPED" LEGEND PLATE.	16.17.8. STEM LENGTHS OF ALL PENDANT FIXTURES SHALL BE AS DIRECTED BY THE OWNER'S REPRESENTATIVE.
9.19.9. ATTACHMENTS TO WOOD STRUCTURAL MEMBERS: PROVIDE BOLTS INSTALLED THROUGH MEMBERS.			14.5. PANEL BOARD ENCLOSURE	15.12.3.1. PACKAGED AIR CONDITION EQUIPMENT	16.18. LIGHTING CONTROL DEVICES
9.20. MOUNTING TO NEW CONCRETE: PROVIDE CHANNEL-TYPE CONCRETE INSERTS AND BOLT TO INSERTS, OR PROVIDE EXPANSION ANCHORS FOR APPLICATIONS WHERE INSERTS ARE NOT PRACTICAL.	11. BOXES AND FITTINGS		14.5.1. PROVIDE GALVANIZED SHEET STEEL CABINET TYPE ENCLOSURES, IN SIZES AND NEUMA TYPES AS INDICATED, CODE GAUGE MINIMUM 16-GAUGE THICKNESS.	15.12.3.2. OTHER EQUIPMENT WHENEMETER IN OTHER SECTIONS TO BE PROVIDED WITH INTEGRAL STARTERS.	16.18.1. PROVIDE WATSTOPPER DW-100 WALL SWITCH (OR EQUIVALENT) AND CONFIGURE AS MANUAL ON, AUTO OFF (VACANCY SENSOR) UNLESS OTHERWISE SPECIFIED ON DRAWINGS. PROVIDE WITH TIME DELAY AS SPECIFIED ON DRAWINGS. IF NO TIME DELAY IS SPECIFIED, PROGRAM TO 10 MINUTES.
9.21. MOUNTING TO EXISTING CONCRETE: EXPANSION ANCHOR FASTENERS. INSTEAD OF EXPANSION ANCHORS, POWDER/CAS-ACTUATED DRIVEN THREADED STUDS PROVIDED WITH LOCK WASHERS AND NUTS MAY BE USED IN EXISTING STANDARD-HIGHT CONCRETE 4 INCHES THICK OR GREATER. DO NOT USE FOR ANCHORAGE TO LIGHTWIGHT-AGGREGATE CONCRETE OR FOR SLABS LESS THAN 4 INCHES THICK. DO NOT USE FOR WORK ANCHORED TO NEWLY INSTALLED CONCRETE. ONLY USE THIS METHOD WHERE OTHER METHODS CANNOT OR SHOULD NOT BE USED, AND ONLY AFTER RECEIVING CASE-BY-CASE PERMISSION FROM OWNER AND DESIGN PROFESSIONALS.	11.1. EXTENT OF ELECTRICAL BOX AND ASSOCIATED FITTING WORK IS GENERALLY INDICATED BY DRAWINGS AND SCHEDULES.		14.5.2. ENCLOSURES SHALL BE SUFFICIENT MOUNTED UNLESS OTHERWISE INDICATED.	15.14. EQUIPMENT CONNECTION COORDINATION	16.18.2. OCCUPANCY SENSORS, DUAL TECHNOLOGY CEILING SENSORS:
9.21.1. LUMBER SHALL BE PRESERVATIVE TREATED IN ACCORDANCE WITH ANPB LP-2, AND KILN DRIED TO A MOISTURE CONTENT OF NOT MORE THAN 19 PERCENT.	11.2. PROVIDE BOXES AND FITTINGS FOR ALL WORK.		14.5.3. PANEL BOARDS AND ENCLOSURE CABINETS SHALL CONFORM TO STANDARDS ESTABLISHED BY UNDERWRITERS' LABORATORIES, INC., AND REQUIREMENTS OF THE N.E.C.	15.14.1. COORDINATE EXACT LOCATION OF OUTLETS, EQUIPMENT CONNECTIONS, AND REQUIREMENTS PRIOR TO ROUGH-IN AND INSTALLATION OF DEVICES. AS DETERMINED BY THE ACTUAL EQUIPMENT AND FURNITURE LAY OUT. VERIFY WITH FIXTURE PLAN AND EQUIPMENT INSTALLER.	16.18.3. PROVIDE WATSTOPPER DT-300 CEILING MOUNTED OCCUPANCY SENSOR (OR EQUIVALENT) PROVIDE WITH TIME DELAY AS SPECIFIED ON DRAWINGS. IF NO TIME DELAY IS SPECIFIED, PROGRAM TO 20 MINUTES. ADJUST SENSITIVITY BASED ON FIELD CONDITIONS AND OCCUPANCY OF ROOM TO PROVIDE 100% COVERAGE WITHOUT NUISANCE TRIPPING.
9.21.2. PROVIDE MARINE GRADE PRODUCTS WHERE SUBJECT TO MOISTURE CONDITIONS.	11.3. PROVIDE WIRING DEVICES WHICH ARE U.L. LISTED AND LABELED.		14.5.4. CONSTRUCT WITH MULTIPLE KNOCKOUTS AND WIRING GUTTERS.	15.14.2. COORDINATE ELECTRICAL REQUIREMENTS OF EQUIPMENT NOT SHOWN ON DETAILS, I.e. ROOF-TOP UNITS, UNIT HEATS, FANS, ETC., AND EQUIPMENT/DEVICES REQUIRING AN ELECTRICAL CIRCUIT AND CONTROL RATING (SCCR) INDICATED.	16.18.4. PROVIDE WATSTOPPER DW-300 UNIVERSAL VOLTAGE PACK(S) AS REQUIRED TO PROPERLY POWER ALL OCCUPANCY SENSORS AND PROVIDE SWITCHING PER THE DESIGN INTENT.
9.21.3. PROVIDE SIMPSON STRONG TIE (OR EQUAL) EXPANSION SCREW ANCHORS.	11.4. DO NOT SHARE NEUTRALS WHEN AMONGST MULTIPLE BRANCH CIRCUITS OR WITH MULTI-WIRE BRANCH CIRCUITS.		14.5.5. PROVIDE FRONTS WITH ADJUSTABLE TRIM CLAMPS, AND DOORS WITH FLUSH LOCKS AND KEYS, ALL PANEL BOARD ENCLOSURES KEYS ALIKE WITH CONCEALED PLUMB DOOR HINGES.	15.15. UNLESS OTHERWISE NOTED OR SPECIFIED IN INDIVIDUAL SECTION, ALL 3-PHASE MOTORS SHALL BE STANDARD NEUMA CONTINUOUS DUTY "B" TYPE, WITH CLASS B INSULATION, OPEN DRIP-PROOF FRAME FOR INDOOR SERVICE, TERC FOR OUTDOOR SERVICES AND A SERVICE FACTOR OF 1.15. ALL MOTORS 5 HP AND LARGER SHALL BE U.S. MOTORS HI-EFFICIENCY MODEL OR RELIANCE XC HI-EFFICIENCY MODEL.	16.18.5. IN AREAS WHERE MULTIPLE OCCUPANCY SENSORS CONTROL A SINGLE ZONE TOGETHER, INTERLOCK OCCUPANCY SENSORS/POWER PACKS PER MANUFACTURER INSTRUCTIONS TO MEET CONTROL INTENT.
9.21.4. CUT, FIT, AND PLACE WOOD RODS, NAILERS, BLOCKING, AND ANCHORAGE ACCURATELY IN LOCATION, ALIGNMENT, AND ELEVATION TO SUPPORT AND ANCHOR ELECTRICAL MATERIALS AND EQUIPMENT.	11.5. UNLESS SPECIFICALLY INDICATED OTHERWISE ON DRAWINGS, PROVIDE GROUNDING (NEUTRAL) CONDUCTORS THAT ARE AT LEAST COMPARABLY-SIZED WITH CORRESPONDING PHASE CONDUCTORS FOR ALL APPLICATIONS.		14.5.6. EQUIP WITH INTERIOR CIRCUIT DIRECTORY FRAME, AND CARD WITH CLEAR PLASTIC COVERING.	15.16. MANUAL CONTROL SWITCHES SHALL BE FURNISHED AND/OR INSTALLED BY THE ELECTRICAL CONTRACTOR AS INDICATED.	16.19. OCCUPANCY ADJUSTMENTS: WHEN REQUESTED WITHIN 12 MONTHS OF DATE OF SUBSTANTIAL COMPLETION, PROVIDE ON-SITE ASSISTANCE IN ADJUSTING AVAILABLE LUMINAIRES TO SUIT ACTUAL OCCUPIED CONDITIONS. PROVIDE UP TO TWO VISITS TO PROJECT DURING OTHER-THAN-NORMAL OCCUPANCY HOURS FOR THIS PURPOSE. SOME OF THIS WORK MAY BE REQUIRED AFTER DARK. ADJUST AVAILABLE LUMINAIRES IN THE PRESENCE OF OWNER'S REPRESENTATIVE AND DESIGN PROFESSIONALS.
9.21.5. PROVIDE MARINE GRADE PRODUCTS WHERE SUBJECT TO MOISTURE CONDITIONS.	11.6. DO NOT SHARE NEUTRALS WHEN AMONGST MULTIPLE BRANCH CIRCUITS OR WITH MULTI-WIRE BRANCH CIRCUITS.		14.5.7. PROVIDE BAKED GRAY ENAMEL FINISH OVER A RUST INHIBITOR COATING.	15.17. VARIABLE FREQUENCY DRIVERS (VFD) SHALL BE FURNISHED BY THE MECHANICAL CONTRACTOR.	
9.21.6. MAKE TIGHT CONNECTIONS BETWEEN MEMBERS. INSTALL FASTENERS WITHOUT SPLITTING WOOD MEMBERS. ATTACH TO SUBSTRATES AS REQUIRED TO SUPPORT APPLIED LOADS.	11.7. MAKE TERMINATIONS SO THERE IS NO BARE CONDUCTOR AT THE TERMINAL.		14.5.8. PROVIDE ENCLOSURES FABRICATED BY SAME MANUFACTURER AS PANELBOARDS.	15.18. FURNISH AND INSTALL A DISCONNECT FOR EACH MOTOR (UNLESS PROVIDED INTEGRAL BY MANUFACTURER).	
9.21.7. MAKE TIGHT CONNECTIONS BETWEEN MEMBERS. INSTALL FASTENERS WITHOUT SPLITTING WOOD MEMBERS. ATTACH TO SUBSTRATES AS REQUIRED TO SUPPORT APPLIED LOADS.	11.8. NORMAL SYSTEM POWER FEEDERS AND BRANCH CIRCUITS SHALL BE INSTALLED IN SEPARATE RACEWAYS FROM EMERGENCY SYSTEM POWER. ALL WIRING FOR DIFFERENT POWER VOLTAGES SHALL BE INSTALLED IN RACEWAY SYSTEMS SEPARATE FROM EACH OTHER. ALL WIRING FOR THE VARIOUS ELECTRICAL SYSTEMS SHALL BE INSTALLED IN RACEWAY SYSTEMS SEPARATE FROM EACH OTHER.		14.5.9. MINIMUM DEPTH OF 5-3/4" & MINIMUM WIDTH OF 20" FOR PANELS.	15.19. DISCONNECTS SHALL BE FUSED OR UNFUSED SAFETY SWITCHES AS REQUIRED. UPON COMPLETION OF MOTOR INSTALLATION WORK, CONTRACTOR SHALL ENSURE PHASES ARE CORRECTLY CONNECTED BY CHECKING ROTATION OF MOTOR.	
9.21.8. MAKE TIGHT CONNECTIONS BETWEEN MEMBERS. INSTALL FASTENERS WITHOUT SPLITTING WOOD MEMBERS. ATTACH TO SUBSTRATES AS REQUIRED TO SUPPORT APPLIED LOADS.	11.9. EXTENT OF ELECTRICAL BOX AND ASSOCIATED FITTING WORK IS GENERALLY INDICATED BY DRAWINGS AND SCHEDULES.		14.6. BALANCE AMPERES OF CIRCUITS AFTER ACTIVE LOADS ARE ENERGIZED TO WITHIN 10% MAXIMUM AND PROVIDE TYPEWRITTEN PANEL BOARD SCHEDULE WITH CORRECTED LOADS.	15.2. PROVIDE PANEL BOARD WITH CIRCUIT BREAKERS HAVING THE SHORT CIRCUIT RATING (SCCR) INDICATED.	
9.21.9. MAKE TIGHT CONNECTIONS BETWEEN MEMBERS. INSTALL FASTENERS WITHOUT SPLITTING WOOD MEMBERS. ATTACH TO SUBSTRATES AS REQUIRED TO SUPPORT APPLIED LOADS.	11.10. GANG TYPE OUTLET BOXES SHALL NOT BE USED.		14.7. PROVIDE PANEL BOARD WITH CIRCUIT BREAKERS HAVING THE SHORT CIRCUIT RATING (SCCR) INDICATED.	15.1. PROVIDE ALL POWER WIRING AND CONNECTIONS FROM SOURCE TO STARTER, STARTER TO DISCONNECT, AND DISCONNECT TO MOTOR OR DEVICE, EXCEPT WHERE SUCH WIRING IS PROVIDED BY EQUIPMENT MANUFACTURER.	
9.21.10. MAKE TIGHT CONNECTIONS BETWEEN MEMBERS. INSTALL FASTENERS WITHOUT SPLITTING WOOD MEMBERS. ATTACH TO SUBSTRATES AS REQUIRED TO SUPPORT APPLIED LOADS.	11.11. THE OUTLET BOX LOCATIONS INDICATED ON DRAWINGS SHALL BE CONSIDERED APPROXIMATE, AND THEREFORE, IT SHALL BE INCUMBENT UPON CONTRACTOR TO STUDY THE GENERAL CONSTRUCTION WITH RELATION TO SPACES AND EQUIPMENT SURROUNDING EACH OUTLET.		14.8. BALANCE AMPERES OF CIRCUITS AFTER ACTIVE LOADS ARE ENERGIZED TO WITHIN 10% MAXIMUM AND PROVIDE TYPEWRITTEN PANEL BOARD SCHEDULE WITH CORRECTED LOADS.	15.2. ALL AUTOMATIC TEMPERATURE CONTROL WIRING SHALL BE FURNISHED AND INSTALLED UNDER DIVISION 23 – MECHANICAL, UNLESS INDICATED OR SPECIFIED OTHERWISE. HOWEVER, ELECTRICAL CONTRACTOR SHALL PROVIDE AND INSTALL ALL STARTERS AND MAKE ALL POWER CONNECTIONS.	
9.21.11. MAKE TIGHT CONNECTIONS BETWEEN MEMBERS. INSTALL FASTENERS WITHOUT SPLITTING WOOD MEMBERS. ATTACH TO SUBSTRATES AS REQUIRED TO SUPPORT APPLIED LOADS.	11.12. ALL OUTLET, SWITCH, AND JUNCTION BOXES SHALL BE MADE OF CODE COMPLIANT #12 GAUGE STEEL, COMPLETE WITH RINGS AND REMOVABLE FRONT FASTENED SCREW COVER PLATES WITH COUNTER SINK HEAD SCREWS.		14.9. EQUIP WITH INTERIOR CIRCUIT DIRECTORY FRAME, AND CARD WITH CLEAR PLASTIC COVERING.	15.3. MANUAL CONTROL SWITCHES SHALL BE FURNISHED AND/OR INSTALLED BY THE ELECTRICAL CONTRACTOR AS INDICATED.	
9.21.12. MAKE TIGHT CONNECTIONS BETWEEN MEMBERS. INSTALL FASTENERS WITHOUT SPLITTING WOOD MEMBERS. ATTACH TO SUBSTRATES AS REQUIRED TO SUPPORT APPLIED LOADS.	11.13. OUTLET BOXES		14.10. SEE PANEL SCHEDULE NOTES FOR SPECIFIED ALTERNATE SPECIAL CONDITIONS.	15.4. VARIABLE FREQUENCY DRIVERS (VFD) SHALL BE FURNISHED BY THE MECHANICAL CONTRACTOR.	
9.21.13. MAKE TIGHT CONNECTIONS BETWEEN MEMBERS. INSTALL FASTENERS WITHOUT SPLITTING WOOD MEMBERS. ATTACH TO SUBSTRATES AS REQUIRED TO SUPPORT APPLIED LOADS.	11.13.1. WHERE CONDUIT IS CONCEALED, BOXES SHALL NOT BE LESS THAN 4" SQUARE, X 1-1/2" DEEP.		14.11. PROVIDE CIRCUIT BREAKERS HAVING THE AC RATING INDICATED AND IF "THE PANEL BOARD IS "SCREWS RATED", MANUFACTURER SHALL PROVIDE LABELING REQUIRED BY N.E.C. RELATIVE TO THIS RATING-AND SHALL SUBMIT DATA WITH THE PANEL BOARD AND CIRCUIT BREAKER SUBMITTAL.	15.5. FURNISH AND INSTALL A DISCONNECT FOR EACH MOTOR (UNLESS PROVIDED INTEGRAL BY MANUFACTURER).	
9.21.14. MAKE TIGHT CONNECTIONS BETWEEN MEMBERS. INSTALL FASTENERS WITHOUT SPLITTING WOOD MEMBERS. ATTACH TO SUBSTRATES AS REQUIRED TO SUPPORT APPLIED LOADS.	11.13.2. CEILING: 4" SQUARE, 2-1/8" DEEP FOR EXPOSED OR FLURRED WORK, 3" DEEP FOR BOXES POURED IN CONCRETE.		14.12. COMPLIANT WITH UL 1917	15.6. DISCONNECTS SHALL BE FUSED OR UNFUSED SAFETY SWITCHES AS REQUIRED. UPON COMPLETION OF MOTOR INSTALLATION WORK, CONTRACTOR SHALL ENSURE PHASES ARE CORRECTLY CONNECTED BY CHECKING ROTATION OF MOTOR.	
9.21.15. MAKE TIGHT CONNECTIONS BETWEEN MEMBERS. INSTALL FASTENERS WITHOUT SPLITTING WOOD MEMBERS. ATTACH TO SUBSTRATES AS REQUIRED TO SUPPORT APPLIED LOADS.	11.13.2.1. PROVIDE CONCRETE FOUR BOXES OF THE TYPE SPECIALLY DESIGNED FOR THE APPLICATION.		14.13. CONTINUOUSLY ADJUSTABLE SLIDER, TOGGLE SWITCH, OR ROTARY KNOB, 1/5 A OR 1.5 A.	15.7. DISCONNECTS SHALL BE FUSED OR UNFUSED SAFETY SWITCHES AS REQUIRED. UPON COMPLETION OF MOTOR INSTALLATION WORK, CONTRACTOR SHALL ENSURE PHASES ARE CORRECTLY CONNECTED BY CHECKING ROTATION OF MOTOR.	
9.21.16. MAKE TIGHT CONNECTIONS BETWEEN MEMBERS. INSTALL FASTENERS WITHOUT SPLITTING WOOD MEMBERS. ATTACH TO SUBSTRATES AS REQUIRED TO SUPPORT APPLIED LOADS.	11.13.2.2. PROVIDE PLASTER RINGS OR EXTENSION RINGS WHERE REQUIRED.		14.14. CONTINUOUSLY ADJUSTABLE SLIDER, TOGGLE SWITCH, OR ROTARY KNOB, 1/5 A OR 1.5 A.	15.8. DRAWINGS INDICATE MINIMUM MOTOR HORSEPOWER RATINGS.	
9.21.17. MAKE TIGHT CONNECTIONS BETWEEN MEMBERS. INSTALL FASTENERS WITHOUT SPLITTING WOOD MEMBERS. ATTACH TO SUBSTRATES AS REQUIRED TO SUPPORT APPLIED LOADS.	11.13.2.3. PROVIDE DEEPER BOXES WHERE REQUIRED.		14.15. CONTINUOUSLY ADJUSTABLE SLIDER, TOGGLE SWITCH, OR ROTARY KNOB, 1/5 A OR 1.5 A.	15.9. MOTOR CHARACTERISTICS, SUCH AS INSULATION CLASS, SPEED/TORQUE CURVE, BEARING DESIGN LIFE, SHAFT MATERIAL, BODY MATERIAL, ETC., SHALL BE SELECTED BY THE EQUIPMENT MANUFACTURER BASED UPON THE INTENDED SERVICE OF THE EQUIPMENT.	
9.21.18. MAKE TIGHT CONNECTIONS BETWEEN MEMBERS. INSTALL FASTENERS WITHOUT SPLITTING WOOD MEMBERS. ATTACH TO SUBSTRATES AS REQUIRED TO SUPPORT APPLIED LOADS.	11.13.3. WALL 4" SQUARE, 2-1/8" DEEP BOXES.		14.16. COMPLIANT WITH UL 1917	15.10. PROVIDE INHERENT THERMAL PROTECTION FOR ALL FRACTIONAL HORSEPOWER MOTORS.	
9.21.19. MAKE TIGHT CONNECTIONS BETWEEN MEMBERS. INSTALL FASTENERS WITHOUT SPLITTING WOOD MEMBERS. ATTACH TO SUBSTRATES AS REQUIRED TO SUPPORT APPLIED LOADS.	11.13.3.1. PROVIDE EXTENSION RINGS OR COVERS OF SUFFICIENT DEPTH TO BRING COVERS FLUSH WITH THE FINISHED SURFACE.		14.17. CONTINUOUSLY ADJUSTABLE SLIDER, TOGGLE SWITCH, OR ROTARY KNOB, 1/5 A OR 1.5 A.	15.11. DISCONNECT SWITCHES AND SAFETY SWITCHES	
9.21.20. MAKE TIGHT CONNECTIONS BETWEEN MEMBERS. INSTALL FASTENERS WITHOUT SPLITTING WOOD MEMBERS. ATTACH TO SUBSTRATES AS REQUIRED TO SUPPORT APPLIED LOADS.	11.13.4. MASONRY: FOR FLUSH MOUNTED BOXES IN EXPOSED MASONRY OR TILE.		14.18. CONTINUOUSLY ADJUSTABLE SLIDER, TOGGLE SWITCH, OR ROTARY KNOB, 1/5 A OR 1.5 A.	15.11.1. SQUARE D COMPANY	
9.21.21. MAKE TIGHT CONNECTIONS BETWEEN MEMBERS. INSTALL FASTENERS WITHOUT SPLITTING WOOD MEMBERS. ATTACH TO SUBSTRATES AS REQUIRED TO SUPPORT APPLIED LOADS.	11.13.4.1. PROVIDE COVERS WITH SQUARE CORNERS ON THE BASED PORTION AND WITH SUFFICIENT DEPTH TO TRIM OUT FLUSH WITH FINISHED SURFACE.		14.19. CONTINUOUSLY ADJUSTABLE SLIDER, TOGGLE SWITCH, OR ROTARY KNOB, 1/5 A OR 1.5 A.	15.11.2. EATON-CUTLER HAMMER	
9.21.22. MAKE TIGHT CONNECTIONS BETWEEN MEMBERS. INSTALL FASTENERS WITHOUT SPLITTING WOOD MEMBERS. ATTACH TO SUBSTRATES AS REQUIRED TO SUPPORT APPLIED LOADS.	11.13.5. OUTLET BOXES SHALL BE 4" OCTAGONAL FOR LIGHTING FIXTURES WITH APPROVED PLASTER RINGS AND PLATES.		14.20. COMPLIANT WITH UL 1917	15.11.3. SIEMENS	
9.21.23. MAKE TIGHT CONNECTIONS BETWEEN MEMBERS. INSTALL FASTENERS WITHOUT SPLITTING WOOD MEMBERS. ATTACH TO SUBSTRATES AS REQUIRED TO SUPPORT APPLIED LOADS.	11.14. EXPOSED AND WET OR DAMP AREA AT EQUIPMENT, SUCH AS EXTERIOR, ROOFS OR KITCHENS WHEN PRESENT, PROVIDE IPX5 OR IPX6 BOXES WITH SUITABLE WEATHERPROOF COVERS.		14.21. MODULAR, 120-V, FULL-WAVE, SOLID-STATE UNITS WITH INTEGRAL, QUIET ON-OFF SWITCHES, WITH ADJUSTABLE FREQUENCY AND EM/RFI SUPPRESSION FILTERS.	15.12. FURNISH AND INSTALL SAFETY SWITCHES WHERE INDICATED AND AS REQUIRED FOR MOTOR OUTLETS OR OTHER EQUIPMENT. SWITCHES SHALL	
9.21.24. MAKE TIGHT CONNECTIONS BETWEEN MEMBERS. INSTALL FASTENERS WITHOUT SPLITTING WOOD MEMBERS. ATTACH TO SUBSTRATES AS REQUIRED TO SUPPORT APPLIED LOADS.	11.15. PROVIDE COVERS OF "IN USE" TYPE COVERS WHERE CORDS/PLUGS ARE INSTALLED AND MUST REMAIN AS "IN USE".		14.22. CONTINUOUSLY ADJUSTABLE SLIDER, TOGGLE SWITCH, OR ROTARY KNOB, 1/5 A OR 1.5 A.	15.13. DISCONNECT SWITCHES AND SAFETY SWITCHES	
9.22. MOUNTING TO MASONRY: APPROVED TOGGLE-TYPE BOLTS ON HOLLOW MASONRY UNITS AND EXPANSION ANCHOR FASTENERS ON SOLID MASONRY UNITS.	11.16. PULL AND JUNCTION BOXES		14.23. OCCUPANCY SENSORS	15.14. VARIABLE FREQUENCY DRIVERS (VFD) SHALL BE FURNISHED BY THE MECHANICAL CONTRACTOR.	
9.23. MOUNTING TO STEEL: WELDED THREADED STUDS COMPLYING WITH AWS D11/D1 TM, WITH LOCK WASHERS AND NUTS, OR BEAM CLAMPS (MSS TYPE 19, 21, 23, 25, OR 27) COMPLYING WITH MSS SP-69, CLAMPED TO FLANGES OF BEAMS OR IN UPON TRUSS CHORDS OF BAR JOISTS.	11.16.1. PROVIDE BOXES WHERE REQUIRED TO FACILITATE THE PULLING OF WIRES OR CABLES.		14.24.1. PASSIVE-INFRARED TYPE, 120/277 V, ADJUSTABLE TIME DELAY UP TO 30 MINUTES, 180-DEGREE FIELD OF VIEW, WITH A MINIMUM COVERAGE AREA OF 900 SQ. FT.	15.5. FURNISH AND INSTALL A DISCONNECT FOR EACH MOTOR (UNLESS PROVIDED INTEGRAL BY MANUFACTURER).	
9.24. MOUNTING TO CONCRETE: EXPANSION ANCHOR FASTENERS. INSTEAD OF EXPANSION ANCHORS, POWDER/CAS-ACTUATED DRIVEN THREADED STUDS PROVIDED WITH LOCK WASHERS AND NUTS MAY BE USED IN EXISTING STANDARD-HIGHT CONCRETE 4 INCHES THICK OR GREATER. DO NOT USE FOR ANCHORAGE TO LIGHTWIGHT-AGGREGATE CONCRETE OR FOR SLABS LESS THAN 4 INCHES THICK. DO NOT USE FOR WORK ANCHORED TO NEWLY INSTALLED CONCRETE. ONLY USE THIS METHOD WHERE OTHER METHODS CANNOT OR SHOULD NOT BE USED, AND ONLY AFTER RECEIVING CASE-BY-CASE PERMISSION FROM OWNER AND DESIGN PROFESSIONALS.	11.16.2. BOXES SHALL BE IN ACCORDANCE WITH ARTICLE 370 OF N.E.C.		14.24.2. ADAPTIVE-TECHNOLOGY TYPE, 120/277 V, ADJUSTABLE TIME DELAY UP TO 20 MINUTES, 180-DEGREE FIELD OF VIEW, WITH A MINIMUM COVERAGE AREA OF 900 SQ. FT.	15.6. DISCONNECTS SHALL BE FUSED OR UNFUSED SAFETY SWITCHES AS REQUIRED. UPON COMPLETION OF MOTOR INSTALLATION WORK, CONTRACTOR SHALL ENSURE PHASES ARE CORRECTLY CONNECTED BY CHECKING ROTATION OF MOTOR.	
9.25. MOUNTING TO LIFT STEEL: SHEET METAL SCREWS.	11.17. ACCESSORIES		14.25.1. TELEPHONE OUTLET – SINGLE RJ-45 JACK FOR TERMINATING 100-OHM, BALANCED, FOUR-PAIR UTP; 11A/DIA-568-B-1 COMPLYING WITH CATEGORY 5E. COMPLY WITH UL 1663.	15.7. DISCONNECTS SHALL BE FUSED OR UNFUSED SAFETY SWITCHES AS REQUIRED. UPON COMPLETION OF MOTOR INSTALLATION WORK, CONTRACTOR SHALL ENSURE PHASES ARE CORRECTLY CONNECTED BY CHECKING ROTATION OF MOTOR.	
9.26. ITEMS MOUNTED ON HOLLOW WALLS AND NONSTRUCTURAL BUILDING SURFACES: MOUNT CABINETS, PANELBOARDS, DISCONNECT SWITCHES, CONTROL ENCLOSURES, PULL AND JUNCTION BOXES, TRANSFORMERS, AND OTHER DEVICES ON SLOTTED-CHANNEL RACKS ATTACHED TO SUBSTRATE.	11.17.1. PROVIDE CORROSION-RESISTANT KNOCKOUT CLOSURES, CONDUIT LOCKOUTS AND WALLBOARD NON CONDUIT BUSHINGS, OFFSET CONNECTORS, OF TYPES AND SIZES, TO SUIT RESPECTIVE INSTALLATION REQUIREMENTS AND APPLICATIONS.		14.25.2. COMBINATION TIE AND TELEPHONE OUTLET – SINGLE RJ-45 JACK FOR 100-OHM, BALANCED, FOUR-PAIR UTP; 11A/DIA-568-B-1; COMPLYING WITH CATEGORY 5E, AND ONE TYPE F COAXIAL CABLE CONNECTOR.	15.8. DRAWINGS INDICATE MINIMUM MOTOR HORSEPOWER RATINGS.	
9.27. FABRICATED METAL EQUIPMENT SUPPORT ASSEMBLIES: WELDED OR BOLTED, STRUCTURAL-STEEL SHAPES, SHOP OR FIELD FAB					